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# Contents

# 1 SETS

## MCQs with One Correct Answer

**1.** If A and B are non-empty sets such that $A \supset B$, then
(a) $B' - A' = A - B$ (b) $B' - A' = B - A$
(c) $A' - B' = A - B$ (d) $A' \cap B' = B - A$

**2.** Let $A = \{\theta : \sin(\theta) = \tan(\theta)\}$ and $B = \{\theta : \cos(\theta) = 1\}$ be two sets. Then :
(a) $A = B$
(b) $A \not\subset B$
(c) $B \not\subset A$
(d) $A \subset B$ and $B - A \neq \phi$

**3.** A set S contains 3 elements, the number of subsets of which of following sets is 256.
(a) S (b) P(S)
(c) P(P(S)) (d) None of these

**4.** In a college of 300 students every student reads 5 newspapers and every newspaper is read by 60 students. The number of newpapers is
(a) at least 30 (b) at most 20
(c) exactly 25 (d) None of these

**5.** Let A and B be two sets then
$(A \cup B)' \cup (A' \cap B)$ is equal to
(a) $A'$ (b) A
(c) $B'$ (d) None of these

**6.** The set $(A \setminus B) \cup (B \setminus A)$ is equal to
(a) $[A \setminus (A \cap B)] \cap [B \setminus (A \cap B)]$
(b) $(A \cup B) \setminus (A \cap B)$
(c) $A \setminus (A \cap B)$
(d) $\overline{A \cap B} \setminus A \cup B$

**7.** Let $A = \{x : x \in \mathbf{R}, |x| < 1\}$
$B = \{x : x \in \mathbf{R}, |x - 1| \geq 1\}$ and
$A \cup B = \mathbf{R} - D$, then the set D is :
(a) $\{x : 1 < x \leq 2\}$ (b) $\{x : 1 \leq x < 2\}$
(c) $\{x : -2 \leq x \leq 2\}$ (d) None of these

**8.** If $aN = \{ax : x \in \mathbf{N}\}$ and $bN \cap cN = dN$, where b, $c \in \mathbf{N}$ are relatively prime, then
(a) $d = bc$ (b) $c = bd$
(c) $b = cd$ (d) None of these

**9.** The set $(A \cup B \cup C) \cup (A \cap B' \cap C')' \cap C'$ is equal to
(a) $B \cap C'$ (b) $A \cap C$
(c) $B' \cap C'$ (d) None of these

**10.** A dinner party is to be fixed for a group of 100 persons. In this party, 50 persons do not prefer fish, 60 prefer chicken and 10 do not prefer either chicken or fish. The number of persons who prefer both fish and chicken is
(a) 20 (b) 22
(c) 25 (d) None of these

**11.** Let U be the universal set and $A \cup B \cup C = U$. Then $\{(A - B) \cup (B - C) \cup (C - A)\}'$ is equal to:
(a) $A \cup B \cup C$ (b) $A \cup (B \cap C)$
(c) $A \cap B \cap C$ (d) $A \cap (B \cup C)$

**12.** If $n(A) = 1000$, $n(B) = 500$ and if $n(A \cap B) \geq 1$ and $n(A \cup B) = p$, then
(a) $500 \leq p \leq 1000$ (b) $1001 \leq p \leq 1498$
(c) $1000 \leq p \leq 1498$ (d) $1000 \leq p \leq 1499$

**13.** In a battle 70% of the combatants lost one eye, 80% an ear, 75% an arm, 85% a leg, $x$ % lost all the four limbs. The minimum value of $x$ is
(a) 10 (b) 12
(c) 15 (d) None of these

**14.** The number of students who take both the subjects mathematics and chemistry is 30. This represents 10% of the enrolment in mathematics and 12% of the enrolment in chemistry. How many students take at least one of these two subjects?

(a) 520 (b) 490
(c) 560 (d) 480

**15.** At a certain conference of 100 people, there are 29 Indian women and 23 Indian men. Of these Indian people 4 are doctors and 24 are either men or doctors. There are no foreign doctors. How many foreigners and women doctors are attending the conference?

(a) 48, 1 (b) 34, 3
(c) 46, 4 (d) 42, 2

**16.** Each student in a class of 40, studies at least one of the subjects English, Mathematics and Economics. 16 study English, 22 Economics and 26 Mathematics, 5 study English and Economics, 14 Mathematics and Economics and 2 study all the three subjects. The number of students who study English and Mathematics but not Economics is

(a) 7 (b) 5
(c) 10 (d) 4

**17.** In a market research project, 20% opted for 'Nirma' detergent whereas 60% opted for 'Surf blue' detergent. The remaining individuals were not certain. If the difference between those who opted for 'Surf blue' and those who were uncertain was 720, how many respondents were covered in the survey

(a) 1100 (b) 1150
(c) 1800 (d) None of these

**18.** In a school 80 students like chocolate, 40 like coffee if the number of students doesn't like any of them is equal to the number of students who like both of them then what is the total number of students in the school?

(a) 115 (b) 90
(c) 120 (d) None of these

**19.** In a school there are 100 students 60 of them don't like Chocolate and 50 don't like Biscuit and 10 of them like none then how many of them like both?

(a) 20 (b) 30
(c) 40 (d) None of these

**20.** A survey was conducted of 100 people whether they have read recent issues of 'Golmal', a monthly magazine. Summarized information is presented below:

Only September: 18
September but not August: 23
September and July: 8
September: 28
July: 48
July and August: 10
None of the three months: 24

What is the number of surveyed people who have read exactly for two consecutive months?

(a) 7 (b) 9
(c) 12 (d) 14

## Numeric Value Answer

**21.** An investigator interviewed 100 students to determine their preferences for the three drinks : milk ($M$), coffee ($C$) and tea ($T$). He reported the following : 10 students had all the three drinks $M$, $C$ and $T$; 20 had $M$ and $C$; 30 had $C$ and $T$; 25 had $M$ and $T$; 12 had $M$ only; 5 had $C$ only; and 8 had $T$ only. If number of students who did not take any of the three drinks is n, then $\frac{n}{5}$ is

**22.** In a class of 80 students numbered 1 to 80, all odd numbered students opt of Cricket, students whose numbers are divisible by 5 opt for Football and those whose numbers are divisible by 7 opt for Hockey. If the number of students who do not opt any of the three games is n, then $\frac{n}{4}$ is equal to

**23.** A survey shows that 61%, 46% and 29% of the people watched "3 idiots", "Rajneeti" and "Avatar" respectively. 25% people watched exactly two of the three movies and 3% watched none. What percentage of people watched all the three movies?

**24.** If n(A) = 4 and n(B) = 7, then the difference between maximum and minimum value of $n(A \cup B)$ is

**25.** Two finite sets have $m$ and $n$ elements. The number of subsets of the first set is 112 more than that of the second set. The value of $m - n$ is

**ANSWER KEY**

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (a) | **4** | (c) | **7** | (b) | **10** | (a) | **13** | (a) | **16** | (b) | **19** | (d) | **22** | (7) | **25** | (3) |
| **2** | (b) | **5** | (a) | **8** | (a) | **11** | (c) | **14** | (a) | **17** | (c) | **20** | (b) | **23** | (7) | | |
| **3** | (c) | **6** | (b) | **9** | (a) | **12** | (d) | **15** | (a) | **18** | (c) | **21** | (4) | **24** | (4) | | |

# 2

# RELATIONS & FUNCTIONS-1

## MCQs with One Correct Answer

1. If $f(x) \cdot f(y) = f(x) + f(y) + f(xy) - 2 \ \forall \ x, y \in R$ and if $f(x)$ is not a constant function, then the value of $f(a)$ is
(a) 1 (b) 2 (c) 0 (d) – 1

2. The domain of the function
$f(x) = \sqrt{x^{14} - x^{11} + x^6 - x^3 + x^2 + 1}$ is
(a) $(-\infty, \infty)$ (b) $[0, \infty)$
(c) $(-\infty, 0]$ (d) $R - [0,1]$

3. If $f(x) = \cos(\log x)$ then
$f(x)f(y) - \frac{1}{2}\left\{f\left(\frac{x}{y}\right) + f(xy)\right\}$ is equal to :
(a) 0 (b) 1
(c) –1 (d) None of these

4. The domain of the function
$f(x) = \sqrt{10 - \sqrt{x^4 - 21x^2}}$ is
(a) $[5, \infty)$
(b) $\left[-\sqrt{21}, \sqrt{21}\right]$
(c) $\left[-5, -\sqrt{21}\right] \cup \left[\sqrt{21}, 5\right] \cup \{0\}$
(d) $(-\infty, -5)$

5. The function f satisfies the functional equation
$3f(x) + 2f\left(\frac{x+59}{x-1}\right) = 10x + 30$ for all real $x \neq 1$.
The value of $f(7)$ is
(a) 8 (b) 4 (c) –8 (d) 11

6. If $af(x+1) + bf\left(\frac{1}{x+1}\right) = x$, $x \neq -1$, $a \neq b$, then $f(2)$ is equal to
(a) $\frac{2a+b}{2(a^2-b^2)}$ (b) $\frac{a}{a^2-b^2}$
(c) $\frac{a+2b}{a^2-b^2}$ (d) None of these

7. The domain of $f(x) = \sqrt{\cos(\sin x)} + \sqrt{\log_x \{x\}}$ ; $\{.\}$ denote the fractional part, is
(a) $[1, \pi)$ (b) $(0, 2\pi) - [1, \pi)$
(c) $\left(0, \frac{\pi}{2}\right) - \{1\}$ (d) $(0, 1)$

8. Let $f(x) = \frac{1}{2} - \tan\left(\frac{\pi x}{2}\right), -1 < x < 1$ & $g(x) = \sqrt{3 + 4x - 4x^2}$, then dom $(f + g)$ is given by
(a) $\left[\frac{1}{2}, 1\right]$ (b) $\left[\frac{1}{2}, -1\right)$
(c) $\left[-\frac{1}{2}, 1\right)$ (d) $\left[-\frac{1}{2}, -1\right]$

9. The domain of function
$f(x) = \log_4[\log_5\{\log_3(18x - x^2 - 77)\}]$ is:
(a) (7, 11) (b) (8, 10)
(c) (8, 11) (d) (7, 10)

10. Let $f(x) = a^x (a > 0)$ be written as $f(x) = g(x) + h(x)$, where $g(x)$ is an even function and $h(x)$ is an odd function. Then the value of $g(x + y) + g(x - y)$ is
(a) $2g(x)\,g(y)$
(b) $2g(x + y)\,g(x - y)$
(c) $2g(x)$
(d) None of these

**11.** Let $f$ be a real valued function such that for any real $x$, $f(\lambda+x)=f(\lambda-x)$ and $f(2\lambda+x)=-f(2\lambda-x)$ for some $\lambda>0$. Then
(a) $f$ is even and non-periodic
(b) $f$ is odd and periodic
(c) $f$ is odd and non-periodic
(d) $f$ is even and periodic

**12.** Let $f(x)=([a]^2-5[a]+4)x^3-(6\{a\}^2-5\{a\}+1)x-(\tan x)\,\text{sgn}\,x$, be an even function for all $x\in R$, then sum of all possible values of '$a$' is (where [ ] and { } denote greatest integer function and fractional part functions respectively)
(a) $\frac{17}{6}$ (b) $\frac{53}{6}$ (c) $\frac{31}{3}$ (d) $\frac{35}{3}$

**13.** The set of all integer values of $n$ for which the function $f(x)=\cos nx\,.\,\sin\frac{5x}{n}$ is periodic with period $2\pi$ is equal to
(a) $\{1,5,10\}$ (b) $\{1,5\}$
(c) $\{\pm1,\pm5\}$ (d) None of these

**14.** If $f:\mathbb{R}\to\mathbb{R}$ & $g:\mathbb{R}\to\mathbb{R}$ be two given functions, then $2\min\{f(x)-g(x),0\}$ equals
(a) $f(x)+g(x)-|g(x)-f(x)|$
(b) $f(x)+g(x)+|g(x)-f(x)|$
(c) $f(x)-g(x)+|g(x)-f(x)|$
(d) $f(x)-g(x)-|g(x)-f(x)|$

**15.** The domain of f(x) is (0, 1), therefore the domain of $y=f(e^x)+f(\ln|x|)$ is :
(a) $\left(\frac{1}{e},1\right)$ (b) $(-e,-1)$
(c) $\left(-1,-\frac{1}{e}\right)$ (d) $(-e,-1)\cup(1,e)$

**16.** Suppose that $f$ is a periodic function with period $\frac{1}{2}$ and that $f(2)=5$ and $f\left(\frac{9}{4}\right)=2$ then $f(-3)-f\left(\frac{1}{4}\right)$ has the value equal to
(a) 2 (b) 3 (c) 5 (d) 7

**17.** Let $f(x)=\begin{bmatrix}2x+3\ ; & x\le1\\ a^2x+1; & x>1\end{bmatrix}$. If the range of f(x) = R (set of real numbers) then number of integral value(s), which $a$ may take is
(a) 2 (b) 3 (c) 4 (d) 5

**18.** Let $f(x)=\sin x-\cos x$ and $g(x)=\log\sqrt{5x}$; then the range of $g(\sqrt{2}f(x)+3)$ is
(a) [0, 1] (b) [0, 2]
(c) $\left[0,\frac{3}{2}\right]$ (d) None of these

**19.** Let $f(x)=1+\frac{1}{4\sqrt{x}}$ and $g(x,y)=\log y$, then the domain of $g\left(\frac{1}{2},-g(2,f(x))-1\right)$ is
(a) $0<x<1$ (b) $0<x\le1$
(c) $x\ge1$ (d) Null set

**20.** Let $f(x)$ be defined as
$$f(x)=n\begin{cases}|x| & 0\le x<1\\ |x-1|+|x-2| & 1\le x<2\\ |x-3| & 2\le x<3\end{cases}$$
The range of function $g(x)=\sin(7(f(x)))$ is :
(a) [0, 1] (b) [–1, 0]
(c) $\left[-\frac{1}{2},\frac{1}{2}\right]$ (d) [–1, 1]

## Numeric Value Answer

**21.** If $\phi(x)=\frac{1}{1+e^{-x}}$ and $S=\phi(5)+\phi(4)+\phi(3)+....+\phi(-3)+\phi(-4)+\phi(-5)$, then the value of S is

**22.** If the period of $f(x)$ satisfying the condition:
$f(x+p)=1+\{1-3f(x)+3f^2(x)-f^3(x)\}^{1/3}$ is $\lambda p$, then evaluate $\lambda$.

**23.** If f(x) is an odd function, f(1) = 3, and f(x + 2) = f(x) + f(2), then the value of f(3) is

**24.** Let $f(x, y)$ be a function satisfying the functional equation: $f(x,y)=f(2x+2y,2y-2x)$ for all real numbers $x, y$. Define $g(x)$ by $g(x)=f(2^x,0)$. Also given that $g(x)$ is a periodic function with period $k$, then find value of $\frac{k}{2}$.

**25.** Number of elements in the range set of $f(x)=\left[\frac{x}{15}\right]\left[-\frac{15}{x}\right]\ \forall\ x\in(0,90)$; (where [.] denotes greatest integer function)

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | (b) | 4 | (c) | 7 | (d) | 10 | (a) | 13 | (c) | 16 | (b) | 19 | (d) | 22 | (2) | 25 | (6) |
| 2 | (a) | 5 | (b) | 8 | (c) | 11 | (b) | 14 | (d) | 17 | (c) | 20 | (d) | 23 | (9) | | |
| 3 | (a) | 6 | (a) | 9 | (b) | 12 | (d) | 15 | (b) | 18 | (b) | 21 | (5.5 | 24 | (6) | | |

# 3

# TRIGONOMETRIC FUNCTIONS

## MCQs with One Correct Answer

**1.** If $\sin^{16}\alpha = \frac{1}{5}$, then the value of

$$\frac{1}{\cos^2\alpha}+\frac{1}{1+\sin^2\alpha}+\frac{2}{1+\sin^4\alpha}+\frac{4}{1+\sin^8\alpha}$$

is equal to

(a) 2 (b) 4 (c) 6 (d) 10

**2.** If a, b, g, d are the smallest positive angles in ascending order of magnitude which have their sines equal to the positive quantity $k$, then the value of $4\sin\frac{\alpha}{2}+3\sin\frac{\beta}{2}+2\sin\frac{\gamma}{2}+\sin\frac{\delta}{2}$ is equal to

(a) $2\sqrt{1-k}$ (b) $2\sqrt{1+k}$

(c) $2\sqrt{k}$ (d) none of these

**3.** If $\cos\alpha = \frac{2\cos\beta-1}{2-\cos\beta}$ $(0<\alpha<\beta<\pi)$, then

$\frac{\tan\alpha/2}{\tan\beta/2}$ is equal to

(a) 1 (b) $\sqrt{2}$ (c) $\sqrt{3}$ (d) $\frac{1}{\sqrt{3}}$

**4.** Let $0 \le \alpha, \beta, \gamma, \delta \le \pi$ whrere $\beta$ and $\gamma$ are not complementary such that
$2\cos\alpha + 6\cos\beta + 7\cos\gamma + 9\cos\delta = 0$ and $2\sin\alpha - 6\sin\beta + 7\sin\gamma - 9\sin\delta = 0$

If $\frac{\cos(\alpha+\delta)}{\cos(\beta+\gamma)} = \frac{m}{n}$ where m and n are relatively prime positive numbers, then the value of $(m + n)$ is equal to:

(a) 11 (b) 10 (c) 9 (d) 7

**5.** If $\sin\theta + \sin 2\theta + \sin 3\theta = \sin\alpha$ and $\cos\theta + \cos 2\theta + \cos 3\theta = \cos\alpha$, then $\theta$ is equal to

(a) $\alpha/2$ (b) $\alpha$ (c) $2\alpha$ (c) $\alpha/6$

**6.** The value of the expression

$$\frac{\sin 7\alpha + 6\sin 5\alpha + 17\sin 3\alpha + 12\sin\alpha}{\sin 6\alpha + 5\sin 4\alpha + 12\sin 2\alpha}$$ where

$\alpha = \frac{\pi}{5}$ is equal to:

(a) $\frac{\sqrt{5}-1}{4}$ (b) $\frac{\sqrt{5}+1}{4}$

(c) $\frac{\sqrt{5}+1}{2}$ (d) $\frac{\sqrt{5}-1}{2}$

**7.** $\frac{\sin\theta}{\cos(3\theta)}+\frac{\sin(3\theta)}{\cos(9\theta)}+\frac{\sin(9\theta)}{\cos(27\theta)}+\frac{\sin(27\theta)}{\cos(81\theta)} =$

(a) $\frac{\sin(81\theta)}{2\cos(80\theta)\cos\theta}$ (b) $\frac{\sin(80\theta)}{2\cos(81\theta)\cos\theta}$

(c) $\frac{\sin(81\theta)}{\cos(80\theta)\cos\theta}$ (d) $\frac{\sin(80\theta)}{\cos(81\theta)\cos\theta}$

**8.** If $\operatorname{cosec}\theta = \frac{p+q}{p-q}$, then $\cot\left(\frac{\pi}{4}+\frac{\theta}{2}\right) =$

(a) $\sqrt{\frac{p}{q}}$ (b) $\sqrt{\frac{q}{p}}$ (c) $\sqrt{pq}$ (d) pq

**9.** If $\sin 2\theta + \sin 2\phi = 1/2$ and $\cos 2\theta + \cos 2\phi = 3/2$, then $\cos^2(\theta - \phi) =$

(a) 3/8 (b) 5/8 (c) 3/4 (d) 5/4

**10.** If $u = (1 + \cos\theta)(1 + \cos 2\theta) - \sin\theta \cdot \sin 2\theta$, $v = \sin\theta(1 + \cos 2\theta) + \sin 2\theta(1 + \cos\theta)$, then $u^2 + v^2 =$

(a) $4(1 + \cos\theta)(1 + \cos 2\theta)$

(b) $4(1 + \sin\theta)(1 + \sin 2\theta)$

(c) $4(1 - \cos\theta)(1 - \cos 2\theta)$

(d) $4(1 - \sin\theta)(1 - \sin 2\theta)$

**11.** The product

$$\cos\left\{\frac{2\pi}{2^{64}-1}\right\}\cos\left\{\frac{2^2\pi}{2^{64}-1}\right\}\ldots\cos\left\{\frac{2^{64}\pi}{2^{64}-1}\right\}$$

(a) $\frac{1}{16^{16}}$ (b) $\frac{1}{8^8}$ (c) $\frac{1}{32^{32}}$ (d) $\frac{1}{64^{64}}$

**12.** Let x, y $\in$ R satisfy the condition such that $\sin x \sin y + 3\cos y + 4\sin y \cos x = \sqrt{26}$. The value of $\tan^2 x + \cot^2 y$ is equal to

(a) $9 \times 17$ (b) 205

(c) $\frac{1}{16}+\frac{9}{17}$ (d) None of these

**13.** The product

$$\left(\cos\frac{x}{2}\right)\cdot\left(\cos\frac{x}{4}\right)\cdot\left(\cos\frac{x}{8}\right)\ldots\ldots\cdot\left(\cos\frac{x}{256}\right)$$

is equal to:

(a) $\frac{\sin x}{128\sin\frac{x}{256}}$ (b) $\frac{\sin x}{256\sin\frac{x}{256}}$

(c) $\frac{\sin x}{128\sin\frac{x}{128}}$ (d) $\frac{\sin x}{512\sin\frac{x}{512}}$

**14.** The maximum value of $\log_{20}(3\sin x - 4\cos x + 15)$ is equal to:

(a) 1 (b) 2 (c) 3 (d) 4

**15.** The value of

$$\frac{\cos 25°}{\sin 70° \sin 85°}+\frac{\cos 70°}{\sin 25° \sin 85°}+\frac{\cos 85°}{\sin 25° \sin 70°}$$

is

(a) 1/2 (b) 1 (c) 2 (d) 3/2

**16.** The number of solutions of the equation

$3^{2\sec^2 x} + 1 = 10.3^{\tan^2 x}$ in the interval $[0, 2\pi]$ is

(a) 8 (b) 6 (c) 4 (d) 2

**17.** Number of solutions of the equation

$4(\cos^2 2x + \cos 2x + 1) + \tan x(\tan x - 2\sqrt{3}) = 0$ in $[0, 2\pi]$ is

(a) 0 (b) 1 (c) 2 (d) 3

**18.** The number of solutions of the equation

$$\left|2\sin x - \sqrt{3}\right|^{2\cos^2 x - 3\cos x + 1} = 1 \text{ in } [0, \pi] \text{ is}$$

(a) 2 (b) 3 (c) 4 (d) 5

**19.** The number of solutions of the equation

$$\left(2\sin\left(\frac{\sin x}{2}\right)\right)\left(\cos\left(\frac{\sin x}{2}\right)\right)$$

$$\left(\sin\left(2\tan\frac{x}{2}\cos^2\frac{x}{2}\right)-3\right)+2=0 \text{ in } [0, 2\pi] \text{ is:}$$

(a) 0 (b) 1 (c) 2 (d) 4

**20.** If the equation $x^2 + 12 + 3\sin(a + bx) + 6x = 0$ has atleast one real solution, where $a, b \in [0, 2\pi]$, then the value of $a - 3b$ is $(n \in Z)$

(a) $2n\pi$ (b) $(2n + 1)\pi$

(c) $(4n - 1)\frac{\pi}{2}$ (d) $(4n + 1)\frac{\pi}{2}$

## Numeric Value Answer

**21.** The number of solutions of the equation

$$5^{1/2} + 5^{1/2+\log_5(\sin x)} = 15^{\frac{1}{2}+\log_{15}\cos x)} \text{ for}$$

$x \in [0, 100\pi]$ is

**22.** Number of solution(s) of the equation

$$\frac{\sin x}{\cos 3x}+\frac{\sin 3x}{\cos 9x}+\frac{\sin 9x}{\cos 27x}=0 \text{ in the interval}$$

$\left(0, \frac{\pi}{4}\right)$ is _______.

**23.** Number of integral value(s) of m for which the equation $\sin x - \sqrt{3}\cos x = \frac{4m-6}{4-m}$ has solutions, $x \in [0, 2\pi]$, is _______.

**24.** The number of solutions of the equation

$$\cos^2\left(x+\frac{\pi}{6}\right)+\cos^2 x - 2$$

$$\cos\left(x+\frac{\pi}{6}\right)\cdot\cos\frac{\pi}{6}=\sin^2\frac{\pi}{6} \text{ in interval}$$

$\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ is _______.

**25.** The number of solutions of the equation $1 + \cos x + \cos 2x + \sin x + \sin 2x + \sin 3x = 0$, which satisfy the condition $\frac{\pi}{2} < \left|3x - \frac{\pi}{2}\right| \le \pi$ is ___________.

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (d) | **4** | (b) | **7** | (b) | **10** | (a) | **13** | (b) | **16** | (c) | **19** | (a) | **22** | (6) | **25** | (2) |
| **2** | (b) | **5** | (a) | **8** | (b) | **11** | (a) | **14** | (a) | **17** | (c) | **20** | (c) | **23** | (4) | | |
| **3** | (c) | **6** | (c) | **9** | (b) | **12** | (c) | **15** | (c) | **18** | (b) | **21** | (50) | **24** | (2) | | |

# Principle of Mathematical Induction

# 4

## MCQs with One Correct Answer

**1.** If P(n) = 2 + 4 + 6 + .....+ 2n, $n \in \mathbf{N}$, then $P(k) = k(k+1)+2$

$\Rightarrow P(k+1) = (k+1)(k+2)+2$ for all $k \in \mathbf{N}$. So we can conclude that $P(n) = n(n+1)+2$ for

(a) all $n \in \mathbf{N}$
(b) $n > 1$
(c) $n > 2$
(d) nothing can be said

**2.** The greatest positive integer, which divides $n(n+1)(n+2)(n+3)$ for all $n \in \mathbf{N}$, is

(a) 2 (b) 6 (c) 24 (d) 120

**3.** For any $n \in \mathbf{N}$, the value of the expression

$$\underbrace{\sqrt{2+\sqrt{2+.....+\sqrt{2}}}}_{n-\text{roots}} \text{ is}$$

(a) $2\cos\left(\dfrac{\pi}{2^{n+1}}\right)$ (b) $2\sin\left(\dfrac{\pi}{2^{n+1}}\right)$

(c) $\sqrt{2}\cos(2^{n+1}\pi)$ (d) none of these

**4.** For a positive integer $n$,

Let $a(n) = 1 + \dfrac{1}{2} + \dfrac{1}{3} + \dfrac{1}{4} + \ldots + \dfrac{1}{(2^n)-1}$. Then

(a) $a(100) \le 100$ (b) $a(100) > 100$
(c) $a(200) \le 100$ (d) $a(200) < 100$

**5.** If $n \in \mathbf{N}$, then the result

$$\frac{1}{n}+\frac{1}{n+1}+\frac{1}{n+2}+......+\frac{1}{2n-1}$$

$= 1-\dfrac{1}{2}+\dfrac{1}{3}-\dfrac{1}{4}+......+\dfrac{1}{2n-1}$ holds for

(a) all $n \in \mathbf{N}$
(b) for even values of n
(c) for odd values of n
(d) not true for any n

**6.** For all $n \ge 1$, find

$$\frac{1}{1.2}+\frac{1}{2.3}+\frac{1}{3.4}+.....+\frac{1}{n(n+1)}$$

(a) $\dfrac{n}{n+1}$ (b) $\dfrac{1}{n+1}$

(c) $\dfrac{1}{n(n+1)}$ (d) None of these

**7.** For all natural numbers $n$, find

$$\left(1+\frac{3}{1}\right)\left(1+\frac{5}{4}\right)\left(1+\frac{7}{9}\right).....\left(1+\frac{2n+1}{n^2}\right)$$

(a) $(n+1)^2$ (b) $(n-1)^2$
(c) $n(n+1)$ (d) None of these

**8.** $2^n > n^2$ when $n \in$ N such that

(a) $n > 2$ (b) $n > 3$ (c) $n < 5$ (d) $n \ge 5$

**9.** If $\dfrac{4^n}{n+1} < \dfrac{(2n)!}{(n!)^2}$, then P($n$) is true for

(a) $n \ge 1$ (b) $n > 0$ (c) $n < 0$ (d) $n \ge 2$

**10.** If $P(n): 3^n < n!$, $n \in N$, then $P(n)$ is true

(a) for $n \geq 6$ (b) for $n \geq 7$

(c) for $n \geq 3$ (d) for all $n$

**11.** If $n$ is a positive integer, then $2 \cdot 4^{2n+1} + 3^{3n+1}$ is divisible by :

(a) 2 (b) 7 (c) 11 (d) 27

**12.** For every natural number n, $n(n^2 - 1)$ is divisible by

(a) 4 (b) 6

(c) 10 (d) None of these

**13.** If $p$ is a prime number, then $n^p - n$ is divisible by $p$ when $n$ is a

(a) Natural number greater than 1

(b) Irrational number

(c) Complex number

(d) Odd number

**14.** If $49^n + 16n + \lambda$ is divisible by 64 for all $n \in N$, then the least negative value of $\lambda$ is

(a) –2 (b) –1 (c) –3 (d) – 4

**15.** By mathematical induction,

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \ldots.. + \frac{1}{n(n+1)(n+2)}$$ is equal to

(a) $\frac{n(n+1)}{4(n+2)(n+3)}$ (b) $\frac{n(n+3)}{4(n+1)(n+2)}$

(c) $\frac{n(n+2)}{4(n+1)(n+3)}$ (d) None of these

**16.** For all $n \in N$, $3.5^{2n+1} + 2^{3n+1}$ is divisible by

(a) 19 (b) 17 (c) 23 (d) 25

**17.** For all $n \in N$, $1.3 + 2.3^2 + 3.3^3 + \ldots.. + n.3^n$ is equal to

(a) $\frac{(2n+1)3^{n+1}+3}{4}$ (b) $\frac{(2n-1)3^{n+1}+3}{4}$

(c) $\frac{(2n+1)3^{n}+3}{4}$ (d) $\frac{(2n-1)3^{n+1}+1}{4}$

**18.** For all $n \in N$,

$$1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \ldots.. + \frac{1}{1+2+3+\ldots..+n}$$

is equal to

(a) $\frac{3n}{n+1}$ (b) $\frac{n}{n+1}$ (c) $\frac{2n}{n-1}$ (d) $\frac{2n}{n+1}$

**19.** When $2^{301}$ is divided by 5, the least positive remainder is

(a) 4 (b) 8 (c) 2 (d) 6

**20.** By the principle of induction $\forall$ $n \in N$, $3^{2n}$ when divided by 8, leaves remainder

(a) 2 (b) 3 (c) 7 (d) 1

## Numeric Value Answer

**21.** If $n \in N$, then $11^{n+2} + 12^{2n+1}$ is divisible by ____________.

**22.** For all $n \in N$, $41^n - 14^n$ is a multiple of ____________.

**23.** If m, n are any two odd positive integers with n < m, then the largest positive integer which divides all the numbers of the type $m^2 - n^2$ is ____________.

**24.** For every natural number $n$, $3^{2n+2} - 8n - 9$ is divisible by ____________.

**25.** The remainder when $5^{99}$ is divided by 13, is ____________.

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | (d) | 4 | (a) | 7 | (a) | 10 | (b) | 13 | (a) | 16 | (b) | 19 | (c) | 22 | (27) | 25 | (8) |
| 2 | (c) | 5 | (a) | 8 | (d) | 11 | (c) | 14 | (b) | 17 | (b) | 20 | (d) | 23 | (8) | | |
| 3 | (a) | 6 | (a) | 9 | (d) | 12 | (b) | 15 | (b) | 18 | (d) | 21 | (133) | 24 | (16) | | |

# 5

# COMPLEX NUMBERS AND QUADRATIC EQUATIONS

## MCQs with One Correct Answer

1. Let $x_1$ and $y_1$ be real numbers. If $z_1$ and $z_2$ are complex numbers such that $|z_1| = |z_2| = 4$, then $|x_1 z_1 - y_1 z_2|^2 + |y_1 z_1 + x_1 z_2|^2 =$
   (a) $32(x_1^2 + y_1^2)$ (b) $16(x_1^2 + y_1^2)$
   (c) $4(x_1^2 + y_1^2)$ (d) 32
2. Let $z$ and $\omega$ be two non-zero complex numbers such that $|z| = |\omega|$ and $\text{Arg}\, z + \text{Arg}\, \omega = \pi$, then $z$ equals
   (a) $\omega$ (b) $-\omega$ (c) $\overline{\omega}$ (d) $-\overline{\omega}$
3. If $|z-1| + |z+3| \le 8$, then the range of values of $|z-4|$ is
   (a) $(0, 7)$ (b) $(1, 8)$ (c) $[1, 9]$ (d) $[2, 5]$
4. Which of the following is/are value of $\sin \ln (i^i)^{i^2} + \cos \ln (i^i)^{i^2}$ ?
   (a) $-1$ (b) 1
   (c) 0 (d) None of these
5. If $n_1$, $n_2$ are positive integers, then $(1+i)^{n_1} + (1+i^3)^{n_1} + (1+i^5)^{n_2} + (1+i^7)^{n_2}$ is real number if and only if:
   (a) $n_1 = n_2 + 1$
   (b) $n_1 + 1 = n_2$
   (c) $n_1 = n_2$
   (d) $n_1, n_2$ are any positive integers
6. Let $r_k > 0$ and $z_k = r_k(\cos \alpha_k + i \sin \alpha_k)$ for $k = 1$, 2, 3 be such that $\dfrac{1}{z_1} + \dfrac{1}{z_2} + \dfrac{1}{z_3} = 0$ Let $A_k$ be the point in the complex plane given by $w_k = \dfrac{\cos 2\alpha_k + i \sin 2\alpha_k}{z_k}$ for k = 1, 2, 3. The origin, $O$ is the
   (a) incentre of $\Delta A_1 A_2 A_3$
   (b) orthocentre of $\Delta A_1 A_2 A_3$
   (c) circumcentre of $\Delta A_1 A_2 A_3$
   (d) centroid of $\Delta A_1 A_2 A_3$
7. If $a$ point $z_1$ is the reflection of a point $z_2$ through the line $b\,\overline{z} + \overline{b}\, z = c, b \ne 0$ in the argand plane, then $\overline{b}\, z_2 + b\, \overline{z}_1$ is equal to
   (a) $4c$ (b) $2c$
   (c) $c$ (d) None of these
8. $z_1$ and $z_2$ lie on a circle with centre at the origin. The point of intersection $z_3$ of the tangents at $z_1$ and $z_2$ is given by
   (a) $\dfrac{1}{2}(\overline{z}_1 + \overline{z}_2)$ (b) $\dfrac{2 z_1 z_2}{z_1 + z_2}$
   (c) $\dfrac{1}{2}\left(\dfrac{1}{z_1} + \dfrac{1}{z_2}\right)$ (d) $\dfrac{z_1 + z_2}{\overline{z}_1 + \overline{z}_2}$
9. If $z_1$, $z_2$ are two complex numbers such that $\left|\dfrac{z_1 - z_2}{z_1 + z_2}\right| = 1$ and $i\, z_1 = K z_2$, where $K \in R$, then the angle between $z_1 - z_2$ and $z_1 + z_2$ is
   (a) $\tan^{-1}\left(\dfrac{2K}{K^2 + 1}\right)$ (b) $\tan^{-1}\left(\dfrac{2K}{1 - K^2}\right)$
   (c) $-2 \tan^{-1} K$ (d) $2 \tan^{-1} K$

**10.** Let '$z$' be a complex number and '$a$' be a real parameter such that $z^2 + az + a^2 = 0$, then which is of the following is not true?
(a) locus of $z$ is a pair of staight lines
(b) $|z| = |a|$
(c) $\arg(z) = \pm\frac{2\pi}{3}$
(d) None of these

**11.** Let $P$ denotes a complex number $z = r(\cos\theta + i\sin\theta)$ on the Argand's plane, and $Q$ denotes a complex number $\sqrt{2|z|^2}\left(\cos\left(\theta+\frac{\pi}{4}\right)+i\sin\left(\theta+\frac{\pi}{4}\right)\right)$.
If '$O$' is the origin, then $\Delta OPQ$ is
(a) isosceles but not right angled
(b) right angled but not isosceles
(c) right isosceles
(d) equilateral

**12.** If $\omega \neq 1$ and $\omega^3 = 1$, then
$\frac{a\omega + b + c\omega^2}{a\omega^2 + b\omega + c} + \frac{a\omega^2 + b + c\omega}{a + b\omega + c\omega^2}$ is equal to
(a) 2 (b) $\omega$ (c) $2\omega$ (d) $2\omega^2$

**13.** If $z + \frac{1}{z} = 1$ and $a = z^{2017} + \frac{1}{z^{2017}}$ and b is the last digit of the number $2^{2^n} - 1$, when the integer $n > 1$, the value of $a^2 + b^2$ is
(a) 23 (b) 24 (c) 26 (d) 27

**14.** If $y_1 = \max ||z - \omega| - |z - \omega^2||$, where $|z| = 2$ and $y_2 = \max ||z - \omega| - |z - \omega^2||$, where $|z| = \frac{1}{2}$ and $\omega$ and $\omega^2$ are complex cube roots of unity, then
(a) $y_1 = \sqrt{3}\,; y_2 = \sqrt{3}$
(b) $y_1 < \sqrt{3}\,; y_2 = \sqrt{3}$
(c) $y_1 = \sqrt{3}\,; y_2 < \sqrt{3}$
(d) $y_1 > 3; y_2 < \sqrt{3}$

**15.** If $\alpha$ and $\beta$ are the roots of the equation $ax^2 + bx + c = 0$, $(c \neq 0)$, then the equation whose roots are $\frac{1}{a\alpha + b}$ and $\frac{1}{a\beta + b}$ is
(a) $acx^2 - bx + 1 = 0$ (b) $x^2 - acx + bc + 1 = 0$
(c) $acx^2 + bx - 1 = 0$ (d) $x^2 + acx - bc + 11 = 0$

**16.** If the roots of the equation $x^2 + px + c = 0$ are 2, –2 and the roots of the equation $x^2 + bx + q = 0$ are –1, –2, then the roots of the equation $x^2 + bx + c = 0$ are
(a) –3, –2 (b) –3, 2 (c) 1, –4 (d) –5, 1

**17.** If $x \in \boldsymbol{R}$, then the maximum value of
$y = 2(a - x)\ \left(x + \sqrt{x^2 + b^2}\right)$ is
(a) $a^2 + b^2$ (b) $a^2 - b^2$
(c) $a^2 + 2b^2$ (d) None of these

**18.** If $\alpha, \beta$ are the roots of $x^2 + px + q = 0$, and $x^{2n} + p^n x^n + q^n = 0$ and $\frac{\alpha}{\beta}$ is a root of $x^n + 1 + (x + 1)^n = 0$, $\alpha^n \neq \beta^n$, the $n$ must be
(a) any integer (b) an even integer
(c) an odd integer (d) None of these

**19.** If $0 < \alpha < \beta < \gamma < \pi/2$, then the equation $(x - \sin\beta)(x - \sin\gamma) + (x - \sin\alpha)(x - \sin\gamma) + (x - \sin\alpha)(x - \sin\beta) = 0$ has
(a) real and unequal roots.
(b) non-real roots.
(c) real and equal roots.
(d) real and unequal roots greater than 2.

**20.** The set of values of $a$ for which inequation $(a - 1)x^2 - (a + 1)x + a - 1 \geq 0$ is true for all $x \geq 2$
(a) $\left[1, \frac{7}{3}\right]$ (b) $(-\infty, 1)$
(c) $\left[\frac{7}{3}, \infty\right)$ (d) None of these

## Numeric Value Answer

**21.** $a, b, c$ are integers, not all simultaneously equal and $\omega$ is cube root of unity $(\omega \neq 1)$, then minimum value of $|a + b\omega + c\omega^2|$ is

**22.** If $2 - i$ is a root of the equation $ax^2 + 12x + b = 0$ (where a and b are real), then the value of ab is equal to

**23.** If the equations $ax^2 + bx + c = 0$ and $cx^2 + bx + a = 0, a \neq c$ have a negative common root, then the value of $a - b + c$ is

**24.** If $\alpha$, $\beta$ are the roots the quadratic equation $x^2 - (3+2^{\sqrt{\log_2 3}} - 3^{\sqrt{\log_3 2}})x - 2(3^{\log_3 2} - 2^{\log_2 3}) = 0$, then the value of $\alpha^2 + \alpha\beta + \beta^2$ is equal to

**25.** Let $f(x) = x^4 + ax^3 + bx^2 + cx + d$ be a polynomial with real coefficients and real zeroes. If $|f(i)| = 1$, (where $i = \sqrt{-1}$) then $a + b + c + d$ is equal to

**26.** If $\omega$ and $\omega^2$ be the non-real cube roots of unity and $\frac{1}{a+\omega} + \frac{1}{b+\omega} + \frac{1}{c+\omega} = 2\omega^2$ and $\frac{1}{a+\omega^2} + \frac{1}{b+\omega^2} + \frac{1}{c+\omega^2} = 2\omega$, where $a, b, c$ are real then the value of $\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1}$ is equal to :

**27.** For any real $x$, the maximum value of $\frac{2}{k^2}(x-k)\left(x + \sqrt{x^2 + k^2}\right)$ is equal to

**28.** If $z$ and $w$ are two complex numbers having non-negative imaginary parts such that $\arg\left(\frac{z-2}{z+2}\right) = \arg\left(\frac{w-1}{w+1}\right) = \pi/2$, then $|w-z| < k$; evaluate $k$. (Here $k$ is least upper bond)

**29.** If $z = \frac{\pi}{4}(1+i)^4\left(\frac{1-\sqrt{\pi}i}{\sqrt{\pi}+i} + \frac{\sqrt{\pi}-i}{1+\sqrt{\pi}i}\right)$, where $i = \sqrt{-1}$, then $\left(\frac{|z|}{\text{amp}(z)}\right)$ equals to

**30.** If $p, q, r$ are positive and are in A.P., then the roots of the quadratic equation $px^2 + qx + r = 0$ are real for $\left|\frac{r}{p} - 7\right| \geq k\sqrt{3}$ then find the value of $k$.

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (a) | **5** | (d) | **9** | (d) | **13** | (c) | **17** | (a) | **21** | (1) | **25** | (0) | **29** | (4) | | | | |
| **2** | (d) | **6** | (d) | **10** | (d) | **14** | (c) | **18** | (b) | **22** | (45) | **26** | (2) | **30** | (4) | | | | |
| **3** | (c) | **7** | (c) | **11** | (c) | **15** | (a) | **19** | (a) | **23** | (0) | **27** | (2) | | | | | | |
| **4** | (b) | **8** | (b) | **12** | (d) | **16** | (c) | **20** | (c) | **24** | (7) | **28** | (3) | | | | | | |

# 6

# LINEAR INEQUALITIES

## MCQs with One Correct Answer

**1.** The set of all $x$ satisfying the inequality

$$\frac{2}{x^2-x+1}-\frac{1}{x+1}-\frac{2x-1}{x^3+1}\geq 0$$

(a) $(-\infty, 2]$ (b) $[1, 2]$
(c) $(-\infty, -1)\cup(-1, 2]$ (d) $(2, \infty]$

**2.** The set of all x satisfying the inequality $(2x+1)(x-3)(x+7)<0$

(a) $(-\infty, 7)$ (b) $\left(-\frac{1}{2}, 3\right)$
(c) $(-\infty, 7)\cup\left(-\frac{1}{2}, 3\right)$ (d) $(-\infty, 3)$

**3.** The set of real values of x satisfying $|x-1|\leq 3$ and $|x-1|\geq 1$ is

(a) $[2, 4]$
(b) $(-\infty, 2]\cup[4, +\infty)$
(c) $[-2, 0]\cup[2, 4]$
(d) none of these

**4.** The system of equation

$|x-1|+3y=4,\ x-|y-1|=2$ has

(a) No solution
(b) A unique solution
(c) Two solutions
(d) More than two solutions

**5.** The number of real roots of the equation

$|2-|1-|x|||=1$ is

(a) 1 (b) 3 (c) 5 (d) 6

**6.** The solution set of the inequality

$|x+2|-|x-1|<x-\frac{3}{2}$ is

(a) $\left(\frac{9}{2}, \infty\right)$ (b) $\left(-\infty, \frac{3}{2}\right)$
(c) $\left(-2, -\frac{3}{2}\right)$ (d) $\left(-1, \frac{3}{2}\right)$

**7.** If $\left|\frac{12x}{4x^2+9}\right|\geq 1$ for all real values of x, the inequality being satisfied only if $|x|$ is equal to

(a) $\frac{3}{2}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) $\frac{1}{2}$

**8.** The equation $||x-1|+a|=4$ can have real solutions for x if 'a' belongs to the interval

(a) $(-\infty, +\infty)$ (b) $(-\infty, 4]$
(c) $(4, +\infty)$ (d) $[-4, 4]$

**9.** The interval(s) that satisfy the equation

$\left|\frac{x^2-8x+12}{x^2-10x+21}\right|=-\frac{x^2-8x+12}{x^2-10x+21}$ is /are

(a) $(-\infty, 2]$ (b) $[2, \infty)$
(c) $[6, 7)$ (d) $[3, 6]\cup[7, \infty)$

**10.** The set of all real x satisfying the inequality

$\frac{3-|x|}{4-|x|}\geq 0$ is

(a) $[-3, 3]\cup(-\infty, -4)\cup(4, \infty)$
(b) $(-\infty, -4)\cup(4, \infty)$
(c) $(-\infty, -3)\cup(4, \infty)$
(d) $(-\infty, -3)\cup(3, \infty)$

**11.** If x, y, z be any three positive real number and

$$A=\frac{x+y}{x^2+y^2}+\frac{y+z}{y^2+z^2}+\frac{x_1+z}{x^2+z^2}$$

and $B=\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$;

then which of the following is true?
(a) $A \geq B$ (b) $A \leq B$ (c) $A = B$ (d) $A < B$

**12.** If x, y, z are arbitrary real numbers satisfiying the condition $xy + yz + zx < 0$ and if $u = \dfrac{x^2 + y^2 + z^2}{xy + yz + zx}$ then only one of the following statements is always correct. Which one is it?
(a) $-1 \leq u < 0$
(b) u takes all negative real values
(c) $-2 < u \leq -1$
(d) $u \leq -2$

**13.** If $a_1, a_2, ....., a_n$ are positive real numbers whose product is a fixed number c, then the minimum value of $a_1 + a_2 + ....... + a_{n-1} + 2a_n$ is
(a) $n(2c)^{1/n}$ (b) $(n+1)c^{1/n}$
(c) $2nc^{1/n}$ (d) $(n+1)(2c)^{1/n}$

**14.** The set of real values of x for which $\log_{0.2}\dfrac{x+2}{x} \leq 1$ is
(a) $\left(-\infty, -\dfrac{5}{2}\right] \cup (0, \infty)$ (b) $\left[\dfrac{5}{2}, \infty\right)$
(c) $(-\infty, -2) \cup [0, \infty)$ (d) None of these

**15.** If $\log_{1/2}\dfrac{x^2 + 6x + 9}{2(x+1)} < -\log_2(x+1)$, then x lies in the interval
(a) $(-1, -1+2\sqrt{2})$ (b) $(1-2\sqrt{2}, 2)$
(c) $(-1, \infty)$ (d) None of these

**16.** The least integer a, for which $1 + \log_5(x^2 + 1) \leq \log_5(ax^2 + 4x + a)$ is true for all $x \in R$ is
(a) 6 (b) 7 (c) 10 (d) 1

**17.** The set of all real numbers x for which $x^2 - [x+2] + x > 0$, is
(a) $(-\infty, -2) \cup (2, \infty)$
(b) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
(c) $(-\infty, -1) \cup (1, \infty)$
(d) $(\sqrt{2}, \infty)$

**18.** Solution set of the in equality $\log_{10^2} x - 3(\log_{10} x)(\log_{10}(x-2)) + 2\log_{10^2}(x-2) < 0$, is :
(a) $(0, 4)$ (b) $(-\infty, 1)$
(c) $(4, \infty)$ (d) $(2, 4)$

**19.** The solution set of $\log_{|\sin x|}(x^2 - 8x + 23) > \dfrac{3}{\log_2|\sin x|}$ contains
(a) $x \in (3, \pi) \cup \left(\pi, \dfrac{3\pi}{2}\right) \cup \left(\dfrac{3\pi}{2}, 5\right)$
(b) $x \in (3, \pi) \cup (\pi, 5)$
(c) $x \in \left(3, \dfrac{5\pi}{2}\right)$
(d) None of these

**20.** If x is real, the maximum value of $\dfrac{3x^2 + 9x + 17}{3x^2 + 9x + 7}$ is
(a) $\dfrac{1}{4}$ (b) 41 (c) 1 (d) $\dfrac{17}{7}$

## Numeric Value Answer

**21.** If 'k' any integer such that set of all real values of x, satisfy the equation $(\sin 3^x)(\cos 3^x) \leq \dfrac{3x + 3^{-x}}{k}$; then find the minimum value k.

**22.** For any $x, y \in R$, $xy > 0$. Then the minimum value of $\dfrac{2x}{y^3} + \dfrac{x^3 y}{3} + \dfrac{4y^2}{9x^4}$ is.

**23.** Number of integeral solution satisfying $x + \sqrt{3-x} \geq \sqrt{3-x} + 3$ is

**24.** Solve for x, $\dfrac{|x+3| + x}{x+2} > 1$ and then find modulus of smallest integral member of the interval.

**25.** The number of integers satisfying the equation $|x| + \left|\dfrac{4-x^2}{x}\right| = \left|x + \dfrac{4-x^2}{x}\right|$ is

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (c) | **4** | (b) | **7** | (a) | **10** | (a) | **13** | (a) | **16** | (b) | **19** | (a) | **22** | (2) | **25** | (4) |
| **2** | (c) | **5** | (c) | **8** | (b) | **11** | (b) | **14** | (a) | **17** | (b) | **20** | (b) | **23** | (1) | | |
| **3** | (c) | **6** | (a) | **9** | (c) | **12** | (d) | **15** | (a) | **18** | (c) | **21** | (4) | **24** | (4) | | |

# 7

# PERMUTATIONS AND COMBINATIONS

## MCQs with One Correct Answer

1. Ten different letters of an alphabet are given words with five letters are formed from three given letters. Then the number of words which have at least one letter repeated are
(a) 69760 (b) 30240
(c) 99748 (d) None of these

2. How many different nine digit numbers can be formed from the number 223355888 by rearranging its digits so that the odd digits occupy even positions ?
(a) 16 (b) 36 (c) 60 (d) 180

3. The letters of the word COCHIN are permuted and all the permutations are arranged in an alphabetical order as in an English dictionary. The number of words that appear before the word COCHIN is
(a) 360 (b) 192 (c) 96 (d) 48

4. The total number of 5-digit numbers of different digits in which the digit in the middle is the largest is
(a) $\sum_{n=4}^{9} {}^{n}p_4$ (b) 4563
(c) 2688 (d) 5292

5. All possible 120 permutations of WDSMC are arranged in dictionary order, as if each were an ordinary five-letter word. The last letter of the $86^{th}$ word in the list, is :
(a) W (b) D (c) M (d) C

6. If $m$ be the number of different words that can be formed with the letters of the word BHARAT in which $B$ and $H$ are never together and $n$ be number of different words that can be formed with the letters of the words BHARAT in which words always begin with $B$ and end with $T$. Then $m/n$ is
(a) 10 (b) 20 (c) 1 (d) 2

7. Anil have tiled his square bathroom wall with congruent square tiles. All the tiles are red, except those along the two diagonals, which are all blue. If he used 121 blue tiles, then the number of red tiles used are
(a) 900 (b) 1800 (c) 3600 (d) 7200

8. The number of distinct natural numbers up to a maximum of four digits and divisible by 5, which can be formed with the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, each digit not occurring more than once in each number, is
(a) 1246 (b) 952
(c) 1106 (d) None of these

9. 6 white and 6 black balls are distributed among ten identical urns, so that there is atleast one ball in each urn. Balls are all alike except for the colour and each box can hold any number of balls. The number of different distributions of the balls is:
(a) 26250 (b) 132 (c) 12 (d) 10

10. Number of ways in which two Americans, two British, one Chinese, one Dutch and one Egyptian can sit on a round table so that persons of the same nationality are separated is
(a) 48 (b) 240
(c) 336 (d) None of these

11. In an examination of 9 papers a candidate has to pass in more papers than the number of papers in which he fails in order to be successful. The number of ways in which he can be unsuccessful is
(a) 255 (b) 256 (c) 193 (d) 319

**12.** During a draw of lottery, tickets bearing numbers 1, 2, 3, ..., 40, 6 tickets are drawn out and then arranged in the descending order of their numbers. In how many ways, it is possible to have 4th ticket bearing number 25?
(a) ${}^{15}C_3 \times {}^{24}C_2$ (b) ${}^{12}C_3 \times {}^{20}C_2$
(c) ${}^{15}C_3 + {}^{24}C_2$ (d) None of these

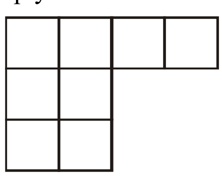
**13.** A teacher takes 3 children from her class to the zoo at a time as often as she can, but she does not take the same three children to the zoo more than once he finds than she goes to the zoo 84 times more than a particular child goes to the zoo. The number of children in her class is
(a) 12 (b) 10
(c) 60 (d) None of these

**14.** Messages are conveyed by arranging four white, one blue, and three red flags on a pole. Flags of the same colour are alike. If a message is transmitted by the order in which the colours are arranged, the total number of messages that can be transmitted if exactly six flags are used is
(a) 45 (b) 65 (c) 125 (d) 185

**15.** There were two women participating in a chess tournament. Every participant played two games with the other participants. The number of games that the men played between themselves proved to exceed by 66 the number of games that the men played with the women. The number of participants is
(a) 6 (b) 11
(c) 13 (d) None of these

**16.** In a class tournament, all participants were to play different games with one another. Two players fell ill after having played three games each. If the total number of games played in the tournament is equal to 84, the total number of participants in the beginning was equal to
(a) 10 (b) 15 (c) 12 (d) 14

**17.** The number of ways in which 5 X's can be placed in the squares of the figure so that no row remains empty is



(a) 97 (b) 44 (c) 100 (d) 126

**18.** There are three coplanar parallel lines. If any $p$ points are taken on each the lines, the maximum number of triangles with vertices at these points is
(a) $3p^2(p-1)+1$ (b) $3p^2(p-1)$
(c) $p^2(4p-3)$ (d) None of these

**19.** From the vertices of a regular polygon of 10 sides, the number of ways of selecting three vertices such that no two vertices are consecutive is
(a) 10 (b) 30 (c) 50 (d) 40

**20.** 5 different objects are to be distributed among 3 persons such that no two persons get the same number of objects. Number of ways this can be done, is
(a) 60 (b) 90 (c) 120 (d) 150

## Numeric Value Answer

**21.** If $a, b, c$ are three natural numbers in $AP$ such that $a+b+c=21$ and if possible number of ordered triplet $(a, b, c)$ is then the value of $(\lambda - 5)$ is

**22.** In a single correct match the column question, column I contain 10 questions and Column II contain 10 answers written in some arbitrary order. If the number ways a student can answer this question so that exactly 6 of his matching are correct is k, then (sum of digits of k)/2 is equal to

**23.** There are 720 permutations of the digits 1, 2, 3, 4, 5, 6. Suppose these permutations are arranged from smallest to largest numerical values, beginning from 1 2 3 4 5 6 and ending with 6 5 4 3 2 1. Then the digit in unit place of number at 267th position is ...... .

**24.** If $N$ is the number of ways in which a person can walk up a stairway which has 7 steps if he can take 1 or 2 steps up the stairs at a time, then the value of $N/3$ is ...... .

**25.** In an international convention participants from 10 different countries were arranged in a row such that all the participants from the same country were together. Each country has different number of participants with maximum 10 participants from a country. If K is the number of ways that they can be arranged in a row then find the highest power of 10 in K.

**ANSWER KEY**

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (a) | **4** | (d) | **7** | (c) | **10** | (c) | **13** | (b) | **16** | (b) | **19** | (c) | **22** | (9) | **25** | (9) |
| **2** | (c) | **5** | (b) | **8** | (c) | **11** | (b) | **14** | (d) | **17** | (b) | **20** | (b) | **23** | (6) | | |
| **3** | (c) | **6** | (b) | **9** | (d) | **12** | (a) | **15** | (c) | **18** | (c) | **21** | (8) | **24** | (7) | | |

# 8

# BINOMIAL THEOREM

## MCQs with One Correct Answer

**1.** $2^{60}$ when divided by 7 leaves the remainder
(a) 1 (b) 6 (c) 5 (d) 2

**2.** If 7 divides $32^{32^{32}}$, the remainder is
(a) 1 (b) 0 (c) 4 (d) 6

**3.** $\sum\limits_{r=1}^{n}\left(\sum\limits_{p=1}^{r-1} {}^{n}C_{r}\,{}^{r}C_{p}\,2^{p}\right)$ is equal to
(a) $4^n - 3^n + 1$ (b) $4^n - 3^n - 1$
(c) $4^n - 3^n + 2$ (d) $4^n - 3^n$

**4.** If $(1 + x - 2x^2)^6 = 1 + a_1 x + a_2 x^2 + ..... + a_{12} x^{12}$, then the value of $a_2 + a_4 + ..... + a_{12}$ is :
(a) 1024 (b) 64
(c) 32 (d) 31

**5.** The last term in the binomial expansion of $(2^{1/3} - 1/\sqrt{2})^n$ is $(1/3.9^{1/3})\log_3 8$. Then the $5^{th}$ term from the beginning is
(a) ${}^{10}C_6$ (b) $2.{}^{10}C_4$
(c) $1/2.{}^{10}C_4$ (d) None

**6.** If $(1+ax)^n = 1 + 8x + 24x^2 + .........$; than $\dfrac{a-n}{a+n}$ is equal to ($n$ being a positive integer)
(a) 3 (b) $-3$
(c) $-\dfrac{1}{3}$ (d) $\dfrac{1}{3}$

**7.** The unit digit of $17^{2009} + 11^{2009} - 7^{2009}$ is
(a) 1 (b) 2 (c) 3 (d) 0

**8.** Let $N = 2^{1224} - 1$, $\alpha = 2^{153} + 2^{77} + 1$ and $\beta = 2^{408} - 2^{204} + 1$. Then which of the following statement is correct ?
(a) $\alpha$ divides $N$ but $\beta$ does not
(b) $\beta$ divides $N$ but $\alpha$ does not
(c) $\alpha$ and $\beta$ both divides $N$
(d) neither $\alpha$ nor $\beta$ divides $N$

**9.** The number $N = {}^{20}C_7 - {}^{20}C_8 + {}^{20}C_9 - {}^{20}C_{10} + ..... - {}^{20}C_{20}$ is not divisible by :
(a) 3 (b) 7 (c) 11 (d) 19

**10.** Let $x$ be the $7^{th}$ term from the beginning and $y$ be the $7^{th}$ term from the end in the expansion of $\left(3^{1/3} + \dfrac{1}{4^{1/3}}\right)^n$. If $y = 12x$ then the value of $n$ is :
(a) 9 (b) 8 (c) 10 (d) 11

**11.** The coefficient of the term independent of $x$ in the expansion of $\left(\dfrac{x+1}{x^{\frac{2}{3}} - x^{\frac{1}{3}} + 1} - \dfrac{x-1}{x - x^{\frac{1}{2}}}\right)^{10}$ is
(a) 70 (b) 112 (c) 105 (d) 210

**12.** If the middle term of $(1+x)^{2n}$ ($x > 0$, $n \in N$) is the greatest term of the expansion. Then the interval in which $x$ lies, is
(a) $\left[\dfrac{n+1}{n}, \dfrac{n+2}{n}\right]$ (b) $\left[\dfrac{n-1}{n}, \dfrac{n+1}{n}\right]$
(c) $\left[\dfrac{n}{n+1}, \dfrac{n+1}{n}\right]$ (d) None of these

**13.** The number of irrational terms in the expansion of $\left(\sqrt[8]{5}+\sqrt[6]{2}\right)^{100}$ is

(a) 97 (b) 98 (c) 96 (d) 99

**14.** The greatest value of the term independent of $x$ in the expansion of $(x\sin\alpha + x^{-1}\cos\alpha)^{10}$, $\alpha \in \boldsymbol{R}$, is

(a) $2^5$ (b) $\frac{10!}{(5!)^2}$

(c) $\frac{1}{2^5}\cdot\frac{10!}{(5!)^2}$ (d) None of these

**15.** The largest term in the expansion of $(2+3x)^{25}$ where $x = 2$ is its.

(a) $13^{th}$ term (b) $19^{th}$ term

(c) $20^{th}$ term (d) $26^{th}$ term

**16.** If the middle term in the expansion of $\left(\frac{1}{x}+x\sin x\right)^{10}$ equals to $7\frac{7}{8}$ then $x$ is equal to $(n \in I)$

(a) $2n\pi \pm \frac{\pi}{6}$ (b) $n\pi + \frac{\pi}{6}$

(c) $n\pi + (-1)^n\frac{\pi}{6}$ (d) $n\pi + (-1)^n\frac{5\pi}{6}$

**17.** The sum of the coefficients of all the integral powers of $x$ in the expansion of $(1+2\sqrt{x})^{40}$ is

(a) $3^{40}+1$ (b) $3^{40}-1$

(c) $\frac{1}{2}(3^{40}-1)$ (d) $\frac{1}{2}(3^{40}+1)$

**18.** If $\pi$ (n) denotes product of all binomial coefficients in $(1+x)^n$, then ratio of $\pi$(2002) to $\pi$(2001) is

(a) 2002 (b) $\frac{(2002)^{2001}}{(2001)!}$

(c) $\frac{(2001)^{2002}}{(2002)!}$ (d) 2001

**19.** The value of $\frac{C_1}{2}+\frac{C_3}{4}+\frac{C_5}{6}+.......$ is equal to

(a) $\frac{2^n+1}{n+1}$ (b) $\frac{2^n}{n+1}$

(c) $\frac{2^n+1}{n-1}$ (d) $\frac{2^n-1}{n+1}$

**20.** If $I$ is integral part of $(2+\sqrt{3})^n$ and $f$ is its fractional part. Then $(I+f)(1-f)$ is

(a) $I+1$ (b) 1 (c) $n$ (d) $2^n$

## Numeric Value Answer

**21.** The largest real value for x such that $\sum_{k=0}^{4}\left(\frac{3^{4-k}}{(4-k)!}\right)\left(\frac{x^k}{k!}\right)=\frac{32}{3}$ is

**22.** Find the coefficient of $x^{13}$ in the expansion of $(1-x)^5(1+x+x^2+x^3)^4$ is

**23.** If $(1+x)^n = C_0 + C_1x + C_2x^2 + ..... + C_nx^n$, then find the value of $3C_0 - 5C_1 + 7C_2 + ... + (-1)^n(2n+3)C_n$

**24.** If $\frac{1}{1!10!}+\frac{1}{3!8!}+\frac{1}{5!6!}+...+\frac{1}{11!1!}=\frac{2^k}{11!}$ then $\frac{k}{2}$ is equal to

**25.** $(1-2x+5x^2-10x^3)(1+x)^n = 1 + a_1x + a_2x^2 + ...$ and that $a_1^2 = 2a_2$, then the value of $n$ is____

| ANSWER KEY | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (a) | **4** | (d) | **7** | (a) | **10** | (a) | **13** | (a) | **16** | (c) | **19** | (d) | **22** | (4) | **25** | (6) |
| **2** | (c) | **5** | (a) | **8** | (c) | **11** | (d) | **14** | (c) | **17** | (d) | **20** | (b) | **23** | (0) | | |
| **3** | (d) | **6** | (c) | **9** | (c) | **12** | (c) | **15** | (c) | **18** | (b) | **21** | (1) | **24** | (5) | | |

# 9

# SEQUENCES AND SERIES

## MCQs with One Correct Answer

1. Sum to n terms of the series

$1^3 + 3.2^3 + 3^3 + 3.4^3 + 5^3 + ................$ is (n is even)

(a) $\frac{n(n^2+1)(2n+1)}{3}$

(b) $\frac{n(n^3+4n^2+10n+8)}{8}$

(c) $\frac{n(n^3+1)}{8}$

(d) $\frac{n^2(2n^2+6n+5)}{4}$

2. If $1, \log_9(3^{1-x}+2), \log_3(4.3^x-1)$ are in A.P., then $x$ equals

(a) $\log_3 4$ (b) $1-\log_3 4$

(c) $1-\log_4 3$ (d) $\log_4 3$

3. In the sum of first $n$ terms of an A.P. is $cn^2$, then the sum of squares of these $n$ terms is

(a) $\frac{n(4n^2-1)c^2}{6}$ (b) $\frac{n(4n^2+1)c^2}{3}$

(c) $\frac{n(4n^2-1)c^2}{3}$ (d) $\frac{n(4n^2+1)c^2}{6}$

4. If $a_1, a_2, ..., a_n$ are in A.P. with common difference $d \neq 0$, then $(\sin d)$ [$\sec a_1 \sec a_2 + \sec a_2 \sec a_3 + ... + \sec a_{n-1} \sec a_n$] is equal to

(a) $\cot a_n - \cot a_1$ (b) $\cot a_1 - \cot a_n$

(c) $\tan a_n - \tan a_1$ (d) $\tan a_n - \tan a_{n-1}$

5. If $x = \frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+..., y=\frac{1}{1^2}+\frac{3}{2^2}+\frac{1}{3^2}+\frac{3}{4^2}$ $+ ...$ and $z = \frac{1}{1^2}-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+...$, then

(a) $x, y, z$ are in A.P. (b) $\frac{y}{6},\frac{x}{3},\frac{z}{2}$ are in A.P.

(c) $\frac{y}{6},\frac{x}{3},\frac{z}{2}$ are in A.P. (d) $6y, 3x, 2z$ are in A.P.

6. For a, b, c $\in$ R $- \{0\}$, let $\frac{a+b}{1-ab}, b, \frac{b+c}{1-bc}$ are in A.P. If $\alpha, \beta$ are the roots of the quadratic equation $2ac\,x^2 + 2abc\,x + (a + c) = 0$, then the value of $(1 + \alpha)(1 + \beta)$ is

(a) 0 (b) 1 (c) $-1$ (d) 2

7. An A.P. consist of even number of terms $2n$ having middle terms equal to 1 and 7 respectively. If n is the maximum value which satisfy $t_1 t_{2n} + 713 \geq 0$, then the value of the first term of the series is

(a) 17 (b) $-15$ (c) 21 (d) $-23$

8. If $a, b, c$ are in G. P., $x$ and $y$ be the A. M.'s between $a, b$ and $b, c$ respectively, then $\left(\frac{a}{x}+\frac{c}{y}\right)\left(\frac{b}{x}+\frac{b}{y}\right)$ is equal to

(a) $-2$ (b) $-4$ (c) 2 (d) 4

9. Suppose $a, b, c$ are in A.P. and $a^2, b^2, c^2$ are in GP. if $a<b<c$ and $a+b+c=\frac{3}{2}$, then the value of $a$ is

(a) $\frac{1}{2\sqrt{2}}$ (b) $\frac{1}{2\sqrt{3}}$

(c) $\frac{1}{2}-\frac{1}{\sqrt{3}}$ (d) $\frac{1}{2}-\frac{1}{\sqrt{2}}$

10. An infinite G.P. has first term '$x$' and sum '5', then $x$ belongs to

(a) $x<-10$ (b) $-10<x<0$

(c) $0<x<10$ (d) $x>10$

11. In the quadratic equation $ax^2 + bx + c = 0$, $\Delta = b^2 - 4ac$ and $\alpha + \beta, \alpha^2 + \beta^2, \alpha^3 + \beta^3$, are in G.P. where $\alpha, \beta$ are the root of $ax^2 + bx + c = 0$, then

(a) $\Delta \neq 0$ (b) $b\Delta = 0$ (c) $c\Delta = 0$ (d) $\Delta = 0$

**12.** The sum of an infinite geometric series is 2 and the sum of the geometric series made from the cubes of this infinite sereis is 24. Then the series is

(a) $3+\frac{3}{2}-\frac{3}{4}+\frac{3}{8}-....$ (b) $3+\frac{3}{2}+\frac{3}{4}+\frac{3}{8}+....$

(c) $3-\frac{3}{2}+\frac{3}{4}-\frac{3}{8}+...$ (d) None of these

**13.** If $a$, $b$, $c$ are in G. P. and $\log a - \log 2b$, $\log 2b - \log 3c$ and $\log 3c - \log a$ are in A. P., then $a$, $b$, $c$ are the sides of a triangle which is

(a) Acute angled (b) Obtuse angled

(c) Right angled (d) None of these

**14.** $A_r; r = 1, 2, 3, ..........., n$ are $n$ points on the parabola $y^2 = 4x$ in the first quadrant.

If $A_r = (x_r, y_r)$, where $x_1, x_2, x_3, ..............., x_n$ are in G. P. and $x_1 = 1$, $x_2 = 2$, then $y_n$ is equal to

(a) $-2^{\frac{n+1}{2}}$ (b) $2^{n+1}$ (c) $(\sqrt{2})^{n+1}$ (d) $2^{\frac{n}{2}}$

**15.** If three successive terms of a G..P. with common ratio $r$ $(r > 1)$ form the sides of a $\Delta$ ABC and [r] denotes greatest integer function, then [r] + [– r] =

(a) 0 (b) 1

(c) – 1 (d) None of these

**16.** If $a$, $b$, $c$, are in A.P. and $p$, $p'$ are respectively A.M. and G.M. between $a$ and $b$ while $q$, $q'$ are respectively AM. and G.M. between b and c, then

(a) $p^2 + q^2 = p'^2 + q'^2$

(b) $pq = p'q'$

(c) $p^2 - q^2 = p'^2 - q'^2$

(d) $p^2 + p'^2 = q^2 + q'^2$

**17.** The sum of the series

$1+2.2+3.2^2+4.2^3+5.2^4+...+100.2^{99}$ is

(a) $99.2^{100}-1$ (b) $100.2^{100}$

(c) $99.2^{100}$ (d) $99.2^{100}+1$

**18.** If $a$, $b$, $c$, $d$ are positive real number such that $a+b+c+d=2$, then $M=(a+b)(c+d)$ satisfies the relation:

(a) $0 < M \leq 1$ (b) $1 \leq M \leq 2$

(c) $2 \leq M \leq 3$ (d) $3 \leq M \leq 4$

**19.** The sum of $\frac{3}{1.2}\cdot\frac{1}{2}+\frac{4}{2.3}\cdot\left(\frac{1}{2}\right)^2+\frac{5}{3.4}\cdot\left(\frac{1}{2}\right)^3$ + ......... to $n$ terms is equal to

(a) $1-\frac{1}{(n+1)2^n}$ (b) $n-\frac{1}{2^{n+1}}$

(c) $1-\frac{1}{n.2^{n+1}}$ (d) None of these

**20.** Let $ax^2+\frac{b}{x} \geq c$ for all positive $x$, where a < 0 and $b < 0$. The value of the expression $27ab^2$ cannot be less than

(a) $4c^3$ (b) $4c^2$ (c) $8c^3$ (d) $c^3$

## Numeric Value Answer

**21.** Sum of infinite number of terms of GP is 20 and sum of their square is 100. The common ratio of GP is

**22.** Three numbers $a$, $b$, $c$ are in GP. If $a$, $b$, $c - 64$ are in AP and $a$, $b - 8$, $c - 64$ are in GP, then the sum of the numbers may be

**23.** $a$, $b$, $c$ are positive integers forming an increasing G.P. and $b - a$ is a perfect cube and $\log_6 a + \log_6 b + \log_6 c = 6$, then $a + b + c =$

**24.** The sum to infinite term of the series

$1+\frac{2}{3}+\frac{6}{3^2}+\frac{10}{3^3}+\frac{14}{3^4}+...$ is

**25.** The 20th term of the series 2 + 3 + 5 + 9 + 16 +.......is

**26.** Two consecutive numbers from 1, 2, 3,.........., $n$ are removed. If the arithmetic mean of the remaining numbers is 105/4 then $\frac{n}{10}$ is equal to

**27.** Let a, b, c, d be four distinct real numbers in A.P. Then half of the smallest positive value of k satisfying $2(a-b)+k(b-c)^2+(c-a)^3 = 2(a-d)+(b-d)^2+(c-d)^3$ is ..... .

**28.** Let $x_1, x_2, ... \in (0, \pi)$ denote the of values of $x$ satisfying the equation

$27^{(1+|\cos x|+\cos^2 x+|\cos x|^3+....\text{upto}\infty)} = 9^3$, find the value of $\frac{1}{\pi}(x_1+x_2+...)$

**29.** For $a, b > 0$, let $5a - b$, $2a + b$, $a + 2b$ be in A.P. and $(b+1)^2$, $ab+1$, $(a-1)^2$ are in G.P., then the value of $(a^{-1} + b^{-1})$ is ..... .

**30.** If one geomteric mean G and two Arithmetic means P and q be inserted between two quantities, then $G^2 = (kp - q)(kq - p)$ then find $k$.

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (d) | **4** | (c) | **7** | (d) | **10** | (c) | **13** | (b) | **16** | (c) | **19** | (a) | **22** | (124) | **25** | (990) | **28** | (1) |
| **2** | (b) | **5** | (b) | **8** | (d) | **11** | (c) | **14** | (c) | **17** | (d) | **20** | (a) | **23** | (189) | **26** | (5) | **29** | (6) |
| **3** | (c) | **6** | (b) | **9** | (d) | **12** | (c) | **15** | (c) | **18** | (a) | **21** | (0.60) | **24** | (3) | **27** | (8) | **30** | (2) |

# 10

# STRAIGHT LINES

## MCQs with One Correct Answer

**1.** Through the point $P(\alpha, \beta)$, where $\alpha\beta > 0$, the straight line $\frac{x}{a}+\frac{y}{b}=1$ is drawn so as to form with coordinate axes a triangle of area *S*. If $ab > 0$, then least value of *S* is

(a) $2\alpha\beta$ (b) $\frac{1}{2}\alpha\beta$

(c) $\alpha\beta$ (d) None of these

**2.** The vertices of a triangle are $\left(ab, \frac{1}{ab}\right)$, $\left(bc, \frac{1}{bc}\right)$ and $\left(ca, \frac{1}{ca}\right)$ where *a*, *b*, *c* are the roots of the equation $x^3 - 3x^2 + 6x + 1 = 0$. The coordinates of its centroid are.

(a) $(1, 2)$ (b) $(2, -1)$ (c) $(1, -1)$ (d) $(2, 3)$

**3.** Consider points $A$ (3, 4) and $B$ (7, 13). If *P* be a point on the line $y = x$ such that $PA + PB$ is minimum, then coordinates of *P* are

(a) $\left(\frac{12}{7}, \frac{12}{7}\right)$ (b) $\left(\frac{13}{7}, \frac{13}{7}\right)$

(c) $\left(\frac{31}{7}, \frac{31}{7}\right)$ (d) $(0, 0)$

**4.** If the straight lines $2x + 3y - 1 = 0$, $x + 2y - 1 = 0$ and $ax + by - 1 = 0$ form a triangle with origin as orthocentre, then $(a, b)$ is given by

(a) $(6, 4)$ (b) $(-3, 3)$ (c) $(-8, 8)$ (d) $(0, 7)$

**5.** The line $\frac{x}{a}+\frac{y}{b}=1$ meets the axis of *x* and *y* at *A* and *B* respectively and the line $y = x$ at *C* so that area of the trinagle *AOC* is twice the area of the triangle *BOC*, *O* being the origin, then one of the positions of *C* is

(a) $(a, a)$ (b) $\left(\frac{2a}{3}, \frac{2a}{3}\right)$

(c) $\left(\frac{b}{3}, \frac{b}{3}\right)$ (d) $\left(\frac{2b}{3}, \frac{2b}{3}\right)$

**6.** The range of values of $\beta$ such that $(0, \beta)$ lie on or inside the triangle formed by the lines $y + 3x + 2 = 0$, $3y - 2x - 5 = 0$, $4y + x - 14 = 0$ is

(a) $5 < \beta \le 7$ (b) $\frac{1}{2} \le \beta \le 1$

(c) $\frac{5}{3} \le \beta \le \frac{7}{2}$ (d) None of these

**7.** The intercepts on the straight line $y = mx$ by the lines $y = 2$ and $y = 6$ is less than 5, then *m* belongs to

(a) $\left(-\frac{4}{3}, \frac{4}{3}\right)$

(b) $\left(\frac{4}{3}, \frac{3}{8}\right)$

(c) $\left(-\infty, -\frac{4}{3}\right) \cup \left(\frac{4}{3}, \infty\right)$

(d) $\left(\frac{4}{3}, \infty\right)$

**8.** If three distinct points $A$, $B$, $C$ are given in the 2-dimensional coordinate plane such that the ratio of the distance of each one of them from the point (1, 0) to the distance from (– 1, 0) is equal to $\frac{1}{2}$, then the circumcentre of the triangle $ABC$ is at the point

(a) $\left(\frac{5}{3}, 0\right)$ (b) (0, 0)

(c) $\left(\frac{1}{3}, 0\right)$ (d) (3, 0)

**9.** Let A (–3, 2) and B (–2, 1) be the vertices of a triangle ABC. If the centroid of this triangle lies on the line $3x + 4y + 2 = 0$, then the vertex $C$ lies on the line :

(a) $4x + 3y + 5 = 0$ (b) $3x + 4y + 3 = 0$

(c) $4x + 3y + 3 = 0$ (d) $3x + 4y + 5 = 0$

**10.** The circumcentre of a triangle lies at the origin and its centroid is the mid point of the line segment joining the points $(a^2 + 1, a^2 + 1)$ and $(2a, – 2a)$, $a \neq 0$. Then for any a, the orthocentre of this triangle lies on the line:

(a) $y – 2ax = 0$

(b) $y – (a^2 + 1)x = 0$

(c) $y + x = 0$

(d) $(a – 1)^2x – (a + 1)^2y = 0$

**11.** The line parallel to the $x$- axis and passing through the intersection of the lines $ax + 2by + 3b = 0$ and $bx – 2ay – 3a = 0$, where $(a, b) \neq (0, 0)$ is

(a) below the $x$ - axis at a distance of $\frac{3}{2}$ from it

(b) below the $x$ - axis at a distance of $\frac{2}{3}$ from it

(c) above the $x$ - axis at a distance of $\frac{3}{2}$ from it

(d) above the $x$ - axis at a distance of $\frac{2}{3}$ from it

**12.** The straight line $y = x – 2$ rotates about a point where it cuts the x-axis and becomes perpendicular to the straight line $ax + by + c = 0$. Then its equation is

(a) $ax + by + 2a = 0$ (b) $ax – by – 2a = 0$

(c) $bx + ay – 2b = 0$ (d) $ay – bx + 2b = 0$

**13.** If the point $(a, 2)$ lies between the lines $x – y – 1 = 0$ and $2 (x – y) + 5 = 0$, then the set of values of '$a$' is

(a) $(– \infty , 3) \cup \left(\frac{9}{2}, \infty\right)$

(b) $\left(3, \frac{9}{2}\right)$

(c) $(– \infty , 3)$

(d) $\left(-\frac{1}{2}, 3\right)$

**14.** If two vertices of a triangle are (5, –1) and (–2, 3) and its orthocentre is at (0, 0), then the third vertex is

(a) (4, – 7) (b) (– 4, – 7)

(c) (– 4, 7) (d) (4, 7)

**15.** The base of an equilateral triangle is along the line given by $3x + 4y = 9$. If a vertex of the triangle is (1, 2), then the length of a side of the triangle is:

(a) $\frac{2\sqrt{3}}{15}$ (b) $\frac{4\sqrt{3}}{15}$

(c) $\frac{4\sqrt{3}}{5}$ (d) $\frac{2\sqrt{3}}{5}$

**16.** The equation of bisector of that angle between the lines $x + y + 1 = 0$ and $2x – 3y – 5 = 0$ which contains the point (10, – 20) is

(a) $x(\sqrt{13} + 2\sqrt{2}) + y(\sqrt{13} - 3\sqrt{2}) + (\sqrt{13} - 5\sqrt{2}) = 0$

(b) $x(\sqrt{13} - 2\sqrt{2}) + y(\sqrt{13} + 3\sqrt{2}) + (\sqrt{13} + 5\sqrt{2}) = 0$

(c) $x(\sqrt{13} + 2\sqrt{2}) + y(\sqrt{13} + 3\sqrt{2}) + (\sqrt{13} + 5\sqrt{2}) = 0$

(d) None of these

**17.** The bisector of the acute angle formed between the lines $4x-3y+7=0$ and $3x-4y+14=0$ has the equation :

(a) $x+y+3=0$ (b) $x-y-3=0$

(c) $x-y+3=0$ (d) $3x+y-7=0$

## Numeric Value Answer

**18.** A straight line through the origin $O$ meets the parallel lines $4x + 2y = 9$ and $2x + y + 6 = 0$ at points $P$ and $Q$ respectively. If the point $O$ divides the segment $PQ$ in the ratio $\frac{m}{n}$, then m + n is ________.

**19.** The vertex of an equilateral triangle is (2, –1), and the equation of its base is $x + 2y = 1$. If the length of its sides is $2/\sqrt{K}$, then value of K is ____.

**20.** If $(\sin\theta, \cos\theta)$, $\theta \in [0, 2\pi]$ and (1, 4) lie on the same side or on the line $\sqrt{3}x - y + 1 = 0$, then the maximum value of $\sin\theta$ will be ______.

**21.** The straight lines $(3\sec\theta+5\operatorname{cosec}\theta)x$ $+(7\sec\theta-3\operatorname{cosec}\theta)y+11(\sec\theta-\operatorname{cosec}\theta)=0$ always pass through a fixed point $P$ for all possible values of $\theta$. If the maximum value of the difference of distances of $P$ and $B$ (3, 4) from a point on the line $x-y+3=0$ is $k$ then $\frac{k^2}{10}$ is equal to .

**22.** The straight line $L \equiv x+y+1=0$ and $L_1 \equiv x+2y+3=0$ are intersecting. $m$ is the slope of the straight line $L_2$ such that $L$ is the bisector of the anlge between $L_1$ and $L_2$. The unit digit of $812m^2+3$ is equal to

**23.** If $\tan\alpha$, $\tan\beta$, $\tan\lambda$ are the roots of the equation $t^3 - 12t^2 + 15t - 1 = 0$; then the centroid of triangle having vertices $(\tan\alpha, \cot\alpha)$; $(\tan\beta, \cot\beta)$; $(\tan\lambda, \cot\lambda)$ is given by $G(h, k)$; then evaluate $(h + k)/(k - h)$.

**24.** Consider a $\Delta ABC$ whose sides $AB$, $BC$, and $CA$ are represented by the straight lines $2x + y = 0$, $x + py = q$, and $x - y = 3$, respectively. The point $P(2, 3)$ is the orthocenter. The value of $(p + q)/10$ is ..... .

**25.** In $\Delta ABC$, the vertex $A = (1, 2)$, $y = x$ is the perpendicular bisector of the side $AB$ and $x - 2y + 1 = 0$ is the equation of the internal angle bisector of $\underline{|L}$. If the equation of the side $BC$ is $ax + by - 5 = 0$, then the value of $a - b$ is ..... .

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (a) | **4** | (c) | **7** | (c) | **10** | (d) | **13** | (d) | **16** | (a) | **19** | (15) | **22** | (1) | **25** | (4) |
| **2** | (b) | **5** | (d) | **8** | (a) | **11** | (a) | **14** | (b) | **17** | (c) | **20** | (0) | **23** | (9) | | |
| **3** | (c) | **6** | (c) | **9** | (b) | **12** | (d) | **15** | (b) | **18** | (7) | **21** | (4) | **24** | (5) | | |

# Conic Sections

# 11

## MCQs with One Correct Answer

1. The line $4x+3y-4=0$ divides the circumference of the circle centered at (5, 3), in the ratio 1 : 2. Then the equation of the circle is
   (a) $x^2+y^2-10x-6y-66=0$
   (b) $x^2+y^2-10x-6y+100=0$
   (c) $x^2+y^2-10x-6y+66=0$
   (d) $x^2+y^2-10x-6y-100=0$
2. Let $A(-4, 0)$ and $B(4, 0)$. Then the number of points $C=(x, y)$ on the circle $x^2+y^2=16$ lying in first quadrant such that the area of the triangle whose vertices are $A$, $B$ and $C$ is a integer is
   (a) 14 (b) 15
   (c) 16 (d) None of these
3. If $(\alpha, \beta)$ is a point on the circle whose centre is on the x-axis and which touches the line $x+y=0$ at (2, –2), then the greatest value of $\alpha$ is
   (a) $4-\sqrt{2}$ (b) 6
   (c) $4+2\sqrt{2}$ (d) $4+\sqrt{2}$
4. The set of values of '$c$' so that the equations $y=|x|+c$ and $x^2+y^2-8|x|-9=0$ have no solution is
   (a) $(-\infty, -3)\cup(3, \infty)$
   (b) $(-3, 3)$
   (c) $(-\infty, -\sqrt{2})\cup(5\sqrt{2}, \infty)$
   (d) $(5\sqrt{2}-4, \infty)$
5. Tangents are drawn from $O$ (origin) to touch the circle $x^2+y^2+2gx+2fy+c=0$ at points $P$ and $Q$. The equation of the circle circumscribing triangle $OPQ$ is
   (a) $2x^2+2y^2+gx+fy=0$
   (b) $x^2+y^2+gx+fy=0$
   (c) $x^2+y^2+2gx+2fy=0$
   (d) None of these
6. A ray of light incident at the point $(-2, -1)$ gets reflected from the tangent at $(0, -1)$ to the circle $x^2+y^2=1$. The reflected ray touches the circle. The equation the line along which the incident ray moved, is
   (a) $4x-3y+11=0$ (b) $4x+3y+11=0$
   (c) $3x+4y+11=0$ (d) $4x+3y+7=0$
7. If the line $y=mx+1$ meets the circle $x^2+y^2+3x=0$ in two points equidistant from and on opposite sides of $x$-axis, then
   (a) $3m+2=0$ (b) $3m-2=0$
   (c) $2m+3=0$ (d) $2m-3=0$
8. If a circle passes through the point $(a, b)$ and cuts the circle $x^2+y^2=4$ orthogonally, then the locus of its centre is
   (a) $2ax-2by-(a^2+b^2+4)=0$
   (b) $2ax+2by-(a^2+b^2+4)=0$
   (c) $2ax-2by+(a^2+b^2+4)=0$
   (d) $2ax+2by+(a^2+b^2+4)=0$

**9.** The set of all real values of $\lambda$ for which exactly two common tangents can be drawn to the circles $x^2 + y^2 - 4x - 4y + 6 = 0$ and $x^2 + y^2 - 10x - 10y + \lambda = 0$ is the interval:

(a) $(12, 32)$ (b) $(18, 42)$
(c) $(12, 24)$ (d) $(18, 48)$

**10.** A circle bisects the circumference of the circle $x^2 + y^2 - 2y - 3 = 0$ and touches the line $x = y$ and the point (1, 1). Its radius is :

(a) $\frac{3}{\sqrt{2}}$ (b) $\frac{9}{\sqrt{2}}$ (c) $4\sqrt{2}$ (d) $3\sqrt{2}$

**11.** Let $L_1$ be the length of the common chord of the curves $x^2 + y^2 = 9$ and $y^2 = 8x$, and $L_2$ be the length of the latus rectum of $y^2 = 8x$, then:

(a) $L_1 > L_2$ (b) $L_1 = L_2$
(c) $L_1 < L_2$ (d) $\frac{L_1}{L_2} = \sqrt{2}$

**12.** If the tangent at the point $P(x_1, y_1)$ to the parabola $y^2 = 4ax$ meets the parabola $y^2 = 4a(x + b)$ at $Q$ and $R$, then the mid-point of $QR$ is

(a) $(x_1 + b, y_1 + b)$ (b) $(x_1 - b, y_1 - b)$
(c) $(x_1, y_1)$ (d) $(x_1 + b, y_1 - b)$

**13.** Tangent to the curve $y = x^2 + 6$ at a point (1, 7) touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at a point $Q$. Then the coordinates of $Q$ are

(a) $(-6, -11)$ (b) $(-9, -13)$
(c) $(-10, -15)$ (d) $(-6, -7)$

**14.** A circle is drawn with centre at the focus $S$ of the parabola $y^2 = 4x$ so that a common chord of the parabola and the circle is equidistant from the focus and the vertex. Then the equation of the circle is

(a) $(x - 1)^2 + y^2 = \frac{9}{4}$ (b) $(x - 1)^2 = \frac{9}{16} - y^2$
(c) $(x - 1)^2 + x^2 = \frac{9}{4}$ (d) $(y - 1)^2 + x^2 = \frac{9}{16}$

**15.** Locus of all such points so that sum of its distances from (2, – 3) and (2, 5) is always 10, is

(a) $\frac{(x-2)^2}{25} + \frac{(y-1)^2}{9} = 1$
(b) $\frac{(x-2)^2}{25} + \frac{(y-1)^2}{16} = 1$
(c) $\frac{(x-2)^2}{16} + \frac{(y-1)^2}{25} = 1$
(d) $\frac{(x-2)^2}{9} + \frac{(y-1)^2}{25} = 1$

**16.** The radius of the circle passing through the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, and having its centre at (0, 3) is

(a) 4 (b) 3 (c) $\sqrt{\frac{1}{2}}$ (d) $\frac{7}{2}$

**17.** Equation of the line passing through the points of intersection of the parabola $x^2 = 8y$ and the ellipse $\frac{x^2}{3} + y^2 = 1$ is :

(a) $y - 3 = 0$ (b) $y + 3 = 0$
(c) $3y + 1 = 0$ (d) $3y - 1 = 0$

**18.** Equation of the largest circle with centre (1, 0) that can be inscribed in the ellipse $x^2 + 4y^2 = 16$, is

(a) $2x^2 + 2y^2 - 4x + 7 = 0$
(b) $x^2 + y^2 - 2x + 5 = 0$
(c) $3x^2 + 3y^2 - 6x - 8 = 0$
(d) None of these

**19.** The normal at $\left(2, \frac{3}{2}\right)$ to the ellipse, $\frac{x^2}{16} + \frac{y^2}{3} = 1$ touches a parabola, whose equation is

(a) $y^2 = -104x$ (b) $y^2 = 14x$
(c) $y^2 = 26x$ (d) $y^2 = -14x$

**20.** The angle subtended by the common tangent of the two ellipse $\frac{(x-4)^2}{25} + \frac{y^2}{4} = 1$ and $\frac{(x+1)^2}{1} + \frac{y^2}{4} = 1$ at the origin is

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$

## Numeric Value Answer

**21.** Two equal chords $AB$ and $AC$ of the circle $x^2 + y^2 - 6x - 8y - 24 = 0$ are drawn from the point $A\left(\sqrt{33}+3,0\right)$. Another chord PQ is drawn intersecting $AB$ and $AC$ at points $R$ and $S$, respectively given that $AR = SC = 7$ and $RB = AS = 3$. The value of $PR/QS$ is

**22.** If p and q be the longest and the shortest distance respectively of the point (–7, 2) from any point (α, β) on the curve whose equation is $x^2 + y^2 - 10x - 14y - 51 = 0$ and G.M. of p and q is $2\sqrt{k}$, then value k is _______.

**23.** The straight line $y = mx + c$ $(m > 0)$ touches the parabolas $y^2 = 8(x+2)$ then the minimum value taken by $c$ is

**24.** Two tangents are drawn from a point (–2, –1) to the curve, $y^2 = 4x$. If α is the angle between them, then |tan α| is equal to:

**25.** Tangents are drawn to the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$ at ends of latus rectum. The area of quadrilateral so formed is

**26.** A trapezium is inscribed in the parabola $y^2 = 4x$ such that its diagonal pass through the point (1, 0) and each has length $\frac{25}{4}$. If the area of trapezium be $P$ then $\left[\frac{P}{4}\right]$ is equal to

**27.** $S_1$ and $S_2$ be the foci of the hyperbola whose transverse axis length is 4 and conjugate axis length is 6, $S_3$ and $S_4$ be the foci of the conjugate hyperbola. If the area of the quadrilateral $S_1 S_3 S_2 S_4$ is A, then find $\frac{A}{13}$.

**28.** If the ratio of the area of equilateral triangles made of the common chord of the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 8x + 4 = 0$ and their respective pairs of tangents drawn from points on the positive x- axis is $57+24\sqrt{3}:k$ then $k$ is ________.

**29.** $P(a,b)$ is a points in the first quadrant. Circles are drawn through $P$ touching the coordinate axes, such that the length of common chord of these circle is maximum. If possible values of $a/b$ is $k_1 \pm k_2\sqrt{2}$ then $k_1 + k_2$ is equal to______.

**30.** $C$ is the centre of the hyperbola $\frac{x^2}{4}-\frac{y^2}{1}=1$, and '$A$' is any point on it. The tangent at A to the hyperbola meets the line $x-2y=0$ and $x+2y=0$ at $Q$ and $R$ respectively. The value of $CQ.CR$ is equal to

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | (a) | 5 | (b) | 9 | (b) | 13 | (d) | 17 | (d) | 21 | (1) | 25 | (27) | 29 | (5) | | |
| 2 | (b) | 6 | (b) | 10 | (b) | 14 | (a) | 18 | (c) | 22 | (11) | 26 | (4) | 30 | (5) | | |
| 3 | (c) | 7 | (b) | 11 | (c) | 15 | (d) | 19 | (a) | 23 | (4) | 27 | (2) | | | | |
| 4 | (d) | 8 | (b) | 12 | (c) | 16 | (a) | 20 | (a) | 24 | (3) | 28 | (9) | | | | |

# LIMITS & DERIVATIVES

12

## MCQs with One Correct Answer

**1.** Given a real valued function '$f$' such that

$$f(x)=\begin{cases}\dfrac{\tan^2\{x\}}{x^2-[x]^2} & \text{for } x>0\\ 1 & \text{for } x=0\\ \sqrt{\{x\}\cot\{x\}} & \text{for } x<0\end{cases}$$

then

(a) LHL = 1

(b) RHL = $\sqrt{\cot 1}$

(c) $\lim_{x\to 0} f(x)$ exist

(d) $\lim_{x\to 0} f(x)$ does not exists

**2.** $\lim_{x\to\infty}\left(\dfrac{x^2+5x+3}{x^2+x+2}\right)^x$

(a) $e^4$ (b) $e^2$ (c) $e^3$ (d) 1

**3.** If $3-\dfrac{x^2}{12}\le f(x)\le 3+\dfrac{x^3}{9}$ for all $x\ne 0$, then the value of $\lim_{h\to 0} f(x)$ is equal to

(a) 1/3 (b) 3 (c) –3 (d) –1/3

**4.** Given that

$$\lim_{n\to\infty}\sum_{r=1}^{n}\frac{\log(n+r)-\log n}{n}=2\left(\log 2-\frac{1}{2}\right),$$

then $\lim_{n\to\infty}\dfrac{1}{n^k}[(n+1)^k(n+2)^k...(n+n)^k]^{1/n}$ is equal to

(a) $\dfrac{4k}{e}$ (b) $\left(\dfrac{4}{e}\right)^{1/k}$ (c) $\left(\dfrac{4}{e}\right)^k$ (d) $\left(\dfrac{e}{4}\right)^k$

**5.** The integer n for which

$\lim_{x\to 0}\dfrac{(\cos x-1)(\cos x-e^x)}{x^n}$ is a finite non-zero number is

(a) 1 (b) 2 (c) 3 (d) 4

**6.** The value of $\lim_{x\to 0}\left((\sin x)^{1/x}+(1/x)^{\sin x}\right)$, where $x>0$ is

(a) 0 (b) –1 (c) 1 (d) 2

**7.** The limit $\lim_{x\to a}\left(2-\dfrac{a}{x}\right)^{\tan\left(\frac{\pi x}{2a}\right)}$ is equal to

(a) $e^{-\frac{a}{\pi}}$ (b) $e^{-\frac{2a}{\pi}}$ (c) $e^{-\frac{2}{\pi}}$ (d) 1

**8.** If α and β are the roots of the quadratic equation $ax^2+bx+c=0$, then

$$\lim_{x\to\frac{1}{\alpha}}\sqrt{\frac{1-\cos(cx^2+bx+a)}{2(1-\alpha x)^2}}=$$

(a) $\left|\dfrac{c}{2\alpha}\left(\dfrac{1}{\alpha}-\dfrac{1}{\beta}\right)\right|$ (b) $\left|\dfrac{c}{2\beta}\left(\dfrac{1}{\alpha}-\dfrac{1}{\beta}\right)\right|$

(c) $\left|\dfrac{c}{\alpha\beta}\left(\dfrac{1}{\alpha}-\dfrac{1}{\beta}\right)\right|$ (d) None of these

**9.** $\lim_{x\to 0}\dfrac{\cos^2(1-\cos^2(1-\cos^2(............-\cos^2(x))))..........)}{\sin\left\{\pi\left(\dfrac{\sqrt{x+4}-2}{x}\right)\right\}}$

is equal to

(a) $\dfrac{\pi}{4}$ (b) $\dfrac{4}{\pi}$ (c) $\dfrac{\pi}{2}$ (d) $\dfrac{2}{\pi}$

**10.** The value of
$$\lim_{x\to\pi/2}[1^{1/\cos^2 x}+2^{1/\cos^2 x}+.....+n^{1/\cos^2 x}]^{\cos^2 x}$$
is

(a) 0 (b) $n$ (c) $\infty$ (d) $\frac{n(n+1)}{2}$

**11.** $\lim_{n\to\infty}\{\log_{n-1}(n)\log_n(n+1)\log_{n+1}(n+2)...$
$...\log_{n^k-1}(n^x)\}$ is equal to

(a) $n$ (b) $k$
(c) $\infty$ (d) None of these

**12.** If $[x]$ denotes the greatest integer $\le x$, then $\lim_{n\to\infty}\frac{1}{n^3}\{[1^2x]+[2^2x]+[3^2x]+...+[n^2x]\}$ equals

(a) $x/2$ (b) $x/3$ (c) $x/6$ (d) 0

**13.** If $f(x)=\begin{cases}\dfrac{(e^{(x+3)\ln 27})^{\frac{x}{27}}-9}{3^x-27}; & x<3\\ \lambda\dfrac{1-\cos(x-3)}{(x-3)\tan(x-3)}; & x>3\end{cases}$

If $\lim_{x\to3}f(x)$ exist, then $\lambda=$

(a) $\frac{9}{2}$ (b) $\frac{2}{9}$ (c) $\frac{2}{3}$ (d) None

**14.** The value of
$$\lim_{x\to0}\left\{\sin^2\left(\frac{\pi}{2-ax}\right)\right\}^{\sec^2\frac{\pi}{2-bx}}$$ is equal to

(a) $e^{-a/b}$ (b) $e^{-a^2/b^2}$
(c) $a^{2a/b}$ (d) $e^{4a/b}$

**15.** If $\lim_{x\to0}\frac{f(x)}{x^2}=a$ and $\lim_{x\to0}\frac{f(1-\cos x)}{g(x)\sin^2 x}=b$ (where $b\ne0$), then $\lim_{x\to0}\frac{g(1-\cos 2x)}{x^4}$ is

(a) $\frac{4a}{b}$ (b) $\frac{a}{4b}$ (c) $\frac{a}{b}$ (d) None

**16.** Let $f(x)$ be a polynomial function of second degree. If $f(1)=f(-1)$ and $a_1,a_2,a_3$ are in A.P. then $f'(a_1),f'(a_2),f'(a_3)$ are in

(a) A. P. (b) G.P.
(c) H. P. (d) None of these

**17.** Let $f(x)$ be a polynomial function satisfying $f(x)\cdot f\left(\frac{1}{x}\right)=f(x)+f\left(\frac{1}{x}\right)$. If $f(4)=65$ and $l_1,l_2,l_3$ are in GP, then $f'(l_1),f'(l_2),f'(l_3)$ are in

(a) AP (b) GP
(c) HP (d) None of these

## Numeric Value Answer

**18.** If $f(a)=2$, $f'(a)=1$, $g(a)=-1$, $g'(a)=2$, then the value of $\lim_{x\to a}\frac{g(x)f(a)-g(a)f(x)}{x-a}$ is

**19.** The value of $\lim_{x\to\frac{\pi}{2}}(\sin x)^{\tan x}$ is

**20.** The value of $\lim_{x\to\infty}\left[\sqrt{x+\sqrt{x+\sqrt{x}}}-\sqrt{x}\right]$ is

**21.** Let $f(x)$ be a function such that $\lim_{x\to0}\frac{f(x)}{x}=1$ and $\lim_{x\to0}\frac{x(1+a\cos x)-b\sin x}{\{f(x)^3\}}=1$, then $b-3a$ is equal to

**22.** The largest value of non-negative integer $a$ for which $\lim_{x\to1}\left\{\frac{-ax+\sin(x-1)+a}{x+\sin(x-1)-1}\right\}^{\frac{1-x}{1-\sqrt{x}}}=\frac{1}{4}$ is

**23.** The value of $\lim_{x\to\infty}\left\{\cfrac{x}{x+\cfrac{\sqrt[3]{x}}{x+\cfrac{\sqrt[3]{x}}{x+\sqrt[3]{x}}....\infty}}\right\}$ is

**24.** If $\alpha,\beta$ are two distinct real roots of the equation $ax^3+x-1-a=0$, $(a\ne-1,0)$, none of which is equal to unity, then the value of $\lim_{x\to(1/\alpha)}\frac{(1+\alpha)x^3-x^2-\alpha}{(e^{1-\alpha x}-1)(x-1)}$ is $\frac{al(k\alpha-\beta)}{\alpha}$. Find the value of $kl$.

**25.** $\lim_{x\to2}\frac{3^x+3^{3-x}-12}{3^{-x/2}-3^{1-x}}$ is equal to ________.

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | (d) | 4 | (c) | 7 | (c) | 10 | (b) | 13 | (c) | 16 | (a) | 19 | (1) | 22 | (2) | 25 | (36) |
| 2 | (a) | 5 | (c) | 8 | (a) | 11 | (b) | 14 | (a) | 17 | (b) | 20 | (0.50) | 23 | (1) | | |
| 3 | (b) | 6 | (c) | 9 | (b) | 12 | (b) | 15 | (c) | 18 | (5) | 21 | (6) | 24 | (1) | | |

# 13

# MATHEMATICAL REASONING

## MCQs with One Correct Answer

1. For the statement “17 is a real number or a positive integer”, the “or” is
   (a) Inclusive (b) Exclusive
   (c) Only (a) (d) None of these
2. Let $p$ and $q$ be any two logical statements and $r : p \to (\sim p \vee q)$. If $r$ has a truth value $F$, then the truth values of $p$ and $q$ are respectively :
   (a) F, F (b) T, T (c) T, F (d) F, T
3. If $p$ : Ashok works hard
   $q$ : Ashok gets good grade
   The verbal form for $(\sim p \to q)$ is
   (a) If Ashok works hard then gets good grade
   (b) If Ashok does not work hard then he gets good grade
   (c) If Ashok does not work hard then he does not get good grade
   (d) Ashok works hard if and only if he gets grade
4. If $p$ is false and $q$ is true, then
   (a) $p \wedge q$ is true (b) $p \vee \sim q$ is true
   (c) $q \wedge p$ is true (d) $p \Rightarrow q$ is true
5. $\sim p \wedge q$ is logically equivalent to
   (a) $p \to q$ (b) $q \to p$
   (c) $\sim (p \to q)$ (d) $\sim (q \to p)$
6. Which of the following is a contradiction?
   (a) $(p \wedge q) \wedge \sim (p \vee q)$ (b) $p \vee (\sim p \wedge q)$
   (c) $(p \Rightarrow q) \Rightarrow p$ (d) None of these
7. $(p \wedge \sim q) \wedge (\sim p \wedge q)$ is
   (a) A tautology
   (b) A contradiction
   (c) Both a tautology and a contradiction
   (d) Neither a tautology nor a contradiction
8. The false statement in the following is
   (a) $p \wedge (\sim p)$ is contradiction
   (b) $(p \Rightarrow q) \Leftrightarrow (\sim q \Rightarrow \sim p)$ is a contradiction
   (c) $\sim (\sim p) \Leftrightarrow p$ is a tautology
   (d) $p \vee (\sim p) \Leftrightarrow p$ is a tautology
9. The conditional $(p \wedge q) \Rightarrow p$ is
   (a) A tautology
   (b) A fallacy *i.e.*, contradiction
   (c) Neither tautology nor fallacy
   (d) None of these
10. If $p$ and $q$ are two statements, then $(p \Rightarrow q) \Leftrightarrow (-q \Rightarrow \sim p)$ is a
    (a) contradiction (b) tautology
    (c) neither (a) nor (b) (d) None of these
11. Which of the following is false?
    (a) $p \vee \sim p$ is a tautology
    (b) $\sim (\sim p) \leftrightarrow p$ is a tautology
    (c) $p \wedge \sim p$ is a contradiction
    (d) $((p \wedge q) \to q) \to p$ is a tautology
12. In the truth table for the statement $(p \to q) \leftrightarrow (\sim p \vee q)$, the last column has the truth value in the following order is
    (a) TTFF (b) FFFF
    (c) TTTT (d) FTFT

**13.** If $p \Rightarrow (\sim p \vee q)$ is false, then truth values of $p$ and $q$ are respectively

(a) F. T (b) F, F

(c) T, T (d) T, F

**14.** Negation of "2 + 3 = 5 and 8 < 10" is

(a) $2+3 \neq 5$ and $< 10$ (b) $2+3=5$ and $8 \nless 10$

(c) $2+3 \neq 5$ or $8 \nless 10$ (d) None of these

**15.** If the compound statement $p \to (\sim p \vee q)$ is false then the truth value of $p$ and $q$ are respectively

(a) T, T (b) T, F (c) F, T (d) F, F

**16.** The contrapositive of $p \to (\sim q \to \sim r)$ is

(a) $(\sim q \wedge r) \to \sim p$ (b) $(q \to r) \to \sim p$

(c) $(q \vee \sim r) \to \sim p$ (d) None of these

**17.** The negation of the compound proposition $p \vee (\sim p \vee q)$ is

(a) $(p \wedge \sim q) \wedge \sim p$ (b) $(p \wedge \sim q) \vee \sim p$

(c) $(p \vee \sim q) \vee \sim p$ (d) None of these

**18.** The inverse of the statement $(p \wedge \sim q) \to r$ is

(a) $\sim (p \vee \sim q) \to \sim r$ (b) $(\sim p \wedge q) \to \sim r$

(c) $(\sim p \vee q) \to \sim r$ (d) None of these

**19.** $\sim ((\sim p) \wedge q)$ is equal to

(a) $p \vee (\sim q)$ (b) $p \vee q$

(c) $p \wedge (\sim q)$ (d) $\sim p \wedge \sim q$

**20.** Which of the following is true?

(a) $p \Rightarrow q \equiv \sim p \Rightarrow \sim q$

(b) $\sim (p \Rightarrow \sim q) \equiv \sim p \wedge q$

(c) $\sim (\sim p \Rightarrow \sim q) \equiv \sim p \wedge q$

(d) $\sim (\sim p \Leftrightarrow q) \equiv [\sim (p \Rightarrow q) \wedge \sim (q \Rightarrow p)]$

**21.** The negation of $(p \vee q) \wedge (p \vee \sim r)$ is

(a) $(\sim p \wedge \sim q) \vee (q \wedge \sim r)$

(b) $(\sim p \wedge \sim q) \vee (\sim q \wedge r)$

(c) $(\sim p \wedge \sim q) \vee (\sim q \wedge r)$

(d) $(p \wedge q) \vee (\sim q \wedge \sim r)$

**22.** Let p, q and r be any three logical statements. Which of the following is true?

(a) $\sim [p \wedge (\sim q)] \equiv (\sim p) \wedge q$

(b) $\sim [(p \vee q) \wedge (\sim r) \equiv (\sim p) \vee (\sim q) \vee (\sim r)$

(c) $\sim [p \vee (\sim q)] \equiv (\sim p) \wedge q$

(d) $\sim [p \vee (\sim q)] \equiv (\sim p) \wedge \sim q$

**23.** Identify the false statements

(a) $\sim [p \vee (\sim q)] \equiv (\sim p) \vee q$

(b) $[p \vee q] \vee (\sim p)$ is a tautology

(c) $[p \wedge q) \wedge (\sim p)$ is a contradiction

(d) $\sim [p \vee q] \equiv (\sim p) \vee (\sim q)$

**24.** Negation of the statement $(p \wedge r) \to (r \vee q)$ is

(a) $\sim (p \wedge r) \to \sim (r \vee q)$

(b) $(\sim p \vee \sim r) \vee (r \vee q)$

(c) $(p \wedge r) \wedge (r \wedge q)$

(d) $(p \wedge r) \wedge (\sim r \wedge \sim q)$

**25.** Let $A$, $B$, $C$ and $D$ be four non-empty sets. The contrapositive statement of "If $A \subseteq B$ and $B \subseteq D$, then $A \subseteq C$" is:

(a) If $A \not\subseteq C$, then $A \subseteq B$ and $B \subseteq D$

(b) If $A \subseteq C$, then $B \subset A$ or $D \subset B$

(c) If $A \not\subseteq C$, then $A \not\subseteq B$ and $B \subseteq D$

(d) If $A \not\subseteq C$, then $A \not\subseteq B$ or $B \not\subseteq D$

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (a) | **4** | (d) | **7** | (b) | **10** | (b) | **13** | (d) | **16** | (a) | **19** | (a) | **22** | (c) | **25** | (d) |
| **2** | (c) | **5** | (d) | **8** | (b) | **11** | (b) | **14** | (c) | **17** | (a) | **20** | (c) | **23** | (d) | | |
| **3** | (b) | **6** | (a) | **9** | (a) | **12** | (c) | **15** | (b) | **18** | (c) | **21** | (c) | **24** | (d) | | |

# 14

# STATISTICS

## MCQs with One Correct Answer

**1.** The weighted mean of first $n$ natural numbers whose weights are equal to the number of selections out of $n$ natural numbers of corresponding numbers is

(a) $\frac{n.2^{n-1}}{2^n-1}$ (b) $\frac{3n(n+1)}{2(2n+1)}$

(c) $\frac{(n+1)(2n+1)}{6}$ (d) $\frac{n(n+1)}{2}$

**2.** The A.M. of $n$ observations is M. If the sum of $n-4$ observation is $a$, then the mean of remaining 4 observation is

(a) $\frac{nM-a}{4}$ (b) $\frac{nM+a}{2}$

(c) $\frac{nM-a}{2}$ (d) $nM+a$

**3.** The mean income of a group of persons is ₹ 400. Another group of persons has mean income ₹ 480. If the mean income of all the persons in the two groups together is ₹ 430, then ratio of the number of persons in the groups is

(a) $\frac{n_1}{n_2}=\frac{5}{3}$ (b) $\frac{n_1}{n_2}=\frac{2}{5}$

(c) $\frac{n_1}{n_2}=\frac{7}{4}$ (d) None of these

**4.** The mean of six numbers is 30. If one number is excluded, the mean of the remaining numbers is 29. The excluded number is

(a) 29 (b) 30 (c) 35 (d) 45

**5.** The mean of n items is $\overline{X}$. If the first item is increased by 1, second by 2 and so on, the new mean is :

(a) $\overline{X}+\frac{x}{2}$ (b) $\overline{X}+x$

(c) $\overline{X}+\frac{n+1}{2}$ (d) none of these

**6.** The median of 100 observations grouped in classes of equal width is 25. If the median class interval is 20 - 30 and the number of observations less than 20 is 45, then the frequency of median class is

(a) 10 (b) 20 (c) 15 (d) 12

**7.** If the mean deviation of the numbers 1, $1+d$, $1+2d$, .... $1+100d$ from their mean is 255, then $d$ is equal to :

(a) 20.0 (b) 10.1 (c) 20.2 (d) 10.0

**8.** In a series of $2n$ observations, half of them equal $a$ and remaining half equal $-a$. If the standard deviation of the observations is 2, then $|a|$ equals

(a) $\frac{\sqrt{2}}{n}$ (b) $\sqrt{2}$ (c) 2 (d) $\frac{1}{n}$

**9.** The standard deviations of two sets containing 10 and 20 members are 2 and 3 respectively measured from their common mean 5. The SD for the whole set of 30 members is

(a) $\frac{2}{\sqrt{3}}$ (b) $\sqrt{6}$

(c) $\sqrt{\left(\frac{22}{3}\right)}$ (d) $\sqrt{3}$

**10.** If M. D. is 12, the value of S.D. will be
(a) 15 (b) 12
(c) 24 (d) None of these

**11.** Suppose values taken by a variable $x$ are such that $a \le x_i \le b$, where $x_i$ denotes the value of $x$ in $i^{th}$ case for $i = 1, 2, \dots n$. Then
(a) $a \le \text{Var}(x) \le b$ (b) $a^2 \le \text{Var}(x) \le b^2$
(c) $\frac{a^2}{4} \le \text{Var}(x)$ (d) $(b-a)^2 \ge \text{Var}(x)$

**12.** If the median and the range of four numbers $\{x, y, 2x+y, x-y\}$, where $0 < y < x < 2y$, are 10 and 28 respectively, then the mean of the numbers is :
(a) 18 (b) 10 (c) 5 (d) 14

**13.** Let $\overline{X}$ and M.D. be the mean and the mean deviation about $\overline{X}$ of n observations $x_i$, i = 1, 2, ........, n. If each of the observations is increased by 5, then the new mean and the mean deviation about the new mean, respectively, are :
(a) $\overline{X}$, M.D. (b) $\overline{X}+5$, M.D.
(c) $\overline{X}$, M.D. + 5 (d) $\overline{X}+5$, M.D. + 5

**14.** The mean and variance for first n natural numbers are respectively
(a) mean $= \frac{n+1}{2}$, variance $= \frac{n^2-1}{12}$
(b) mean $= \frac{n-1}{2}$, variance $= \frac{n^2+1}{12}$
(c) mean $= \frac{n^2-1}{12}$, variance $= \frac{n+1}{2}$
(d) mean $= \frac{n^2+1}{2}$, variance $= \frac{n-1}{2}$

**15.** The variance of 20 observations is 5. If each observation is multiplied by 2, then the new variance of the resulting observation is
(a) $2^3 \times 5$ (b) $2^2 \times 5$ (c) $2 \times 5$ (d) $2^4 \times 5$

**16.** All the students of a class performed poorly in Mathematics. The teacher decided to give grace marks of 10 to each of the students. Which of the following statistical measures will not change even after the grace marks were given ?
(a) mean (b) median
(c) mode (d) variance

**17.** Coefficient of variation of two distribution are 60 and 70, and their standard deviations are 21 and 16, respectively. What are their arithmetic means?
(a) 35, 22.85 (b) 22.85, 35.28
(c) 36, 22.85 (d) 35.28, 23.85

## Numeric Value Answer

**18.** Variance of the data 2, 4, 5, 6, 8, 17 is 23.33. Then, variance of 4, 8, 10, 12, 16, 34 will be

**19.** Coefficient of variation of two distributions are 50 and 60 and their arithmetic means are 30 and 25, respectively. Then, difference of their standard deviations is

**20.** In an experiment with 15 observations on $x$, the following results were available:
$$\sum x^2 = 2830, \sum x = 170$$
One observation that was 20 was found to be wrong and was replaced by the correct value 30. The corrected variance is

**21.** Consider the following data
1, 2, 3, 4, 5, 6, 7, 8, 9, 10
If 1 is added to each number, then variance of the numbers so obtained is

**22.** Coefficient of variation of two distribution are 50% and 60% and their standard deviation are 10 and 15, respectively. Then, difference of their arithmetic means is

**23.** The mean of 5 observation is 4.4 and their variance is 8.24. If three of the observations are 1, 2 and 6, then difference of the other two observations is

**24.** If the variance of the first $n$ natural numbers is 10 and the variance of the first $m$ even natural numbers is 16, then $m + n$ is equal to _______.

**25.** If the mean and variance of eight numbers 3, 7, 9, 12, 13, 20, $x$ and $y$ be 10 and 25 respectively, then $x \times y$ is equal to ____.

**ANSWER KEY**

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | (a) | 4 | (c) | 7 | (b) | 10 | (a) | 13 | (b) | 16 | (d) | 19 | (0) | 21 | (8.25) | 24 | (18) |
| 2 | (a) | 5 | (c) | 8 | (c) | 11 | (d) | 14 | (a) | 17 | (a) | 20 | (78) | 22 | (5) | 25 | (52) |
| 3 | (a) | 6 | (a) | 9 | (c) | 12 | (d) | 15 | (b) | 18 | (93.32) | | | 23 | (5) | | |

# 15

# PROBABILITY-1

## MCQs with One Correct Answer

**1.** If two numbers $p$ and $q$ are choosen randomly from the set $\{1, 2, 3, 4\}$ with replacement, then the probability that $p^2 \geq 4q$ is equal to

(a) $\frac{1}{4}$ (b) $\frac{3}{16}$ (c) $\frac{1}{2}$ (d) $\frac{7}{16}$

**2.** A natural number x is chosen at random from the first 100 natural numbers. Then the probability, for the equation $x + \frac{100}{x} > 50$ is

(a) $\frac{1}{20}$ (b) $\frac{11}{20}$ (c) $\frac{1}{3}$ (d) $\frac{3}{20}$

**3.** A person throws two dice, one the common cube and the other regular tetrahedron with numbers 1, 2, 3, 4 on its faces, the number on the lowest face being taken in the case of a tetrahedron.The chance that the sum of numbers thrown is not less than 5 is

(a) $\frac{1}{4}$ (b) $\frac{3}{4}$ (c) $\frac{4}{5}$ (d) $\frac{5}{6}$

**4.** $A$ is a set containing n element. $A$ subset $P$ of $A$ is chosen at random, and the set $A$ is reconstrued by replacing the elements of $P$. Another subset $Q$ of $A$ is now chosen at random. The probability that $P \cup Q$ contains exactly $r$ elements, $1 \leq r \leq n$, is

(a) $\frac{{}^nC_r 3^r}{4^n}$ (b) $\frac{3^n}{4^n}$

(c) $\frac{{}^nC_r 3^{n-r}}{4^n}$ (d) $\frac{{}^nC_r 3^n}{4^n}$

**5.** 10 persons sit around a circular table with 10 numbered chairs. The probability that the two particular persons $A$ and $B$ are always together is

(a) $\frac{2}{9}$ (b) $\frac{1}{5}$ (c) $\frac{1}{9}$ (d) $\frac{2}{5}$

**6.** 5 different games are to be distributed among 4 children randomly. The probability that each child gets atleast one game is

(a) $\frac{1}{4}$ (b) $\frac{15}{64}$ (c) $\frac{21}{64}$ (d) $\frac{1}{4}$

**7.** If three squares are chosen at random on a chess board, then find the probability that they should be in a diagonal line

(a) 9/743 (b) 11/743

(c) 13/743 (d) None of these

**8.** If the letters of the word MATHEMATICS are arranged arbitrarily, the probability that $C$ comes before $E$, $E$ before $H$, $H$ before $I$ and $I$ before $S$ is

(a) $\frac{1}{75}$ (b) $\frac{1}{24}$ (c) $\frac{1}{120}$ (d) $\frac{1}{720}$

**9.** In a game of chance a player throws a pair of dice and scores points equal to the difference between the numbers on the two dice. Winner is the person who scores exactly 5 points more than his opponent. If two players are playing this game only one time, then the probability that neither of them wins is

(a) $\frac{1}{54}$ (b) $\frac{1}{108}$ (c) $\frac{53}{54}$ (d) $\frac{107}{108}$

**10.** A box contains 2 fifty paise coins, 5 twenty five paise coins and 15 ten paise coins. Five coins are taken out of the box at random. Probability that the value of these five coins is less than one rupee and fifity paise is

(a) $\frac{170}{{}^{22}C_5}$ (b) $1 - \frac{150}{{}^{22}C_5}$

(c) $1 - \frac{170}{{}^{22}C_5}$ (d) None of these

**11.** Two distinct numbers $a$ and $b$ are chosen randomly from the set $\{2, 2^2, 2^3, .... 2^{25}\}$. Then the probability that $\log_a b$ is an integer is

(a) $\frac{131}{300}$ (b) $\frac{31}{300}$

(c) $\frac{21}{200}$ (d) $\frac{62}{300}$

**12.** There are two vans each having numbered seats, 3 in the front and 4 at the back. There are 3 girls and 9 boys to be seated in the vans. The probability of 3 girls sitting together in a back row on adjacent seats, is

(a) $\frac{1}{13}$ (b) $\frac{1}{39}$ (c) $\frac{1}{65}$ (d) $\frac{1}{91}$

**13.** For the three events $A$, $B$, and $C$, $P$ (exactly one of the events $A$ or $B$ occurs) = $P$ (exactly one of the two events $B$ or $C$ occurs) = $P$(exactly one of the events $C$ or $A$ occurs) = $p$ and $P$ (all the three events occur simultaneously) = $p^2$, where $0 < p < 1/2$. Then the probability of at least one of the three events $A$, $B$ and $C$ occurring is

(a) $\frac{3p+2p^2}{2}$ (b) $\frac{p+3p^2}{4}$

(c) $\frac{p+3p^2}{2}$ (d) $\frac{3p+2p^2}{4}$

**14.** There are three events $E_1$, $E_2$ and $E_3$. one of which must, and only one can happen. The odds are 7 to 4 against $E_1$ and 5 to 3 against $E_2$. The odds against $E_3$ is

(a) 4 : 11 (b) 3 : 8 (c) 23 : 88 (d) 65 : 23

**15.** The chance of an event happening is the square of the chance of a second event but the odds against the first are the cube of the odds against the second. The chances of the events are

(a) $\frac{1}{9}, \frac{1}{3}$ (b) $\frac{1}{16}, \frac{1}{4}$

(c) $\frac{1}{4}, \frac{1}{2}$ (d) None of these

**16.** Ashmit and Bishmit are two students who appeared in Class X exam, Probability that Ashmit will pass is 0.5 while probability that both of them will pass is at least 0.1 and at most 0.3, and probability that Bishmit will pass is P(B) then find the range of P(B)

(a) (0.2, 0.8) (b) (0.15, 0.25)

(c) (0.1, 0.8) (d) None of these

**17.** From a well-shuffled pack of 52 cards, four cards are selected at random. The probability of drawing exactly 2 spades and exactly 2 aces is

(a) $\frac{1494}{270725}$ (b) $\frac{1594}{270725}$

(c) $\frac{1296}{270725}$ (d) $\frac{1396}{270725}$

## Numeric Value Answer

**18.** The probability that in the random arrangement of the letters of the word 'UNIVERSITY', the two I's does not come together is

**19.** A four digit number is formed by the digits 1, 2, 3, 4 with no repetition. The probability that the number is odd, is

**20.** Two numbers $x$ and $y$ are chosen at random without replacement from amongst the numbers 1, 2, 3, .........,$3n$. The probability that $x^3 + y^3$ is divisible by 3 is

**21.** 5 girls and 10 boys sit at random in a row having 15 chairs numbered as 1 to 15. If the probability that the end seats are occupied by the girls and between any two girls odd number of boys take seat is $\frac{20}{n}$, then $\frac{n}{1001}$ is equal to

**22.** A quadratic equation is chosen from the set of all quadratic equations which are unchanged by squaring their roots. If the chance that the chosen equation has equal roots, is $\frac{p}{q}$, then p + q =

**23.** In a multiple choice question, there are five alternative answers of which one or more than one are correct. A candidate will get marks on the question, if he ticks all the correct answers. If he decides to tick answers at random, then the least number of choice should he be allowed so that the probability of his getting marks on the question exceeds $\frac{1}{8}$ is

**24.** A die is rolled three times, if $p$ be the probability of getting a large number than the previous number, then the value of $54p$ is

**25.** A class contains 20 boys and 20 girls of which half the boys and half the girls have cat eyes. If one student is selected from the class & if the probability that either the student is a boy or has cat eyes is $\frac{a}{b}$, then $\sqrt{a^2+b^2}$ =

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (d) | **4** | (b) | **7** | (d) | **10** | (c) | **13** | (a) | **16** | (c) | **19** | (0.50) | **22** | (3) | **25** | (5) |
| **2** | (b) | **5** | (a) | **8** | (c) | **11** | (b) | **14** | (d) | **17** | (a) | **20** | (0.33) | **23** | (4) | | |
| **3** | (b) | **6** | (b) | **9** | (c) | **12** | (d) | **15** | (a) | **18** | (0.80) | **21** | (3) | **24** | (5) | | |

# RELATIONS & FUNCTIONS-2

16

## MCQs with One Correct Answer

**1.** Let P be the relation defined on the set of all real numbers such that P = {(a, b) : $\sec^2 a - \tan^2 b = 1$}. Then P is:
(a) reflexive and symmetric but not transitive.
(b) reflexive and transitive but not symmetric.
(c) symmetric and transitive but not reflexive.
(d) an equivalence relation.

**2.** Consider the following relations. $R = \{(x, y) | x, y$ are real numbers and $x = wy$ for some rational number $w\}$ $S = \{\left(\frac{m}{n}, \frac{p}{q}\right) |$ $m, n, p$ and $q$ are integers such that, $q \neq 0, n \neq 0$ and $qm = pn\}$. Then
(a) Neither $R$ nor $S$ is an equivalence relation
(b) $S$ is an equivalence relation but $R$ is not an equivalence relation
(c) $R$ and $S$ both are equivalence relations
(d) $R$ is an equivalence relation but $S$ is not an equivalence relation

**3.** For real numbers $x$ and $y$, we define $xRy$ iff $x - y + \sqrt{5}$ is an irrational number. Then, relation $R$ is
(a) reflexive (b) symmetric
(c) transitive (d) None of these

**4.** If $f : R \to S$, defined by $f(x) = \sin x - \sqrt{3}\cos x + 1$, is onto, then the interval of $S$ is
(a) $[-1, 3]$ (b) $[-1, 1]$
(c) $[0, 1]$ (d) $[0, 3]$

**5.** If the function f : $(-\infty, \infty) \to$ B defined by $f(x) = -x^2 + 6x - 8$ is bijective, then B =
(a) $[1, \infty)$ (b) $(-\infty, 1]$
(c) $(-\infty, \infty)$ (d) None of these

**6.** Let $f : f : \{x, y, z\} \to \{1, 2, 3\}$ be a one-one mapping such that only one of the following three statements is true and remaining two are false : $f(x) \neq 2, f(y) = 3, f(z) \neq 1$, then –
(a) $f(x) > f(y) > f(z)$ (b) $f(x) < f(y) < f(z)$
(c) $f(y) < f(x) < f(z)$ (d) $f(y) < f(z) < f(x)$

**7.** Let $A \equiv \{1,2,3,4\}$ $B \equiv \{a, b, c\}$, then number of functions from $A \to B$, which are not onto is
(a) 8 (b) 24 (c) 45 (d) 6

**8.** A quadratic polynomial maps from [–2, 3] onto [0, 3] and touches $X$-axis at $x = 3$, then the polynomial is
(a) $\frac{3}{16}(x^2 - 6x + 16)$ (b) $\frac{3}{25}(x^2 - 6x + 9)$
(c) $\frac{3}{25}(x^2 - 6x + 16)$ (d) $\frac{3}{16}(x^2 - 6x + 9)$

**9.** Let $f : R^+ \to \{-1, 0, 1\}$ defined by $f(x) = \text{sgn}(x - x^4 + x^7 - x^8 - 1)$ where sgn denotes signum function. Then $f(x)$ is
(a) many-one and onto
(b) many-one and into
(c) one-one and onto
(d) one-one and into

**10.** Let $f : R \to R$ be a function defined by $f(x) = x + \sqrt{x^2}$, then $f$ is
(a) injective (b) surjective
(c) bijective (d) None of these

**11.** If $f(x) = \sin x + \cos x$, $g(x) = x^2 - 1$, then $g(f(x))$ is invertible in the domain
(a) $\left[0, \frac{\pi}{2}\right]$ (b) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$
(c) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (d) $[0, \pi]$

**12.** Let $g: R \to R$ be given by $g(x) = 3 + 4x$.

If $g^n(x) = gogo.......og\ (x)$, then $g^{-n}(x) =$

(where $g^{-n}(x)$ denotes inverse of $g^n(x)$)

(a) $(4^n - 1) + 4^n x$ (b) $(x+1)4^{-n} - 1$

(c) $(x+1)4^n - 1$ (d) $(4^{-n} - 1)x + 4^n$

**13.** If $f(x) = \sqrt{3|x| - x - 2}$ and $g(x) = \sin x$, then domain of definition of fog $(x)$ is

(a) $\left\{2n\pi + \frac{\pi}{2}\right\}_{n \in I}$

(b) $\bigcup_{n \in I}\left(2n\pi + \frac{7\pi}{6}, 2n\pi + \frac{11\pi}{6}\right)$

(c) $\left\{2n\pi + \frac{7\pi}{6}\right\}_{n \in I}$

(d) $\{(4m+1)\frac{\pi}{2} : m \in I\} \bigcup_{n \in I}\left[2n\pi + \frac{7\pi}{6}, 2n\pi + \frac{11\pi}{6}\right]$

**14.** If $f(x) = \sin^2 x + \sin^2\left(x + \frac{\pi}{3}\right) + \cos x \cos\left(x + \frac{\pi}{3}\right)$ and $g\left(\frac{5}{4}\right) = 1$, then graph of $y = g[f(x)]$ is

(a) a circle (b) a straight line

(c) a parabola (d) None of these

**15.** Let $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in R$. Then the set of all $x$ satisfying (f o g o g o f) (x) = (g o g o f) $(x)$, where (f o g) $(x) = f(g(x))$, is

(a) $\pm\sqrt{n\pi}, n \in \{0, 1, 2, ....\}$

(b) $\pm\sqrt{n\pi}, n \in \{1, 2, ....\}$

(c) $\frac{\pi}{2} + 2n\pi, n \in \{... -2, -1, 0, 1, 2....\}$

(d) $2n\pi, n \in \{... -2, -1, 0, 1, 2, ....\}$

**16.** Let $f: [0, 1] \to [-1, 1]$ and $g: [-1, 1] \to [0, 2]$ be two functions such that g is injective and $gof: [0, 1] \to [0, 2]$ is surjective. Then,

(a) $f$ must be injective but need not be surjective

(b) $f$ must be surjective but need not be injective

(c) $f$ must be bijective

(d) $f$ must be a constant function

**17.** Let $f: [0, 1] \to R$ be an injective continuous function that satisfies the condition $-1 < f(0) < f(1) < 1$.
Then, the number of functions $g: [-1, 1] \to [0, 1]$ such that $(gof)(x) = x$ for all $x \in [0, 1]$ is

(a) 0

(b) 1

(c) more than 1, but finite

(d) infinite

## Numeric Value Answer

**18.** Let E = {1, 2, 3, 4} and F = {1, 2}. Then the number of onto functions from E to F is

**19.** If for $x > 0, f(x) = (a - x^n)^{1/n}$,

$g(x) = x^2 + px + q; p, q \in R$

and the equation $g(x) - x = 0$ has imaginary roots, then number of real roots of equation $g(g(x)) - f(f(x)) = 0$ is

**20.** Let $g(x) = 1 + x - [x]$ and $f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$.

Then for all $x$, $f(g(x))$ is equal to

**21.** Let $g(x) = \frac{e^x - e^{-x}}{2}$ and $g(f(x)) = x$, then

$f\left(\frac{e^{22} - 1}{2e^{11}}\right) - 5 =$.

**22.** If $f(x) = \begin{cases} -x + 1, & x \le 0 \\ -(x-1)^2, & x \ge 1 \end{cases}$, then the number of solutions of $f(x) - f^{-1}(x) = 0$ is

**23.** If $f(x) = \cos^2 x + \cos^2\left(\frac{\pi}{3} + x\right) - \cos x \cos\left(\frac{\pi}{3} + x\right)$ and $g(3/4) = 2$, then find the value of $(gof)$ (1).

**24.** If a non-zero function $f(x)$ is symmetrical about $y = x$, then the value of $p$ (constant) such that

$f^2(x) = (f^{-1}(x))^2 - px \cdot f(x) \cdot f^{-1}(x) + 2x^2 f(x)$

for all $x \in R^+$ is ....... .

**25.** Let $f$ be defined on the natural numbers as follows: $f(1) = 1$ and for $n > 1$, $f(n) = f[f(n-1)] + f[n - f(n-1)]$, the value of $\frac{1}{30}\sum_{r=1}^{20} f(r)$ is

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (d) | **4** | (a) | **7** | (c) | **10** | (d) | **13** | (d) | **16** | (b) | **19** | (0) | **22** | (4) | **25** | (7) |
| **2** | (b) | **5** | (b) | **8** | (b) | **11** | (b) | **14** | (b) | **17** | (d) | **20** | (1) | **23** | (2) | | |
| **3** | (a) | **6** | (c) | **9** | (b) | **12** | (b) | **15** | (a) | **18** | (14) | **21** | (6) | **24** | (2) | | |

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# Inverse Trigonometric Functions

## MCQs with One Correct Answer

**1.** If $ax + b(\sec(\tan^{-1} x)) = c$ and $ay + b(\sec(\tan^{-1} y)) = c$, then $\dfrac{x+y}{1-xy} =$

(a) $\dfrac{ac}{a^2+c^2}$ (b) $\dfrac{2ac}{a-c}$

(c) $\dfrac{2ac}{a^2-c^2}$ (d) $\dfrac{a+c}{1-ac}$

**2.** The range of the function

$f(x) = \sin^{-1}\left[x^2 + \dfrac{1}{2}\right] + \cos^{-1}\left[x^2 - \dfrac{1}{2}\right]$, where [.] is the greatest integer function, is

(a) $\left\{\dfrac{\pi}{2}, \pi\right\}$ (b) $\left\{0, -\dfrac{1}{2}\right\}$

(c) $\{\pi\}$ (d) $\left(0, \dfrac{\pi}{2}\right)$

**3.** If $\sum_{r=1}^{k} \cos^{-1}\beta_r = \dfrac{k\pi}{2}$ for any $k \geq 1$ where $\beta_r \geq 0 \forall r$ and $A = \sum_{r=1}^{k} (\beta_r)^r$. Then

$\lim_{x \to A} \dfrac{(1+x^2)^{1/3} - (1-2x)^{1/4}}{x + x^2} =$

(a) $\dfrac{1}{2}$ (b) 0 (c) $\dfrac{A}{2}$ (d) $\dfrac{\pi}{2}$

**4.** The range of the function

$f(x) = \sin^{-1}(\log[x]) + \log(\sin^{-1}[x])$; (where [.] denotes the greatest integer function) is

(a) R (b) $[1, 2)$

(c) $\left\{\log\dfrac{\pi}{2}\right\}$ (d) $\{-\sin 1\}$

**5.** If $x_1, x_2, x_3, x_4$ are roots of the equation $x^4 - x^3 \sin 2\beta + x^2 \cos 2\beta - x\cos\beta - \sin\beta = 0$, then

$\sum_{i=1}^{4} \tan^{-1} x_i$ is equal to

(a) $\pi - \beta$ (b) $\pi - 2\beta$

(c) $\dfrac{\pi}{2} - \beta$ (d) $\dfrac{\pi}{2} - 2\beta$

**6.** The maximum value of f(x)

$= \tan^{-1}\left(\dfrac{\left(\sqrt{12}-2\right)x^2}{x^4 + 2x^2 + 3}\right)$ is

(a) 18º (b) 36º

(c) 22.5º (d) 15º

**7.** The number of solutions of the equation

$\left|\tan^{-1}|x|\right| = \sqrt{\left(x^2+1\right)^2 - 4x^2}$ is

(a) 2 (b) 3

(c) 4 (d) none of these

**8.** $S=\tan^{-1}\left(\frac{1}{n^2+n+1}\right)+\tan^{-1}\left(\frac{1}{n^2+3n+3}\right)+\ldots$
$+\tan^{-1}\left(\frac{1}{1+(n+19)(n+20)}\right)$, then $\tan S$ is equal to :

(a) $\frac{20}{401+20n}$ (b) $\frac{n}{n^2+20n+1}$
(c) $\frac{20}{n^2+20n+1}$ (d) $\frac{n}{401+20n}$

**9.** The number of roots of the equation

$\sin^{-1}x-\frac{1}{\sin^{-1}x}=\cos^{-1}x-\frac{1}{\cos^{-1}x}$ is

(a) 0 (b) 1
(c) 2 (d) 3

**10.** The minimum integral value of $\alpha$ for which the quadratic equation $(\cot^{-1}\alpha)x^2-(\tan^{-1}\alpha)^{3/2}x+2(\cot^{-1}\alpha)^2=0$ has both positive roots

(a) 1 (b) 2
(c) 3 (d) 4

**11.** The sum of series

$\sec^{-1}\sqrt{2}+\sec^{-1}\frac{\sqrt{10}}{3}+\sec^{-1}\frac{\sqrt{50}}{7}+\ldots+$

$\sec^{-1}\sqrt{\frac{(n^2+1)(n^2-2n+2)}{(n^2-n+1)^2}}$ is

(a) $\tan^{-1}1$ (b) $\tan^{-1}n$
(c) $\tan^{-1}(n+1)$ (d) $\tan^{-1}(n-1)$

**12.** The value of x satisfying the equation

$(\sin^{-1}x)^3-(\cos^{-1}x)^3+$

$(\sin^{-1}x)(\sin^{-1}x-\cos^{-1}x)=\frac{\pi^3}{16}$

is:

(a) $\cos\frac{\pi}{5}$ (b) $\cos\frac{\pi}{4}$
(c) $\cos\frac{\pi}{8}$ (d) $\cos\frac{\pi}{12}$

**13.** $\sin^{-1}\left(a-\frac{a^2}{3}+\frac{a^3}{9}+\ldots\right)$

$+\cos^{-1}(1+b+b^2+\ldots)=\frac{\pi}{2}$

when

(a) $a=-3$ & $b=1$ (b) $a=1$ & $b=-\frac{1}{3}$
(c) $a=\frac{1}{6}$ & $b=\frac{1}{2}$ (d) None of these

**14.** If $S_n$ denotes the sum to $n$ terms of the series

$\cot^{-1}\frac{7}{4}+\cot^{-1}\frac{19}{4}+\cot^{-1}\frac{39}{4}+\ldots$ then

(a) $S_n=\tan^{-1}\frac{n}{2n+5}$

(b) $S_n=\cot^{-1}\frac{n+5}{2n}$

(c) $S_n=\cot^{-1}\frac{4n}{2n+5}$

(d) $S_\infty=\cot^{-1}\frac{1}{2}$

**15.** Find : $\tan^{-1}(1/3)+\tan^{-1}(1/7)+\tan^{-1}(1/13)$

$+\ldots+\tan^{-1}\left(\frac{1}{n^2+n+1}\right)+\ldots\infty$

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

**16.** If the equation $x^3+bx^2+cx+1=0$ $(b<c)$ has only one real root $\alpha$.
Then the value of $2\tan^{-1}(\operatorname{cosec}\alpha)+\tan^{-1}(2\sin\alpha\sec^2\alpha)$ is:

(a) $-\frac{\pi}{2}$ (b) $-\pi$
(c) $\frac{\pi}{2}$ (d) $\pi$

**17.** Solution of equation $\cos^{-1}x\sqrt{3}+\cos^{-1}x=\frac{\pi}{2}$ is

(a) $x=\frac{1}{2}$ (b) $x=-\frac{1}{2}$
(c) $x=\pm\frac{1}{2}$ (d) None of these

**18.** The sum of infinite series

$$\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)+\sin^{-1}\left(\frac{\sqrt{2}-1}{\sqrt{6}}\right)+\sin^{-1}\left(\frac{\sqrt{3}-\sqrt{2}}{2\sqrt{2}}\right)$$

$$+...+\sin^{-1}\left(\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}}\right)+... \text{ is}$$

(a) $\frac{\pi}{2}$ (b) $\pi$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{4}$

**19.** The value of

$$\frac{\alpha^3}{2}\operatorname{cosec}^2\left(\frac{1}{2}\tan^{-1}\frac{\alpha}{\beta}\right)+\frac{\beta^3}{2}\sec^2\left(\frac{1}{2}\tan^{-1}\left(\frac{\beta}{\alpha}\right)\right)$$

is equal to

(a) $(\alpha-\beta)(\alpha^2+\beta^2)$ (b) $(\alpha+\beta)(\alpha^2-\beta^2)$

(c) $(\alpha+\beta)(\alpha^2+\beta^2)$ (d) None of these

**20.** $\cot^{-1}(2.1^2)+\cot^{-1}(2.2^2)+\cot^{-1}(2.3^2)+...$ is equal to

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{5}$

## Numeric Value Answer

**21.** $\lim_{x\to\infty}\left(\frac{\pi}{2}-\tan^{-1}x\right)^{1/x}$ equals

**22.** Find the value of

$$\tan^{-1}\left(\frac{1}{2}(\tan 2A)+\tan^{-1}(\cot A)+\tan^{-1}(\cot^3 A)\right),$$

when $\frac{\pi}{4}<A<\frac{\pi}{2}$

**23.** The number of ordered triplets (x, y, z) that satisfy the equation

$$(\sin^{-1}x)^2=\frac{\pi^2}{4}+(\sec^{-1}y)^2+(\tan^{-1}z)^2 \text{ is}$$

**24.** Find the number of solution of $\cos(2\sin^{-1}(\cot(\tan^{-1}(\sec(6\operatorname{cosec}^{-1}x)))))+1=0$; where $x>0$.

**25.** If the domain of the function

$f(x)=\sqrt{3\cos^{-1}(4x)-\pi}$ is [a, b], then the value of (4a + 64b) is _______.

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (c) | **4** | (c) | **7** | (c) | **10** | (b) | **13** | (b) | **16** | (b) | **19** | (c) | **22** | (0) | **25** | (7) |
| **2** | (c) | **5** | (c) | **8** | (c) | **11** | (b) | **14** | (d) | **17** | (a) | **20** | (a) | **23** | (2) | | |
| **3** | (a) | **6** | (d) | **9** | (c) | **12** | (c) | **15** | (c) | **18** | (a) | **21** | (1) | **24** | (3) | | |

# 18

# MATRICES

## MCQs with One Correct Answer

1. The product of the matrices

$$A=\begin{bmatrix}\cos^2\theta & \cos\theta\sin\theta\\ \cos\theta\sin\theta & \sin^2\theta\end{bmatrix} \text{ and}$$

$$B=\begin{bmatrix}\cos^2\phi & \cos\phi\sin\phi\\ \cos\phi\sin\phi & \sin^2\phi\end{bmatrix} \text{ is a null matrix if}$$

$\theta-\phi=$

(a) $(2n+1)\frac{\pi}{2}$ (b) $n\pi$

(c) $2n\pi$ (d) $n\frac{\pi}{2}$

2. If $\begin{bmatrix}2 & 1\\ 3 & 2\end{bmatrix}$ A $\begin{bmatrix}-3 & 2\\ 5 & -3\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, then the matrix A equals

(a) $\begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix}$ (b) $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$

(c) $\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$ (d) $\begin{bmatrix}0 & 1\\ 1 & 1\end{bmatrix}$

3. If matrix $P=\begin{bmatrix}0 & -\tan(\theta/2)\\ \tan\theta/2 & 0\end{bmatrix}$, then find

$(I-P)\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$

(a) $I+2p$ (b) $2I+P$

(c) $I+P$ (d) None of these

4. For each real $x: -1<x<1$. Let $A(x)$ be the matrix

$(1-x)^{-1}\begin{bmatrix}1 & -x\\ -x & 1\end{bmatrix}$ and $z=\dfrac{x+y}{1+xy}$ then

(a) $A(z)=A(x)+A(y)$

(b) $A(z)=A(x)[A(y)]^{-1}$

(c) $A(z)=A(x)A(y)$

(d) $A(z)=A(x)-A(y)$

5. If $A=\begin{bmatrix}1 & 0\\ 1/2 & 1\end{bmatrix}$, then $A^{50}$ is

(a) $\begin{bmatrix}1 & 0\\ 0 & 50\end{bmatrix}$ (b) $\begin{bmatrix}1 & 0\\ 50 & 1\end{bmatrix}$

(c) $\begin{bmatrix}1 & 25\\ 0 & 1\end{bmatrix}$ (d) None of these

6. If $A=[a_{ij}]_{m\times n}$ and $a_{ij}=(i^2+j^2-ij)(j-i)$, $n$ is odd, then which of the following is not the value of $Tr(A)$

(a) 0 (b) $|A|$

(c) $2|A|$ (d) None of these

**7.** If $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$ and $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q =$ $PAP^T$ and $x = P^TQ^{2005}P$ then $x$ is equal to

(a) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 4+2005\sqrt{3} & 6015 \\ 2005 & 4-2005\sqrt{3} \end{bmatrix}$

(c) $\frac{1}{4}\begin{bmatrix} 2+\sqrt{3} & 1 \\ -1 & 2-\sqrt{3} \end{bmatrix}$

(d) $\frac{1}{4}\begin{bmatrix} 2005 & 2-\sqrt{3} \\ 2+\sqrt{3} & 2005 \end{bmatrix}$

**8.** The value of $a$, $b$, $c$ when $\begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$ is orthogonal, are

(a) $\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{6}}, \pm\frac{1}{\sqrt{2}}$

(b) $\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{6}}$

(c) $\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{6}}, \pm\frac{1}{\sqrt{3}}$

(d) $\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}$

**9.** If $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix}$ then $A^3 - rA^2 - qA =$

(a) $pI$ (b) $qI$
(c) $rI$ (d) None of these

**10.** Matrix $A$ is such that $A^2 = 2A - I$ when $I$ is unit matrix then $A^n$ is equal to

(a) $nA - (n-1)I$ (b) $nA - I$
(c) $2^{n-1}A - I$ (d) none of these

**11.** The matrices

$$P = \begin{bmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{bmatrix}; Q = \frac{1}{9}\begin{bmatrix} 2 & 2 & 1 \\ 13 & -5 & m \\ -8 & 1 & 5 \end{bmatrix}$$

are such that PQ = I, an identity matrix. Solving the equation $\begin{bmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$, the value of y comes out to be –3. Then the value of m is equal to

(a) 27 (b) 7 (c) –27 (d) –7

**12.** If A, B are two square matrices such that AB = A and BA = B, then

(a) only B is idempotent
(b) A, B are idempotent
(c) only A is idempotent
(d) None of these

**13.** If $A_1, A_3, ..., A_{2n-1}$ are n skew – symmetric matrices of same order, then

$X = A_1 + 3A_3^3 + ...(2n-1)(A_{2n-1})^{2n-1}$ will be

(a) symmetric
(b) skew-symmetric
(c) neither symmetric nor skew symmetric
(d) depends on 'n' is even or odd

**14.** For $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$, $a, b, c, d, e, f, g, h, i \in C$, we say $A^\theta = \begin{bmatrix} \bar{a} & \bar{d} & \bar{g} \\ \bar{b} & \bar{e} & \bar{h} \\ \bar{c} & \bar{f} & \bar{i} \end{bmatrix}$ and we say that $A$ is the Hermitian matrix if $A = A^\theta$. Suppose $A$ is the Hermitian matrix such that $A^2 = O$ then

(a) $A = -A'$ (b) $A = A'$
(c) $A = O$ (d) $A = I$

**15.** $A, B, C$ are three matrices of the same order such that any two are symmetric and the 3rd one is skew symmetric. If $X = ABC + CBA$ and $Y = ABC - CBA$, then $(XY)^T$ is
(a) symmetric (b) skew symmetric
(c) $I - XY$ (d) $-YX$

**16.** If $A = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$, $C = ABA^T$, then $A^T C^n A$ equals to $(n \in I^+)$

(a) $\begin{bmatrix} -n & 1 \\ 1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & -n \\ 0 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 1 \\ 1 & -n \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 \\ -n & 1 \end{bmatrix}$

**17.** Let $A$ be a $3 \times 3$ matrix given by $A = (a_{ij})_{3\times 3}$. If for every column vector $X$ satisfies $X'AX = 0$ and $a_{12} = 2008$, $a_{13} = 1010$ and $a_{23} = -2012$. Then the value of $a_{21} + a_{31} + a_{32} =$
(a) $-6$ (b) 2006
(c) $-2006$ (d) 0

## Numeric Value Answer

**18.** If B, C are square matrices of order n and if $A = B + C$, $BC = CB$, $C^2 = 0$, then for any positive integer N, $A^{N+1} = B^K[B + (N+1)C]$, then K/N is

**19.** $A$ and $B$ are two square matrices such that $A^2B = BA$ and if $(AB)^{10} = A^k B^{10}$, then $k$ is

**20.** If the product of $n$ matrices
$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\cdots\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ is equal to the matrix $\begin{bmatrix} 1 & 378 \\ 0 & 1 \end{bmatrix}$, then the value of $n$ is equal to

**21.** If matrix $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ where $a, b, c$ are real positive numbers, $abc = 1$ and $A^T A = I$, then find the value of $a^3 + b^3 + c^3$.

**22.** Let $z = \frac{-1+\sqrt{3}i}{2}$, where $i = \sqrt{-1}$, and $r, s \in \{1, 2, 3\}$. Let $P = \begin{bmatrix} (-z)^r & z^{2s} \\ z^{2s} & z^r \end{bmatrix}$ and I be the identity matrix of order 2. Then the total number of ordered pairs (r, s) for which $P^2 = -I$ is

**23.** Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and $I$ be the identity matrix of order 3. If $Q = [q_{ij}]$ is a matrix such that $P^{50} - Q = I$, then $\frac{q_{31} + q_{32}}{103\, q_{21}}$ equals

**24.** Let $X$ be the solution set of the equation $A^x = I$; where $A = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}$ and $I$ is the corresponding unit matrix and $x \in \mathbb{N}$, then find the minimum value of $\Sigma(\cos^x\theta + \sin^x\theta), \theta \in \mathbb{R} - \left\{\frac{n\pi}{2}; n \in \mathbb{Z}\right\}$.

**25.** Let $A = \begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix}$ and $(A + I)^{50} - 50A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, find $a + b + c + d$.

| ANSWER KEY | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (a) | **4** | (c) | **7** | (a) | **10** | (a) | **13** | (b) | **16** | (d) | **19** | (1023) | **22** | (1) | **25** | (2) |
| **2** | (a) | **5** | (d) | **8** | (c) | **11** | (d) | **14** | (c) | **17** | (c) | **20** | (27) | **23** | (1) | | |
| **3** | (c) | **6** | (d) | **9** | (a) | **12** | (b) | **15** | (d) | **18** | (1) | **21** | (4) | **24** | (2) | | |

# 19 DETERMINANTS

## MCQs with One Correct Answer

**1.** The value of the determinant

$$\begin{vmatrix} 1 & a & a^2 \\ \cos(n-1)x & \cos nx & \cos(n+1)x \\ \sin(n-1)x & \sin nx & \sin(n+1)x \end{vmatrix}$$ is zero, if

(a) $\sin x = 0$ (b) $\cos x = 0$

(c) $a = 0$ (d) $\cos x = \dfrac{1+a^2}{2a}$

**2.** If $a$, $b$, $c$, are the sides of a triangle $ABC$ such that $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 0$, then

(a) $\Delta ABC$ is non-isosceles right-angled triangle
(b) $\Delta\, ABC$ is an equilateral
(c) $\Delta\, ABC$ is an acute angled triangle with no two angles being equal
(d) $\Delta\, ABC$ is an isosceles

**3.** If $a$, $b$, $c$, are the sides of a triangle $A\,B\,C$ and $A$, $B$, $C$ are angles opposite to $a$, $b$, $c$, and

$$\Delta = \begin{vmatrix} a^2 & b\sin A & c\sin A \\ b\sin A & 1 & \cos A \\ c\sin A & \cos A & 1 \end{vmatrix}$$ then,

(a) $\Delta =$ area of triangle
(b) $\Delta =$ perimeter of the triangle
(c) $\Delta = \Sigma a^2$
(d) None of these

**4.** If $P$, $Q$ and $R$ represent the angles of an acute angled triangle, no two of them being equal then the value of $\begin{vmatrix} 1 & 1+\sin P & \sin P(1+\sin P) \\ 1 & 1+\sin Q & \sin Q(1+\sin Q) \\ 1 & 1+\sin R & \sin R(1+\sin R) \end{vmatrix}$ is

(a) 0 (b) positive
(c) negative (d) can not be determined

**5.** Consider three points

$P = (-\sin(\beta-\alpha), -\cos\beta)$,

$Q = (\cos(\beta-\alpha), \sin\beta)$ and

$R = (\cos(\beta-\alpha+\theta), \sin(\beta-\theta))$, where

$0 < \alpha, \beta, \theta < \dfrac{\pi}{4}$. Then,

(a) $P$ lies on the line segment $RQ$
(b) $Q$ lies on the line segment $PR$
(c) $R$ lies on the line segment $QP$
(d) $P$, $Q$, $R$ are non-collinear

**6.** If $\Delta_r = \begin{vmatrix} r & 2r-1 & 3r-2 \\ \dfrac{n}{2} & n-1 & a \\ \dfrac{1}{2}n(n-1) & (n-1)^2 & \dfrac{1}{2}(n-1)(3n-4) \end{vmatrix}$

then the value of $\sum_{r=1}^{n-1} \Delta_r$

(a) depends only on a
(b) depends only on n
(c) depends both on a and n
(d) is independent of both a and n

**7.** If $a^2 + b^2 + c^2 + ab + bc + ca \le 0\ \forall\ a, b, c \in R$, then value of the determinant,

$$\begin{vmatrix} (a+b+2)^2 & a^2+b^2 & 1 \\ 1 & (b+c+2)^2 & b^2+c^2 \\ c^2+a^2 & 1 & (c+a+2)^2 \end{vmatrix}$$

equals

(a) 65 (b) $a^2+b^2+c^2+31$
(c) $4(a^2+b^2+c^2)$ (d) 0

**8.** If $A$ is matrix of order 3 such that $|A| = 5$ and $B =$ adj $A$, then the value of $\left||A^{-1}|(AB)^T\right|$ is equal to (where $|A|$ denotes determinant of matrix $A$. $A^T$ denotes transpose of matrix $A$, $A^{-1}$ denotes inverse of matrix $A$. adj $A$ denotes adjoint of matrix $A$)

(a) 5 (b) 1 (c) 25 (d) $\frac{1}{25}$

**9.** If $f(x)$, $h(x)$ are polynomials of degree 4 and $\begin{vmatrix} f(x) & g(x) & h(x) \\ a & b & c \\ p & q & r \end{vmatrix} = mx^4 + nx^3 + rx^2 + sx + t$ be an identity in $x$, then $\begin{vmatrix} f'''(0) - f''(0) & g'''(0) - g''(0) & h'''(0) - h''(0) \\ a & b & c \\ p & q & r \end{vmatrix}$ is

(a) $2(3n - r)$ (b) $2(2n - 3r)$
(c) $3(n - 2r)$ (d) None of these

**10.** If $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$,

where a, $b$, $c$ are all different, then the determinant

$$\begin{vmatrix} 1 & 1 & 1 \\ (x-a)^2 & (x-b)^2 & (x-c)^2 \\ (x-b)(x-c) & (x-c)(x-a) & (x-a)(x-b) \end{vmatrix}$$

vanishes when

(a) $a + b + c = 0$ (b) $x = \frac{1}{3}(a+b+c)$

(c) $x = \frac{1}{2}(a+b+c)$ (d) $x = a + b + c$

**11.** If $t$ is real and $\lambda = \frac{t^2 - 3t + 4}{t^2 + 3t + 4}$, then the number of solutions of the system of equations $3x - y + 4z = 3, x + 2y - 3z = -2, 6x + 5y + \lambda z = -3$ is :

(a) one (b) two (c) zero (d) infinite

**12.** The set of homogeneous equations :

$tx + (t+1)y + (t-1)z = 0;$
$(t+1)x + ty + (t+2)z = 0;$
$(t-1)x + (t+2)y + tz = 0$

has non-trivial solution for

(a) three values of $t$ (b) two values of $t$
(c) one value of $t$ (d) no value of $t$

**13.** If $a$, $b$, $c$ are non-zeros, then the system of equations :

$(\alpha + a)x + \alpha y + \alpha z = 0;\ \alpha x + (\alpha + b)y + \alpha z = 0;$
$\alpha x + \alpha y + (\alpha + c)z = 0$ has a non-trivial solution if :

(a) $\alpha^{-1} = -(a^{-1} + b^{-1} + c^{-1})$
(b) $\alpha^{-1} = a + b + c$
(c) $\alpha + a + b + c = 1$
(d) None of these

**14.** If the system of equations ax + y + z = 0, x + by + z = 0 and x + y + cz = 0 (a, b, c $\neq$ 1) has a non-trivial solution, then the value of $\frac{1}{1-\text{a}} + \frac{1}{1-\text{b}} + \frac{1}{1-\text{c}}$ is

(a) $-1$ (b) 0
(c) 1 (d) None of these

**15.** If $a$, $b$, $c$ are non-zeros, then the system of equations $(\alpha + a)x + \alpha y + \alpha z = 0$, $\alpha x + (\alpha + b)y + \alpha z = 0$, $\alpha x + \alpha y + (\alpha + c)z = 0$ has a non-trivial solution if

(a) $\alpha^{-1} = -(a^{-1} + b^{-1} + c^{-1})$
(b) $\alpha^{-1} = a + b + c$
(c) $\alpha + a + b + \text{c} = 1$
(d) None of these

**16.** If $c < 1$ and the system of equations $x + y - 1 = 0$, $2x - y - c = 0$, and $-bx + 3by - c = 0$ is consistent, then the possible real values of $b$ are

(a) $b \in \left(-3, \frac{3}{4}\right)$ (b) $b \in \left(-\frac{3}{2}, 4\right)$

(c) $b \in \left(-\frac{3}{4}, 3\right)$ (d) None of these

**17.** If f(x) = $\begin{vmatrix} x^{n-1} & \cos x & \frac{1}{x+3} \\ 0 & \cos\frac{n\pi}{2} & \frac{(-1)^n n!}{3^{n+1}} \\ \alpha & \alpha^3 & \alpha^5 \end{vmatrix}$ then $\frac{d^n}{dx^n}[f(x)]_{x=0} =$

(a) 1 (b) $-1$
(c) 0 (d) None of these

**18.** Suppose $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and

$$D' = \begin{vmatrix} a_1 + pb_1 & b_1 + qc_1 & c_1 + ra_1 \\ a_2 + pb_2 & b_2 + qc_2 & c_2 + ra_2 \\ a_3 + pb_3 & b_3 + qc_3 & c_3 + ra_3 \end{vmatrix}. \text{ Then}$$

(a) $D' = D$ (b) $D' = D(1 - pqr)$
(c) $D' = D(1+p+q+r)$ (d) $D' = D(1 + pqr)$

**19.** If $A = \begin{bmatrix} x & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}; \left( x \neq \frac{-11}{3} \right)$ and det $(adj(adj\ A)) = (14)^4$. Then the value of $x$ is

(a) 2 (b) $-2$
(c) 0 (d) None of these

**20.** Matrix $A = \begin{bmatrix} x & 3 & 2 \\ 1 & y & 4 \\ 2 & 2 & z \end{bmatrix}$, if $xyz = 60$ and $8x + 4y + 3z = 20$, then $A(\text{adj } A)$ is equal to

(a) $\begin{bmatrix} 64 & 0 & 0 \\ 0 & 64 & 0 \\ 0 & 0 & 64 \end{bmatrix}$ (b) $\begin{bmatrix} 88 & 0 & 0 \\ 0 & 88 & 0 \\ 0 & 0 & 88 \end{bmatrix}$

(c) $\begin{bmatrix} 68 & 0 & 0 \\ 0 & 68 & 0 \\ 0 & 0 & 68 \end{bmatrix}$ (d) $\begin{bmatrix} 34 & 0 & 0 \\ 0 & 34 & 0 \\ 0 & 0 & 34 \end{bmatrix}$

## Numeric Value Answer

**21.** The value of the determinant

$$\begin{vmatrix} \sin^2\left(x+\frac{3\pi}{2}\right) & \sin^2\left(x+\frac{5\pi}{2}\right) & \sin^2\left(x+\frac{7\pi}{2}\right) \\ \sin\left(x+\frac{3\pi}{2}\right) & \sin\left(x+\frac{5\pi}{2}\right) & \sin\left(x+\frac{7\pi}{2}\right) \\ \sin\left(x-\frac{3\pi}{2}\right) & \sin\left(x-\frac{5\pi}{2}\right) & \sin\left(x-\frac{7\pi}{2}\right) \end{vmatrix}$$ is

**22.** Let $S = \left\{ \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} : a_{ij} \in \{0,1,2\}, a_{11} = a_{22} \right\}$

Then the number of non-singular matrices in the set S is :

**23.** Let $A = [a_{ij}]_{3\times 3}$ be such that $a_{ij} = \begin{bmatrix} 3; \text{when} & \hat{i} = \hat{j} \\ 0; & \hat{i} \neq \hat{j} \end{bmatrix}$,

then $\left\{ \frac{\det(\text{adj}(\text{adj } A))}{5} \right\}$ equals: (where $\{\cdot\}$ denotes fractional part function)

**24.** If $a_1, a_2, a_3, \ldots, a_n$ are in G.P. and $a_i > 0$ for each $i$, then the determinant

$$\Delta = \begin{vmatrix} \log a_n & \log a_{n+2} & \log a_{n+4} \\ \log a_{n+6} & \log a_{n+8} & \log a_{n+10} \\ \log a_{n+12} & \log a_{n+14} & \log a_{n+16} \end{vmatrix}$$ is equal to:

**25.** Area of triangle whose vertices are $(a, a^2)$ $(b, b^2)$ $(c, c^2)$ is $\frac{1}{2}$, and area of another triangle whose vertices are $(p, p^2)$, $(q, q^2)$ and $(r, r^2)$ is 4, then the value of $\begin{vmatrix} (1+ap)^2 & (1+bp)^2 & (1+cp)^2 \\ (1+aq)^2 & (1+bq)^2 & (1+cq)^2 \\ (1+ar)^2 & (1+br)^2 & (1+cr)^2 \end{vmatrix}$ is

**26.** If $f(x) = \begin{vmatrix} x & \sin x & \cos x \\ x^2 & -\tan x & -x^3 \\ 2x & \sin 2x & 5x \end{vmatrix}$ then $\lim_{x\to 0} \frac{f'(x)}{x}$ is equal to

**27.** If $\alpha, \beta$ are the roots of the equation $x^2 - 2x + 4 = 0$, find the value of

$$\begin{vmatrix} \alpha+\beta & \alpha^2+\beta^2 & \alpha^3+\beta^3 \\ \alpha^2+\beta^2 & \alpha^3+\beta^3 & \alpha^4+\beta^4 \\ \alpha^3+\beta^3 & \alpha^4+\beta^4 & \alpha^5+\beta^5 \end{vmatrix}$$

**28.** Find the co-efficient of $x$ in the expansion of the determinant

$$\begin{vmatrix} (1+x)^2 & (1+x)^4 & (1+x)^6 \\ (1+x)^3 & (1+x)^6 & (1+x)^9 \\ (1+x)^4 & (1+x)^8 & (1+x)^{12} \end{vmatrix}.$$

**29.** Suppose $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is a real matrix with non-zero entries, $ad - bc = 0$ and $A^2 = A$. Then, $a + d$ equal to

**30.** The number of integers $x$ satisfying

$$-3x^4 + \det \begin{bmatrix} 1 & x & x^2 \\ 1 & x^2 & x^4 \\ 1 & x^3 & x^6 \end{bmatrix} = 0$$ is equal to

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | (a) | 5 | (d) | 9 | (a) | 13 | (a) | 17 | (c) | 21 | (0) | 25 | (16) | 29 | (1) | | |
| 2 | (d) | 6 | (d) | 10 | (b) | 14 | (c) | 18 | (d) | 22 | (20) | 26 | (4) | 30 | (2) | | |
| 3 | (d) | 7 | (a) | 11 | (a) | 15 | (a) | 19 | (d) | 23 | (0.20) | 27 | (0) | | | | |
| 4 | (d) | 8 | (b) | 12 | (c) | 16 | (c) | 20 | (c) | 24 | (0) | 28 | (0) | | | | |

# CONTINUITY AND DIFFERENTIABILITY

20

## MCQs with One Correct Answer

1. The function $f(x)=(\sin 2x)^{\tan^2 2x}$ is not defined at $x=\frac{\pi}{4}$. The value of $f\left(\frac{\pi}{4}\right)$ so that f is continuous at $x=\frac{\pi}{4}$ is
   (a) $\sqrt{e}$ (b) $\frac{1}{\sqrt{e}}$
   (c) 2 (d) None of these

2. If $f(x)=\begin{cases}\frac{1-[x]}{1+x}, & x\neq -1\\ 1, & x=-1\end{cases}$, then the value of $f(|2k|)$ will be (where [] shows the greatest integer function]
   (a) Continuous at $x=-1$
   (b) Continuous at $x=0$
   (c) Discontinuous at $x=\frac{1}{2}$
   (d) All of these

3. Suppose '$f$' is continuous function from $R$ to $R$ and $f(f(a))=a$ for some $a\in R$ then the equation $f(x)=x$ has
   (a) no solution
   (b) exactly one solution
   (c) at most one solution
   (d) atleast three solutions

4. Let $g(x)$ be a polynomial of degree one and $f(x)$ be defined by $f(x)=\begin{cases}g(x), & x\le 0\\ |x|^{\sin x}, & x>0\end{cases}$. If $f(x)$ is continuous satisfying $f'(1)=f(-1)$, then $g(x)$ is
   (a) $(1+\sin 1)x+1$ (b) $(1-\sin 1)\,x+1$
   (c) $(1-\sin 1)x-1$ (d) $(1+\sin 1)x-1$

5. Let $f(x)$ be defined as follows :
$$f(x)=\begin{cases}(\cos x-\sin x)^{\operatorname{cosec} x} & , -\frac{\pi}{2}<x<0\\ a & , x=0\\ \frac{e^{1/x}+e^{2/x}+e^{3/x}}{ae^{2/x}+be^{3/x}} & , 0<x<\frac{\pi}{2}\end{cases}$$
   If $f(x)$ is continuous at $x=0$, then $(a, b)=$
   (a) $\left(e,\frac{1}{e}\right)$ (b) $\left(\frac{1}{e},e\right)$
   (c) $(e,e)$ (d) $(e^{-1},e^{-1})$

6. The function $f$ defined by
$$f(x)=\lim_{t\to\infty}\left\{\frac{(1+\sin\pi x)^t-1}{(1+\sin\pi x)^t+1}\right\}\text{ is}$$
   (a) every where continuous
   (b) discontinuous at all integer values of $x$
   (c) continuous at $x=0$
   (d) None of these

7. Let $f:R\to R$ be a function defined by $f(x)=\min\{x+1,|x|+1\}$, then which of the following is true?
   (a) $f(x)$ is differentiable everywhere
   (b) $f(x)$ is not differentiable at $x=0$
   (c) $f(x)\ge 1$ for all $x\in R$
   (d) $f(x)$ is not differentiable at $x=1$

8. Let $g(x)=\dfrac{(x-1)^n}{\log\cos^m(x-1)}$; $0<x<2$, m and n are integers, $m\neq 0$, $n>0$, and let $p$ be the left hand derivative of $|x-1|$ at $x=1$. If $\lim\limits_{x\to 1^+} g(x)=p$, then
(a) $n=1, m=1$ (b) $n=1, m=-1$
(c) $n=2, m=2$ (d) $n>2, m=n$

9. Let f: R $\to$ R be a function such that $|f(x)|\le x^2$, for all $x\in R$. Then, at $x=0$, f is:
(a) continuous but not differentiable.
(b) continuous as well as differentiable.
(c) neither continuous nor differentiable.
(d) differentiable but not continuous.

10. If $f(x)=\cot^{-1}\left(\dfrac{x^x-x^{-x}}{2}\right)$, then $f'(1)$ is
(a) $-1$ (b) 1
(c) log 2 (d) $-\log 2$

11. If g is the inverse of f and f' $(x)=\dfrac{1}{1+x^3}$, then find g' $(x)$.
(a) $1+[g(x)]^2$ (b) $1-[g(x)]^2$
(c) $1+[g(x)]^3$ (d) $1-[g(x)]^3$

12. If $f(x)=\sin\left(\dfrac{\pi}{3}[x]-x^2\right)$, where $2<x<3$ and [ . ] represetns greatest integer function, then $f'\left(\sqrt{\dfrac{5\pi}{3}}\right)$ is equal to
(a) $\sqrt{\dfrac{5\pi}{3}}$ (b) $2\sqrt{\dfrac{5\pi}{3}}$
(c) $-2\sqrt{\dfrac{5\pi}{3}}$ (d) $-\sqrt{\dfrac{5\pi}{3}}$

13. If $\sqrt{1-x^{2n}}+\sqrt{1-y^{2n}}=a(x^n-y^n)$, then $\sqrt{\dfrac{1-x^{2n}}{1-y^{2n}}}\dfrac{dy}{dx}$ is equal to
(a) 1 (b) $x/y$
(c) $\dfrac{x^{n-1}}{y^{n-1}}$ (d) None of these

14. If $\sqrt{(1-x^6)}+\sqrt{(1-y^6)}=a(x^3-y^3)$, and $\dfrac{dy}{dx}$ $=f(x,y)\sqrt{\left(\dfrac{1-y^6}{1-x^6}\right)}$, then
(a) $f(x,y)=y/x$ (b) $f(x,y)=y^2/x^2$
(c) $f(x,y)=2y^2/x^2$ (d) $f(x,y)=x^2/y^2$

15. If $f(x)=(1+x)^n$, then the value of
$$f(0)+f'(0)+\frac{f''(0)}{2!}+....+\frac{f^n(0)}{n!}$$ is
(a) n (b) $2^n$
(c) $2^{n-1}$ (d) None of these

16. If $\alpha$ and $\beta$ are any two roots of equation $e^x\cos x=1$, then the equation $e^x\sin x-1=0$ has
(a) exactly one root in $(\alpha,\beta)$
(b) exactly two roots in $(\alpha,\beta)$
(c) at least one root in $(\alpha,\beta)$
(d) no root in $(\alpha,\beta)$

17. Let $f(x)=x|\sin x|, x\in R$. Then
(a) $f$ is differentiable for all $x$, except at $x=n\pi, n=1,2,3,....,$
(b) $f$ is differentiable for all $x$, except at $x=n\pi, n=\pm1,\pm2,\pm3,...$
(c) $f$ is differentiable for all $x$, except at $x=n\pi, n=0,1,2,3$
(d) $f$ is differentiable for all $x$, except at $x=n\pi, n=0,\pm1,\pm2,\pm3,...$

18. $f:[-1,1]\to R$ be a function defined by
$$f(x)=\begin{cases} x^2\left|\cos\left(\dfrac{\pi}{x}\right)\right| & \text{for } x\neq 0 \\ 0 & \text{for } x=0\end{cases}$$
The set of points where $f$ is not differentiable is
(a) $\{x\in[-1,1]:x\neq 0\}$
(b) $\left\{x\in[-1,1]:x=0 \text{ or } x=\dfrac{2}{2n+1}, n\in Z\right\}$
(c) $\left\{x\in[-1,1]:x=\dfrac{2}{2n+1}, n\in Z\right\}$
(d) $[-1,1]$

19. Let $f:\mathbf{R}\to\mathbf{R}$ be a continuous function such that $f(x^2)=f(x^3)$ for all $x\in\mathbf{R}$. Consider the following statements.

I. $f$ is an odd function.
II. $f$ is an even function.
III. $f$ is differentiable everywhere

Then

(a) I is true and III is false
(b) II is true and III is false
(c) both I and III are true
(d) both II and III are true

**20.** Let $f: \mathbf{R} \to \mathbf{R}$ be function defined by

$$f(x) = \begin{cases} \dfrac{\sin(x^2)}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0, \end{cases}$$

Then, at $x = 0$, $f$ is

(a) not continuous
(b) continuous but not differentiable
(c) differentiable and the derivative is not continuous
(d) differentiable and the derivative is continuous

## Numeric Value Answer

**21.** If $f$ is a real valued differentiable function satisfying $|f(x) - f(y)| \le (x-y)^2, x, y \in R$ and $f(0) = 0$, then $f(1)$ is equal to

**22.** If $x = \operatorname{cosec}\theta - \sin\theta$ ; $y = \operatorname{cosec}^n\theta - \sin^n\theta$, and $(x^2+4)\left(\dfrac{dy}{dx}\right)^2 - n^2y^2 = kn^2$, then value of $k$ is

**23.** If $f(x) = \cos x \cos 2x \cos 2^2 x \cos 2^3 x \ldots\ldots \cos 2^{n-1} x$ and $n > 1$, then $f'\left(\dfrac{\pi}{2}\right)$ is

**24.** If $f''(x) = -f(x)$ and $g(x) = f'(x)$ and

$$F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2 \text{ and given that}$$

$F(5) = 5$, then $F(10)$ is equal to

**25.** Number of functions $f: [0, 1] \to [0, 1]$ satisfying $|f(x) - f(y)| = |x - y|$ for all $x, y$ in $[0, 1]$ is

**26.** If $y = (1 + 1/x)^x$ then find the value of

$$\frac{2\sqrt{y_2(2) + 1/8}}{\log 3/2 - 1/3}$$

**27.** If $y = \sqrt{(x - \sin x) + \sqrt{(x - \sin x) + \ldots}},$ then

$$\left| \left|\frac{dx}{dy}\right|^2_{x=\frac{\pi}{2}} - 2\pi \right| = \ldots\ldots\ldots$$

**28.** Let $f(x) = \begin{cases} \dfrac{\alpha \cot x}{x} + \dfrac{\beta}{x^2}, & 0 < |x| \le 1 \\ \dfrac{1}{3} & x = 0 \end{cases}$

If $f(x)$ is continuous at $x = 0$, then the value of $\alpha^2 + \beta^2$ is

**29.** Number of points where function $f(x)$ defined as $f: [0, 2\pi] \to R, f(x)$

$$= \begin{cases} 3 - \left|\cos x - \dfrac{1}{\sqrt{2}}\right|, & |\sin x| < \dfrac{1}{\sqrt{2}} \\ 2 + \left|\cos x + \dfrac{1}{\sqrt{2}}\right|, & |\sin x| \ge \dfrac{1}{\sqrt{2}} \end{cases} \text{ is}$$

non-differentiable is

**30.** If the derivative of the function

$$f(x) = \cos^{-1}\left\{\frac{1}{\sqrt{13}}(2\cos x - 3\sin x)\right\} + \sin^{-1}\left\{\frac{1}{\sqrt{13}}(2\cos x + 3\sin x)\right\}$$

w.r.t. $\sqrt{1+x^2}$ at $x = \dfrac{3}{4}$ is $\dfrac{10}{b}$, then $b =$

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (b) | **5** | (b) | **9** | (b) | **13** | (c) | **17** | (b) | **21** | (0) | **25** | (2) | **29** | (4) | | |
| **2** | (d) | **6** | (b) | **10** | (a) | **14** | (d) | **18** | (c) | **22** | (4) | **26** | (3) | **30** | (3) | | |
| **3** | (d) | **7** | (a) | **11** | (c) | **15** | (b) | **19** | (d) | **23** | (1) | **27** | (3) | | | | |
| **4** | (b) | **8** | (c) | **12** | (b) | **16** | (c) | **20** | (d) | **24** | (5) | **28** | (2) | | | | |

# APPLICATION OF DERIVATIVES

21

## MCQs with One Correct Answer

1. The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in

(a) $\left(0, \frac{\pi}{2}\right)$ (b) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(c) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ (d) $\left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$

2. The greatest of the numbers

$1, 2^{1/2}, 3^{1/3}, 4^{1/4}, 5^{1/5}, 6^{1/6}$ and $7^{1/7}$ is

(a) $2^{1/2}$ (b) $3^{1/3}$

(c) $7^{1/4}$ (d) All but 1 are equal

3. The interval of decrease of the function

$f(x) = x^2 \log 27 - 6x \log 27$

$+(3x^2 - 18x + 24)\log(x^2 - 6x + 8)$ is

(a) $(3-\sqrt{1+1/3e}, 2) \cup (4, 3+\sqrt{1+1/3e})$

(b) $(3-\sqrt{1+1/3e},\ 3+\sqrt{1+1/3e})$

(c) $(-\infty,\ 3-\sqrt{1+1/3e}) \cup (3,\ 4+\sqrt{1+1/3e})$

(d) None of these.

4. The set of positive values of the parameter '$a$' for each of which the function

$f(x) = \sin 2x - 8(a+1)\sin x - (4a^2 + 8a - 14)x$

is monotonic increasing in $R$ and has no critical points are

(a) $(0, \sqrt{6}-2)$ (b) $(-2-\sqrt{6}, \sqrt{6}-2)$

(c) $(-2-\sqrt{6}, 0)$ (d) None of these

5. If $f(x) = \frac{x}{\sin x}$ and $g(x) = \frac{x}{\tan x}$, where $0 < x \le 1$, then in this interval

(a) both $f(x)$ and $g(x)$ are increasing functions

(b) both $f(x)$ and $g(x)$ are decreasing functions

(c) $f(x)$ is an increasing function

(d) $g(x)$ is an increasing function.

6. Let the function $g : (-\infty, \infty) \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be given by $g(u) = 2\tan^{-1}(e^u) - \frac{\pi}{2}$. Then, g is

(a) even and is strictly increasing in $(0, \infty)$

(b) odd and is strictly decreasing in $(-\infty, \infty)$

(c) odd and is strictly increasing in $(-\infty, \infty)$

(d) neither even nor odd, but is strictly increasing in $(-\infty, \infty)$

7. The largest term in the sequence,

$a_n = \frac{n^2}{n^3 + 200}$ is

(a) $a_6$ (b) $a_7$

(c) $a_8$ (d) None of these

8. Let $f(x)$ be a function given by $f(x) = \frac{x^2+1}{[x]}$ for all $x \in [1, 4]$, where [.] denotes the greatest integer function. then, $f(x)$ is monotonically.

(a) increasing on [1, 4)

(b) decreasing on [1, 4)

(c) increasing on [1, 2)

(d) decreasing on [2, 3)

9. If $A > 0$, $B > 0$ and $A + B = \pi/3$, then the maximum value of tan A tan B is

(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{3}$ (c) 3 (d) $\sqrt{3}$

**10.** The set of values for which the function

$f(x) = (4a-3)(x+\ln 5) + 2(a-7)\cot\frac{x}{2}\sin^2\frac{x}{2}$

does not possess critical points is

(a) $\left(-\infty, -\frac{4}{3}\right) \cup (2, \infty)$

(b) $(-\infty, 2)$

(c) $[1, \infty)$

(d) $(1, \infty)$

**11.** A given right circular cone has a volume $p$, and the largest right circular cylinder that can be inscribed in the cone has a volume $q$. then $p:q$ is

(a) $9:4$ (b) $8:3$

(c) $7:2$ (d) None of these

**12.** $f(x) = \begin{cases} 2-|x^2+5x+6|; & x \neq -2 \\ a^2+1 \quad ; & x = -2 \end{cases}$

then the range of $a$ is so, that $f(x)$ has maxima at $x = -2$ is

(a) $|a| \geq 1$ (b) $|a| < 1$

(c) $a > 1$ (d) $a < 1$

**13.** Let $f(x) = \sin^3 x + \lambda \sin^2 x,\ -\frac{\pi}{2} < x < \frac{\pi}{2}$.

In order that $f(x)$ has exactly one minimum, $\lambda$ should belong to

(a) $(-1, 1)$ (b) $\left(-\frac{3}{2}, 0\right) \cup \left(0, \frac{3}{2}\right)$

(c) $(0, \infty)$ (d) $\left(-\frac{3}{2}, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, \frac{3}{2}\right)$

**14.** The function $f(x) = 1 + x(\sin x)[\cos x], 0 < x \leq \frac{\pi}{2}$ (where [ . ] is G.I.F.)

(a) is continuous on $\left(0, \frac{\pi}{2}\right)$

(b) is strictly increasing in $\left(0, \frac{\pi}{2}\right)$

(c) is strictly decreasing in $\left(0, \frac{\pi}{2}\right)$

(d) has global maximum value 2

**15.** Two men $P$ and $Q$ start with velocities $v$ at the same time from the junction of two roads inclined at 45° to each other. If they travel by different roads, the rate at which they are being separated is

(a) $v\sqrt{2}$ (b) $v\sqrt{2+\sqrt{2}}$

(c) $v\sqrt{2-\sqrt{2}}$ (d) $v/\sqrt{2}$

**16.** The general value of $\alpha$ such that the line $x\cos\alpha + y\sin\alpha = p$ is a normal to the curve $(x+a)y = c^2$ is

(a) $\left(2n\pi+\frac{\pi}{2}, (2n+1)\pi\right) \cup \left(2n\pi+\frac{3\pi}{2}, (2n+2)\pi\right)$

(b) $\left(2n\pi+\pi, 2n\pi+\frac{3\pi}{2}\right)$

(c) $\left(2n\pi+\frac{3\pi}{2}, (2n+2)\pi\right)$

(d) $\left(2n\pi, 2n\pi+\frac{\pi}{2},\right) \cup$

$\left((2n+1)\pi, 2n\pi+\frac{3\pi}{2},\right) (n \in I)$

**17.** Let $P$ be a point on the hyperbola $x^2 - y^2 = a^2$, where $a$ is a parameter, such that $P$ is nearest to the line $y = 2x$. Then the locus of $P$ is

(a) $y = 2x$ (b) $y = x$

(c) $2y = x$ (d) $x + y = 0$

**18.** For the curve $y = 3\sin\theta\cos\theta$, $x = e^{\theta}\sin\theta$, $0 \leq \theta \leq \pi$, the tangent is parallel to x-axis when $\theta$ is:

(a) $\frac{3\pi}{4}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

**19.** Let $C$ be the curve $y^3 - 3xy + 2 = 0$. If $H$ is the set of points on the curve $C$, where the tangent is parallel to $x$-axis and $V$ is the set of points on $C$ where the tangent is parallel to $y$-axis, then

(a) $H=\{(x,y): y=0, x\in R\}$, $V=\{(1,1)\}$
(b) $H=\{(x,y): x=0, y\in R\}$, $V=\{(1,1)\}$
(c) $H=\phi$, $V=\{(1,1)\}$
(d) $H=\{(1,1)\}$, $V=\{(x,y): y=0, x\in R\}$

**20.** The number of polynomials $p: R\to R$ satisfying $p(0)=0, p(x)>x^2$ for all $x\neq 0$ and $p''(0)=\frac{1}{2}$ is

(a) 0
(b) 1
(c) more than 1, but finite
(d) infinite

## Numeric Value Answer

**21.** The fuel charges for running a train are proportional to the square of the speed generated at 16 miles per hour and costs ₹ 48 per hour. The most economical speed if the fixed charges i.e. salaries etc. amount to ₹ 300 per hour is

**22.** The curve $x+y-\ell n(x+y)=2x+5$ has a vertical tangent at the point $(\alpha,\beta)$. Then $\alpha+\beta$ is equal to

**23.** Two ships A and B are sailing straightaway from a fixed point O along routes such that $\angle AOB$ is always 120°. At a certain instance, OA = 8 km, OB = 6 km and the ship A is sailing at the rate of 20 km/hr while the ship B sailing at the rate of 30 km/hr. If the distance between A and B is changing at the rate $\frac{k}{\sqrt{37}}$ km/hr, then value of k is _______.

**24.** A rectangle with its sides parallel to the X-axis and Y-axis is inscribed in the region bounded by the curves $y = x^2 - 4$ and $2y = 4 - x^2$. The maximum possible area of such a rectangle is

**25.** Let $P$ be a non-zero polynomial such that $P(1+x)=P(1-x)$ for all real $x$, and $P(1)=0$. Let $m$ be the largest integer such that $(x-1)^m$ divides $P(x)$ for all such $P(x)$. Then $m$ is equal to

**26.** The integral value of $b$ for which the function

$$f(x)=(b^2-3b+2)(\cos^2 x-\sin^2 x)+(b-1)x+\sin(b+1)$$

does not possess stationary point is

**27.** If the curves $ax^2 + by^2 = 1$ and $a_1x^2 + b_1y^2 = 1$ may cut each other orthogonally such that $\frac{1}{a}-\frac{1}{a_1}=\lambda\left(\frac{1}{b}-\frac{1}{b_1}\right)$ then $\lambda$ is equal to

**28.** Tangent at $P_1(2,3)$ on the curve $3y=x^3+1$ meets the curve again at $P_2$. The tangent at $P_2$ meets the curve at $P_3$ and so on. If the sum of the ordinates for $P_1, P_2, P_3, .....P_{60}$ be $S$ then $S+\left(\frac{2^{183}-8}{27}\right)$ is equal to 5k, when k is equal to

**29.** A conical vessel is to be prepared out of a circular sheet of metal of unit radius. In order that the vessel has maximum volume, the sectorial area that must be removed from the sheet is $A_1$ and the area of the given sheet is $A$. If $\frac{A}{A_1}=m+\sqrt{n}$, then $m+n$ is equal to

**30.** An arch way is in the shape of semi-ellipse, the road level being the major axis. If the breadth of the road is 30 m and the height of the arch is 6 m at a distance of 2 m from the side, the greatest height of the arch in metres, is

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (d) | **4** | (a) | **7** | (b) | **10** | (a) | **13** | (b) | **16** | (a) | **19** | (c) | **22** | (1) | **25** | (2) | **28** | (4) |
| **2** | (b) | **5** | (c) | **8** | (c) | **11** | (a) | **14** | (a) | **17** | (c) | **20** | (a) | **23** | (260) | **26** | (2) | **29** | (9) |
| **3** | (a) | **6** | (c) | **9** | (b) | **12** | (a) | **15** | (c) | **18** | (c) | **21** | (40) | **24** | (9.22) | **27** | (1) | **30** | (6) |

# INDEFINITE INTEGRATION

22

## MCQs with One Correct Answer

**1.** If $\int \frac{\sin x}{\sin(x-\alpha)} dx = Ax + B \log \sin(x-\alpha) + C,$ then value of $(A, B)$ is

(a) $(-\cos\alpha, \sin\alpha)$ (b) $(\cos\alpha, \sin\alpha)$

(c) $(-\sin\alpha, \cos\alpha)$ (d) $(\sin\alpha, \cos\alpha)$

**2.** If $f(x) = \lim\limits_{n\to\infty} [2x + 4x^3 + ......... + 2nx^{2n-1}]$; $(0 < x < 1)$, then $\int (f(x))\, dx$ is equal to

(a) $-\sqrt{1-x^2} + c$ (b) $\frac{1}{\sqrt{1-x^2}} + c$

(c) $\frac{1}{x^2-1} + c$ (d) $\frac{1}{1-x^2} + c$

**3.** $\int \left(\frac{f(x)g'(x) - f'(x)g(x)}{f(x)g(x)}\right)(\log(g(x)) - \log(f(x)))\, dx$ is equal to

(a) $\log\left(\frac{g(x)}{f(x)}\right) + C$

(b) $\frac{1}{2}\left(\frac{g(x)}{f(x)}\right)^2 + C$

(c) $\frac{1}{2}\left(\log\left(\frac{g(x)}{f(x)}\right)\right)^2 + C$

(d) $\log\left(\left(\frac{g(x)}{f(x)}\right)^2\right) + C$

**4.** $\int e^x \frac{1 + nx^{n-1} - x^{2n}}{(1-x^n)\sqrt{1-x^{2n}}} dx =$

(a) $e^x\left(\frac{1-x^n}{1+x^n}\right) + C$ (b) $e^x\left(\frac{1+x^n}{1-x^n}\right) + C$

(c) $e^x\left(\sqrt{\frac{1+x^n}{1-x^n}}\right) + C$ (d) $e^x\left(\sqrt{\frac{1-x^n}{1+x^n}}\right) + C$

**5.** If $f(x) = \lim\limits_{n\to\infty} n^2(x^{1/n} - x^{1/(n+1)})$, x > 0 then $\int xf(x)$ dx is equal to

(a) $\frac{x^2}{2} + \ln x + C$ (b) $-\frac{x^2}{4}\ln x + \frac{x^2}{2} + C$

(c) $\frac{x^3}{3} + x\ln x + C$ (d) $\frac{x^2}{2}\ln x - \frac{x^2}{4} + C$

**6.** If $f(x) = \lim\limits_{n\to\infty} \frac{x^n - x^{-n}}{x^n + x^{-n}}, 0 < x < 1$, $n \in N$ then $\int (\sin^{-1} x) f(x) dx$ is equal to

(a) $-[x\sin^{-1} x + \sqrt{1-x^2}] + C$

(b) $x\sin^{-1} x + \sqrt{1-x^2} + C$

(c) $\frac{x^2}{2} + C$

(d) $\frac{1}{2}(\sin^{-1} x)^2 + C$

**7.** $\int x\left(\frac{\ln a^{a^{x/2}}}{3a^{5x/2}b^{3x}}+\frac{\ln b^{b^x}}{2a^{2x}b^{4x}}\right)dx$ (where $a, b \in R^+$) is equal to

(a) $\frac{1}{6\ln a^2b^3}a^{2x}b^{3x}\ln\frac{a^{2x}b^{3x}}{e}+k$

(b) $\frac{1}{6\ln a^2b^3}\frac{1}{a^{2x}b^{3x}}\ln\frac{1}{ea^{2x}b^{3x}}+k$

(c) $\frac{1}{6\ln a^2b^3}\frac{1}{a^{2x}b^{3x}}\ln(a^{2x}b^{3x})+k$

(d) $-\frac{1}{6\ln a^2b^3}\frac{1}{a^{2x}b^{3x}}\ln(a^{2x}b^{3x})+k$

**8.** If $I_n=\int(\sin x+\cos x)^n dx$

$=\frac{(\sin x+\cos x)^{n-1}}{n}(\sin x-\cos x)+2\frac{(n-1)}{n}I_{n-2}$,

then $I_5$ equals ($t = (\sin x + \cos x)$)

(a) $\frac{(\sin x-\cos x)}{15}[t^5-3t^3+8t-32]+C$

(b) $\frac{(\sin x-\cos x)}{15}[t^5+3t^4+8t^3+32t+1]+C$

(c) $\frac{(\sin x-\cos x)}{15}[3t^4+8t^2+32]+C$

(d) None of these

**9.** $\int\cos 2\theta\log_e\left(\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\right)d\theta$

$=\frac{\sin 2\theta}{2}\times\log_e\left(\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\right)-\frac{1}{2}f(x)+c$

where $f(x)$ is

(a) $\sec 2\theta$ (b) $\log_e(\sec 2\theta)$
(c) $2\tan^{-1}\theta$ (d) $\tan 2\theta$

**10.** $\int\frac{1}{x}\{\log_{ex}e\cdot\log_{e^2x}e\cdot\log_{e^3x}e\}\,dx$ is equal to

(a) $\frac{1}{2}\log\{\log_e e^2x\}-\log\{\log_e e^3x\}$
$+\log\{\log_e e^4x\}+C$

(b) $\frac{1}{2}\log\{\log_e x\}-\log\{\log_e x\}$
$+\log\{\log_e x\}+C$

(c) $\frac{1}{2}\log\{\log_e ex\}-\log\{\log_e e^2x^2\}$
$+\log\{\log_e e^3x^3\}+C$

(d) $\frac{1}{2}\log\{\log_e ex\}-\log\{\log_e e^2x\}$
$+\frac{1}{2}\log\{\log_e e^3x\}+C$

**11.** $\int\frac{dx}{(x-\beta)\sqrt{(x-\alpha)(\beta-x)}}$ is

(a) $\frac{2}{\alpha-\beta}\sqrt{\frac{x-\alpha}{\beta-x}}+C$

(b) $\frac{2}{\alpha-\beta}\sqrt{(x-\alpha)(\beta-x)}+C$

(c) $\frac{\alpha-\beta}{2}(x-\alpha)\sqrt{\beta-x}+C$

(d) None of these

**12.** If $\int\frac{\tan\left(\frac{\pi}{4}-x\right)}{\cos^2 x\sqrt{\tan^3 x+\tan^2 x+\tan x}}$ dx.

$= -2\tan^{-1}u + c$ then $u$ is equal to

(a) $\sqrt{1+\tan x+\cot x}$ (b) $\sqrt{1+\tan x+\tan^2 x}$

(c) $\sqrt{\tan x+\cot x}$ (d) $\tan^{-1}(\tan x+\cot x)$

**13.** If $\int\frac{dx}{(a+bx^2)\sqrt{b-ax^2}}=\mathrm{K}\tan^{-1}(L\tan\theta)+M$,

M being constant of integration then KL is equal to

(a) 1 (b) $\frac{1}{a}$ (c) $\frac{1}{\sqrt[a]{a}}$ (d) $\sqrt{a}$

**14.** If $\int f(x)\sin x\cos x\,dx=\frac{1}{2(b^2-a^2)}\ln f(x)+c$,

then $f(x)$ is

(a) $\frac{1}{a\sin x+b\cos x}$

(b) $\frac{1}{a^2\sin^2 x+b^2\cos^2 x}$

(c) $\frac{1}{a^2\sin x+b^2\cos x}$

(d) $\frac{1}{a\sin^2 x+b\cos^2 x}$

**15.** The integral

$$\int\left(\frac{\sec 6\alpha}{\operatorname{cosec} 2\alpha}+\frac{\sec 18\alpha}{\operatorname{cosec} 6\alpha}+\frac{\sec 54\alpha}{\operatorname{cosec} 18\alpha}\right)d\alpha$$

is equal to

(a) $\frac{\ln|\sec 54\alpha|}{108}-\frac{\ln|\sec 2\alpha|}{4}+c$

(b) $\frac{\ln|\sec 6\alpha|}{6}+\frac{\ln|\sec 18\alpha|}{18}+\frac{\ln|\sec 54\alpha|}{54}+c$

(c) $\frac{1}{6}\ln\left|\frac{\sec 6\alpha}{\operatorname{cosec} 2\alpha}\right|+\frac{1}{18}\ln\left|\frac{\sec 18\alpha}{\operatorname{cosec} 6\alpha}\right|+\frac{1}{54}\ln\left|\frac{\sec 54\alpha}{\operatorname{cosec} 18\alpha}\right|+c$

(d) None of these

**16.** If $I_n=\int(\sin x+\cos x)^n dx\ (n\ge 2)$, then $nI_n-2(n-1)I_{n-2}$ is equal to

(a) $(\sin x+\cos x)^n(\sin x-\cos x)$

(b) $(\sin x+\cos x)^{n-1}(\sin x-\cos x)$

(c) $(\sin x+\cos x)^{n+1}(\sin x-\cos x)$

(d) $(\sin x-\cos x)^{n-1}(\sin x+\cos x)$

**17.** Let $\ell$n x denote the logarithm of $x$ with respect to the base e. Let $S\subset R$ be the set of all points where the function $\ell n\,(x^2-1)$ is well-defined. Then, the number of functions $f: S\to R$ that are differentiable, satisfy $f'(x)=\ell n\,(x^2-1)$ for all $x\in S$ and $f(2)=0$, is

(a) 0 (b) 1

(c) 2 (d) infinite

## Numeric Value Answer

**18.** If $\lim\limits_{x\to 0}\frac{f(x)}{x^2}$ exists finitely

$\lim\limits_{x\to 0}\left(1+x+\frac{f(x)}{x}\right)^{1/x}=e^3$, and $\int f(x)\log_e x\,dx$ is equal to $ax^3(\log_e x-b)+c$, then value of $a+b$ is _______.

**19.** If $\int\frac{\cos^2 x+\sin 2x}{(2\cos x-\sin x)^2}dx$

$=\frac{\cos x}{2\cos x-\sin x}+ax+b\,ln\,|2\cos x-\sin x|+c,$

then value of |a + b| is _______.

**20.** If $\int\frac{1-(\cot x)^{2010}}{\tan x+(\cot x)^{2011}}dx$

$=\frac{1}{k}\log_e|(\sin x)^k+(\cos x)^k|+c$

then $k$ is equal to

**21.** If $\int\frac{\sin^3\frac{\theta}{2}\,d\theta}{\cos\frac{\theta}{2}\sqrt{\cos^3\theta+\cos^2\theta+\cos\theta}}$

$=\tan^{-1}\sqrt{f(\theta)}+c$

then the least value of f(θ) for allowable values of θ is equal to

**22.** Let $f(x)$ be a function such that $f(xy)=f(x)\cdot f(x)$ $\forall\ x,y\in\mathbf{R}$. If $f(1+x)=1+x(1+g(x))$ where $\lim\limits_{t\to 0}g(x)=0$. Further if $\int\frac{f(x)}{f'(x)}dx=\frac{x^k}{k}+C,$ then evaluate the numerical quantity $k$.

**23.** $\int\{\sin(101x)\cdot\sin^{99}x\}dx=\frac{\sin(100x)(\sin x)^{\lambda}}{\mu},$

then $\frac{\lambda}{\mu}$ is equal to ..........

**24.** If $\int\frac{dx}{\cos^3 x-\sin^3 x}$

$=A\tan^{-1}f(x)+B\ln\left|\frac{\sqrt{2}+f(x)}{\sqrt{2}-f(x)}\right|+C$

where $f(x)=\sin x+\cos x$ find the value of $(12A+9\sqrt{2}B)-3$.

**25.** If $\int\frac{\sin^8 x}{\cos x}dx=-\frac{\sin^7 x}{a}+\frac{\sin^5 x}{b}-\frac{\sin^3 x}{c}+\frac{\sin x}{d}+\ln\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right|+C,$

then evaluate $(a+b+c+d)$.

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | (b) | 4 | (c) | 7 | (b) | 10 | (d) | 13 | (c) | 16 | (b) | 19 | (0.60) | 22 | (2) | 25 | (4) |
| 2 | (d) | 5 | (d) | 8 | (c) | 11 | (a) | 14 | (b) | 17 | (d) | 20 | (2012) | 23 | (1) | | |
| 3 | (c) | 6 | (a) | 9 | (b) | 12 | (a) | 15 | (a) | 18 | (1) | 21 | (3) | 24 | (8) | | |

23

# DEFINITE INTEGRATION

## MCQs with One Correct Answer

**1.** Let $p(x)$ be a polynomial of least degree whose graph has three points of inflection (–1, –1), (1, 1) and a point with abscissa 0 at which the curve is inclined to the axis of abscissa at an angle of 60°. Then $\int_0^1 p(x)dx$ is equal to

(a) $\frac{3\sqrt{3}+4}{14}$ (b) $\frac{3\sqrt{3}}{7}$

(c) $\frac{\sqrt{3}+\sqrt{7}}{14}$ (d) $\frac{\sqrt{3}+2}{7}$

**2.** $\int_{-\pi}^{\pi} (\cos px - \sin qx)^2\, dx$; where $p, q$ are integers; is equal to

(a) $-\pi$ (b) 0 (c) $\pi$ (d) $2\pi$

**3.** The value of the definite integral $\int_0^{2\pi} e^{\cos\theta} \cos\theta(\sin\theta)d\theta$ is

(a) 0 (b) $\pi$ (c) $2\pi$ (d) $e^{\pi}$

**4.** If $p, q, r, s$ are in arithmetic progression and

$$f(x) = \begin{vmatrix} p+\sin x & q+\sin x & p-r+\sin x \\ q+\sin x & r+\sin x & -1+\sin x \\ r+\sin x & s+\sin x & s-q+\sin x \end{vmatrix}$$

such that $\int_0^2 f(x)dx = -4$, then the common difference of the progession is

(a) $\pm 1$ (b) $\frac{1}{2}$

(c) $\pm 2$ (d) None of these

**5.** If $\int_0^{\pi/2} \frac{dx}{a^2\cos^2 x + b^2\sin^2 x} = \frac{\pi}{16}$. Then minimum value of $a\cos x + b\sin x$ is

(a) $-4$ (b) $-8$

(c) $-2$ (d) $-2\sqrt{2}$

**6.** Let $I = \int_{\pi/4}^{\pi/3} \frac{\sin x}{x} dx$, then $I$ belongs to

(a) $\left(\frac{\sqrt{3}}{8}, \frac{\sqrt{2}}{6}\right)$ (b) $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}\right)$

(c) $\left(\frac{1}{2}, \frac{\sqrt{2}}{2}\right)$ (d) None of these

**7.** Let $f(x)$ be positive, continuous and differentiable on the interval $(a, b)$ and $\lim_{x\to a^+} f(x) = 1$, $\lim_{x\to b^-} f(x) = 3^{1/4}$. If $f'(x) \geq f^3(x) + \frac{1}{f(x)}$ then the greatest value of $b - a$ is

(a) 1 (b) $3^{1/4}$

(c) $(3^{1/4} - 1)\frac{\pi}{24}$ (d) $\frac{\pi}{24}$

**8.** Let $a$, $b$, $c$ be non-zero real numbers such that

$$\int_0^1 (1+\cos^8 x)(ax^2+bx+c)\,dx$$

$$=\int_0^2 (1+\cos^8 x)(ax^2+bx+c)\,dx.$$

Then the quadratic equation

$ax^2+bx+c=0$ has

(a) no root in (0, 2)
(b) at least one root in (0, 2)
(c) a double root in (0, 2)
(d) two imaginary roots

**9.** Let $f(x)=\int_x^{x^2}(t-1)dt$. Then the value of $|f'(\omega)|$ where $\omega$ is a complex cube root of unity is

(a) $\sqrt{3}$ (b) $2\sqrt{3}$ (c) $3\sqrt{3}$ (d) $4\sqrt{3}$

**10.** $\lim\limits_{x\to\frac{\pi}{4}} \dfrac{\int_2^{\sec^2 x} f(t)dt}{x^2-\frac{\pi^2}{16}}$ equals

(a) $\frac{8}{\pi}f(2)$ (b) $\frac{2}{\pi}f(2)$

(c) $\frac{2}{\pi}f\left(\frac{1}{2}\right)$ (d) $4f(2)$

**11.** The value of the integral $\int_0^{\pi}(1-|\sin 8x|)dx$ is

(a) 0 (b) $\pi-1$
(c) $\pi-2$ (d) $\pi-3$

**12.** Let S be the set of real numbers p such that there is no non-zero continuous function f : R → R satisfying $\int_0^x f(t)dt = p\,f(x)$ for all $x \in R$. Then, $S$ is

(a) the empty set
(b) the set of all rational numbers
(c) the set of all irrational numbers
(d) the whole set R.

**13.** Suppose the limit L = $\lim\limits_{n\to\infty}\sqrt{n}\int_0^1 \frac{1}{(1+x^2)^n}dx$ exists and is larger than $\frac{1}{2}$. Then,

(a) $\frac{1}{2}<L<2$ (b) $2<L<3$
(c) $3<L<4$ (d) $L\geq 4$

**14.** Suppose a continuous function $f$ : [0, ∞) → **R** satisfies

$$f(x)=2\int_0^x tf(t)dt+1 \text{ for all } x\geq 0.$$

Then $f(1)$ equals

(a) $e$ (b) $e^2$ (c) $e^4$ (d) $e^6$

**15.** Let $J=\int_0^1 \frac{x}{1+x^8}dx.$

Consider the following assertions:

I. $J>\frac{1}{4}$ II. $J<\frac{\pi}{8}$

Then
(a) only I is true
(b) only II is true
(c) both I and II are true
(d) neither I nor II is true

**16.** For $x \in$ **R**, let $f(x)=|\sin x|$ and $g(x)=\int_0^x f(t)\,dt.$

Let $p(x)=g(x)-\frac{2}{\pi}x$. Then

(a) $p(x+\pi)=p(x)$ for all $x$
(b) $p(x+\pi)\neq p(x)$ for at least one but finitely many $x$
(c) $p(x+\pi)\neq p(x)$ for infinitely many $x$
(d) $p$ is a one-one function

**17.** Let $f$: [0, 1] → [0, 1] be a continuous function such that

$x^2+(f(x))^2\leq 1$ for all $x\in[0,1]$ and $\int_0^1 f(x)\,dx=\frac{\pi}{4}.$

Then $\int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}}\frac{f(x)}{1-x^2}dx$ equals

(a) $\frac{\pi}{12}$ (b) $\frac{\pi}{15}$

(c) $\frac{\sqrt{2}-1}{2}\pi$ (d) $\frac{\pi}{10}$

## Numeric Value Answer

**18.** Let $f: R \to R$ be a differentiable function and $f(1) = 4$. Then the value of $\lim_{x\to1}\frac{\int_4^{f(x)} 2t\,dt}{x-1}$, if $f'(1)=2$ is

**19.** If $f(x)=\begin{vmatrix}\sin x & \sec x & x^2-1\\ \text{cosec } x & x\sin x & \cos x\\ \tan x & x\tan x & x^2+1\end{vmatrix}$ then $\int_{-\pi/3}^{\pi/3} f(x)dx =$

**20.** $\lim_{x\to0}\left[\frac{1}{x^5}\int_0^x e^{-t^2}dt-\frac{1}{x^4}+\frac{1}{3x^2}\right]=$

**21.** If $I(n)=\int_0^{\pi/2}\theta\sin^n\theta d\theta,\ n\in N,\ n>3,$ then $\frac{1}{1005}[2010\,I(2010)\ -2009\,I(2008)]^{-1}$ is equal to

**22.** Let $f:(0,\infty)\to R$ be a differentiable function such that

$$x\int_0^x(1-t)f(t)dt=\int_0^x t\,f(t)dt\ \forall x\in(0,\infty)\text{ and}$$

$f(1)=1$. The value of the limit $\lim_{x\to\infty} f(x)$ is equal to

**23.** If $p$ and $q$ are different roots of the equation $\tan x = x$ then $\int_0^1 2\sin(pt)\sin(qt)dt$ is equal to

**24.** Find the value $\int_0^3\left[\sqrt{x}\right]dx$, when $[x]$ is the greatest integral value of $x$.

**25.** Let $f: \mathbb{R}\to\mathbb{R}$ be a continuous odd function, which vanishes exactly at one point and $f(1)=\frac{1}{2}$. Suppose that $F(x)=\int_{-1}^x f(t)dt$ for all $x\in[-1, 2]$ and $G(x)=\int_{-1}^x t\,|f(f(t))|\,dt$ for all $x\in[-1, 2]$. If $\lim_{x\to1}\frac{F(x)}{G(x)}=\frac{1}{14}$, then the value of $f\left(\frac{1}{2}\right)$ is

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (a) | **4** | (a) | **7** | (d) | **10** | (a) | **13** | (a) | **16** | (a) | **19** | (0) | **22** | 0 | **25** | (7) |
| **2** | (d) | **5** | (a) | **8** | (b) | **11** | (c) | **14** | (a) | **17** | (a) | **20** | (0.30) | **23** | 0 | | |
| **3** | (c) | **6** | (a) | **9** | (c) | **12** | (d) | **15** | (a) | **18** | (16) | **21** | (2) | **24** | (2) | | |

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# APPLICATION OF INTEGRALS

## MCQs with One Correct Answer

**1.** The area bounded by the x-axis, the curve y = f(x) and the lines x = 1, x = b, is equal to $\sqrt{b^2+1}-\sqrt{2}$ for all b > 1, then f(x) is

(a) $\sqrt{x-1}$ (b) $\sqrt{x+1}$

(c) $\sqrt{x^2+1}$ (d) $\dfrac{x}{\sqrt{1+x^2}}$

**2.** Area of region bounded by $[x]^2=[y]^2$ if $x \in [1, 5]$ (where [.] represents the greatest integer function), is

(a) 10 sq. unit (b) 8 sq. unit

(c) 6 sq. unit (d) 5 sq. unit

**3.** If a point $P$ moves such that its distance from line $y=\sqrt{3}x-7$ is same as its distance from $(2\sqrt{3},-1)$, then area of the curve, described by $P$, enclosed between the coordinate axes is

(a) $\dfrac{\sqrt{3}}{2}$ (b) $2\sqrt{3}$ (c) 6 (d) $\sqrt{3}$

**4.** The area bounded by the curve $f(x)=||\tan x+\cot x|-|\tan x-\cot x||$ between the lines $x=0$, $x=\dfrac{\pi}{2}$ and the x-axis, is

(a) $ln\sqrt{2}$ (b) $2\,ln\,2$

(c) $ln\,4$ (d) $\sqrt{2}ln2$

**5.** If the line $y=mx+2$ cuts the parabola $2y=x^2$ at points $(x_1,y_1)$ and $(x_2,y_2)$ where $(x_1<x_2)$, then the value of $m$ for which $\displaystyle\int_{x_1}^{x_2}\left(mx+2-\frac{x^2}{2}\right)dx$ is minimum, is

(a) $\sqrt{2}$ (b) $\dfrac{8}{3}$ (c) $\dfrac{1}{\sqrt{3}}$ (d) 0

**6.** The area bounded by the curve $f(x)=x+\sin x$ and its inverse function between the ordinates $x=0$ to $x=2\pi$ is

(a) $4\pi$ square units (b) $8\pi$ square units

(c) 4 square units (d) 2 square units

**7.** The area bounded by the curve $y^2(2a-x)=x^3$ and the line $x=2a$ is

(a) $3\pi a^2$ sq. unit (b) $\dfrac{3\pi a^2}{2}$ sq. unit

(c) $\dfrac{3\pi a^2}{4}$ sq. unit (d) $\dfrac{6\pi a^2}{5}$ sq. unit

**8.** The area bounded by the curves $y=xe^x$, $y=xe^{-x}$ and the line $x=1$ is

(a) $\dfrac{2}{e}$ (b) $1-\dfrac{2}{e}$ (c) $\dfrac{1}{e}$ (d) $1-\dfrac{1}{e}$

**9.** Area enclosed by the curve $|x+y-1|+|2x+y-1|=1$ is

(a) 2 sq. units (b) 3 sq. units

(c) 6 sq. units (d) 7 sq. units

**10.** The area bounded by the curve

$f(x) = \max.\{4 - x^2, |x-2|, (x-2)^{1/3}\}$

for $x \in [-2, 4]$ and x-axis is

(a) $\frac{45}{4}$ sq. units (b) $\frac{23}{4}$ sq. units

(c) $\frac{59}{4}$ sq. units (d) $\frac{53}{4}$ sq. units

**11.** The area of the region between the curves $y = \sqrt{\frac{1+\sin x}{\cos x}}$ and $y = \sqrt{\frac{1-\sin x}{\cos x}}$ bounded by the lines $x = 0$ and $x = \frac{\pi}{4}$ is

(a) $\int_0^{\sqrt{2}-1} \frac{t}{(1+t^2)\sqrt{1-t^2}} dt$

(b) $\int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$

(c) $\int_0^{\sqrt{2}+1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$

(d) $\int_0^{\sqrt{2}+1} \frac{t}{(1+t^2)\sqrt{1-t^2}} dt$

**12.** The area enclosed by the curves $y = x^2$, $y = x^3$, $x = 0$ and $x = p$, where $p > 1$, is 1/6. Then p equals

(a) 8/3 (b) 16/3 (c) 2 (d) 4/3

**13.** If a straight line $y - x = 2$ divides the region $x^2 + y^2 \le 4$ into two parts, then the ratio of the area of the smaller part to the area of the greater part is

(a) $3\pi - 8 : \pi + 8$ (b) $\pi - 3 : 3\pi + 3$

(c) $3\pi - 4 : \pi + 4$ (d) $\pi - 2 : 3\pi + 2$

**14.** The area of the region above the x-axis bounded by the curve $y = \tan x$, $0 \le x \le \frac{\pi}{2}$ and the tangent to the curve at $x = \frac{\pi}{4}$ is:

(a) $\frac{1}{2}\left(\log 2 - \frac{1}{2}\right)$ (b) $\frac{1}{2}\left(\log 2 + \frac{1}{2}\right)$

(c) $\frac{1}{2}(1 - \log 2)$ (d) $\frac{1}{2}(1 + \log 2)$

**15.** Area of the region

$\left\{(x, y) \in \mathbb{R}^2 : y \ge \sqrt{|x+3|}, 5y \le x + 9 \le 15\right\}$

is equal to

(a) $\frac{1}{6}$ (b) $\frac{4}{3}$ (c) $\frac{3}{2}$ (d) $\frac{5}{3}$

**16.** The maximum possible area bounded by the parabola $y = x^2 + x + 10$ and a chord of the parabola of length 1 is

(a) $\frac{1}{12}$ (b) $\frac{1}{6}$ (c) $\frac{1}{3}$ (d) $\frac{1}{2}$

**17.** The area of the region bounded by the lines $x = 1$, $x = 2$ and the curves $x(y - e^x) = \sin x$ and $2xy = 2\sin x + x^3$ is

(a) $e^2 - e - \frac{1}{6}$ (b) $e^2 - e - \frac{7}{6}$

(c) $e^2 - e + \frac{1}{6}$ (d) $e^2 - e + \frac{7}{6}$

## Numeric Value Answer

**18.** If $[x]$ is the greatest integrer $\le x$, then

$\int_{-2}^{2} \min\{x - [x], -x - [-x]\} dx =$

**19.** Let for all $n \in N$, $n - 2 \ge 2\log_e n$ then the minimum value of the area bounded by the curve $|y| + \frac{1}{n} \le e^{-|x|}$ is

**20.** If $f(x) = \begin{cases} \sqrt{\{x\}}, & x \notin I \\ 1, & x \in I \end{cases}$ and $g(x) = \{x\}^2$, (where {x} is fractional part of $x$), then area bounded between $f(x)$ and $g(x)$ $\forall x \in [0, 10]$ is .

**21.** $P(x, y)$ is a point, which moves in the $xy$ plane such that $2[y] = 3[x]$ where $[\,.\,]$ denotes the greatest integer function and $-2 \le x \le 5$ and $-3 \le y \le 6$. The area of the region containing the point $P(x, y)$ is equal to

**22.** If the value of $a$ ($a \ge 1$) for which the area of the figure bounded by pair of straight lines $y^2 - 3y + 2 = 0$ and the curves $y = [a]x^2$, $y = \frac{1}{2}[a]x^2$ is greatest, is $[a, b)$, then $(b - a) =$ where [.] denotes the greatest integer function.

**23.** If the area bounded by $y = f(x)$ and the curve $y = \frac{2}{1+x^2}$ where $f$ is a continuous function satisfying the conditions $f(x) \cdot f(y) = f(xy)$. $\forall$ $x, y, \in R$ and $f'(1) = 2$, $f(1) = 1$ is $\left(\pi - \frac{a}{3}\right)$, then $a =$

**24.** Let $f(x)$ be a polynomial of degree 3 such that the curve $y = f(x)$ has relative extremes at $x = \pm 2/\sqrt{3}$ and passes through (0, 0) and $(1, -2)$ dividing the circle $x^2 + y^2 = 4$ in two parts. Find out the difference of areas of these two parts.

**25.** Area of the region bounded by the curve $y = \{x^2\}$, where $\{.\}$ denotes fractional part function $\forall\, x \in [-2, 2]$ is $p\left(\sqrt{q} + \sqrt{r} - \frac{7}{3}\right)$. Find the value of $p + q + r$.

| ANSWER KEY | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (d) | **4** | (c) | **7** | (b) | **10** | (c) | **13** | (d) | **16** | (b) | **19** | (2) | **22** | (1) | **25** | (7) |
| **2** | (b) | **5** | (d) | **8** | (a) | **11** | (b) | **14** | (a) | **17** | (b) | **20** | (3.33) | **23** | (2) | | |
| **3** | (a) | **6** | (d) | **9** | (a) | **12** | (d) | **15** | (c) | **18** | (1) | **21** | (4) | **24** | (0) | | |

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# DIFFERENTIAL EQUATION AND ITS APPLICATIONS

## MCQs with One Correct Answer

1. Through any point $(x, y)$ of a curve which passes through the origin, lines are drawn parallel to the co-ordinate axes. The curve, given that it divides the rectangle formed by the two lines and the axes into two areas, one of which is twice the other, represents a family of
   (a) circles
   (b) pair of straight lines
   (c) parabolas
   (d) rectangular hyperbolas

2. Lef $f(x)$ be a positive, continuous and differentiable function on the interval $(a, b)$. If $\lim_{x\to a^+} f(x)=1$ and $\lim_{x\to b^-} f(x)=3^{1/4}$. Also $f'(x)\ge f^3(x)+\dfrac{1}{f(x)}$ then
   (a) $b-a\ge\dfrac{\pi}{4}$  (b) $b-a\le\dfrac{\pi}{4}$
   (c) $b-a\le\dfrac{\pi}{24}$  (d) None of these

3. Let $f$ be a non-negative function defined on the interval $[0, 1]$.
   If $\int_0^x \sqrt{1-(f'(t))^2}\,dt=\int_0^x f(t)\,dt,\ 0\le x\le 1$ and $f(0)=0$, then
   (a) $f(1/2)<1/2$ and $f(1/3)>1/3$
   (b) $f(1/2)>1/2$ and $f(1/3)>1/3$
   (c) $f(1/2)<1/2$ and $f(1/3)<1/3$
   (d) $f(1/2)>1/2$ and $f(1/3)<1/3$

4. The solution of the differential equation $x^3\dfrac{dy}{dx}=y^3+y^2\sqrt{y^2-x^2}$ is
   (a) $y+\sqrt{y^2-x^2}=cxy$
   (b) $y-\sqrt{y^2-x^2}=cxy$
   (c) $y\sqrt{y^2-x^2}=cx+y$
   (d) $x\sqrt{y^2-x^2}=cx+y$

5. The solution of the differential equation, $2x^2y\dfrac{dy}{dx}=\tan(x^2y^2)-2xy^2$ given $y(1)=\sqrt{\dfrac{\pi}{2}}$ is
   (a) $\sin x^2y^2=e^{x-1}$  (b) $\sin(x^2y^2)=x$
   (c) $\cos x^2y^2+x=0$  (d) $\sin(x^2y^2)=e.e^x$

6. Solution of the differental equation $x\cos\left(\dfrac{y}{x}\right)(ydx+xdy)=y\sin\left(\dfrac{y}{x}\right)(xdy-ydx)$ is
   (a) $y=cx\cos\left(\dfrac{x}{y}\right)$  (b) $\sec\left(\dfrac{y}{x}\right)=cxy$
   (c) $\left(\dfrac{y}{x}\right)\sec\left(\dfrac{y}{x}\right)=c$  (d) None of these

7. The family of curves whose tangents form an angle of $\dfrac{\pi}{4}$ with the hyperbolas $xy=C$ are
   (a) pair of straight lines
   (b) $y^2-xy-x^2=C$
   (c) $y^2-2xy-x^2=C$
   (d) $y^2+2xy-x^2=C$

**8.** Solution of the differential equation

$$\left\{\frac{1}{x}-\frac{y^2}{(x-y)^2}\right\}dx+\left\{\frac{x^2}{(x-y)^2}-\frac{1}{y}\right\}dy=0 \text{ is}$$

(a) $\ln\left|\frac{x}{y}\right|+\frac{xy}{x-y}=c$

(b) $\ln|xy|+\frac{xy}{(x-y)}=c$

(c) $\frac{xy}{(x-y)}=ce^{x/y}$

(d) $\frac{xy}{(x-y)}=c\,e^{xy}$

(where $c$ is arbitrary constant)

**9.** The equation of the curve passing through the points $(3a, a)$ $(a>0)$ in the form $x=f(y)$ which satisfy the differential equation;

$$\frac{a^2}{xy}\cdot\frac{dx}{dy}=\frac{x}{y}+\frac{y}{x}-2, \text{ is}$$

(a) $x=y+a\left(\frac{1+e^{y-k}}{1-2e^{y-k}}\right)$

(b) $x=y+a\left(\frac{1+e^{y-k}}{1-e^{y-k}}\right)$

(c) $y=x+a\left(\frac{1+e^{y-k}}{1-e^{y-k}}\right)$

(d) None of these

**10.** The solution of the differential equation

$$(y+x\sqrt{xy}(x+y))dx+(y\sqrt{xy}(x+y)-x)dy=0,$$

is

(a) $\frac{x^2+y^2}{2}+2\tan^{-1}\sqrt{\frac{x}{2y}}=C$

(b) $\frac{x^2+y^2}{2}+2\tan^{-1}\sqrt{\frac{x}{y}}=C$

(c) $\frac{x^2+y^2}{\sqrt{2}}+2\tan^{-1}\sqrt{\frac{x}{y}}=C$

(d) None of these

**11.** Solution of $\frac{dy}{dx}=\frac{y^3}{e^{2x}+y^2}$ is

(a) $e^{-2x}y^2+2\ln|y|=c$

(b) $e^{2x}y^2-2\ln|y|=c$

(c) $e^x+\ln|y|=c$

(d) None of these

**12.** The orthogonal trajectory of the curve $a^{n-1}y=x^n$ is

(a) $n^2+x^2=$ constant (b) $ny^2+x^2=$ constant

(c) $x^2+y^2=$ constant (d) $y^2+nx^2=$ constant

**13.** A curve $f(x)$ passes through the point $P(1, 1)$. The normal to the curve at point $P$ is $a(y-1)+(x-1)=0$. If the slope of the tangent at any point on the curve is proportional to the ordinate at that point, then the equation of the curve is

(a) $y=e^{ax}-1$ (b) $y-1=e^{ax}$

(c) $y=e^{a(x-1)}$ (d) $y-a=e^{ax}$

**14.** A tangent and a normal to a curve at any point $P$ meet the $x$ and $y$ axes at $A, B$ and $C, D$ respectively. If the centre of circle through $O$, $C$, $P$ and $B$ lies on the line $y=x$ ($O$ is the origin) then the differential equation of all such curves is :

(a) $\frac{dy}{dx}=\frac{y-x}{y+x}$ (b) $\frac{dy}{dx}=\frac{y^2-x^2}{y^2+x^2}$

(c) $\frac{dy}{dx}=\frac{x-y}{xy}$ (d) None of these

**15.** If $\frac{dy}{dx}-y\log_e 2=2^{\sin x}(\cos x-1)\log_e 2$, then $y=$

(a) $2^{\sin x}+c2^x$ (b) $2^{\cos x}+c2^x$

(c) $2^{\sin x}+c2^{-x}$ (d) $2^{\cos x}+c2^{-x}$

**16.** Solution of differential equation

$$\frac{dt}{dx}=\frac{t\left(\frac{d}{dx}(g(x))\right)-t^2}{g(x)} \text{ is}$$

(a) $t=\frac{g(x)}{x}+c$ (b) $t=\frac{g(x)}{x^2}+c$

(c) $t=\frac{g(x)}{x+c}$ (d) $t=g(x)+x+c$

**17.** Let $f(x)$ be continuously differentiable on the interval $(0, \infty)$ such that $f(1) = 1$, and $\lim_{t\to x}\frac{t^2 f(x)-x^2 f(t)}{t-x}=1$ for each $x > 0$, then $f(x)$ is

(a) $\frac{1}{3x}+\frac{2x^2}{3}$ (b) $-\frac{1}{3x}+\frac{4x^2}{3}$

(c) $-\frac{1}{x}+\frac{2}{x^2}$ (d) $\frac{1}{x}$

## Numeric Value Answer

**18.** Find the sum of the order 'O' and degree D of the differential equation

$$y=1+x\left(\frac{dy}{dx}\right)+\frac{x^2}{2!}\left(\frac{dy}{dx}\right)^2+.....+\frac{x^n}{n!}\left(\frac{dy}{dx}\right)^n$$

$+........\infty$.

**19.** Let $y = f(x)$ be a curve passing through $(e, e^e)$, which satisfy the differential equation $(2ny + xy \log_e x)\, dx - x \log_e x\, dy = 0$, $x > 0$, $y > 0$. If $g(x)=\lim_{n\to\infty} f(x)$, then $\int_{1/e}^{e} g(x)\,dx$ equals to.

**20.** If the solution of the differential equation $\frac{dy}{dx}=\frac{1}{xy[x^2\sin y^2+1]}$ is $x^2(\cos y^2-\sin y^2-2Ce^{-y^2})=k$, then value of $k$ is ______.

**21.** If the equation of a curve $y = y(x)$ satisfies the differential equation $x\int_0^x y(t)dt=(x+1)\int_0^x ty(t)dt,\ x>0$, and $y(1) = e$, then $y\left(\frac{1}{2}\right)$ is equal to

**22.** If the solution of the differential equation

$$\frac{x\,dx-y\,dy}{x\,dy-y\,dx}=\sqrt{\left(\frac{1+x^2-y^2}{x^2-y^2}\right)}$$ be

$$\sqrt{f(x,y)}+\sqrt{1+f(x,y)}=c\left(\frac{x+y}{\sqrt{f(x,y)}}\right)$$

where $c$ is an arbitrary constant then $f(3, 2)$ is equal to

**23.** A curve $y = f(x)$ is such that $f(x)\geq 0$ and $f(0)=0$ and bounds a curvilinear trapezoid with the base $[0, x]$ whose area is proportional to $(n+1)^{\text{th}}$ power of $f(x)$. If $f(1) = 1$, then $\{f(10)\}^n$ is equal to 10 k, where k equals

**24.** Let $y = f(x)$ is a polynomial function satisfying

$$x^2 f'(x)f\left(\frac{1}{x}\right)-f(x)f'\left(\frac{1}{x}\right)=x^2 f'(x)-f'\left(\frac{1}{x}\right).$$

If $f(1) = 2$ and $f(5) = 26$, then find $f(6) - 30$.

**25.** If the area bounded by $y = f(x)$, $x=\frac{1}{2}$, $x=\frac{\sqrt{3}}{2}$ and the $X$-axis is $A$ sq. units where $f(x)=x+\frac{2}{3}x^3+\frac{2}{3}\cdot\frac{4}{5}x^5+\frac{2}{3}\cdot\frac{4}{5}\cdot\frac{6}{7}x^7+$ $....\infty$, $|x|<1$, then the value of $[4A]$ is (where [.] is G.I.F)

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (c) | **4** | (a) | **7** | (c) | **10** | (b) | **13** | (c) | **16** | (c) | **19** | (0) | **22** | (5) | **25** | (1) |
| **2** | (c) | **5** | (a) | **8** | (a) | **11** | (a) | **14** | (a) | **17** | (a) | **20** | (2) | **23** | (1) | | |
| **3** | (c) | **6** | (b) | **9** | (b) | **12** | (b) | **15** | (a) | **18** | (2) | **21** | (8) | **24** | (7) | | |

# 26

# VECTOR ALGEBRA

## MCQs with One Correct Answer

1. The vectors $\vec{a}(x) = \cos x\hat{i} + \sin x\hat{j}$ and $\vec{b}(x) = x\hat{i} + \sin x\hat{j}$ are collinear for
(a) Unique value of $x$, $0 < x < \frac{\pi}{6}$
(b) Unique value of $\frac{\pi}{6} < x < \frac{\pi}{3}$
(c) No value of $x$
(d) Infinitely many values of $x$, $0 < x < \frac{\pi}{2}$

2. If $\cos\alpha \neq 1$, $\cos\beta \neq 1$ and $\cos\gamma \neq 1$, then the vector $\vec{a} = \hat{i}\cos\alpha + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j}\cos\beta + \hat{k}$, $\vec{c} = \hat{i} + \hat{j} + \hat{k}\cos\gamma$ are
(a) coplanar vectors
(b) coplanar vectors if $\cos\alpha = \cos\beta = \cos\gamma \neq 1$
(c) coplanar vectors if os $\alpha \neq \cos\beta \neq \cos\gamma$
(d) never coplanar

3. Let $a$, $b$, $c$ be distinct non-negative numbers. If the vectors $a\hat{i} + a\hat{j} + c\hat{k}, \hat{i} + \hat{k}$ and $c\hat{i} + c\hat{j} + b\hat{k}$ lie in a plane, then $c$ is
(a) the Arithmetic Mean of $a$ and $b$
(b) the Geometric Mean of $a$ and $b$
(c) the Harmonic Mean of $a$ and $b$
(d) equal to zero

4. Let $\vec{a}$ and $\vec{b}$ are two vectors making angle $\theta$ with each other, then which of the following represents unit vectors along bisector of $\vec{a}$ and $\vec{b}$ is:
(a) $\pm\frac{\hat{a}+\hat{b}}{2}$
(b) $\pm\frac{\hat{a}+\hat{b}}{2\cos\theta}$
(c) $\pm\frac{\hat{a}+\hat{b}}{2\cos\theta/2}$
(d) $\frac{(\hat{a}+\hat{b})}{|\hat{a}+\hat{b}|}$

5. In a parallelogram $ABCD$, $|\overrightarrow{AB}| = a$, $|\overrightarrow{AD}| = b$ and $|\overrightarrow{AC}| = c$, the value of $\overrightarrow{DB}.\overrightarrow{AB}$ is
(a) $\frac{3a^2 + b^2 - c^2}{2}$
(b) $\frac{a^2 + 3b^2 - c^2}{2}$
(c) $\frac{a^2 - b^2 + 3c^2}{2}$
(d) $\frac{a^2 + 3b^2 + c^2}{2}$

6. $\vec{a}$ and $\vec{c}$ are unit vectors and $|\vec{b}| = 4$. The angle between $\vec{a}$ and $\vec{c}$ is $\cos^{-1}\left(\frac{1}{4}\right)$. If $\vec{b} - 2\vec{c} = \lambda\vec{a}$, then $\lambda$ is equal to
(a) $3, -4$
(b) $\frac{1}{4}, \frac{3}{4}$
(c) $-3, 4$
(d) $-\frac{1}{4}, \frac{3}{4}$

7. The angles of a triangle, two of whose side are represented by the vectors $\sqrt{3}(\hat{a} \times \vec{b})$ and $\vec{b} - (\hat{a}.\vec{b}).\hat{a}$ where $\vec{b}$ is a non-zero vector and $\hat{a}$ is a unit vector, are
(a) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right); \tan^{-1}\left(\frac{1}{2}\right); \tan^{-1}\left(\frac{\sqrt{3}+2}{1-2\sqrt{3}}\right)$
(b) $\tan^{-1}(\sqrt{3}); \tan^{-1}\left(\frac{1}{\sqrt{3}}\right); \cot^{-1}(0)$
(c) $\tan^{-1}(\sqrt{3}); \tan^{-1}(2); \tan^{-1}\left(\frac{\sqrt{3}+2}{2\sqrt{3}-1}\right)$
(d) $\tan^{-1}(\sqrt{3}); \tan^{-1}(2); \tan^{-1}\left(\frac{\sqrt{2}+3}{3\sqrt{2}-1}\right)$

**8.** If the vectors $\overrightarrow{a} = (c\log_2 x)\hat{i} - 6\hat{j} + 3\hat{k}$ and $\overrightarrow{b} = (\log_2 x)\hat{i} + 2\hat{j} + (2c\log_2 x)\hat{k}$ make an obtuse angle for any $x \in (0, \infty)$, then the interval to which '$c$' belongs is

(a) $(0, \infty)$ (b) $(-\infty, 0)$

(c) $(-\frac{4}{3}, 0)$ (d) $(-1, 0) \cup (0, \frac{2}{3})$

**9.** Let $\vec{p} = a\hat{i} + b\hat{j} + c\hat{k}$ and $\vec{q} = b\hat{i} + c\hat{j} + a\hat{k}$, where a, b, $c \in R$. If 'θ' be the angle between $\vec{p}$ and $\vec{q}$ then,

(a) $\theta \in (0, \pi/2)$ (b) $\theta \in [0, 2\pi/3]$

(c) $\theta \in (2\pi/3, \pi]$ (d) $\theta \in [\pi/2, \pi]$

**10.** The value of '$a$' so that the volume of parallelopiped formed by $\hat{i} + a\hat{j} + \hat{k}$, $\hat{j} + a\hat{k}$ and $a\hat{i} + \hat{k}$ becomes minimum is

(a) $-3$ (b) $3$ (c) $1/\sqrt{3}$ (d) $\sqrt{3}$

**11.** If $\vec{a}, \vec{b}, \vec{c}$ are three non-zero, non-coplanar vectors and

$$\vec{b}_1 = \vec{b} - \frac{\vec{b}\cdot\vec{a}}{|\vec{a}|^2}\vec{a},\quad \vec{b}_2 = \vec{b} + \frac{\vec{b}\cdot\vec{a}}{|\vec{a}|^2}\vec{a},$$

$$\vec{c}_1 = \vec{c} - \frac{\vec{c}\cdot\vec{a}}{|\vec{a}|^2}\vec{a} + \frac{\vec{b}\cdot\vec{c}}{|\vec{c}|^2}\vec{b}_1,$$

$$\vec{c}_2 = \vec{c} - \frac{\vec{c}\cdot\vec{a}}{|\vec{a}|^2}\vec{a} - \frac{\vec{b}_1\cdot\vec{c}}{|\vec{b}_1|^2}\vec{b}_1,$$

$$\vec{c}_3 = \vec{c} - \frac{\vec{c}\cdot\vec{a}}{|\vec{c}|^2}\vec{a} + \frac{\vec{b}\cdot\vec{c}}{|\vec{c}|^2}\vec{b}_1,$$

$$\vec{c}_4 = \vec{c} - \frac{\vec{c}.\vec{a}}{|\vec{c}|^2}\vec{a} = \frac{\vec{b}.\vec{c}}{|\vec{b}|^2}\vec{b}_1,$$

then the set of orthogonal vectors is

(a) $(\vec{a}, \vec{b}_1, \vec{c}_3)$ (b) $(\vec{a}, \vec{b}_1, \vec{c}_2)$

(c) $(\vec{a}, \vec{b}_1, \vec{c}_1)$ (d) $(\vec{a}, \vec{b}_2, \vec{c}_2)$

**12.** A unit vector which is perpendicular to the vector $2\hat{i} - \hat{j} + 2\hat{k}$ and is coplanar with the vectors $\hat{i} + \hat{j} - \hat{k}$ and $2\hat{i} + 2\hat{j} - \hat{k}$ is

(a) $\frac{2\hat{j} + \hat{k}}{\sqrt{5}}$ (b) $\frac{3\hat{i} + 2\hat{j} - 2\hat{k}}{\sqrt{17}}$

(c) $\frac{3\hat{i} + 2\hat{j} + 2\hat{k}}{\sqrt{17}}$ (d) $\frac{2\hat{i} + 2\hat{j} - 2\hat{k}}{3}$

**13.** If $a$, $b$ and $c$ are $p$th, $q$th, $r$th terms of HP and $\vec{u} = (q-r)\hat{i} + (r-p)\hat{j} + (p-q)\hat{k}$, $\vec{v} = \frac{\hat{i}}{a} + \frac{\hat{j}}{b} + \frac{\hat{k}}{c}$, then

(a) $\vec{u}$ and $\vec{v}$ are parallel vectors

(b) $\vec{u}$ and $\vec{v}$ are orthogonal vectors

(c) $\vec{u}\cdot\vec{v} = 1$

(d) $\vec{u}\times\vec{v} = \hat{i} + \hat{j} + \hat{k}$

**14.** If the two adjacent sides of two rectangles are represented by vectors $\vec{p} = 5\vec{a} - 3\vec{b}$; $\vec{q} = -\vec{a} - 2\vec{b}$ and $\vec{r} = -4\vec{a} - \vec{b}$; $\vec{s} = -\vec{a} + \vec{b}$, respectively, then the angle between the vectors $\vec{x} = \frac{1}{3}(\vec{p} + \vec{r} + \vec{s})$ and $\vec{y} = \frac{1}{5}(\vec{r} + \vec{s})$ is

(a) $-\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$

(b) $\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$

(c) $\pi\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$

(d) Cannot be evaluated

**15.** $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-coplanar vectors and $\vec{d}$ be a non-zero vector, which is perpendicular to $(\vec{a} + \vec{b} + \vec{c})$. Now if $\vec{d} = \sin x(\vec{a}\times\vec{b}) + \cos y(\vec{b}\times\vec{c}) + 2(\vec{c}\times\vec{a})$, then minimum value of $x^2 + y^2$ is equal to

(a) $\pi^2$ (b) $0$ (c) $\pi^2/4$ (d) $5\pi^2/4$

**16.** The vectors $\vec{a} = 2\lambda^2\hat{i} + 4\lambda\hat{j} + \hat{k}$ and $\vec{b} = 7\hat{i} - 2\hat{j} + \lambda\hat{k}$ make an obtuse angle whereas the angle between $\vec{b}$ and $\hat{k}$ is acute and less than $\pi/6$,

(a) $0 < \lambda < \frac{1}{2}$ (b) $\lambda > \sqrt{159}$

(c) $-\frac{1}{2} < \lambda < 0$ (d) null set

**17.** The angle $\theta$ between two non-zero vectors $\vec{a}$ and $\vec{b}$ satisfies the relation

$$\cos\theta = (\vec{a}\times\hat{i})\cdot(\vec{b}\times\hat{i}) + (\vec{a}\times\hat{j})\cdot(\vec{b}\times\hat{j}) + (\vec{a}\times\hat{k})\cdot(\vec{b}\times\hat{k}),$$

then the least value of $|a| + |b|$ is equal to (where $\theta \neq 90°$)

(a) $\frac{1}{2}$ (b) 2

(c) $\sqrt{2}$ (d) 4

## Numeric Value Answer

**18.** If the sum of two unit vectors is a unit vector, and the magnitude of their difference is $\sqrt{k}$, then value of $k$ is

**19.** If $\hat{x}, \hat{y}$ and $\hat{z}$ are three unit vectors in three-dimensional space, then the minimum value of

$$|\hat{x}+\hat{y}|^2 + |\hat{y}+\hat{z}|^2 + |\hat{z}+\hat{x}|^2$$

**20.** Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that $\hat{a}\cdot\hat{b} = \frac{1}{3}$ and $\hat{a}\times\hat{b} = \hat{c}$. Also $\vec{F} = \alpha\hat{a} + \beta\hat{b} + \gamma\hat{c}$, where $\alpha$, $\beta$, $\gamma$ are scalars. If $\alpha = k_1(\hat{F}\cdot\hat{a}) - k_2(\hat{F}\cdot\hat{b})$, then the value of $2(k_1 + k_2)$ is

**21.** Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = x_1\hat{i} + x_2\hat{j} + x_3\hat{k}$, where $x_1, x_2, x_3 \in \{-3, -2, -1, 0, 1, 2\}$.

If the number of possible vectors $\vec{b}$ such that $\vec{a}$ and $\vec{b}$ are mutually perpendicular is $t$, then $t/5 =$

**22.** Find the absolute value of parameter $t$ for which the area of the triangle whose vertices are $A(-1, 1, 2); B(1, 2, 3)$ and $C(t, 1, 1)$ is minimum.

**23.** Given a vector $\vec{A}$ defined as

$$\vec{A} = (\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d}) + (\vec{a}\times\vec{c})\times(\vec{b}\times\vec{d}) + (\vec{a}\times\vec{d})\times(\vec{b}\times\vec{c})$$, then find the value of $|\vec{A}\times\vec{a}|$.

**24.** Let $\vec{v}_0$ be a fixed vector and $\vec{v}_0 = \begin{bmatrix}1\\0\end{bmatrix}$. Then for $n \geq 0$ a sequence is defined

$$\vec{v}_{n+1} = \vec{v}_n + \left(\frac{1}{2}\right)^{n+1}\begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}^{n+1}\vec{v}_0 \text{ then}$$

$\lim_{n\to\infty}\vec{v}_n = \begin{bmatrix}\alpha\\\beta\end{bmatrix}$. Find $\frac{\alpha}{\beta}$.

**25.** If $A_1, A_2, ...., A_8$ are vertices of a regular octagon, then $\sum_{j=1}^{7}(\overrightarrow{OA}_j \times \overrightarrow{OA}_{j+1}) = K(\overrightarrow{OA}_1 \times \overrightarrow{OA}_2)$ where the value of $K$ is ........

## ANSWER KEY

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (b) | **4** | (c) | **7** | (b) | **10** | (c) | **13** | (b) | **16** | (d) | **19** | (3) | **22** | (2) | **25** | (7) |
| **2** | (d) | **5** | (a) | **8** | (c) | **11** | (b) | **14** | (b) | **17** | (c) | **20** | (3) | **23** | (0) | | |
| **3** | (b) | **6** | (a) | **9** | (b) | **12** | (d) | **15** | (d) | **18** | (3) | **21** | (5) | **24** | (2) | | |

# THREE DIMENSIONAL GEOMETRY

27

## MCQs with One Correct Answer

**1.** The angle between the lines whose direction cosines are given by the equations

$3l + m + 5n = 0$, $6nm - 2nl + 5lm = 0$ is

(a) $\cos^{-1}\left(\frac{1}{6}\right)$ (b) $\cos^{-1}\left(-\frac{1}{6}\right)$

(c) $\cos^{-1}\left(\frac{2}{3}\right)$ (d) $\cos^{-1}\left(-\frac{5}{6}\right)$

**2.** A line in the 3-dimensional space makes an angle $\theta\left(0 < \theta \le \frac{\pi}{2}\right)$ with both the x and y axes. Then the set of all values of $\theta$ is the interval:

(a) $\left(0, \frac{\pi}{4}\right]$ (b) $\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$

(c) $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$ (d) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right]$

**3.** If $l_1, m_1, n_1$ and $l_2, m_2, n_2$ are DCs of the two lines inclined to each other at an angle $\theta$, then the DCs of the internal bisector of the angle between these lines are

(a) $\frac{l_1 + l_2}{2\sin\theta/2}, \frac{m_1 + m_2}{2\sin\theta/2}, \frac{n_1 + n_2}{2\sin\theta/2}$

(b) $\frac{l_1 + l_2}{2\cos\theta/2}, \frac{m_1 + m_2}{2\cos\theta/2}, \frac{n_1 + n_2}{2\cos\theta/2}$

(c) $\frac{l_1 - l_2}{2\sin\theta/2}, \frac{m_1 - m_2}{2\sin\theta/2}, \frac{n_1 - n_2}{2\sin\theta/2}$

(d) $\frac{l_1 - l_2}{2\cos\theta/2}, \frac{m_1 - m_2}{2\cos\theta/2}, \frac{n_1 - n_2}{2\cos\theta/2}$

**4.** If the straight lines

$x = 1 + s, y = -3 - \lambda s, z = 1 + \lambda s$

and $x = \frac{t}{2}, y = 1 + t, z = 2 - t,$ with parameters s and t respectively, are co-planar, then $\lambda$ equals.

(a) 0 (b) –1 (c) $-\frac{1}{2}$ (d) –2

**5.** $L_1$ and $L_2$ are two lines whose vector equations are $L_1 : \vec{\mathbf{r}} = \lambda[(\cos\theta + \sqrt{3})\hat{\mathbf{i}} +$

$(\sqrt{2}\sin\theta)\hat{\mathbf{j}} + (\cos\theta - \sqrt{3})\hat{\mathbf{k}}]$

and $L_2 : \vec{\mathbf{r}} = \mu(a\hat{\mathbf{i}} + b\hat{\mathbf{j}} + c\hat{\mathbf{k}})$ where, $\lambda$ and $\mu$ are scalars and $\alpha$ is the acute angle between $L_1$ and $L_2$. If the angle $\alpha$ is independent of $\theta$, then the value of $\alpha$ is

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

**6.** The line $\frac{x+6}{5} = \frac{y+10}{3} = \frac{z+14}{8}$ is the hypotenuse of an isosceles right angled triangle whose opposite vertex is (7, 2, 4). Then which of the following is not the side of the triangle?

(a) $\frac{x-7}{2}=\frac{y-2}{-3}=\frac{z-4}{6}$

(b) $\frac{x-7}{3}=\frac{y-2}{6}=\frac{z-4}{2}$

(c) $\frac{x-7}{3}=\frac{y-2}{5}=\frac{z-4}{-1}$

(d) None of these

**7.** The vertex $A$ of the triangle $ABC$ is on the line $\vec{r}=\hat{i}+\hat{j}+\lambda\hat{k}$ and the vertices $B$ and $C$ have respective position vectors $\hat{i}$ and $\hat{j}$. Let $\Delta$ be the area of the triangle and $\Delta\in\left[\frac{3}{2},\frac{\sqrt{33}}{2}\right]$, then the range of values $\lambda$ corresponding to '$A$' is

(a) $[-8,-4]\cup[4,8]$ (b) $[-4,4]$

(c) $[-2,2]$ (d) $[-4,-2]\cup[2,4]$

**8.** A line with positive direction cosines passes through the point $P(2,-1,2)$ and makes equal angles with the coordinate axes. The line meets the plane $2x+y+z=9$ at point $Q$. The length of the line segment $PQ$ equals

(a) 1 (b) $\sqrt{2}$ (c) $\sqrt{3}$ (d) 2

**9.** Through a point $P(h,k,l)$ a plane is drawn at right angles to $OP$ to meet the co-ordinate axes in $A$, $B$ and $C$. If $OP=p$, then the area of $\Delta ABC$ is :

(a) $\frac{p^2hk}{l^2}$ (b) $\frac{p^3l}{3hk}$

(c) $\frac{p^2t^2}{2hk}$ (d) $\frac{p^5}{2hkl}$

**10.** Projection of the line $x+y+z-3=0=2x+3y+4z-6$ on the plane $z=0$ is

(a) $\frac{x}{-2}=\frac{y-6}{1}=\frac{z}{0}$ (b) $\frac{x}{1}=\frac{y-6}{-2}=\frac{z}{0}$

(c) $\frac{x}{1}=\frac{y-6}{2}=\frac{z}{0}$ (d) none of these

**11.** A variable plane at a distance of the one unit from the origin cuts the coordinates axes at $A$, $B$ and $C$. If the centroid $D(x,y,z)$ of triangle $ABC$ satisfies the relation $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=k$, then the value $k$ is

(a) 3 (b) 1 (c) $\frac{1}{3}$ (d) 9

**12.** The plane containing the line $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{3}$ and parallel to the line $\frac{x}{1}=\frac{y}{1}=\frac{z}{4}$ passes through the point:

(a) $(1,-2,5)$ (b) $(1,0,5)$

(c) $(0,3,-5)$ (d) $(-1,-3,0)$

**13.** The position vectors of points $a$ and $b$ are $\hat{i}-\hat{j}+3\hat{k}$ and $3\hat{i}+3\hat{j}+3\hat{k}$ respectively. The equation of a plane is $r\cdot(5\hat{i}+2\hat{j}-7\hat{k})+9=0$. The points $a$ and $b$

(a) lie on the plane

(b) are on the same side of the plane

(c) are on the opposite side of the plane

(d) None of the above

**14.** The angle between the pair of planes represented by equation

$2x^2-2y^2+4z^2+6xz+2yz+3xy=0$ is

(a) $\cos^{-1}\left(\frac{1}{3}\right)$ (b) $\cos^{-1}\left(\frac{4}{21}\right)$

(c) $\cos^{-1}\left(\frac{4}{9}\right)$ (d) $\cos^{-1}\left(\frac{7}{\sqrt{84}}\right)$

**15.** Let $P=(-3,1,1)$ and $Q=(3,4,2)$. $R$ divides $\overline{PQ}$ in the ratio $PR:PQ=1:3$. Then, the equation of the plane perpendicular to $\overrightarrow{PQ}$ at $R$ is

(a) $18x+9y+3z=8$ (b) $18x+9y+3z=4$

(c) $9x+18y+3z=4$ (d) $3x+9y+18z=8$

**16.** Let $\sigma_1$, $\sigma_2$, $\sigma_3$ be planes passing through the origin. Assume that $\sigma_1$ is perpendicular to the vector $(1,1,1)$, $\sigma_2$ is perpendicular to a vector $(a,b,c)$, and $\sigma_3$ is perpendicular to the vector $(a^2,b^2,c^2)$.

What are all the positive values of $a$, $b$ and $c$ so that $\sigma_1 \cap \sigma_2 \cap \sigma_3$ is a single point?

(a) Any positive value of $a$, $b$, and $c$ other than 1

(b) Any positive values of $a$, $b$ and $c$ where either $a \neq b$, $b \neq c$ or $a \neq c$

(c) Any three distinct positive values of $a$, $b$, and $c$

(d) There exist no such positive real numbers $a$, $b$ and $c$

**17.** Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = 2\hat{i} + 2\hat{j} + \hat{k}$, and $\vec{c} = 5\hat{i} + \hat{j} - \hat{k}$ be three vectors. The area of the region formed by the set of points whose position vectors $\vec{r}$ satisfy the equations $\vec{r}.\vec{a} = 5$ and $|\vec{r} - \vec{b}| + |\vec{r} - \vec{c}| = 4$ is closest to the integer

(a) 4 (b) 9

(c) 14 (d) 19

## Numeric Value Answer

**18.** A line with direction ratios (2, 1, 2) intersects the lines $\vec{r} = -\hat{j} + \lambda(\hat{i} + \hat{j} + \hat{k})$ and $\vec{r} = -\hat{i} + \mu(2\hat{i} + \hat{j} + \hat{k})$ at $A$ and $B$, respectively, then length of $AB$ is equal to

**19.** The shortest distance between the z-axis and the line, $x + y + 2z - 3 = 0$, $2x + 3y + 4z - 4 = 0$ is

**20.** The distance of the point (1, 0, –3) from the plane $x - y - z = 9$ measured parallel to the line $\frac{x-2}{2} = \frac{y+2}{3} = \frac{z-6}{-6}$ is

**21.** Let the equation of the plane containing line $x - y - z - 4 = 0 = x + y + 2z - 4$ and parallel to the line of intersection of the planes $2x + 3y + z = 1$ and $x + 3y + 2z = 2$ be $x + Ay + Bz + C = 0$. Then the values of $|A + B + C - 4|$ is ........ .

**22.** Let $f$ be a one-one function with domain {–2, 1, 0} and range {1, 2, 3} such that exactly one of the following statements is true :

$f(-2) = 1, f(1) \neq 1, f(0) \neq 2$ and the remaining two are false. If the area of the triangle formed by (–2, 1, 0) and ($f$(–2), $f$(1), $f$(0)) and the origin is given by $\frac{\sqrt{k}}{2}$; then sum of digits of $k$ is.

**23.** If the equation of the plane through the intersection of the planes $x + 2y + 3z - 4 = 0$ and $2x + y - z + 5 = 0$ and perpendicular to the plane $5x + 3y + 6z + 8 = 0$ is $ax + by + cz + 173 = 0$, then $b - 9(a + c)$ is equal to ........

**24.** If a line is passing through ($a$, $b$, $c$) and intersecting $y = 0$, $z^2 = 4\alpha x$ lies on the surface $(bz - cy)^2 = k\alpha(b - y)(bx - ay)$; then find the value of $k$.

**25.** If the volume enclosed by the equation $|x| \leq 8, |y| \leq 8, |z| \leq 8$ and $|x + y + z| \leq 8$ is $t$, then $\frac{t}{512} =$

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (b) | **4** | (d) | **7** | (d) | **10** | (b) | **13** | (c) | **16** | (c) | **19** | (2) | **22** | (7) | **25** | (4) |
| **2** | (c) | **5** | (a) | **8** | (c) | **11** | (d) | **14** | (c) | **17** | (a) | **20** | (b) | **23** | (6) | | |
| **3** | (b) | **6** | (c) | **9** | (d) | **12** | (b) | **15** | (b) | **18** | (3) | **21** | (7) | **24** | (4) | | |

# 28

# PROBABILITY-2

1. Let $A$ and $B$ be the sets $\{1, 2, \ldots 10\}$ and $\{1, 2, \ldots 20\}$ respectively. A function is selected randomly from $A$ to $B$ the probability that the function is non-decreasing is

(a) $\frac{^{29}C_{10}}{(20)^{10}}$ (b) $\frac{^{29}C_{20}}{(20)^{10}}$

(c) $\frac{^{29}C_{19}}{(20)^{10}}$ (d) None of these

2. Raj and Sanchita are playing game in which they throw two dice alternately till one of them gets 9. Which one of the following could be the probability that Sanchita win the game?

(a) 7/15 or 8/15 (b) 6/11 or 5/11

(c) 8/17 or 9/17 (d) None of these

3. A boy whose hobby is tossing a fair coin is to score one point for every tail and 2 points for every head. The boy goes on tossing the coin, till his score reaches n or exceeds n where $n \geq 2$. If $p_n$ is the probability that his score attains exactly n, then $p_n$ is equal to

(a) $p_{n-1} + p_{n-2}$ (b) $\frac{1}{2}p_{n-1} + p_{n-2}$

(c) $\frac{2}{3} + \frac{(-1)^n}{2^n}$ (d) $\frac{2^{n+1} + (-1)^n}{3.2^n}$

4. Entries of a $2 \times 2$ determinant are chosen from the set $\{-1, 1\}$. The probability that determinant has zero value is

(a) $\frac{1}{4}$ (b) $\frac{1}{3}$

(c) $\frac{1}{2}$ (d) None of these

5. The odds in favour of a book reviewed by three independent critics are, respectively, 5 : 2, 4 : 3 and 3 : 4. The probability that majority of the critics give favourable remark is

(a) $\frac{210}{343}$ (b) $\frac{209}{343}$ (c) $\frac{211}{343}$ (d) $\frac{205}{343}$

6. Matrices of order $3 \times 3$ are formed by using the elements of the set $A = \{-3, -2, -1, 0, 1, 2, 3\}$, then probability that matrix is either symmetric or skew symmetric is

(a) $\frac{1}{7^6} + \frac{1}{7^3}$ (b) $\frac{1}{7^9} + \frac{1}{7^3} - \frac{1}{7^6}$

(c) $\frac{1}{7^3} + \frac{1}{7^9}$ (d) $\frac{1}{7^3} + \frac{1}{7^6} - \frac{1}{7^9}$

7. If $a \in [-20, 0]$, then probability that the graph of the function $y = 16x^2 + 8(a + 5)x - 7a - 5$ is strictly above the $x$-axis is

(a) $\frac{7}{20}$ (b) $\frac{13}{20}$ (c) $\frac{17}{20}$ (d) $\frac{3}{20}$

8. In a Competitive test, a candidate guesses, copies or knows the answer to a multiple choice question with four choices. The probability that he makes a guess is 1/3 and the probability that he copies the answer is 1/6. The probability that his answer is correct given that he copied it is 1/8. Find the probability that he knew the answer to the question, given that he answered it correctly.

(a) 24/29 (b) 26/29

(c) 22/29 (d) None of these

**9.** A wire of length l is cut into three pieces. Then the probability that the three pieces form a triangle is

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$

(c) $\frac{2}{3}$ (d) None of these

**10.** Given that the sum of two non-negative quantities is 200, the probability that their product is not less than $\frac{3}{4}$ times their greatest product value is

(a) $\frac{7}{16}$ (b) $\frac{101}{201}$ (c) $\frac{9}{16}$ (d) $\frac{10}{16}$

**11.** On each evening a boy either watches DOORDARSHAN channel or TEN SPORTS. The probability that he watches TEN SPORTS is $\frac{4}{5}$. If he watches DOORDARSHAN, there is a chance of $\frac{3}{4}$ that he will be asleep, while it is $\frac{1}{4}$ when he watches TEN SPORTS. On one day, the boy is found to be asleep. The probability that the boy watched DOORDARSHAN is

(a) $\frac{5}{7}$ (b) $\frac{2}{7}$ (c) $\frac{3}{7}$ (d) $\frac{4}{7}$

**12.** Let the probability $P_n$ that a family has exactly n children be $\alpha p^n$, when $n \geq 1$ and $P_0 = 1 - \alpha p (1 + p + p^2 + ...)$. Suppose that all sex distributions of n children have the same probability. If $k \geq 1$, then the probability that a family contains exactly k boys is

(a) $\frac{2\alpha}{(2-p)^{k+1}}$ (b) $\frac{p^k}{(2-p)^{k+1}}$

(c) $\frac{2\alpha \, . \, p^k}{(2-p)^{k+1}}$ (d) None of these

**13.** A die is thrown 2n + 1 times, $n \in N$. The probability that faces with even numbers show odd numbers of times, is

(a) $\frac{2n+1}{4n+3}$ (b) less than $\frac{1}{2}$

(c) greater than $\frac{1}{2}$ (d) None of these

**14.** A book writer writes a good book with probability $\frac{1}{2}$. If it is a good book, the probability that it will be published is $\frac{2}{3}$, otherwise it is $\frac{1}{4}$. If he writes 2 books, the probability that at least one book will be published is

(a) $\frac{407}{576}$ (b) $\frac{411}{576}$ (c) $\frac{405}{576}$ (d) $\frac{307}{576}$

**15.** If X follows a binomial distribution with parameters n = 8 and $p = \frac{1}{2}$, then $p(|x-4| \leq 2)$ is equal to

(a) $\frac{121}{128}$ (b) $\frac{119}{128}$ (c) $\frac{117}{128}$ (d) $\frac{115}{128}$

**16.** Ravi and Rashmi are each holding 2 red cards and 2 black cards (all four red and all four black cards are identical). Ravi picks a card at random from Rashmi, and then Rashmi picks a card at random from Ravi. This process is repeated a second time. Let *p* be the probability that both have all 4 cards of the same colour. Then p satisfies

(a) $p \leq 5\%$ (b) $5\% < p \leq 10\%$

(c) $10\% < p \leq 15\%$ (d) $15\% < p$

**17.** The probability of men getting a certain disease is $\frac{1}{2}$ and that of women getting the same disease is $\frac{1}{5}$. The blood test that identifies the disease gives the correct result with probability $\frac{4}{5}$. Suppose a person is chosen at random from a group of 30 males and 20 females, and the blood test of the person is found to be positive. What is the probability that the chosen person is a man?

(a) $\frac{75}{107}$ (b) $\frac{3}{5}$ (c) $\frac{15}{19}$ (d) $\frac{3}{10}$

## Numeric Value Answer

**18.** *A* and *B* play a game of tennis. The situation of the game is as follows : if one scores two consecutive points after a deuce, he wins, if loss of a point is followed by win of a point, it is deuce. The chance of a server to win a point is $\frac{2}{3}$. The game is at deuce and *A* is serving. Probability that *A* will win the match is, (serves are changed after each game).

**19.** The probablilty that the length of a randomly chosen chord of a circle lies between $\frac{2}{3}$ and $\frac{5}{6}$ of its diameter is

**20.** In a hurdle race, a runner has probability p of jumping over a specific hurdle. Given that in 5 trials, the runner succeeded 3 times, the conditional probability that the runner had succeeded in the first trial, is

**21.** An artillery target may be either at point I with probability $\frac{8}{9}$ or at the point (II) with probability $\frac{1}{9}$. There are 21 shells each of which can be fired either at point I or II. Each shell may hit the target independently of the other shell with probability $\frac{1}{2}$. Minimum number of shells that must be fired at point I to hit the target with maximum probability is equal to 2k. Then value of k is.

**22.** A number x is selected from the set of first 9 natural numbers (i.e., x = 1, 2, 3, ......., 9). If the probability that $f(f(x)) = x$ where $f(x) = x^2 - 3x + 3$ is $\frac{m}{9}$, then m is equal to _________.

**23.** A special die is so constructed that the probabilities of throwing 1, 2, 3, 4, 5 and 6 are $(1 - k) / 6$, $(1 + 2k) / 6$, $(1 - k) / 6$, $(1 + k) / 6$, $(1 - 2k) / 6$ and $(1 + k) / 6$, respectively. If two such dice are thrown and the probability of getting a sum equal to 9 lies between $\frac{1}{9}$ and $\frac{2}{9}$, then the integral value of k is.

**24.** A sportsman's chance of shooting an animal at a distance r > a ("a" is constant) is given to be $\frac{a^2}{r^2}$. He fires at r = 2a and if he misses, then again fires at r = 3a. He repeats the same process at r = 4a, 5a and 6a. When he misses at r = 6a, the animal escapes into the jungle. If the odds against the sportsman are p : q, then q – p is _______.,

**25.** A determinant of the second order is made with the elements 0 and 1. If $\frac{m}{n}$ be the probability that the determinant made is non-negative, where m and n are relative primes, then the value of n – m is

**ANSWER KEY**

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (c) | **4** | (c) | **7** | (b) | **10** | (b) | **13** | (d) | **16** | (a) | **19** | (0.25) | **22** | (2) | **25** | (3) |
| **2** | (c) | **5** | (b) | **8** | (a) | **11** | (c) | **14** | (a) | **17** | (a) | **20** | (0.60) | **23** | (0) | | |
| **3** | (d) | **6** | (d) | **9** | (b) | **12** | (c) | **15** | (b) | **18** | (0.50) | **21** | (6) | **24** | (5) | | |

# 29

# PROPERTIES OF TRIANGLE

## MCQs with One Correct Answer

**1.** If the sines of the angles $A$ and $B$ of a triangle $ABC$ satisfy the equation $c^2x^2 - c(a+b)x + ab = 0$, then the triangle is ($a,b,c$, are sides of D)
(a) acute angled
(b) obtuse angled
(c) right angled
(d) no such triangle is possible

**2.** In $\Delta ABC,\ a \geq b \geq c$. If
$$\frac{a^3+b^3+c^3}{\sin^3 A+\sin^3 B+\sin^3 C}=7,$$ then the maximum possible value of $a$ is
(a) 7 (b) 49 (c) $\sqrt[3]{7}$ (d) $\frac{1}{\sqrt{7}}$

**3.** If in a triangle $ABC$, $\sin A$, $\sin B$, $\sin C$ are in $A.P.$, then
(a) the altitudes are in $A.P.$
(b) the reciprocals of altitudes are in $A.P.$
(c) the altitudes are in $G.P.$
(d) the medians are in $A.P.$

**4.** Let $a$, $b$ and $A$ are given and $c_1$ and $c_2$ are two values of $c$, then the value of $c_1^2 + c_2^2 - 2c_1c_2 \cos 2A =$
(a) $a^2\sin^2 A$ (b) $a^2\cos^2 A$
(c) $2(a^2 + b^2)$ (d) $4a^2\cos^2 A$

**5.** If $a$, $b$, $c$ be the sides of a triangle and
$$P=\frac{(a+b+c)^2}{ab+bc+ca},$$ then
(a) $P \in [1, 2]$ (b) $P \in [3, 4)$
(c) $P \in (2, 4]$ (d) None of these

**6.** If $a$, $b$, $c$ are the sides of a triangle such that $b.\ c = \lambda^2$, for some positive $\lambda$, then
(a) $a \geq 2\lambda \sin\frac{A}{2}$ (b) $b \geq 2\lambda \sin\frac{B}{2}$
(c) $c \geq 2\lambda \sin\frac{C}{2}$ (d) all are correct

**7.** In $\Delta ABC$, $a^2(s-a) + b^2(s-b) + c^2(s-c) =$
(a) $4R(\cos A + \cos B + \cos C)$
(b) $4R\Delta(\sin A + \sin B + \sin C)$
(c) $4R\Delta\left(1+4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)$
(d) None of these

**8.** In an obtuse angled triangle, the obtuse angle is $\frac{3\pi}{4}$ and the other two angles are equal to two values of $\theta$ satisfying $a\tan\theta + b\sec\theta = c$, where $|b| \leq \sqrt{a^2+c^2}$, then $a^2 - c^2$ is equal to
(a) $ac$ (b) $2ac$
(c) $\frac{a}{c}$ (d) None of these

**9.** In a triangle $ABC$, $2ac\sin\frac{1}{2}(A-B+C)=$

(a) $a^2+b^2-c^2$ (b) $c^2+a^2-b^2$
(c) $b^2-c^2-a^2$ (d) $c^2-a^2-b^2$

**10.** The sides of a triangle are $\sin\alpha$, $\cos\alpha$ and $\sqrt{1+\sin\alpha\cos\alpha}$ for some $0<\alpha<\frac{\pi}{2}$. Then the greatest angle of the triangle is

(a) 150° (b) 90° (c) 120° (d) 60°

**11.** Given a triangle ΔABC such that $\sin^2 A+\sin^2 C = 1001 \times \sin^2 B$. Then the value of $\frac{2(\tan A+\tan C)\cdot\tan^2 B}{\tan A+\tan B+\tan C}$

(a) $\frac{1}{2000}$ (b) $\frac{1}{1000}$ (c) $\frac{1}{500}$ (d) $\frac{1}{250}$

**12.** In ΔABC, if

$$\frac{\sin A}{c\sin B}+\frac{\sin B}{c}+\frac{\sin C}{b}=\frac{c}{ab}+\frac{b}{ac}+\frac{a}{bc},$$ then

the value of angle A is

(a) 120° (b) 90° (c) 60° (d) 30°

**13.** If the hypotenuse of a right-angled triangle is four times the length of the perpendicular drawn from the opposite vertex to it, then the difference of the two acute angles will be

(a) 60° (b) 15° (c) 75° (d) 30°

**14.** In a ΔABC, $\angle B=\frac{\pi}{3}$ and $\angle C=\frac{\pi}{4}$ let D divide BC internally in the ratio 1 : 3, then $\frac{\sin(\angle BAD)}{\sin(\angle CAD)}$ is equal to:

(a) $\frac{1}{\sqrt{6}}$ (b) $\frac{1}{3}$ (c) $\frac{1}{\sqrt{3}}$ (d) $\frac{\sqrt{2}}{3}$

**15.** In a triangle ABC, a = 5, b = 4 and cos (A – B) $=\frac{31}{32}$, then the third side is equal to:

(where symbols used have usual meanings)

(a) $\sqrt{6}$ (b) $6\sqrt{6}$
(c) 6 (d) $(216)^{1/4}$

**16.** In triangle ABC if A : B : C = 1 : 2 : 4, then $(a^2-b^2)(b^2-c^2)(c^2-a^2)=\lambda a^2b^2c^2$, where $\lambda$ = (where notations have their usual meaning)

(a) 1 (b) 2 (c) 4 (d) 9

**17.** If in a right angle triangle $ABC$, $4\sin A\cos B-1=0$ and $\tan A$ is real, then

(a) angles are in $A.P.$ (b) $B^2=AC$
(c) $B=\frac{2AC}{A+C}$ (d) None of these

**18.** A tower of height b subtends an angle at a point O on the level of the foot of the tower and at a distance $a$ from the foot of the tower. If a pole mounted on the tower also subtends an equal angle at O, the height of the pole is

(a) $b\left(\frac{a^2-b^2}{a^2+b^2}\right)$ (b) $b\left(\frac{a^2+b^2}{a^2-b^2}\right)$

(c) $a\left(\frac{a^2-b^2}{a^2+b^2}\right)$ (d) $a\left(\frac{a^2+b^2}{a^2-b^2}\right)$

**19.** An observer on the top of a tree, finds the angle of depression of a car moving towards the tree to be 30°. After 3 minutes this angle becomes 60°. After how much more time, the car will reach the tree?

(a) 4 min. (b) 4.5 m
(c) 1.5 min (d) 2 min.

**20.** Two flagstaffs stand on a horizontal plane. $A$ and $B$ are two points on the line joining their feet and between them. The angles of elevation of the tops of the flagstaffs as seen from $A$ are 30° and 60° and as seen from $B$ are 60° and 45°. If $AB$ is 30m, then the distance between the flagstaffs in metres is

(a) $30+15\sqrt{3}$ (b) $45+15\sqrt{3}$
(c) $60-15\sqrt{3}$ (d) $60+15\sqrt{3}$

## Numeric Value Answer

**21.** Points D, E are taken on the side BC of an acute angled triangle ABC, such that BD = DE = EC. If $\angle BAD = x$, $\angle DAE = y$ and $\angle EAC = z$ then the value of $\dfrac{\sin(x+y)\sin(y+z)}{\sin x \sin z}$ is _________.

**22.** In triangle ABC, $\sin A \sin B + \sin B \sin C + \sin C \sin A = 9/4$ and $a = 2$, then the value of $\sqrt{3}\Delta$, where $\Delta$ is the area of triangle, is _________.

**23.** Triangle ABC is right angled at A. The points P and Q are on hypotenuse BC such that BP = PQ = QC. If AP = 3 and AQ = 4, if $BC = a\sqrt{b}$ then $|a-b| =$

**24.** AD, BE, CF are internal angular bisectors of $\Delta ABC$ and I is the incentre. If $a(b+c)\sec\dfrac{A}{2}$ ID + $b(a+c)\sec\dfrac{B}{2}$ IE + $c(a+b)\sec\dfrac{C}{2}$ IF = kabc, then the value of k is

**25.** A man from the top of a 100 metres high tower see a car moving towards the tower at an angle of depression of 30°. After some time, the angle of depression becomes 60°. If the distance (in metres) travelled by the car during this time is $\dfrac{a\sqrt{b}}{3}$, then $\left(\dfrac{a}{20}-b\right) =$

**ANSWER KEY**

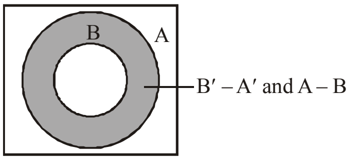
| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | (c) | **4** | (d) | **7** | (c) | **10** | (c) | **13** | (a) | **16** | (a) | **19** | (c) | **22** | (3) |
| **2** | (c) | **5** | (b) | **8** | (b) | **11** | (d) | **14** | (a) | **17** | (a) | **20** | (d) | **23** | (2) |
| **3** | (b) | **6** | (a) | **9** | (b) | **12** | (b) | **15** | (c) | **18** | (b) | **21** | (4) | **24** | (2) |

| | |
|---|---|
| **25** | (7) |
| | |
| | |

# Hints & Solutions

## CHAPTER 1 Sets

**1. (a)**




**2. (b)** Let A = {θ : sin θ = tan θ}
and B = {θ : cos θ = 1}

$\therefore \quad A = \left\{\theta : \sin\theta = \frac{\sin\theta}{\cos\theta}\right\}$

$= \{\theta : \sin\theta(\cos\theta - 1) = 0\} = \{\theta = 0, \pi, 2\pi, 3\pi, .....\}$

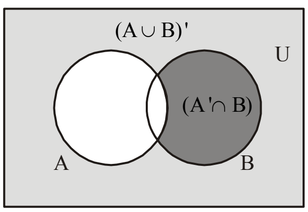
For B : $\cos\theta = 1 \Rightarrow \theta = \pi, 2\pi, 4\pi, ......$

This shows that A is not contained in B. i.e. $A \not\subset B$.
but $B \subset A$.

**3. (c)** $n(P(S)) = 2^3 = 8$ elements.
$n(P(P(S))) = 2^8 = 256$ elements.

**4. (c)** Let $n$ be the number of newspapers which are read. Then $60\,n = (300)(5) \Rightarrow n = 25$

**5. (a)** From Venn-Euler's Diagram.




$\therefore (A \cup B)' \cup (A' \cap B) = A'$

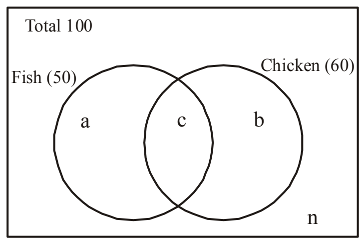
**6. (b)** Given set can be written as
$(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$
(By definition of symmetric difference)
Hence, $(A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$

**7. (b)** $A = \{x : |x| < 1\} = (-1, 1)$
Since, $|x| < 1 \Rightarrow -1 < x < 1$
$B = \{x : |x-1| \geq 1\} = (-\infty, 0] \cup [2, \infty)$
Since, $|x - 1| \geq 1 \Rightarrow x - 1 \leq -1$ or $x - 1 \geq 1$
$\Rightarrow x \leq 0$ or $x \geq 2$
$\therefore A \cup B = (-\infty, 0] \cup [2, \infty) \cup (-1, 1)$
$= (-\infty, 1) \cup [2, \infty) = \mathbf{R} - [1, 2)$
$\therefore \ D = [1, 2) = \{x : 1 \leq x < 2\}$

**8. (a)** $bN = \{bx : x \in \mathbf{N}\}$; $cN = \{cx : x \in \mathbf{N}\}$
$\therefore bN \cap cN = \{x : x$ is multiple of b and c both$\}$
= { x: x is multiple of l.c.m. of b and c }
= { x : x is multiple of b c}
[given b and c are relatively prime ∴ l.c.m. of b and c = bc]
$\therefore bN \cap cN = \{bc\,x : x \in \mathbf{N}\} = d\mathbf{N}$ (Given)
$\therefore$ d = bc.

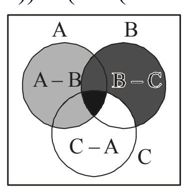
**9. (a)** $(A \cup B \cup C) \cup (A \cap B' \cap C')' \cap C'$
$= (A \cup B \cup C) \cap (A' \cup B \cup C) \cap C'$
$= [(A \cap A') \cup (B \cup C)] \cap C' = (\phi \cup B \cup C) \cap C'$
$= (B \cup C) \cap C' = (B \cap C') \cup (C \cap C')$
$= (B \cap C') \cup \phi = B \cap C'$

**10. (a)** Total number of persons = $a + b + c + n = 100$




Do not prefer fish $b + n = 50$
60 prefer chicken hence $b + c = 60$
Do not like fish and chicken is $n = 10$
On solving these equations we will get a = 30, b = 40, c = 20
The number of persons who prefer both fish and chicken is = c = 20

**11. (c)** $\{(A - B) \cup (B - C) \cup (C - A)\}'$
$= (A - B)' \cap (B - C)' (C - A)'$
$= [(U - (A - B)) \cap (U - (B - C) \cap (U - (C - A))]$

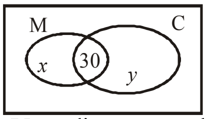



By Venn - diagram = $(A \cap B \cap C)$

**12.** **(d)** $n(A) = 1000, n(B) = 500,$
$n(A \cap B) \geq 1$ & $n(A \cup B) = p$
Also, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$\Rightarrow p = 1000 + 500 - n(A \cap B)$
$\because 1 \leq n(A \cap B) \leq 500$
Hence $p \leq 1499$ and $p \geq 1000 \Rightarrow 1000 \leq p \leq 1499$

**13.** **(a)** Minimum value of $n$
$= 100 - (30 + 20 + 25 + 15) = 100 - 90 = 10$

**14.** **(a)** Let the number of students who take only Math be $x$ and only Chemistry be $y$.



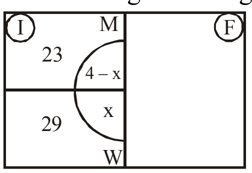

So, from the Venn diagram, we have total number of students who take Math = $x + 30$ and take Chemistry $= y + 30$.
According to question, we have

$$30 = \frac{10}{100}(x + 30)$$

$$\Rightarrow \quad x = 270 \text{ and } 30 = \frac{12}{100}(30 + y)$$

$\Rightarrow \quad y = 220$
$x + y + 30 = 270 + 220 + 30 = 520.$

**15.** **(a)** See the following Venn diagram



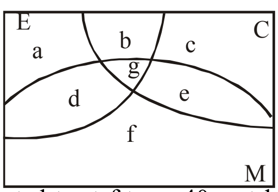

$n(I) = 29 + 23 = 52; n(F) = 100 - 52 = 48$

$n(m \cup D) = n(m) + n(D) - n(m \cap D)$

$24 = 23 + 4 - n(m \cap D) \quad \therefore n(m \cap D) = 3$

$\therefore n(W \cap D) = 4 - 3 = 1$

**16.** **(b)** C stands for set of students taking economics




$a + b + c + d + e + f + g = 40; \; a + b + d + g = 16$
$b + c + e + g = 22; \; d + e + f + g = 26$
$b + g = 5; \; e + g = 14; \; g = 2$
Go by backward substitution
$e = 12, b = 3, d + f = 12, \; c + e = 17 \Rightarrow c = 5;$
$a + d = 11$
$a + d + f = 18 \Rightarrow f = 7 \qquad \therefore d = 12 - 7 = 5$

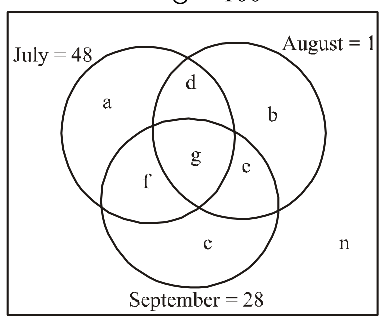
**17.** **(c)** Let those who opted for Nirma = a and those who opted Surf Blue = b and those who opted for none is n.
Hence a = 20k and b = 60k, then
$n = 100k - 20k - 60k = 20k$
The difference between those who opted for 'Surf blue' and those who were uncertain
= 60k – 20k = 40k = 720 hence, k = 18,
Hence total number of persons covered in survey
= 100k = 1800

**18.** **(c)** From the given information (C) = 80, (F) = 40, and $(C \cap F) = (n)$
Hence $(U) = (C) + (F) - (C \cap F) + (n)$
Or $80 + 40 - x + x = 120$

**19.** **(d)** If number of students who like chocolate = a + c
Number of students who like Biscuit = b + c
Number of students who like Both = c
Number of students who like none = n = 10
From the given condition 100 = a + b + c + n
Since 60 of them don't like Chocolate, hence b + n = 60 or b = 50
And 50 of them don't like Biscuit hence a + n = 50, a = 40
Hence 100 = 40 + 50 + c + 10 or c = 0

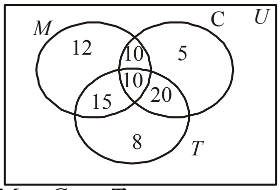
**20.** **(b)** The given information can be represented as follows.

U = 100




Now from the given information we can frame following equations-
c = 18, f + c = 23, f + g = 8,
c + f + g + e = 28, a + d + f + g = 48,
d + g = 10, n = 24
People who read exactly two consecutive months is represented by d and e.
f + c = 23 and c = 18 $\quad \therefore$ f = 5
f + g = 8 and f = 5 $\quad \therefore$ g = 3
d + g = 10 and g = 3 $\quad \therefore$ d = 7
c + f + g + e = 28, c = 18, f = 5 and g = 3
$\therefore$ e = 2 or $\therefore$ d + e = 9

**21.** **(4)** We have
$n(U) = 100$, where $U$ stands for universal set
$n(M \cap C \cap T) = 10; n(M \cap C) = 20;$
$n(C \cap T) = 30; n(M \cap T) = 25;$
$n(\text{only } M) = 12; n(\text{only } C) = 5; n(\text{only } T) = 8$
Filling all the entries we obtain the Venn diagram as shown below :



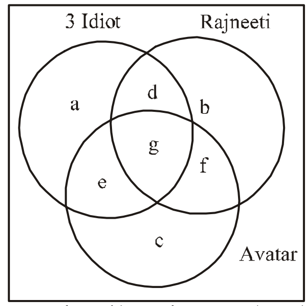

$\therefore \quad n(M \cup C \cup T)$
$= 12 + 10 + 5 + 15 + 10 + 20 + 8 = 80$
$\therefore \quad n(M \cup C \cup T)' = 100 - 80 = 20 = \text{n}.$

So, $\frac{\text{n}}{5} = \frac{20}{5} = 4$

**22.** **(7)** Numbers which are divisible by 5 are 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80 they are 16 in numbers. Now, Numbers which are divisible by 7 are 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77 they are 11 in numbers.
Also, total odd numbers = 40
Let C represents the students who opt. for cricket, F for football and H for hockey.
$\therefore$ we have $n(C) = 40$, $n(F) = 16$, $n(H) = 11$
Now, $C \cap F$ = Odd numbers which are divisible by 5.
$C \cap H$ = Odd numbers which are divisible by 7.
$F \cap H$ = Numbers which are divisible by both 5 and 7.
$n(C \cap F)$, 8, $n(C \cap H) = 6$,
$n(F \cap H) = 2$, $n(C \cap F \cap H) = 1$
We Know
$n(C \cup F \cup H) = n(C) + n(F) + n(H)$
$\qquad - n(C \cap F) - n(C \cap H)$
$\qquad - n(F \cap H) + n(C \cap H \cap F)$
$n(C \cup F \cup H) = 67 - 16 + 1 = 52$
$\therefore n(C' \cap F' \cap H')$
$= \text{Total students} - n(C \cup F \cup H)$

$n(C' \cap F' \cap H') = 80 - 52 = 28$, So $\frac{\text{n}}{4} = \frac{28}{4} = 7$

**23.** **(7)** The given condition is as follows-




We know that $\{(a + d + e + g) + (b + d + f + g) + (c + e + f + g)\} - (d + e + f) - 2g$
$= a + b + c + d + e + f + g$
or $\quad 61x + 46x + 29x - 25x - 2g = 97x$
or $\quad 2g = 14x$ or $g = 7x$. So 7% of people watched all the three movies)

**24.** **(4)** $n(A \cup B)$ is minimum when $A \subseteq B$. In this case, $(A \cup B) = B$ and hence minimum value of $n(A \cup B) = n(B) = 7$.
$n(A \cup B)$ is maximum when A and B are disjoint.
$\therefore$ Maximum value of $n(A \cup B) = 4 + 7 = 11$.
So $11 - 7 = 4$

**25.** **(3)** $2^m - 2^n = 112 \Rightarrow 2^n(2^{m-n} - 1) = 16.7$
$\therefore \quad 2^n(2^{m-n} - 1) = 2^4(2^3 - 1)$
Comparing, we get $n = 4$ and $m - n = 3$
$\Rightarrow \quad n = 4$ and $m = 7$
So $m - n = 7 - 4 = 3$

# CHAPTER 2 Relations & Functions-1

**1.** **(b)** Put $x = y = 1$, $(f(1))^2 = 3f(1) - 2$

$\Rightarrow f(1) = 1$ or $2$

Let $f(1) = 1$, then put $y = 1$

$f(x).f(1) = f(x) + f(1) + f(x) - 2$

$\Rightarrow f(x) = 1$ constant function $\therefore f(1) \neq 1$,

hence $f(1) = 2$

**2.** **(a)** Given

$$f(x) = \sqrt{x^{14} - x^{11} + x^6 - x^3 + x^2 + 1}$$

for f(x) to be defined,

$x^{14} - x^{11} + x^6 - x^3 + x^2 + 1 \geq 0$

**Case 1 : $x \geq 1$**

$x^{14} - x^{11} + x^6 - x^3 + x^2 + 1$

$$= (x^{14} - x^{11}) + (x^6 - x^3) + (x^2 + 1) > 0$$

**Case 2 :** $0 \leq x \leq 1$

$x^{14} - x^{11} + x^6 - x^3 + x^2 + 1$

$$= x^{14} - \{(x^{11} - x^6) + (x^3 - x^2)\} + 1 > 0$$

$\{\because x^{11} - x^6 \leq 0, x^3 - x^2 \leq 0\}$

**Case 3 : $x < 0$**

$x^{14} - x^{11} + x^6 - x^3 + x^2 + 1 > 0$

$(\because x^{11} < 0, x^3 < 0, x^{14}, x^6, x^2 > 0)$

Thus, for all real x,

$x^{14} - x^{11} + x^6 - x^3 + x^2 + 1 \geq 0$

Hence, the domain of $f(x) = \text{R} = (-\infty, \infty)$

**3.** **(a)** Given $f(x) = \cos(\log x)$

$\therefore\ f(xy) = \cos(\log xy)$

$f(xy) = \cos[\log x + \log y]$ ....(1)

And $f\left(\frac{x}{y}\right) = \cos\left(\log\frac{x}{y}\right)$

$f\left(\frac{x}{y}\right) = \cos(\log x - \log y)$ ....(2)

Adding (1) and (2), we get

$f(xy) + f\left(\frac{x}{y}\right)$

$= \cos(\log x + \log y) + \cos(\log x - \log y)$

$= 2\cos(\log x).\cos(\log y)$

$\Rightarrow f(xy) + f\left(\frac{x}{y}\right) = 2f(x).f(y)$

Then the value of $f(x)f(y)$

$-\frac{1}{2}\left\{f\left(\frac{x}{y}\right) + f(xy)\right\}$

$= f(x)\ f(y) - \frac{1}{2}.2\{f(x).f(y)\} = 0$

**4.** **(c)** We must $x^4 - 21x^2 \geq 0$ and

$10 - \sqrt{x^4 - 21x^2} \geq 0$

$\Rightarrow\ x^2\left(x^2 - 21\right) \geq 0$ ...(1)

and $100 \geq x^4 - 21x^2$ ...(2)

Eq. (1) gives $x = 0$ or $x \leq -\sqrt{21}$ or $x \geq \sqrt{21}$

Eq. (2) $\Rightarrow x^4 - 21x^2 - 100 \leq 0$

$\Rightarrow\ \left(x^2 - 25\right)\left(x^2 + 4\right) \leq 0$

$\Rightarrow\ x^2 - 25 \leq 0$ (as $x^2 + 4 > 0$ always)

$\Rightarrow\ -5 \leq x \leq 5$

Domain is given by

$\left[-5, -\sqrt{21}\right] \cup \left[\sqrt{21}, 5\right] \cup \{0\}$

**5.** **(b)** $3f(x) + 2f\left(\frac{x+59}{x-1}\right) = 10x + 30$

For $x = 7$, $3f(7) + 2f(11) = 70 + 30 = 100$

For $x = 11$, $3f(11) + 2f(7) = 140$

$\frac{f(7)}{-20} = \frac{f(11)}{-220} = \frac{-1}{9-4}$ or $f(7) = 4$

**6.** **(a)** $af(x+1) + bf\left(\frac{1}{x+1}\right) = (x+1) - 1$ ...(1)

Replacing $x + 1$ by $\frac{1}{x+1}$, we get

$af\left(\frac{1}{x+1}\right) + bf(x+1) = \frac{1}{x+1} - 1$ ...(2)

Eq. (1) × a – Eq. (2) × b

$\Rightarrow\ (a^2 - b^2)f(x+1) = a(x+1) - a - \frac{b}{x+1} + b$

Putting $x = 1$, $(a^2 - b^2)f(2) = 2a - a - \frac{b}{2} + b$

$= a + \frac{b}{2} = \frac{2a+b}{2}$

**7.** **(d)** $f(x) = \sqrt{\cos(\sin x)} + \sqrt{\log_x\{x\}}$

Domain $\cos(\sin x) \geq 0$ $\{x\} > 0,\ x > 0,\ x \neq 1,$

$\log_x\{x\} \geq 0$

(i) $\cos(\sin x) \geq 0$ for all $x$, $x \in R[-1, 1]$

(ii) $\{x\} > 0$, $x \notin$ Int. (iii) $x > 0$, $x \in (0, \infty)$

(iv) $x \neq 1$

(v) $\log_x\{x\} \geq 0 \Rightarrow 1 > f(x) \geq 0$,

so $1 > x \geq 0$ $\log_x f(x)$ is positive $x \in [0, 1)$

$\Rightarrow x \in (0, 1)$

**8.** **(c)** Given that,

$f(x) = \frac{1}{2} - \tan\left(\frac{\pi x}{2}\right), -1 < x < 1$

Here, domain of $f(x) = (-1, 1)$ and

$g(x) = \sqrt{3 + 4x - 4x^2} = \sqrt{-(2x-3)(2x+1)}$

$\Rightarrow (2x-3)(2x+1) \leq 0 \Rightarrow -\frac{1}{2} \leq x \leq \frac{3}{2}$

$\therefore$ Domain of $g(x) = \left[-\frac{1}{2}, \frac{3}{2}\right]$

Hence, domain of $(f + g)$ = intersection of their

domains $= \left[-\frac{1}{2}, 1\right)$.

**9.** **(b)** We have f(x)

$= \log_4[\log_5\{\log_3(18x - x^2 - 77)\}]$

Since, $\log_a x$ is defined for all $x > 0$, f(x) is defined if

$\log_5\{\log_3(18x - x^2 - 77)\} > 0$ and $18x - x^2 - 77 > 0$

or $\log_3(18x - x^2 - 77) > 5^0$ and $x^2 - 18x + 77 < 0$

or $\log_3(18x - x^2 - 77) > 1$ and $(x - 11)(x - 7) < 0$

or $18x - x^2 - 77 > 3^1$ and $7 < x < 11$

or $18x - x^2 - 80 > 0$ and $7 < x < 11$

or $x^2 - 18x + 80 < 0$ and $7 < x < 11$

or $(x-10)(x-8)<0$ and $7<x<11$
or $8<x<10$ and $7<x<11$ or $8<x<10$
or $x \in (8, 10)$
Hence, the domain of f(x) is (8, 10).

**10.** **(a)** Clearly, $g(x) = \frac{1}{2}(a^x + a^{-x})$ and

$h(x) = \frac{1}{2}(a^x - a^{-x})$. Now $g(x+y)+g(x-y)$

$$= \frac{1}{2}(a^{x+y} + a^{-(x+y)}) + \frac{1}{2}(a^{x-y} + a^{-x+y})$$

$$= \frac{1}{2}(a^x a^y + a^x a^{-y} + a^{-x} a^y + a^{-x} a^{-y})$$

$$= \frac{1}{2}(a^x (a^y + a^{-y}) + a^{-x}(a^y + a^{-y}))$$

$$= 2\left(\frac{1}{2}(a^x + a^{-x})\right)\left(\frac{1}{2}(a^y + a^{-y})\right) = 2g(x)g(y)$$

**11.** **(b)** Given $f(\lambda+x) = f(\lambda-x)$ ...(1)
$f(2\lambda+x) = -f(2\lambda-x)$ ...(2)
for $\lambda>0$
Replacing $x$ by $\lambda-x$ in $\lambda-x$ in (1), we get
$f(2\lambda-x) = f(x)$ ...(3)
$\therefore$ From (2) and (3), $f(x) = -f(2\lambda+x)$
$\Rightarrow f(x) = -[-f(2\lambda+2\lambda+x)]$
$\Rightarrow f(x) = f(x+4\lambda)$ ...(4)
$\Rightarrow f(x)$ is periodic with period $4\lambda$.
Further from (3), replacing $x$ by $-x$, we get
$f(2\lambda+x) = f(-x)$ ...(5)
From (2), (3) and (5), we have
$f(-x) = f(2\lambda+x) = -f(2\lambda-x) = -f(x)$
i.e. $f(-x) = -f(x)$
$\Rightarrow f(x)$ is odd function
Thus, $f$ is odd and periodic function.

**12.** **(d)** $f(x) = \alpha x^3 - \beta x - (\tan x)\,\text{sgn}\, x$

Since $f(-x) = f(x)$

$\Rightarrow -\alpha x^3 + \beta x - \tan x\, \text{sgn}\, x$

$= \alpha x^3 - \beta x - (\tan x)(\text{sgn}\, x)$

$\Rightarrow 2(\alpha x^2 - \beta)x = 0\ \forall\ x \in R \Rightarrow \alpha = 0$ and $\beta = 0$

$\therefore [a]^2 - 5[a] + 4 = 0$

and $6\{a\}^2 - 5\{a\} + 1 = 0$
$\Rightarrow ([a]-1)([a]-4) = 0$ and
$(3\{x\}-1)(2\{x\}-1) = 0$

$\Rightarrow [a] = 1, 4$ and $\{a\} = \frac{1}{3}, \frac{1}{2}$

$\therefore a = 1+\frac{1}{3}, 1+\frac{1}{2}, 4+\frac{1}{3}, 4+\frac{1}{2}$

Sum of all possible values of $a = \frac{35}{3}$

**13.** **(c)** $f(x) = \cos nx \,.\, \sin\left(\frac{5x}{n}\right)$;

Period of $\cos nx = \frac{2\pi}{|n|}$

Period of $\sin \frac{5x}{n} = \frac{2\pi}{\left|\frac{5}{n}\right|} = \frac{2|n|\pi}{5}$

$\therefore$ Period of $f(x)$ = L.C.M. $\left(\frac{2\pi}{|n|}, \frac{2|n|\pi}{5}\right) = 2\pi$ (given)

$\Rightarrow$ L.C.M.$\left(\frac{1}{|n|}, \frac{|n|}{5}\right) = 1 = \frac{\text{L.C.M.}(1, |n|)}{\text{H.C.F.}(|n|, 5)} = 1$

$\Rightarrow \frac{|n|}{\text{H.C.F.}(|n|, 5)} = 1 \Rightarrow$ H.C.F. $(|n|, 5) = |n|$
If g.c.d. $(|n|, 5) = 1 \Rightarrow |n| = 1 \Rightarrow n = 1$
If g.c.d. $(|n|, 5) \neq 1 \Rightarrow |n| = 5m; m \in \mathbb{N}$
$\Rightarrow$ g.c.d. $(5m, 5) = 1$
$\Rightarrow |n| = 5 \Rightarrow n = \pm 5 \therefore n \in \{\pm 1, \pm 5\}$

**14.** **(d)** $f: \mathbb{R} \to \mathbb{R}, g; \mathbb{R} \to \mathbb{R}$
We know that min. $\{f_1(x), f_2(x)\}$

$$= \frac{(f_1(x) + f_2(x)) - |f_1(x) - f_2(x)|}{2}$$

$\therefore$ min $\{f(x) - g(x), 0\}$

$$= \frac{(f(x) - g(x) + 0) - |f(x) - g(x) - 0|}{2}$$

$$= \frac{(f(x) - g(x)) - |f(x) - g(x)|}{2}$$

**15.** **(b)** $y = f(e^x) + f(|\ln|\,|x|)$ Domain $f(x) = (0, 1)$
$\Rightarrow 0 < e^x < 1 \Rightarrow x < 0$ ...(1)
and $0 < \ln|x| < 1 \Rightarrow 1 < |x| < e$
$\Rightarrow x \in \{-e, -1\} \cup (1, e)$ ... (2)
Taking intersection $x \in (-e, -1)$

**16.** **(b)** $f\left(x+\frac{1}{2}\right) = f(x); f(2) = 5;$

$f\left(\frac{9}{4}\right) = 2, f(-3) - f\left(\frac{1}{4}\right) = ?$

$\because$ $f(x)$ is periodic with period $\frac{1}{2}$

$\Rightarrow$ $f(x) = f\left(x + \frac{n}{2}\right) \ \forall\ x \in \mathbb{N}$

$\Rightarrow$ $f(-3) = f\left(-3 + \frac{10}{2}\right) = f(2) = 5$

and $f\left(\frac{1}{4}\right) = f\left(\frac{1}{4} + 4\left(\frac{1}{2}\right)\right) = f\left(\frac{9}{4}\right) = 2$

$\therefore$ $f(-3) - f\left(\frac{1}{4}\right) = 5 - 2 = 3$

**17. (c)** $f(x) = \begin{cases} 2x+3 & x \le 1 \\ a^2x+1 & x > 1 \end{cases}$

For $x \le 1$; $f(x) \le 5$

So for range of f(x) to be R.

$\Rightarrow a^2 + 1 \le 5$ and $a \ne 0$ $\Rightarrow a \in [-2, 2]$

Hence, $a = \{-2, -1, 1, 2\}$

**18. (b)** $g(\sqrt{2}f(x) + 3)$

$= \log_{\sqrt{5}}(\sqrt{2}\ (\sin x - \cos x) + 3)$

We know that

$-\sqrt{2} \le (\sin x - \cos x) \le \sqrt{2}\ \forall\ x \in \mathbb{R}$

$\left[\because -\sqrt{a^2+b^2} \le a \sin x + b \cos x \le \sqrt{a^2+b^2}\right]$

$\Rightarrow$ $-2 \le \sqrt{2}(\sin x - \cos x) \le 2$

$\Rightarrow$ $1 \le \sqrt{2}(\sin x - \cos x) + 3 \le 5$

$\Rightarrow$ $0 \le \log_{\sqrt{5}}(\sqrt{2}\ (\sin x - \cos x) + 3) \le 2$

($\because$ $\log_a x$ is increasing for $a > 1$)

Hence, range of $g(\sqrt{2}f(x) + 3)$ is $[0, 2]$.

**19. (d)** $-g(2, f(x)) = -\log_2\left(1 + \frac{1}{4\sqrt{x}}\right)$

$\Rightarrow$ $-g(2, f(x)) - 1 = -\log_2\left(1 + \frac{1}{4\sqrt{x}}\right) - 1$

$\therefore$ $g\left(\frac{1}{2}, -g(2, f(x)) - 1\right)$

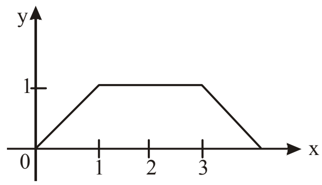
$= \log_{1/2}\left(-\log_2\left(1 + \frac{1}{4\sqrt{x}}\right) - 1\right)$

$\Rightarrow$ $\log_2\left(1 + \frac{1}{4\sqrt{x}}\right) + 1 < 0$

$\Rightarrow$ $0 < 1 + \frac{1}{4\sqrt{x}} < 2^{-1}$

$\Rightarrow$ $-1 < \frac{1}{4\sqrt{x}} < -\frac{1}{2}$ $\Rightarrow x \in \phi$ (null set)

**20. (d)**




$0 \le f(x) \le 1$ $\Rightarrow 0 \le 7\,f(x) \le 7$

$\Rightarrow -1 \le \sin(7\,f(x)) \le 1$

**21. (5.5)** Here $\phi(-x) = \frac{1}{1+e^x}$

So, $\phi(x) + \phi(-x) = \frac{1}{1+e^{-x}} + \frac{1}{1+e^x}$

$= \frac{e^x}{e^x+1} + \frac{1}{1+e^x} = \frac{e^x+1}{e^x+1} = 1$

$\therefore S = \{\phi(5) + \phi(-5)\} + \ldots + \{\phi(1) + \phi(-1)\} + \phi(0)$

$= 1 + 1 + 1 + 1 + 1 + \phi(0) = 5 + \frac{1}{1+e^0} = 5 + \frac{1}{2} = \frac{11}{2}$

**22. (2)** $f(x+p) = 1 + \{1 - 3f(x) + 3f^2(x) - f^3(x)\}^{1/3}$

$\Rightarrow$ $f(x+p) = 1 + (1 - f(x)) = 2 - f(x)$

$\Rightarrow$ $f(x+p) = 2 - [2 - f(x-p)]$

$\Rightarrow f(x+p) = f(x-p)$

$\Rightarrow$ $f(x) = f(x+2p)$

$\Rightarrow$ Period of $f(x) = 2p = \lambda p$ (given)

$\Rightarrow$ $\lambda = 2$

**23. (9)** Given $f(x+2) = f(x) + f(2)$

Put $x = -1$. Then $f(1) = f(-1) + f(2)$

or $f(1) = -f(1) + f(2)$ [as f(x) is an odd function]

or $f(2) = 2f(1) = 6$

Now, put $x = 1$

We have $f(3) = f(1) + f(2) = 3 + 6 = 9$

**24. (6)** Given $f(x, y) = f(2x + 2y, 2y - 2x)\ \forall\ x, y \in \mathbb{R}$,

$f(x) = f(2^x, 0)$ and $f(x)$ is periodic with period $k$.

$\Rightarrow f(x) = f(2^x, 0) = f(2.2^x + 2(0), 2(0) - 2.2^x)$

$= f(2^{x+1}, 2^{x+1})$

$= f(2.2^{x+1} - 2.2^{x+1}, -2.2^{x+1} - 2.2^{x+1})$

$= f(0, -2^{x+3})$

$= f(2.(-2^{x+3}), -2.2^{x+3}) = f(-2^{x+4}, -2^{x+4})$

$= f(-2^{x+6}, 0)$

$= f(-2^{x+7}, 2^{x+7}) = f(0, 2^{x+9})$

$= f(2^{x+10}, 20^{x+10}) = f(2^{x+12}, 0) = f(x+12)$

$\Rightarrow f(x)$ is periodic with period $12 \Rightarrow k = 12$.

**25.** **(6)** $f(x)=\left[\frac{x}{15}\right]\left[-\frac{15}{x}\right]$ $\quad x\in(0,90)$

$0\le x<15$ $\quad f(x)=0$
$15\le x<30$ $\quad f(x)=-1$
$30\le x<45$ $\quad f(x)=-2$
$45\le x<60$ $\quad f(x)=-3$
$60\le x<75$ $\quad f(x)=-4$
$75\le x<90$ $\quad f(x)=-5$

Total integers in range $f(x)=\{0,-1,-2,-3,-4,-5\}$

CHAPTER 3

# Trigonometric Functions

**1.** **(d)** $\frac{1}{\cos^2\alpha}+\frac{1}{1+\sin^2\alpha}+\frac{2}{1+\sin^4\alpha}+\frac{4}{1+\sin^8\alpha}$

$=\frac{2}{1-\sin^4\alpha}+\frac{2}{1+\sin^4\alpha}+\frac{4}{1+\sin^8\alpha}$

$=\frac{4}{1-\sin^8\alpha}+\frac{4}{1+\sin^8\alpha}=\frac{8}{1-\sin^{16}\alpha}$

$=\frac{8}{1-\frac{1}{5}}=\frac{40}{4}=10\quad\left(\because\ \sin^{16}\alpha=\frac{1}{5}\right)$

**2.** **(b)** $\alpha<\beta<\gamma<\delta$ and

$\sin\alpha=\sin\beta=\sin\gamma=\sin\delta=k$

$\Rightarrow\beta=\pi-\alpha,\ \gamma=2\pi+\alpha,\ \delta=3\pi-\alpha$.

So that the given expression is equal to

$4\sin\frac{\alpha}{2}+3\sin\left(\frac{\pi-\alpha}{2}\right)+2\sin\frac{2\pi+\alpha}{2}+\sin\frac{3\pi-\alpha}{2}$

$=4\sin\frac{\alpha}{2}+3\cos\frac{\alpha}{2}-2\sin\frac{\alpha}{2}-\cos\frac{\alpha}{2}$

$=2\left(\sin\frac{\alpha}{2}+\cos\frac{\alpha}{2}\right)=2\sqrt{1+2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}}=2\sqrt{1+k}$

**3.** **(c)** $1+\cos\alpha=1+\frac{2\cos\beta-1}{2-\cos\beta}=\frac{2-\cos\beta+2\cos\beta-1}{2-\cos\beta}$

$=\frac{1+\cos\beta}{2-\cos\beta}\Rightarrow\cos^2\frac{\alpha}{2}=\frac{\cos^2\frac{\beta}{2}}{1+2\sin^2\frac{\beta}{2}}$ $\quad\dots$(i)

$\Rightarrow 1-\cos^2\frac{\alpha}{2}=1-\frac{\cos^2\frac{\beta}{2}}{1+2\sin^2\frac{\beta}{2}}$

$=\frac{1+2\sin^2\frac{\beta}{2}-\cos^2\frac{\beta}{2}}{1+2\sin^2\frac{\beta}{2}}=\frac{1+2\sin^2\frac{\beta}{2}-[1-\sin^2\frac{\beta}{2}]}{1+2\sin^2\frac{\beta}{2}}$

$\Rightarrow\sin^2\frac{\alpha}{2}=\frac{3\sin^2\frac{\beta}{2}}{1+2\sin^2\frac{\beta}{2}}$ $\quad\dots$(ii)

Divide equation (ii) by (i)

$\tan^2\frac{\alpha}{2}=3\tan^2\frac{\beta}{2}\Rightarrow\frac{\tan\frac{\alpha}{2}}{\tan\frac{\beta}{2}}=\sqrt{3}$

**4.** **(b)** Gives equations can be written as

$2\cos\alpha+9\cos\delta=-6\cos\beta-7\cos\gamma$ ..(i)

$2\sin\alpha-9\sin\delta=6\sin\beta-7\sin\gamma$ ...(ii)

Square and add equations (i) and (ii),

$\Rightarrow\ 4+36+36\,[\cos\alpha\cos\delta-\sin\delta\sin\alpha]$
$=36+49+84\,[\cos\beta\cos\gamma-\sin\beta\sin\gamma]$

$\Rightarrow\ 36\,[\cos(\alpha+\delta)]=84\,[\cos(\beta+\gamma)]$

$\frac{\cos(\alpha+\delta)}{\cos(\beta+\gamma)}=\frac{84}{36}=\frac{7}{3}=\frac{m}{n};\quad m+n=10$

**5.** **(a)** $\sin\theta+\sin3\theta+\sin2\theta=\sin\alpha$

$\Rightarrow\ 2\sin2\theta\cos\theta+\sin2\theta=\sin\alpha$

$\Rightarrow\ \sin2\theta(2\cos\theta+1)=\sin\alpha$ ...(i)

Now $\cos\theta+\cos3\theta+\cos2\theta=\cos\alpha$

$2\cos2\theta\cos\theta+\cos2\theta=\cos\alpha$

$\cos2\theta\,(2\cos\theta+1)=\cos\alpha$ ...(ii)

From (i) and (ii),

$\tan2\theta=\tan\alpha\ \Rightarrow\ 2\theta=\alpha\ \Rightarrow\ \theta=\alpha/2$

**6.** **(c)** $\frac{(\sin7\alpha+\sin5\alpha)+5(\sin5\alpha+\sin3\alpha)+12(\sin3\alpha+\sin\alpha)}{\sin6\alpha+5\sin4\alpha+12\sin2\alpha}$

$=\frac{2\sin6\alpha\cos\alpha+5(2\sin4\alpha\cos\alpha)+12(2\sin2\alpha\cos\alpha)}{\sin6\alpha+5\sin4\alpha+12\sin2\alpha}$

$=2\cos\alpha=\frac{\sqrt{5}+1}{2}$

**7.** **(b)**

$$\frac{1}{2}\left[\frac{2\sin\theta\cos\theta}{\cos\theta\cos3\theta}+\frac{2\sin3\theta\cos3\theta}{\cos9\theta\cos3\theta}+\frac{2\sin9\theta\cos9\theta}{\cos9\theta\cos27\theta}+\frac{2\sin27\theta\cos27\theta}{\cos27\theta\cos81\theta}\right]$$

$$=\frac{1}{2}\left[\frac{\sin(3\theta-\theta)}{\cos\theta\cos3\theta}+\frac{\sin(9\theta-3\theta)}{\cos3\theta\cos9\theta}+\frac{\sin(27\theta-9\theta)}{\cos9\theta\cos27\theta}+\frac{\sin(81\theta-27\theta)}{\cos27\theta\cos81\theta}\right]$$

$$=\frac{1}{2}[\tan 81\theta-\tan\theta]=\frac{1}{2}\left[\frac{\sin 80\theta}{\cos\theta\cos 81\theta}\right]$$

**8.** **(b)** $\text{cosec}\,\theta=\frac{p+q}{p-q}\Rightarrow\frac{1}{\sin\theta}=\frac{p+q}{p-q}$

Apply componendo and dividendo

$$\frac{1+\sin\theta}{1-\sin\theta}=\frac{p+q+p-q}{p+q-p+q}$$

$$\Rightarrow\left\{\frac{\cos\frac{\theta}{2}+\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}-\sin\frac{\theta}{2}}\right\}^2=\frac{p}{q}\Rightarrow\left\{\frac{1+\tan\frac{\theta}{2}}{1-\tan\frac{\theta}{2}}\right\}^2=\frac{p}{q}$$

$$\Rightarrow\tan^2\left(\frac{\pi}{4}+\frac{\theta}{2}\right)=\frac{p}{q}\Rightarrow\cot^2\left(\frac{\pi}{4}+\frac{\theta}{2}\right)=\frac{q}{p}$$

**9.** **(b)** Given, $\sin 2\theta+\sin 2\phi=1/2$ ...(i)

and $\cos 2\theta+\cos 2\phi=3/2$ ...(ii)

Square and adding,

$\therefore$ $(\sin^2 2\theta+\cos^2 2\theta)+(\sin^2 2\phi+\cos^2 2\phi)$ $+2[\sin 2\theta\sin 2\phi+\cos 2\theta\cos 2\phi]=1/4+9/4$

$\Rightarrow$ $\cos 2\theta\cos 2\phi+\sin 2\theta\sin 2\phi=1/4$

$\Rightarrow$ $\cos(2\theta-2\phi)=1/4\Rightarrow\cos^2(\theta-\phi)=5/8$

**10.** **(a)** $u=1+\cos\theta+\cos 2\theta+\cos(\theta+2\theta)$

$=(1+\cos 3\theta)+(\cos\theta+\cos 2\theta)$

$$=2\cos^2\frac{3\theta}{2}+2\cos\frac{3\theta}{2}+\cos\frac{3\theta}{2}$$

$$=2\cos\frac{3\theta}{2}\left[\cos\frac{3\theta}{2}+\cos\frac{\theta}{2}\right]$$

Similarly $v=2\sin\frac{3\theta}{2}\left[\cos\frac{3\theta}{2}+\cos\frac{\theta}{2}\right]$

$$\therefore\quad u^2+v^2=\left(\cos\frac{3\theta}{2}+\cos\frac{\theta}{2}\right)^2$$

$$=4\left(2\cos\theta\cdot\cos\frac{\theta}{2}\right)^2=16\cos^2\theta\cdot\cos^2\frac{\theta}{2}$$

$=4(1+\cos\theta)(1+\cos 2\theta)$

**11.** **(a)** Let

$$P=\cos\left\{\frac{2\pi}{2^{64}-1}\right\}\cos\left\{\frac{2^2\pi}{2^{64}-1}\right\}...\cos\left\{\frac{2^{64}\pi}{2^{64}-1}\right\}$$

$$\Rightarrow P=\frac{1}{2\sin\left(\frac{2\pi}{2^{64}-1}\right)}\sin\left(\frac{2^2\pi}{2^{64}-1}\right)\cos\left(\frac{2^2\pi}{2^{64}-1}\right)...\cos\left(\frac{2^{64}\pi}{2^{64}-1}\right)$$

$$\Rightarrow P=\frac{1}{2\sin\left(\frac{2\pi}{2^{64}-1}\right)}\sin\left(\frac{2^3\pi}{2^{64}-1}\right)...\cos\left(\frac{2^{64}\pi}{2^{64}-1}\right)$$

$$\Rightarrow P=\frac{1}{2^{64}\sin\left(\frac{2\pi}{2^{64}-1}\right)}\sin\left(\frac{2^{65}\pi}{2^{64}-1}\right)=\frac{1}{2^{64}}$$

**12.** **(c)** $\sin x\sin y+3\cos y+4\sin y\cos x=\sqrt{26}$

$\therefore$ $3\cos y+(\sin x+4\cos x)\sin y=\sqrt{26}$

$\therefore$ $3\cos y+(\sin x+4\cos x)\sin y$

$\leq\sqrt{9+(\sin x+4\cos x)^2}\leq\sqrt{9+1+16}=\sqrt{26}$

$\therefore$ $\sin x\sin y+3\cos y+4\sin y\cos x=\sqrt{26}$

$$\Rightarrow\quad \sin x\sin y=\frac{\cos y}{3}=\frac{\sin y\cos x}{4}$$

$\Rightarrow$ $3\tan y=\text{cosec}\,x$ and $\tan x=1/4$

$\Rightarrow$ $9\tan^2 y=\text{cosec}^2 x=(1+\cot^2 x)=17$

$$\Rightarrow\quad \tan^2 x+\cot^2 y=\frac{1}{16}+\frac{9}{17}$$

**13.** **(b)** $\cos\cdot\frac{x}{256}\cdot\cos\cdot\frac{x}{128}\cdot\cos\cdot\frac{x}{64}\cdot.....\cos\cdot\frac{x}{4}\cdot\cos\cdot\frac{x}{2}$

$$=\frac{\sin x}{256\sin\left(\frac{x}{256}\right)}$$

**14.** **(a)** $-5\leq 3\sin x-4\cos x\leq 5$

$10\leq 3\sin x-4\cos x+15\leq 20$

$\log_{20}10\leq\log_{20}(3\sin x-4\cos x+15)\leq\log_{20}20$

**15.** **(c)** Given expression

$$= \frac{\sin 50° + \sin 140° + \sin 170°}{2 \sin 25° \sin 70° \sin 85°}$$

$= \frac{4}{2}$ (Using sin 2A + sin 2B + sin 2C = 4 sin A sin B sin C, where A + B + C = 180°)
= 2

**16.** **(c)** $3^{2(1+\tan^2 x)} + 1 - 10 \cdot 3^{\tan^2 x} = 0$

$\Rightarrow \quad 3^2(3^{\tan^2 x})^2 + 1 - 10.3^{\tan^2 x} = 0$.

Put $3^{\tan^2 x} = t$; $\quad 9t^2 - 10t + 1 = 0$

$\Rightarrow \quad t = \frac{1}{9}$ and $t = 1$

$\Rightarrow 3^{\tan^2 x} = 3^{-2}$ and $3^{\tan^2 x} = 1$

$\Rightarrow$ $\tan^2 x = -2$ gives no solution and $\tan^2 x = 1$ gives four solutions.

**17.** **(c)** $(4\cos^2 2x + 4\cos^2 x + 1)$

$+ (\tan^2 x - 2\sqrt{3}\tan x + 3) = 0$

$\Rightarrow \quad (2\cos 2x + 1)^2 + (\tan x - \sqrt{3})^2 = 0$

$\Rightarrow \quad \cos 2x = -\frac{1}{2}$ and $\tan x = \sqrt{3}$

$\Rightarrow \quad 2\cos^2 x - 1 = -\frac{1}{2}$ and $\tan x = \sqrt{3}$

$\Rightarrow \quad \cos x = \pm -\frac{1}{2}$ and $\tan x = \sqrt{3}$

$\Rightarrow \quad x = \frac{\pi}{3}, \frac{4\pi}{3}$

$\therefore$ Number of solutions of equation = 2.

**18.** **(b)** $\left|2\sin x - \sqrt{3}\right|^{2\cos^2 x - 3\cos x + 1} = 1$

**Case I:** $2\cos^2 x - 3\cos x + 1 = 0$

$\cos x = \frac{1}{2}, 1$; $\; x = 0, \frac{\pi}{3}$

But at $x = \frac{\pi}{3}$ L.H.S. = 0°

$\therefore \quad x = \frac{\pi}{3}$ (rejected) $\qquad \therefore$ x = 0 is a solution

**Case II:** $2\sin x - \sqrt{3} = 1, -1$

$\therefore \quad \sin x = \frac{\sqrt{3}+1}{2}, \frac{\sqrt{3}-1}{2}$ $\quad \sin x = \frac{\sqrt{3}-1}{2}$

$\therefore$ x has 2 values in $[0, \pi]$.

$\therefore$ Total number of solutions = 2 + 1 = 3.

**19.** **(a)** $\sin^2(\sin x) - 3\sin(\sin x) + 2 = 0$

$\{\sin(\sin x) - 2\}\{\sin(\sin x) - 1\} = 0$

Equation has no solution.

**20.** **(c)** $x^2 + 12 + 3\sin(a + bx) + 6x = 0$

$\Rightarrow \quad (x + 3)^2 + 3 + 3\sin(a + bx) = 0$

$\Rightarrow \quad (x + 3)^2 + 3 = -3\sin(a + bx)$

L.H.S. $\geq 3$ but R.H.S. $\leq 3$

L.H.S. = R.H.S. = 3

$\therefore \quad x = -3$ and $\sin(a + bx) = -1$

$\Rightarrow \quad \sin(a - 3b) = -1$

or $\quad a - 3b = (4n - 1)\frac{\pi}{2}$ $n \in Z$.

**21.** **(50)** $5^{\frac{1}{2}} + 5^{\frac{1}{2} + \log_5(\sin x)} = 15^{\frac{1}{2} + \log_5 \cos x}$

$\Rightarrow \quad 5^{\frac{1}{2}} + 5^{\frac{1}{2}} \cdot 5^{\log_5(\sin x)} = 15^{1/2} \cdot 15 \log 15 \cos x$

$\Rightarrow \quad 5^{1/2} + 5^{1/2} \cdot \sin x = 15^{1/2} \cdot \cos x$

$\Rightarrow \quad 1 + \sin x = \sqrt{3}\cos x$

$\Rightarrow \quad \frac{\sqrt{3}}{2}\cos x - \frac{\sin x}{2} = \frac{1}{2}$

$\Rightarrow \quad \cos\left(x + \frac{\pi}{6}\right) = \cos\frac{\pi}{3}$

$\Rightarrow \quad x + \frac{\pi}{6} = 2n\pi \pm \frac{\pi}{3}, n \in Z$

$\Rightarrow \quad x = 2n\pi - \frac{\pi}{2}, 2n\pi + \frac{\pi}{6}, n \in Z$.

But we must have $\sin x, \cos x > 0$

$\therefore \quad x = 2n\pi + \pi/6, n \in Z$.

**22.** **(6)** $\frac{\sin x}{\cos 3x} + \frac{\sin 3x}{\cos 9x} + \frac{\sin 9x}{\cos 27x} = 0$

or $\quad \frac{2\sin x\cos x}{2\cos 3x\cos x} + \frac{2\sin 3x\cos 3x}{2\cos 9x\cos 3x} + \frac{2\sin 9x\cos 9x}{2\cos 27x\cos 9x} = 0$

or $\quad \frac{\sin(3x - x)}{2\cos 3x\cos x} + \frac{\sin(9x - 3x)}{2\cos 9x\cos 3x} + \frac{\sin(27x - 9x)}{2\cos 27x\cos 9x} = 0$

or $\quad (\tan 3x - \tan x) + (\tan 9x - \tan 3x) + (\tan 27x - \tan 9x) = 0$

or $\quad \tan 27x - \tan x = 0$ or $\tan x = \tan 27x$

$\Rightarrow \quad 27x = n\pi + x, n \in I$ or $x = \frac{n\pi}{26}, n \in I$

or $\quad \frac{\pi}{26}, \frac{2\pi}{26}, \frac{3\pi}{26}, \frac{4\pi}{26}, \frac{5\pi}{26}, \frac{6\pi}{26}$

Hence, there are six solutions.

**23.** **(4)** Since $-2 \le \sin x - \sqrt{3}\cos x \le 2$, we have

$$-2 \le \frac{4m-6}{4-m} \le 2 \text{ or } -1 \le \frac{2m-3}{4-m} \le 1$$

If $\frac{2m-3}{4-m} \le 1$, we have

$$\frac{(2m-3)-(4-m)}{4-m} \le 0 \text{ or } \frac{3m-7}{m-4} \ge 0. \quad \text{...(i)}$$

Also, $-1 \le \frac{2m-3}{4-m} \Rightarrow \frac{m+1}{m-4} \le 0$ ...(ii)

From Eqs. (i) and (ii), we get $m \in \left[-1, \frac{7}{3}\right]$.

Therefore, the possible integers are – 1, 0, 1, 2.

**24.** **(2)** We have,

$$\left(\cos\left(x+\frac{\pi}{6}\right)-\cos\left(\frac{\pi}{6}\right)\right)^2 = \sin^2 x$$

$$\Rightarrow 4\sin^2\left(\frac{x}{2}\right)\cdot\sin^2\left(\frac{x}{2}+\frac{\pi}{6}\right) = 4\sin^2\left(\frac{x}{2}\right)\cdot\cos^2\left(\frac{x}{2}\right)$$

$$\therefore \sin\frac{x}{2} = 0 \text{ or } \sin^2\left(\frac{x}{2}+\frac{\pi}{6}\right) = \cos^2\frac{x}{2}$$

$$\Rightarrow x = 2m\pi, m \in Z \text{ or } \frac{\pi}{2}-\left(\frac{x}{2}+\frac{\pi}{6}\right) = n\pi + \frac{x}{2}, n \in I$$

$$\therefore x = 0 \text{ or } x = \frac{\pi}{3}.$$

**25.** **(2)** Given $\frac{\pi}{2} < \left|3x - \frac{\pi}{2}\right| \le \pi$

$$\Rightarrow \frac{\pi}{2} < \left(3x-\frac{\pi}{2}\right) \le \pi \text{ or } -\pi \le \left(3x-\frac{\pi}{2}\right) < \frac{-\pi}{2}$$

$$\Rightarrow \pi < 3x = \frac{3\pi}{2} \text{ or } \frac{-\pi}{6} \le x < 0$$

$$\therefore x \in \left[\frac{-\pi}{6}, 0\right) \cup \left(\frac{\pi}{3}, \frac{\pi}{2}\right]$$

Now, $1 + \cos x + \cos 2x + \sin x + \sin 2x + \sin 3x = 0$

$\Rightarrow 2\cos^2 x + \cos x + \sin 2x + 2\sin 2x \cos x = 0$

$\Rightarrow \cos x(2\cos x + 1) + \sin 2x(2\cos x + 1) = 0$

$\Rightarrow (\cos x + \sin 2x)(2\cos x + 1) = 0$

$\Rightarrow \cos x(1 + 2\sin x)(2\cos x + 1) = 0$

$\Rightarrow \cos x = 0$ or $\sin x = \frac{-1}{2}$

(as for given interval, $\cos x > 0$)

$\Rightarrow x = \frac{\pi}{2}$ or $x = \frac{-\pi}{6}$.

Hence, there are 2 solutions.

CHAPTER 4

# Principle of Mathematical Induction

**1.** **(d)** We note that $P(1) = 2$ and hence, $P(n) = n(n+1) + 2$ is not true for $n = 1$. So the principle of mathematical induction is not applicable and nothing can be said about the validity of the statement $P(n) = n(n+1) + 2$.

**2.** **(c)** The product of r consecutive integers is divisible by r ! . Thus $n(n+1)(n+2)(n+3)$ is divisible by $4! = 24$.

**3.** **(a)** Let $f(n) = \sqrt{2+\sqrt{2+.....+\sqrt{2}}}$ (number of roots is n)

Then $f(1) = \sqrt{2} = 2\cos\frac{\pi}{4}$ or $2\sin\frac{\pi}{4}$

$\therefore$ f (1) may be true for (a) as well as for (b)

Again $f(2) = \sqrt{2+\sqrt{2}} = 2 \times \frac{\sqrt{2+\sqrt{2}}}{2} = 2\cos\frac{\pi}{8}$

$\therefore$ f (2) is true for (a).

We check it for any integer.

Let $\underbrace{\sqrt{2+\sqrt{2+.....\sqrt{2}}}}_{k-\text{roots}} = 2\cos\left(\frac{\pi}{2^{k+1}}\right)$ for some $k \ge 1$ ...(i)

Now,

$$\underbrace{\sqrt{2+\sqrt{2+\sqrt{2+......\sqrt{2}}}}}_{(k+1)\text{ terms}} = \sqrt{2+\underbrace{\sqrt{2+\sqrt{2+......\sqrt{2}}}}_{k\text{ terms}}}$$

$$= \sqrt{2+2\cos\left(\frac{\pi}{2^{k+1}}\right)} \quad \text{[From (i)]}$$

$$= \sqrt{2\left[1+\cos\frac{\pi}{2^{k+1}}\right]} = \sqrt{2.2\cos^2\frac{\pi}{2^{k+2}}} = 2\cos\frac{\pi}{2^{k+2}}$$

$\therefore$ The result is true for $n = k + 1$. Hence, by the principle of mathematical induction

$$\underbrace{\sqrt{2+\sqrt{2+....+\sqrt{2}}}}_{n-\text{roots}} = 2\cos\left(\frac{\pi}{2^{n+1}}\right) \text{ for all } n \in \mathbf{N}.$$

**4.** **(a)** It can be proved with the help of mathematical induction that $\frac{n}{2} < a(n) \le n$.

$\therefore \quad \frac{200}{2} < a(200)$

$\Rightarrow \quad a(200) > 100$ and $a(100) \le 100.$

**5.** **(a)** Let $U_n = \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + .... + \frac{1}{2n-1}$

$V_n = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + ..... + \frac{1}{2n-1}$

Let T(n) be the statement $U_n = V_n$

Then T(1) is true. For $U_1 = 1$ and $V_1 = 1$, so that $U_1 = V_1$

Let T(k) be true for some positive integer K. Now,

$$U_{k+1} - U_k = \left[\frac{1}{k+1} + \frac{1}{k+2} + .... + \frac{1}{2k+1}\right] - \left[\frac{1}{k} + \frac{1}{k+1} + .... + \frac{1}{2k-1}\right]$$

$$= \frac{1}{2k} + \frac{1}{2k+1} - \frac{1}{k} = -\frac{1}{2k} + \frac{1}{2k+1} \quad ...(i)$$

Also,

$$V_{k+1} - V_k = \left[1 - \frac{1}{2} + \frac{1}{3} - ......+ \frac{1}{2k+1}\right] - \left[1 - \frac{1}{2} + \frac{1}{3} - ..... + \frac{1}{2k-1}\right]$$

$$= -\frac{1}{2k} + \frac{1}{2k+1} \quad ...(ii)$$

From (i) and (ii), we find that

$U_{k+1} - U_k = V_{k+1} - V_k$

Since, $U_k = V_k$, therefore, it follows that $U_{k+1} = V_{k+1}$

$\therefore$ T(n) is true for all $n \in \mathbf{N}$.

**6.** **(a)** Let us write the statement

$$P(n): \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + ..... + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

we note that P(1) : $\frac{1}{1.2} = \frac{1}{1+1} \Rightarrow \frac{1}{2} = \frac{1}{2}$ is true

thus P(n) is true for n = 1

Suppose that P(k) is true for some natural number 'k'

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + ..... + \frac{1}{k(k+1)} = \frac{k}{k+1} ......(1)$$

Now,

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + ..... + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)}$$

$$= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} \quad \text{[From (1)]}$$

$$= \frac{k(k+2)+1}{(k+1)(k+2)}$$

$$= \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2} = \frac{k+1}{(k+1)+1}$$

Thus P(k + 1) is true whenever P(k) is true. Hence. by the principle of mathematical induction P(n) is true for all natural numbers.

**7.** **(a)** Let us write the statement

$$P(n): \left(1+\frac{3}{1}\right)\left(1+\frac{5}{4}\right)\left(1+\frac{7}{9}\right)......\left(1+\frac{2n+1}{n^2}\right) = (n+1)^2$$

We note that

$P: \left(1+\frac{3}{1}\right) = 4 = (1+1)^2$, which is true

Thus P (n) is true for n = 1

Suppose that P(k) is true for some natural number 'k' i.e.,

$$\left(1+\frac{3}{1}\right)\left(1+\frac{5}{4}\right)\left(1+\frac{7}{9}\right).....\left(1+\frac{2k+1}{k^2}\right) = (k+1)^2 \quad ....(1)$$

Now,

$$\left(1+\frac{3}{1}\right)\left(1+\frac{5}{4}\right)\left(1+\frac{7}{9}\right).....\left(1+\frac{2k+1}{k^2}\right)\left\{1+\frac{2k+3}{(k+1)^2}\right\}$$

$$= (k+1)^2\left\{1+\frac{2k+3}{(k+1)^2}\right\} \quad \text{[Using (1)]}$$

$$= (k+1)^2\left\{\frac{(k+1)^2+2k+3}{(k+1)^2}\right\} = k^2+2k+1+2k+3$$

$$= k^2+4k+4 = (k+2)^2 = \{(k+1)+1\}^2$$

Thus , P(k + 1) is true whenever P(k) is true. Hence by the principle of mathematical induction P(n) is true for all natural numbers.

**8.** **(d)** $\because 2^1 > 1^2. 2^2 = 2^2, 2^3 < 3^2, 2^4 = 4^2$

But $2^5 > 5^2$, $2^6 > 6^2$ and so on.

**9.** **(d)** Let $P(n): \frac{4^n}{n+1} < \frac{(2n)!}{(n!)^2}$

For $n = 2$,

$$P(2): \frac{4^2}{2+1} < \frac{4!}{(2)^2} \Rightarrow \frac{16}{3} < \frac{24}{4}$$

which is true.
Let for $n = m \geq 2$, $P(m)$ is true.

i.e. $\frac{4^m}{m+1} < \frac{(2m)!}{(m!)^2}$

$$\text{Now, } \frac{4^{m+1}}{m+2} = \frac{4^m}{m+1} \cdot \frac{4(m+1)}{m+2} < \frac{(2m)!}{(m!)^2} \cdot \frac{4(m+1)}{(m+2)}$$

$$= \frac{(2m)!(2m+1)(2m+2)4(m+1)(m+1)^2}{(2m+1)(2m+2)(m!)^2(m+1)^2(m+2)}$$

$$= \frac{[2(m+1)]!}{[(m+1)!]^2} \cdot \frac{2(m+1)^2}{(2m+1)(m+2)} < \frac{[2(m+1)]!}{[(m+1)!]^2}$$

Hence, for $n \geq 2$, P(n) is true.

**10.** **(b)** Given that, $P(n): 3^n < n!$
Now, $P(7): 3^7 < 7!$ is true
Let $P(k): 3^k < k!$
$\Rightarrow P(k+1): 3^{k+1} = 3.3^k < 3.k! < (k+1)!$
$(\because \; k+1 > 3)$

**11.** **(c)** Let $P(n) = 2 . 4^{2n+1} + 3^{3n+1}$
Then $P(1) = 2 . 4^3 + 3^4 = 209$, which is divisible by 11 but not divisible by 2, 7 or 27.
Further, let $P(k) = 2 . 4^{2k+1} + 3^{3k+1}$ is divisible by 11, that is,

$2.4^{2k+1} + 3^{3k+1} = 11q$ for some integer $q$.

Now $P(k+1) = 2 . 4^{2k+3} + 3^{3k+4}$

$= 2 . 4^{2k+1}.4^2 + 3^{3k+1}.3^3 = 16 . 2.4^{2k+1} + 27.3^{3k+1}$

$= 16 . 2 . 4^{2k+1} + (16+11) . 3^{3k+1}$

$= 16[2. 4^{2k+1} + 3^{3k+1}] + 11 . 3^{3k+1}$

$= 16 . 11q + 11 . 3^{3k+1}$

$= 11(16q + 3^{3k+1}) = 11m$

where $m = 16q + 3^{3k+1}$ is another integer.

$\therefore P(k+1)$ is divisible by 11.

$\therefore P(n) = 2 . 4^{2n+1} + 3^{3n+1}$ is divisible by 11 for all $n \in \mathrm{N}$.

**12.** **(b)** $n(n^2 - 1) = (n-1)(n)(n+1)$
It is product of three consecutive natural numbers, so according to Langrange's theorem it is divisible by 3! i.e., 6

**13.** **(a)** $n^p - n$ is divisible by $p$ for any natural number greater than 1.
**Trick :** Let $n = 4$ and $p = 2$
Then $(4)^2 - 4 = 16 - 4 = 12$, it is divisible by 2. So, it is true for any natural number greater than 1

**14.** **(b)** For $n = 1$, we have
$49^n + 16n + \lambda = 49 + 16 + \lambda = 65 + \lambda$
$= 64 + (\lambda + 1)$, which is divisible by 64 if $\lambda = -1$
For $n = 2$, we have
$49^n + 16n + \lambda = 49^2 + 16 \times 2 + \lambda = 2433 + \lambda$
$= 64 \times 38 + (\lambda + 1)$, which is divisible by 64 if $\lambda = -1$
Hence, $\lambda = -1$

**15.** **(b)** Let P(n) :

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + ..... + \frac{1}{n(n+1)(n+2)}$$

$$= \frac{n(n+3)}{4(n+1)(n+2)}$$

For n = 1, $\quad$ L.H.S.$= \frac{1}{1 \cdot 2 \cdot 3} = \frac{1}{6}$ and

R.H.S.$= \frac{1(1+3)}{4(1+1)(1+2)} = \frac{1}{6}$

$\therefore$ P(1) is true. Let P(k) is true, then P(k) :

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + ..... + \frac{1}{k(k+1)(k+2)}$$

$$= \frac{k(k+3)}{4(k+1)(k+2)} \quad \text{... (i)}$$

For n = k + 1,

$$P(k+1): \frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + .....$$

$$+ \frac{1}{k(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)}$$

$$= \frac{(k+1)(k+4)}{4(k+2)(k+3)}$$

$$\text{L.H.S.} = \frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + .....$$

$$+ \frac{1}{k(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)}$$

$$= \frac{k(k+3)}{4(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)}$$

[from (i)]

$$= \frac{(k+1)^2(k+4)}{4(k+1)(k+2)(k+3)} = \frac{(k+1)(k+4)}{4(k+2)(k+3)}$$

= R.H.S. Hence, P(k + 1) is true.

Hence, by principle of mathematical induction for all $n \in N$, P(n) is true.

**16.** **(b)** $3.5^{2n+1} + 2^{3n+1}$

Put n = 1, we get

$(3 \times 5^3) + 2^4 = 391$, which is divisible by 17.

Put n = 2, we get

$(3 \times 5^5) + 2^7 = 9503$, which is divisible by 17 only.

**17.** **(b)** Let the statement P(n) be defined as

$P(n) = 1.3 + 2.3^2 + 3.3^3 + ..... + n.3^n$

$$= \frac{(2n-1)3^{n+1}+3}{4}$$

**Step I :** For n = 1,

$$P(1) : 1.3 = \frac{(2.1-1)3^{1+1}+3}{4} = \frac{3^2+3}{4}$$

$$= \frac{9+3}{4} = \frac{12}{4} = 3 = 1.3, \text{ which is true.}$$

**Step II :** Let it is true for n = k,

i.e. $1.3 + 2.3^2 + 3.3^3 + ..... + k.3^k$

$$= \frac{(2k-1)3^{k+1}+3}{4} \quad ...\text{(i)}$$

**Step III :** For n = k + 1,

$(1.3 + 2.3^2 + 3.3^3 + ..... + k.3^k) + (k + 1)3^{k+1}$

$$= \frac{(2k-1)3^{k+1}+3}{4} + (k+1)3^{k+1}$$

[Using equation (i)]

$$= \frac{(2k-1)3^{k+1}+3+4(k+1)3^{k+1}}{4}$$

$$= \frac{3^{k+1}(2k-1+4k+4)+3}{4}$$

[taking $3^{k+1}$ common in first and last term of numerator part]

$$= \frac{3^{k+1}(6k+3)+3}{4} = \frac{3^{k+1}\cdot 3(2k+1)+3}{4}$$

[taking 3 common in first term of numerator part]

$$= \frac{3^{(k+1)+1}[2k+2-1]+3}{4}$$

$$= \frac{[2(k+1)-1]3^{(k+1)+1}+3}{4}$$

Therefore, P(k + 1) is true when P(k) is true.

Hence, from the principle of mathematical induction, the statement is true for all natural numbers n.

**18.** **(d)** Let the statement P(n) be defined as

$$P(n) : 1 + \frac{1}{1+2} + \frac{1}{1+2+3} + ..... + \frac{1}{1+2+3+.....+n} = \frac{2n}{n+1}$$

$$\text{i.e. } P(n) : 1 + \frac{1}{1+2} + \frac{1}{1+2+3} + ..... + \frac{2}{n(n+1)} = \frac{2n}{n+1}$$

$$\left[\because \text{ sum of natural numbers} = \frac{n(n+1)}{2}\right]$$

**Step I :** For n = 1,

$$P(1) : 1 = \frac{2 \times 1}{1+1} = \frac{2}{2} = 1, \text{ which is true.}$$

**Step II :** Let it is true for n = k,

$$\text{i.e. } 1 + \frac{1}{1+2} + \frac{1}{1+2+3} + ..... + \frac{2}{k(k+1)} = \frac{2k}{k+1} \quad ...\text{(i)}$$

**Step III :** For n = k + 1,

$$\left(1 + \frac{1}{1+2} + \frac{1}{1+2+3} + ..... + \frac{2}{k(k+1)}\right) + \frac{2}{(k+1)(k+2)}$$

$$= \frac{2k}{k+1} + \frac{2}{(k+1)(k+2)} \quad \text{[using equation (i)]}$$

$$= \frac{2k(k+2)+2}{(k+1)(k+2)} = \frac{2[k^2+2k+1]}{(k+1)(k+2)}$$

[taking 2 common in numerator part]

$$= \frac{2(k+1)^2}{(k+1)(k+2)} = \frac{2(k+1)}{k+2} = \frac{2(k+1)}{(k+1)+1}$$

Therefore, P(k + 1) is true, when P(k) is true.

Hence, from the principle of mathematical induction, the statement is true for all natural numbers n.

**19.** **(c)** $2^4 \equiv 1 \pmod 5 \Rightarrow (2^4)^{75} \equiv (1)^{75} \pmod 5$

i.e. $2^{300} \equiv 1 \pmod 5 \Rightarrow 2^{300} \times 2 \equiv (1.2) \pmod 5$

$\Rightarrow$ $2^{301} \equiv 2 \pmod 5$

$\therefore$ Least positive remainder is 2.

**20.** **(d)** Let P(n) be the statement given by

P(n) : $3^{2n}$ when divided by 8, the remainder is 1.

or P(n) : $3^{2n} = 8\lambda + 1$ for some $\lambda \in N$

For n = 1,

P(1) : $3^2 = (8 \times 1) + 1 = 8\lambda + 1$, where $\lambda = 1$

$\therefore$ P(1) is true.
Let P(k) be true.
Then, $3^{2k} = 8\lambda + 1$ for some $\lambda \in N$ ... (i)
We shall now show that P(k + 1) is true, for which we have to show that $3^{2(k+1)}$ when divided by 8, the remainder is 1.
Now, $3^{2(k+1)} = 3^{2k} \cdot 3^2 = (8\lambda + 1) \times 9$ [Using (i)]
$= 72\lambda + 9 = 72\lambda + 8 + 1 = 8(9\lambda + 1) + 1$
$= 8\mu + 1$, where $\mu = 9\lambda + 1 \in N$
$\Rightarrow$ P(k + 1) is true.
Thus, P(k + 1) is true, whenever P(k) is true. Hence, by the principle of mathematical induction P(n) is true for all $n \in N$.

**21.** **(133)** Putting $n = 1$ in $11^{n+2} + 12^{2n+1}$
We get, $11^{1+2} + 12^{2\times1+1} = 11^3 + 12^3 = 3059$, which is divisible by 133.

**22.** **(27)** Let P(n) be the statement given by
P(n) : $41^n - 14^n$ is a multiple of 27
For n = 1,
i.e. $P(1) = 41^1 - 14^1 = 27 = 1 \times 27$, which is a multiple of 27.
$\therefore$ P(1) is true.
Let P(k) be true, i.e. $41^k - 14^k = 27\lambda$ ... (i)
For n = k + 1,
$41^{k+1} - 14^{k+1} = 41^k\, 41 - 14^k\, 14$
$= (27\lambda + 14^k)\, 41 - 14^k\, 14$ [using (i)]
$= (27\lambda \times 41) + (14^k \times 41) - (14^k \times 14)$
$= (27\lambda \times 41) + 14^k (41 - 14) = (27\lambda \times 41) + (14^k \times 27)$
$= 27(41\lambda + 14^k)$,
which is a multiple of 27.
Therefore, P(k + 1) is true when P(k) is true. Hence,from the principle of mathematical induction, the statement is true for all natural numbers n.

**23.** **(8)** Let m = 2k + 1, n = 2k – 1 ($k \in N$)
$\therefore$ $m^2 - n^2$
$= 4k^2 + 1 + 4k - 4k^2 + 4k - 1 = 8k$
Hence, all the numbers of the form $m^2 - n^2$ are always divisible by 8.

**24.** **(16)** $3^{2n+2} - 8n - 9, \forall n \in N$
Putting $n = 2$
$\Rightarrow 3^{2\times2+2} - 8\times2 - 9 = 729 - 16 - 9 = 704$
It is divisible by 16.

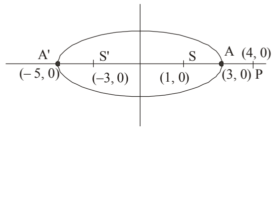
**25.** **(8)** $5^{99} = (5)(5^2)^{49} = 5(25)^{49} = 5(26-1)^{49}$
$= 5\times(26)\times(\text{Positive terms}) - 5$, So when it is divided
by 13 it gives the remainder –5 or (13 – 5) i.e., 8.

## CHAPTER 5 Complex Numbers and Quadratic Equations

**1.** **(a)** $|x_1z_1 - y_1z_2|^2 + |y_1z_1 + x_1z_2|^2$
$= |x_1z_1|^2 + |y_1z_2|^2 - 2\mathrm{Re}(x_1y_1z_1z_2) + |y_1z_1|^2 + |x_1z_2|^2 + 2\mathrm{Re}(x_1y_1z_1z_2)$
$= x_1^2|z_1|^2 + y_1^2|z_2|^2 + y_1^2|z_1|^2 + x_1^2|z_2|^2$
$= x_1^2|z_1|^2 + y_1^2|z_2|^2 + y_1^2|z_1|^2 + x_1^2|z_2|^2$
$= 2(x_1^2 + y_1^2)(4^2) = 32(x_1^2 + y_1^2)$

**2.** **(d)** $\because |z| = |\omega|$ and $\arg z = \pi - \arg \omega$
Let $\omega = re^{i\theta}$, then $z = re^{i(\pi-\theta)}$
$\Rightarrow z = re^{i\pi} . e^{-i\theta} = (re^{-i\theta})(\cos\pi + i\sin\pi)$
$= \bar{\omega}(-1) = -\bar{\omega}$

**3.** **(c)** $\because |z-1| + |z+3| \le 8$
$\therefore$ $z$ lies inside or on the ellipse whose foci are (1, 0) and (– 3, 0) and vertices are (– 5, 0) and (3, 0). Clearly the minimum and maximum values of $|z-4|$ are 1 and 9 respectively, representing the distances $PA$ and $PA'$. Thus, $1 \le |z-4| \le 9$.




**4.** **(b)** Let $z = \sin \ln (i^i)^{i^2} + \cos \ln (i^i)^{i^2}$
$= \sin\left\{\ln\left(e^{\frac{\pi}{2}}\right)^{-1}\right\} + \cos\left\{\ln\left(e^{-\frac{\pi}{2}}\right)^{-1}\right\}$
$= \sin\frac{\pi}{2} + \cos\frac{\pi}{2} = 1$

**5.** **(d)** Let $z = (1+i)^{n_1} + (1-i)^{n_1} + (1+i)^{n_2} + (1-i)^{n_2}$
Observe that $(1+i)^{n_1} + (1-i)^{n_1}$
$= \left(\sqrt{2}\right)\left\{2\cos\frac{n_1\pi}{4}\right\}$,
always real $\forall n_1 \in \mathbb{N}$
Similarly $(1+i)^{n_2} + (1-i)^{n_2}$ is always real $\forall n_2 \in \mathbb{N}$
$\Rightarrow$ $n_1, n_2$ may be any positive integers

**6.** **(d)** $w_k = \dfrac{(\cos\alpha_k + i\sin\alpha_k)^2}{z_k}$

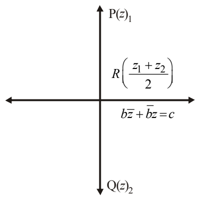
$= \dfrac{z_k^{\,2}}{r_k^2 z_k} = \dfrac{z_k^2}{z_k \bar{z}_k z_k} = \dfrac{1}{\bar{z}_k}$ We have

$w_1 + w_2 + w_3 = \dfrac{1}{\bar{z}_1} + \dfrac{1}{\bar{z}_2} + \dfrac{1}{\bar{z}_3}$

$= \overline{\left(\dfrac{1}{z_1} + \dfrac{1}{z_2} + \dfrac{1}{z_3}\right)} = 0$

$\therefore$ The origin $O$ is the centroid of $\Delta A_1A_2A_3$.

**7.** **(c)** If P($z_1$) is the reflection of Q($z_2$) through the line $b\bar{z} + \bar{b}z = c$ in the argand plane. Then,




$R\left(\dfrac{z_1+z_2}{2}\right)$ lies on the line.

$b\left(\dfrac{\bar{z}_1+\bar{z}_2}{2}\right) + \bar{b}\left(\dfrac{z_1+z_2}{2}\right) = c$

$\Rightarrow \quad b\bar{z}_1 + \bar{b}z_1 + b\bar{z}_2 + \bar{b}z_2 = 2c$ ... (i)

Since, PQ is perpendicular to the line $b\bar{z} + \bar{b}z = c$. Therefore

Slope of PQ + Slope of the line = 0

$\Rightarrow \left(\dfrac{z_2 - z_1}{\bar{z}_2 - \bar{z}_1}\right) + \left(\dfrac{-b}{\bar{b}}\right) = 0$

$\Rightarrow \bar{b}(z_2 - z_1) - b(\bar{z}_2 - \bar{z}_1) = 0$

$\Rightarrow \quad \bar{b}z_2 - \bar{b}z_1 - b\bar{z}_2 + b\bar{z}_1 = 0$

Adding Eqs. (i) and (ii), we get

$2(b\bar{z}_1 + \bar{b}z_2) = 2c \Rightarrow b\bar{z}_1 + \bar{b}z_2 = c$

**8.** **(b)** As $\Delta OAC$ is a right triangle with right angle at $A$, $|z_1|^2 + |z_3 - z_1|^2 = |z_3|^2$

$\Rightarrow \quad 2|z_1|^2 - \bar{z}_3 z_1 - \bar{z}_1 z_3 = 0$

$\Rightarrow \quad 2\bar{z}_1 - \bar{z}_3 - \dfrac{\bar{z}_1}{z_1} z_3 = 0$ ... (1)

Similarly, $2\bar{z}_2 - \bar{z}_3 - \dfrac{\bar{z}_2}{z_2} z_3 = 0$ ... (2)

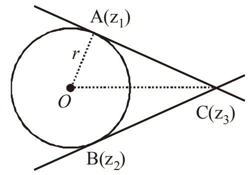
Subtracting (2) from (1) we get

$2(\bar{z}_2 - \bar{z}_1) = z_3\left(\dfrac{\bar{z}_1}{z_1} - \dfrac{\bar{z}_2}{z_2}\right)$

$\Rightarrow \quad \dfrac{2r^2(z_1 - z_2)}{z_1 z_2} = z_3 r^2\left(\dfrac{z_2^2 - z_1^2}{z_1^2 z_2^2}\right)$

$[\because |z_1|^2 = |z_2|^2 = r^2]$

$\Rightarrow \quad z_3 = \dfrac{2z_1z_2}{z_1 + z_2}$




**9.** **(d)** $\dfrac{z_1 - z_2}{z_1 + z_2} = \cos\alpha + i\sin\alpha$

$\Rightarrow \dfrac{2z_1}{-2z_2} = \dfrac{\cos\alpha + i\sin\alpha + 1}{\cos\alpha - 1 + i\sin\alpha}$

$= \dfrac{2\cos^2\alpha/2 + 2i\sin\alpha/2\cos\alpha/2}{2i\sin\alpha/2\cos\alpha/2 - 2\sin^2\alpha/2}$

$= \dfrac{2\cos\alpha/2[\cos\alpha/2 + i\sin\alpha/2]}{2i\sin\alpha/2[\cos\alpha/2 + i\sin\alpha/2]}$

$\Rightarrow \dfrac{z_1}{z_2} = i\cot\dfrac{\alpha}{3} \Rightarrow$ Given $\dfrac{z_1}{z_2} = \dfrac{K}{1}$

$\therefore \quad \tan\alpha/2 = -1/K \quad \Rightarrow \tan\alpha = \dfrac{2\tan\alpha/2}{1 - \tan^2\alpha/2}$

$\Rightarrow \quad \dfrac{-2/K}{1 - 1/K^2} \Rightarrow \dfrac{-2K}{K^2 - 1}$

$\alpha = \tan^{-1}\left(\dfrac{2K}{1 - K^2}\right) \Rightarrow 2\tan^{-1}(K)$

**10.** **(d)** $z^2 + az + a^2 = 0$

$\Rightarrow \quad z = a\omega, a\omega^2$

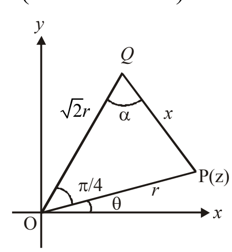
(where 'ω' is non-real root of cube unity)

$\Rightarrow$ Locus of $z$ is a pair of straight lines and $\arg(z) = \arg(a) + \arg(\omega)$ or $\arg(a) + \arg(\omega^2)$

$\Rightarrow \quad \arg(z) = \pm\dfrac{2\pi}{3}$

Also, $|z| = |a||\omega|$ or $|a||\omega^2| \quad \Rightarrow |z| = |a|$.

**11.** **(c)** $Z_p = r(\cos\theta + i\sin\theta)$



$$Z_Q = \sqrt{2|z|^2}\left(\cos\left(\theta+\frac{\pi}{4}\right)+i\sin\left(\theta+\frac{\pi}{4}\right)\right)$$

$$= \sqrt{2}r\left[\cos\left(\theta+\frac{\pi}{4}\right)+i\sin\left(\theta+\frac{\pi}{4}\right)\right]$$

From the figure,

$$\cos\frac{\pi}{4}=\frac{2r^2+r^2-x^2}{2\cdot\sqrt{2}r\cdot r}=\frac{3r^2-x^2}{2\sqrt{2}r^2} \quad \therefore 1=\frac{3r^2-x^2}{2r^2}$$

$\Rightarrow \quad r^2=x^2 \Rightarrow x=r \Rightarrow$ Triangle is right isosceles.

**12.** **(d)** $\dfrac{a\omega+b+c\omega^2}{a\omega^2+b\omega^2+c}+\dfrac{a\omega^2+b+c\omega}{a+b\omega+c\omega^2}$

$$=\frac{a\omega^2+b\omega+c\omega^3}{\omega(a\omega^2+b\omega+c)}+\frac{a\omega^3+b\omega+c\omega^2}{\omega(a+b\omega+c\omega^2)}$$

$$=\frac{1}{\omega}\frac{(a\omega^2+b\omega+c)}{(a\omega^2+b\omega+c)}+\frac{(a+b\omega+c\omega^2)}{\omega(a+b\omega+c\omega^2)} \quad (\because \omega^3=1)$$

$$=\frac{1}{\omega}+\frac{1}{\omega}=\frac{2}{\omega}=\frac{2\omega^2}{\omega^3}=2\omega^2$$

**13.** **(c)** $\because \quad z+\dfrac{1}{z}=1 \Rightarrow z^2-z+1=0$

$\therefore \quad z=\dfrac{-(-1)\pm\sqrt{(1-4)}}{2}=-\omega,-\omega^2$

[$\omega$ is cube root of unity]

and $z^{2017}=(-\omega)^{2017}=-\omega$,

$z^{2017}=(-\omega^2)^{2017}=-\omega^2$

$\therefore \quad a=z^{2017}+\dfrac{1}{z^{2017}}=-\left(\omega+\dfrac{1}{\omega}\right)$

$=-(\omega+\omega^2)=1$

and $2^{2^n}=2^{4.2^{n-4}}=16^{2^{n-4}}$ has last digit 6.

$\therefore \quad b=6-1=5$

Hence, $a^2+b^2=1^2+5^2=26$

**14.** **(c)** We have $||z_1|-|z_2||\le|z_1-z_2|$ and equality holds only when $\arg z_1=\arg z_2$

$\Rightarrow \quad ||z-\omega|-|z-\omega^2||\le|\omega^2-\omega|\le\sqrt{3}$ and equality can hold only when $|z|=2$ and not when $|z|=\dfrac{1}{2}$.

**15.** **(a)** $\alpha$ and $\beta$ are roots of $ax^2+bx+c=0$

$\Rightarrow \alpha+\beta=-\dfrac{b}{a}$ and $\alpha\beta=\dfrac{c}{a}=\dfrac{1}{a\alpha+b}+\dfrac{1}{a\beta+b}$

$$=\frac{a\beta+b+a\alpha+b}{(a\alpha+b)(a\beta+b)}$$

$$=\frac{a\times\left(\frac{-b}{a}\right)+2b}{a^2\cdot\frac{c}{a}+ab\cdot\left(\frac{-b}{a}\right)+b^2}=\frac{b}{ac}$$

$$\frac{1}{(a\alpha+b)(a\beta+b)}=\frac{1}{ac}$$

$\therefore$ Required quadratic equation

$x^2-$ (sum of roots) x + product of roots = 0

$\Rightarrow acx^2-bx+1=0$

**16.** **(c)** $c=-4,-b=-3$

So, $x^2+bx+c=0$ becomes $x^2+3x-4=0$

or $(x+4)(x-1)=0$

This gives 1 and – 4 as its roots.

**17.** **(a)** Let $t=x+\sqrt{x^2+b^2}$

$$\Rightarrow \frac{1}{t}=\frac{1}{x+\sqrt{x^2+b^2}}=\frac{\sqrt{x^2+b^2}-x}{b^2}$$

$$\Rightarrow t-\frac{b^2}{t}=2x \;\&\; t+\frac{b^2}{t}=2\sqrt{x^2+b^2}$$

$$\therefore \quad 2(a-x)(x+\sqrt{x^2+b^2})$$

$$=\left(2a-t+\frac{b^2}{t}\right)(t)=2at-t^2+b^2$$

$$\Rightarrow a^2+b^2-(a^2-2at+t^2)$$

$$=a^2+b^2-(a-t)^2\le a^2+b^2$$

**18.** **(b)** $\alpha,\beta$ are the roots of

$x^2+px+q=0 \Rightarrow \alpha+\beta=-p$ and $\alpha\beta=q$ ...(i)

Again, $\alpha,\beta$ are roots of $x^{2n}+p^nx^n+q^n=0$

$\Rightarrow \quad \alpha^{2n}+p^n\alpha^n+q^n=0$ and

$\beta^{2n}+p^n\beta^n+q^n=0$

$\Rightarrow \quad (\alpha^{2n}-\beta^{2n})+p^n(\alpha^n-\beta^n)=0$

$\Rightarrow \alpha^n+\beta^n=-p^n$ ...(ii)

Now $\dfrac{\alpha}{\beta}$ is a root of $(x^n+1)+(x+1)^n=0$

$$\Rightarrow \left(\frac{\alpha^n}{\beta^n}+1\right)+\left(\frac{\alpha}{\beta}+1\right)^n=0$$

$$\Rightarrow (\alpha^n+\beta^n)+(\alpha+\beta)^n=0$$

$$\Rightarrow \quad -p^n+(-p)^n=0,$$

Which holds only if n is an even integer.

**19.** **(a)** Let $f(x) = (x - \sin\beta)(x - \sin\gamma) + (x - \sin\alpha)(x - \sin\gamma) + (x - \sin\alpha)(x - \sin\beta)$

Now, $f(\sin\alpha) = (\sin\alpha - \sin\beta)(\sin\alpha - \sin\gamma)$
$= (-)(-) =$ positive

$f(\sin\beta) = (\sin\beta - \sin\alpha)(\sin\beta - \sin\gamma) = (+)(-)$
$=$ negative

$f(\sin\gamma) = (\sin\gamma - \sin\alpha)(\sin\gamma - \sin\beta) = (+)(+)$
$=$ positive

$\Rightarrow$ Roots of $f(x) = 0$ are real and distinct.

**20.** **(c)** $(a-1)x^2 - (a+1)x + (a-1) \geq 0\ \forall\ x \geq 2$

For $a = 1, -2x \geq 0\ \forall\ x \leq 2 \Rightarrow a \neq 1$

$\Rightarrow\ a > 1$ and $f'(x) \geq 0\ \forall\ x \geq 2, f(2) \geq 0$

$\Rightarrow\ 2(a-1)x - (a+1) \geq 0\ \forall\ x \geq 2, f(2) \geq 0$

$\Rightarrow\ 2(a-1)x \geq (a+1)\ \forall\ x \geq 2, 3a - 7 \geq 0$

$$\Rightarrow\ x \geq \frac{a+1}{2(a-1)}\ \forall x \geq 2,\ x \geq \frac{7}{3}$$

$$\Rightarrow\ \frac{a+1}{2(a-1)} \leq 2; a \geq \frac{7}{3}$$

$$\Rightarrow\ a + 1 \leq 4a - 4; a \geq \frac{7}{3}; a > 1$$

$$\Rightarrow\ a \geq \frac{5}{3}; a \geq \frac{7}{3}; a > 1\ \Rightarrow\ a \in \left[\frac{7}{3}, \infty\right)$$

**21.** **(1)** Given that $a, b, c$ are integers not all equal, $\omega$ is cube root of unity $\neq 1$, then

$|a + b\omega + c\omega^2|$

$$= \left|a + b\left(\frac{-1+i\sqrt{3}}{2}\right) + c\left(\frac{-1-i\sqrt{3}}{2}\right)\right|$$

$$= \left|\left(\frac{2a-b-c}{2}\right) + i\left(\frac{b\sqrt{3} - c\sqrt{3}}{2}\right)\right|$$

$$= \frac{1}{2}\sqrt{(2a-b-c)^2 + 3(b-c)^2}$$

$$= \frac{1}{2}\sqrt{4a^2 + b^2 + c^2 - 4ab + 2bc - 4ac + 3b^2 + 3c^2 - 6bc}$$

$$= \sqrt{a^2 + b^2 + c^2 - ab - bc - ca}$$

$$= \sqrt{\frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]}$$

R.H.S. will be min. when $a = b = c$, but we cannot take $a = b = c$ as per question.

$\therefore$ The min value is obtained when any two are zero and third is a minimum magnitude integer i.e. 1.

Thus $b = c = 0, a = 1$ gives us the minimum value 1.

**22.** **(45)** If $2 - i$ is the root then $2 + i$ is also the root sum of roots $= 4 \Rightarrow a = -3$

Product of roots $= \frac{b}{a} = (2-i)(2+i)$

$\Rightarrow\ b = -15\quad \therefore\ ab = 45$

**23.** **(0)** As the coefficients of two equations are in reverse order, if the roots of $ax^2 + bx + c = 0$ are $\alpha$ and $\beta$ then the roots of second equation are $\frac{1}{\alpha}, \frac{1}{\beta}$. Given that one negative root is common, two possibilities may arise.

Either $\alpha = \frac{1}{\alpha} < 0 \Rightarrow \alpha = -1$ or $\alpha = \frac{1}{\beta} < 0$

$\Rightarrow \alpha\beta = 1 \Rightarrow \frac{c}{a} = 1 \Rightarrow c = a$ (not possible)

$\therefore\ \alpha = -1$ is the common root. Put $\alpha = -1$ in any of the equations, we get $a - b + c = 0$.

**24.** **(7)** $x^2 - (3 + 2^{\sqrt{\log_2 3}} - 3^{\sqrt{\log_3 2}})x$

$$-2(3^{\log_3 2} - 2^{\log_2 3}) = 0$$

$\Rightarrow\ x^2 - 3(x) - 2(2-3) = 0 \Rightarrow x^2 - 3x + 2 = 0$

$\Rightarrow\ \alpha = 2, \beta = 1 \Rightarrow \alpha^2 + \alpha\beta + \beta^2 = 4 + 2 + 1 = 7$

**25.** **(0)** Let $f(x) = (x - x_1)(x - x_2)(x - x_3)(x - x_4)$

$\Rightarrow\ |f(i)| = \sqrt{1 + x_1^2}\sqrt{1 + x_2^2}\sqrt{1 + x_3^3}\sqrt{1 + x_4^4} = 1$

$\Rightarrow\ x_1 = x_2 = x_3 = x_4 = 0$

$\Rightarrow$ All four roots are zero.

$\Rightarrow\ f(x) = x^4 \Rightarrow a + b + c + d = 0$

**26.** **(2)** The given relation can be rewritten as

$$\frac{1}{a+\omega} + \frac{1}{b+\omega} + \frac{1}{c+\omega} = \frac{2}{\omega}$$

and $$\frac{1}{a+\omega^2} + \frac{1}{b+\omega^2} + \frac{1}{c+\omega^2} = \frac{2}{\omega^2}$$

$\Rightarrow \omega$ and $\omega^2$ are roots of

$$\frac{1}{a+x} + \frac{1}{b+x} + \frac{1}{c+x} = \frac{2}{x}$$

$$\Rightarrow\ \frac{3x^2 + 2(a+b+c)x + bc + ca + ab}{(a+x)(b+x)(c+x)} = \frac{2}{x}$$

$$\Rightarrow\ x^3 - (bc + ca + ab)x - 2abc = 0 \qquad \text{...(i)}$$

Two roots of the equation (i) are $\omega$ and $\omega^2$. Let the third root be $\alpha$, then

$\alpha + \omega + \omega^2 = 0 \Rightarrow \alpha = -\omega - \omega^2 = 1.$

$\therefore\ \alpha = 1$ will satisfy equation (i)

$$\Rightarrow\ \frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1} = 2$$

**27.** **(2)** Let $y = x + \sqrt{x^2 + k^2} = \dfrac{k^2}{\sqrt{x^2 + k^2} - x}$

$\Rightarrow \dfrac{k^2}{y} = \sqrt{x^2 + k^2} - x \;\therefore\; 2x = y - \dfrac{k^2}{y}$ ...(i)

Let $z = 2(k - x)\left(x + \sqrt{x^2 + k^2}\right)$

$= \left(2k - y + \dfrac{k^2}{y}\right)y$

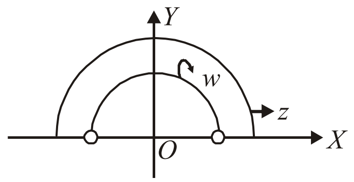
$= k^2 - y^2 + 2ky = 2k^2 - (k - y)^2 \le 2k^2$

$\therefore \dfrac{z}{k^2} \le 2$

**28.** **(3)** $Arg\left(\dfrac{z-2}{z+2}\right) = \dfrac{\pi}{2}$ gives semi-circle with radius $r = 2$.

$Arg\left(\dfrac{w-1}{w+1}\right) = \dfrac{\pi}{2}$ gives semi circle with radius $r = 1$.

Consider $z = 2^- + 0_i^+$ and $w = -1^+ + 0^+ i$




$\Rightarrow |z - w| < 3$ so $k = 3$ (as $|z - w| < k$ and $k$ is least upper bond).

**29.** **(4)** $\because (1+i)^4 = [(1+i)^2]^2$

$= (1 + i^2 + 2i)^2 = (1 - 1 + 2i)^2$
$= 4i^2 = -4$ ...(i)

& $\dfrac{1 - \sqrt{\pi}i}{\sqrt{\pi} + i} + \dfrac{\sqrt{\pi} - i}{1 + \sqrt{\pi}i}$

$= \dfrac{(1 - \sqrt{\pi}i)(\sqrt{\pi} - i)}{\pi + 1} + \dfrac{(\sqrt{\pi} - i)(1 - \sqrt{\pi}i)}{1 + \pi}$

$= \dfrac{\sqrt{\pi} - i - \pi i - \sqrt{\pi} + \sqrt{\pi} - \pi i - i - \sqrt{\pi}}{\pi + 1}$

$= \dfrac{-2\pi i - 2i}{\pi + 1} = -2i$ ...(ii)

Given, $z = \dfrac{\pi}{4}(1+i)^4\left(\dfrac{1 - \sqrt{\pi}i}{\sqrt{\pi} + i} + \dfrac{\sqrt{\pi} - i}{1 + \sqrt{\pi}i}\right)$

$= \dfrac{\pi}{4}(-4)(-2i) = 2\pi i$

[from Eqs. (i) and (ii)]:

Now, $\left(\dfrac{|z|}{\text{amp}(z)}\right) = \dfrac{2\pi}{\pi/2} = 4.$

**30.** **(4)** If $p, q, r$ are in A.P., $p + r = 2q$

Since $D \ge 0 \Rightarrow \dfrac{p^2 + r^2 + 2pr}{4} - 4pr \ge 0$

$\Rightarrow p^2 + r^2 - 14pr > 0$

$\Rightarrow \dfrac{p}{r} + \dfrac{r}{p} > 14 \Rightarrow \left|\dfrac{r}{p} - 7\right| \ge 4\sqrt{3} \Rightarrow k = 4$

CHAPTER 6

# Linear Inequalities

**1.** **(c)** $\dfrac{2}{x^2 - x + 1} - \dfrac{1}{x+1} - \dfrac{2x - 1}{x^3 + 1} \ge 0$

or $\dfrac{2(x+1) - (x^2 - x + 1) - (2x - 1)}{(x+1)(x^2 - x + 1)} \ge 0$

or $\dfrac{-(x^2 - x - 2)}{(x+1)(x^2 - x + 1)} \ge 0$

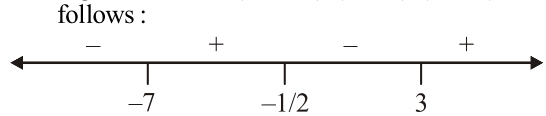
or $\dfrac{-(x-2)(x+1)}{(x+1)(x^2 - x + 1)} \ge 0$

or $\dfrac{2 - x}{x^2 - x + 1} \ge 0$, where $x \ne -1$

or $2 - x \ge 0, x \ne -1$ (as $x^2 - x + 1 > 0$ for $\forall\ x \in R$)

$\Rightarrow\; x \le 2, x \ne -1 \;\Rightarrow x \in (-\infty, -1) \cup (-1, 2]$

**2.** **(c)** $(2x + 1)(x - 3)(x + 7) < 0$
Sign scheme of $(2x + 1)(x - 3)(x + 7)$ is as follows :




Hence, solution is $(-\infty, 7) \cup (-1/2, 3)$

**3.** **(c)** $|x - 1| \le 3 \Rightarrow -3 \le x - 1 \le 3 \Rightarrow -2 \le x \le 4$

and $|x - 1| \ge 1 \Rightarrow x - 1 \le -1$

or $x-1\ge 1\Rightarrow x\le 0$ or $x\ge 2$
Taking the common values of x, we get
$x\in[-2,\ 0]\cup[2,\ 4]$

**4.** **(b)** The given equations are

$$|x-1|+3y=4\Rightarrow\begin{cases} x+3y=5,\ x\ge 1 & \cdots(i)\\ -x+3y=3,\ x<1 & \cdots(ii)\end{cases}$$

and

$$x-|y-1|=2\Rightarrow\begin{cases} x-y=1,\ \ y\ge 1 & \cdots(iii)\\ x+y=3,\ \ y<1 & \cdots(iv)\end{cases}$$

Solving (i) and (iii), we get $x=2, y=1$
Solving (i) and (iv), we get $x=2, y=1$
no solution $(\because x\ge 1 \text{ and } y<1)$
Solving (ii) and (iii), we get $x=3, y=2$
no solution $(\because x<1,\ y\ge 1)$

solving (ii) and (iv) we get, $x=\dfrac{5}{2},\ y=\dfrac{3}{2}$

no solution $(\because x<1,\ \ y<1)$
Hence solution is $x=2, y=1$ (a unique solution)

**5.** **(c)** $\left|2-|1-|x||\right|=1\Rightarrow 2-|1-|x||=\pm 1$

$\Rightarrow |1-|x||=1$ or $3$

If $\left|1-|x|\right|=1\Rightarrow 1-|x|=\pm 1\Rightarrow |x|=0$

or $2\Rightarrow x=0$ or $\pm 2$

If $\left|1-|x|\right|=3\Rightarrow 1-|x|=\pm 3\Rightarrow |x|=-2$ or $4$

$\Rightarrow |x|=4\Rightarrow x=\pm 4$ $\qquad [\because |x|\ne -2]$
$\therefore$ Solution set is $\{-4, -2, 0, 2, 4\}$, hence 5 real roots in all.

**6.** **(a)** The inequality is $|x+2|-|x-2|<x-\dfrac{3}{2}$.
Dividing the problem into three intervals :
(i) If $x<-2$, then

$$-(x+2)+(x-1)<x-\frac{3}{2}\Rightarrow x>-\frac{3}{2}$$

But $-\dfrac{3}{2}>-2$, hence no common values

$\Rightarrow x\in\phi$

(ii) If $-2\le x<1$, then

$$(x+2)+(x-1)<x-\frac{3}{2}\Rightarrow x<-\frac{5}{2}$$

But $-\dfrac{5}{2}<-2$, hence no common values

$\Rightarrow x\in\phi$

(iii) If $x\ge 1$, then

$$(x+2)-(x-1)<x-\frac{3}{2}\Rightarrow x>\frac{9}{2}$$

$\because \dfrac{9}{2}>1.\Rightarrow$ common solution is

$$x>\frac{9}{2}\Rightarrow x\in\left(\frac{9}{2},\infty\right)$$

$\therefore$ Solution set is $x\in\left(\dfrac{9}{2},\ \infty\right)$.

**7.** **(a)** $\left|\dfrac{12x}{4x^2+9}\right|\ge 1\Rightarrow\dfrac{12|x|}{4x^2+9}\ge 1$ $\qquad \because 4x^2+9>0$

$\Rightarrow 4x^2-12|x|+9\le 0\ \Rightarrow 4|x|^2-12|x|+9\le 0$

$\Rightarrow (2|x|-3)^2=0\Rightarrow |x|=\dfrac{3}{2}$

**8.** **(b)** $\left||x-1|+a\right|=4\Rightarrow |x-1|+a=\pm 4$

$\Rightarrow |x-1|=-a\pm 4$
The above equation holds if $-a+4\ge 0$
or $-a-4\ge 0$
$\Rightarrow a\le 4$ or $a\le -4\ \Rightarrow a\in(-\infty,\ 4]\cup(-\infty,\ -4]$
$\Rightarrow a\in(-\infty,\ 4]$

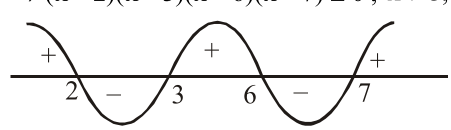
**9.** **(c)** We know that $|f(x)|=-f(x)$ if $f(x)\le 0$

$$\therefore \left|\frac{x^2-8x+12}{x^2-10x+21}\right|=-\frac{x^2-8x+12}{x^2-10x+21}$$

$$\Rightarrow\frac{x^2-8x+12}{x^2-10x+21}\le 0$$

$$\Rightarrow\frac{(x-2)(x-6)}{(x-3)(x-7)}\le 0,\ \ x\ne 3,\ 7$$

$\Rightarrow (x-2)(x-3)(x-6)(x-7)\le 0$, $x\ne 3,\ 7$




$\Rightarrow 2\le x<3$ or $6\le x<7$
$\Rightarrow x\in[2,\ 3)\cup[6,\ 7)$

**10.** **(a)** **Case (i) :** When $x\ge 0$
$\therefore\ \ |x|=x$

$\frac{3-x}{4-x} \geq 0 \Rightarrow \frac{(3-x)(4-x)}{(4-x)^2} \geq 0$

$\Rightarrow$ $(x-3)(x-4) \geq 0$ and $x \neq 4$ $\Rightarrow x \leq 3$ or $x > 4$

but $x \geq 0$ so $x \in [0, 3] \cup (4, \infty)$ ...(i)

**Case (ii) :** When $x < 0$

$\therefore$ $|x| = -x$ $\therefore$ $\frac{3+x}{4+x} \geq 0 \Rightarrow \frac{(x+3)(x+4)}{(x+4)^2} \geq 0$

$\Rightarrow$ $x < -4$ or $x \geq -3$

but $x < 0$ so $x \in (-\infty, -4) \cup [-3, 0)$ ...(ii)

So union of (i) and (ii) gives

$[-3,3] \cup (-\infty,-4) \cup (4,\infty)$

**11.** **(b)** $\frac{x+y}{x^2+y^2} - \frac{1}{2}\left(\frac{1}{x}+\frac{1}{y}\right)$

$\frac{x+y}{x^2+y^2} - \frac{x+y}{2xy} = (x+y)\left(\frac{1}{x^2+y^2} - \frac{1}{2xy}\right)$

$= (x+y)\left(\frac{2xy-(x^2+y^2)}{(x^2+y^2)(2xy)}\right)$

and $(x+y)\left(\frac{-(x-y)^2}{(x^2+y^2)(2xy)}\right) \leq 0$

Similarly, $\frac{y+z}{y^2+z^2} \leq \frac{1}{2}\left(\frac{1}{y}+\frac{1}{z}\right)$ and

$\frac{x+z}{x^2+z^2} \leq \frac{1}{2}\left(\frac{1}{x}+\frac{1}{z}\right)$

$\therefore$ Adding; we get

$\frac{x+y}{x^2+y^2} + \frac{y+z}{y^2+z^2} + \frac{z+x}{z^2+x^2} \leq \frac{1}{x}+\frac{1}{y}+\frac{1}{z}$

and Hence $A \leq B$

**12.** **(d)** $xy + yz + zx < 0$ and $\mu = \frac{x^2+y^2+z^2}{xy+yz+zx}$

$\therefore$ $(x+y+z)^2 = x^2+y^2+z^2+2(xy+yz+zx)$

$\therefore$ $(x+y+z)^2 \geq 0$

$\Rightarrow$ $x^2+y^2+z^2+2(xy+yz+zx) \geq 0$

$\Rightarrow$ $x^2+y^2+z^2 \geq -2(xy+yz+zx)$

$\Rightarrow$ $\frac{x^2+y^2+z^2}{xy+yz+zx} \leq -2$

**13.** **(a)** We have

$\frac{(a_1+a_2+...+a_{n-1}+2a_n)}{n} \geq (a_1 a_2 ... a_{n-1} 2a_n)^{1/n}$

[Using A.M. $\geq$ G.M.]

$\Rightarrow$ $a_1+a_2+a_3+.....+a_{n-1}+2a_n \geq n(2c)^{1/n}$

**14.** **(a)** The inequality is $\log_{0.2}\frac{x+2}{x} \leq 1$. The L.H.S is valid if $\frac{x+2}{x} > 0$

$\Rightarrow x(x+2) > 0 \Rightarrow x < -2$ or $x > 0$.

Solving the inequality, we get (note that base < 1)

$\frac{x+2}{x} \geq 0.2 = \frac{1}{5} \Rightarrow \frac{x+2}{x} - \frac{1}{5} \geq 0 \Rightarrow \frac{4x+10}{5x} \geq 0$

$x(2x+5) \geq 0 \Rightarrow x \leq -\frac{5}{2}$ or $x \geq 0$.

Taking the intersection, we get

$x \leq -\frac{5}{2}$ or $x > 0$

$\Rightarrow x \in \left(-\infty, -\frac{5}{2}\right] \cup (0, \infty)$

**15.** **(a)** The log functions are defined if

$\frac{x^2+6x+9}{2(x+1)} > 0$ and $x+1 > 0 \Rightarrow \frac{(x+3)^2}{2(x+1)} > 0$

and $x+1 > 0 \Rightarrow x > -1$

Now the inequality is

$\log_{2^{-1}}\frac{x^2+6x+9}{2(x+1)} < -\log_2(x+1)$

$\Rightarrow -\log_2\frac{x^2+6x+9}{2(x+1)} < -\log_2(x+1)$

$\Rightarrow \log_2\frac{x^2+6x+9}{2(x+1)} > \log_2(x+1)$

$\Rightarrow \frac{x^2+6x+9}{2(x+1)} > (x+1)$ $\Rightarrow \frac{-x^2+2x+7}{2(x+1)} > 0$

$\Rightarrow (x+1)(x^2-2x-7) < 0$

$\Rightarrow x^2-2x-7 < 0$ $[\because x+1 > 0]$

$\Rightarrow -1-2\sqrt{2} < x < -1+2\sqrt{2}$,

but $x > -1 \Rightarrow -1 < x < -1+2\sqrt{2}$

**16.** **(b)** For the validity of inequality $ax^2+4x+a > 0$, which is possible if $a > 0$ and $16-4a^2 < 0$

$\Rightarrow$ $a > 2$ ...(i)

Further, the inequality can be rewritten as

$\log_5 5 + \log_5(x^2+1) \leq \log_5(ax^2+4x+a)$

$\Rightarrow 5(x^2+1) \leq ax^2+4x+a$

$\Rightarrow (a-5)x^2+4x+(a-5)\geq 0$.

It holds if $a-5>0$ and $16-4(a-5)^2\leq 0$

$\Rightarrow a>5$ and $a\leq 3$ or $a\geq 7 \Rightarrow a\geq 7$ ...(ii)

Combining the results of (i) and (ii) for common values, we get $a\in[7, \infty)$

**17.** **(b)** For $x\geq -2$, $x^2-x-2+x>0$

$\Rightarrow x^2>2 \Rightarrow x\in(-\infty,-\sqrt{2})\cup(\sqrt{2},\infty)$

$\Rightarrow x\in[-2,-\sqrt{2})\cup(\sqrt{2},\infty)$

For $x<-2$

$x^2+x+2+x>0$ or $x^2+2x+2>0$

which is true for all x.

Hence $x\in(-\infty,-\sqrt{2})\cup(\sqrt{2},\infty)$

**18.** **(c)** $\log_{10} x = A$

$x>0$

$\log_{10}(x-2)=B,\ x-2>0\Rightarrow x>2$

$\Rightarrow A^2-3AB+2B^2<0 \Rightarrow (A-2B)(A-B)<0$

$\Rightarrow (\log x - 2\log(x-2)(\log x-\log(x-2))<0$

**Case - I :** $\log x-2\log(x-2)<0$

and $\log x-\log(x-2)>0$

**Case - II :** $\log x-2\log(x-2)>0$

and $\log x-\log(x-2)<0$

From (1) & (2), $x\in(4,\infty)$

**19.** **(a)** $x\neq(2n+1)\pi/2$, $n\pi n\in I$. The given inequality can be written as

$$\frac{\log_2(x^2-8x+23)}{\log_2|\sin x|}>\frac{3}{\log_2|\sin x|}$$

As $\log_2|\sin x|<0$, we get; $\log(x^2-8x+23)<3$

$\Rightarrow x^2-8x+23<2^3=8 \Rightarrow x^2-8x+15<0$

$\Rightarrow (x-5)(x-3)<0 \Rightarrow 3<x<5$

For $x\in(3,5)$, $x\neq\pi, \frac{\pi}{2}, \frac{3\pi}{2}$. Hence

$$x\in(3,\pi)\cup\left(\pi,\frac{3\pi}{2}\right)\cup\left(\frac{3\pi}{2},5\right).$$

**20.** **(b)** $y=\dfrac{3x^2+9x+17}{3x^2+9x+7}$

$3x^2(y-1)+9x(y-1)+7y-17=0$

$D\geq 0 \because$ x is real

$81(y-1)^2-4\times 3(y-1)(7y-17)\geq 0$

$\Rightarrow (y-1)(y-41)\leq 0 \Rightarrow 1\leq y\leq 41$

$\therefore$ Max value of $y$ is 41

**21.** **(4)** Consider the L.H.S of the equation,

$(\sin 3^x)(\cos 3^x)$

$$=\frac{2\sin 3^x.\cos 3^x}{2}=\frac{\sin(2.3^x)}{2}\leq\frac{1}{2}$$

$\Rightarrow \dfrac{3^x+3^x}{k}\leq\dfrac{1}{2}$ but $(3^x+3^{-x})\geq 2$

$\Rightarrow$ Minimum value of $k=4$

**22.** **(2)** As x, y $\in$ R, and xy > 0, so x and y will be of same sign.

Therefore, all the quantities $\dfrac{2x}{y^3}, \dfrac{x^3y}{3}, \dfrac{4y^2}{9x^4}$

are positive.

Now, A.M. $\geq$ G.M.

$$\Rightarrow \frac{2x}{y^3}+\frac{x^3y}{3}+\frac{4y^2}{9x^4}\geq 3\left(\left(\frac{2x}{y^3}\right)\left(\frac{x^3y}{3}\right)\left(\frac{4y^2}{9x^4}\right)\right)^{1/3}$$

$$=3\times\frac{2}{3}=2$$

**23.** **(1)** $x+\sqrt{3-x}\geq\sqrt{3-x}+3 \Rightarrow x\geq 3$

But $3-x\geq 0 \Rightarrow x\leq 3$

Hence, $x=3$ is the only integral solution.

**24.** **(4)** We have $\dfrac{|x+3|+x}{x+2}>1$

$\Rightarrow \dfrac{|x+3|+x}{x+2}-1>0 \Rightarrow \dfrac{|x+3|-2}{x+2}>0$

Now two cases arise :

**Case I :** When $x+3\geq 0$, i.e., $x\geq -3$. Then

$\dfrac{|x+3|-2}{x+2}>0 \Rightarrow \dfrac{x+3-2}{x+2}>0$

$\Rightarrow \dfrac{x+1}{x+2}>0 \Rightarrow \{(x+1)>0 \text{ and } x+2>0\}$

or $\{x+1<0 \text{ and } x+2<0\}$

$\Rightarrow \{x>-1 \text{ and } x>-2\}$

or $\{x<-1 \text{ and } x<-2\} \Rightarrow x>-1$ or $x<-2$

$\Rightarrow x\in(-1,\infty)$ or $x\in(-\infty,-2)$

$\Rightarrow x\in(-3,-2)\cup(-1,\infty)$ [Since $x\geq -3$ ]...(i)

**Case II :** When $x+3<0$, i.e., $x<-3$

$\dfrac{|x+3|-2}{x+2}>0 \Rightarrow \dfrac{-x-3-2}{x+2}>0$

$\Rightarrow \dfrac{-(x+5)}{x+2}>0 \Rightarrow \dfrac{x+5}{x+2}<0$

$\Rightarrow (x+5<0 \text{ and } x+2>0)$

or $(x+5>0 \text{ and } x+2<0)$

$\Rightarrow (x<-5 \text{ and } x>-2)$

or $(x>-5 \text{ and } x<-2)$. It is not possible.

$\Rightarrow x\in(-5,-2)$ ...(ii)

Combining (i) and (ii), the required solution is
$x \in (-5, -2) \cup (-1, \infty)$,
smallest integral member $= -4$
$\therefore$ Modulus of smallest integral member $= |-4| = 4$

**25.** **(4)** $|x| + \left|\dfrac{4-x^2}{x}\right| = \left|x + \dfrac{4-x^2}{x}\right|$

As $|a| + |b| = |a+b|$

$\Rightarrow ab \geq 0$

$\therefore x\left(\dfrac{4-x^2}{x}\right) \geq 0,\ x \neq 0$

$\Rightarrow x^2 - 4 \leq 0$
$\Rightarrow x \in [-2, 2] - \{0\}$
Number of integral values $= 4$.

# CHAPTER 7 Permutations and Combinations

**1.** **(a)** Total number of words that can be formed using 5 letters out of 10 given different letters
$= 10 \times 10 \times 10 \times 10 \times 10$ (as letters can repeat)
$= 1,00,000$
Number of words that can be formed using 5 different letters out of 10 different letters
$= {}^{10}P_5$ (none can repeat)

$= \dfrac{10!}{5!} = 30,240$

$\therefore$ Number of words in which at least one letter is repeated
= total words-words with none of the letters repeated
$= 1,00,000 - 30,240 = 69760$

**2.** **(c)** $X-X-X-X-X$. The four digits 3, 3, 5, 5 can be arranged at (–) places in $\dfrac{4!}{2!2!} = 6$ ways.
The five digits 2, 2, 8, 8, 8 can be arranged at ($X$) places in $\dfrac{5!}{2!3!} = 10$ ways.
Total no. of arrangements $= 6 \times 10 = 60$ ways.

**3.** **(c)** The letter of word *COCHIN* in alphabetic order are $C, C, H, I, N, O$.
Fixing first letter $C$ and keeping $C$ at second place, rest 4 can be arranged in 4! ways.
Similarly the words starting with *CH*, *CI*, *CN* are 4! in each case.
Then fixing first two letters as *CO* next four places when filled in alphabetic order give the word *COCHIN*.
$\therefore$ Numbers of words coming before *COCHIN* are $4 \times 4! = 4 \times 24 = 96$

**4.** **(d)** The smallest number, which can occur in the middle is 4.
The number of numbers with 4 in the middle
$= {}^4P_4 - {}^3P_3$
($\because$ The other four places are to be filled by 0, 1, 2 and 3, and a number cannot begin with 0)
Similarly, the number of numbers with 5 in the middle
$= {}^5P_4 - {}^4P_3$, etc.
$\therefore$ The required number of numbers
$= ({}^4P_4 - {}^3P_3) + ({}^5P_4 - {}^4P_3) + ({}^6P_4 - {}^5P_3) +$
$\ldots\ldots\ldots\ldots + ({}^9P_4 - {}^8P_3) = \sum_{n=4}^{9} ({}^nP_4 - {}^{n-1}P_3)$

**5.** **(b)** $C\ldots\ldots = 4! = 24$
$D\ldots\ldots = 4! = 24$
$M\ldots\ldots = 4! = 24$
$S\,C\ldots\ldots = 3! = 6$
$S\,D\ldots\ldots = 3! = 6$
$S\,M\,C\,D\,W = 1$
$S\,M\,C\,W\,D = 1$

**6.** **(b)** $m$ = Number of permutations of letters of the word BHARAT, when $B$ and $H$ are never together
$= \dfrac{6!}{2!} - \dfrac{5!}{2!} \times 2!$ (keeping (BH) together and permuting with $A, R, A, T$)
$= 360 - 120 = 240$ and $n$ = number of permutations of the letters of the word BHARAT when each word begins with $B$ and ends with
$T = \dfrac{4!}{2!} = 12 \therefore \dfrac{m}{n} = \dfrac{240}{12} = 20$

**7.** **(c)** Let the square has dimensions $n \times n$.
If $n$ is even, total no. of blue (diagonal) tiles $= n + n = 2n$
If $n$ is odd, then total no. of blue tiles
$= n + n - 1 = 2n - 1$
(the middle tile is be counted twice).
Hence, $2n - 1 = 121$
$\Rightarrow n = 61$
So, number of red tiles $= 61 \times 61 - 121 = 3600$

**8.** **(c)**

| Number of digits | Numbers ending with 0 | Numbers ending with 5 | Total |
|---|---|---|---|
| 1 | 0 | 1 | 1 |
| 2 | 8 | 9 | 17 |
| 3 | 9.8 = 72 | 8.8 = 64 | 136 |
| 4 | 9.8.7 = 504 | 8.8.7 = 448 | 952 |
| | | Total | 1106 |

**9.** **(d)** 10 identical boxes can contain 12 balls such that each box contains at least one ball in the following cases:

**Case (i) :** 8 boxes contains single ball each + 2 boxes contains two balls each

$$\overbrace{2W, 6B;}^{\text{Single ball boxes}} \quad \overbrace{(WW)(WW)}^{\text{Boxes with double ball}}$$
$$3W, 5B; \quad (WW)(WB)$$
$$4W, 4B; \quad (WW)(BB); \quad (WB)(WB)$$
$$5W, 3B; \quad (WB)(BB)$$
$$6W, 2B; \quad (BB)(BB)$$

**Case (ii):** 9 boxes single ball each + 1 box containing 3 balls

$$\overbrace{3W+6B;}^{\text{Single ball boxes}} \quad \overbrace{(WWW)}^{\text{Triple ball boxes}}$$
$$4W+5B; \quad (WWB)$$
$$5W+4B; \quad (WBB)$$
$$6W+3B; \quad (BBB)$$

$\therefore$ Total 10 ways are there

**10.** **(c)** Required number of ways = (Total number of ways of arranging 7 persons) – (Number of ways where two American are together + Number of ways when two Britishian are together – Number of ways when both American and both Britishian are together)

$= 6! - [2 \times 5! + 2 \times 5! - 4! \times 2! \times 2!)]$

$= 720 - [480 - 96] = 240 + 96 = 336$

**11.** **(b)** The candidate is unsuccessful if he fails in 9 or 8 or 7 or 6 or 5 papers.

$\therefore$ The number of ways to be unsuccessful

$$= {}^9C_9 + {}^9C_8 + {}^9C_7 + {}^9C_6 + {}^9C_5$$
$$= {}^9C_0 + {}^9C_1 + {}^9C_2 + {}^9C_3 + {}^9C_4$$
$$= \frac{1}{2}({}^9C_0 + {}^9C_1 + \text{.........} + {}^9C_9)$$
$$= \frac{1}{2} \cdot 2^9 = 2^8 = 256.$$

**12.** **(a)** Arrangement does not matter because of descending order there should be only one arrangement

$\therefore$ required number of ways $= {}^{15}C_3 \times {}^{24}C_2$

$$\underbrace{\boxed{1}\boxed{\cdots\cdots\cdots}\boxed{24}}_{\text{Selecting any 2 out of } 1 \text{ to } 24 = {}^{24}C_2}\boxed{25}\underbrace{\boxed{26}\boxed{\cdots\cdots\cdots}\boxed{40}}_{\text{Selecting any 3 from } 26 \text{ to } 40 = {}^{15}C_3}$$

**13.** **(b)** Let the number of children in the class = $n$.

Number of groups of 3 children = ${}^nC_3$

Each child would go to zoo ${}^{n-1}C_2$ times.

$\therefore$ According the question: ${}^nC_3 = 84 + {}^{n-1}C_2$

$$\Rightarrow \frac{n(n-1)(n-2)}{6} = 84 + \frac{(n-1)(n-2)}{2}$$

$\Rightarrow n(n-1)(n-2) = 504 + 3(n-1)(n-2)$

$\Rightarrow (n-1)(n-2)(n-3) = 504$

$\Rightarrow (n-1)(n-2)(n-3) = 7 \times 8 \times 9 \Rightarrow x = 10$

**14.** **(d)** We will consider the following cases:

| Case | Flags | No. of signals |
|---|---|---|
| 4 alike and 2 others alike | 4 white and 2 red | $\frac{6!}{4!2!} = 15$ |
| 4 alike and 2 others different | 4 white, 1 red and 1 blue | $\frac{6!}{4!} = 30$ |
| 3 alike and 3 others alike | 3 white, 3 red | $\frac{6!}{3!3!} = 20$ |
| 3 alike and 2 others alike and 1 different | 3 white, 1 blue, 2 red or 3 red, 1 blue, 2 white | ${}^2C_1 \times \frac{6!}{3!2!} = 120$ |
| | Total | 185 |

**15.** **(c)** Let the number of men = n

Number of women = 2

$\therefore$ Total participants = $(n + 2)$

Now number of games played by men among the m selves = $2({}^nC_2)$

Number of games men played with women = $4n$

According to question $2({}^nC_2) = 4n + 66$

$\Rightarrow n(n-1) = 4n + 66 \Rightarrow n^2 - 5n - 66 = 0$

$\Rightarrow (n-11)(n+6) = 0 \Rightarrow n = 11$ or $n = -6$

$\therefore$ Total number of participants = 11 + 2 = 13

**16.** **(b)** Suppose there n players in the beginning. The total number of games to be played was equal to ${}^nC_2$ and each player would have played $n-1$ games.

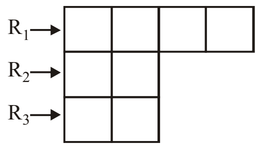
Let us assume that $A$ and $B$ fell ill. Now the total number of games of $A$ and $B$ is $(n-1)+(n-1)-1 = 2n - 3$. But they have played 3 games each. Then their remaining number of games is

$2n-3-6=2n-9$. Given,

${}^nC_2-(2n-9)=84$

$\Rightarrow \quad n^2-5n-150=0$ or $n=15$

**17.** **(b)** Required ways = ${}^8C_5$ – (Number of ways when $R_2$ or $R_3$ are unused)




$= {}^8C_5-({}^6C_5+{}^6C_5)={}^8C_5-(12)=56-12=44$

**18.** **(c)** The number of triangles with vertices on different lines $= {}^pC_1 \times {}^pC_1 \times {}^pC_1 = p^3$.

The number of triangles with 2 vertices on one line and the third vertex on any one of the other two lines

$$= {}^3C_1\left\{{}^pC_2 \times {}^{2p}C_1\right\} = 6p\,\frac{p(p-1)}{2}$$

$\therefore$ The required number of triangles

$=p^3+3p^2(p-1)$.

**19.** **(c)** Let $A_1, A_2, ..., A_{10}$ be vertices of a regular polygon of 10 sides.

The number of ways of selecting 3 vertices is ${}^{10}C_3$.

The number of ways of selecting 3 consecutive vertices is (i.e., $A_1A_2A_3, A_2A_3A_4, ..., A_{10}A_1A_2$) $=10$.

The number of ways of selecting three vertices such that two vertices are consecutive = (First select 2 consecutive vertices, leave their neighbouring two vertices and select one more from the remaining 6 vertices) is

$10 \times {}^6C_1 = 60$ The total number of required selections is

${}^{10}C_3-10-60=120-70=50$

**20.** **(b)** Division of objects can be (0, 1, 4) or (0, 2, 3)

So total number of distribution ways

$$=\frac{5!}{0!.1!.4!}\times 3! + \frac{5!}{0!.2!.3!}\times 3! = 30+60=90 \text{ ways}$$

**21.** **(8)** $a+b+c=21 \Rightarrow 3b=21 \Rightarrow b=7$

$[\because a+c=2b]$

$\Rightarrow \quad a+b+c=21 \Rightarrow a+c=14$

$\Rightarrow \lambda = {}^{14-1}C_{2-1} = {}^{13}C_1 = 13$

Hence $\lambda-5=13-5=8$

**22.** **(9)** If exactly 6 of his matching are correct, then the remaining 4 statements from column I must be incorrectly matched with 4 statements from column II

$\therefore$ Number of ways

$$={}^{10}C_6\cdot 4!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}\right]$$

$$=\frac{10!}{6!4!}\cdot 4!\left[\frac{1}{2}-\frac{1}{6}+\frac{1}{24}\right]$$

$$=(10\times 9\times 8\times 7)\left[\frac{12-4+1}{24}\right]=1890$$

**23.** **(6)** Number of numbers beginning with 1 = 120

| 1 | | | | | |
|---|---|---|---|---|---|

Number of numbers beginning with 2 = 120

| 2 | | | | | |
|---|---|---|---|---|---|

Starting with 31 ................................ = 24

| 3 | 1 | | | | |
|---|---|---|---|---|---|

Starting with 3214 ................................ = 2

| 3 | 2 | 1 | 4 | | |
|---|---|---|---|---|---|

Finally = 1

| 3 | 2 | 1 | 5 | 4 | 6 |
|---|---|---|---|---|---|

Hence, unit place digit of 267th number is 6.

**24.** **(7)** If $x$ denotes the number of times he can take unit step and $y$ denotes the number of times he can take 2 steps, then $x+2y=7$.

We must have $x=1, 3, 5$

If $x=1$, the steps will be 1 2 2 2

$\Rightarrow$ Number of ways $=\frac{4!}{3!}=4$

If $x=3$, the steps will be 1 1 1 2 2

$\Rightarrow$ Number of ways $=\frac{5!}{2!.3!}=10$

If $x=5$, the steps will be 1 1 1 1 1 2

$\Rightarrow$ Number of ways $={}^6C_1=6$

If $x=7$, the steps will be 1 1 1 1 1 1 1

$\Rightarrow \quad {}^7C_0=1$

Hence total number of ways $=N=21$

$\Rightarrow \quad N/3=7$

**25.** **(9)** In this case number of participants from 10 different countries are 1, 2, 3… 10. number of ways that they can be arranged in a row such that all the participants from the same country were together is (10!)(1!)(2!)(3!)(4!)…. (10!)

Hence K = (10!)(1!)(2!)(3!)(4!)…. (10!) highest power of 10 or that of 5 can be given by only (5!)(6!)(7!)(8!) (9!)(10!)(10!) which is

$1+1+1+1+1+2+2=9$

CHAPTER 8

# Binomial Theorem

**1.** **(a)** $2^{60} = (1+7)^{20}$

$= {}^{20}C_0 + {}^{20}C_1 . 7 + {}^{20}C_2 . 7^2 + \ldots\ldots\ldots\ldots + {}^{20}C_{20} . 7^{20}$

$\therefore$ the remainder $= {}^{20}C_0 = 1$

**2.** **(c)** $32 = 2^5 \Rightarrow (32)^{32} = (2^5)^{32}$

$= 2^{160} = (3-1)^{160} = 3m+1, m \in N$

$\therefore (32)^{32^{32}} = (32)^{3m+1} = 2^{5(3m+1)}$

$= 2^{3(5m+1)}\, 2^2 = 4.8^{5m+1}$

$4(7+1)^{5m+1} = 4(7n+1), n \in N = 28n+4$

$\therefore$ When 7 divides $(32)^{32^{32}}$, remainder $= 4$

**3.** **(d)** $\sum_{r=1}^{n}\left(\sum_{p=0}^{r-1} {}^nC_r . {}^rC_p . 2^p\right)$

$= \sum_{r=1}^{n} {}^nC_r \left(\sum_{p=0}^{r-1} {}^rC_p . 2^p\right)$

$= \sum_{r=1}^{n} {}^nC_r \{(1+2)^r - 2^r\} = \sum_{r=1}^{n} {}^nC_r 3^r - {}^nC_r 2^r$

$= (4^n - 1) - (3^n - 1) = 4^n - 3^n$

**4.** **(d)** Given expression is $(1+x-2x^2)^6$

$= 1 + a_1x + a_2x^2 + \ldots\ldots + a_{12}x^{12}$

Put $x = 1$ both side;

$0 = 1 + a_1 + a_2 + \ldots. + a_{12}$ …(i)

Put $x = -1$ both sides;

$(-2)^6 = 1 - a_1 + a_2 - \ldots. + a_{12}$ …(ii)

Adding (i) and (ii),

$2 + 2a_2 + 2a_4 + \ldots. + 2a_{12} = 64$

$\Rightarrow a_2 + a_4 + \ldots. + a_{12} = \dfrac{64-2}{2} = 31$

**5.** **(a)** The last term $= {}^nC_n . (-1/\sqrt{2})^n$

$= (1/3. \sqrt[3]{9})^{\log_3 8}$.

$\therefore (-1)^n . (1/2)^{n/2} = \left(\dfrac{1}{3^{5/3}}\right)^{\log_3 8.}$

$= 3^{-\frac{5}{3} . 3\log_3 2} = 3^{-5\log_3 2} = 2^{-5}$

So, $t_5 = {}^{10}C_4\, (2^{1/3})^6\, (-1/\sqrt{2})^4 = {}^{10}C_4 . = {}^{10}C_{10-4}$.

**6.** **(c)** ${}^nC_0 + {}^nC_1(ax)^1 + {}^nC_2(ax)^2 + \ldots\ldots..$

$= 1 + 8x + 24x^2 +$ ... (i)

Comparing coefficients of $x^0, x^1, x^2$ in equation (i),

${}^nC_0 = 1 \Rightarrow 1 = 1$ ...(ii)

${}^nC_1 a = 8 \Rightarrow na = 8$ ...(iii)

and ${}^nC_2 . a^2 = 24 \Rightarrow \dfrac{n(n-1)}{2}a^2 = 24$ ... (iv)

Put $n = \dfrac{8}{a}$ in equation (iv), from equation (iii)

$\Rightarrow \left[\left(\dfrac{8}{a}\right)^2 - \dfrac{8}{a}\right]a^2 = 48 \Rightarrow 64 - 8a = 48$

$\Rightarrow a = 2$ $\therefore n = 4$

$\therefore \dfrac{a-n}{a+n} = \dfrac{2-4}{2+4} = -\dfrac{1}{3}$

**7.** **(a)** $17^{2009} + 11^{2009} - 7^{2009} =$

$(17^{2009} - 7^{2009}) + 11^{2009} = (17-7)$

$(17^{2008} + 17^{2007}.7^1 + \ldots.. + 7^{2008}) + 11^{2009}$

Cleary last digit in $17^{2009} - 7^{2009}$ is 0.

Also, last digit in $11^{2009}$ is 1.

Last digit of $17^{2009} + 11^{2009} - 7^{2009}$ is $0 + 1 = 1$.

**8.** **(c)** Let $x = 2^{\frac{153}{2}}$ $\therefore \alpha = x^2 + \sqrt{2}x + 1$

Let $y = 2^{204}$ $\therefore \beta = y^2 - y + 1$

$\Rightarrow N = x^{16} - 1 \Rightarrow N = (x^4 - 1)(x^4 + 1)(x^8 + 1)$

$\Rightarrow N = (x^4 - 1)(x^2 + \sqrt{2}x + 1)(x^2 - \sqrt{2}x + 1)(x^8 + 1)$

$\Rightarrow N = y^6 - 1 = (y^3 - 1)(y^3 + 1)$

$\Rightarrow (y^3 - 1)(y + 1)(y^2 - y + 1)$

**9.** **(c)** $N = {}^{20}C_7 - {}^{20}C_8 + {}^{20}C_9 - {}^{20}C_{10} + \ldots.. - {}^{20}C_{20}$

$= ({}^{20}C_7 + {}^{20}C_9 + {}^{20}C_{11} + \ldots.. + {}^{20}C_{19}) - ({}^{20}C_8 - {}^{20}C_{10} + \ldots.. + {}^{20}C_{20})$

$= ({}^{20}C_0 + {}^{20}C_2 + {}^{20}C_4 + {}^{20}C_6) - ({}^{20}C_1 + {}^{20}C_3 + {}^{20}C_5)$

$= (1 + 190 + 4845 + 38760) - (20 + 1140 + 15504)$

$= 43796 - 16664 = 27132 = 3 \times 4 \times 7 \times 19 \times 17$

**10.** **(a)** $x = T_7 = {}^nC_6(3^{1/3})^{n-6}.(4^{-1/3})^6$
$y = T_{n-5} = {}^nC_{n-6}(3^{1/3})^6.(4^{-1/3})^{n-6}$
$\Rightarrow y = 12x$
${}^nC_{n-6}(3^{1/3})^6\,(4^{-1/3})^{n-6}$
$= 12.\,{}^nC_6(3^{1/3})^{n-6}.(4^{-1/3})^6$
$\Rightarrow \quad 12 = (12^{1/3})^{12-n} \Rightarrow n = 9$

**11.** **(d)** Given expression

$$= \frac{(x^{1/3})^3 + (1)^3}{x^{2/3} - x^{1/3} + 1} - \frac{(x-1)}{x^{1/2}(x^{1/2}-1)}$$

$$= \frac{(x^{1/3}+1)\,(x^{2/3} - x^{1/3}+1)}{(x^{2/3}-x^{1/3}+1)}$$

$$- \frac{(x^{1/2}+1)\,(x^{1/2}-1)}{x^{1/2}(x^{1/2}-1)}$$

$$= (x^{1/3}+1) - (1 + x^{-1/2}) = x^{1/3} - x^{-1/2}$$

$$\Rightarrow \left(\frac{x+1}{x^{2/3}-x^{1/3}+1} - \frac{x-1}{x - x^{1/2}}\right)^{10}$$

$= (x^{1/3} - x^{-1/2})^{10}$

$T_{r+1}$ in $(x^{1/3} - x^{-1/2})^{10}$ is

$${}^{10}C_r\,(x^{1/3})^{10-r}.(-1)^r.(x^{-1/2})^r$$

$= (-1)^r\,{}^{10}C_r\,x^{\left(\frac{10-r}{3} - \frac{r}{2}\right)}$ which is independent of $x$ if $\left(\frac{10-r}{3} - \frac{r}{2}\right) = 0 \Rightarrow r = 4$

Hence required coefficient $= {}^{10}C_4\,(-1)^4 = 210$.

**12.** **(c)** Middle term $= {}^{2n}C_n.x^n = t_{n+1}$

If $t_{n+1}$ is the greatest term also, then

$t_{n+1} \geq t_n$ ....(i)

$t_{n+1} \geq t_{n+2}$ .....(ii)

From (i), ${}^{2n}C_n.\,x^n \geq {}^{2n}C_{n-1}.\,x^{n-1}$

$$\Rightarrow \frac{(2n)!}{n!\,n!} \geq \frac{(2n)!}{(n-1)!(n+1)!} \Rightarrow (n+1)\,x \geq n$$

$\Rightarrow x \geq \dfrac{n}{n+1}$ ...(iii)

From (ii),

$${}^{2n}C_n\cdot x^n \geq {}^{2n}C_{n+1}\cdot x^{n+1}$$

$$\Rightarrow \frac{(2n)!}{n!n!} \geq \frac{(2n)!}{(n-1)!(n+1)!}\,x$$

$\Rightarrow x \leq \dfrac{n+1}{n}$ ...(iv)

So, from (iii) and (iv), $x \in \left[\dfrac{n}{n+1}, \dfrac{n+1}{n}\right]$

**13.** **(a)** $t_{r+1} = {}^{100}C_r\left(\sqrt[8]{5}\right)^{100-r}.\left(\sqrt[6]{2}\right)^r$. As 2 and 5 are coprime, $t_{r+1}$ will be rational if $100 - r$ is a multiple of 8 and r is a multiple of 6.
Also $0 \leq r \leq 100$.
$\therefore r = 0, 6, 12, \ldots\ldots\ldots\ldots, 96$
$\Rightarrow 100 - r = 4, 10, 16, \ldots\ldots\ldots\ldots, 100$
But $100 - r$ is to be a multiple of 8.
So, $100 - r = 0, 8, 16, 24, \ldots 96$.
The common terms in (1) and (2) are 16, 40, 64 and 88.
$\therefore r = 84, 60, 36, 12$ give rational terms.
$\therefore$ The number of irrational terms
$= 101 - 4 = 97$.

**14.** **(c)** $t_{r+1} = {}^{10}C_r\,(x\sin\alpha)^{10-r}.\left(\dfrac{\cos\alpha}{x}\right)^r$.

It is independent of $x$ if $r = 5$

$\therefore$ The term independent of $x = {}^{10}C_5.\sin^5\alpha$.

$$\cos^5\alpha = {}^{10}C_5.\frac{1}{2^5}\,(\sin 2\alpha)^5 \leq {}^{10}C_5\,\frac{1}{2^5}$$

$$= \frac{(10)!}{(5!)^2}.\frac{1}{2^5}$$

**15.** **(c)** If $T_{r+1} \geq T_r$, then $\dfrac{T_{r+1}}{T_r} \geq 1$

$$\Rightarrow \frac{25 - r + 1}{r}\cdot\frac{3x}{2} \geq 1$$

$$\Rightarrow \frac{26-r}{r}.3 \geq 1 \qquad (\because x = 2)$$

$$\Rightarrow 78 - 3r \geq r \quad \therefore\ r \leq \frac{78}{4} \Rightarrow r \leq 19\frac{1}{2}$$

$\Rightarrow r = 19$
Largest term $= (r+1)^{\text{th}}$ term $= 20^{\text{th}}$ term

**16.** **(c)** Middle term in the expansion is

$\left(\dfrac{10}{2}+1\right)^{\text{th}}$ i.e., $6^{\text{th}}$ term.

$$\text{Thus } T_6 = 7\frac{7}{8} \Rightarrow {}^{10}C_5\frac{1}{x^5}.x^5\sin^5 x = \frac{63}{8}$$

$$\Rightarrow 252.\sin^5 x = \frac{63}{8} \Rightarrow \sin^5 x = \frac{1}{32}$$

$$\Rightarrow \sin x = \frac{1}{2}\ \therefore\ x = n\pi + (-1)^n\frac{\pi}{6}$$

**17.** **(d)** The cofficients of the integral powers of $x$ are

$^{40}C_0$, $^{40}C_2 \cdot 2^2$, $^{40}C_4 \cdot 2^4$, ....., $^{40}C_{40} \cdot 2^{40}$

$\Rightarrow \quad (1+2)^{40} = {}^{40}C_0 + {}^{40}C_1 2 + {}^{40}C_2 \cdot 2^2 + ......$
$+ {}^{40}C_{40} \cdot 2^{40}$

$\Rightarrow \quad (1-2)^{40} = {}^{40}C_0 - {}^{40}C_1 \cdot 2$
$+ {}^{40}C_2 \cdot 2^2 - ......+ {}^{40}C_{40} \cdot 2^{40}$

Adding, $3^{40} + 1 = 2 \times$ (required sum)

**18.** **(b)**

$$\frac{\pi(n)}{\pi(n+1)} = \frac{{}^nC_0 . {}^nC_1 . {}^nC_2 ..... {}^nC_n}{{}^{n+1}C_0 . {}^{n+1}C_1 . {}^{n+1}C_2 ...... {}^{n+1}C_{n+1}}$$

$$= \frac{1}{{}^{n+1}C_0}\left(\frac{{}^nC_0}{{}^{n+1}C_1}\right)\left(\frac{{}^nC_1}{{}^{n+1}C_2}\right)......\left(\frac{{}^nC_n}{{}^{n+1}C_{n+1}}\right)$$

$$= \frac{1}{1}\left(\frac{1}{n+1}\right)\left(\frac{2}{n+1}\right)........\left(\frac{n+1}{n-1}\right)$$

$$\left[\because \frac{{}^nC_r}{{}^{n+1}C_{r+1}} = \frac{r+1}{n+1}\right]$$

$$= \frac{(n+1)!}{(n+1)^{n+1}} = \frac{n!}{(n+1)^n} \therefore \frac{\pi(2002)}{\pi(2001)} = \frac{(2002)^{2001}}{(2001)!}$$

**19.** **(d)**

$$\frac{C_1}{2} + \frac{C_3}{4} + \frac{C_5}{6} + ..... = \frac{n}{2} + \frac{n(n-1)(n-2)}{4!}$$
$$+ \frac{n(n-1)(n-2)(n-3)(n-4)}{6!} + ......$$

$$= \frac{1}{n+1}\left[\frac{(n+1)n}{2!} + \frac{(n+1)n(n-1)(n-2)}{4!} + \frac{(n+1)n(n-1)(n-2)(n-3)(n-4)}{6\,!} + .....\right]$$

$$= \frac{1}{n+1}[{}^{n+1}C_2 + {}^{n+1}C_4 + {}^{n+1}C_6 + .....]$$

$$= \frac{1}{n+1}\left[2^{n+1-1} - {}^{n+1}C_0\right] = \frac{2^n - 1}{n+1}$$

**20.** **(b)** Given $(2+\sqrt{3})^n = I + f$, where $I$ is integer and $0 \le f < 1$. We note that $(2+\sqrt{3})\ (2-\sqrt{3})$ = 1. So let us assume that $F = (2-\sqrt{3})^n$. Clearly $0 < F < 1$. Now,

$I + f + F = (2+\sqrt{3})^n + (2-\sqrt{3})^n$

$= 2[{}^nC_0 2^n + {}^nC_2 2^{n-2}.3 + {}^nC_4 .2^{n-4}.3^2 + ............]$

$= 2 \times$ Integer = Integer

$\because$ I + $f$ + $F$ is integer $\Rightarrow$ $f$ + $F$ must be integer.

$\therefore \ 0 \le f < 1$ and $0 < F < 1 \Rightarrow 0 < f + F < 2$

$\Rightarrow f + F = 1 \Rightarrow F = 1 - f \ \therefore (I + f)(1 - f)$

$= (I + f)\, F = (2+\sqrt{3})^n \ (2-\sqrt{3})^n = 1$

**21.** **(1)** $$\sum_{k=0}^{4}\left(\frac{3^{4-k}}{(4-k)!}\right)\left(\frac{x^k}{k!}\right)$$

$$= \sum_{k=0}^{4}\left(\frac{3^{4-k}}{(4-k)!}\right)\left(\frac{x^k}{k!}\right)\frac{4!}{4!}$$

$$= \sum_{k=0}^{4}\frac{{}^4C_k .3^{4-k}\, x^k}{4!} = \frac{(3+x)^4}{4!}$$

According to the question.

$$\frac{(3+x)^4}{4!} = \frac{32}{3}$$

or $(3+x)^4 = 256$ or $x + 3 = 4$ or $x = 1$

**22.** **(4)** Expression $= (1-x)^5 .(1+x)^4 (1+x^2)^4$
$= (1-x)(1-x^2)^4 (1+x^2)^4 = (1-x)(1-x^4)^4$
$\therefore$ so the coefficient of $x^{13} = -{}^4C_3 (-1)^3 = 4$

**23.** **(0)** We have $3C_0 - 5C_1 + 7C_2 + ...$
$+ (-1)^n (2n+3)C_n$
$= 3C_0 - 3C_1 + 3C_2 + ... + (-1)^n 3C_n - 2C_1 + 4C_2 + ...$
$+ (-1)^n 2nC_n$
$= 3\,[C_0 - C_1 + C_2 + .... + (-1)^n C_n]$
$-2[C_1 - 2C_2 + .... (-1)^n nC_n]$
$= 3 \times 0 - 2 \times 0 = 0$

**24.** **(5)** We have

$$11!\left[\frac{1}{1!10!} + \frac{1}{3!8!} + \frac{1}{5!6!} + ... + \frac{1}{11!1!}\right]$$

$$= \frac{11}{1!10!} + \frac{11}{3!8!} + \frac{11}{5!6!} + ... + \frac{11}{11!}$$

$= {}^{11}C_1 + {}^{11}C_3 + {}^{11}C_5 + {}^{11}C_7 + {}^{11}C_9 + {}^{11}C_{11}$

= sum of the even coefficient in the expansion of $(1+x)^{11} = 2^{10}$

Therefore $K = 10$

**25.** **(6)** $(1 - 2x + 5x^2 - 10x^3)\,[C_0 + C_1 x + C_2 x^2 + ...]$
$= 1 + a_1 x + a_2 x^2 + ...$

$$\Rightarrow \quad a_1 = n - 2 \quad \text{and} \quad a_2 = \frac{n(n-1)}{2} - 2n + 5$$

$a_1^2 = 2a_2$. Therefore, we have

$(n-2)^2 = n(n-1) - 4n + 10$

$\Rightarrow \ n^2 - 4n + 4 = n^2 - 5n + 10$

$\therefore \ n = 6$

CHAPTER 9

# Sequences and Series

**1.** **(d)** We have

$S_n = 1^3 + 3.2^3 + 3^3 + 3.4^3 + 5^3 + ...........$

Let $n = 2m$

Then $S_{2m} = (1^3 + 3^3 + 5^3 + ......$ to $m$ terms$)$

$+ 3(2^3 + 4^3 + 6^3 + ...$to $m$ terms$)$

$= \{1^3 + 2^3 + 3^3 + 4^3 + ...... + (2m-1)^3 + (2m)^3\}$
$- \{2^3 + 4^3 + ....+(2m)^3\} + 3\{2^3 + 4^3 + 6^3 + ..... + (2m)^3\}$

$$= \left[\frac{2m(2m+1)}{2}\right]^2 + 8 \times 2\{1^3 + 2^3 + 3^3 + ... + m^3\}$$

$$= m^2(2m+1)^2 + 16.\frac{m^2(m+1)^2}{4}$$

$$= \frac{n^2(2n^2+6n+5)}{4} \quad [\text{Put } m = \frac{n}{2}]$$

**2.** **(b)** $1, \log_9(3^{1-x}+2), \log_3(4.3^x - 1)$ are in A.P.

$\Rightarrow 2\log_9(3^{1-x}+2) = 1 + \log_3(4.3^x - 1)$

$\Rightarrow \log_3(3^{1-x}+2) = \log_3 3 + \log_3(4.3^x - 1)$

$\Rightarrow \log_3(3^{1-x}+2) = \log_3[3(4.3^x - 1)]$

$\Rightarrow 3^{1-x} + 2 = 3(4.3^x - 1)$

Put $3^x = t \Rightarrow \frac{3}{t} + 2 = 12t - 3$ or $12t^2 - 5t - 3 = 0$;

Hence $t = -\frac{1}{3}, \frac{3}{4}$

$\Rightarrow 3^x = \frac{3}{4}$ (as $3^x \neq -ve$)

$\Rightarrow x = \log_3\left(\frac{3}{4}\right)$ or $x = \log_3 3 - \log_3 4$

$\Rightarrow x = 1 - \log_3 4$

**3.** **(c)** Given that for an A.P, $S_n = cn^2$

Then, $T_n = S_n - S_{n-1} = cn^2 - c(n-1)^2$

$= (2n-1)c$

$\therefore$ Sum of squares of $n$ terms of this A.P

$$= \sum T_n^2 = \sum (2n-1)^2.c^2$$

$$= c^2\left[4\sum n^2 - 4\sum n + n\right]$$

$$= c^2\left[\frac{4n(n+1)(2n+1)}{6} - \frac{4n(n+1)}{2} + n\right]$$

$$= c^2 n\left[\frac{2(2n^2+3n+1) - 6(n+1) + 3}{3}\right]$$

$$= c^2 n\left[\frac{4n^2-1}{3}\right] = \frac{n(4n^2-1)c^2}{3}$$

**4.** **(c)** $d = a_2 - a_1 = a_3 - a_2 = ... = a_n - a_{n-1}$

$\therefore \sin d\,[\sec a_1 \sec a_2 + \sec a_2 \sec a_3 + ... + \sec a_{n-1} \sec a_n]$

$$= \frac{\sin(a_2 - a_1)}{\cos a_1 \cos a_2} + \frac{\sin(a_3 - a_2)}{\cos a_2 \cos a_3} + ... + \frac{\sin(a_n - a_{n-1})}{\cos a_n \cos a_{n-1}}$$

$$= \frac{\sin a_2 \cos a_1 - \cos a_2 \sin a_1}{\cos a_1 \cos a_2} + \frac{\sin a_3 \cos a_2 - \cos a_3 \sin a_2}{\cos a_2 \cos a_3} + ...$$

$= \tan a_2 - \tan a_1 + \tan a_3 - \tan a_2 + ... + \tan a_n - \tan a_{n-1} = \tan a_n - \tan a_1$

**5.** **(b)** $y - x = 3\left(\frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{6^2} + ...\right)$

$$x - z = \left(\frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{6^2} + ...\right)$$

$\therefore (y-x) = 3(x-z) \Rightarrow 4x = y + 3z$

$\Rightarrow 2\frac{x}{3} = \frac{y}{6} + \frac{z}{2}$

**6.** **(b)** Given $\frac{a+b}{1-ab}, b, \frac{b+c}{1-bc}$ are in A.P.

$$\Rightarrow b - \frac{a+b}{1-ab} = \frac{b+c}{1-bc} - b$$

$\Rightarrow \dfrac{-a(b^2+1)}{1-ab}=\dfrac{c(b^2+1)}{1-bc}$

$\Rightarrow \quad a+c=2abc$

Now, given quadratic equation is

$2ac\,x^2+2abc\,x+2abc=0$

(Substituting $a+c$

$=2abc$ and then cancelling $2ac$)

$\Rightarrow \quad x^2+bx+b=0$

**7.** **(d)** Given mid terms $t_n=1$ and $t_{n+1}=7$

$\therefore \quad t_n+t_{n+1}=8=t_1+t_{2n}$

and $t_{n+1}-t_n=6=d$ (common difference of A.P.)

$t_n+t_{n+1}=8$

$\therefore \quad a+(n-1)d+a+nd=8 \quad \therefore a+6n=7$

Now $4t_1t_{2n}=[(t_1+t_{2n})^2-(t_{2n}-t_1)^2]$

$=64-36(2n-1)^2$ [as $t_{2n}-t_1=(2n-1)\times 6$]

$\therefore\ t_1t_{2n}=16-9(2n-1)^2$

$\therefore\ 16-9(2n-1)^2+713\geq 0$

$\therefore -4\leq n\leq 5 \ \therefore\ n=5$

Hence, from $a+6n=7, a=-23$

**8.** **(d)** Given, $b^2=ac,\ x=\dfrac{a+b}{2},\ y=\dfrac{b+c}{2}$

Now, $\dfrac{a}{x}+\dfrac{c}{y}=\dfrac{2a}{a+b}+\dfrac{2c}{b+c}$

$=\dfrac{2(ab+ac+ac+bc)}{ab+ac+b^2+bc}=2 \qquad [\because b^2=ac]$

Again, $\dfrac{b}{x}+\dfrac{b}{y}=2b\left[\dfrac{1}{a+b}+\dfrac{1}{b+c}\right]$

$=\dfrac{2b(b+c+a+b)}{ab+ac+b^2+bc}=2$

$\therefore \left(\dfrac{a}{x}+\dfrac{c}{y}\right)\left(\dfrac{b}{x}+\dfrac{b}{y}\right)=4.$

**9.** **(d)** Given that $a, b, c$ are in A.P.

$\Rightarrow \quad 2b=a+c$

but given $a+b+c=3/2 \ \Rightarrow \ 3b=3/2$

$\Rightarrow \quad b=1/2$ and then $a+c=1$

Again $a^2, b^2, c^2$, are in G.P. $\Rightarrow\ b^4=a^2c^2$

$\Rightarrow \quad b^2=\pm\, ac \Rightarrow ac=\dfrac{1}{4}\ or\ -\dfrac{1}{4}$

and $\ a+c=1$ ....(i)

Considering $a+c=1$ and $ac=1/4$

$\Rightarrow \quad (a-c)^2=1-1=0 \ \Rightarrow\ a=c$ but

$a\neq c$ as given that $a<b<c$

$\therefore \quad$ We consider $a+c=1$ and $ac=-1/4$

$\Rightarrow \quad (a-c)^2=1+1=2 \Rightarrow a-c=\pm\sqrt{2}$

but $a<c \Rightarrow a-c=-\sqrt{2}$ ....(ii)

Solving (i) and (ii) we get $a=\dfrac{1}{2}-\dfrac{1}{\sqrt{2}}$

**10.** **(c)** $\dfrac{x}{1-r}=5 \Rightarrow r=1-\dfrac{x}{5}$

Since G.P. contains infinite terms

$\therefore \quad -1<r<1$

$\Rightarrow \quad -1<1-\dfrac{x}{5}<1 \Rightarrow -2<-\dfrac{x}{5}<0$

$\Rightarrow \quad -10<x<0. \Rightarrow 0<\dfrac{x}{5}<2 \Rightarrow 0<x<10.$

**11.** **(c)** In the quadratic equation $ax^2+bx+c=0$

$\Delta=b^2-4ac$ and $\alpha+\beta=-\dfrac{b}{a}, \alpha\beta=\dfrac{c}{a}$

$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$

$=\dfrac{b^2}{a^2}-\dfrac{2c}{a}=\dfrac{b^2-2ac}{a^2}$

and $\quad \alpha^3+\beta^3=-\dfrac{b^3}{a^3}-\dfrac{3c}{a}\left(-\dfrac{b}{a}\right)$

$=-\left(\dfrac{b^3-3abc}{a^3}\right)$

ATQ $\ \alpha+\beta, \alpha^2+\beta^2,\ \alpha^3+\beta^3$ are in G.P.

$\Rightarrow \quad -\dfrac{b}{a}, \dfrac{b^2-2ac}{a^2}, -\dfrac{(b^3-3\,abc)}{a^3}$ are in G.P.

$\Rightarrow \quad \left(\dfrac{b^2-2\,ac}{a^2}\right)^2=\dfrac{b}{a}\left(\dfrac{b^3-3\,abc}{a^3}\right)$

$\Rightarrow \quad b^4+4a^2c^2-4ab^2c=b^4-3ab^2c$

$\Rightarrow \quad 4a^2c^2-ab^2c=0 \Rightarrow ac\,\Delta=0$

$\Rightarrow \quad c\,\Delta=0 \quad (\because$ In quadratic $a\neq 0)$

**12.** **(c)** Let first term = a, common ratio = r, where $-1<r<1$

Then, $\dfrac{a}{1-r}=2$ and $\dfrac{a^3}{1-r^3}=24$

$\therefore \quad \dfrac{1-r^3}{(1-r)^3} = \dfrac{1}{3}$

i.e., $1-2r+r^2 = 3(1+r+r^2)$ or $2r^2+5r+2=0$

$\therefore \quad r = -2$ or $\dfrac{-1}{2}$

As $-1 < r < 1$ $\therefore$ we have $r = -\dfrac{1}{2}$

$\therefore$ The series is $3-\dfrac{3}{2}+\dfrac{3}{4}-\dfrac{3}{8}+...$

**13.** **(b)** Given $b^2 = ac$ ($\because$ $a, b, c$ are in G. P.) and $2(\log 2b - \log 3c) = \log a - \log 2b + \log 3c - \log a$

[$\because$ given terms are in A. P.]

$\Rightarrow \quad \log\left(\dfrac{2b}{3c}\right)^2 = \log\left(\dfrac{3c}{2b}\right) \Rightarrow b = \dfrac{3c}{2}$

Now, $a = \dfrac{b^2}{c} = \dfrac{3b}{2} = \dfrac{9c}{4}$

$\therefore$ $a$ is the largest side.

Now, $\cos A = \dfrac{b^2+c^2-a^2}{2bc}$

$= \dfrac{\dfrac{9c^2}{4}+c^2-\dfrac{81}{16}c^2}{2\times\dfrac{3c}{2}\times c}$ = negative

$\therefore$ $A > 90°$ $\therefore$ triangle is obtuse.

**14.** **(c)** $y = 2\sqrt{x}$, being in the first quadrant. The sequence of $x$-coordinate 1, 2, 4, 8, ................

$\therefore$ the sequence of $y$-coordinate

$2, 2\sqrt{2}, 2\sqrt{4}, 2\sqrt{8}$, ................ is a G.P. where the common ratio is $\sqrt{2}$.

$y_n = 2(\sqrt{2})^{n-1} = (\sqrt{2})^{n+1}$

**15.** **(c)** Let sides of triangle be $a$, $ar$, $ar^2$

Since $r > 1$ $\therefore$ $ar^2$ is greatest side

$\therefore$ $a + ar > ar^2 \Rightarrow r^2 - r - 1 < 0$

$\dfrac{1-\sqrt{5}}{2} < r < \dfrac{1+\sqrt{5}}{2} \Rightarrow 1 < r < \dfrac{1+\sqrt{5}}{2}$

($\because$ $r > 1$)

$\therefore \quad [r] = 1$. Also $-\dfrac{1+\sqrt{5}}{2} < -r < -1$

$\therefore$ $[-r] = -2$

$[r] + [-r] = 1 - 2 = -1.$

**16.** **(c)** We have $2b = a + c$ and $a, p, b, q, c$ are in A.P.

$\Rightarrow \quad p = \dfrac{a+b}{2}, q = \dfrac{b+c}{2}$

Again, $p' = \sqrt{ab}$ and $q' = \sqrt{bc}$

$\therefore \quad p^2 - q^2 = \dfrac{(a+b)^2-(b+c)^2}{4}$

$= \dfrac{(a-c)(a+c+2b)}{4} = (a-c)b = p'^2 - q'^2$

**17.** **(d)** Let $S = 1 + 2.2 + 3.2^2 + 4.2^3 + 5.2^4 + .... + 100.2^{99}$

$\therefore \quad 2S = 1.2 + 2.2^2 + 3.2^3 + ... + 99.2^{99} + 100.2^{100}$

Subtracting, we get

$-S = 1 + 1.2 + 1.2^2 + .... + 1.2^{99} - 100.2^{100}$

$= (1 + 2 + 2^2 + ... + 2^{99}) - 100.2^{100}$

$= \dfrac{1(2^{100}-1)}{2-1} - 100.2^{100} = 2^{100} - 1 - 100.2^{100}$

$\therefore \quad S = 100.2^{100} - 2^{100} + 1 = 99.2^{100} + 1$

**18.** **(a)** Consider two real numbers $(a + b)$ and $(c + d)$. Using G. M. $\le$ A. M.

We get $\sqrt{(a+b)(c+d)} \le \dfrac{(a+b)+(c+d)}{2}$

$\Rightarrow \sqrt{M} \le \dfrac{2}{2} \Rightarrow M \le 1$

Also $a, b, c, d$ are positive. Therefore

$M = (a + b)(c + d) > 0$. $\therefore$ $0 < M \le 1$

**19.** **(a)** $t_n = \dfrac{n+2}{n(n+1)}\cdot\left(\dfrac{1}{2}\right)^n$

$= \dfrac{2(n+1)-n}{n(n+1)}\cdot\left(\dfrac{1}{2}\right)^n$

$$= \frac{1}{n}\left(\frac{1}{2}\right)^{n-1} - \frac{1}{n+1}\cdot\left(\frac{1}{2}\right)^{n}.$$

$$S_n = \sum_{n=1}^{n} t_n = \left\{\frac{1}{1}\left(\frac{1}{2}\right)^{\circ} - \frac{1}{2}\left(\frac{1}{2}\right)^{1}\right\}$$

$$+\left\{\frac{1}{2}\left(\frac{1}{2}\right)^{1} - \frac{1}{3}\left(\frac{1}{2}\right)^{2}\right\} + \ldots\ldots\ldots$$

$$+\left\{\frac{1}{n}\left(\frac{1}{2}\right)^{n-1} - \frac{1}{n+1}\left(\frac{1}{2}\right)^{n}\right\} = 1 - \frac{1}{(n+1)2^n}$$

**20. (a)** As A.M. $\geq$ G.M., we get

$$\frac{1}{3}\left(ax^2 + \frac{b}{2x} + \frac{b}{2x}\right) \geq \left(ax^2\cdot\frac{b}{2x}\cdot\frac{b}{2x}\right)^{1/3}$$

$$\Rightarrow \quad \frac{1}{3}\left(ax^2 + \frac{b}{x}\right) \geq \left(\frac{ab}{4}\right)^{1/3}$$

But the least value of $ax^2 + \frac{b}{x}$ is $c$, therefore

$$3\left(\frac{ab^2}{4}\right)^{1/3} \geq c \Rightarrow 27ab^2 \geq 4c^3$$

**21. (0.60)** Let $a$ = first term of G.P. and $r$ = common ratio of G.P.; Then G.P. is $a, ar, ar^2$

Given $S_\infty = 20 \Rightarrow \frac{a}{1-r} = 20$

$\Rightarrow a = 20(1-r)$ ... (i)

Also $a^2 + a^2r^2 + a^2r^4 + \ldots$ to $\infty = 100$

$\Rightarrow \frac{a^2}{1-r^2} = 100$

$\Rightarrow a^2 = 100(1-r)(1+r)$ ... (ii)

From (i), $a^2 = 400(1-r)^2$;

From (ii), we get $100(1-r)(1+r) = 400(1-r)^2$

$\Rightarrow 1 + r = 4 - 4r \Rightarrow 5r = 3 \Rightarrow r = 3/5 = 0.60$

**22. (124)** Let $b = ar$ and $c = ar^2$. Given that $a, ar, ar^2 - 4$ are in AP. Therefore

$a + (ar^2 - 64) = 2(ar)$ ...(i)

$\Rightarrow a(r^2 - 2r + 1) = 64$

Again $a, ar - 8, ar^2 - 64$ are in GP. Therefore

$a(ar^2 - 64) = (ar - 8)^2$ ...(ii)

$\Rightarrow a(16r - 64) = 54$

From Eqs. (i) and (ii), we get

$r^2 - 2r + 1 = 16r - 64 \Rightarrow r^2 - 18r + 65 = 0$

$(r-5)(r-13) = 0$

$r = 5$ or $13$

If $r = 5$, then $a(80 - 64) = 64$ and hence $a = 4$. In this case the numbers are 4,20,100 and their sum is 124.

**23. (189)** $\log_6(abc) = 6 \Rightarrow (abc) = 6^6$

Let $a = \frac{b}{r}$ and $c = br \Rightarrow b = 36$ and $a = \frac{36}{r}$

$\Rightarrow r = 2, 3, 4, 6, 9, 12, 18$

Also $36\left(1 - \frac{1}{r}\right)$ is a perfect cube. $\Rightarrow r = 4$

$\Rightarrow a + b + c = 9 + 36 + 144 = 189.$

**24. (3)** We have

$$S = 1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \frac{14}{3^4} + \ldots\ldots.\infty \qquad \ldots.(i)$$

Multiplying both sides by $\frac{1}{3}$, we get

$$\frac{1}{3}S = \frac{1}{3} + \frac{2}{3^2} + \frac{6}{3^3} + \frac{10}{3^4} + \ldots\ldots..\infty \qquad \ldots.(ii)$$

Subtracting eqn. (ii) from eqn. (i), we get

$$\frac{2}{3}S = 1 + \frac{1}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \frac{4}{3^4} + \ldots\ldots..\infty$$

$$\Rightarrow \frac{2}{3}S = \frac{4}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \frac{4}{3^4} + \ldots\ldots..\infty$$

$$\Rightarrow \frac{2}{3}S = \frac{\frac{4}{3}}{1 - \frac{1}{3}} = \frac{4}{3}\times\frac{3}{2} \Rightarrow S = 3$$

**25. (990)** Let $S_1 = 2 + 3 + 5 + 9 + 16 + \ldots\ldots\ldots\ldots + x_n$

$$S_1 = \quad 2 + 3 + 5 + 9 + \ldots\ldots\ldots x_{n-1} + x_n$$

$$0 = 2 + [1 + 2 + 4 + 7 + \ldots\ldots$$
$$+ to\ (n-1)\ term] - x_n$$

$\therefore x_n = 2 + [1 + 2 + 4 + 7 + \ldots..$to $(n-1)$ terms]

Again let

$S_2 = 1 + 2 + 4 + 7 + \ldots\ldots.. + t_{n-1}$

$$S_2 = \quad 1 + 2 + 4 + 7\ldots.. + t_{n-2} + t_{n-1}$$

$$0 = 1 + [1 + 2 + 3 + \ldots\ldots. + (n-2)\text{ term}] - t_{n-1}$$

$$t_{n-1} = 1 + \frac{(n-2)(n-1)}{2} = \frac{n^2 - 3n + 4}{2}$$

$$\therefore S_2 = \sum_{n=1}^{n-1} t_{n-1} = \frac{1}{2}\Sigma n^2 - \frac{3}{2}\Sigma n + 2\Sigma 1$$

$$= \frac{1}{2}\frac{(n-1)\,n(2n-1)}{6} - \frac{3}{2}\frac{n(n-1)}{2} + 2(n-1)$$

$$= (n-1)\left[\frac{2n^2 - n}{12} - \frac{3n}{4} + 2\right]$$

$$= \frac{n-1}{12}\left[2n^2 - n - 9n + 24\right]$$

$$= \frac{n-1}{6}\left[n^2 - 5n + 12\right] = \frac{n^3 - 6n^2 + 17n - 12}{6}$$

$$\therefore x_n = 2 + S_2 = 2 + \frac{n^3 - 6n^2 + 17n - 12}{6}$$

$$= \frac{n^3 - 6n^2 + 17n}{6}$$

So, $x_{20} = 990$

**26.** **(5)** Let $k$ and $k+1$ be removed. Then,

$$\frac{\frac{n}{2}(n+1) - 2k - 1}{n-2} = \frac{105}{4}$$

$\Rightarrow 2n^2 - 103n - 8k + 206 = 0$

Since $n$ and $k$ are integers, so n must be even, say $n = 2m$, then $k = \frac{4m^2 + 103(1-m)}{4}$. Clearly $(1-m)$ must be divisible by 4.

Let $m = 1 + 4t$, then we get $k = 16t^2 - 95t + 1$ and $1 \le k < n$

$\Rightarrow 1 \le 16t^2 - 95t + 1 < 8t + 2$

$\Rightarrow t = 6$ and so, $n = 50$

**27.** **(8)** Since a, b, c, d are in A.P., we have

$b - a = c - b = d - c = D$ (let common difference)

or $d = a + 3D$

$\Rightarrow$ $a - d = -3d$ and $d = b + 2D$

or $b - d = -2D$

Also $c = a + 2D$ or $c - a = 2D$

$\therefore$ Given equation $2(a - b) + k(b - c)^2 + (c - a)^3 = 2(a - d) + (b - d)^2 + (c - d)^3$

becomes $-2D + kD^2 + (2D)^3 = -6D + 4D^2 - D^3$

$\Rightarrow$ $9D^2 + (k-4)D + 4 = 0$

Since $D$ is real, we have $(k-4)^2 - 4(4)(9) \ge 0$

or $k^2 - 8k - 128 \ge 0$ or $(k-16)(k+8) \ge 0$

$\therefore$ $k \in (-\infty, -8] \cup [16, \infty)$

Hence, the smallest positive value of $k = 16$.

**28.** **(1)** $x \in (0, \pi)$, implies $|\cos x| < 1$. Thus,

$S_\infty = 1 + |\cos x| + \cos^2 x + |\cos^3 x| + \ldots$ up to $\infty$

$= 1 + |\cos x| + |\cos x|^2 + |\cos x|^3 + \ldots$ up to $\infty$

$$= \frac{1}{1 - |\cos x|}$$

$$\therefore\ 3^{3/(1 - |\cos x|)} = 3^6 \Rightarrow \frac{3}{1 - |\cos x|} = 6$$

$$\Rightarrow\ |\cos x| = \frac{1}{2} \Rightarrow \cos x = \pm\frac{1}{2}$$

$$\Rightarrow\ x = \frac{\pi}{3}, \frac{2\pi}{3} \Rightarrow \frac{1}{\pi}(x_1 + x_2) = 1.$$

**29.** **(6)** For the given A.P., we have

$2(2a + b) = (5a - b) + (a + 2b)$

$\Rightarrow b = 2a$ ...(1)

Also for the given G.P., we have $(ab + 1)^2 = (a - 1)^2 (b + 1)^2$ ...(2)

Putting $b = 2a$ from (1) in (2), we get $a = 0, -2$, or $\frac{1}{4}$

But $a > 0$, so $a = \frac{1}{4}$ and $b = 2a = \frac{1}{2}$

Hence, $(a^{-1} + b^{-1}) = 2 + 4 = 6$.

**30.** **(2)** Let the two quantities be a and b

$$G = \sqrt{ab}\ ;\ p = a + \frac{b-a}{3}\ ;\ q = a + \frac{2(b-a)}{3}$$

$$(kp - q) = ka + \frac{k(b-a)}{3} - a$$

$$-2\frac{(b-a)}{3} = a(k-1) + \frac{b-a}{3}(k-2)$$

$$kq - p = ak + \frac{2k(b-a)}{3} - a - \frac{(b-a)}{3}$$

$$\Rightarrow a(k-1) + \frac{b-a}{3}(2k-1)$$

$\therefore (kp - q)(kq - p) = ab$ $\qquad \therefore k = 2$

CHAPTER 10

# Straight Lines

**1.** **(a)** Given $P \equiv (\alpha, \beta)$

Given line is $\frac{x}{a}+\frac{y}{b}=1$ ...(i)

If line (i) cuts $x$ and $y$ axes at $A$ and $B$ respectively, then $A=(a, 0)$ and $B=(0,b)$.

Also the area of $\Delta OAB = S$ i.e. $\frac{1}{2}\ ab = S$

$\Rightarrow ab = 2S$

Since line (i) passes through P($\alpha$, $\beta$)

$\frac{\alpha}{a}+\frac{\beta}{b}=1 \Rightarrow \frac{\alpha}{a}+\frac{a\beta}{2S}=1$

$\Rightarrow a^2\beta-(2S)a+2\alpha S=0$

Since $a$ is real, $4S^2-8\alpha\beta S \ge 0 \Rightarrow S \ge 2\alpha\beta$

Hence the least value of $S = 2\alpha\beta$.

**2.** **(b)** $a$, $b$, $c$ are the roots of equation

$x^3-3x^2+6x+1=0.$

So, $a+b+c=3$, $ab+bc+ca=6$ and $abc=-1$

Now, the centroid of the triangle is

$$\left(\frac{ab+bc+ca}{3}, \frac{\frac{1}{ab}+\frac{1}{bc}+\frac{1}{ca}}{3}\right)$$

i.e. $\left(\frac{ab+bc+ca}{3}, \frac{a+b+c}{3abc}\right)$

$=\left(\frac{6}{3}, \frac{3}{-3}\right)$ or $(2,-1)$

**3.** **(c)** Consider a point $A'$, the image of $A$ in $y=x$

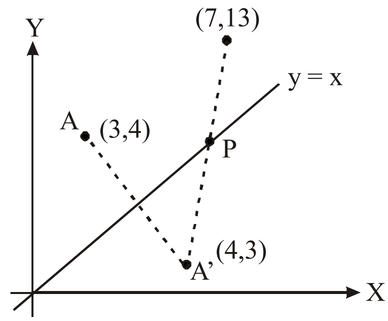
$\therefore$ Coordinates of $A'=(4, 3)$

[Notice that $A$ and $B$ lie to the same side with respect to $y=x$].

Then $PA=PA'$. Thus, $PA+PB$ is minimum, if $PA'+PB$ is minimum, i.e. if $P, A', B$ are collinear.

Now, $AB$ is $y-3=\frac{13-3}{7-4}(x-4)$

$\Rightarrow 3y-10x+31=0.$



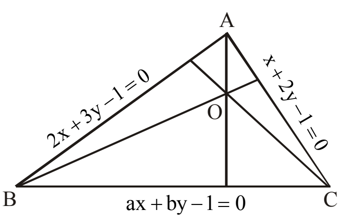

It intersects $y=x$ at $\left(\frac{31}{7}, \frac{31}{7}\right)$, which is the required point $P$.

**4.** **(c)** Equation of $AO$ is $2x+3y-1+\lambda(x+2y-1)=0$

Where $\lambda=-1$, since the line passes through the origin. So, $x+y=0$

Since $AO$ is perpendicular to $BC$, $(-1)\left(-\frac{a}{b}\right)=-1$

$\therefore a=-b$




Similarly, $(2x+3y-1)+\mu(ax-ay-1)=0$ will be the equation of $BO$ for $\mu=-1$.

$BO$ is perpendicular to $AC$

$\Rightarrow \left\{-\frac{(2-a)}{3+a}\right\}\left(-\frac{1}{2}\right)=-1. \therefore a=-8, b=8.$

**5.** **(d)** The coordinates of $A$ are $(a, 0)$ and of $B$ are $(0, b)$. If the coordinates of $C$ are $(x_1, x_1)$ then area of the

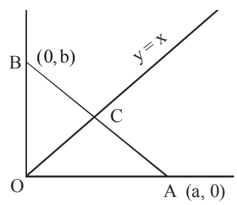
$\Delta AOC = \frac{1}{2} OA \times x_1$ and the area of the

$\Delta BOC = \frac{1}{2} OB \times x_1$.

According to the given condition,

$\frac{1}{2} OA \times x_1 = 2 \times \frac{1}{2} OB \times x_1 \Rightarrow OA = 2OB$

$\Rightarrow a = 2b.$




Equation of $AB$ is therefore $\frac{x}{2b} + \frac{y}{b} = 1$

or $x + 2y = 2b$

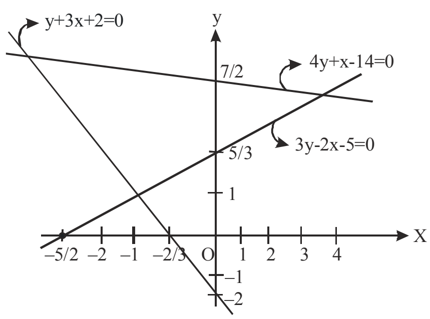
Since $C$ lies on it, $x_1 + 2x_1 = 2b$

$\Rightarrow x_1 = \frac{2b}{3} = \frac{a}{3}$

Hence the coordinates of $C$ are

$\left(\frac{2b}{3}, \frac{2b}{3}\right)$ or $\left(\frac{a}{3}, \frac{a}{3}\right)$.

**6.** **(c)**




From diagram it is clear that $\frac{5}{3} \le \beta \le \frac{7}{2}$.

**7.** **(c)** $y = 2$ intersects $y = mx$ at $P\left(\frac{2}{m}, 2\right)$

$y = 6$ intersects $y = mx$ at $Q\left(\frac{6}{m}, 6\right)$

Thus $PQ = \sqrt{\left(\frac{6}{m} - \frac{2}{m}\right)^2 + (6-2)^2} < 5$

$\Rightarrow \left(\frac{4}{m}\right)^2 + 16 < 25$

$\Rightarrow \left(\frac{4}{m}\right)^2 < 9 \Rightarrow -3 < \frac{4}{m} < 3 \Rightarrow -\frac{3}{4} < \frac{1}{m} < \frac{3}{4}$

$\therefore \quad m \in \left(-\infty, -\frac{4}{3}\right) \cup \left(\frac{4}{3}, \infty\right)$

**8.** **(a)** Let $P(1, 0)$ and $Q(-1, 0)$, $A(x, y)$

Given: $\frac{AP}{AQ} = \frac{BP}{BQ} = \frac{CP}{CQ} = \frac{1}{2}$

$\Rightarrow \quad 2AP = AQ \Rightarrow 4(AP)^2 = AQ^2$

$\Rightarrow \quad 4[(x-1)^2 + y^2] = (x+1)^2 + y^2$

$\Rightarrow \quad 4(x^2 + 1 - 2x) + 4y^2 = x^2 + 1 + 2x + y^2$

$\Rightarrow \quad 3x^2 + 3y^2 - 8x - 2x + 4 - 1 = 0$

$\Rightarrow \quad 3x^2 + 3y^2 - 10x + 3 = 0$

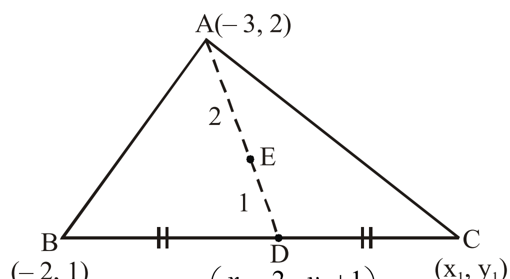
$\Rightarrow \quad x^2 + y^2 - \frac{10}{3}x + 1 = 0 \qquad ...(1)$

$\therefore$ A lies on the circle given by (1). As $B$ and $C$ also follow the same condition.

$\therefore$ Centre of circumcircle of $\Delta ABC$ = centre of circle given by (1)

$= \left(\frac{5}{3}, 0\right)$.

**9.** **(b)** Let $C = (x_1, y_1)$




$D\left(\frac{x_1 - 2}{2}, \frac{y_1 + 1}{2}\right)$

Centroid, $E = \left(\frac{x_1 - 5}{3}, \frac{y_1 + 3}{3}\right)$

Since centroid lies on the line

$3x+4y+2=0$

$\therefore \quad 3\left(\frac{x_1-5}{3}\right)+4\left(\frac{y_1+3}{3}\right)+2=0$

$\Rightarrow \quad 3x_1+4y_1+3=0$

Hence vertex $(x_1, y_1)$ lies on the line $3x+4y+3=0$

**10.** **(d)** Circumcentre = (0, 0)

Centroid = $\left(\frac{(a+1)^2}{2}, \frac{(a-1)^2}{2}\right)$

We know the circumcentre (O), Centroid (G) and orthocentre (H) of a triangle lie on the line joining the O and G.

Also, $\frac{HG}{GO}=\frac{2}{1} \Rightarrow$ Coordinate of orthocentre

$=\frac{3(a+1)^2}{2}, \frac{3(a-1)^2}{2}$

Now, these coordinates satisfies eqn given in option (d)

Hence, required eqn of line is

$(a-1)^2 x-(a+1)^2 y=0$

**11.** **(a)** The line passing through the intersection of lines $ax+2by+3b=0$ and $bx-2ay-3a=0$ is

$ax+2by+3b+\lambda(bx-2ay-3a)=0$

$\Rightarrow \quad (a+b\lambda)x+(2b-2a\lambda)y+3b-3\lambda a=0$

As this line is parallel to $x$-axis.

$\therefore \; a+b\lambda=0 \Rightarrow \lambda=-a/b$

$\Rightarrow \quad ax+2by+3b-\frac{a}{b}(bx-2ay-3a)=0$

$\Rightarrow \quad ax+2by+3b-ax+\frac{2a^2}{b}y+\frac{3a^2}{b}=0$

$y\left(2b+\frac{2a^2}{b}\right)+3b+\frac{3a^2}{b}=0$

$y\left(\frac{2b^2+2a^2}{b}\right)=-\left(\frac{3b^2+3a^2}{b}\right)$

$y=\frac{-3(a^2+b^2)}{2(b^2+a^2)}=\frac{-3}{2}$

So it is 3/2 units below $x$-axis.

**12.** **(d)** Slope of the line in the new position is $\frac{b}{a}$, since it is $\perp$ to the line $ax+by+c=0$ and it cuts the x-axis at (2,0). Hence, the required line passes through (2, 0) and its slope is $\frac{b}{a}$. Required eq. is

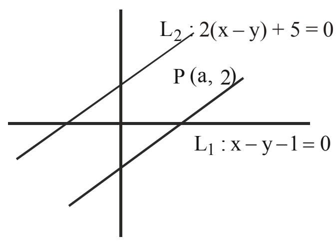
$y-0=\frac{b}{a}(x-2)$

$\Rightarrow ay=bx-2b \Rightarrow ay-bx+2b=0$

**13.** **(d)** Let $L_1(x,y)=x-y-1$ and $L_2(x,y)=2(x-y)+5$

Then, from figure $P(a,2)$ lies below $L_2$,

So $\frac{2a-4+5}{-2}<0 \Rightarrow a>-\frac{1}{2}$



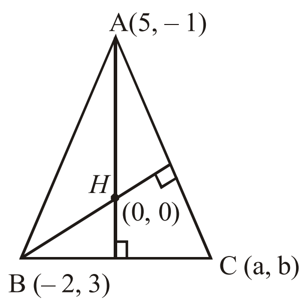

Also $P(a,2)$ lies above $L_1$,

So $\frac{a-2-1}{-1}>0 \Rightarrow a<3$

Taking intersection, we get $-\frac{1}{2}<a<3$

$\Rightarrow a \in \left(-\frac{1}{2}, 3\right)$.

**14.** **(b)**



Let the third vertex of $\Delta ABC$ be $(a, b)$.
Orthocentre $= H(0, 0)$
Let $A\ (5, -1)$ and $B\ (-2, 3)$ be other two vertices of $\Delta ABC$.
Now, (Slope of $AH$) × (Slope of $BC$) = – 1

$$\Rightarrow \quad \left(\frac{-1-0}{5-0}\right)\left(\frac{b-3}{a+2}\right) = -1$$

$\Rightarrow \quad b - 3 = 5\,(a + 2)$ ...(1)

Similarly,
(Slope of $BH$) × (Slope of $AC$) = – 1

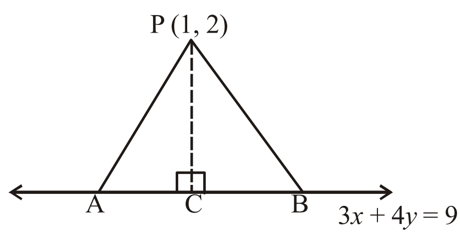
$$\Rightarrow \quad -\left(\frac{3}{2}\right) \times \left(\frac{b+1}{a-5}\right) = -1$$

$\Rightarrow \quad 3b + 3 = 2a - 10$

$\Rightarrow \quad 3b - 2a + 13 = 0$ ...(2)

On solving equations (1) and (2) we get
$a = -4,\ b = -7$
Hence, third vertex is $(-4, -7)$.

**15.** **(b)**




Shortest distance of a point $(x_1, y_1)$ from line

$$ax + by = c \text{ is } d = \left|\frac{ax_1 + by_1 - c}{\sqrt{a^2 + b^2}}\right|$$

Now shortest distance of P (1, 2) from $3x + 4y = 9$ is

$$\text{PC} = d = \left|\frac{3(1) + 4(2) - 9}{\sqrt{3^2 + 4^2}}\right| = \frac{2}{5}$$

Given that $\Delta$APB is an equilateral triangle
Let '$a$' be its side

then PB $= a$, CB $= \dfrac{a}{2}$

Now, In $\Delta$PCB, $(\text{PB})^2 = (\text{PC})^2 + (\text{CB})^2$
(By Pythagoras theoresm)

$$\Rightarrow \quad a^2 = \left(\frac{2}{5}\right)^2 + \frac{a^2}{4}$$

$$\Rightarrow \quad a^2 - \frac{a^4}{4} = \frac{4}{25} \Rightarrow \frac{3a^2}{4} = \frac{4}{25}$$

$$a^2 = \frac{16}{75} \Rightarrow a = \sqrt{\frac{16}{75}} = \frac{4}{5\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{4\sqrt{3}}{15}$$

$\therefore$ Length of Equilateral triangle $(a) = \dfrac{4\sqrt{3}}{15}$

**16.** **(a)** Let, $L_1\,(x, y) = x + y + 1$

and $L_2\,(x, y) = 2x - 3y - 5$

$\therefore \quad L_1\ (10, -20) = 10 - 20 + 1 = -9 \Rightarrow$ '–'ve

and $L_2\ (10, -20) = 20 + 60 - 5 = 75 \Rightarrow$ '+'ve

$\therefore$ Equation of the bisector will be

$$\frac{x + y + 1}{\sqrt{2}} = -\left(\frac{2x - 3y - 5}{\sqrt{13}}\right)$$

$$\Rightarrow x\,(\sqrt{13} + 2\sqrt{2}) + y\,(\sqrt{13} - 3\sqrt{2}) + (\sqrt{13} - 5\sqrt{2}) = 0$$

**17.** **(c)** If a point is equidistant from the two intersecting lines, then the locus of this point is the angle bisector of those lines.
Now, let $(h, k)$ be the point which is equidistant from the lines $4x - 3y + 7 = 0$ and $3x - 4y + 14 = 0$

Then $\dfrac{4h - 3k + 7}{\sqrt{4^2 + (-3)^2}} = \pm \dfrac{3h - 4k + 14}{\sqrt{3^2 + (-4)^2}}$

$\Rightarrow 4h - 3k + 7 = \pm(3h - 4k + 14)$

$\Rightarrow \quad h + k - 7 = 0$ and $7h - 7k + 21 = 0$

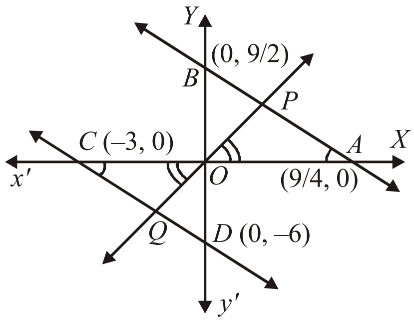
Hence locus of $(h, k)$ is $x + y - 7 = 0$
and $x - y + 3 = 0$

**18.** **(7)** The given lines are

$2x + y = 9/2$ ...(i)

and $2x + y = -6$ ...(ii)

Signs of constants on R.H.S. show that two lines lie on opp. sides of origin. Let any line through origin meets these lines in $P$ and $Q$ respectively then req. ratio is $OP : OQ$




Now in $\Delta OPA$ and $\Delta OQC$,

$\angle POA = \angle QOC$ (ver. opp. $\angle$'s)

$\angle PAO = \angle OCQ$ (alt. int. $\angle$'s)

$\therefore$ $\Delta OPA \sim \Delta OQC$ (by $AA$ similarity)

$\therefore$ $\dfrac{OP}{OQ} = \dfrac{OA}{OC} = \dfrac{9/4}{3} = \dfrac{3}{4}$

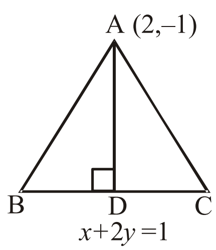
$\therefore$ Req. ratio is 3 : 4.

$$= OP : OQ = \left|\frac{9}{\sqrt{20}}\right| : \left|\frac{-6}{\sqrt{5}}\right| = 3:4 = \frac{m}{n}$$

$\Rightarrow$ $m + n = 7$.

**19.** **(15)** Let ABC be an equilateral triangle with base BC.

So, AD $\perp$ BC




$$\text{Now, length AD} = \left|\frac{2 + 2(-1) - 1}{\sqrt{(1)^2 + (2)^2}}\right|$$

$$= \left|\frac{2 - 2 - 1}{\sqrt{5}}\right| = \frac{1}{\sqrt{5}}$$

$$\left(\begin{array}{l}\because \text{ Perpendicular distance of} \\ \text{ax + by = c from } (x_1, y_1) \text{ is} \\ \quad p = \left|\dfrac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}\right|\end{array}\right)$$

Let $AB = BC = AC = x$

So, in $\Delta ABD$, $AD^2 + BD^2 = AB^2$

$$\Rightarrow \left(\frac{1}{\sqrt{5}}\right)^2 + \frac{x^2}{4} = x^2 \qquad \left(\because BD = \frac{BC}{2}\right)$$

$$\Rightarrow x^2 = \frac{4}{15} \Rightarrow x = \frac{2}{\sqrt{15}} = \frac{2}{\sqrt{K}}$$

$\Rightarrow$ K = 15

**20.** **(0)** Since $\sqrt{3}.1 - 4 + 1 < 0$,

so $\sqrt{3}\sin\theta - \cos\theta + 1 \le 0$

$$\Rightarrow \frac{\sqrt{3}}{2}\sin\theta - \frac{1}{2}\cos\theta \le -\frac{1}{2}$$

$$\Rightarrow \sin\left(\theta - \frac{\pi}{6}\right) \le -\frac{1}{2}$$

$$\Rightarrow \frac{7\pi}{6} \le \theta - \frac{\pi}{6} \le \frac{11\pi}{6} \Rightarrow \frac{4\pi}{3} \le \theta \le 2\pi$$

$\Rightarrow$ maximum value of $\sin\theta$ is 0.

**21.** **(4)** Family of lines can be written as

$$(3x + 7y + 11)\sec\theta + \operatorname{cosec}\theta(5x - 3y - 11) = 0$$

$$\Rightarrow (5x - 3y - 11) + \tan\theta(3x + 7y + 11) = 0$$

So, for any value of $\theta$ this line passes through the point of intersection of lines

$5x - 3y - 11 = 0$ and $3x + 7y + 11 = 0$ i.e from $P(1, -2)$.

Let $A(x, y)$ be a point on the line

$x - y + 3 = 0$

Applying cosines rule in $\Delta PAB$

$$PB^2 = PA^2 + AB^2 - 2PA.AB\cos\theta \geq PA^2 + AB^2 - 2PA.AB = (PA - AB)^2$$

$\Rightarrow \quad (PA - AB)^2 \leq PB^2$

So, maximum value of

$| PA - PB |$ is $PB = \sqrt{4+36} = 2\sqrt{10}.$

$\Rightarrow k^2 = 40$

**22.** **(1)** Let the equation of $L_2$ be $L_1 + \lambda L = 0$

$\Rightarrow \quad (1+\lambda)x + (2+\lambda)y + 3 + \lambda = 0$

Slopes of $L_2$, $L$ and $L_1$ are $-\dfrac{1+\lambda}{2+\lambda}, -1, -1/2$

Since $L$ is the bisector of the angle between $L_1$ and $L_2$

$$\therefore \quad \frac{-\dfrac{1+\lambda}{2+\lambda}+1}{1+\dfrac{1+\lambda}{2+\lambda}} = \frac{-1+1/2}{1+1/2} \Rightarrow \lambda = -3$$

So the equation of $L_2$ is $y + 2x = 0$

$\Rightarrow m = -2$ and $812m^2 + 3 = 812 \times 4 + 3 = 3251$

**23.** **(9)** $\tan\alpha, \tan\beta, \tan\lambda$ are the roots of equation $t^3 - 12t^2 + 15t - 1 = 0$ (given).

So, $\tan\alpha, \tan\beta, \tan\lambda = 12$ and $\tan\alpha, \tan\beta, \tan\lambda = 1$. Also $\Sigma\tan\alpha\tan\beta = 15$

When $A(\tan\alpha, \cot\alpha)$, $B(\tan\beta, \cot\beta)$ and $C(\tan\lambda, \cot\lambda)$

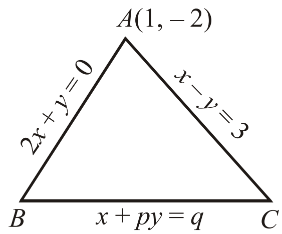
The centroid $G(h, k)$

$$= \left(\frac{\tan\alpha+\tan\beta+\tan\lambda}{3}, \frac{\cot\alpha+\cot\beta+\cot\lambda}{3}\right)$$

$$= \left(\frac{12}{3}, \frac{\Delta\tan\alpha.\tan\beta}{3(\tan\alpha\tan\beta\tan\lambda)}\right) = (4, 5)$$

$$\Rightarrow \quad \frac{k+h}{k-h} = \frac{5+4}{5-4} = 9$$

**24.** **(5)**




$\left(\dfrac{q}{2p \quad 1}, \dfrac{2q}{2p \quad 1}\right) \qquad \left(\dfrac{3p+q}{p+1}, \dfrac{q \quad 3}{p+1}\right)$

$P$ is the orthocenter. Therefore, $AP \perp BC$

or $\left(-\dfrac{1}{p}\right)\left(\dfrac{3+2}{2-1}\right) = -1$ or $\dfrac{5}{p} = 1$ or $p = 5$

Since $BP \perp AC$, we have $\dfrac{27-2q}{18+q} = -1$

or $\quad q = 27 + 18$ or

$q = 45$

$\therefore \quad p + q = 5 + 45 = 50$

**25.** **(4)** Since $y = x$ is the perpendicular bisector of the side $AB$ and $A = (1, 2)$, we have $B = (2, 1)$. Since the image $A'(x, y)$ of $A$ in the angular bisector $x - 2y + 1 = 0$ lies on the line $BC$, we have

$$\frac{x-1}{1} = \frac{y-2}{-2} = \frac{-2(1-2(2)+1)}{1^2+2^2} = \frac{4}{5}$$

Therefore, $A' = \left(\dfrac{9}{5}, \dfrac{2}{5}\right)$

Since equation of $BC$ is the equation of $BA'$, we have the equation of $BC$ as

$$y - 1 = \frac{1-(2/5)}{2-(9/5)}(x-2)$$

$\Rightarrow \quad y - 1 = 3(x - 2)$

$\Rightarrow 3x - y - 5 = 0$

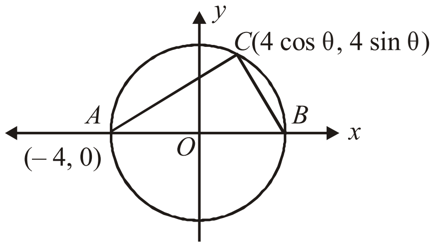
So that $a = 3$, $b = -1$. Hence, $a - b = 4$.

CHAPTER 11

# Conic Sections

**1.** **(a)** Since $4x + 3y - 4 = 0$ is dividing the circumference in the ratio 1 : 2, angle subtended at the centre $= 2\pi/3$.
Also the perpendicular distance from the centre of the given line is 5
$\Rightarrow$ Radius = 10 $\Rightarrow$ Equation of the circle is
$x^2 + y^2 - 10x - 6y - 66 = 0.$

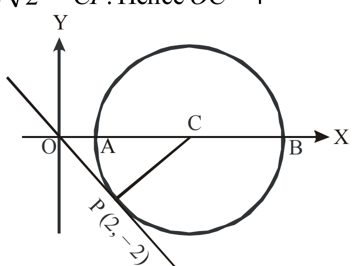
**2.** **(b)**




$A = \frac{1}{2} . 8 . 4 \sin\theta = |16 \sin \theta|$

Now, $\sin\theta$ can be equal to $\frac{1}{16}, \frac{2}{16}, ... \frac{15}{16}$
i.e. there are 15 points in each quadrant.

**3.** **(c)** If $(a, 0)$ is the centre $C$ and $P$ is $(2, -2)$, then $\angle COP = 45°$.
Since the equation of $OP$ is $x + y = 0$.
$\therefore$ $OP = 2\sqrt{2} = CP$. Hence $OC = 4$

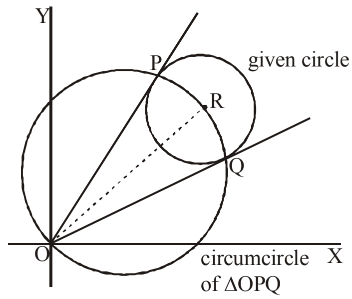



The point on the circle with the greatest $x$ coordinate is $B$.
$\alpha = OB = OC + CB = 4 + 2\sqrt{2}$ .

**4.** **(d)** Since $y = |x| + c$ and $x^2 + y^2 - 8|x| - 9 = 0$ both are symmetrical about $y$-axis we consider the case $x > 0$, when the equations become $y = x + c$ and $x^2 + y^2 - 8x - 9 = 0$. Equation of tangent to circle $x^2 + y^2 - 8x - 9 = 0$ parallel to $y = x + c$ is $y = (x - 4) + 5\sqrt{1+1}$

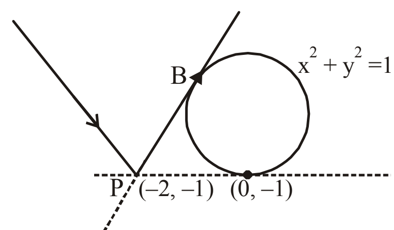
$\Rightarrow$ $y = x + (5\sqrt{2} - 4)$
For no solution $c > 5\sqrt{2} - 4$
$\therefore c \in (5\sqrt{2} - 4, \infty)$.

**5.** **(b)** Equation of the given circle is
$$x^2 + y^2 + 2gx + 2fy + c = 0 \qquad ......(i)$$




Equation of the chord of contact $PQ$, drawn from the origin (0, 0) to the given circle will be
$$gx + fy + c = 0 \qquad .....(ii)$$
Eq. of any circle passing through the intersection points of the given circle and the chord $PQ$ can be written as
$$(x^2 + y^2 + 2gx + 2fy + c) + \lambda (gx + fy + c) = 0 \qquad ....(iii)$$
If this circle passes through the origin, then we have,
$$c + \lambda c = 0 \text{ gives } \lambda = -1$$
Putting the above value of $\lambda$ in equation (iii) gives the equation of the required circle as
$$x^2 + y^2 + gx + fy = 0$$

**6.** **(b)**




Any line through $(-2, -1)$ is $y + 1 = m(x + 2)$

It touches the circle if $\left|\frac{2m - 1}{\sqrt{1 + m^2}}\right| = 1$

$\Rightarrow$ $m = 0, \frac{4}{3}$

$\therefore$ Equation of PB is $y+1=\frac{4}{3}(x+2)$

$\Rightarrow 4x-3y+5=0$

A point on PB is (– 5, – 5), (we can choose some other point as well)

Its image by the line $y = -1$ is P′ ( – 5 , 3).

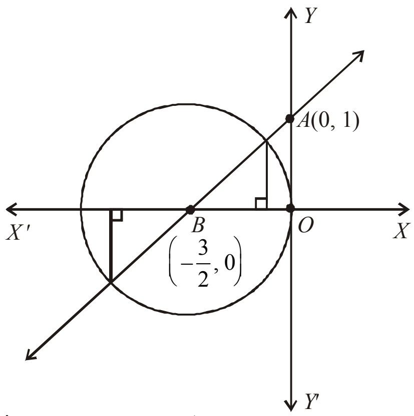
Hence equation of incident ray PP′ is

$y-3=\frac{3+1}{-5+2}(x+5) \Rightarrow \quad 4x+3y+11=0$

**7.** **(b)** Circle : $x^2+y^2+3x=0$,

Centre, $B=\left(-\frac{3}{2},0\right)$

Radius $=\frac{3}{2}$ units.




Line : $y=mx+1$

y-intercept of the line = 1

$\therefore$ A = (0, 1)

Slope of line, $m=\tan\theta=\frac{OA}{OB}$

$\Rightarrow m=\frac{1}{\frac{3}{2}}=\frac{2}{3} \Rightarrow 3m-2=0$

**8.** **(b)** Let the variable circle is

$x^2+y^2+2gx+2fy+c=0$ ...(i)

It passes through $(a, b)$

$\therefore a^2+b^2+2ga+2fb+c=0$ ...(ii)

(i) cuts $x^2+y^2=4$ orthogonally

$\therefore 2(g\times 0+f\times 0)=c-4 \Rightarrow c=4$

$\therefore$ from (ii) $a^2+b^2+2ga+2fb+4=0$

$\therefore$ Locus of centre $(-g,-f)$ is

$a^2+b^2-2ax-2by+4=0$

or $2ax+2by=a^2+b^2+4$

**9.** **(b)** The equations of the circles are

$x^2+y^2-10x-10y+\lambda=0$ ...(i)

and $x^2+y^2-4x-4y+6=0$ ...(ii)

$C_1$ = centre of (i) = (5, 5)

$C_2$ = centre of (ii) = (2, 2)

$d$ = distance between centres = $C_1C_2$

$=\sqrt{9+9}=\sqrt{18}$

$r_1=\sqrt{50-\lambda}$, $r_2=\sqrt{2}$

For exactly two common tangents we have

$r_1-r_2<C_1C_2<r_1+r_2$

$\Rightarrow \sqrt{50-\lambda}-\sqrt{2}<3\sqrt{2}<\sqrt{50-\lambda}+\sqrt{2}$

$\Rightarrow \sqrt{50-\lambda}-\sqrt{2}<3\sqrt{2}$ or $3\sqrt{2}<\sqrt{50-\lambda}+\sqrt{2}$

$\Rightarrow \sqrt{50-\lambda}<4\sqrt{2}$ or $2\sqrt{2}<\sqrt{50-\lambda}$

$\Rightarrow 50-\lambda<32$ or $8<50-\lambda$

$\Rightarrow \lambda>18$ or $\lambda<42$

Required interval is (18, 42)

**10.** **(b)** Equation of required circle :

$S:(x-1)^2+(y-1)^2+\lambda(x-y)=0$

$S':x^2+y^2-2y-3=0$

Common chord of $S = 0$ and $S' = 0$ is $S - S' = 0$

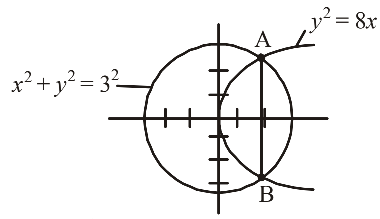
$(\lambda-2)x-(\lambda+4)y+5=0$

Centre of $S'$ : (0, – 1) lies on common chord

$\Rightarrow \lambda=-9$

$S:(x-1)^2+(y-1)^2-9(x-y)=0 \Rightarrow r=\frac{9}{\sqrt{2}}$

**11.** **(c)**




We have : $x^2 + (8x) = 9$

$\Rightarrow x^2 + 9x - x - 9 = 0$

$\Rightarrow x(x + 9) - 1(x + 9) = 0$

$\Rightarrow (x + 9)(x - 1) = 0 \Rightarrow x = -9, 1$

for $x = 1$, $y=\pm 2\sqrt{2x}=\pm 2\sqrt{2}$

$L_1$ = Length of AB

$=\sqrt{(2\sqrt{2}+2\sqrt{2})^2+(1-1)^2}=4\sqrt{2}$

$L_2$ = Length of latus rectum = $4a = 4 \times 2 = 8$

$L_1 < L_2$

**12.** **(c)** Equation of the tangent at $P(x_1, y_1)$ to $y^2=4ax$ is $yy_1-2ax-2ax_1=0$ ...(i)

Equation of the chord of $y^2 = 4a(x + b)$ whose mid-point is $(x', y')$ is

$yy' - 2ax - 2ax' - 4ab$
$= y'^2 - 4a\,x' - 4ab$ or $yy' - 2ax - (y'^2 - 2a\,x') = 0$ ...(ii)

Equation (i) and (ii) represent the same line.

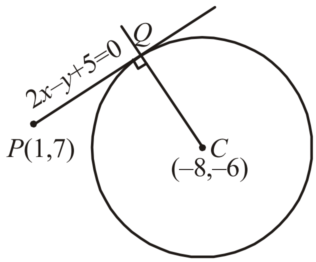
$$\therefore \quad \frac{y_1}{y'} = \frac{2a}{2a} = -\frac{2ax}{y'^2 - 2ax'}$$

This gives $y' = y_1$ and then $2ax_1 = y'^2 - 2ax'$
$= y_1^2 - 2ax' = 4ax_1 - 2ax'$ $\therefore$ $x' = x_1$
$\therefore$ mid-point $(x', y') = (x_1, y_1)$.

**13. (d)** The given curve is $y = x^2 + 6$
Equation of tangent at (1, 7) is

$$\frac{1}{2}(y+7) = x.1 + 6 \Rightarrow 2x - y + 5 = 0 \quad \text{...(i)}$$

As given this tangent (1) touches the circle
$x^2 + y^2 + 16x + 12y + c = 0$ at $Q$
Centre of circle = (– 8, – 6).




Then equation of $CQ$ which is perpendicular to (1) and passes through (– 8, – 6) is

$$y + 6 = -\frac{1}{2}(x+8) \Rightarrow x + 2y + 20 = 0 \quad \text{...(ii)}$$

Now $Q$ is pt. of intersection of (i) and (ii)
$\therefore$ Solving eqs (i) & (ii) we get ; $x = -6, y = -7$
$\therefore$ Req. pt. is (– 6, – 7).

**14. (a)** Since $S = (a, 0) = (1, 0)$, the circle is of the form $(x-1)^2 + y^2 = r^2$
Suppose $AB$ is a common chord. Since this is equidistant from the focus and the vertex. $M(1/2,0)$ lies on $AB$ and $AB$ is double ordinate of the parabola, let $A = (1/2, y)$ so that

$$y^2 = 4\left(\frac{1}{2}\right) \Rightarrow y = \pm\sqrt{2}$$

$$\Rightarrow \quad A = \left(\frac{1}{2}, \sqrt{2}\right) \text{ and } B = \left(\frac{1}{2}, -\sqrt{2}\right)$$

Since $\Delta AMS$ is right-angled triangle, we have

$$SA^2 = SM^2 + MA^2 = \frac{1}{4} + 2 = \frac{9}{4} = (\text{Radius})^2$$

Hence, the equation of the circle is

$$(x-1)^2 + y^2 = \frac{9}{4}$$

**15. (d)** As per the definition, the locus must be an ellipse, with given points as foci and 10 as its major axis. Since the line segment joining (2, –3) and (2, 5) is parallel to y-axis, therefore, ellipse is vertical.

$$\therefore \quad 2\,be = 8 \text{ and } 2b = 10 \Rightarrow b = 5 \text{ and } e = \frac{4}{5}$$

$\therefore$ $a^2 = b^2(1 - e^2) = 9$ and centre of the ellipse is (2, 1)
$\therefore$ Equation of the required ellipse is

$$\frac{(x-2)^2}{9} + \frac{(y-1)^2}{25} = 1$$

**16. (a)** For ellipse $\frac{x^2}{4^2} + \frac{y^2}{3^2} = 1$
$a = 4, b = 3$

$$\Rightarrow \quad e = \sqrt{1 - \left(\frac{3}{4}\right)^2} = \frac{\sqrt{7}}{4}$$

$\therefore$ Foci are $\left(\sqrt{7}, 0\right)$ and $\left(-\sqrt{7}, 0\right)$
Centre of circle is at (0, 3) and it passes through $\left(\pm\sqrt{7}, 0\right)$, therefore radius of circle

$$= \sqrt{\left(\sqrt{7}\right)^2 + (3)^2} = 4$$

**17. (d)** $x^2 = 8y$ ...(i)

$$\frac{x^2}{3} + y^2 = 1 \quad \text{...(ii)}$$

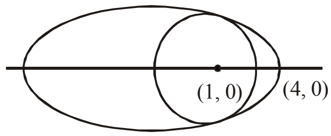
From (i) and (ii), $\frac{8y}{3} + y^2 = 1 \Rightarrow y = -3, \frac{1}{3}$
When $y = -3$, then $x^2 = -24$, which is not possible.
When $y = \frac{1}{3}$, then $x = \pm\frac{2\sqrt{6}}{3}$
Point of intersection are

$$\left(\frac{2\sqrt{6}}{3}, \frac{1}{3}\right) \text{ and } \left(-\frac{2\sqrt{6}}{3}, \frac{1}{3}\right)$$

Required equation of the line, $y - \frac{1}{3} = 0$
$\Rightarrow 3y - 1 = 0$

**18. (c)** Given ellipse is $\frac{x^2}{16} + \frac{y^2}{4} = 1$ ... (i)
Equation of a circle centered at (1, 0) can be written as $(x-1)^2 + y^2 = r^2$ ... (ii)
The abscissae of the intersection points of the circle and the ellipse is given by the equation

$$(x-1)^2 + \frac{16-x^2}{4} = r^2$$




i.e. $4(x^2 - 2x + 1) + 16 - x^2 = 4r^2$

i.e. $3x^2 - 8x + 20 - 4r^2 = 0$

If the circle lies inside the ellipse, then the roots of the above equation must be imaginary or equal

i.e. $D \le 0$ i.e. $64 + 12(4r^2 - 20) \le 0$

$\Rightarrow r = \le \sqrt{\dfrac{11}{3}}$

Hence, greatest value of $r = \sqrt{\dfrac{11}{3}}$ and the equation of required circle is

$$(x-1)^2 + y^2 = \frac{11}{3}$$

i.e. $3(x^2 + y^2) - 6x - 8 = 0.$

**19.** **(a)** Ellipse is $\dfrac{x^2}{16} + \dfrac{y^2}{3} = 1$

Now, equation of normal at (2, 3/2) is

$$\frac{16x}{2} - \frac{3y}{3/2} = 16 - 3$$

$\Rightarrow 8x - 2y = 13 \Rightarrow y = 4x - \dfrac{13}{2}$

Let $y = 4x - \dfrac{13}{2}$ touches a parabola

$y^2 = 4ax.$

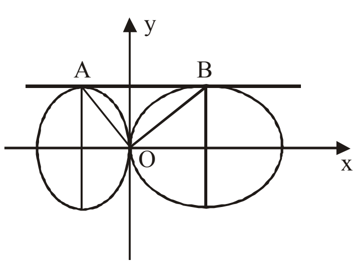
We know, a straight line $y = mx + c$ touches a parabola $y^2 = 4ax$ if $a - mc = 0$

$\therefore \quad a - (4)\left(-\dfrac{13}{2}\right) = 0 \Rightarrow a = -26$

Hence, required equation of parabola is

$y^2 = 4(-26)x = -104x$

**20.** **(a)**




See figure $B(4, 2)$ is one end of the minor axis of the ellipse $\dfrac{(x-4)^2}{25} + \dfrac{y^2}{4} = 1$ and $(-1, 2)$ is one end of the major axis of the second ellipse.

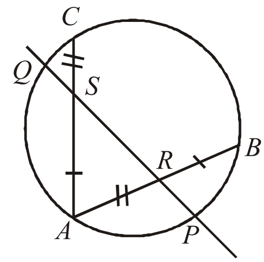
Therefore,

$AB = 5$, $OB = \sqrt{16+4} = \sqrt{20}$, $OA = \sqrt{1+4} = \sqrt{5}$

We have $(OA)^2 + (OB)^2 = 25 = (AB)^2$

Therefore $\angle AOB = \dfrac{\pi}{2}$

**21.** **(1)**




Point $A(\sqrt{33} + 3, 0)$ lies on the given circle,

$x^2 + y^2 - 6x - 8y - 24 = 0$

$PQ$ and $AB$ intersect inside the circle.

Let $PR = a$, $RS = b$, $QS = c$

Since $PR \times RQ = AR \times RB \Rightarrow a(b+c) = 3 \times 7$

Also, $QS \times SP = 3 \times 7 \Rightarrow c(a+b) = 3 \times 7$

$\Rightarrow a = c \;\therefore\; PR/QS = 1$

**22.** **(11)** The centre $C$ of the circle = ( 5,7 ) and the radius

$= \sqrt{5^2 + 7^2 + 51} = 5\sqrt{5}$

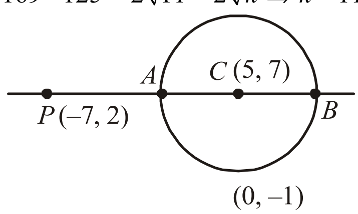
$PC = \sqrt{12^2 + 5^2} = 13 \Rightarrow q = PA = 13 - 5\sqrt{5}$

and $p = PB = 13 + 5\sqrt{3}$

$\therefore$ G.M. of $p$ and $q$

$= \sqrt{pq} = \sqrt{(13 - 5\sqrt{5}(13 + 5\sqrt{5}))}$

$= \sqrt{169 - 125} = 2\sqrt{11} = 2\sqrt{k} \Rightarrow k = 11.$




**23.** **(4)** Tangent to $y^2 = 8(x+2)$ is

$$y = m(x+2) + \frac{2}{m}$$

$$c = 2m + \frac{2}{m} \Rightarrow \frac{c}{2} = \left(m + \frac{1}{m}\right)$$

$\because \quad m+\frac{1}{m} \geq 2 \;\Rightarrow\; \frac{c}{2} \geq 2 \;\Rightarrow c \geq 4$

$\Rightarrow$ The minimum value of $c = 4$.

**24.** **(3)** The locus of the point of intersection of tangents to the parabola $y^2 = 4ax$ inclined at an angle $\alpha$ to each other is $\tan^2\alpha\,(x + a)^2 = y^2 - 4ax$

Given equation of Parabola $y^2 = 4x$ $\{a = 1\}$

Point of intersection $(-2, -1)$

$\tan^2\alpha\,(-2 + 1)^2 = (-1)^2 - 4 \times 1 \times (-2)$

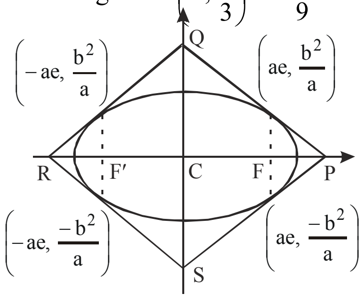
$\Rightarrow \tan^2\alpha = 9 \Rightarrow \tan\alpha = \pm 3 \Rightarrow |\tan\alpha| = 3$

**25.** **(27)** $\frac{x^2}{9} + \frac{y^2}{5} = 1 \Rightarrow e^2 = 1 - \frac{5}{9} = \frac{4}{9}$

$\Rightarrow e = \frac{2}{3}$

One end of latusrectum is $\left(2, \frac{5}{3}\right)$

Equation of tangent at $\left(2, \frac{5}{3}\right)$ is $\frac{2x}{9} + \frac{y}{3} = 1$




Area of $\Delta CPQ = \frac{1}{2} \times \frac{9}{2} \times 3 = \frac{27}{4}$

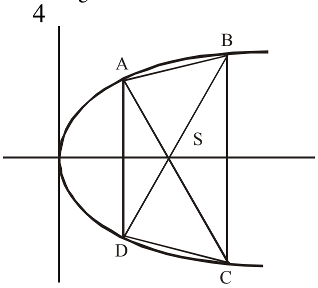
$\therefore$ Area of quadrilateral $PQRS = 4 \times \frac{27}{4} = 27.$

**26.** **(4)** Focus of the parabola $y^2 = 4x$ is $(1, 0)$

So diagonals are focal chord.

$AS = 1 + t^2 = c$ (say)

$\because \frac{1}{c} + \frac{1}{\frac{25}{4} - c} = 1 \qquad \left[\because \frac{1}{AS} + \frac{1}{CS} = \frac{1}{a}\right]$




$\frac{25}{4} = \frac{25}{4}c - c^2 \Rightarrow 4c^2 - 25c + 25 = 0$

$\Rightarrow c = \frac{5}{4}, 5$

For $c = \frac{5}{4}, 1 + t^2 = \frac{5}{4} \Rightarrow t^2 = \frac{1}{4} \Rightarrow t = \pm\frac{1}{2}$

For $c = 5, 1 + t^2 = 5 \Rightarrow t = \pm 2$

$A \equiv \left(\frac{1}{4}, 1\right)$, $B \equiv (4, 4)$, $C \equiv (4, -4)$ and

$D \equiv \left(\frac{1}{4}, -1\right)$

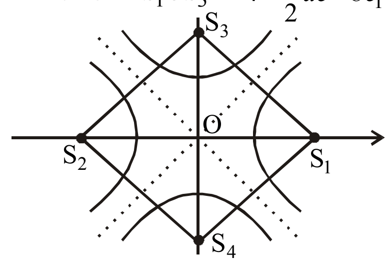
$AD = 2$ and $BC = 8$, distance between $AD$ and $BC$ $= \frac{15}{4}$

$\therefore$ Area of trapezium $ABCD$

$= \frac{1}{2}(2 + 8) \times \frac{15}{4} = \frac{75}{4}$ sq. units.

**27.** **(2)** Due to symmetry the desired area

$= 4 \times \text{area of } \Delta S_1OS_3 = 4 \times \frac{1}{2} ae \times be_1$




Where $e_1$ is eccentricity of conjugate hyperbola

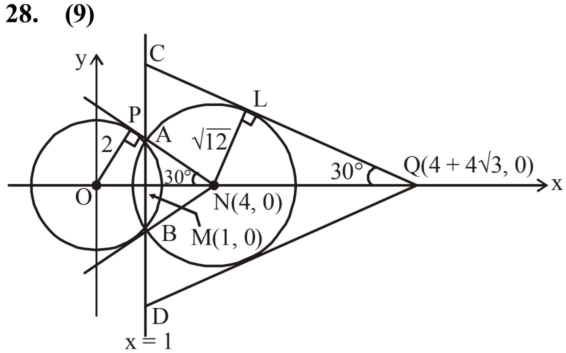
$= 2 \times 2e \times 3e_1 = 12ee_1$

Now $b^2 = a^2(e^2 - 1) \Rightarrow e^2 = 13/4$

and $\frac{1}{e^2} + \frac{1}{e_1^2} = 1 \Rightarrow e_1^2 = \frac{13}{9}$

$\therefore$ Required area $= 12 \times \frac{\sqrt{13}}{2} \times \frac{\sqrt{13}}{3} = 26$

**28.** **(9)**



Common chord of both the circles is $x = 1$.
Now, we have to find the ratio of areas of equilateral triangles $ANB$ and $CQD$.
Now in triangle $OPN$,
$ON = OP \operatorname{cosec} 30° = 2 \times 2 = 4$.
Area of triangle $NAB$

$\frac{1}{2} MN.AB = MN.AM = MN.\ MN \tan 30°$
$= (ON - OM)^2 \tan 30°$

$= (4-1)^2 \frac{1}{\sqrt{3}} = \frac{9}{\sqrt{3}}$ sq. units.

Now in triangle $NLQ$,
$NQ = NL \operatorname{cosec} 30° = 4\sqrt{3}$.

Since area of triangle $CQD = \frac{1}{2} QM.CD$

$= QM.CM$

QM. QM $\tan 30° = (MN + NQ)^2 \tan 30°$

$= (3+4\sqrt{3})^2 \frac{1}{\sqrt{3}} = \frac{57+24\sqrt{3}}{\sqrt{3}}$ sq. units.

So, ratio of area of trianlges $= \frac{57+24\sqrt{3}}{9}$.

**29.** **(5)** Let $r$ be the radius of the circle. Its equation is $x^2 + y^2 - 2r(x+y) + r^2 = 0$. Since it passes through $P(a,b)$

$a^2 + b^2 - 2r(a+b) + r^2 = 0$

Solving $r_1 = a + b + \sqrt{2ab}$ ...(1)

$r_2 = a + b - \sqrt{2ab}$

Now, the equations of two circles are

$x^2 + y^2 - 2r_1(x+y) + r_1^2 = 0$ and

$x^2 + y^2 - 2r_2(x+y) + r_2^2 = 0$

The common chord is $S_1 - S_2 = 0$

$\Rightarrow 2(r_2 - r_1)(x+y) + r_1^2 - r_2^2 = 0$

$\Rightarrow 2(x+y) = r_1 + r_2$

For maximum length of the common chord, it must pass through the centre of the smaller circle $(r_2, r_2)$, so

$4r_2 = r_1 + r_2 \Rightarrow \frac{r_1}{r_2} = 3$

$\Rightarrow \frac{a+b+\sqrt{2ab}}{a+b-\sqrt{2ab}} = 3 \Rightarrow 2(a+b) = 4\sqrt{2ab}$

$\Rightarrow (a+b)^2 = 8ab \Rightarrow a^2 - 6ab + b^2 = 0$

$\Rightarrow a = \frac{6b \pm \sqrt{36b^2 - 4b^2}}{2} = (3 \pm 2\sqrt{2})b$

$\Rightarrow \frac{a}{b} = 3 \pm 2\sqrt{2}$

**30.** **(5)** The tangent at any point $A(2\sec\theta, \tan\theta)$ is given by $\frac{x\sec\theta}{2} - \frac{y\tan\theta}{1} = 1$.

It meets the line $x - 2y = 0$

$\Rightarrow \frac{x\sec\theta}{2} - \frac{x\tan\theta}{2} = 1 \Rightarrow x = \frac{2}{\sec\theta - \tan\theta}$

$\Rightarrow Q \equiv \left(\frac{2}{\sec\theta - \tan\theta}, \frac{1}{\sec\theta - \tan\theta}\right)$ ...(1)

Also, the tangent meets the line $x + 2y = 0$ at $R$, so

$\Rightarrow \frac{x}{2}\sec\theta + \frac{x}{2}\tan\theta = 1$

$\Rightarrow x = \frac{2}{\sec\theta + \tan\theta}$

$\Rightarrow R \equiv \left(\frac{2}{\sec\theta + \tan\theta}, \frac{-1}{\sec\theta + \tan\theta}\right)$ ...(2)

Now,

$CQ.CR = \sqrt{\frac{2^2+1^2}{(\sec\theta - \tan\theta)^2}} \sqrt{\frac{2^2+1^2}{(\sec\theta + \tan\theta)^2}}$

$= 2^2 + 1^2$

$\Rightarrow CQ.CR = 5$

CHAPTER 12

# Limits & Derivatives

**1.** **(d)** $\lim_{x\to 0^-} f(x) = \lim_{h\to 0} \sqrt{\{-h\}\cot\{-h\}}$

$= \lim_{h\to 0} \sqrt{(1-h)\cot(1-h)} = \sqrt{\cot 1}$

$\lim_{x\to 0^+} f(x) = \lim_{h\to 0} \frac{\tan^2\{h\}}{h^2 - [h]^2} = \lim_{h\to 0} \frac{\tan^2 h}{h^2} = 1$

$\therefore \lim_{x\to 0} f(x)$ does not exist.

**2.** **(a)** $\lim_{x\to\infty} \left(\frac{x^2+5x+3}{x^2+x+2}\right)^x = \lim_{x\to\infty} \left(1 + \frac{4x+1}{x^2+x+2}\right)^x$

$$= \lim_{x\to\infty}\left[\left(1+\frac{4x+1}{x^2+x+2}\right)^{\frac{x^2+x+2}{4x+1}}\right]^{\frac{(4x+1)x}{x^2+x+2}}$$

$$= e^{\lim_{x\to\infty}\frac{4x^2+x}{x^2+x+2}} = e^{\lim_{x\to\infty}\frac{4+\frac{1}{x}}{1+\frac{1}{x}+\frac{2}{x^2}}} = e^4$$

**3.** **(b)** According to the question

$$\lim_{x\to0}\left(3-\frac{x^2}{12}\right) \le \lim_{x\to0} f(x) \le \lim_{x\to0}\left(3+\frac{x^3}{9}\right)$$

$\Rightarrow$ $(3-0) \le \lim_{x\to0} f(x) \le (3+0)$

Hence $\lim_{x\to0} f(x) = 3$ (from Sandwitch Theorem)

**4.** **(c)** Let $y$

$$= \lim_{n\to\infty}\frac{1}{n^k}((n+1)^k(n+2)^k\ldots(n+n)^k)^{1/n}$$

$$\Rightarrow ln\ y = \lim_{n\to\infty}\frac{1}{n}$$

$$\left(ln\left(\frac{n+1}{n}\right)^k + ln\left(\frac{n+2}{n}\right)^k + \ldots + ln\left(\frac{n+n}{n}\right)^k\right)$$

$$= \lim_{n\to\infty}\frac{k}{n}\left(ln\left(\frac{n+1}{n}\right) + ln\left(\frac{n+2}{n}\right) + \ldots + ln\left(\frac{n+n}{n}\right)\right)$$

$$= \lim_{n\to\infty} k.\sum_{r=1}^{n}\left(\frac{ln(n+r)-ln\ n}{n}\right)$$

$$= k.\lim_{n\to\infty}\sum_{r=1}^{n}\left(\frac{ln(n+r)-ln\ n}{n}\right)$$

$$= k.\left(2\left(ln\ 2-\frac{1}{2}\right)\right) = ln\left(\frac{4}{e}\right)^k. \Rightarrow y = \left(\frac{4}{e}\right)^k.$$

**5.** **(c)** Given that,

$$\lim_{x\to0}\frac{(\cos x-1)(\cos x-e^x)}{x^n} = \text{finite non-zero}$$
number

$$= \lim_{x\to0}\frac{(1-\cos x)(1+\cos x)(e^x-\cos x)}{x^n(1+\cos x)}$$

$$= \lim_{x\to0}\left(\frac{\sin^2 x}{x^2}\right).\left(\frac{e^x-\cos x}{x^{n-2}}\right).\left(\frac{1}{1+\cos x}\right)$$

$$= \lim_{x\to0} 1^2.\frac{e^x-\cos x}{x^{n-2}}.\frac{1}{2}$$

$$= \frac{1}{2}\lim_{x\to0}\frac{e^x-\sin x}{(x-2)x^{n-3}} \text{ [using L' Hospital's rule]}$$

For this limit to be finite, n – 3 = 0

$\Rightarrow$ n = 3

**6.** **(c)** $\lim_{x\to0}[(\sin x)^{1/x} + (1/x)^{\sin x}]$

$$= \lim_{x\to0}(\sin x)^{1/x} + \lim_{x\to0}\left(\frac{1}{x}\right)^{\sin x}$$

$$= 0 + e^{\lim_{x\to0}\sin x\log\left(\frac{1}{x}\right)}$$

[$\because |\sin x| < 1$ when $x\to0$]

$$= e^{\lim_{x\to0}\frac{-\log x}{\operatorname{cosec} x}} = e^{\lim_{x\to0}\frac{-1/x}{-\operatorname{cosec} x\cot x}}$$

[Using L' Hospital rule]

$$= e^{\lim_{x\to0}\frac{\sin x}{x}.\tan x} = e^0 = 1$$

**7.** **(c)** Limit is of the form $1^\infty$, so

$$\lim_{x\to a}\left(2-\frac{a}{x}\right)^{\tan\frac{\pi x}{2a}} = \lim_{x\to a}\left\{1+\left(1-\frac{a}{x}\right)\right\}^{\tan\frac{\pi x}{2a}}$$

$$= e^{\lim_{x\to a}\left(1-\frac{a}{x}\right)\tan\frac{\pi x}{2a}} = e^{\lim_{x\to a}\left(\frac{x-a}{x}\right)\tan\frac{\pi x}{2a}}$$

Let $x-a=h$ we get the limit

$$= e^{\lim_{h\to0}\left(\frac{h}{a+h}\right)\tan\frac{\pi}{2a}(a+h)}$$

$$= e^{\lim_{h\to0}\frac{h}{a+h}\tan\left(\frac{\pi}{2}+\frac{\pi h}{2a}\right)}$$

$$= e^{\lim_{h\to0}\frac{h}{a+h}\left(-\cot\left(\frac{\pi h}{2a}\right)\right)}$$

$$= e^{\lim_{h\to0}\frac{-h}{(a+h)\tan\left(\frac{\pi h}{2a}\right)}.\frac{\pi}{2a}.\frac{1}{\pi/2a}}$$

$$= e^{\lim_{h\to0}\frac{-2a}{\pi(a+h)}.\frac{\pi h/2a}{\tan(\pi h/2a)}} = e^{-\frac{2a}{\pi a}} = e^{-\frac{2}{\pi}}$$

**8.** **(a)** $\because$ $ax^2+bx+c=0$ has roots $\alpha$ and $\beta$

then $\frac{a}{x^2}+\frac{b}{x}+c=0$

i.e., $cx^2+bx+a=0$ has roots $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.

$$\Rightarrow\ cx^2+bx+a=c\left(x-\frac{1}{\alpha}\right)\left(x-\frac{1}{\beta}\right)$$

Now $\lim\limits_{x\to\frac{1}{\alpha}}\sqrt{\left(\dfrac{1-\cos(cx^2+bx+a)}{2(1-\alpha x)^2}\right)}$

$$=\lim_{x\to 1/\alpha}\left\{\frac{\sin^2\left(\dfrac{cx^2+bx+a}{2}\right)}{(1-\alpha x)^2}\right\}^{\frac{1}{2}}$$

$$=\lim_{x\to 1/\alpha}\left|\frac{\sin\left(\dfrac{cx^2+bx+a}{2}\right)}{(1-\alpha x)}\right|$$

$$=\lim_{x\to 1/\alpha}\left|\frac{\sin\left\{\dfrac{c}{2}\left(x-\dfrac{1}{\alpha}\right)\left(x-\dfrac{1}{\beta}\right)\right\}}{-\alpha\left(x-\dfrac{1}{\alpha}\right)}\right|$$

$$=\left|\lim_{x\to 1/\alpha}\frac{\sin\left(\dfrac{c}{2}\left(x-\dfrac{1}{\alpha}\right)\left(x-\dfrac{1}{\beta}\right)\right)}{\dfrac{c}{2}\left(x-\dfrac{1}{\alpha}\right)\left(x-\dfrac{1}{\beta}\right)}.\lim_{x\to 1/\alpha}\frac{\dfrac{c}{2}\left(x-\dfrac{1}{\beta}\right)}{-\alpha}\right|$$

$$=\left|1.\frac{\dfrac{c}{2}\left(\dfrac{1}{\alpha}-\dfrac{1}{\beta}\right)}{-\alpha}\right|=\left|\frac{c}{2\alpha}\left(\frac{1}{\alpha}-\frac{1}{\beta}\right)\right|$$

**9.** **(b)** Let $P$

$$=\lim_{x\to 0}\frac{\cos^2(1-\cos^2(1-\cos^2(......\cos^2(x))......}{\sin\left\{\pi\left(\dfrac{\sqrt{(x+4)}-2}{x}\right)\right\}}$$

$$=\lim_{x\to 0}\frac{\cos^2\sin^2(1-\cos^2(......\cos^2(x))......}{\sin\left\{\pi\left(\dfrac{\sqrt{(x+4)}-2}{x}\right)\right\}}$$

$$=\lim_{x\to 0}\frac{\cos^2(\sin^2(\sin^2(......(\sin^2(x))......)}{\dfrac{\sin\left\{\pi\left(\dfrac{\sqrt{x+4}-2}{x}\right)\right\}}{\left\{\pi\left(\dfrac{\sqrt{x+4}-2}{x}\right)\right\}}\left\{\pi\left(\dfrac{\sqrt{x+4}-2}{x}\right)\right\}}$$

$$=\lim_{x\to 0}\frac{\cos^2(\sin^2(\sin^2).........(\sin^2(x)).....)}{\dfrac{\sin\left\{\pi\left(\dfrac{\sqrt{(x+4)}-2}{x}\right)\right\}}{\left(\pi\dfrac{\sqrt{(x+4)}-2}{x}\right)}}.\frac{(\sqrt{x+4}+2)}{\pi}$$

$$=\frac{\cos^2 0}{1}.\frac{(2+2)}{\pi}=\frac{4}{\pi}$$

**10.** **(b)** $\lim\limits_{x\to\pi/2}\left[1^{1/\cos^2 x}+1^{2/\cos^2 x}+.....+n^{1/\cos^2 x}\right]^{\cos^2 x}$

$$=\lim_{t\to\infty}(1^t+2^t+.......n^t)^{1/t}$$

$$\left[\text{Putting } \frac{1}{\cos^2 x}=t\ge 1\right]$$

$$=\lim_{t\to\infty}(n^t)^{1/t}\left[\left(\frac{1}{n}\right)^t+\left(\frac{2}{n}\right)^t+......+\left(\frac{n}{n}\right)^t\right]^{1/t}$$

$$=n\lim_{t\to\infty}\left[\left(\frac{1}{n}\right)^t+\left(\frac{2}{n}\right)^t+......+\left(\frac{n}{n}\right)^t\right]^{1/t}$$

$$=n\,(0+0+......+1)^0=n$$

**11.** **(b)** $\lim\limits_{n\to\infty}\left\{\dfrac{\log_e n}{\log_e(n-1)}\times\dfrac{\log_e(n+1)}{\log_e n}\times\dfrac{\log_e(n+2)}{\log_e(n+1)}\times....\times\dfrac{\log_e n^k}{\log_e(n^k-1)}\right\}$

$$\lim_{n\to\infty}\left\{\frac{\log_e n^k}{\log_e(n-1)}\right\}=k\lim_{n\to\infty}\frac{\log_e n}{\log_e(n-1)}$$

Let $n=\dfrac{1}{h}$, then

$$=k\lim_{h\to 0}\frac{\log_e\left(\dfrac{1}{h}\right)}{\log_e\left(1-\dfrac{1}{h}\right)}=k\lim_{h\to 0}\frac{-\log_e(h)}{\log_e(1-h)-\log_e(h)}$$

$$=k\lim_{h\to 0}\frac{1}{\left\{1-\dfrac{\log_e(1-h)}{\log_e(h)}\right\}}=k\cdot\frac{1}{(1-0)}=k$$

**12.** **(b)** $\because\ \lim\limits_{n\to\infty}\dfrac{1}{n^3}\{[1^2x]+[2^2x]+[3^2x]+...+[n^2x]\}$

$$=\lim_{n\to\infty}\left\{\frac{\sum\limits_{r=1}^{n}[r^2x]}{n^3}\right\}=\lim_{n\to\infty}\left(\frac{\sum\limits_{r=1}^{n}r^2x-\{r^2x\}}{n^3}\right)$$

$$= \lim_{n\to\infty}\left(\frac{x.\dfrac{n(n+1)(2n+1)}{6}}{n^3} - \sum_{r=1}^{n}\frac{\{r^2x\}}{n^3}\right)$$

$$= x.\frac{(1)(1)(2)}{6} - 0 = \frac{x}{3}$$

**13. (c)** $\lim_{x\to3^-} f(x) = \lim_{x\to3^+} f(x)$

$$\Rightarrow \lim_{x\to3^-}\frac{(27)^{\frac{(x+3)x}{27}}-9}{3^x-27} = \lim_{x\to3^+}\lambda\frac{1-\cos(x-3)}{(x-3)^2}$$

$$\Rightarrow \lim_{x\to3^-}\frac{3^2\left(3^{\frac{x^2+3x-18}{9}}-1\right)}{3^3(3^{x-3}-1)} = \frac{\lambda}{2}$$

$$\Rightarrow \lim_{x\to3}\frac{1}{3}\frac{x^2+3x-18}{9(x-3)} = \frac{\lambda}{2} \Rightarrow \frac{1}{27}\cdot 9 = \frac{\lambda}{2} \Rightarrow \lambda = \frac{2}{3}.$$

**14. (a)** Here, $\lim_{x\to0}\left(\sin^2\left(\frac{\pi}{2-ax}\right)\right)^{\sec^2\left(\frac{\pi}{2-bx}\right)}$

$$= \lim_{x\to0}\left\{1-\cos^2\left(\frac{\pi}{2-ax}\right)\right\}^{\sec^2\left(\frac{\pi}{2-bx}\right)}$$

$$= e^{\lim_{x\to0} -\left\{\cos^2\left(\frac{\pi}{2-ax}\right)\cdot\frac{1}{\cos^2\left(\frac{\pi}{2-bx}\right)}\right\}}$$

$$= e^{\lim_{x\to0} -\left\{\frac{2\sin\left(\frac{\pi}{2-ax}\right)\cos\left(\frac{\pi}{2-ax}\right)\times\frac{\pi a}{(2-ax)^2}}{2\sin\left(\frac{\pi}{2-bx}\right)\cos\left(\frac{\pi}{2-bx}\right)\times\frac{\pi b}{(2-bx)^2}}\right\}}$$

[using L'Hospital's rule]

$$= e^{-\lim_{x\to0}\frac{\sin\left(\frac{2\pi}{2-ax}\right)}{\sin\left(\frac{2\pi}{2-bx}\right)}\cdot\frac{a}{b}\cdot\frac{(2-bx)^2}{(2-ax)^2}}$$

$$= e^{-\lim_{x\to0}\frac{a}{b}\cdot\frac{(2-bx)^3}{(2-ax)^3}} = e^{-\frac{a}{b}}.$$

**15. (c)** $\lim_{x\to0}\frac{f(1-\cos x)}{g(x)\sin^2 x}$

$$= \lim_{x\to0}\frac{f\left(2\sin^2\frac{x}{2}\right)}{g(x)\left(2\sin^2\frac{x}{2}\right)}\times\frac{\left(2\sin^2\frac{x}{2}\right)}{4\left(\sin^2\frac{x}{2}\right)\left(\cos^2\frac{x}{2}\right)}$$

$$= \lim_{x\to0}\frac{a}{g(x)}\times\tan^2\frac{x}{2} = a\lim_{x\to0}\frac{\left(\frac{x}{2}\right)^2}{g(x)}\times\frac{\tan^2\frac{x}{2}}{\left(\frac{x}{2}\right)^2}$$

$$= a\lim_{x\to0}\frac{x^2}{4g(x)} \quad \therefore \lim_{x\to0}\frac{x^2}{g(x)} = \frac{4b}{a}$$

Now, $\lim_{x\to0}\frac{g(1-\cos 2x)}{x^4} = \lim_{x\to0}\frac{g(2\sin^2 x)}{x^4}$

$$= \lim_{x\to0}\frac{g(2\sin^2 x)}{(2\sin^2 x)^2}\times\frac{(2\sin^2 x)^2}{x^4} = \frac{a}{4b}\times4 = \frac{a}{b}.$$

**16. (a)** Let $f(x) = lx^2 + mx + n$

$\Rightarrow f'(x) = 2\ell x + m$

Now, $f(1) = f(-1) \Rightarrow \ell + m + n = \ell - m + n$

$\Rightarrow m = 0$

$\therefore f'(x) = 2lx$

$\therefore f'(a_1) = 2\ell a, f'(a_2) = 2\ell a_2, f'(a_3) = 2\ell a_3$

As $a_1, a_2, a_3$ are in A.P.

$\therefore f'(a_1), f'(a_2), f'(a_3)$ are in A. P.

**17. (b)** Since, $f(x)$ is a polynomial function satisfying

$$f(x)\cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right),$$

$\therefore f(x) = x^n + 1$ or $f(x) = -x^n + 1$

If $f(x) = -x^n + 1$, then $f(4) = -4^n + 1 \neq 65$

So, $f(x) = x^n + 1$ Since, $f(4) = 65$

$\therefore 4^n + 1 = 65$

$\Rightarrow n = 3 \therefore f(x) = x^3 + 1 \Rightarrow f'(x) = 3x^2$

$\therefore f'(l_1) = 3l_1^2, f'(l_2) = 3l_2^2, f'(l_3) = 3l_3^2$

Since, $l_1, l_2, l_3$ are in GP.

$\therefore f'(l_1), f'(l_2), f'(l_3)$ are also in GP.

**18. (5)** $\lim_{x\to a}\frac{g(x)f(a)-g(a)f(x)}{x-a}$

$$= \lim_{h\to0}\frac{g(a+h)f(a)-g(a)f(a+h)}{h}$$

[For x = a + h]

$$= \lim_{h\to0}\frac{g(a+h)f(a)-g(a)f(a)+g(a)f(a)-g(a)f(a+h)}{h}$$

$$= \lim_{h\to 0} f(a)\left[\frac{g(a+h)-g(a)}{h}\right] - \lim_{h\to 0} g(a)\left[\frac{f(a+h)-f(a)}{h}\right]$$

$$= f(a)\, g'(a) - g(a)\, f'(a) = 2\times 2 - (-1)\times 1 = 5$$

**19.** **(1)** $\lim_{x\to\frac{\pi}{2}} (\sin x)^{\tan x} = e^{\lim_{x\to\frac{\pi}{2}}(\tan x(\sin x - 1))}$

$$= e^{\lim_{x\to\frac{\pi}{2}} \frac{\sin^2 x - \sin x}{\cos x}} \quad \left(\frac{0}{0}\text{form}\right)$$

$$= e^{\lim_{x\to\frac{\pi}{2}} \frac{2\sin x\cos x - \cos x}{-\sin x}} = e^0 = 1$$

**20.** **(0.50)** Required limit,

$$l = \lim_{x\to\infty} \frac{\left(x+\sqrt{x+\sqrt{x}}\right) - x}{\sqrt{x+\sqrt{x+\sqrt{x}}} + \sqrt{x}}$$

Dividing numerator & denominator by $\sqrt{x}$, we get

$$l = \lim_{x\to\infty} \frac{\sqrt{1+\frac{\sqrt{x}}{x}}}{\sqrt{1+\sqrt{\frac{x+\sqrt{x}}{x^2}}}+1}$$

$$= \lim_{x\to\infty} \frac{\sqrt{1+\frac{1}{\sqrt{x}}}}{\sqrt{1+\sqrt{\frac{1}{x}+\frac{1}{x^{3/2}}}}+1} = \frac{1}{1+1} = \frac{1}{2} = 0.50$$

**21.** **(6)** $\lim_{x\to 0} \frac{x(1+a\cos x) - b\sin x}{\{f(x)\}^3} = 1$

$$\Rightarrow \lim_{x\to 0} \frac{x + ax\left\{1-\frac{x^2}{2!}+\frac{x^4}{4!}-......\right\} - b\left\{x - \frac{x^3}{3!}+\frac{x^5}{5!}-......\right\}}{\{f(x)\}^3} = 1$$

$$\Rightarrow \lim_{x\to 0} \frac{\frac{1+a-b}{x^2} + \left(-\frac{a}{2!}+\frac{b}{3!}\right) + x^2\left(\frac{a}{4!}-\frac{b}{5!}\right) + .......}{\left\{\frac{f(x)}{x}\right\}^3} = 1$$

$$\Rightarrow 1+a-b = 0 \text{ and } -\frac{a}{2!}+\frac{b}{3!} = 1$$

$$\Rightarrow a = -\frac{5}{2} \text{ and } b = -\frac{3}{2}. \text{ Thus } b - 3a = 6$$

**22.** **(2)** $\lim_{x\to 1}\left\{\frac{-ax+\sin(x-1)+a}{x+\sin(x-1)-1}\right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$

$$\Rightarrow \lim_{x\to 1}\left\{\frac{a(1-x)+\sin(x-1)}{(x-1)+\sin(x-1)}\right\}^{1+\sqrt{x}}$$

$$\Rightarrow \lim_{x\to 1}\left\{\frac{-a+\frac{\sin(x-1)}{x-1}}{1+\frac{\sin(x-1)}{x-1}}\right\}^{1+\sqrt{x}}$$

$$\Rightarrow \left(\frac{-a+1}{2}\right)^2 = \frac{1}{4} \Rightarrow a = 0 \text{ or } 2$$

$\therefore$ Largest value of $a$ is 2.

**23.** **(1)** Let

$$y = \frac{x}{x+\frac{\sqrt[3]{x}}{x+\sqrt[3]{x}....\infty}} = \frac{x}{x+\frac{1}{x^{2/3}}\frac{x}{x+\sqrt[3]{x}....\infty}}$$

$$= \frac{x^{5/3}}{x^{5/3}+y} = \frac{x}{x+\frac{y}{x^{2/3}}}$$

or $y^2 + (x^{5/3})y - x^{5/3} = 0$

$$\therefore y = \frac{-x^{5/3} \pm \sqrt{x^{10/3}+4x^{5/3}}}{2}$$

$$= \frac{-x^{5/3} + \sqrt{x^{10/3}+4x^{5/3}}}{2} \qquad (\because y > 0)$$

$$= \frac{4x^{5/3}}{2(\sqrt{(x^{10/3}+4x^{5/3})}+x^{5/3})}$$

$$= \frac{2}{\sqrt{\left(1+\frac{4}{x^{5/3}}\right)}+1}$$

$$\therefore \lim_{x\to\infty} y = \frac{2}{\sqrt{1+0}+1} = \frac{2}{2} = 1.$$

**24.** **(1)** $a(x^3-1)+(x-1) = 0$

or $(x-1)(ax^2+ax+a+1) = 0$

$\alpha, \beta \neq 1$ so, $\alpha, \beta$ are roots of

$ax^2+ax+a+1 = 0$

$\alpha+\beta = -1,\ \alpha\beta = \frac{a+1}{a}$

$$\lim_{x\to\frac{1}{\alpha}} \frac{(1+a)x^3-x^2-a}{(e^{1-ax}-1)(x-1)} = \lim_{x\to\frac{1}{\alpha}} \frac{(x^3-x^2)+a(x^3-1)}{(e^{1-ax}-1)(x-1)}$$

$$= \lim_{x \to \frac{1}{\alpha}} \frac{[x^2 + a(x^2 + x + 1)]}{(e^{1-ax} - 1)} = \lim_{x \to \frac{1}{\alpha}} \frac{(1+a)x^2 + ax + a}{\left(\frac{e^{1-ax} - 1}{1 - ax}\right)(1 - ax)}$$

$$= \lim_{x \to \frac{1}{\alpha}} a \frac{\left[\left(\frac{1+a}{a}\right)x^2 + (1)x + 1\right]}{(1 - ax)} = \lim_{x \to \frac{1}{\alpha}} a \frac{(\alpha\beta x^2 - (\alpha + \beta)x + 1)}{(1 - \alpha x)}$$

$$= \lim_{x \to \frac{1}{\alpha}} a \frac{(1 - (\alpha)x)(1 - (\beta)x)}{(1 - \alpha x)} = \frac{a(\alpha - \beta)}{\alpha}.$$

**25.** **(36)** Let $3^x = t^2$

$$\lim_{t \to 3} \frac{t^2 + \frac{27}{t^2} - 12}{\frac{1}{t} - \frac{3}{t^2}} = \lim_{t \to 3} \frac{t^4 - 12t^2 + 27}{t - 3}$$

$$= \lim_{t \to 3} \frac{(t^2 - 3)(t + 3)(t - 3)}{t - 3}$$

$= (3^2 - 3)(3 + 3) = 36.$

CHAPTER 13

# Mathematical Reasoning

**1.** **(a)** Inclusive "or". 17 is a real number or a positive integer or both.

**2.** **(c)** $p \to (\sim p \vee q)$ has truth value F.
It means $p \to (\sim p \vee q)$ is false.
It means $p$ is true and $\sim p \vee q$ is false.
$\Rightarrow$ $p$ is true and both $\sim p$ and $q$ are false.
$\Rightarrow$ $p$ is true and $q$ is false.

**3.** **(b)** $\sim p$ : Ashok does not work hard
Use '$\to$' symbol for then
($\sim p \to q$) mean = If Ashok does not work hard then he gets good grade.

**4.** **(d)** When $p$ is false and $q$ is true, then $p \wedge q$ is false, $p \vee \sim q$ is false.
($\because$ both $p$ and $\sim q$ are false)
and $q \Rightarrow p$ is also false,
only $p \Rightarrow q$ is true.

**5.** **(d)** $\sim p \wedge q = \sim (q \to p)$

**6.** **(a)**

| p | q | $p \wedge q$ | $p \vee q$ | $\sim(p \vee q)$ | $(p \wedge q) \wedge \sim(p \vee q)$ |
|---|---|---|---|---|---|
| T | T | T | T | F | F |
| T | F | F | T | F | F |
| F | T | F | T | F | F |
| F | F | F | F | T | F |

$\therefore (p \wedge q) \wedge (\sim (p \vee q))$ is a contradiction.

**7.** **(b)** $(p \wedge \sim q) \wedge (\sim p \wedge q) = (p \wedge \sim q) \wedge (\sim q \wedge q)$
$= f \wedge f = f$
(By using associative laws and commutative laws)
$\therefore (p \wedge \sim q) \wedge (\sim p \wedge q)$ is a contradiction.

**8.** **(b)** $p \Rightarrow q$ is logically equivalent to $\sim q \Rightarrow \sim p$
$\therefore$ $(p \Rightarrow q) \Leftrightarrow (\sim q \Rightarrow \sim p)$ is a tautology but not a contradiction.

**9.** **(a)**

| $p$ | $q$ | $p \wedge q$ | $(p \wedge q) \Rightarrow p$ |
|---|---|---|---|
| $T$ | $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ | $T$ |
| $F$ | $T$ | $F$ | $T$ |
| $F$ | $F$ | $F$ | $T$ |

$\therefore (p \wedge q) \Rightarrow p$ is a tautology.

**10.** **(b)**

| $p \Rightarrow q$ | $\sim q \Rightarrow \sim p$ | $p \Rightarrow q \Leftrightarrow \sim q \Rightarrow \sim p$ |
|---|---|---|
| T | T | T |
| F | F | T |
| T | T | T |
| T | T | T |

**11.** **(b)** The truth value of $\sim(\sim p) \leftrightarrow p$ as follow

| $p$ | $\sim p$ | $\sim(\sim p)$ | $\sim(\sim p) \to p$ | $p \to \sim(\sim p)$ | $\sim(\sim p) \leftrightarrow p$ |
|---|---|---|---|---|---|
| T | F | T | T | T | T |
| F | T | F | T | T | T |

Since last column of above truth table contains only T.
Hence $\sim(\sim p) \to p$ is a tautology.

**12.** **(c)**

| **p** | **q** | **$p \to q$** | **$\sim p$** | **$\sim p \vee q$** | **$(p \to q) \leftrightarrow \sim (p \vee q)$** |
|---|---|---|---|---|---|
| T | T | T | F | T | T |
| T | F | F | F | F | T |
| F | T | T | T | T | T |
| F | F | T | T | T | T |

**13.** **(d)** $p \Rightarrow (\sim p \vee q)$ is false means p is true and $\sim p \vee q$ is false.
$\Rightarrow p$ is true and both $\sim p$ and $q$ are false. $\Rightarrow p$ is true and $q$ is false

**14.** **(c)** Let $p: 2+3=5, q: 8<10$
Given proposition is : $p \wedge q$.
Its negation is $\sim(p \wedge q) = \sim p \vee \sim q$
$\therefore$ we have $2+3 \neq 5$ or $8 \not< 10$.

**15.** **(b)** We know that $p \to q$ is false only when $p$ is true and $q$ is false.
So $p \to (\sim p \vee q)$ is false only when $p$ is true and $(\sim p \vee q)$ is false.
But $(\sim p \vee q)$ is false if $q$ is false because $\sim p$ is false.
Hence $p \to (\sim p \vee q)$ is false when truth value of $p$ and $q$ are T and F respectively.

**16.** **(a)** We know that the contrapositive of $p \to q$ is $\sim q \to \sim p$.
So contrapositive of $p \to (\sim q \to \sim r)$ is
$\sim(\sim q \to \sim r) \to \sim p \equiv \sim q \wedge [\sim(\sim r)] \sim p$
$\because \sim(p \to q) \equiv p \wedge \sim q \equiv \sim q \wedge r \to \sim p$

**17.** **(a)** $\sim[p \vee (\sim p \vee q)] \equiv \sim p \wedge \sim(\sim p \vee q)$
$\equiv \sim p \wedge (\sim(\sim p) \wedge \sim q) \ \equiv \sim p \wedge (p \wedge \sim q)$.

**18.** **(c)** The inverse of the proposition $(p \wedge \sim q) \to r$ is $\sim (p \wedge \sim q) \to \sim r$
$\equiv \sim p \vee \sim(\sim q) \to \sim r \equiv \sim p \vee q \to \sim r$

**19.** **(a)** $\sim((\sim p) \wedge q) \equiv \sim(\sim p) \vee \sim q \equiv p \vee (\sim q)$

**20.** **(c)** $\sim(p \Rightarrow q) \equiv p \wedge \sim q$
$\therefore \sim(\sim p \Rightarrow \sim q) \equiv \sim p \wedge \sim(\sim q) \equiv \sim p \wedge q$.
Thus $\sim(\sim p \Rightarrow \sim q) \equiv \sim p \wedge q$

**21.** **(c)** $\sim[(p \vee q) \wedge (q \vee \sim r)]$
$\equiv \sim(p \vee q) \vee \sim(q \vee \sim r)$
$\equiv (\sim p \wedge \sim q) \vee (\sim q \wedge r)$

**22.** **(c)** Statement given in option (c) is correct.
$\sim[p \vee (\sim q)] = (\sim p) \wedge \sim(\sim q) = (\sim p) \wedge q$.

**23.** **(d)** Since $\sim(p \vee q) \equiv \sim p \wedge \sim q$
(By De-Morgans' law)
$\therefore \sim(p \vee q) \neq \sim p \vee \sim q$
$\therefore$ (d) is the false statement.

**24.** **(d)** We know that $\sim(p \to q) \equiv p \wedge \sim q$
$\therefore \sim((p \wedge r) \to (r \vee q)) \equiv (p \wedge r) \wedge [\sim(r \vee q)]$
$\equiv (p \wedge r) \wedge (\sim r \wedge \sim q)$

**25.** **(d)** Let $P = A \subseteq B, Q = B \subseteq D, R = A \subseteq C$
Contrapositive of $(P \wedge Q) \to R$ is $(\to \sim R \to \sim (P \wedge Q))$.
$\sim R \to \sim P \vee \sim Q$

# CHAPTER 14 Statistics

**1.** **(a)** $\bar{X} = \dfrac{1.\,{}^nC_1 + 2.\,{}^nC_2 + 3.\,{}^nC_3 + .... + n.\,{}^nC_n}{{}^nC_1 + {}^nC_2 + .... + {}^nC_n}$

$$\bar{X} = \frac{\sum_{r=1}^{n} r.\,{}^nC_r}{\sum_{r=1}^{n} {}^nC_r} = \frac{n\sum_{r=1}^{n} {}^{n-1}C_{r-1}}{\sum_{r=1}^{n} {}^nC_r} = \frac{n.2^{n-1}}{2^n - 1}.$$

**2.** **(a)** Let the mean of the remaining 4 observations be $\bar{x}_1$.

Then, $M = \dfrac{a + 4\bar{x}_1}{(n-4)+4} \Rightarrow \bar{x}_1 = \dfrac{nM - a}{4}$.

**3.** **(a)** $\bar{x} = \dfrac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

$\because \quad \bar{x}_1 = 400, \bar{x}_2 = 480, \bar{x} = 430$

$\therefore \quad 430 = \dfrac{n_1(400) + n_2(480)}{n_1 + n_2} \Rightarrow 30n_1 = 50n_2$

$\Rightarrow \dfrac{n_1}{n_2} = \dfrac{5}{3}$

**4.** **(c)** Sum of 6 numbers $= 30 \times 6 = 180$
Sum of remaining 5 numbers $= 29 \times 5 = 145$
$\therefore$ Excluded number $= 180 - 145 = 35$.

**5.** **(c)** Let the items be $a_1, a_2, ........, a_n$.
then $\bar{X} = \dfrac{a_1 + a_2 + ........ + a_n}{n}$.
Now, according to the given condition:

$$\bar{X} = \frac{(a_1+1) + (a_2+2) + ........ + (a_n+n)}{n}$$

$$= \bar{X} + \frac{1+2+3+........+n}{n} = \bar{X} + \frac{n(n+1)}{2n}$$

(using sum of n natural nos.)

$$= \bar{X} + \frac{n+1}{2}.$$

**6.** **(a)** Median is given as

$$M = l + \frac{\frac{N}{2} - F}{f} \times C$$

where

$l$ = lower limit of the median - class
$f$ = frequency of the median class
$N$ = total frequency
$F$ = cumulative frequency of the class just before the median class
$C$ = length of median class.

Now, given, $M = 25, N = 100, F = 45$,
$C = 20 - 30 = 10,\ l = 20$.

$\therefore$ By using formula, we have

$$25 = 20 + \frac{50-45}{f} \times 10$$

$$25 - 20 = \frac{50}{f} \Rightarrow 5 = \frac{50}{f} \Rightarrow f = 10.$$

**7.** **(b)** $\text{Mean} = \frac{101 + d(1+2+3+......+100)}{101}$

$$= 1 + \frac{d \times 100 \times 101}{101 \times 2} = 1 + 50\text{ d}$$

$\because$ Mean deviation from the mean = 255

$$\Rightarrow \frac{1}{101}[|1-(1+50d)| + |(1+d)-(1+50d)| + |(1+2d)| - (1+50d)| + .... + |(1+100d)-(1+50d)|] = 255$$

$$\Rightarrow 2d[1+2+3+...+50] = 101 \times 255$$

$$\Rightarrow 2d \times \frac{50 \times 51}{2} = 101 \times 255$$

$$\Rightarrow d = \frac{101 \times 255}{50 \times 51} = \frac{25755}{2550} = 10.1$$

**8.** **(c)** Clearly mean $A = 0$.

Standard deviation $\sigma = \sqrt{\frac{\sum (x-A)^2}{2n}}$

$$2 = \sqrt{\frac{(a-0)^2 + (a-0)^2 + ...(0-a)^2 + ...}{2n}}$$

$$= \sqrt{\frac{a^2.2n}{2n}} = |a|. \text{ Hence } |a| = 2.$$

**9.** **(c)** Using,

$$\sigma = \sqrt{\frac{\mathrm{n_1}(\sigma_1^2 + \mathrm{d}_1^2) + \mathrm{n_2}(\sigma_2^2 + \mathrm{d}_2^2)}{\mathrm{n_1} + \mathrm{n_2}}} = \sqrt{\frac{22}{3}}.$$

**10.** **(a)** We know that Q.D

$$= \frac{5}{6} \times M.D. = \frac{5}{6} \times 12 = 10$$

$$\therefore \text{S.D} = \frac{3}{2} \times Q.D. = \frac{3}{2} \times 10 \Rightarrow S.D. = 15.$$

**11.** **(d)** Since S.D. $\leq$ Range $= b - a$

$\therefore\ Var(x) \leq (b-a)^2$ or $(b-a)^2 \geq Var(x)$.

**12.** **(d)** Since $0 < y < x < 2y$

$$\therefore\ y > \frac{x}{2} \Rightarrow x - y < \frac{x}{2}$$

$$\therefore\ x - y < y < x < 2x + y.$$

Hence median $= \frac{y+x}{2} = 10$

$\Rightarrow x + y = 20.$ ...(i)

And range $= (2x+y) - (x-y) = x + 2y.$

But range $= 28\ \therefore\ x + 2y = 28$ ...(ii)

From equations (i) and (ii), $x = 12, y = 8$

$$\therefore \text{Mean} = \frac{(x-y) + y + x + (2x+y)}{4} = \frac{4x+y}{4}$$

$$= x + \frac{y}{4} = 12 + \frac{8}{4} = 14.$$

**13.** **(b)** Let $x_i$ be $n$ observations, $i = 1, 2, ...n$

Let $\overline{X}$ be the mean and M.D be the mean deviation about $\overline{X}$.

If each observation is increased by 5 then new mean will be $\overline{X} + 5$ and new M.D. about new mean will be M.D. $\left(\because \text{Mean} = \sum_{i=1}^{n} \frac{x_i}{n}\right)$

**14.** **(a)** The first $n$ natural numbers are 1, 2, 3, ...............$n$

Their mean, $\bar{x} = \frac{1+2+3+4+...+n}{n}$

$$= \frac{n(n+1)}{2n} = \frac{n+1}{2}.$$

[$\because$ The sum of Ist n natural numbers is $\frac{n(n+1)}{2}$]

Now, Variance $= \sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$

$$= \frac{1}{n}\left[\sum (x_i^2 - 2\bar{x}x_i + \bar{x}^2)\right]$$

$$= \frac{\sum x_i^2}{n} - 2\bar{x}\frac{\sum x_i}{n} + \frac{\bar{x}^2.n}{n}$$

$$= \frac{\sum x_i^2}{n} - 2\bar{x}^2 + \bar{x}^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2$$

[Since frequency of each variate is one]

$$\because \sum x_i^2 = \frac{n(n+1)(2n+1)}{6}.$$

$$\therefore \text{ Variance } = \frac{n(n+1)(2n+1)}{6n} - \left(\frac{(n+1)}{2}\right)^2$$

$$= \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4}$$

$$= (n+1)\left(\frac{2n+1}{6} - \frac{n+1}{4}\right) = \frac{(n+1)(n-1)}{12}$$

$$= \frac{n^2-1}{12}.$$

**15.** **(b)** Let the observations be $x_1, x_2, ...., x_{20}$ and $\bar{x}$ be their mean. Given that, variance = 5 and n = 20. We know that,

$$\text{Variance } (\sigma^2) = \frac{1}{n}\sum_{i=1}^{20}(x_i - \bar{x})^2$$

$$\text{i.e. } 5 = \frac{1}{20}\sum_{i=1}^{20}(x_i - \bar{x})^2$$

$$\text{or } \sum_{i=1}^{20}(x_i - \bar{x})^2 = 100 \quad ...(i)$$

If each observation is multiplied by 2 and the new resulting observations are $y_i$, then

$$y_i = 2x_i \text{ i.e., } x_i = \frac{1}{2}y_i.$$

$$\text{Therefore, } \bar{y} = \frac{1}{n}\sum_{i=1}^{20} y_i = \frac{1}{20}\sum_{i=1}^{20} 2x_i$$

$$= 2.\frac{1}{20}\sum_{i=1}^{20} x_i$$

$$\text{i.e., } \bar{y} = 2\bar{x} \text{ or } \bar{x} = \frac{1}{2}\bar{y}.$$

On substituting the values of $x_i$ and $\bar{x}$ in eq. (i), we get

$$\sum_{i=1}^{20}\left(\frac{1}{2}y_i - \frac{1}{2}\bar{y}\right)^2 = 100$$

$$\text{i.e., } \sum_{i=1}^{20}(y_i - \bar{y})^2 = 400.$$

Thus, the variance of new observations

$$= \frac{1}{20} \times 400 = 20 = 2^2 \times 5.$$

**16.** **(d)** If initially all marks were $x_i$ then

$$\sigma_i^2 = \frac{\sum_i (x_i - \bar{x})^2}{N}.$$

Now each is increased by 10

$$\sigma_i^2 = \frac{\sum_i [(x_i+10)-(\bar{x}+10)]^2}{N} = \frac{\sum_i (x_i-\bar{x})^2}{N} = \sigma_i^2$$

Hence, variance will not change even after the grace marks were given.

**17.** **(a)** C.V. (1st distribution) = 60, $\sigma_1 = 21$

C.V. (2nd distribution) = 70, $\sigma_2 = 16$

Let $\bar{x}_1$ and $\bar{x}_2$ be the means of 1st and 2nd distribution, respectively, Then

$$\text{C.V. (1st distribution)} = \frac{\sigma_1}{\bar{x}_1} \times 100$$

$$\therefore 60 = \frac{21}{\bar{x}_1} \times 100 \quad \text{or} \quad \bar{x}_1 = \frac{21}{60} \times 100 = 35$$

$$\text{and C.V. (2nd distribution)} = \frac{\sigma_2}{\bar{x}_2} \times 100$$

$$\text{i.e., } 70 = \frac{16}{\bar{x}_2} \times 100 \quad \text{or,} \quad \bar{x}_2 = \frac{16}{70} \times 100 = 22.85$$

**18.** **(93.32)** When each observation is multiplied by 2, then variance is also multiplied by 2.

We are given, 2, 4, 5, 6, 8, 17.

When each observation multiplied by 2, we get 4, 8, 10, 12, 16, 34.

$\therefore$ Variance of new series = $2^2$ × Variance of given data = 4 × 23.33 = 93.32.

**19.** **(0)** We know that,

$$\text{Coefficient of variation} = \frac{\sigma}{\bar{x}} \times 100$$

$$\therefore \text{ CV of 1st distribution} = \frac{\sigma_1}{30} \times 100$$

$$\Rightarrow \quad 50 = \frac{\sigma_1}{30} \times 100 \quad \text{[CV of 1st distribution = 50 (given)]}$$

$$\Rightarrow \quad \sigma_1 = 15$$

$$\text{Also, CV of 2nd distribution} = \frac{\sigma_2}{25} \times 100$$

$$\Rightarrow \quad 60 = \frac{\sigma_2}{25} \times 100$$

$$\Rightarrow \quad \sigma_2 = \frac{60 \times 25}{100} \Rightarrow \sigma_2 = 15$$

Thus, $\sigma_1 - \sigma_2 = 15 - 15 = 0.$

**20.** **(78)** $\sum x = 170, \sum x^2 = 2830$ increase in $\sum x = 10$, then $\sum x' = 170 + 10 = 180$

Increase in $\sum x^2 = 900 - 400 = 500$, then

$$\sum x'^2 = 2830 + 500 = 3330$$

$$\text{Variance} = \frac{1}{n}\sum x'^2 - \left(\frac{1}{n}\sum x'\right)^2$$

$$= \frac{1}{15} \times 3330 - \left(\frac{1}{15} \times 180\right)^2 = 222 - 144 = 78.$$

**21.** **(8.25)**

$$\sigma^2 = \frac{\frac{11 \cdot 12.23}{6} - 1}{10} - \left(\frac{\frac{11.12}{2} - 1}{10}\right)^2 = 8.25$$

**22.** **(5)** We have,
CV of 1st distribution $(CV_1) = 50$.
CV of 2nd distribution $(CV_2) = 60$
$\sigma_1 = 10$ and $\sigma_2 = 15$.

We know that, $CV = \frac{\sigma}{\overline{x}} \times 100$

$$\therefore CV_1 = \frac{\sigma_1}{\overline{x}_1} \times 100 \Rightarrow 50 = \frac{10}{\overline{x}_1} \times 100$$

$$\Rightarrow \overline{x}_1 = \frac{10 \times 100}{50} = 20.$$

Also, $CV_2 = \frac{\sigma_2}{\overline{x}_2} \times 100$

$$\Rightarrow 60 = \frac{15 \times 100}{\overline{x}_2}$$

$$\Rightarrow \overline{x}_2 = \frac{15 \times 100}{60} \Rightarrow \overline{x}_2 = 25.$$

Thus, $\overline{x}_2 - \overline{x}_1 = 25 - 20 = 5$.

**23.** **(5)** Let the other two observations be x and y.
Therefore, the series is 1, 2, 6, x, y.

Now, mean $(\overline{x}) = 4.4 = \frac{1 + 2 + 6 + x + y}{5}$

or $22 = 9 + x + y$.
Therefore, $x + y = 13$ …(i)

Also, variance $(\sigma^2) = 8.24 = \frac{1}{n}\sum_{i=1}^{5}(x_i - \overline{x})^2$

i.e., $8.24 = \frac{1}{5}[(3.4)^2 + (2.4)^2 + (1.6)^2 + x^2 + y^2 - 2 \times 4.4(x + y) + 2 \times (4.4)^2]$

or $41.20 = 11.56 + 5.76 + 2.56 + x^2 + y^2 - 8.8 \times 13 + 38.72$.

Therefore, $x^2 + y^2 = 97$ …(ii)

But from eq. (i), we have

$x^2 + y^2 + 2xy = 169$ …(iii)

From eqs. (ii) and (iii), we have

$2xy = 72$ …(iv)

On subtracting eq, (iv) from eq. (ii), we get

$x^2 + y^2 - 2xy = 97 - 72$

i.e. $(x - y)^2 = 25$ or $x - y = \pm 5$ …(v)

So, from eqs. (i) and (v), we get
$x = 9, y = 4$ when $x - y = 5$
or $x = 4, y = 9$ when $x - y = -5$.
Thus, the remaining observations are 4 and 9.
Required difference = 5.

**24.** **(18)** $\text{Var}(1, 2, ....., n) = 10$

$$\Rightarrow \frac{1^2 + 2^2 + ..... + n^2}{n} - \left(\frac{1 + 2 + ..... + n}{n}\right)^2 = 10$$

$$\Rightarrow \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2 = 10$$

$\Rightarrow n^2 - 1 = 120$

$\Rightarrow n = 11$

$\text{Var}(2, 4, 6, ....., 2m) = 16 \Rightarrow \text{Var}(1, 2, ....., m) = 4$

$\Rightarrow m^2 - 1 = 48 \Rightarrow m = 7$

$\Rightarrow m + n = 18$

**25.** **(52)** Mean

$$= \overline{x} = \frac{3 + 7 + 9 + 12 + 13 + 20 + x + y}{8} = 10$$

$\Rightarrow x + y = 16$ ...(i)

$$\text{Variance} = \sigma^2 = \frac{\Sigma(x_i)^2}{8} - (\overline{x})^2 = 25$$

$$\sigma^2 = \frac{9 + 49 + 81 + 144 + 169 + 400 + x^2 + y^2}{8} - 100 = 25$$

$\Rightarrow x^2 + y^2 = 148$ ...(ii)

From eqn. (i), $(x + y)^2 = (16)^2$

$\Rightarrow x^2 + y^2 + 2xy = 256$

Using eqn. (ii), $148 + 2xy = 256$

$\Rightarrow xy = 52$

CHAPTER 15

# Probability-1

1. **(d)** Total number of outcomes
S = {(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2,4), (3, 1), (3, 2), (3, 3), (3, 4), (4, 1), (4, 2), (4, 3), (4, 4)}
n(S) = 16
Number of favourable outcomes
E = {(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3), (4, 4)}
n(E) = 7
Required probability $= \frac{n(E)}{n(S)} = \frac{7}{16}$

2. **(b)** Given equation
$x + \frac{100}{x} > 50 \Rightarrow x^2 - 50x + 100 > 0$
$\Rightarrow (x - 25)^2 > 525$
$\Rightarrow x - 25 < -\sqrt{(525)}$ or $x - 25 > \sqrt{(525)}$
$\Rightarrow x < 25 - \sqrt{(525)}$ or $x > 25 + \sqrt{(525)}$
As x is positive integer and $\sqrt{(525)} = 22.91$, we must have $x \le 2$ or $x \ge 48$
Let E be the event for favourable cases and S be the sample space.
$\therefore E = \{1, 2, 48, 49, ......100\}$
$\therefore n(E) = 55$ and $n(S) = 100$
Hence the required probability
$P(E) = \frac{n(E)}{n(S)} = \frac{55}{100} = \frac{11}{20}$.

3. **(b)** Total number of combinations of numbers on the cube and the tetrahedron = 6 × 4 = 24
Favourable number of ways of getting a sum not less than 5
= sum of coefficients of $x^6, x^7, .... x^{10}$ in the product
$= (x + x^2 + x^3 + x^4 + x^5 + x^6)(x + x^2 + x^3 + x^4)$
$= (x^2 + 2x^3 + 3x^4 + 4x^5 + 4x^6 + 4x^7 + 3x^8 + 2x^9 + x^{10})$
$= 4 + 4 + 4 + 3 + 2 + 1 = 18$
$\therefore$ Required probability $= \frac{18}{6 \times 4} = \frac{3}{4}$

4. **(b)** Let $A = \{a_1, a_2, a_3, ........... a_n\}$. For any $a_i \in A$, $1 \le i \le n$, we have following choices
(1) $a_i \in P$ and $a_i \in Q$ (2) $a_i \in P$ and $a_i \notin Q$
(3) $a_i \notin P$ and $a_i \in Q$ (4) $a_i \notin P$ and $a_i \notin Q$
So, for each $a_i \in A$, there are four possibilities.
$\therefore$ Total no. of cases $= 4 \times 4 \times ............ \times 4$ ($n$ times) $= 4^n$
Further out of above four possibilities, first three satisfy $a_i \in P \cup Q$
So, the number of cases, when exactly $r$ element of $A$ belong to $P \cup Q = {}^nC_r(3)^r$
$\therefore$ Required probability $= \frac{{}^nC_r(3)^r}{4^n}$

5. **(a)** Since the chairs are numbered, so for counting of total number of cases it is equivalent to linear permutation. Hence, total number of cases = 10!
If two particular persons A and B sit together then the total number of linear arrangements = 2! 9!. Consider one of such arrangements in which the arrangement started at chair 1 ($C_1$) and ends at chair 10 ($C_{10}$).
$C_1 - C_2 - C_3 - ......... - C_9 - C_{10}$
If two persons sit at $C_1$ and $C_{10}$ then it will lead to 2! 8! new arrangements. So the favourable number of cases = 2! 9! + 2! 8! = 2!8! (10)
$\therefore$ Probability $= \frac{2!8!(10)}{10!} = \frac{2}{9}$

6. **(b)** Total ways of distribution $= 4^5 \Rightarrow n(S) = 4^5$
Total ways of distribution so that each child get atleast one game $= 4^5 - {}^4C_1 3^5 + {}^4C_2 2^5 - {}^4C_3$
$= 1024 - 4 \times 243 + 6 \times 32 - 4 = 240$
$n(E) = 240$
required probablity $= \frac{n(E)}{n(S)} = \frac{240}{4^5} = \frac{15}{64}$.

7. **(d)** A chess board is a square divided into 64 equal squares.
In 1st diagonal we have only 1 square
In 2nd diagonal we have only 2 squares
In 3rd diagonal we have 3 squares so selection can be done in ${}^3C_3$ ways
In 4 diagonal we have 4 squares so selection can be done in $C_3{}^4$ ways
And so on
Hence the total number of ways in which 3 squares can be chosen
$2({}^3C_3 + {}^4C_3 + {}^5C_3 + {}^6C_3 + {}^6C_3 + {}^7C_3) + {}^8C_3$
[Note that we do not have $2 \cdot {}^8C_3$]
Hence the total number of favourable ways $m$

$= 4({}^3C_3 + {}^4C_3 + {}^5C_3 + {}^6C_3 + {}^7C_3) + 2 \cdot {}^8C_3 = 392.$

And total number of ways $= \dfrac{64 \cdot 63 \cdot 62}{1 \cdot 2 \cdot 3}\, {}^{64}C_3$

$= 32 \cdot 21 \cdot 64$

Hence the required probability

$= \dfrac{m}{n} = \dfrac{392}{32 \cdot 21 \cdot 62} = \dfrac{7}{744}$

**8.** **(c)** The total numbers of arrangements is

$\dfrac{11!}{2!2!2!} = \dfrac{11!}{8}$

The number of arrangements in which *C, E, H, I, S* appear in that order ${}^{11}C_5 \dfrac{6!}{2!2!2!} = \dfrac{11!}{8 \cdot 5!}$

$\therefore$ Required Probability $= \dfrac{\frac{11!}{8 \cdot 5!}}{\frac{11!}{8}} = \dfrac{1}{120}$

**9.** **(c)** Player $A$ can win if $A$ throws (1, 6) or (6, 1) and $B$ throws ((1, 1), (2, 2), (3, 3), (4, 4), (5, 5) or (6, 6)). Thus the number of ways is 12.

Similarly the number of ways in which $B$ can win is 12.

Total number of ways in which either $A$ wins or $B$ wins = 24.

Thus the number of ways in which none of the two wins = $6^4 - 24$.

$\therefore$ The required probability $= \dfrac{6^4 - 24}{6^4} = \dfrac{53}{54}$.

**10.** **(c)**

| Value | 50*p*. | 25*p*. | 10*p*. |
|---|---|---|---|
| No. | 2 | 5 | 15 |

5 coins out of 22 can be chosen in ${}^{22}C_2$ ways. Let $x$ be the no. of 50 $P$ coins selected. $y$ be the no. of 25*p*. coins selected and $z$ be the no. of 10 *p*. coins selected. We desire that $50x + 25y + 10z < 150$ or $10x + 5y + 2z < 30$ where $0 \le x \le 2$; or $0 \le y \le 5$, $0 \le z \le 14$ and $x + y + z = 5$. Thus following cases are impossible.

**Case (i) :** $x = 2, y = 2, z = 1$

No. of ways of such selections

$= {}^2C_2 \times {}^5C_2 \times {}^{15}C_1 = 1 \times 10 \times 15 = 150.$

**Case (ii) :** $x = 1, y = 4, z = 0$

No. of ways of such selections $= {}^2C_1 \times {}^5C_4 = 10.$

**Case (iii) :** $x = 2, y = 3, z = 0$. No. of ways of such selection $= {}^2C_2 \times {}^5C_3 \times {}^{15}C_0 = 1 \times 10 = 10.$

$\therefore$ Total $= 150 + 10 + 10 = 170$ ways are to be rejected.

$\therefore$ Required probability $= 1 - \dfrac{170}{{}^{22}C_5}$.

**11.** **(b)** $\log_a b = \dfrac{\log b}{\log a}$

Let $b = 2^m$ and $a = 2^n$ where $m$ and $n$ denote the exponents on the base 2 in the given set then

$\log_a b = \dfrac{m}{n}$

Therefore, $\log_a b$ is an integer only if $n$ divides $m$.

Now, total no. of ways $m$ and $n$ can be chosen $= 25 \times 24 = 600$.

For favourable cases

Let $n = 1$, So $m$ can take values $1, 2, 3, 4, 5, 6, ..., 24$ $= 24$

if $n = 2$ $m = 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24$ $= 11$

$n = 3$ $m = 6, 9, 12, 15, 18, 21, 24$ $= 7$

$n = 4$ $m = 8, 12, 16, 20, 24$ $= 5$

$n = 5$ $m = 10, 15, 20, 25$ $= 4$

$n = 6$ $m = 12, 18, 24$ $= 3$

$n = 7$ $m = 14, 21$ $= 2$

$n = 8$ $m = 16, 24$ $= 2$

$n = 9, 10, 11, 12$ $m = 1$ for each $= 4$

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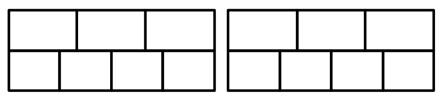
Hence, required probability $= \dfrac{62}{600} = \dfrac{31}{300}$.

**12.** **(d)**



$n(S)$ = Number of total ways

$= {}^{14}P_{12} = \dfrac{14!}{2!} = 7 \times 13!$

The girls can be seated together in the back seats leaving a corner seat in $4 \times 3! = 24$ ways and the boys can be seated in the remaining 11 seats in

${}^{11}P_9 = \dfrac{11!}{2!} = \dfrac{1}{2} \times 11!$ ways

$\therefore$ $n(E)$ = Number of favourable ways

$= 24 \times \dfrac{1}{2} \times 11! = 12!$

The required probability $= \dfrac{n(E)}{n(S)} = \dfrac{12!}{7 \times 13!} = \dfrac{1}{91}$

**13.** **(a)** We know that $P$ (exactly one of $A$ or $B$ occurs) $= P(A) + P(B) - 2P(A \cap B)$.

Therefore, $P(A) + P(B) - 2P(A \cap B) = p$ ...(i)

Similarly, $P(B) + P(C) - 2P(B \cap C) = p$ ...(ii)

and $P(C) + P(A) - 2P(C \cap A) = p$ ...(iii)

Adding (i), (ii) and (iii) we get

$$2[P(A)+P(B)+P(C)-P(A\cap B)-P(B\cap C)-P(C\cap A)]=3p$$

$$\Rightarrow P(A)+P(B)+P(C)-P(A\cap B)-P(B\cap C)-P(C\cap A)=3p/2 \ldots \text{(iv)}$$

We are also given that,

$$P(A\cap B\cap C)=p^2 \quad \ldots \text{(v)}$$

Now, $P$ (at least one of $A$, $B$ and $C$)

$$=P(A)+P(B)+P(C)-P(A\cap B)-P(B\cap C)-P(C\cap A)+P(A\cap B\cap C)$$

$$=\frac{3p}{2}+p^2=\frac{3p+2p^2}{2} \quad \text{[using (iv) and (v)]}$$

**14.** **(d)** Since, one and only one of the three events $E_1$, $E_2$ and $E_3$ can happen, therefore $P(E_1)+P(E_2)+P(E_3)=1$ .....(i)

$\because$ Odds against $E_1$ are 7 : 4

$$\Rightarrow P(E_1)=\frac{4}{4+7}=\frac{4}{11} \quad \text{.....(ii)}$$

$\because$ Odds against $E_2$ are 5 : 3

$$\Rightarrow P(E_2)=\frac{3}{3+5}=\frac{3}{8} \quad \text{......(iii)}$$

From (i), (ii) and (iii), we have,

$$\frac{4}{11}+\frac{3}{8}+P(E_3)=1.$$

$$\Rightarrow P(E_3)=1-\frac{4}{11}-\frac{3}{8}=\frac{88-32-33}{88}$$

$$=\frac{23}{88}=\frac{23}{23+65}$$

Hence odds against $E_3$ are 65 : 23.

**15.** **(a)** Let $p_1$, $p_2$ be the chances of happening of the first and second event, respectively; then, according to the given conditions, we have

$$p_1=p_2^2 \text{ and } \frac{1-p_1}{p_1}=\left(\frac{1-p_2}{p_2}\right)^3$$

$$\Rightarrow \frac{1-p_2^2}{p_2^2}=\left(\frac{1-p_2}{p_2}\right)^3 \Rightarrow p_2(1+p_2)=(1-p_2)^2$$

$$\Rightarrow 3p_2=1 \Rightarrow p_2=\frac{1}{3} \text{ and so } p_1=\frac{1}{9}.$$

**16.** **(c)** Let P(A) = probability that Ashmit will pass, and P(B) = probability that Bishmit will pass, then from the given information

$P(A)=0.5,\ 0.1\le P(A\cap B)\le 0.3$.

We have to find the value of P(B).

Since $P(A\cup B)=P(A)+P(B)-P(A\cap B)$

Or $P(B)=P(A\cup B)+P(A\cap B)-P(A)$

For maximum value of P(B) both $P(A\cup B)$ and $P(A\cap B)$ should be maximum, maximum value of $P(A\cup B)$ is 1 and maximum value of $P(A\cap B)$ = is 0.3.

Hence $P(B)=1+0.3-0.5=0.8$.

Similarly For minimum value of P(B) both $P(A\cup B)$ and $P(A\cap B)$ should be minimum, minimum value of $P(A\cup B)=P(A)=0.5$ and minimum value of $P(A\cap B)$ = is 0.1.

Hence $P(B)=0.5+0.1-0.5=0.1$.

So required range is (0.1, 0.8).

**17.** **(a)** Total number of outcomes $={}^{52}C_4=270725$

$E$ = Event of exactly two spade cards and exists two aces.

$A$ = Event of 1 spade ace, 1 non-spade ace and 1 spade card and 1 non-spade card

$B$ = Event of 2 non-spade aces and 2 spade cards.

$$P(A)\frac{1({}^3C_1)({}^{12}C_1)({}^{36}C_1)}{{}^{52}C_4}=\frac{3\times 12\times 36}{270725}=\frac{1296}{270725}$$

$$P(B)=\frac{{}^3C_2\times{}^{12}C_2}{{}^{52}C_4}=\frac{3\times 66}{270725}=\frac{198}{270725}$$

Now, $E=A\cup B$ and $A\cap B=\phi$. This implies

$$P(E)=P(A\cup B)=P(A)+P(B)=\frac{1494}{270725}.$$

**18.** **(0.80)** Total no. of arrangements of the letters of the word UNIVERSITY is $\frac{10!}{2!}$.

No. of arrangements when both I's are together = 9!

So. the no. of ways in which 2 I's do not together

$$=\frac{10!}{2!}-9!$$

$$\therefore \text{ Required probability}=\frac{\frac{10!}{2!}-9!}{\frac{10!}{2!}}=\frac{10!-9!\,2!}{10!}$$

$$=\frac{10\times 9!-9!.2!}{10!}=\frac{9![10-2]}{10\times 9!}=\frac{8}{10}=\frac{4}{5}=0.80$$

**19.** **(0.50)** Total number of numbers = 4! = 24

For odd nos. 1 or 3 has to be at unit's place

If 1 is at unit place, then total number of numbers = 3! = 6

And if 3 is at unit place, then total number of numbers = 3! = 6

$\therefore$ Total number of odd number = 6 + 6 = 12

$$\therefore \text{ Required probability}=\frac{12}{24}=\frac{1}{2}=0.50$$

**20.** **(0.33)** Total number of cases = ${}^{3n}C_2$

$= \frac{3n(3n-1)}{2}$

Now $x^3 + y^3 = (x+y)\ (x^2 - xy + y^2)$

$x^3 + y^3$ is divisible by 3 if 3 divdes $x + y$ or divides $x^2 - xy + y^2$ we arrange the $3n$ numbers in 3 sequences.

$A : \{1, 4, 7, ............, 3n-2\}$

$B : \{2, 5, 8, .............., 3n-1\}$

$C : \{3, 6, 9, ............., 3n\}$

Clearly we must choose either one number from the first sequence and other number from the second sequence or both numbers from the third sequence only.

$\therefore$ Number of favourable cases

$= n \times n + {}^nC_2 = n^2 + \frac{n(n-1)}{2} = \frac{n}{2}(3n-1)$

$\therefore$ Required probability $= \dfrac{\frac{n}{2}(3n-1)}{\frac{3n(3n-1)}{2}} = \frac{1}{3} = 0.33$

**21.** **(3)** Total no. of arrangements = 15!

Extreme chairs are occupied by girls, thus there are four gaps among 5 girls where boys can be seated. Let the number of boys in these four gaps be $2x+1, 2y+1, 2z+1$ and $2t+1$, then

$2x + 1 + 2y + 1 + 2z + 1 + 2t + 1 = 10$

$\Rightarrow\ x + y + z + t = 3$

Where $x, y, z, t$ are integers and

$0 \le x \le 3,\ 0 \le y \le 3,\ 0 \le z \le 3,\ 0 \le t \le 3$

$\therefore$ The number of ways of selecting positions for boys

= coefficient of $x^3$ in $(1 + x + x^2 + x^3)^4$

= coefficient of $x^3$ in $\left(\frac{1-x^4}{1-x}\right)^4$

= coefficent of $x^3$ in $(1 - x^4)^4 (1 - x)^{-4} = {}^6C_3 = 20$

$\therefore$ Number of arrangements of boys and girls with given condition = 20 × 10 ! × 5 !

$\therefore$ Required probability $= \frac{20 \times 10! \times 5!}{15!} = \frac{20}{3003}$

$\Rightarrow \frac{\text{n}}{1001} = \frac{3003}{1001} = 3$

**22.** **(3)** Let $\alpha$ and $\beta$ be the roots of the quadratic equation. According to question,

$\alpha + \beta = \alpha^2 + \beta^2$ and $\alpha\beta = \alpha^2\beta^2$

$\Rightarrow \alpha\beta(\alpha\beta - 1) = 0 \ \Rightarrow \alpha\beta = 1$ or $\alpha\beta = 0$

$\Rightarrow \alpha = 1, \beta = 1;\ \alpha = \omega,\ \beta = \omega^2$

[Cube roots and unity]

$\alpha = 1,\ \beta = 0,\ \alpha = 0,\ \beta = 0$

$\therefore n(S) =$ Number of quadratic equations which are unchanged by squaring their roots = 4

and $n(E)$ = Number of quadratic equations have equal roots = 2.

$\therefore$ Required probability $= \frac{n(E)}{n(S)} = \frac{2}{4} = \frac{1}{2} = \frac{p}{q}$.

$\Rightarrow\ \ p + q = 3.$

**23.** **(4)** The number of ways to answer a question $= 2^5 - 1 = 31.$

i.e. In 31 ways only one correct.

Let number of choices = $n$

Now, according to the question $\frac{n}{31} > \frac{1}{8}$

$\Rightarrow n > \frac{31}{8} \Rightarrow n > 3.8$ $\quad \therefore$ Least value of $n = 4$.

**24.** **(5)** Let $S$ be the sample space and $E$ be the event of getting a large number than the previous number.

$\therefore n(S) = 6 \times 6 \times 6 = 216$

Now, we count the number of favourable ways. Obviously, the second number has to be greater than 1. If the second number is $i (i > 1)$, then the number of favourable ways = $(i - 1) \times (6 - i)$

$\therefore\ n(E)$ = Total number of favourable ways

$= \sum_{i=1}^{6} (i-1) \times (6-i)$

$= 0 + 1 \times 4 + 2 \times 3 + 3 \times 2 + 4 \times 1 + 5 \times 0$

$= 4 + 6 + 6 + 4 = 20.$

Therefore, the required probability,

$P(E) = \frac{n(E)}{n(S)} = \frac{20}{216} = \frac{5}{54} = p$ (given)

$\therefore 54p = 5$

**25.** **(5)** Let $A$ be the event of a boy and $B$ the event of having cat eyes. So

$P(A) = \frac{20}{40} = \frac{1}{2}$ and $P(B) = \frac{20}{40} = \frac{1}{2}$

Now, $P(A \cap B) = \frac{10}{40} = \frac{1}{4}$

$\therefore\ \ P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4} = \frac{a}{b}$

$= \sqrt{a^2 + b^2} = \sqrt{3^2 + 4^2} = 5$

CHAPTER 16

# Relations & Functions-2

**1.** **(d)** P = $\{(a,b): \sec^2 a - \tan^2 b = 1\}$

**For reflexive :**

$\sec^2 a - \tan^2 a = 1$ (true $\forall\, a$)

**For symmetric :**

$\sec^2 b - \tan^2 a = 1$

L.H.S. $1 + \tan^2 b - (\sec^2 a - 1)$

$= 1 + \tan^2 b - \sec^2 a + 1$

$= -(\sec^2 a - \tan^2 b) + 2 = -1 + 2 = 1$

So, Relation is symmetric.

**For transitive :**

if $\sec^2 a - \tan^2 b = 1$ and $\sec^2 b - \tan^2 c = 1$

$\sec^2 a - \tan^2 c = (1 + \tan^2 b) - (\sec^2 b - 1)$

$= -\sec^2 b + \tan^2 b + 2 = -1 + 2 = 1$

So, relation is transitive.

Hence, relation P is an equivalence relation.

**2.** **(b)** **Reflexivity :** $xRx$ as $x = 1 \,.\, x$ and 1 is a rational number.

$\frac{m}{n} S \frac{m}{n}$ iff $m \,.\, n = m \,.\, n$, which is true.

Thus, $R$ and $S$ are reflexive relations.

**Symmetricity :** Let $xRy$

$\Rightarrow x = wy$ for some rational number $w$

$\Rightarrow y = \frac{1}{w} x$

$\Rightarrow yRx$ provided $w \neq 0$, but if $x = 0$ and $y$ is any non-zero integer, then $x = 0y$ and there exist no rational number $w$ for which $y = wx$.

$\Rightarrow y \not R x$ thus $R$ is not symmetric.

Now, let $\frac{m}{n} S \frac{p}{q}$

$\Rightarrow mq = pn \Rightarrow pn = mq \Rightarrow \frac{p}{q} S \frac{m}{n}$

$\Rightarrow S$ is a symmetric relation.

**Transitivity :** Let $xRy$ and $yRz$.

$\Rightarrow x = w_1 y$ and $y = w_2 z$ for some $w_1, w_2 \in \mathbf{Q}$

$\Rightarrow x = w_1(w_2 z) = (w_1 \cdot w_2) z$

$\Rightarrow xRz$ as $w_1 \cdot w_2 \in \mathbf{Q}$

Now let $\frac{m}{n} S \frac{p}{q}$ and $\frac{p}{q} S \frac{r}{t}$; where $n, q, t \neq 0$.

**3.** **(a)** $x \in R \Rightarrow x - x + \sqrt{5} = \sqrt{5}$ is an irrational number.

$\therefore (x, x) \in R$

So, $R$ is reflexive.

$(\sqrt{5}, 1) \in R$ because $\sqrt{5} - 1 + \sqrt{5} = 2\sqrt{5} - 1$ which is an irrational number.

But $(1, \sqrt{5}) \notin R$

$\therefore$ $R$ is not symmetric.

We have, $(\sqrt{5}, 1), (1, 2\sqrt{5}) \in R$ because

$\sqrt{5} - 1 + \sqrt{5} = 2\sqrt{5} - 1$

If $1 - 2\sqrt{5} + \sqrt{5} = 1 - \sqrt{5}$ are irrational numbers.

Also, $(\sqrt{5}, 2\sqrt{5}) \in R$ and $\sqrt{5} - 2\sqrt{5} + \sqrt{5} = 0$ which is not an irrational numbers.

$\therefore (\sqrt{5}, 2\sqrt{5}) \notin R$

So, $R$ is not transitive.

**4.** **(a)** $f(x)$ is onto

$\therefore$ $S$ = range of $f(x)$.

Now $f(x) = \sin x - \sqrt{3}\cos x + 1$

$= 2\sin\left(x - \frac{\pi}{3}\right) + 1$

$\because -1 \leq \sin\left(x - \frac{\pi}{3}\right) \leq 1$

$-1 \leq 2\sin\left(x - \frac{\pi}{3}\right) + 1 \leq 3$

$\therefore f(x) \in [-1, 3] = S$.

5. (b)

6. (c) Let $f(x) \neq 2$ be true and $f(y) = 2, f(z) \neq 1$ are false
$\Rightarrow f(x) \neq 2, f(y) \neq 2, f(z) = 1$
$\Rightarrow f(x) = 3, f(y) = 3, f(z) = 1$
but then function is many one, similarly two other cases.

7. (c) Total number of functions from $A \to B$ $= 3^4 = 81$ and number of onto mappings
$=$ coefficient of $x^4$ in $4!\,(e^x - 1)^3$.
$=$ coefficient of $x^4$ in $4!\,(e^{3x} - 3e^{2x} + 3e^{2x} + 3e^x - 1)$
$$= 4!\left(\frac{3^4}{4!} - \frac{3.2^4}{4!} + \frac{3.1^4}{4!} - 0\right)$$
$= 81 - 48 + 3 = 81 - 45 = 36$
$\therefore$ Number of functions from $A \to B$, which are not onto is $81 - 36 = 45$.

8. (b) Let $f(x) = ax^2 + bx + c$ as it touches $X$-axis at $x = 3$
$$\Rightarrow \quad \frac{-b}{2a} = 3$$
$$\Rightarrow \quad b = -6a \qquad ...(1)$$
Also, $9a + 3b + c = 0 \qquad ...(2)$
$4a - 2b + c = 3 \qquad ...(3)$
$$\Rightarrow a = \frac{3}{25},\ b = -\frac{18}{25},\ c = \frac{27}{25}$$
$$\Rightarrow f(x) = \frac{3}{25}(x^2 - 6x + 9)$$

9. (b) $f(x) = \text{sgn}(x - x^4 + x^7 - x^8 - 1)$
For $x \in (0, 1); x - 1 < 0,\ x^7 - x^4 < 0$
$\therefore x - x^4 + x^7 - x^8 - 1 < 0$
For $x \in (1, \infty);\ x < x^4,\ x^7 < x^8$
$\therefore x - x^4 + x^7 - x^8 - 1 < 0$
Also for $x = 1;\ x - x^4 + x^7 - x^8 - 1 = -1$
Thus $x - x^4 + x^7 - x^8 - 1 < 0$ for all $x \in R^+$
Hence $\text{sgn}(x - x^4 + x^7 - x^8 - 1) = -1 \forall x \in R^+$.
Therefore $f(x)$ is many-one and into.

10. (d) We have, $f(x) = x + \sqrt{x^2} = x + |x|$
Clearly, $f$ is not onto as
$$f(-1) = f(-2) = 0 \text{ but } -1 \neq -2$$
Also, $f$ is not onto as $f(x) \geq 0,\ \forall x \in R$.
$\therefore$ Range of $f = (0, \infty) \subset R$.

11. (b) $f(x) = \sin x + \cos x,\ g(x) = x^2 - 1$
$\Rightarrow \quad g(f(x)) = (\sin x + \cos x)^2 - 1 = \sin 2x$
Clearly $g(f(x))$ is invertible in $-\frac{\pi}{2} \leq 2x \leq \frac{\pi}{2}$
[$\because$ $\sin\theta$ is invertible when $-\pi/2 \leq \theta \leq \pi/2$]
$$\Rightarrow \quad -\frac{\pi}{4} \leq x \leq \frac{\pi}{4}.$$

12. (b) Here
$$g^2(x) = gog(x) = g\{g(x)\} = g(3 + 4x)$$
$$= 15 + 4^2 x = (4^2 - 1) + 4^2 x$$
$$g^3(x) = gogog(x) = g(15 + 4^2 x) = 3 + 4(15 + 4^2 x)$$
$$= 63 + 4^3 x = (4^3 - 1) + 4^3 x$$
Generalizing, we get
$$g^n(x) = (4^n - 1) + 4^n x = y \text{ (say)}$$
then $x = (y + 1 - 4^n)4^{-n}$
$$\Rightarrow \quad g^{-n}(y) = (y + 1)4^{-n} - 1$$
$$\therefore \quad g^{-n}(x) = (x + 1)4^{-n} - 1$$

13. (d) $f(x) = \sqrt{3|x| - x - 2}$ and $g(x) = \sin x$
for $\text{fog}(x) = \sqrt{3|\sin x| - \sin x - 2}$ which is defined if
$3|\sin x| - \sin x - 2 \geq 0$
If $\sin x > 0$ then $2\sin x - 2 \geq 0 \Rightarrow \sin x \geq 1$
$$\Rightarrow \sin x = 1 \Rightarrow x = 2n\pi + \frac{\pi}{2}$$
If $\sin x < 0$ then $-4\sin x - 2 \geq 0$
$$\Rightarrow -1 \leq \sin x \leq -\frac{1}{2}$$
$$\Rightarrow x \in \left[2n\pi + \frac{7\pi}{6},\ 2n\pi + \frac{11\pi}{6}\right]$$

$x \in \left[2n\pi + \frac{7\pi}{6}, 2n\pi + \frac{11\pi}{6}\right] \cup \left\{2m\pi + \frac{\pi}{2}\right\}, n, m \in I$ .

**14.** **(b)** $f(x) = \sin^2 x + \left(\frac{\sqrt{3}}{2}\cos x + \frac{1}{2}\sin x\right)^2$

$$+\cos x\left(\frac{1}{2}\cos x - \frac{\sqrt{3}}{2}\sin x\right)$$

$$= \sin^2 x + \frac{3}{4}\cos^2 x + \frac{1}{4}\sin^2 x + \frac{\sqrt{3}}{2}\cos x \sin x$$

$$+\frac{1}{2}\cos^2 x - \frac{\sqrt{3}}{2}\cos x \sin x$$

$$= \frac{5}{4}(\sin^2 x + \cos^2 x) = \frac{5}{4}$$

Now, $y = g(f(x)) = g\left(\frac{5}{4}\right) = 1.$

Clearly $y = 1$ is a straight line.

**15.** **(a)** Given that $f(x) = x^2$ and $g(x) = \sin x$, $\forall x \in R$.
Then (gof) $(x) = \sin x^2$
$\Rightarrow$ (gogof) $(x) = \sin(\sin x^2)$
$\Rightarrow$ (fogogof) $(x) = \sin^2(\sin x^2)$
As given that (fogogof) $(x)$ = (gogof) $(x)$
$\Rightarrow \sin^2(\sin x^2) = \sin(\sin x^2)$
$\Rightarrow \sin(\sin x^2) = 0, 1$

$\Rightarrow \sin x^2 = n\pi$ or $((4n+1)\frac{\pi}{2}$ where $n \in Z$

$\Rightarrow \sin x^2 = 0 \quad \because \sin x^2 \in [-1, 1]$

$\Rightarrow x^2 = n\pi \quad \Rightarrow \quad x = \pm\sqrt{n\pi}$ where $n \in W$

**16.** **(b)** Let $h(x) = g(f(x))$ where, $g(x)$ is injective and $h(x)$ is surjective.
$\therefore$ Codomain of $h(x)$ = Range of $h(x)$.
Range of $h(x) = [0, 2]$
$\Rightarrow$ codomain of $g$ is [0, 2]. So, $g(x)$ must be surjective.
Now, domain of $g = [-1, 1]$ which must be range of $f$.
But codomain of $f = [-1, 1]$
$\Rightarrow f$ must be surjective.

**17.** **(d)** We have, $-1 < f(0) < f(1) < 1$

$g = [-1, 1] \to [0, 1]$

and $gof(x) = x, \forall x \in [0, 1]$

Only condition that $g(x)$ should satisfy for $gof(x) = x, \forall x \in [0, 1]$ is that $g(x)$ should attain all values in [0, 1] when rangte of $f(x)$ a subset of (–1, 1) is used as image for $g(x)$. Thus, there can be infinite such functions $g(x)$ with domain [–1, 1] and range [0, 1]

**18.** **(14)** If set $A$ has $m$ elements and set $B$ has n elements then number of onto functions from $A$ to $B$ is

$$\sum_{r=1}^{n}(-1)^{n-r}\ {}^nC_r\, r^m \quad \text{where } 1 \le n \le m$$

Here $E = \{1, 2, 3, 4\}$, $F = \{1, 2\}$

$m = 4, n = 2$

$\therefore$ No. of onto functions from E to F

$$= \sum_{r=1}^{2}(-1)^{2-r}\ {}^2C_r (r)^4$$

$$= (-1)\ {}^2C_1 + {}^2C_2 (2)^4 = -2 + 16 = 14.$$

**19.** **(0)** $f(x) = (a - x^n)^{1/n}, x > 0$

$\Rightarrow f(f(x)) = [a - \{(a - x^n)^{1/n}\}^n]^{1/n} = x$

Also $g(x) = x^2 + px + q$

$\therefore g(x) - x = 0$ is quadratic equation

$x^2 + (p-1)x + q = 0$

Given that this equation has imaginary roots

$\therefore x^2 + (p-1)x + q > 0$ for all real $x$

[$\therefore$ coefficient of $x^2 = 1 > 0$]

$\therefore g(x) - x > 0$ for all real $x$

$\Rightarrow g(g(x)) - g(x) > 0 \ \forall \ x \in R$

Now $g(g(x)) - f(f(x)) = g(g(x)) - x$

$= [g(g(x)) - g(x)] + [g(x) - x] > 0 \ \forall \ x \in R$

$\therefore$ The equation $g(g(x)) - f(f(x)) = 0$ has no real root.

**20.** **(1)** $g(x) = 1 + x - [x]$;

$$f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$$

For integral values of $x$; $g(x) = 1$.

For $x < 0$; (but not integral value) $x - [x] > 0 \Rightarrow g(x) > 1$.

For $x > 0$; (but not integral value) $x - [x] > 0$

$\Rightarrow g(x) > 1$

$\therefore\ g(x) \ge 1, \forall x \ \therefore\ f(g(x)) = 1, \forall x.$

**21.** **(6)** Let $y = g(x) = \dfrac{e^x - e^{-x}}{2}$

$\Rightarrow e^{2x} - 1 = 2ye^x$

Substituting $e^x = t$, we have $t^2 - 2yt - 1 = 0$

$$\Rightarrow t = \frac{2y \pm \sqrt{4y^2 + 4}}{2} = y \pm \sqrt{y^2 + 1}$$

$$\Rightarrow e^x = y + \sqrt{y^2 + 1} \qquad (\because e^x > 0)$$

$$\Rightarrow x = \ln(y + \sqrt{y^2 + 1})$$

$$\Rightarrow g^{-1}(y) = \ln(y + \sqrt{y^2 + 1})$$

$$\therefore g^{-1}(x) = f(x) = \ln(x + \sqrt{x^2 + 1})$$

$$(\because g(f(x)) = x \Rightarrow g^{-1}(x) = f(x))$$

$$\therefore f\left(\frac{e^{22} - 1}{2e^{11}}\right) = \ln\left(\frac{e^{22} - 1}{2e^{11}} + \sqrt{\left(\frac{e^{22} - 1}{2e^{11}}\right)^2 + 1}\right)$$

$$= \ln\left(\frac{e^{22} - 1}{2e^{11}} + \sqrt{\left(\frac{e^{22} + 1}{2e^{11}}\right)^2}\right) = \ln[e^{11}] = 11.$$

**22.** **(4)** $f^{-1}(x) = \begin{cases} -x + 1, & x \ge 1 \\ 1 + \sqrt{-x}, & x \le 0 \end{cases}$ the solution of $f(x) - f^{-1}(x) = 0$ are $x = -1, 1, 0, 2$.

**23.** **(2)** $f(x) = \cos^2 x + \cos^2\left(\dfrac{\pi}{3} + x\right)$

$$-\cos x \cdot \cos\left(\frac{\pi}{3} + x\right); g\left(\frac{3}{4}\right) = 2;$$

$gof(x) = ?$

$$f(1) = \cos^2 1 + \cos^2\left(\frac{\pi}{3} + 1\right) - \cos 1 \cos\left(\frac{\pi}{3} + 1\right)$$

$$\Rightarrow f(1) = \cos^2 1 + \cos\left(\frac{\pi}{3} + 1\right) \cdot \left[\cos\left(\frac{\pi}{3} + 1\right) - \cos 1\right]$$

$$\Rightarrow f(1) = \cos^2 1 + \cos\left(\frac{\pi}{3} + 1\right) \cdot \left[-2\sin\left(\frac{\pi}{6} + 1\right)\sin\frac{\pi}{6}\right]$$

$$\Rightarrow f(1) = \cos^2 1 - \cos\left(\frac{\pi}{3} + 1\right)\sin\left(\frac{\pi}{6} + 1\right)$$

$$\Rightarrow f(1) = \cos^2 1 - \frac{1}{2}\left[\sin\left(\frac{\pi}{2} + 2\right) + \sin\left(-\frac{\pi}{6}\right)\right]$$

$$\Rightarrow f(1) = \cos^2 1 - \frac{1}{2}\left[\cos 2 - \frac{1}{2}\right]$$

$$= \cos^2 1 - \frac{1}{2}(2\cos^2 1 - 1) + \frac{1}{4} = \frac{3}{4}$$

$$\therefore gof(1) = f(f(1)) = f(3/4) = 2.$$

**24.** **(2)** Since, $f(x)$ is symmetric about $y = x$.

$$\Rightarrow f^{-1}(x) = f(x)$$

$$\therefore f^2(x) = (f^{-1}(x))^2 - px \cdot f(x) f^{-1}(x) + 2x^2 f(x).$$

$$\Rightarrow f^2(x) = f^2(x) - px \cdot f(x) \cdot f(x) + 2x^2 f(x)$$

$$\Rightarrow f(x) \cdot \{2x^2 - px \cdot f(x)\} = 0$$

$$\Rightarrow f(x) = \frac{2x^2}{px} \qquad [\text{as } f(x) \ne 0]$$

$$\Rightarrow f(x) = \frac{2x}{p} \Rightarrow p = 2.$$

**25.** **(7)** Here, $f(1) = 1$

$f(2) = f[f(1)] + f[2 - f(1)]$, using $f(1) = 1$

$f(2) = f(1) + f(1) = 2$.

$f(3) = f[f(3)] + f[3 - f(2)] = f(2) + f(1) = 2 + 1 = 3$.

Thus, $f(n) = n$

$$\therefore \frac{1}{30}\sum_{r=1}^{20} f(r) = \frac{1}{30}[1 + 2 + 3 + .... + 20]$$

$$= \frac{1}{30} \times \frac{20(20 + 1)}{2} = 7.$$

CHAPTER 17

# Inverse Trigonometric Functions

**1.** **(c)** Let $\tan^{-1} x = \alpha$ and $\tan^{-1} y = \beta$

$\Rightarrow \tan\alpha = x, \tan\beta = y$

The given system of equations is :

$a\tan\alpha + b\sec\alpha = c$ and $a\tan\beta + b\sec\beta = c$

$\therefore$ $\alpha$ and $\beta$ are roots of $a\tan\theta + b\sec\theta = c$

$$\Rightarrow (b\sec\theta)^2 = (c - a\tan\theta)^2$$

$$\Rightarrow (a^2 - b^2)\tan^2\theta - 2ac\tan\theta + c^2 - b^2 = 0$$

$$\therefore \tan\alpha + \tan\beta = \frac{2ac}{a^2 - b^2} \text{ and}$$

$$\tan\alpha\tan\beta = \frac{c^2 - b^2}{a^2 - b^2}$$

$$\therefore x + y = \frac{2ac}{a^2 - b^2} \text{ and } xy = \frac{c^2 - b^2}{a^2 - b^2}$$

$$\Rightarrow 1 - xy = \frac{a^2 - c^2}{a^2 - b^2}$$

$$\therefore \frac{x + y}{1 - xy} = \frac{2ac}{a^2 - c^2}$$

**2.** **(c)** $f(x) = \sin^{-1}\left[x^2 + \frac{1}{2}\right] + \cos^{-1}\left[x^2 - \frac{1}{2}\right]$

$$= \sin^{-1}\left[x^2 + \frac{1}{2}\right] + \cos^{-1}\left[x^2 + \frac{1}{2} - 1\right]$$

$$= \sin^{-1}\left[x^2 + \frac{1}{2}\right] + \cos^{-1}\left(\left[x^2 + \frac{1}{2}\right] - 1\right)$$

Since $x^2 + \frac{1}{2} \geq \frac{1}{2}$, $\left[x^2 + \frac{1}{2}\right] = 0$ or 1 as

$\sin^{-1}\left[x^2 + \frac{1}{2}\right]$ is defined only for these two values.

Hence $\left[x^2 + \frac{1}{2}\right] = 0$

$$\Rightarrow f(x) = \sin^{-1} 0 + \cos^{-1}(-1) = \pi \quad \left[x^2 + \frac{1}{2}\right] = 1$$

$$\Rightarrow f(x) = \sin^{-1}(1) + \cos^{-1} 0 = \pi.$$

Therefore range of $f(x) = \{\pi\}$

**3.** **(a)** Given

$$\cos^{-1}\beta_1 + \cos^{-1}\beta_2 + .... + \cos^{-1}\beta_k = k\frac{\pi}{2}$$

We know that $\cos^{-1} x \leq \frac{\pi}{2} \quad \forall x \geq 0$

$$\therefore \cos^{-1}\beta_r \leq \frac{\pi}{2} \quad \forall r = 1, 2, 3, ...., k$$

$$\Rightarrow \sum_{r=1}^{k} \cos^{-1}\beta_r \leq \frac{k\pi}{2}$$

So the given equality holds only if

$$\cos^{-1}\beta_1 = \cos^{-1}\beta_2 = ..... = \cos^{-1}\beta_k = \frac{\pi}{2}$$

$$\Rightarrow \beta_1 = \beta_2 = .... = \beta_k = 0.$$

Thus $A = \sum_{r=1}^{k} (\beta_r)^r = 0$.

$$\text{Required limit} = \lim_{x\to 0} \frac{(1 + x^2)^{1/3} - (1 - 2x)^{1/4}}{x + x^2}$$

$$= \lim_{x\to 0} \frac{\frac{1}{3}(1 + x^2)^{-2/3}(2x) - \frac{1}{4}(1 - 2x)^{-3/4}(-2)}{1 + 2x}$$

($L$' Hospital Rule)

$$= \frac{1}{2}$$

**4.** **(c)** $\sin^{-1}(\log[x])$ is defined if $-1 \leq \log[x] \leq 1$ and $[x] > 0$

$$\Rightarrow \frac{1}{e} \leq [x] \leq e \Rightarrow [x] = 1, 2 \Rightarrow x \in [1, 3)$$

Again, $\log(\sin^{-1}[x])$ is defined if

$\sin^{-1}[x] > 0$ and $-1 \leq [x] \leq 1$

$\Rightarrow [x] > 0$ and $-1 \leq [x] \leq 1 \Rightarrow 0 < [x] \leq 1$

$\Rightarrow x \in [1, 2)$

$\therefore$ Domain of $f(x) = [1, 2)$

For $1 \leq x < 2$, $[x] = 1$

$$\therefore f(x) = \sin^{-1} 0 + \log\frac{\pi}{2} = \log\frac{\pi}{2}, \forall x \in [1, 2)$$

$\therefore$ Range of $f(x) = \left\{\log \dfrac{\pi}{2}\right\}.$

**5.** **(c)** We have

$S_1 = \Sigma x_1 = \sin 2\beta$

$S_2 = \Sigma x_1 x_2 = \cos 2\beta$

$S_3 = \Sigma x_1 x_2 x_3 = \cos\beta$

$S_4 = x_1 x_2 x_3 x_4 = -\sin\beta$

So that $\displaystyle\sum_{i=1}^{4} \tan^{-1} x_i = \tan^{-1} \frac{S_1 - S_3}{1 - S_2 + S_4}$

$$= \tan^{-1} \frac{\sin 2\beta - \cos\beta}{1 - \cos 2\beta - \sin\beta}$$

$$= \tan^{-1} \frac{\cos\beta(2\sin\beta - 1)}{\sin\beta(2\sin\beta - 1)}$$

$$= \tan \cot\beta = \tan^{-1}\left(\tan(\pi/2 - \beta)\right) = \pi/2 - \beta$$

**6.** **(d)** $f(x) = \tan^{-1}\left(\dfrac{(\sqrt{12} - 2)x^2}{x^4 + 2x^2 + 3}\right)$

$$= \tan^{-1}\left(\frac{2(\sqrt{3} - 1)}{x^2 + \dfrac{3}{x^2} + 2}\right)$$

As $x^2 + \dfrac{3}{x^2} \geq 2\sqrt{3}$ [using AM $\geq$ GM]

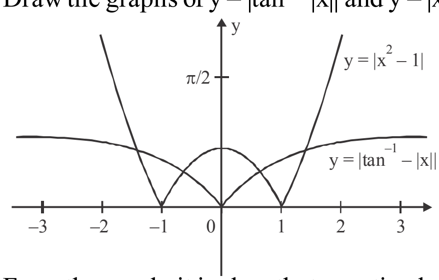
$$\Rightarrow \quad x^2 + \frac{3}{x^2} + 2 \geq 2 + 2\sqrt{3}$$

$$\therefore \quad (f(x))_{\max} = \tan^{-1}\left(\frac{2(\sqrt{3} - 1)}{2(\sqrt{3} + 1)}\right) = \frac{\pi}{12}$$

**7.** **(c)** $\sqrt{(x^2 + 1)^2 - 4x^2} = \sqrt{(x^2 - 1)^2} = |x^2 - 1|$

$\Rightarrow \quad |\tan^{-1}|x|| = |x^2 - 1|$

Draw the graphs of $y = |\tan^{-1}|x||$ and $y = |x^2 - 1|$




From the graph, it is clear that equation has four roots.

**8.** **(c)** We know that,

$$\tan^{-1}\frac{1}{1+2} + \tan^{-1}\frac{1}{1+2\times 3} + \tan^{-1}\frac{1}{1+3\times 4} + \ldots\ldots$$

$$+ \tan^{-1}\frac{1}{1+(n-1)n} + \tan^{-1}\frac{1}{1+n(n+1)} + \ldots\ldots +$$

$$\tan^{-1}\frac{1}{1+(n+19)(n+20)} = \tan^{-1}\frac{n+19}{n+21}$$

$$\Rightarrow \tan^{-1}\frac{n-1}{n+1} + \tan^{-1}\frac{1}{1+n(n+1)}$$

$$+ \tan^{-1}\frac{1}{1+(n+1)(n+2)}$$

$$+ \ldots\ldots + \frac{1}{1+(n+19)(n+20)} = \tan^{-1}\frac{n+19}{n+21}$$

$$\Rightarrow \tan^{-1}\frac{1}{1+n(n+1)} + \tan^{-1}\frac{1}{1+(n+1)(n+2)} + \ldots\ldots +$$

$$\frac{1}{1+(n+19)(n+20)} = \tan^{-1}\frac{n+19}{n+21} - \tan^{-1}\frac{n-1}{n+1}$$

$$\Rightarrow \tan^{-1}\left(\frac{1}{n^2+n+1}\right) + \tan^{-1}\left(\frac{1}{n^2+3n+3}\right) + \ldots\ldots +$$

$$\tan^{-1}\frac{1}{1+(n+19)(n+20)}$$

$$= \tan^{-1}\left(\frac{\dfrac{n+19}{n+21} - \dfrac{n-1}{n+1}}{1 + \dfrac{n+19}{n+21} \times \dfrac{n-1}{n+1}}\right)$$

$$= \tan^{-1}\frac{20}{n^2 + 20n + 1} = S$$

$$\therefore \tan S = \frac{20}{n^2 + 20n + 1}$$

**9.** **(c)** $\sin^{-1} x - \dfrac{1}{\sin^{-1} x} = \cos^{-1} x - \dfrac{1}{\cos^{-1} x}$

$$\Rightarrow (\sin^{-1} x - \cos^{-1} x)\frac{(\sin^{-1} x \,.\, \cos^{-1} x + 1)}{\sin^{-1} x \,.\, \cos^{-1} x} = 0$$

$$\Rightarrow \sin^{-1} x = \cos^{-1} x \text{ or } \sin^{-1} x \cos^{-1} x + 1 = 0$$

$$\Rightarrow x = \frac{1}{\sqrt{2}} \text{ or } \sin^{-1} x\left(\frac{\pi}{2} - \sin^{-1} x\right) + 1 = 0$$

$$\Rightarrow x = \frac{1}{\sqrt{2}} \text{ or } \sin^{-1} x = \frac{\dfrac{\pi}{2} \pm \sqrt{\left(\dfrac{\pi^2}{4} + 4\right)}}{2}$$

$$\Rightarrow x = \frac{1}{\sqrt{2}} \text{ or } \sin^{-1} x = \frac{\pi}{4} - \sqrt{\left(1 + \frac{\pi^2}{16}\right)}$$

**10.** **(b)** $(\cot^{-1}\alpha)x^2 - (\tan^{-1}\alpha)^{3/2} x + 2(\cot^{-1}\alpha)^2 = 0$

Equation has real roots

$\therefore\ D \ge 0 \Rightarrow \left(\tan^{-1}\alpha\right)^3 - 8\left(\cot^{-1}\alpha\right)^3 \ge 0$

$\therefore \tan^{-1}\alpha \ge 2\cot^{-1}\alpha = \pi - 2\tan^{-1}\alpha$

$\Rightarrow \tan^{-1}\alpha \ge \frac{\pi}{3} \Rightarrow \alpha \ge \sqrt{3}$

Sum of roots $> 0 \Rightarrow \frac{\left(\tan^{-1}\alpha\right)^{3/2}}{2\cot^{-1}\alpha} > 0$

$\tan^{-1}\alpha > 0 \Rightarrow \alpha > 0$

Product of roots $> 0 \Rightarrow 2\cot^{-1}\alpha > 0$

$\Rightarrow \alpha \in R \Rightarrow \alpha \ge \sqrt{3}$

**11.** **(b)** $T_n = \sec^{-1}\sqrt{\frac{\left(n^2+1\right)\left(n^2-2n+2\right)}{\left(n^2-n+1\right)^2}}$

$\Rightarrow \sec^2 T_n = \frac{(n^2+1)^2(n^2-2n+2)}{(n^2-n+1)^2}$

$\Rightarrow \sec^2 T_n = \frac{(n^2+1)^2+(n^2+1)-2n(n^2+1)}{(n^2-n+1)^2}$

$\Rightarrow \sec^2 T_n = \frac{1+(n^2+1-n)^2}{(n^2-n+1)^2}$

$\Rightarrow \tan T_n = \frac{1}{n^2-n+1}$

$\Rightarrow \tan T_n = \frac{n-(n-1)}{1+n(n-1)}$

$\Rightarrow \tan^{-1} n - \tan^{-1}(n-1)$

$\therefore\ S = T_1 + T_2 + T_3 + .... + T_n$

$\therefore\ S = \tan^{-1} n$

**12.** **(c)** $(\sin^{-1}x)^3 - (\cos^{-1}x)^3 + (\sin^{-1}x)(\cos^{-1}x)$

$$(\sin^{-1}x - \cos^{-1}x) = \frac{\pi^3}{16}$$

$(\sin^{-1}x - \cos^{-1}x)\ \{(\sin^{-1}x)^2 + (\cos^{-1}x)^2$

$$+ (2\cos^{-1}x \sin^{-1}x)\} = \frac{\pi^3}{16}$$

$(\sin^{-1}x - \cos^{-1}x)(\sin^{-1}x + \cos^{-1}x)^2 = \frac{\pi^3}{16}$

$(\sin^{-1}x - \cos^{-1}x)\frac{\pi^2}{4} = \frac{\pi^3}{16}$

$2\sin^{-1}x - \frac{\pi}{2} = \frac{\pi}{4};\ 2\sin^{-1}x = \frac{3\pi}{4}$

$\Rightarrow \sin^{-1}x = \frac{3\pi}{8} \Rightarrow x = \sin\frac{3x}{8}$ or $\cos\frac{x}{8}$

**13.** **(b)** The given relation is possible when

$a - \frac{a^2}{3} + \frac{a^3}{9} + ... = 1 + b + b^2 + ...$

Also

$-1 \le a - \frac{a^2}{3} + \frac{a^3}{9} + ... \le 1\ \&\ -1 \le 1 + b + b^2 + ... \le 1$

$\Rightarrow |b| < 1 \Rightarrow |a| < 3$ and $\frac{a}{1+\frac{a}{3}} = \frac{1}{1-b}$

$\Rightarrow \frac{3a}{a+3} = \frac{1}{1-b}$, there are infinitely many solution. But in given options it is satisfied only when $a = 1$ and $b = -\frac{1}{3}$.

**14.** **(d)** $T_r = \cot^{-1}\left(\frac{4r^2+3}{4}\right) = \tan^{-1}\frac{1}{1+r^2-\frac{1}{4}}$

$= \tan^{-1}\frac{\left(r+\frac{1}{2}\right)-\left(r-\frac{1}{2}\right)}{1+\left(r+\frac{1}{2}\right)\left(r-\frac{1}{2}\right)}$

$= \tan^{-1}\left(r+\frac{1}{2}\right) - \tan^{-1}\left(r-\frac{1}{2}\right)$

$S_n = \sum T_r = \tan^{-1}\left(n+\frac{1}{2}\right) - \tan^{-1}\frac{1}{2}$

$= \tan^{-1}\left(\frac{4n}{2n+5}\right)$

$S_\infty = \lim_{n\to\infty} S_n = \tan^{-1}2 = \cot^{-1}\frac{1}{2}$.

**15.** **(c)** $T_n = \tan^{-1}\left(\frac{1}{n^2+n+1}\right) = \tan^{-1}\left(\frac{(n+1)-n}{1+(n+1)n}\right)$

$= \tan^{-1}(n+1) - \tan^{-1}n$

$\Rightarrow\ T_1 = \tan^{-1}(1/3) = \tan^{-1}2 - \tan^{-1}1$

$T_2 = \tan^{-1}(1/7) = \tan^{-1}3 - \tan^{-1}2$

..... ........................................

$T_n = \tan^{-1}(n+1) - \tan^{-1}n$

On adding

$T_1 + T_2 + T_3 + ..... + T_n = \tan^{-1}(n+1) - \tan^{-1}1$

$= \tan^{-1}\left(\frac{n}{n+2}\right)$

$\therefore \lim_{n\to\infty}(T_1 + T_2 + T_3 + ..... + T_n)$

$= \lim_{n\to\infty}\tan^{-1}\left(\frac{1}{1+2/n}\right)$

$= \tan^{-1} 1 = \frac{\pi}{4}$

**16.** **(b)** $x^3 + bx^2 + cx + 1 = 0$ ; $f(-1) = b - c < 0$
$f(0) = 1 > 0 \Rightarrow -1 < \alpha < 0$
$\alpha = -B$
$B \in (0, 1)$

$$y = -2\tan^{-1}(\text{cosec } B) - \tan^{-1}\left(\frac{2\sin B}{\cos^2 B}\right)$$

$$= -\left(\pi + \tan^{-1}\frac{2\,\text{cosec } B}{1-\text{cosec}^2 B}\right) - \tan^{-1}\frac{2\sin B}{\cos^2 B} = -\pi$$

**17.** **(a)** $\cos^{-1} x\sqrt{3} + \cos^{-1} x = \frac{\pi}{2}$ is given equation ...(i)

Now $\cos^{-1} x\sqrt{3}, \cos^{-1} x \geq 0$ and their sum is $\frac{\pi}{2}$

$\Rightarrow \cos^{-1} x\sqrt{3}, \cos^{-1} x$ both must belong to $\left[0, \frac{\pi}{2}\right]$

$\Rightarrow x\sqrt{3}, x \in [0,1]$

Now from (i) $\cos^{-1} x\sqrt{3} + \cos^{-1} x = \frac{\pi}{2}$

$\Rightarrow \cos^{-1} x\sqrt{3} = \frac{\pi}{2} - \cos^{-1} x$

$\Rightarrow \cos^{-1} x\sqrt{3} = \sin^{-1} x$

$\Rightarrow \cos^{-1} x\sqrt{3} = \cos^{-1}\sqrt{1-x^2}$

$\Rightarrow x\sqrt{3} = \sqrt{1-x^2}$ as $\cos^{-1}$ x is one-one function

$3x^2 = 1 - x^2 \Rightarrow 4x^2 = 1 \Rightarrow x = \pm\frac{1}{2}$

But $x \in [0, 1] \Rightarrow x = \frac{1}{2}$ is the only possible solution

$\Rightarrow$ (a) is the correct option.

**18.** **(a)** Let $T_n = \sin^{-1}\left[\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}}\right]$

$$= \sin^{-1}\left[\frac{1}{\sqrt{n}}\sqrt{1-\frac{1}{n+1}} - \frac{1}{\sqrt{n+1}}\sqrt{1-\frac{1}{n}}\right]$$

$$T_n = \sin^{-1}\left(\frac{1}{\sqrt{n}}\right) - \sin^{-1}\left(\frac{1}{\sqrt{n+1}}\right)$$

$$\therefore \quad S = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + T_2 + T_3 + T_4 + \ldots$$

$$= \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + \left\{\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) - \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)\right\}$$
$$+ \left\{\sin^{-1}\left(\frac{1}{\sqrt{3}}\right) - \sin^{-1}\left(\frac{1}{\sqrt{4}}\right)\right\}$$
$$+ \left\{\sin^{-1}\left(\frac{1}{\sqrt{4}}\right) - \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)\right\} + \ldots$$

$$= 2\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = 2 \cdot \frac{\pi}{4} = \frac{\pi}{2}$$

**19.** **(c)** $\frac{\alpha^3}{2}\text{cosec}^2\left(\frac{1}{2}\tan^{-1}\frac{\alpha}{\beta}\right) + \frac{\beta^3}{2}\sec^2\left(\frac{1}{2}\tan^{-1}\frac{\beta}{\alpha}\right)$

$$= \alpha^3 \frac{1}{1-\cos\left(\tan^{-1}\left(\frac{\alpha}{\beta}\right)\right)} + \beta^3 \frac{1}{1+\cos\left(\tan^{-1}\frac{\beta}{\alpha}\right)}$$

$$= \alpha^3 \frac{1}{1-\cos\left(\cos^{-1}\left(\frac{\beta}{\sqrt{\alpha^2+\beta^2}}\right)\right)}$$
$$+\beta^3 \frac{1}{1+\cos\left(\cos^{-1}\frac{\alpha}{\sqrt{\alpha^2+\beta^2}}\right)}$$

$$= \alpha^3 \frac{1}{1-\frac{\beta}{\sqrt{\alpha^2+\beta^2}}} + \beta^3 \frac{1}{1+\frac{\alpha}{\sqrt{\alpha^2+\beta^2}}}$$

$$= \sqrt{\alpha^2+\beta^2}\left(-\frac{\alpha^3}{\sqrt{\alpha^2+\beta^2}-\beta} + \frac{\beta^3}{\sqrt{\alpha^2+\beta^2}+\alpha}\right)$$

$$= \sqrt{\alpha^2+\beta^2}\left(\alpha^3\frac{\left(\sqrt{\alpha^2+\beta^2}+\beta\right)}{\alpha^2} + \beta^3\frac{\left(\sqrt{\alpha^2+\beta^2}-\alpha\right)}{\beta^2}\right)$$

$$= \sqrt{\alpha^2+\beta^2}\left[\alpha\left(\sqrt{\alpha^2+\beta^2}+\beta\right)\right]$$
$$+\beta\left(\sqrt{\alpha^2+\beta^2}-\alpha\right)$$

$$= \sqrt{\alpha^2+\beta^2}\,(\alpha+\beta)\sqrt{\alpha^2+\beta^2}$$

$$= (\alpha+\beta)(\alpha^2+\beta^2)$$

**20.** **(a)** $\cot^{-1}(2.1^2) + \cot^{-1}(2.2^2) + \cot^{-1}(2.3^2) + \ldots$

$$= \sum_{r=1}^{\infty}\cot^{-1}\left(2.r^2\right) = \sum_{r=1}^{\infty}\tan^{-1}\left(\frac{1}{2r^2}\right)$$

$$= \sum_{r=1}^{\infty} \tan^{-1}\left[\frac{(1+2r)+(1-2r)}{1-(1+2r)(1-2r)}\right]$$

$$= \sum_{r=1}^{\infty}\left[\tan^{-1}(1+2r)+\tan^{-1}(1-2r)\right]$$

$$= \tan^{-1}3-\tan^{-1}1+\tan^{-1}5-\tan^{-1}3+\tan^{-1}7 - \tan^{-1}5+....+\tan^{-1}\infty$$

$$= -\frac{\pi}{4}+\frac{\pi}{2}=\frac{\pi}{4}$$

**21.** **(1)** Let $y = \lim_{x\to\infty}\left(\frac{\pi}{2}-\tan^{-1}x\right)^{1/x}$

$$= \lim_{x\to\infty}(\cot^{-1}x)^{1/x}$$

$$\therefore \log y = \lim_{x\to\infty}\frac{\log \cot^{-1}x}{x} \quad \left(\frac{\infty}{\infty}\text{form}\right)$$

$$\log y = \lim_{x\to\infty}\frac{-1}{(1+x^2)\cot^{-1}x} \qquad (0.\ \infty \text{ form})$$

$$\log y = -\lim_{x\to\infty}\frac{(1+x^2)^{-1}}{\cot^{-1}x} \qquad \left(\frac{0}{0}\text{form}\right)$$

$$= -\lim_{x\to\infty}\frac{\frac{-2x}{(1+x^2)^2}}{\frac{-1}{1+x^2}}$$

$$= -2\lim_{x\to\infty}\frac{x}{1+x^2} = 2\lim_{x\to\infty}\frac{1}{2x}=0$$

$$\therefore y=e^0=1$$

**22.** **(0)** We know that $\cot A > 1$ if $0 < A < \frac{\pi}{4}$

and $\cot A < 1$ if $\frac{\pi}{4} < A < \frac{\pi}{2}$

$$\tan^{-1}(\cot A)+\tan^{-1}(\cot^3 A)$$

$$= \pi + \tan^{-1}\frac{\cot A+\cot^3 A}{1-\cot^4 A},$$

If $0 < A < \frac{\pi}{4}$

and $\tan^{-1}(\cot A) + \tan^{-1}(\cot^3 A)$

$$= \tan^{-1}\frac{\cot A+\cot^3 A}{1-\cot^4 A} \text{ if } \frac{\pi}{4}<A<\frac{\pi}{2}$$

Also, $\dfrac{\cot A+\cot^3 A}{1-\cot^4 A} = \dfrac{\cot A\,\mathrm{cosec}^2 A.\sin^4 A}{\sin^4 A-\cos^4 A}$

$$= \frac{\sin A\cos A}{(\sin^2 A+\cos^2 A)(\sin^2 A-\cos^2 A)}$$

$$= -\frac{\sin 2A}{2\cos 2A} = -\frac{1}{2}\tan 2A$$

Hence,

$$\tan^{-1}\left(\frac{1}{2}\tan 2A\right)+\tan^{-1}(\cot A)+\tan^{-1}(\cot^3 A)$$

$$= \begin{cases}\pi & \text{if } 0<A<\frac{\pi}{4}\\ 0 & \text{if } \frac{\pi}{4}<A<\frac{\pi}{2}\end{cases} \quad [\because \tan^{-1}(-x)=-\tan^{-1}x]$$

**23.** **(2)** $(\sin^{-1}x) \in \left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

$$\therefore \left(\sin^{-1}x\right)^2 \le \frac{\pi^2}{4} \Rightarrow (\sec^{-1}y)^2, (\tan^{-1}z)^2 \ge 0$$

$$\therefore \text{RHS} \ge \frac{\pi^2}{4} \quad \therefore (\sin^{-1}x)^2 = \frac{\pi^2}{4}$$

$\therefore \quad (\sec^{-1}y)^2+(\tan^{-1}z)^2=0$

or $\sec^{-1}y = \tan^{-1}z = 0$

$$\therefore \sin^{-1}x = \pm\frac{\pi}{2}, y=1, z=0$$

$\therefore x=\pm1, y=1, z=0$

**24.** **(3)** $\cos(2\sin^{-1}(\cot(\tan^{-1}(\sec(6\,\mathrm{cosec}^{-1}x))))) = -1$

$$\sin^{-1}(\cot(\tan^{-1}(\sec(6\,\mathrm{cosec}^{-1}x)))) = \pm\frac{\pi}{2}$$

$$\cot(\tan^{-1}(\sec(6\,\mathrm{cosec}^{-1}x))) = \pm 1$$

$$\tan^{-1}(\sec(6\,\mathrm{cosec}^{-1}x)) = \pm\frac{\pi}{4}$$

$\sec(6\,\mathrm{cosec}^{-1}x) = \pm1$

$6\,\mathrm{cosec}^{-1}x = \pm3\pi, \pm2\pi, \pm\pi$

$$\mathrm{cosec}^{-1}x = \pm\frac{\pi}{2}, \pm\frac{\pi}{3}, \pm\frac{\pi}{6} \Rightarrow x=1, \frac{2}{\sqrt{3}}, 2$$

$$(\because x>0)$$

**25.** **(7)** $f(x)=\sqrt{3\cos^{-1}(4x)-\pi}$ is defined

If $\cos^{-1}4x \ge \frac{\pi}{3} \Rightarrow 4x \le \frac{1}{2}$ or $x \le \frac{1}{8}$ ...(i)

Also, $-1 \le 4x \le 1$ or $\frac{-1}{4} \le x \le \frac{1}{4}$ ...(ii)

Therefore, from eqs. (i) and (ii), we have domain:

$$x \in \left[\frac{-1}{4},\frac{1}{8}\right] \Rightarrow 4a+64b=7$$

CHAPTER 18

# Matrices

**1.** **(a)** $AB = \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}$

$\begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix}$

$$= \begin{bmatrix} \cos^2\theta\cos^2\phi + \cos\theta\cos\phi\sin\theta\sin\phi & \cos^2\theta\cos\phi\sin\phi + \sin^2\phi\sin\theta\cos\theta \\ \cos^2\phi\cos\theta\sin\theta + \sin^2\theta\sin\phi\sin\phi & \cos\theta\cos\phi\sin\theta\sin\phi + \sin^2\theta\sin^2\phi \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta\cos\phi(\cos\theta\cos\phi + \sin\theta\sin\phi) & \cos\theta\sin\phi(\cos\theta\cos\phi + \sin\theta\sin\phi \\ \sin\theta\cos\phi(\cos\theta\cos\phi + \sin\phi\sin\phi & \sin\theta\cos\phi(\cos\theta\cos\phi + \sin\theta\sin\phi) \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta\cos\phi\cos(\theta-\phi) & \cos\theta\sin\phi\cos(\theta-\phi) \\ \sin\theta\cos\phi\cos(\theta-\phi) & \sin\theta\sin\phi\cos(\theta-\phi) \end{bmatrix}$$

Clearly $AB$ is the zero matrix if $\cos(\theta-\phi) = 0$

i.e. $\theta-\phi$ is an odd multiple of $\frac{\pi}{2}$.

**2.** **(a)** Let $B = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix}$

Given $BAC = I \Rightarrow B^{-1}(BAC) = B^{-1}I$

$\Rightarrow I(AC) = B^{-1} \Rightarrow AC = B^{-1}$

$\Rightarrow ACC^{-1} = B^{-1}C^{-1} \Rightarrow AI = B^{-1}C^{-1}$

$\therefore A = (B^{-1})(C^{-1})$

Now $B^{-1} = \frac{1}{4-3}\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$

$C^{-1} = \frac{1}{9-10}\begin{bmatrix} -3 & -2 \\ -5 & -3 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$

$\therefore (B^{-1})(C^{-1}) = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}\begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

$\therefore A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

**3.** **(c)** $I - P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & -\tan(\theta/2) \\ \tan(\theta/2) & 0 \end{bmatrix}$

$= \begin{bmatrix} 1 & \tan(\theta/2) \\ -\tan(\theta/2) & 1 \end{bmatrix}$

$\therefore (I-P)\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

$= \begin{bmatrix} 1 & \tan(\theta/2) \\ -\tan\theta/2 & 1 \end{bmatrix} . \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

$$= \begin{bmatrix} \cos\theta + \tan(\theta/2)\sin\theta & -\sin\theta + \tan(\theta/2)\cos\theta \\ -\tan(\theta/2)\cos\theta + \sin\theta & \tan(\theta/2)\sin\theta + \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} \begin{matrix} 1-2\sin^2(\theta/2) \\ +2\sin^2(\theta/2) \end{matrix} & \begin{matrix} -2\sin(\theta/2)\cos(\theta/2) \\ +\tan(\theta/2)(2\cos^2(\theta/2)-1) \end{matrix} \\ \begin{matrix} -\tan(\theta/2)(2\cos^2\theta/2-1) \\ +2\sin(\theta/2)\cos(\theta/2) \end{matrix} & \begin{matrix} \tan(\theta/2)(2\sin(\theta/2)\cos(\theta/2)) \\ +(1-2\sin^2(\theta/2)) \end{matrix} \end{bmatrix}$$

$= \begin{bmatrix} 1 & -\tan(\theta/2) \\ \tan\theta/2 & 1 \end{bmatrix} = I + P$

**4.** **(c)** $(1-x)^{-1}(1-y)^{-1}\begin{bmatrix} 1 & -x \\ -x & 1 \end{bmatrix}\begin{bmatrix} 1 & -y \\ -y & 1 \end{bmatrix}$

$= (1+xy-(x+y))^{-1}\begin{bmatrix} 1+xy & -(x+y) \\ -(x+y) & 1+xy \end{bmatrix}$

$$= \left(1 - \frac{x+y}{1+xy}\right)^{-1}\begin{bmatrix} 1 & -\frac{x+y}{1+xy} \\ -\frac{x+y}{1+xy} & 1 \end{bmatrix} = A(z)$$

$\Rightarrow A(x)A(y) = A(z)$

**5.** **(d)** We have

$$A^2 = \begin{bmatrix} 1 & 0 \\ 1/2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1/2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2(1/2) & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix};$$

$$A^8 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$$

In general, by induction $A^n = \begin{bmatrix} 1 & 0 \\ n/2 & 1 \end{bmatrix}$

$$\Rightarrow \quad A^{50} = \begin{bmatrix} 1 & 0 \\ 25 & 1 \end{bmatrix}$$

**6.** **(d)** As $a_{ij} = (i^2 + j^2 - ij)(j - i)$
$a_{ji} = (j^2 + i^2 - ji)(i - j) = -a_{ij}$
$\Rightarrow$ $A$ is skew symmetric $\Rightarrow T_r(A) = 0$.
Also $|A| = 0$.

**7.** **(a)** Given that, $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$

$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = P A P^T$ and $X = P^T Q^{2005} P$

We observe that $Q = P A P^T$

$\Rightarrow$ $Q^2 = (P A P^T)(P A P^T) = P A (P^T P) A P^T$
$= PA (I A) P^T$
$\therefore P A^2 P^T$

Proceeding in the same way, we get
$Q^{2005} = P A^{2005} P^T$

Also $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

and proceeding in the same way

$$A^{2005} = \begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$$

Now, $X = P^T Q^{2005} P$
$= P^T (PA^{2005} P^T) P = (P^T P) A^{2005} (P^T P)$

$$= IA^{2005} I = A^{2005} = \begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$$

**8.** **(c)** Let $A = \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$.

Now $A^T = \begin{bmatrix} 0 & a & a \\ 2b & b & -b \\ c & -c & c \end{bmatrix}$

$\because$ $A$ is orthogonal.

$\therefore AA^T = I$

$$\Rightarrow \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}\begin{bmatrix} 0 & a & a \\ 2b & b & -b \\ c & -c & c \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4b^2 + c^2 & 2b^2 - c^2 & -2b^2 + c^2 \\ 2b^2 - c^2 & a^2 + b^2 + c^2 & a^2 - b^2 - c^2 \\ -2b^2 + c^2 & a^2 - b^2 - c^2 & a^2 + b^2 + c^2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Equating the corresponding elements, we get
$4b^2 + c^2 = 1$ ....(i)
$2b^2 - c^2 = 0$ ....(ii)
$a^2 + b^2 + c^2 = 1$ ....(iii)
$a^2 - b^2 - c^2 = 0$ ....(iv)
From solving (i), (ii) and (iii) we get,

$$a = \pm\frac{1}{\sqrt{2}};\ b = \pm\frac{1}{\sqrt{6}};\ c = \pm\frac{1}{\sqrt{3}};$$

**9.** **(a)** $A^2 = AA = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix}$

$$= \begin{bmatrix} 0 & 0 & 1 \\ p & q & r \\ pr & p + qr & q + r^2 \end{bmatrix}$$

Again

$$A^3 = A^2 A = \begin{bmatrix} 0 & 0 & 1 \\ p & q & r \\ pr & p+qr & q+r^2 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix}$$

$$= \begin{bmatrix} p & q & r \\ pr & p+qr & q+r^2 \\ pq+r^2p & pr+q^2+qr^2 & p+2qr+r^3 \end{bmatrix}$$

$$= \begin{bmatrix} p & 0 & 0 \\ 0 & p & 0 \\ 0 & 0 & p \end{bmatrix} + \begin{bmatrix} 0 & q & 0 \\ 0 & 0 & q \\ pq & q^2 & qr \end{bmatrix}$$

$$+ \begin{bmatrix} 0 & 0 & r \\ pr & qr & r^2 \\ pr^2 & pr+qr^2 & pr+r^3 \end{bmatrix}$$

$$= p\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + q\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix}$$

$$+ r\begin{bmatrix} 0 & 0 & 1 \\ p & q & r \\ pr & p+qr & q+r^2 \end{bmatrix}$$

$= pI + qA + rA^2 \qquad \therefore A^3 - rA^2 - qA = pI.$

**10.** **(a)** Given $A^2 = 2A - I \Rightarrow A^2 A = (2A - I) A$

$\Rightarrow A^3 = 2A^2 - A = 2(2A - I) - A$

$= 4A - 2I - A = 3A - 2I$

Again $A^3 A = (3A - 2I) A = 3A^2 - 2A$

$\Rightarrow A^4 = 3(2A - I) - 2A = 4A - 3I$

........

........

In general $A^n = nA - (n-1) I$

**11.** **(d)** $PQ = I \Rightarrow P^{-1} = Q$

Now the system in matrix notation is $PX = B$

$\therefore X = P^{-1} B = QB$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{9}\begin{bmatrix} 2 & 2 & 1 \\ 13 & -5 & m \\ -8 & 1 & 5 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$$

$\therefore y = \frac{1}{9}(13 - 5 + 5m)$

$\Rightarrow -27 = 8 + 5m$ (Given $y = -3$)

$\therefore m = -7$

**12.** **(b)** We have, $AB = A$ and $BA = B$.

Now, $AB = A \Rightarrow (AB)A = A.A \Rightarrow A(BA) = A^2$

$\Rightarrow AB = A^2 (\because BA = B) \Rightarrow A = A^2 (\because AB = A)$

Again, $BA = B \Rightarrow (BA)B = B^2 \Rightarrow B(AB) = B^2$

$\Rightarrow BA = B^2 (\because AB = A) \Rightarrow B = B^2 (\because BA = B)$

$\therefore$ A and B are idempotent matrices.

**13.** **(b)** Given that

$X = A_1 + 3A_3^3 + \ldots (2n-1)(A_{2n-1})^{2n-1}$

We know that if A is a skew-symmetric matrix then $A^T = -A$

$X^T = -[A_1 + 3A_3^3 + \ldots (2n-1)(A_{2n-1})^{2n-1}]$

$= -X$, so skew-symmetric

**14.** **(c)** $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ $A^2 = A \cdot A = AA^\theta$

[$\because$ Given that $A = A^\theta$]

$$\Rightarrow A^2 = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}\begin{bmatrix} \bar{a} & \bar{d} & \bar{g} \\ \bar{d} & \bar{e} & \bar{h} \\ \bar{c} & \bar{f} & \bar{i} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} |a|^2+|b|^2+|c|^2 & a\bar{d}+b\bar{e}+c\bar{f} & a\bar{g}+b\bar{h}+c\bar{i} \\ d\bar{a}+e\bar{b}+f\bar{c} & |d|^2+|e|^2+|f|^2 & d\bar{g}+e\bar{h}+f\bar{i} \\ g\bar{a}+h\bar{b}+i\bar{c} & g\bar{d}+h\bar{e}+i\bar{f} & |g|^2+|h|^2+|i|^2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} (\because \text{Given } A^2 = O)$$

$\Rightarrow |a|^2 + |b|^2 + |c|^2 = 0$

$\Rightarrow Re(a) = Re(b) = Re(c) = Im(a) = Im(b) = Im(c) = 0$

Similarly

$Re(d) = Re(e) = Re(f) = Im(d) = Im(e) = Im(f) = 0$

and $Re(g) = Re(h) = Re(i) = Im(g) = Im(h) = Im(i) = 0$

$$\therefore A=\begin{bmatrix}0+0i & 0+0i & 0+0i\\ 0+0i & 0+0i & 0+0i\\ 0+0i & 0+0i & 0+0i\end{bmatrix}$$

= Null matrix of order 3 × 3

**15.** **(d)** $(XY)^T = Y^TX^T$

$Y^T=(ABC-CBA)^T=C^TB^TA^T-A^TB^TC^T$

$=-CBA+ABC=Y$

$X^T=(ABC+CBA)^T=C^TB^TA^T+A^TB^TC^T$

$=-CBA-ABC=-X$

**16.** **(d)** $A=\begin{bmatrix}\cos\theta & \sin\theta\\ \sin\theta & -\cos\theta\end{bmatrix}$

$\therefore \quad AA^T=I$ ...(i)

Now, $C=ABA^T$

$\Rightarrow \quad A^TC=BA^T$ ...(ii)

Now $A^TC^nA=A^TCC^{n-1}A=BA^TC^{n-1}A$ (from (ii))

$=BA^TCC^{n-2}A=B^2A^TC^{n-2}A=\ldots$

$=B^{n-1}A^TCA=B^{n-1}BA^TA=B^n=\begin{bmatrix}1 & 0\\ -n & 1\end{bmatrix}$

**17.** **(c)** Let $X=\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$, $(x_1\,x_2\,x_3)$

$$\begin{pmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix}=0$$

$a_{11}x_1^2+a_{22}x_2^2+a_{33}x_3^2+(a_{12}+a_{21})x_1x_2+(a_{13}+a_{31})x_1x_3+(a_{23}+a_{32})x_2x_3=0$

It is true for every $x_1, x_2, x_3$,

then $a_{11}=a_{22}=a_{33}=0$, $a_{12}+a_{21}=0$, $a_{13}+a_{31}=0$, $a_{23}+a_{32}=0$

$\therefore$ $A$ is a skew symmetric matrix

$a_{21}=-2008$; $a_{31}=-2010$

$a_{32}=2012$

**18.** **(1)** We have, BC = CB, and

$A^{N+1}=(B+C)^{N+1}$

$={}^{N+1}C_0B^{N+1}+{}^{N+1}C_1B^NC+{}^{N+1}C_2B^{N-1}C^2+\ldots+{}^{N+1}C_rB^{N+1-r}C^r+\ldots$

But given that $C^2=0\Rightarrow C^3=C^4=\ldots=C^r=0$

Hence, $A^{N+1}={}^{N+1}C_NB^{N+1}+{}^{N+1}C_1B^NC$

$=B^{N+1}+(N+1)B^NC$

$=B^N[B+(N+1)C]$

Thus $K=N\Rightarrow K/N=1$.

**19.** **(1023)** $(AB)\cdot(AB)=A(BA)B=A^3B^2$

$(AB)(AB)(AB)=A^7B^3$

so $(AB)^n=A^{2^n-1}B^n$

so $k=2^{10}-1=1023$

**20.** **(27)** Observe $A_1A_2$

$=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}=\begin{bmatrix}1 & (2+1)\\ 0 & 1\end{bmatrix}$

and $A_1A_2A_3=\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}=\begin{bmatrix}1 & (3+2+1)\\ 0 & 1\end{bmatrix}$

So, in general $A_1A_2A_3\ldots A_n$

$=\begin{bmatrix}1 & n+(n-1)+(n-2)+\ldots+3+2+1\\ 0 & 1\end{bmatrix}$

So $\dfrac{n(n+1)}{2}=378\Rightarrow n=27$

**21.** **(4)** Given that $A^TA=I$

where $A=\begin{bmatrix}a & b & c\\ b & c & a\\ c & a & b\end{bmatrix}$

$\therefore A^T=\begin{bmatrix}a & b & c\\ b & c & a\\ c & a & b\end{bmatrix}$

$\therefore A^TA=I$

$\Rightarrow\begin{bmatrix}a & b & c\\ b & c & a\\ c & a & b\end{bmatrix}\begin{bmatrix}a & b & c\\ b & c & a\\ c & a & b\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$

$$\Rightarrow\begin{bmatrix}a^2+b^2+c^2 & ab+bc+ca & ab+bc+ca\\ ab+bc+ca & a^2+b^2+c^2 & ab+bc+ca\\ ab+bc+ca & ab+bc+ca & a^2+b^2+c^2\end{bmatrix}$$

$$=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$\Rightarrow a^2 + b^2 + c^2 = 1$ ...(i)

and $ab + bc + ca = 0$ ...(ii)

$abc = 1$ (given) ...(iii)

Now $a^3 + b^3 + c^3 - 3abc$

$= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

$\Rightarrow a^3 + b^3 + c^3 = (a + b + c)[1 - 0] + 3 \times 1$

[Using (i), (ii) and (iii)]

$\Rightarrow a^3 + b^3 + c^3 = a + b + c + 3$

Now $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

$\Rightarrow (a + b + c)^2 = 1$

$\Rightarrow (a + b + c) = 1$

[$\because$ $(a + b + c) \neq -1$ as $a, b, c$ all are + ve numbers.]

$\therefore$ We get

$a^3 + b^3 + c^3 = 4$

**22.** **(1)** $z = \dfrac{-1 + i\sqrt{3}}{2} \Rightarrow z^3 = 1$ and $1 + z + z^2 = 0$

$$P^2 = \begin{bmatrix} (-z)^r & z^{2s} \\ z^{2s} & z^r \end{bmatrix}\begin{bmatrix} (-z)^r & z^{2s} \\ z^{2s} & z^r \end{bmatrix}$$

$$= \begin{bmatrix} z^{2r} + z^{4s} & z^{2s}\left((-z)^5 + z^r\right) \\ z^{2s}\left((-z)^r + z^r\right) & z^{4s} + z^{2r} \end{bmatrix}$$

For $P^2 = -I$ we should have

$z^{2r} + z^{4s} = 1$ and $z^{2s}((-z)^r + z^r) = 0$

$\Rightarrow z^{2r} + z^s + 1 = 0$

$\Rightarrow$ r is odd and s = r but not a multiple of 3.

Which is possible when s = r = 1

$\therefore$ only one pair is there.

**23.** **(1)** $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix} = I + \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 16 & 4 & 0 \end{bmatrix} = I + A$

$A^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 16 & 0 & 0 \end{bmatrix}$ and $A^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\therefore A^n = O, \forall\, n \geq 3$

Now $P^{50} = (I + A)^{50} = {}^{50}C_0\, I^{50} + {}^{50}C_1\, I^{49} A + {}^{50}C_2 + I^{48} A^2 + O$

$= I + 50A + 25 \times 49 A^2.$

$\therefore Q = P^{50} - I = 50A + 25 \times 49 A^2.$

$\Rightarrow q_{21} = 50 \times 4 = 200$

$\Rightarrow q_{31} = 50 \times 16 + 25 \times 49 \times 16 = 20400$

$\Rightarrow q_{32} = 50 \times 4 = 200$

$\therefore \dfrac{q_{31} + q_{32}}{q_{21}} = \dfrac{20600}{200} = 103 \Rightarrow \dfrac{q_{31} + q_{32}}{103\, q_{21}} = 1$

**24.** **(2)** $A^2 = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I;$

$x = 2, 4, 6, 8....$

$\Rightarrow$ $\Sigma(\cos^x \theta + \sin^x \theta) = (\cos^2 \theta + \sin^2 \theta) + (\cos^4 \theta + \sin^4 \theta) + (\cos^6 \theta + \sin^6 \theta) + \ldots + \ldots$

$= \dfrac{\cos^2 \theta}{1 - \cos^2 \theta} + \dfrac{\sin^2 \theta}{1 - \sin^2 \theta} = \cot^2 \theta + \tan^2 \theta$

Using $AM \geq GM$,

$\dfrac{1}{2}(\cot^2 \theta + \tan^2 \theta) \geq \sqrt{\cot^2 \theta \tan^2 \theta}$

$\Rightarrow$ $(\cot^2 \theta + \tan^2 \theta) \geq 2$

$\Rightarrow$ So, minimum value = 2

**25.** **(2)** As $A^2 = O, A^k = O \ulcorner k \llcorner 2$.

Thus, $(A + I)^{50} = I + 50A$

$\Rightarrow$ $(A + I)^{50} - 50A = I$

$\therefore$ $a = 1, b = 0, c = 0, d = 1$

CHAPTER 19

# Determinants

**1.** **(a)** Given determinant

$$\begin{vmatrix} 1 & a & a^2 \\ \cos(n-1)x & \cos nx & \cos(n+1)x \\ \sin(n-1)x & \sin nx & \sin(n+1)x \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1+a^2-2a\cos x & a & a^2 \\ 0 & \cos nx & \cos(n+1)x \\ 0 & \sin nx & \sin(n+1)x \end{vmatrix} = 0$$

By applying $C_1 \rightarrow C_1 + C_3 - 2\cos x\, C_2$

By expanding

$(1 + a^2 - 2a\cos x)\,[\cos nx \sin(n+1)x$
$- \sin nx \cos(n+1)x] = 0$

Now, $(1 + a^2 - 2a\cos x)\sin(n+1-n)x = 0$

$\Rightarrow (1+a^2-2a\cos x)\sin x = 0$

$\sin x = 0$ or $\cos x = \dfrac{1+a^2}{2a}$

As $a \neq 1\ \therefore \left(\dfrac{1+a^2}{2a}\right) > 1$

$\Rightarrow \cos x > 1$ It is not possible. $\therefore \sin x = 0$

**2.** **(d)** $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 0$

$$\Rightarrow 4\begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 0$$

(taking $R_2 \rightarrow R_2 - R_3$)

$$\Rightarrow \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1-2a & 1-2b & 1-2c \end{vmatrix} = 0$$

(using $R_3 \rightarrow R_3 - R_1$)

$$\Rightarrow \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix} = 0 \quad (\text{using } R_3 \rightarrow R_3 + 2R_2)$$

$$\Rightarrow \begin{vmatrix} a^2 & b^2-a^2 & c^2-a^2 \\ 1 & b-a & c-a \\ 1 & 0 & 0 \end{vmatrix} = 0 \begin{pmatrix} C_2 \rightarrow C_2 - C_1 \\ C_3 \rightarrow C_3 - C_1 \end{pmatrix}$$

$\Rightarrow (b^2-a^2)(c-a) - (b-a)(c^2-a^2) = 0$

$\Rightarrow (b-a)(c-a)(b-c) = 0$

$\Rightarrow a = b$ or $b = c$ or $c = a$

$\Rightarrow \Delta ABC$ is an isosceles.

**3.** **(d)** $\Delta = \begin{vmatrix} a^2 & b\sin A & c\sin A \\ b\sin A & 1 & \cos A \\ c\sin A & \cos A & 1 \end{vmatrix}$

Using, $\dfrac{\sin A}{a} = \dfrac{\sin B}{b} = \dfrac{\sin C}{c} = k$

$\Rightarrow \sin A = ak,\ \sin B = bk,\ \sin C = ck$ So,

$$\Delta = \begin{vmatrix} a^2 & abk & ack \\ abk & 1 & \cos A \\ ack & \cos A & 1 \end{vmatrix}.$$

Take a common from $C_1$ and $R_1$ both, we get

$$\Delta = a^2\begin{vmatrix} 1 & bk & ck \\ bk & 1 & \cos A \\ ck & \cos A & 1 \end{vmatrix}$$

$$= a^2\begin{vmatrix} 1 & \sin B & \sin C \\ \sin B & 1 & \cos A \\ \sin C & \cos A & 1 \end{vmatrix}$$

Operate

$C_2 \rightarrow C_2 - C_1 \sin B$ and $C_3 \rightarrow C_3 - C_1 \sin C$

$$\Delta = a^2\begin{vmatrix} 1 & 0 & 0 \\ \sin B & 1-\sin^2 B & \cos A - \sin B\sin C \\ \sin C & \cos A - \sin B \sin C & 1-\sin^2 C \end{vmatrix}$$

$$= a^2[(1-\sin^2 C - \sin^2 B + \sin^2 B\sin^2 C) - (\cos A - \sin B\sin C)^2]$$

$$= a^2[\sin^2 A - \sin^2 B - \sin^2 C + 2\sin B\sin C\cos A]$$

The above expression does not represent area or perimeter of the triangle.

**4.** **(d)** $\Delta = \begin{vmatrix} 1 & 1+\sin P & \sin P(1+\sin P) \\ 1 & 1+\sin Q & \sin Q(1+\sin Q) \\ 1 & 1+\sin R & \sin R(1+\sin R) \end{vmatrix}$

$$= \begin{vmatrix} 1 & \sin P & \sin P + \sin^2 P \\ 1 & \sin Q & \sin Q + \sin^2 Q \\ 1 & \sin R & \sin R + \sin^2 R \end{vmatrix} \quad (C_2 \to C_2 - C_1)$$

$$= \begin{vmatrix} 1 & \sin P & \sin^2 P \\ 1 & \sin Q & \sin^2 Q \\ 1 & \sin R & \sin^2 R \end{vmatrix} \quad (C_3 \to C_3 - C_2)$$

$$= \begin{vmatrix} 1 & \sin P & \sin^2 P \\ 0 & \sin Q - \sin P & \sin^2 Q - \sin^2 P \\ 0 & \sin R - \sin P & \sin^2 R - \sin^2 P \end{vmatrix}$$

$(R_2 \to R_2 - R_1,\ R_3 \to R_3 - R_1)$

$= (\sin Q - \sin P)\ (\sin R - \sin P)$

$$\begin{vmatrix} 1 & \sin P & \sin^2 P \\ 0 & 1 & \sin Q + \sin P \\ 0 & 1 & \sin R + \sin P \end{vmatrix}$$

$= (\sin Q - \sin P)\ (\sin R - \sin P)\ (\sin R - \sin Q)$

Now $\begin{matrix} \Delta > 0 & \text{if } P < Q < R \\ \Delta < 0 & \text{if } P > Q > R \end{matrix}$

Hence the sign of $\Delta$ cannot be determined.

**5.** **(d)** The given points are

$P(-\sin(\beta-\alpha), -\cos\beta)$, $Q(\cos(\beta-\alpha), \sin\beta)$

$R(\cos(\beta-\alpha+\theta), \sin(\beta-\theta))$

Where $0 < \alpha, \beta, \theta < \frac{\pi}{4}$

$$\therefore\ \Delta = \begin{vmatrix} 1 & 1 & 1 \\ -\sin(\beta-\alpha) & \cos(\beta-\alpha) & \cos(\beta-\alpha+\theta) \\ -\cos\beta & \sin\beta & \sin(\beta-\theta) \end{vmatrix}$$

Operating $C_3 - C_1\sin\theta - C_2\cos\theta$, we get

$$\Delta = \begin{vmatrix} 1 & 1 & 1-\sin\theta-\cos\theta \\ -\sin(\beta-\alpha) & \cos(\beta-\alpha) & 0 \\ -\cos\beta & \sin\beta & 0 \end{vmatrix}$$

$= (1-\sin\theta-\cos\theta)[\cos\beta\cos(\beta-\alpha) - \sin\beta\sin(\beta-\alpha)]$

$\Rightarrow\ \Delta = [1-(\sin\theta+\cos\theta)]\cos(2\beta-\alpha)$

$\because\ 0 < \alpha, \beta, \theta < \frac{\pi}{4}$ $\quad\therefore\ \sin\theta + \cos\theta \neq 1$

Also $2\beta - \alpha < \frac{\pi}{4} \Rightarrow \cos(2\beta-\alpha) \neq 0$

$\therefore\ \Delta \neq 0 \Rightarrow$ the three points are non-collinear.

**6.** **(d)** $\sum_{r=1}^{n-1} r = 1+2+3+\ldots+(n-1) = \frac{n(n-1)}{2}$

$\sum_{r=1}^{n-1} (2r-1) = 1+3+5+\ldots+[2(n-1)-2]$

$= (n-1)^2$

$\sum_{r=1}^{n-1} (3r-2) = 1+4+7+\ldots+(3n-3-2)$

$= \frac{(n-1)(3n-4)}{2}$

$\therefore\ \sum_{r=1}^{n-1} \Delta_r$

$$= \begin{vmatrix} \Sigma r & \Sigma(2r-1) & \Sigma(3r-2) \\ \frac{n}{2} & n-1 & a \\ \frac{n(n-1)}{2} & (n-1)^2 & \frac{(n-1)(3n-4)}{2} \end{vmatrix}$$

$\sum_{r=1}^{n-1} \Delta_r$ consists of $(n-1)$ determinants in L.H.S. and in R.H.S every constituent of first row consists of (n – 1) elements and hence it can be splitted into sum of $(n-1)$ determinants.

$$\therefore \sum_{r=1}^{n-1} \Delta_r$$

$$= \begin{vmatrix} \frac{n(n-1)}{2} & (n-1)^2 & \frac{(n-1)(3n-4)}{2} \\ \frac{n}{2} & n-1 & a \\ \frac{n(n-1)}{2} & (n-1)^2 & \frac{(n-1)(3n-4)}{2} \end{vmatrix} = 0$$

($\because$ $R_1$ and $R_3$ are identical)

Hence, value of $\sum_{r=1}^{n-1} \Delta_r$ is independent of both '$a$' and '$n$'.

**7.** **(a)** We have $a^2+b^2+c^2+ab+bc+ca \le 0$

$\therefore$ $(a+b)^2+(b+c)^2+(c+a)^2 \le 0$

$\therefore$ $a+b=0, b+c=0$ and $c+a=0$

$\therefore$ $a=b=c=0$

$$\Rightarrow \begin{vmatrix} (a+b+2)^2 & a^2+b^2 & 1 \\ 1 & (b+c+2)^2 & b^2+c^2 \\ c^2+a^2 & 1 & (c+a+2)^2 \end{vmatrix}$$

$$= \begin{vmatrix} 4 & 0 & 1 \\ 1 & 4 & 0 \\ 0 & 1 & 4 \end{vmatrix} = 65$$

**8.** **(b)** $|A^{-1}| = \frac{1}{|A|} = \frac{1}{5}$

$|(AB)^T| = |AB| = |A\cdot(\text{adj}\,A)| = |A|\cdot|\text{adj}\,(A)|$

$= 5 \times 5^2 = 5^3$

$\therefore$ $||A^{-1}|(AB)^T| = |\frac{1}{5}(AB)^T| = \frac{1}{5^3}\ |AB| = 1$

**9.** **(a)** Differentiating given equation w.r.t.$x$, we get

$$\begin{vmatrix} f'(x) & g'(x) & h'(x) \\ a & b & c \\ p & q & r \end{vmatrix} = 4mx^3+3nx^2+2rx+5 \quad ...(1)$$

Again differentiating w.r.t.$x$, we get

$$\begin{vmatrix} f''(x) & g''(x) & h''(x) \\ a & b & c \\ p & q & r \end{vmatrix} = 12mx^2+6nx+2r \quad ...(2)$$

Again differentiating w.r.t.$x$, we get

$$\begin{vmatrix} f'''(x) & g'''(x) & h'''(x) \\ a & b & c \\ p & q & r \end{vmatrix} = 24mx+6n \quad ...(3)$$

putting $x=0$ in (2), we get

$$2r = \begin{vmatrix} f''(0) & g''(0) & h''(0) \\ a & b & c \\ p & q & r \end{vmatrix} \quad ...(4)$$

Putting $x=0$ in (3), we get

$$6n = \begin{vmatrix} f'''(0) & g'''(0) & h'''(0) \\ a & b & c \\ p & q & r \end{vmatrix} \quad ...(5)$$

From (5) and (4), we get

$$\begin{vmatrix} f'''(0)-f''(0) & g'''(0)-g''(0) & h'''(0)-h''(0) \\ a & b & c \\ p & q & r \end{vmatrix}$$

$= 2(3n-r)$

**10.** **(b)** We have

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

...(1)

$$\text{Also, } \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = abc \begin{vmatrix} \frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ 1 & 1 & 1 \\ a^2 & b^2 & c^2 \end{vmatrix}$$

(taking $a, b, c$ common from $R_1, R_2, R_3$)

$$= \begin{vmatrix} bc & ac & ab \\ 1 & 1 & 1 \\ a^2 & b^2 & c^2 \end{vmatrix} \text{ (Multiplying } R_1 \text{ by } abc)$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ bc & ac & ab \end{vmatrix} \text{ then,}$$

$$D=\begin{vmatrix} 1 & 1 & 1 \\ (x-a)^2 & (x-b)^2 & (x-c)^2 \\ (x-b)(x-c) & (x-c)(x-a) & (x-a)(x-b) \end{vmatrix}$$

$= (a-b)(b-c)(c-a)(3x-a-b-c)$

Now given that $a$, $b$, $c$ are all different, then $D = 0$. Therefore, $x = \frac{1}{3}(a+b+c)$

**11.** **(a)** $\lambda = \dfrac{t^2-3t+4}{t^2+3t+4}$

$\Rightarrow (\lambda-1)t^2 + 3(\lambda+1)t + 4(\lambda-1) = 0$

Since, $t$ is real

$\Rightarrow 9(\lambda+1)^2 - 16(\lambda-1)^2 \ge 0$

$\Rightarrow (3\lambda+3-4\lambda+4)(3\lambda+3+4\lambda-4) \ge 0$

$\Rightarrow (7-\lambda)(7\lambda-1) \ge 0 \Rightarrow \frac{1}{7} \le \lambda \le 7$

Now, $\Delta = \begin{vmatrix} 3 & -1 & 4 \\ 1 & 2 & -3 \\ 6 & 5 & \lambda \end{vmatrix}$

[Determinant of coefficients of equations]

$= 7(\lambda+5) \ne 0 \quad \left[\because \frac{1}{7} \le \lambda \le 7\right]$

Hence the given system of equations has a unique solution.

**12.** **(c)** Determinant of coefficients

$$= \begin{vmatrix} t & t+1 & t-1 \\ t+1 & t & t+2 \\ t-1 & t+2 & t \end{vmatrix} = \begin{vmatrix} t & 1 & -1 \\ t+1 & -1 & 1 \\ t-1 & 3 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} t & 1 & -1 \\ 2t+1 & 0 & 0 \\ 2t-1 & 4 & 0 \end{vmatrix} = -4(2t+1)$$

For non-trivial solution $t = -\frac{1}{2}$.

**13.** **(a)** The given system of equations will have a non-trivial solution if,

$$\begin{vmatrix} \alpha+a & \alpha & \alpha \\ \alpha & \alpha+b & \alpha \\ \alpha & \alpha & \alpha+c \end{vmatrix} = 0$$

Operate $R_2 \to R_2 - R_1$; $R_3 \to R_3 - R_1$, then

$$\begin{vmatrix} \alpha+a & \alpha & \alpha \\ -a & b & 0 \\ -a & 0 & c \end{vmatrix} = 0$$

$\Rightarrow \alpha ab + c(\alpha b + ab + a\alpha) = 0$ since $a, b, c \ne 0$

$\Rightarrow \alpha(bc+ca+ab) + abc = 0$

$\therefore \frac{1}{\alpha} = -\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$

**14.** **(c)** $\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$

$$\Rightarrow \begin{vmatrix} a-1 & 0 & 1-c \\ 0 & b-1 & 1-c \\ 1 & 1 & c \end{vmatrix} = 0,$$

$R_1 \to R_1 - R_3, R_2 \to R_2 - R_3$

$\Rightarrow (1-a)(1-b)(1-c)$

$$\begin{vmatrix} -1 & 0 & +1 \\ 0 & -1 & 1 \\ \frac{1}{1-a} & \frac{1}{1-b} & \frac{c}{1-c} \end{vmatrix} = 0,$$

$C_1 \to \left(\frac{1}{1-a}\right)C_1, C_2 \to \left(\frac{1}{1-b}\right)C_2,$

$C_3 \to \left(\frac{1}{1-c}\right)C_3$

$\Rightarrow (1-a)(1-b)(1-c)$

$$\left\{\frac{1}{(1-a)}(0+1) - \frac{1}{(1-b)}(-1+0) + \frac{c}{1-c}(1-0)\right\} = 0$$

$\Rightarrow \frac{1}{1-a} + \frac{1}{1-b} + \frac{c}{1-c} = 0,$

as $a \ne 1, b \ne 1, c \ne 1$ (given)

$\Rightarrow \frac{1}{1-a} + \frac{1}{1-b} + \frac{c-1+1}{1-c} = 0$

$\Rightarrow \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 1$

**15.** **(a)** The given system of equations will have a non-trivial solution if

$$\begin{vmatrix} \alpha+a & \alpha & \alpha \\ \alpha & \alpha+b & \alpha \\ \alpha & \alpha & \alpha+c \end{vmatrix} = 0$$

Operating $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$, we get

$$\begin{vmatrix} \alpha+a & \alpha & \alpha \\ -a & b & 0 \\ -a & 0 & c \end{vmatrix} = 0$$

or $\alpha ab + c(\alpha b + ab + a\alpha) = 0$

or $\alpha(bc + ca + ab) + abc = 0$

or $\dfrac{1}{\alpha} = -\left(\dfrac{1}{a}+\dfrac{1}{b}+\dfrac{1}{c}\right) \qquad (\because a, b, c \neq 0)$

**16.** **(c)** The given system is consistent. Therefore,

$$\Delta = \begin{vmatrix} 1 & 1 & -1 \\ 2 & -1 & -c \\ -b & 3b & -c \end{vmatrix} = 0$$

or $\quad c + bc - 6b + b + 2c + 3bc = 0$

or $\quad 3c + 4bc - 5b = 0 \quad$ or $\quad c = \dfrac{5b}{4b+3}$

Now, $c < 1$

$\Rightarrow \quad \dfrac{5b}{4b+3} < 1 \quad$ or $\quad \dfrac{5b}{4b+3} - 1 < 0$

or $\quad \dfrac{b-3}{4b+3} < 0 \quad \Rightarrow b \in \left(-\dfrac{3}{4}, 3\right)$

**17.** **(c)** We have,

$$\frac{d^n}{dx^n}[f(x)] = \begin{vmatrix} 0 & \cos\left(x+\dfrac{n\pi}{2}\right) & \dfrac{(-1)^n n!}{(x+3)^{n+1}} \\ 0 & \cos\dfrac{n\pi}{2} & \dfrac{(-1)^n n!}{3^{n+1}} \\ \alpha & \alpha^3 & \alpha^5 \end{vmatrix}$$

$$\therefore \quad \frac{d^n}{dx^n}[f(x)]_{x=0} = \begin{vmatrix} 0 & \cos\dfrac{n\pi}{2} & \dfrac{(-1)^n n!}{3^{n+1}} \\ 0 & \cos\dfrac{n\pi}{2} & \dfrac{(-1)^n n!}{3^{n+1}} \\ \alpha & \alpha^3 & \alpha^5 \end{vmatrix} = 0$$

($\because$ $R_1$ and $R_2$ are identical)

**18.** **(d)** $D' = \begin{vmatrix} a_1+pb_1 & b_1+qc_1 & c_1+ra_1 \\ a_2+pb_2 & b_2+qc_2 & c_2+ra_2 \\ a_3+pb_3 & b_3+qc_3 & c_3+ra_3 \end{vmatrix}$

$$= \begin{vmatrix} a_1 & b_1+qc_1 & c_1+ra_1 \\ a_2 & b_2+qc_2 & c_2+ra_2 \\ a_3 & b_3+qc_3 & c_3+ra_3 \end{vmatrix} + \begin{vmatrix} pb_1 & b_1+qc_1 & c_1+ra_1 \\ pb_2 & b_2+qc_2 & c_2+ra_2 \\ pb_3 & b_3+qc_3 & c_3+ra_3 \end{vmatrix}$$

In the first determinant, apply $C_3 \to C_3 - rC_1$ and then $C_2 \to C_2 - qC_3$.

In second determinant take $p$ common from $C_1$ and then apply

$C_2 \to C_2 - C_1$. Then take $q$ common from $C_2$ and apply

$C_3 \to C_3 - C_2$. Finally taking r common from $C_3$, we have ultimately $D' = (1 + pqr)D$.

**19.** **(d)** $\det\{\text{Adj}(\text{Adj } A)\} = (14)^4 = |A|^{(n-1)2}$

$\Rightarrow \quad |A| = 14 = 3x + 11 \Rightarrow x = 1$

**20.** **(c)** $A(\text{Adj } A) = |A| I_n$;

$|A| = xyz - (8x + 4y + 3z) + 28$

$$= 60 - 20 + 28 = 68 \Rightarrow A(\text{Adj } A) = \begin{bmatrix} 68 & 0 & 0 \\ 0 & 68 & 0 \\ 0 & 0 & 68 \end{bmatrix}$$

**21.** **(0)** Apply $C_3 \to C_3 - C_1$

$$\Rightarrow \begin{vmatrix} \sin^2\left(x+\dfrac{3\pi}{2}\right) & \sin^2\left(x+\dfrac{5\pi}{2}\right) & \sin(2x+5\pi)\sin(2\pi) \\ \sin\left(x+\dfrac{3\pi}{2}\right) & \sin\left(x+\dfrac{5\pi}{2}\right) & 2\cos\left(x+\dfrac{5\pi}{2}\right)\sin(\pi) \\ \sin\left(x-\dfrac{3\pi}{2}\right) & \sin\left(x-\dfrac{5\pi}{2}\right) & 2\cos\left(x-\dfrac{5\pi}{2}\right)\sin(-\pi) \end{vmatrix} = 0$$

$\because$ All elements of $C_3$ are zero.

**22.** **(20)** The matrices in the form

$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, $a_{ij} \in \{0, 1, 2\}$, $a_{11} = a_{12}$ are

$$\begin{bmatrix} 0 & 0/1/2 \\ 0/1/2 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0/1/2 \\ 0/1/2 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0/1/2 \\ 0/1/2 & 2 \end{bmatrix}$$

At any place, 0/1/2 means 0, 1 or 2 will be the element at that place.
Hence there are total 27 = 3 × 3 + 3 × 3 + 3 × 3) matrices of the above form. Out of which the matrices which are singular are

$$\begin{bmatrix} 0 & 0/1/2 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1/2 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

Hence there are total 7(= 3 + 2 + 1 + 1) singular matrices.
Therefore number of all non-singular matrices in the given form = 27 – 7 = 20

**23. (0.20)** $A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

det (adj(adj($A$))) = $|A|^4 = 27^4$

$$\left\{\frac{27^4}{5}\right\} = \frac{1}{5} = 0.20$$

**24. (0)**

$$\begin{vmatrix} \log a+(n-1)\log r & \log a+(n+1)\log r & \log a+(n+3)\log r \\ \log a+(n+5)\log r & \log a+(n+7)\log & \log a+(n+9)\log r \\ \log a+(n+11)\log r & \log a+(n+13)\log r & \log a+(n+15\log r \end{vmatrix}$$

$\Delta = C_3 \to C_3 - C_2 \quad C_2 \to C_2 - C_1$

$$\Rightarrow \begin{vmatrix} \log a+(n-1)\log r & 2\log r & 2\log r \\ \log a+(n+5)\log r & 2\log r & 2\log r \\ \log a+(n+11)\log r & 2\log r & 2\log r \end{vmatrix} = 0$$

**25. (16)** $\begin{vmatrix} (1+ap)^2 & (1+bp)^2 & (1+cp)^2 \\ (1+aq)^2 & (1+bq)^2 & (1+cq)^2 \\ (1+ar)^2 & (1+br)^2 & (1+cr)^2 \end{vmatrix}$

$$= \begin{vmatrix} 1 & 2a & a^2 \\ 1 & 2b & b^2 \\ 1 & 2c & c^2 \end{vmatrix} \times \begin{vmatrix} 1 & p & p^2 \\ 1 & q & q^2 \\ 1 & r & r^2 \end{vmatrix} = 2 \times 2\Delta_1 \cdot 2\Delta_2$$

$$= 8\Delta_1\Delta_2 = 8 \times \frac{1}{2} \times 4 = 16$$

**26. (4)** $f'(x) = \begin{vmatrix} 1 & \cos x & -\sin x \\ x^2 & -\tan x & -x^3 \\ 2x & \sin 2x & 5x \end{vmatrix}$

$$+ \begin{vmatrix} x & \sin x & \cos x \\ 2x & -\sec^2 x & -3x^2 \\ 2x & \sin 2x & 5x \end{vmatrix} + \begin{vmatrix} x & \sin x & \cos x \\ x^2 & -\tan x & -x^3 \\ 2 & 2\cos 2x & 5 \end{vmatrix}$$

Operate

$R_2 \to \frac{1}{x}R_2$, $R_3 \to \frac{1}{x}R_3$, $R_2 \to \frac{1}{x}R_2$,
respectively on three determinants

$$\frac{f'(x)}{x} = \begin{vmatrix} 1 & \cos x & -\sin x \\ x & -\frac{\tan x}{x} & -x^2 \\ 2x & \sin 2x & 5x \end{vmatrix}$$

$$+ \begin{vmatrix} x & \sin x & \cos x \\ 2x & -\sec^2 x & -3x^2 \\ 2 & \frac{\sin 2x}{x} & 5 \end{vmatrix} + \begin{vmatrix} x & \sin x & \cos x \\ x & -\frac{\tan x}{x} & -x^2 \\ 2 & 2\cos 2x & 5 \end{vmatrix}$$

$$\underset{x\to 0}{\text{Lt}} \frac{f'(x)}{x} = \begin{vmatrix} 1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 2 & 2 & 5 \end{vmatrix}$$

$$+ \begin{vmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 2 & 2 & 5 \end{vmatrix}$$

$$= 0 + 2(0+1) + 2(0+1) = 4$$

**27. (0)** The two roots of the equation are $1 \pm i\sqrt{3}$, so that we can take

$$\alpha = 1 + i\sqrt{3} = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 2e^{i\pi/3} \text{ and}$$

$$\beta = 1 - i\sqrt{3} = 2\left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = 2e^{-i\pi/3}$$

$$\alpha + \beta = 2, \alpha^2 + \beta^2 = 2^2\left(e^{i2\pi/3} + e^{-i2\pi/3}\right)$$

$$= 2^2 . 2\cos\frac{2\pi}{3} = -2^2,$$

$$\alpha^3 + \beta^3 = 2^3\left(e^{i\pi} + e^{-i\pi}\right) = -2^4,$$

$$\alpha^4 + \beta^4 = 2^4\left(e^{i4\pi/3} + e^{-i4\pi/3}\right) = -2^4$$

and $\alpha^5+\beta^5=2^5\left(e^{i5\pi/3}+e^{-i5\pi/3}\right)=2^5$

$$\text{Determinant}=\begin{vmatrix}2 & -2^2 & -2^4\\ -2^2 & -2^4 & -2^4\\ -2^4 & -2^4 & 2^5\end{vmatrix}$$

$$=-2^8\begin{vmatrix}1 & -2 & -2^2\\ 1 & 2^2 & 2\\ -1 & -1 & 1\end{vmatrix}=-2^8\begin{vmatrix}1 & -2 & -2^2\\ 0 & 6 & 6\\ 0 & -3 & -3\end{vmatrix}=0.$$

**28.** **(0)** Let $f(x)=\begin{vmatrix}(1+x)^2 & (1+x)^4 & (1+x)^6\\ (1+x)^3 & (1+x)^6 & (1+x)^9\\ (1+x)^4 & (1+x)^8 & (1+x)^{12}\end{vmatrix}$

Coefficient of '$x$' is $f'(0)$

$$f'(x)=\begin{vmatrix}2(1+x)^2 & 4(1+x)^3 & 6(1+x)^5\\ (1+x)^3 & (1+x)^6 & (1+x)^9\\ (1+x)^4 & (1+x)^8 & (1+x)^{12}\end{vmatrix}$$

$$+\begin{vmatrix}(1+x)^2 & (1+x)^4 & (1+x)^6\\ 3(1+x)^3 & 6(1+x)^5 & 9(1+x)^8\\ (1+x)^4 & (1+x)^8 & (1+x)^{12}\end{vmatrix}$$

$$+\begin{vmatrix}(1+x)^2 & (1+x)^2 & (1+x)^6\\ (1+x)^3 & (1+x)^6 & (1+x)^9\\ 4(1+x)^3 & 8(1+x)^7 & 12(1+x)^{11}\end{vmatrix}$$

Put $x=0, f'(0)=0$

**29.** **(1)** We have,

$$A=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$$

$$\Rightarrow\quad A^2=\begin{bmatrix}a & b\\ c & d\end{bmatrix}\begin{bmatrix}a & b\\ c & d\end{bmatrix}$$

$$=\begin{bmatrix}a^2+bc & ab+bd\\ ac+cd & bc+d^2\end{bmatrix}$$

Given, $A^2=A$ and ad – bc = 0

$$\therefore\quad \begin{bmatrix}a^2+bc & ab+bd\\ ac+cd & bc+d^2\end{bmatrix}=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$$

$\Rightarrow\; ab+bd=b \;\Rightarrow\; b(a+d)=b$

$\Rightarrow\; a+d=1$

**30.** **(2)** $-3x^4+\begin{vmatrix}1 & x & x^2\\ 1 & x^2 & x^4\\ 1 & x^3 & x^6\end{vmatrix}=0$

$$\Rightarrow -3x^4+x^3\begin{vmatrix}1 & 1 & 1\\ 1 & x & x^2\\ 1 & x^2 & x^4\end{vmatrix}=0$$

Now, apply $R_1\to R_1-R_2$, $R_2\to R_2-R_3$

$$\Rightarrow -3x^4+x^3\begin{vmatrix}0 & 1-x & 1-x^2\\ 0 & x-x^2 & x^2-x^4\\ 1 & x^2 & x^4\end{vmatrix}=0$$

$$\Rightarrow\quad -3x^4+x^3[x^2(1-x^2)(1-x)-x(1-x)(1-x^2)]=0$$

$\Rightarrow\; x^4[-3+(1-x)(1-x^2)(x-1)]=0$

$x=0, x=2$

CHAPTER 20

# Continuity and Differentiability

**1.** **(b)** f is continuous at $x=\pi/4$,

if $\lim_{x\to\pi/4} f(x)=f(\pi/4)$.

Now, $L=\lim_{x\to\pi/4}(\sin 2x)^{\tan^2 2x}$

$\Rightarrow \log L=\lim_{x\to\pi/4}\tan^2 2x\log\sin 2x$

$$=\lim_{x\to\pi/4}\frac{\log\sin 2x}{\cot^2 2x}\left(\frac{\infty}{\infty}\right)$$

$$=\lim_{x\to\pi/4}\frac{2\cot 2x}{-2\cot 2x\,\text{cosec}^2 2x.2}=-\frac{1}{2}$$

or $L=e^{-1/2}$

$\therefore f(\pi/4)=e^{-1/2}=1/\sqrt{e}$

**2.** **(d)** $f(x)=\begin{cases}\dfrac{1-[x]}{1+x}, & x\neq -1\\ 1, & x=-1\end{cases}$ and

$$f(x)=\begin{cases}1, & x<0\\ \dfrac{1-x}{1+x}, & x\geq 0\end{cases}$$

$$f(2x)=\begin{cases}1, & x<0\\ \dfrac{1-[2x]}{1+[2x]}, & x>0\end{cases}$$

$$\Rightarrow f(2x)=\begin{cases}1, & x<0\\ 1, & 0\leq x<\dfrac{1}{2}\\ 0, & \dfrac{1}{2}\leq x\leq 1\\ -\dfrac{1}{3}, & 1\leq x<\dfrac{3}{2}\end{cases}$$

$\Rightarrow f$ (x), for all values of x where $x<\frac{1}{2}$ continuous function and for $x=\frac{1}{2}$ and x = 1 $f(x)$ be a discontinuous function.

**3.** **(d)** If $f(a)=a$, then obviously $x=a$ is the solution.

Let $f(a)>a$ and $g(x)=f(x)-x$ then $g(a)>0$ and $g(f(a))=f(f(a))-f(a)=a-f(a)<0$

Since $g(x)$ is continuous, so at least for one $c\in(a, f(a))$, $g(c)=0$.

Similarly we can argue for $f(a)<a$.

**4.** **(b)** $f(0^+)=\lim_{x\to 0^+}|x|^{\sin x}=e^{\lim_{x\to 0^+}\sin x\ln|x|}$

$$=e^{\lim_{x\to 0^+}\frac{\ln|x|}{\text{cosec}\,x}}=e^{\lim_{x\to 0^+}\frac{1/x}{-\text{cosec}\,x\cot x}}$$

$$=e^{-\lim_{x\to 0^+}\left(\frac{\sin x}{x}\right)\tan x}=e^{-1\times 0}=1$$

$\therefore\ f(0^-)=g(0)=1$

Let $g(x)=ax+b \Rightarrow b=1\Rightarrow g(x)=ax+1$

For

$$x>0, f'(x)=e^{\sin x\ln(|x|)}\left[\cos x\ln(|x|)+\frac{\sin x}{x}\right]$$

$f'(1)=1(0+\sin 1)=\sin 1$

$f(-1)=-a+1 \Rightarrow a=1-\sin 1$

$g(x)=(1-\sin 1)x+1.$

**5.** **(b)** We apply check for continuity at $x=0$

$\text{LHL}=\lim_{x\to 0^-}f(x)=\lim_{h\to 0}f(0-h)$

$=\lim_{h\to 0}(\cos h+\sin h)^{-\text{cosec}\,h}$ ($1^\infty$ from)

$=\exp\{\lim_{h\to 0}(\cos h+\sin h-1)\times -\text{cosec}\,h\}$

$$=\exp\left\{\lim_{h\to 0}\left(-2\sin^2\frac{h}{2}+2\sin\frac{h}{2}\cos\frac{h}{2}\right)\times -\frac{1}{2\sin\frac{h}{2}\cos\frac{h}{2}}\right\}$$

$$= \exp\left\{\lim_{h\to0}\left(\frac{\sin\frac{h}{2}-\cos\frac{h}{2}}{\cos\frac{h}{2}}\right)\right\} = e^{-1}$$

$$\text{RHL} = \lim_{x\to0^+} f(x) = \lim_{h\to0} f(0+h)$$

$$= \lim_{h\to0}\frac{e^{1/h}+e^{2/h}+e^{3/h}}{ae^{2/h}+be^{3/h}}$$

$$= \lim_{h\to0}\frac{e^{3/h}(e^{-2/h}+e^{-1/h}+1)}{e^{3/h}\{ae^{-1/h}+b\}} = \frac{1}{b}$$

$$\left[\because \lim_{h\to0} e^{-1/h} = 0\right]$$

$\therefore$ For continuity at $x = 0$

$$e^{-1} = a = b^{-1} \Rightarrow a = \frac{1}{e}, b = e$$

**6.** **(b)** $f(x) = \lim_{t\to\infty}\left\{\frac{(1+\sin\pi x)^t - 1}{(1+\sin\pi x)^t - 1}\right\}$

$\because \sin\pi x > 0$ (in I and II quadrants)

$\therefore\ 2n\pi < \pi x < (2n+1)\pi$

$\Rightarrow\ 2n < x < 2n+1,\ n \in I$

and $\sin\pi x < 0$ (in III and IV quadrrants)

$\therefore\ (2n+1) < \pi x < (2n+2)\pi$

$\Rightarrow\ 2n+1 < x < 2n+2,\ n \in \text{I}$ and

$\sin\pi x = 0$ if $x = 0, 1, 2, \ldots\ldots\ldots$

$$f(x) = \begin{cases} \lim\limits_{t\to\infty}\dfrac{1-\dfrac{1}{(1-\sin\pi x)^t}}{1+\dfrac{1}{(1+\sin\pi x)^t}}, & 2n < x < 2n+1 \\ \lim\limits_{t\to\infty}\dfrac{(1+\sin\pi x)^t+1}{(1+\sin\pi x)^t+1}, & 2n+1 < x < 2n+2 \\ 0, & x = 0,1,2,\ldots.. \end{cases}$$

$$= \begin{cases} 1, & 2n < x < 2n+1 \\ -1 & 2n+1 < x < 2n+2 \\ 0, & x = 0,1,2\ldots\ldots\ldots. \end{cases}$$

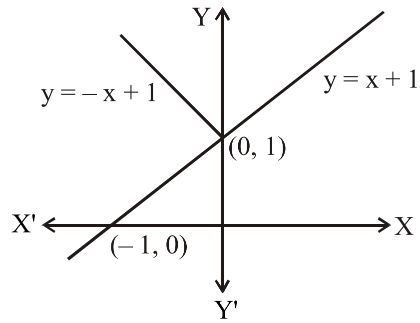
If $k \in I, f(k) = 0$, but $\lim_{x\to k} f(x) = 1$ or $-1$

according as

$k \in (2n, 2n+1)$ or $k \in (2n+1, 2n+2)$

**7.** **(a)** $f(x) = \min\{x+1, |x|+1\}$

$\Rightarrow f(x) = x+1\ \forall\ x \in R$




Hence, $f(x)$ is differentiable everywhere for all $x \in R$.

**8.** **(c)** As per question,

$p$ = left hand derivative of $|x-1|$ at $x = 1$

$\Rightarrow p = -1$

Also $\lim_{x\to1+} g(x) = p$

Where $g(x) = \dfrac{(x-1)^n}{\log\cos^m(x-1)},\ 0 < x < 2,$

$m, n$ are integers, $m \neq 0, n > 0$

$\therefore$ we get,

$$\lim_{x\to1^+}\frac{(x-1)^n}{\log\cos^m(x-1)} = -1$$

$$\Rightarrow \lim_{h\to0}\frac{h^n}{\log\cos^m h} = -1$$

$$\Rightarrow \lim_{h\to0}\frac{h^n}{m(\log\cos h)} = -1$$

[Using L' Hospital's rule]

$$\Rightarrow \lim_{h\to0}\frac{n h^{n-1}\cos h}{m(-\sin h)} = -1$$

[Using L' Hospital's rule]

$$\Rightarrow \lim_{h\to0}\frac{n h^{n-2}\cos h}{m\left(\frac{\sin h}{h}\right)} = 1 \Rightarrow n = 2 \text{ and } m = 2$$

**9.** **(b)** Let $|f(x)| \le x^2,\ \forall x \in R$

Now, at $x = 0$, $|f(0)| \le 0$

$\Rightarrow f(0) = 0$

$$\therefore f'(0) = \lim_{h\to0}\frac{f(h)-f(0)}{h-0} = \lim_{h\to0}\frac{f(h)}{h} \quad \ldots(1)$$

Now, $\left|\frac{f(h)}{h}\right| \le |h|$ $\quad (\because |f(x)| \le x^2)$

$\Rightarrow -|h| \le \frac{f(h)}{h} \le |h| \Rightarrow \lim_{h\to 0} \frac{f(h)}{h} \to 0$ ...(2)

(using sandwich Theorem)

$\therefore$ from (1) and (2), we get $f'(0) = 0$,
i.e. - $f(x)$ is differentiable, at $x = 0$
Since, differentiability $\Rightarrow$ Continuity
$\therefore |f(x)| \le x^2$, for all $x \in R$ is continuous as well as differentiable at $x = 0$.

**10.** **(a)** $f(x) = \cot^{-1}\left(\frac{x^x - x^{-x}}{2}\right) = \cot^{-1}\left(\frac{(x)^{2x}-1}{2x^x}\right)$

$= \tan^{-1}\left(\frac{2x^x}{-1+(x^x)^2}\right) = -2\tan^{-1} x^x$

$\therefore \quad f'(x) = -\frac{2}{1+(x^x)^2} \times x^x(1+\log_e x)$

$\therefore \quad f'(1) = \frac{-2}{2} \times 1 = -1$

**11.** **(c)** We have, $g$ = inverse of $f = f^{-1}$
$\Rightarrow g(x) = f^{-1}(x) \Rightarrow f[g(x)] = x$
Differentiating w.r.t. $x$, we get $f'[g(x)].g'(x) = 1$

$\therefore g'(x) = \frac{1}{f'[g(x)]} = 1 + [g(x)]^3$

$\left[\because f'(x) = \frac{1}{1+x^3} \quad \therefore f'[g(x)] = \frac{1}{1+[g(x)]^3}\right]$

**12.** **(b)** Since, $2 < x < 3$, $\therefore [x] = 2$

$\therefore \quad f(x) = \sin\left(\frac{2\pi}{3} - x^2\right)$

$\therefore f'(x) = \cos\left(\frac{2\pi}{3} - x^2\right)(-2x)$

$\therefore f'\left(\sqrt{\frac{5\pi}{3}}\right) = 2\sqrt{\frac{5\pi}{3}}$

**13.** **(c)** Put $x^n = \cos\alpha$, $y^n = \cos\beta$

$\Rightarrow \tan\frac{\alpha-\beta}{2} = -a$

$\Rightarrow \cos^{-1}(x^n) - \cos^{-1}(y^n) = 2\tan^{-1}(-a)$

$\Rightarrow \frac{y^{n-1}}{\sqrt{1-y^{2n}}}\frac{dy}{dx} = \frac{x^{n-1}}{\sqrt{1-x^{2n}}} = \frac{x^{n-1}}{y^{n-1}}$

**14.** **(d)** Let $x^3 = \cos p$ and $y^3 = \cos q$
Given $\sqrt{(1-x^6)} + \sqrt{(1-y^6)} = a(x^3 - y^3)$

$\Rightarrow \sqrt{(1-\cos^2 p)} + \sqrt{(1-\cos^2 q)}$
$= a(\cos p - \cos q)$

$\Rightarrow \sin p + \sin q = a(\cos p - \cos q)$

$\Rightarrow 2\sin\left(\frac{p+q}{2}\right)\cos\left(\frac{p-q}{2}\right)$

$= -2a\sin\left(\frac{p-q}{2}\right)\sin\left(\frac{p+q}{2}\right)$

$\Rightarrow \tan\left(\frac{p-q}{2}\right) = -\frac{1}{a} \Rightarrow p - q = \tan^{-1}\left(-\frac{1}{a}\right)$

$\Rightarrow \cos^{-1}x^3 - \cos^{-1}y^3 = \tan^{-1}\left(-\frac{1}{a}\right)$

Differentiate w.r.t. x, we have

$-\frac{3x^2}{\sqrt{1-x^6}} + \frac{3y^2}{\sqrt{1-y^6}}\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{x^2}{y^2}\sqrt{\frac{1-y^6}{1-x^6}}$.

Hence, $f(x, y) = x^2/y^2$

**15.** **(b)** $f'(x) = n(1+x)^{n-1}$, $f''(x) = n(n-1)(1+x)^{n-2}$
$\therefore \quad f^n(x) = n!$, $f^n(0) = n!$

$\Rightarrow 1 + n + \frac{n(n-1)}{2!} + ... + \frac{n!}{n!}$
$= {}^nC_0 + {}^nC_1 + {}^nC_2 + ... + {}^nC_n = 2^n$

**16.** **(c)** Given $e^{\alpha}\cos\alpha = 1$ ....(i)

and $e^{\beta}\cos\beta = 1$ .....(ii)

Let $f(x) = e^{-x} - \cos x$, then $f(x)$ is continuous and differentiable.
Also, $f(\alpha) = f(\beta) = 0$ (from (i) and (ii))
Therefore by Rolle's MVT, $f'(x) = 0$ has at least one root in $(\alpha, \beta)$.

$\Rightarrow -e^{-x} + \sin x = 0$ for at least one $x \in (\alpha, \beta)$

**17.** **(b)** We have, $f(x) = x|\sin x|, x \in R$

$$f(x) = \begin{cases} x\sin x, & x \in (2n\pi, (2n+1)\pi) \\ -x\sin x, & x \in (2n+1)\pi, 2\pi n) \end{cases}$$

$f'(n\pi) = \lim_{x\to nx} \frac{f(x) - f(n\pi)}{x - n\pi}$

$= \lim_{x\to nx} \frac{x|\sin x|}{x - n\pi}$

Clearly, $f(x)$ is differentiable for all $x$ except $x = n\pi$, $n = \pm 1, \pm 2, \pm 3, ....$

**18.** **(c)** We have,

$$f(x)=\begin{cases} x^2\left|\cos\left(\frac{\pi}{x}\right)\right| & \text{for } x\neq 0 \\ 0 & \text{for } x=0 \end{cases}$$

For $x=0, f(0)=0$

and $\lim\limits_{x\to 0} x^2\left|\cos\frac{\pi}{n}\right|=0$.

$\Rightarrow$ f(x) is continuous at $x=0$

Now $Lf'(0)=\lim\limits_{x\to 0}\frac{f(0-h)-0}{-h}$

$$=\lim_{x\to 0}\frac{h^2\cos\left(\frac{\pi}{h}\right)-0}{-h}=\lim_{x\to 0}-h\cos\frac{\pi}{h}=0.$$

Simillarly, $Rf'(0)=0$

$\Rightarrow$ differentiable at $x=0$.

Now, $f(x)$ is not differentiable at $\left|\cos\frac{\pi}{x}\right|=0$

$$\cos\frac{\pi}{x}=0\Rightarrow\frac{\pi}{x}=(2n+1)\frac{\pi}{2}$$

$$\Rightarrow\quad x=\frac{2}{2x+1}$$

$\because$ $f(x)$ is not differentiable at

$$x\in[-1,1]: x=\frac{2}{2n+1},\ n\in Z$$

**19.** **(d)** Let $f: R\to R$ be a continuous function such that $f(x^2)=f(x^3)$ ...(i) for all $x\in R$

Put $x=-x$, in eqn. (i), then $f(x^2)=f(-x^3)$

from (i) $f(x^3)=f(-x^3)$

Let $x^3=t$, then $f(t)=f(-t)$

$\Rightarrow f(x)$ is an even function

Now take $x^3=t$, then from (i)

$f(t^{2/3})=f(t)$

Put $t=t^{2/3}$

$\Rightarrow f(t^{(2/3)^2})=f(t^{2/3})$

$\Rightarrow f(t)=f(t^{2/3})=f(t^{(2/3)^2})$

$=f(t^{(2/3)^3})\ldots=f(t^{(2/3)^n})$

This is true for all $t\in R$ and any $n\in I$.

When $n\to\infty$, $\left(\frac{2}{3}\right)^n\to 0$

Then $f(t)=f(t^0)=1$

Hence, $f(x)$ is a constant function and therefore it is differentiable everywhere

**20.** **(d)** $\lim\limits_{x\to 0} f(x)=\lim\limits_{x\to 0}\frac{\sin x^2}{x^2}$

$$=\lim_{x\to 0}\left(\frac{\sin x^2}{x^2}\right)x=(1)(0)=0=f(0)$$

$\Rightarrow f(x)$ is continuous

Now, to check the differentiability of $f(x)$

R.H.D.

$$=\lim_{h\to 0}\frac{f(0+h)-0}{h}=\lim_{h\to 0}\frac{\frac{\sin h^2}{h}-0}{h}=1$$

$$\text{L.H.D.}=\lim_{h\to 0}\frac{f(0-h)-0}{-h}=\lim_{h\to 0}\frac{\frac{\sin h^2}{-h}-0}{-h}=1$$

R.H.D. = L.H.D.

$\Rightarrow f(x)$ is differentiable

Now, differentiate $f(x)$ w.r.t. $x$, we get

$$f'(x)=\frac{x\cos(x^2)2x-\sin(x^2)}{x^2}$$

$$f'(x)=2\cos x^2-\frac{\sin x^2}{x^2}$$

$$f'(x)=\begin{cases}2\cos x^2-\frac{\sin x^2}{x^2}, & x\neq 0\\ 1, & x=0\end{cases}$$

Now, to check the continuity of $f'(x)$

$$\lim_{x\to 0} f'(x)=\lim_{x\to 0}2\cos x^2-\frac{\sin x^2}{x^2}=2-1=1$$

$\Rightarrow f'(x)$ is continuous

**21.** **(0)** $f'(x)=\lim\limits_{h\to 0}\frac{f(x+h)-f(x)}{h}$

$$|f'(x)|=\lim_{h\to 0}\left|\frac{f(x+h)-f(x)}{h}\right|\leq\lim_{h\to 0}\left|\frac{(h)^2}{h}\right|$$

$\Rightarrow\ |f'(x)|\leq 0$

$\Rightarrow\ f'(x)=0$

$\Rightarrow\ f(x)=$ constant

As $f(0)=0\Rightarrow f(1)=0$.

**22.** **(4)** $\because\ x = \text{cosec}\,\theta - \sin\theta$

$\Rightarrow\quad x^2 + 4 = (\text{cosec}\,\theta - \sin\theta)^2 + 4$

$= (\text{cosec}\,\theta + \sin\theta)^2$ ....... (i)

and $\quad y^2 + 4 = (\text{cosec}^n\,\theta - \sin^n\theta)^2 + 4$

$= (\text{cosec}^n\,\theta + \sin^n\theta)^2$ ........(ii)

$$\text{Now},\ \frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)}$$

$$= \frac{n(\text{cosec}^{n-1}\theta)(-\text{cosec}\,\theta\cot\theta) - n\sin^{n-1}\theta\cos\theta}{-\text{cosec}\,\theta\cot\theta - \cos\theta}$$

$$= \frac{n(\text{cosec}^n\theta\cot\theta + \sin^{n-1}\theta\cos\theta)}{(\text{cosec}\,\theta\cot\theta + \cos\theta)}$$

$$= \frac{n\cot\theta(\text{cosec}^n\,\theta + \sin^n\theta)}{\cot\theta(\text{cosec}\,\theta + \sin\theta)}$$

$$= \frac{n(\text{cosec}^n\,\theta + \sin^n\theta)}{(\text{cosec}\,\theta + \sin\theta)} = \frac{n\sqrt{y^2+4}}{\sqrt{x^2+4}}$$

[From (i) and (ii)]

Squaring both side, we get

$$\left(\frac{dy}{dx}\right)^2 = \frac{n^2(y^2+4)}{(x^2+4)}$$

or $\quad (x^2+4)\left(\frac{dy}{dx}\right)^2 - n^2y^2 = 4n^2 = kn^2$

$\Rightarrow k = 4.$

**23.** **(1)** We have,

$f(x) = \cos x \cos 2x \cos 2^2 x \cos 2^3 x$ ..... $\cos 2^{n-1}x$

$$\Rightarrow f(x) = \frac{\sin 2^n x}{2^n \sin x}$$

$$\Rightarrow f'(x) = \frac{2^n \cos 2^n x \sin x - \sin 2^n x \cos x}{2^n \sin^2 x}$$

$$\Rightarrow\quad f'\left(\frac{\pi}{2}\right) = \frac{2^n \cos 2^{n-1}\pi}{2^n} = \cos 2^{n-1}\pi$$

$= (-1)\ 2^{\,n-1} = 1$

**24.** **(5)** $F'(x) = \left[f\left(\frac{x}{2}\right).f'\left(\frac{x}{2}\right) + g\left(\frac{x}{2}\right)g'\left(\frac{x}{2}\right)\right]$

Here, $g(x) = f'(x)$

and $g'(x) = f''(x) = -f(x)$

So, $\quad F'(x) = f\left(\frac{x}{2}\right)g\left(\frac{x}{2}\right) - f\left(\frac{x}{2}\right)g\left(\frac{x}{2}\right) = 0$

$\Rightarrow\quad F(x)$ is constant function

So, $\quad F(10) = 5$

**25.** **(2)** Given, $|f(x) - f(y)| = |x - y|$ for all in [0, 1]

$$\Rightarrow\quad \frac{|f'(x) - f(y)|}{|x-y|} = 1$$

$$\Rightarrow\quad \lim_{x\to y}\frac{|f(x) - f(y)|}{x-y} = 1 \Rightarrow |f'(x)| = 1$$

$\Rightarrow\quad f'(x) = \pm 1 \Rightarrow f(x) = \pm x\ \ x \in [0,1]$

$\therefore\quad f(x)$ has exactly 2 function.

**26.** **(3)** Taking logarithm of both sides, we get

$$\log y = x\left[\log\left(1 + \frac{1}{x}\right)\right]$$

$$\Rightarrow\quad \frac{1}{y}y_1(x) = \frac{x^2}{x+1}\left(-\frac{1}{x^2}\right) + \log\left(1+\frac{1}{x}\right)$$

$$= -\frac{1}{x+1} + \log\left(1+\frac{1}{x}\right) \quad\text{......... (1)}$$

Since $y(2) = (1 + 1/2)^2 = 9/4$

So, $\quad y_1(2) = (9/4)\left(-\frac{1}{3} + \log\frac{3}{2}\right)$

Multiplying (1) by $y$ and then differentiating, we get

$$y_2(x) = y_1(x)\left(-\frac{1}{x+1} + \log\left(1+\frac{1}{x}\right)\right)$$

$$+ y(x).\left(\frac{1}{(x+1)^2} + \frac{x}{x+1}\left(-\frac{1}{x^2}\right)\right)$$

So,

$$y_2(2) = y_1(2)\left(-\frac{1}{3} + \log\frac{3}{2}\right) + y(2)\left(\frac{1}{9} - \frac{1}{6}\right)$$

$$= \left(\frac{9}{4}\right)\left(-\frac{1}{3} + \log\frac{3}{2}\right)^2 - \frac{1}{8}$$

**27.** **(3)** Here,

$$y=\sqrt{\left((x-\sin x)+\sqrt{(x-\sin x)+\ldots\infty}\right)}$$

So, $y=\sqrt{(x-\sin x)+y}$ $\therefore$ $y^2=x-\sin x+y$

Differentiating, we get : $2y\dfrac{dy}{dx}=1-\cos x+\dfrac{dy}{dx}$

At $x=\dfrac{\pi}{2}, y^2-y=\dfrac{\pi}{2}-1\Rightarrow y^2-y+\dfrac{1}{4}=\dfrac{\pi}{2}-\dfrac{3}{4}$

$$\Rightarrow\left(y-\frac{1}{2}\right)^2=\frac{2\pi-3}{4}\Rightarrow(2y-1)=\pm\sqrt{2\pi-3}$$

From Eq. (i), we get $(2y-1)\dfrac{dy}{dx}=1-\cos x$

$$\therefore\quad\left|\frac{dy}{dx}\right|_{x=\frac{\pi}{2}}=\sqrt{2\pi-3}\Rightarrow\left|\left|\frac{dx}{dy}\right|^2_{x=\frac{\pi}{2}}-2\pi\right|=3$$

**28.** **(2)** $\lim\limits_{x\to 0}f(x)=\dfrac{1}{3}$ $\Rightarrow\lim\limits_{x\to 0}\dfrac{x\cdot\alpha\cot x+\beta}{x^2}=\dfrac{1}{3}$

$$\Rightarrow\quad\lim_{x\to 0}\frac{x\alpha+\beta\tan x}{x^2\tan x}=\frac{1}{3}$$

$$\Rightarrow\quad\lim_{x\to 0}\frac{\alpha x+\beta\left(x+\frac{x^3}{3}+\ldots\infty\right)}{x^3\left(\frac{\tan x}{x}\right)}=\frac{1}{3}$$

$$\Rightarrow\quad\lim_{x\to 0}\frac{(\alpha+\beta)x+\left(\frac{\beta}{3}\right)x^3+\ldots\infty}{x^3}=\frac{1}{3}$$

So $\alpha+\beta=0$

Also, $\beta=1\Rightarrow\alpha=-1$ $\Rightarrow\alpha^2+\beta^2=2$

**29.** **(4)** $f(x)=\begin{cases}3-\left|\cos x-\dfrac{1}{\sqrt{2}}\right|, & |\sin x|<\dfrac{1}{\sqrt{2}}\\ 2+\left|\cos x+\dfrac{1}{\sqrt{2}}\right|, & |\sin x|\geq\dfrac{1}{\sqrt{2}}\end{cases}$

$$=\begin{cases}3-\left|\cos x-\dfrac{1}{\sqrt{2}}\right|, & |\cos x|>\dfrac{1}{\sqrt{2}}\\ 2+\left|\cos x+\dfrac{1}{\sqrt{2}}\right|, & |\cos x|\leq\dfrac{1}{\sqrt{2}}\end{cases}$$

$$=\begin{cases}3-\cos x+\dfrac{1}{\sqrt{2}}, & |\cos x|>\dfrac{1}{\sqrt{2}}\\ 2+\cos x+\dfrac{1}{\sqrt{2}}, & |\cos x|\leq\dfrac{1}{\sqrt{2}}\end{cases}$$

Thus, $f(x)$ is discontinuous at

$|\cos x|=\dfrac{1}{\sqrt{2}}$ or $x=\dfrac{\pi}{4},\dfrac{3\pi}{4},\dfrac{5\pi}{4},\dfrac{7\pi}{4}$

**30.** **(3)** Put $\cos\phi=\dfrac{2}{\sqrt{13}};\sin\phi=\dfrac{3}{\sqrt{13}};\tan\phi=\dfrac{3}{2}$

$y=\cos^{-1}\{\cos(x+\phi)\}+\sin^{-1}\{\cos(x-\phi)\}$

$=\cos^{-1}\{\cos(x+\phi)\}+\dfrac{\pi}{2}-\cos^{-1}\{\cos(\phi-x)\}$

$=x+\phi+\dfrac{\pi}{2}-\phi+x$

$y=2x+\dfrac{\pi}{2};z=\sqrt{1+x^2}$

Now, $\dfrac{dy}{dz}=\dfrac{\frac{dy}{dx}}{\frac{dz}{dx}}=\dfrac{2\sqrt{1+x^2}}{x}$

$$\therefore\left(\frac{dy}{dz}\right)_{x=\frac{3}{4}}=\frac{10}{3}$$

CHAPTER 21

# Application of Derivatives

1. **(d)** Given $f(x) = \tan^{-1}(\sin x + \cos x)$

$$f'(x) = \frac{1}{1+(\sin x+\cos x)^2}.(\cos x - \sin x)$$

$$= \frac{\sqrt{2}.\left(\frac{1}{\sqrt{2}}\cos x - \frac{1}{\sqrt{2}}\sin x\right)}{1+(\sin x+\cos x)^2}$$

$$= \frac{\left(\cos\frac{\pi}{4}.\cos x - \sin\frac{\pi}{4}.\sin x\right)}{1+(\sin x+\cos x)^2}$$

$$\therefore \quad f'(x) = \frac{\sqrt{2}\cos\left(x+\frac{\pi}{4}\right)}{1+(\sin x+\cos x)^2}$$

if $f'(x) > 0$ then $f(x)$ is increasing function.

Hence $f(x)$ is increasing, if $-\frac{\pi}{2} < x + \frac{\pi}{4} < \frac{\pi}{2}$

$$\Rightarrow \quad -\frac{3\pi}{4} < x < \frac{\pi}{4}$$

Hence, $f(x)$ is increasing when $n \in \left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$

2. **(b)** If $f(x) = x^{1/x}$, then

$$f'(x) = \frac{1}{x^2}\left[x^{1/x}(1-\ln x)\right]$$

$f$ is decreasing if $x > e$ and $f$ is increasing if $x < e$.
As $e < 3 < 4 < 5 < 6 < 7$

$\therefore \quad$ Max $\{3^{1/3}, 4^{1/4}, 5^{1/5}, 6^{1/6}, 7^{1/7}\} = 3^{1/3}$

Also $1 < 2 < e \quad \therefore$ Max $\{1, 2^{1/2}\} = 2^{1/2}$

But $2^{1/2} = 4^{1/4} < 3^{1/3}$

$\therefore \quad$ the greatest number is $3^{1/3}$

3. **(a)** $f'(x) = 2x\log 27 - 6\log 27$

$$+(6x-18)\log(x^2-6x+8)$$

$$+\frac{(3x^2-18x+24)(2x-6)}{x^2-6x+8}$$

$$= 6(x-3)\left[\log 3 + \log\left(x^2-6x+8\right)+1\right]$$

$$= 6(x-3)\log 3e\left(x^2-6x+8\right)$$

For $f(x)$ to be defined $x^2 - 6x + 8 > 0$
$\Rightarrow x < 2$ or $x > 4$

If $x > 4$ then $f'(x) < 0$ if $\log 3(x^2-6x+8)e < 0$

i.e. $3(x^2-6x+8)e < 1$ i.e. $x^2 - 6x + (8 - 1/3e) < 0$

i.e. $(x-(3+\sqrt{1+1/3e}))(x-(3-\sqrt{1+1/3e}) < 0$

$\Leftrightarrow \quad 3-\sqrt{1+1/3e} < x < (3+\sqrt{1+1/3e})$

Hence $x \in (4, 3+\sqrt{1+1/3e})$
Similarly if $x < 2$, then $f'(x) < 0$,
If $\log 3(x^2-6x+8)\,e > 0$

i.e., $\quad x < 3-\sqrt{1+1/3e}$ or $x > 3+\sqrt{1+1/3e}$

Hence $x \in (3-\sqrt{1+1/3e}, 2)$

4. **(a)** Here $f'(x) > 0 \forall x \in R$

$\Rightarrow \quad 2\cos 2x - 8(a+1)\cos x$

$$-(4a^2+8a-14) > 0 \forall x \in R$$

$\Rightarrow \quad 4\cos^2 x - 8(a+1)\cos x - (4a^2+8a-12) > 0$

$\Rightarrow \quad \cos^2 x - 2(a+1)\cos x - (a^2+2a-3) > 0$

$\Rightarrow \quad (\cos x - \alpha)(\cos x - \beta) > 0$

where $\quad \alpha = (a+1)+\sqrt{2a^2+4a-2}$

and $\quad \beta = (a+1)-\sqrt{2a^2+4a-2}$

$\Rightarrow \quad \cos x - \beta < 0$ since $\cos x - \alpha < 0 \;\forall a > 0$

$\Rightarrow \quad (a+1)-\sqrt{2a^2+4a-2} > \cos x \;\forall x \in R$

$\Rightarrow \quad (a+1)-\sqrt{2a^2+4a-2} > 1$

$\Rightarrow \quad \sqrt{2a^2+4a-2} < a \;\Rightarrow\; a^2+4a-2 < 0$

$\Rightarrow \quad a \in (-2-\sqrt{6}, \sqrt{6}-2)$

Hence $a \in (0, \sqrt{6}-2)$ since $a > 0$.

5. **(c)** We have $f(x) = \dfrac{x}{\sin x}, 0 < x \le 1$

$$\Rightarrow \quad f'(x) = \frac{\sin x - x\cos x}{\sin^2 x}$$

where $\sin^2 x$ is always +ve, when $0 < x \le 1$. But to check Nr., we again let $h(x) = \sin x - x\cos x$

$\Rightarrow \quad h'(x) = x\sin x > 0$ for $0 < x \le 1$

$\Rightarrow \quad h(x)$ is increasing $\Rightarrow h(0) < h(x)$, when $0 < x \le 1$

$\Rightarrow \quad 0 < \sin x - x\cos x$, when $0 < x \le 1$

$\Rightarrow \quad \sin x - x\cos x > 0$, when $0 < x \le 1$

$\Rightarrow \quad f'(x) > 0, x \in (0,1] \Rightarrow f(x)$ is increasing on $(0,1]$

Again $g(x) = \dfrac{x}{\tan x}$

$\Rightarrow \quad g'(x)=\dfrac{\tan x - x\sec^2 x}{\tan^2 x}$, when $0<x\le 1$

Here $\tan^2 x>0$ But to check Nr. we consider $p(x)=\tan x - x\sec^2 x$

$p'(x)=\sec^2 x-\sec^2 x - x.2\sec x.\sec x\tan x$

$\Rightarrow \quad p'(x)=-2x\sec^2 x\tan x<0$ for $0<x\le 1$

$\Rightarrow$ $p(x)$ is decreasing, when $0<x\le 1$

$\Rightarrow$ $p(0)>p(x) \Rightarrow 0>\tan x - x\sec^2 x$

$\therefore \quad g'(x)<0$

Hence $g(x)$ is decreasing when $0<x\le 1$.

**6.** **(c)** Given that $g(u)=2\tan^{-1}(e^u)-\dfrac{\pi}{2}$

$\therefore \quad g(-u)=2\tan^{-1}\left(e^{-u}\right)-\dfrac{\pi}{2}$

$=2\tan^{-1}\left(\dfrac{1}{e^u}\right)-\dfrac{\pi}{2}$

$=2\cot^{-1}\left(e^u\right)-\dfrac{\pi}{2}=2\left[\dfrac{\pi}{2}-\tan^{-1}\left(e^u\right)\right]-\dfrac{\pi}{2}$

$=\pi-2\tan^{-1}(e^u)-\dfrac{\pi}{2}=\dfrac{\pi}{2}-2\tan^{-1}(e^u)=-g(u)$

$\therefore$ $g$ is an odd function.

Also $g'(u)=\dfrac{2e^u}{1+e^{2u}}>0$, $\forall u\in(-\infty,\infty)$

$\therefore$ $g$ is strictly increasing on $(-\infty,\infty)$.

**7.** **(b)** Consider the function $f(x)=\dfrac{x^2}{x^3+200}$ in the interval $[1,\infty)$.

Since the derivative $f'(x)=\dfrac{x(400-x^3)}{(x^3+200)^2}$ is positive at $0<x$ $\sqrt[3]{400}$ and negative at $x>\sqrt[3]{400}$, the function $f(x)$ increases at $0<x<\sqrt[3]{400}<8$ it follows that the largest term in the sequence can be either $a_7$ or $a_8$. Since $a_7=49/534>a_8=8/89$, the largest term in the given sequence is $a_7=\dfrac{49}{543}$.

**8.** **(c)** We have,

$f(x)=\dfrac{x^2+1}{[x]}$, for all $x\in[1,4)$

$$\Rightarrow \quad f(x)=\begin{cases} x^2+1, & \text{for all } x\in[1,2) \\ \dfrac{x^2+1}{2}, & \text{for all } x\in[2,3) \\ \dfrac{x^2+1}{3}, & \text{for all } x\in[3,4) \end{cases}$$

$$\Rightarrow \quad f'(x)=\begin{cases} 2x, & \text{for all } x\in[1,2) \\ x, & \text{for all x } \in[2,3) \\ \dfrac{2x}{3}, & \text{for all } x\in[3,4) \end{cases}$$

Clearly, $f'(x)>0$ in each of the intervals $(1,2)$, $(2,3)$ and $(3,4)$. So, $f(x)$ is increasing in each of these intervals. It is to note here that $f(x)$ is not increasing on $[1, 4]$ because values of $f(x)$ in $(1.5, 2)$ are greater than the values of $f(x)$ in $(2, 3)$.

**9.** **(b)** We have, $A+B=\dfrac{\pi}{3}$

$\therefore B=\dfrac{\pi}{3}-A\Rightarrow \tan B=\dfrac{\sqrt{3}-\tan A}{1+\sqrt{3}\tan A}$

Let Z = tan A. tan B. Then,

$Z=\tan A.\dfrac{\sqrt{3}-\tan A}{1+\sqrt{3}\tan A}=\dfrac{\sqrt{3}\tan A-\tan^2 A}{1+\sqrt{3}\tan A}$

$\Rightarrow Z=\dfrac{\sqrt{3}x-x^2}{1+\sqrt{3}x}$, where $x=\tan A$

$\Rightarrow \dfrac{dZ}{dx}=-\dfrac{(x+\sqrt{3})(\sqrt{3}x-1)}{(1+\sqrt{3}x)^2}$

For max Z, $\dfrac{dZ}{dx}=0\Rightarrow x=\dfrac{1}{\sqrt{3}},-\sqrt{3}.$

$x\ne-\sqrt{3}$ because $A+B=\pi/3$ which implies that $x=\tan A>0$. It can be easily checked that

$\dfrac{d^2Z}{dx^2}<0$ for $x=\dfrac{1}{\sqrt{3}}$. Hence, Z is maximum

for $x=\dfrac{1}{\sqrt{3}}$ i.e. $\tan A=\dfrac{1}{\sqrt{3}}$ or $A=\pi/6$.

For this value of x, $Z=\dfrac{1}{3}$.

**10.** **(a)** $f'(x)=(4a-3)+(a-7)\cos x$

For non-existence of critical point $f'(x)\ne 0$ for any $x\in R$.

$\Rightarrow \quad \cos x\ne\dfrac{3-4a}{a-7}$ for any $x\Rightarrow \left|\dfrac{3-4a}{a-7}\right|>1$

$\Rightarrow \quad |4a-3|-|a-7|>0$

(i) $a\ge 7\Rightarrow 4a-3-a+7>0 \Rightarrow a>-\dfrac{4}{3}$.

Hence $a\ge 7$

(ii) If $\frac{3}{4} \le a < 7 \Rightarrow 4a-3+a-7>0$

$\Rightarrow 5a > 10 \Rightarrow a > 2 \Rightarrow 2 < a < 7$

(iii) If $a < \frac{3}{4} \Rightarrow 3-4a+a-7>0$

$\Rightarrow a < -\frac{4}{3}$

$\therefore a \in \left(-\infty, -\frac{4}{3}\right) \cup (2, \infty)$

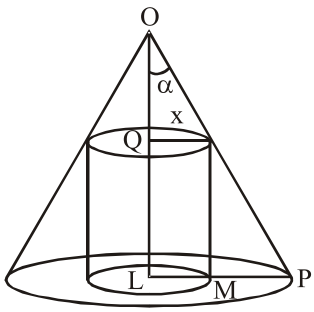
**11.** **(a)** Let $H$ be the height of the cone and $\alpha$ be its semi vertical angle. Suppose that $x$ is the radius of the inscribed cylinder and $h$ be height then

$h = QL = OL - OQ = H - x\cot\alpha$

$V$= volume of cylinder $= \pi x^2 (H - x\cot\alpha)$

Also $p = \frac{1}{3}\pi(H\tan\alpha)^2 H$ ...(i)

$$\frac{dV}{dx} = \pi(2Hx - 3x^2\cot\alpha)$$




So, $\frac{dV}{dx} = 0 \Rightarrow x = 0,\ x = \frac{2}{3}H\tan\alpha;$

$$\left.\frac{d^2V}{dx^2}\right|_{x=\frac{2}{3}H\tan\alpha} = -2\pi H < 0$$

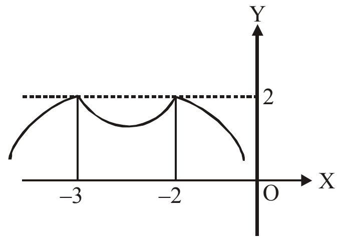
So, $V$ is maximum when $x = \frac{2}{3}H\tan\alpha$ and

$$q = V_{\max} = \pi\frac{4}{9}H^2\tan^2\alpha\frac{1}{3}H$$

$$= \frac{4}{27}\frac{\pi 3p\tan^2\alpha}{\pi\tan^2\alpha} = \frac{4}{9}p \text{ [from (i)]}$$

Hence $p : q = 9 : 4$

**12.** **(a)** The graph of $y = 2 - \left|x^2 + 5x + 6\right|$ is drawn in the adjacent figure. Clearly $f(x)$ will have maxima at $x = -2$ only if $a^2 + 1 \ge 2 \Rightarrow |a| \ge 1$




**13.** **(b)** $f'(x) = \sin x\cos x(3\sin x + 2\lambda)$

For maximum or minimum

$f'(x) = 0 \Rightarrow \sin x = 0$ or $\cos x = 0$ or $\sin x = -\frac{2\lambda}{3}$

$\therefore$ Critical points in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ are $x = 0$, $\sin^{-1}\left(-\frac{2\lambda}{3}\right)$

One of these is point of minima and other is point of maxima, provided $-1 < -\frac{2\lambda}{3} < 1$

$\Rightarrow -\frac{3}{2} < \lambda < \frac{3}{2}$

But if $\lambda = 0$, then $\sin x = 0$, which gives only one critical points

$\therefore \lambda \in \left(-\frac{3}{2}, \frac{3}{2}\right) - \{0\}$

**14.** **(a)** For $0 < x \le \frac{\pi}{2}$; $[\cos x] = 0$

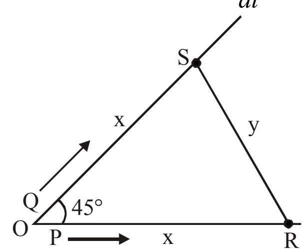
Hence, $f(x) = 1$ for all $\left(0, \frac{\pi}{2}\right]$

Trivially $f(x)$ is continuous on $\left(0, \frac{\pi}{2}\right)$

This function is neither strictly increasing nor strictly decreasing and its global maximum is 1.

**15.** **(c)** Let $R$ and $S$ be the positions of men $P$ and $Q$ at any time $t$. Since velocities are same

$\therefore OR = OS = x$ (say) and given $\frac{dx}{dt} = v$ and let $SR = y$




Now in triangle $ORS$, applying cosine rule, we get

$$y^2 = x^2 + x^2 - 2x \cdot x\cos 45^\circ = 2x^2 - x^2\sqrt{2}$$

$$\therefore\ y = x\sqrt{(2-\sqrt{2})}$$

$\therefore \quad \frac{dy}{dt} = \{\sqrt{(2-\sqrt{2})}\}\frac{dx}{dt} \quad = v\sqrt{(2-\sqrt{2})}$

Hence the required rate at which they are being separated is $v\sqrt{2-\sqrt{2}}.$

**16.** **(a)** Here $y = \frac{c^2}{x+a} \Rightarrow \frac{dy}{dx} = -\frac{c^2}{(x+a)^2}$

Slope of normal is $\Rightarrow \frac{(x+a)^2}{c^2} > 0$ (for all $x$)

$\therefore x\cos\alpha + y\sin\alpha = p$ is normal if,

$-\frac{\cos\alpha}{\sin\alpha} > 0$ or $\cot\alpha < 0$

i.e., $\alpha$ lies in II or IV quadrant.

So,

$\alpha \in \left(2n\pi + \frac{\pi}{2}, (2n+1)\pi\right) \cup \left(2n\pi + \frac{3\pi}{2}, (2n+2)\pi\right)$

**17.** **(c)** Any point on $x^2 - y^2 = a^2$ is $(a\sec\theta, a\tan\theta)$

This point is nearest to $y = 2x$ if the tangent at this point is parallel to $y = 2x$

Now, $x^2 - y^2 = a^2 \Rightarrow \frac{dy}{dx} = \frac{x}{y}$

$\Rightarrow \left(\frac{dy}{dx}\right)_{(a\sec\theta, a\tan\theta)} = \operatorname{cosec}\theta$

$\operatorname{cosec}\theta = 2 \Rightarrow \theta = \frac{\pi}{6}$

Hence the point is $\left(a\sec\frac{\pi}{6}, a\tan\frac{\pi}{6}\right)$ i.e.

$\left(\frac{2a}{\sqrt{3}}, \frac{a}{\sqrt{3}}\right)$, Clearly they lie on the line $2y = x$

**18.** **(c)** Given, $y = 3\sin\theta.\cos\theta$

$\frac{dy}{d\theta} = 3[\sin\theta(-\sin\theta) + \cos\theta(\cos\theta)]$

$\frac{dy}{d\theta} = 3[\cos^2\theta - \sin^2\theta] = 3\cos 2\theta \qquad ...(1)$

and $x = e^\theta \sin\theta$

$\frac{dx}{d\theta} = e^\theta \cos\theta + \sin\theta\, e^\theta$

$\frac{dx}{d\theta} = e^\theta(\sin\theta + \cos\theta) \qquad ...(2)$

Dividing (1) by (2)

$\frac{dy}{dx} = \frac{3\cos 2\theta}{e^\theta(\sin\theta + \cos\theta)} = \frac{3(\cos^2\theta - \sin^2\theta)}{e^\theta(\sin\theta + \cos\theta)}$

$\frac{dy}{dx} = \frac{3(\cos\theta + \sin\theta)(\cos\theta - \sin\theta)}{e^\theta(\sin\theta + \cos\theta)}$

$\frac{dy}{dx} = \frac{3(\cos\theta - \sin\theta)}{e^\theta}$

Given tangent is parallel to $x$-axis then $\frac{dy}{dx} = 0$

$0 = \frac{3(\cos\theta - \sin\theta)}{e^\theta}$

or $\cos\theta - \sin\theta = 0 \Rightarrow \cos\theta = \sin\theta$

$\Rightarrow \tan\theta = 1 \Rightarrow \tan\theta = \frac{\tan\pi}{4} \Rightarrow \theta = \frac{\pi}{4}$

**19.** **(c)** We have,

$y^3 - 3xy + 2 = 0$

$\Rightarrow 3y^2\frac{dy}{dx} - 3\left(x\frac{dy}{dx} + y\right) = 0$

$\Rightarrow \frac{dy}{dx} = \frac{y}{y^2 - x}$

If the tangent is parallel to x-axis, then

$\frac{dy}{dx} = 0 \Rightarrow \frac{y}{y^2 - x} = 0 \Rightarrow y = 0\cdot$

But, $y = 0$ does not satisfy equation (i). So, there is no point on the curve where tangent is parallel to $x$-axis. Therefore, $H = \phi$.

**20.** **(a)** We have,

$p(x) > x^2, p(0) = 0, p''(0) = \frac{1}{2}$

Let $g(x) = p(x) - x^2$

$g(x) > 0, \forall\ x \neq 0$ and $g(0) = p(0) - 0 = 0$

$\Rightarrow$ $x = 0$ should be minima.

$\therefore$ $g''(x)$ should be $\geq 0$ at $x = 0$

Now, $g'(x) = p'(x) - 2x$

$g''(x) = p''(x) - 2$

$g''(0) = p''(0) - 2 = \frac{1}{2} - 2 = -\frac{3}{2}$

But $g''(0) \geq 0 \Rightarrow$ No polynimial exists.

**21.** **(40)** Let the speed of the train be $v$ and distance to be covered be $s$ so that total time taken is $s/v$ hours. Cost of fuel per hour $= kv^2$ (k is constant)

Also $48 = k.\ 16^2$ by given condition $\therefore$ k $= \frac{3}{16}$

$\therefore$ Cost to fuel per hour $\frac{3}{16}v^2$.

Other charges per hour are 300.

Total running cost,

$$C = \left(\frac{3}{16}v^2 + 300\right)\frac{s}{v} = \frac{3s}{16}v + \frac{300s}{v}$$

$$\frac{dC}{dv} = \frac{3s}{16} - \frac{300s}{v^2} = 0 \Rightarrow v = 40$$

$$\frac{d^2C}{dv^2} = \frac{600s}{v^3} > 0 \ \therefore\ v = 40 \text{ results in minimum}$$

running cost

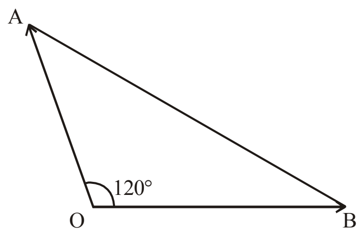
**22.** **(1)** Given $x + y - \ell n\,(x + y) = 2x + 5$

$$\Rightarrow \quad 1 + \frac{dy}{dx} - \frac{1}{x+y}\left(1 + \frac{dy}{dx}\right) = 2$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{x+y+1}{x+y-1}$$

$$\left.\frac{dt}{dx}\right|_{(\alpha,\beta)} = \frac{\alpha+\beta}{\alpha+\beta-1} = \infty \text{ when } \alpha+\beta = 1.$$

**23.** **(260)**




Let OA = $x$ km, OB = $y$ km, AB = R

$(AB)^2 = (OA)^2 + (OB)^2 - 2\,(OA)\,(OB)\cos 120°$

$$R^2 = x^2 + y^2 - 2xy\left(-\frac{1}{2}\right) = x^2 + y^2 + xy \quad ...(1)$$

R at $x$ = 6 km, and $y$ = 8 km

$$R = \sqrt{6^2 + 8^2 + 6 \times 8} = 2\sqrt{37}$$

Differentiating equation (1) with respect to $t$

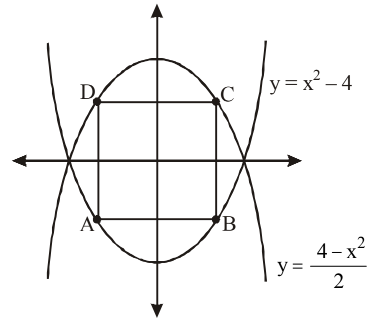
$$2R\frac{dR}{dt} = 2\frac{dx}{dt} + 2y\frac{dy}{dt} + \left(x\frac{dy}{dt} + y\frac{dx}{dt}\right)$$

$$= \frac{1}{2R}[2 \times 8 \times 20 + 2 \times 6 \times 30 + (8 \times 30 + 6 \times 20)]$$

$$\frac{dR}{dt} = \frac{1}{2 \times 2\sqrt{37}}[1040] = \frac{260}{\sqrt{37}} = \frac{k}{\sqrt{37}}$$

**24.** **(9.22)** Equation of parabola

$y = x^2 - 4$ and $2y = 4 - x^2$




Let point $B = \left(h, \frac{4-h^2}{2}\right)$

$A = \left(-h, \frac{4-h^2}{2}\right) \qquad C = (h, h^2 - 4)$

$D = (-h, h^2 - 4)$

$\therefore$ Area of rectangle $ABCD$

$$= AB \times BC = 2h \times \left(\frac{4-h^2}{2} - h^2 + 4\right)$$

$$\Rightarrow \quad A = 12h - 3h^3 \Rightarrow \frac{dA}{dh} = 12 - 9h^2$$

For maxima of minima

$$\frac{dA}{dh} = 0 \Rightarrow 12 - 9h^2 = 0$$

$$h = \pm\frac{2}{\sqrt{3}} \ \therefore\ \text{maximum at } h = \frac{2}{\sqrt{3}}$$

$$\therefore A = 12\left(\frac{2}{\sqrt{3}}\right) - 3\left(\frac{2}{\sqrt{3}}\right)^3 = \frac{24}{\sqrt{3}} - \frac{8}{\sqrt{3}}$$

$$= \frac{16}{\sqrt{3}} = \frac{16\sqrt{3}}{3} \qquad \Rightarrow A = \frac{16 \times 173}{3} = 9.22$$

**25.** **(2)** $\because$ P($x$) is non -zero polynomial and

P$(1 + x)$ = P$(1 - x)$ for all $x$

Now, differentiate w.r.t. $x$, we get

$P'(1 + x) = -P'(1 - x)$

Put $x = 0$, $P'(1) = -P'(1) \Rightarrow P'(1) = 0$

and $P(1) = 0 \Rightarrow P(x)$ touch $x$-axis at $x = 1$

$\therefore P(x) = (x - 1)^2\,Q(x)$

Since, m is the lergest integer such that

$(x - 1)^m$ divides polynomial P($x$).

$\Rightarrow m = 2$ such that $(x - 1)^m$ divides P($x$) for all such P($x$)

**26.** **(2)** $f'(x)=(b^2-3b+2)(-2\sin 2x)+b-1\neq 0$

for any $x\in R$

$\Rightarrow\ (b-1)\{1-(b-2)(2\sin 2x)\}\neq 0$

$\Rightarrow\ b\neq 1$ and

$\left|\dfrac{1}{2(b-2)}\right|>1\Rightarrow b\in\left(\dfrac{3}{2},2\right)\cup\left(2,\dfrac{5}{2}\right)$

When $b=2, f(x)=x+\sin 3\Rightarrow f'(x)=1\neq 0$

$\therefore\ b\in\left(\dfrac{3}{2},\dfrac{5}{2}\right)\Rightarrow$ integral value of b = 2

**27.** **(1)** Equation of given curves are

$ax^2+by^2=1$ ...(i)

and $a_1x^2+b_1y^2=1$ ...(ii)

From (i) $2ax+2by\dfrac{dy}{dx}=0\ \therefore\dfrac{dy}{dx}=-\dfrac{a}{b}\dfrac{x}{y}$ ...(iii)

From (ii) $2a_1x+2b_1y\dfrac{dy}{dx}=0,$

$\therefore\dfrac{dy}{dx}=-\dfrac{a_1}{b_1}\dfrac{x}{y}$ ...(iv)

Curve (i) and (ii) will cut each other at right angles if the product of the values of $\dfrac{dy}{dx}$ for the two curves is $-1$

i.e., $\left(-\dfrac{a}{b}\dfrac{x}{y}\right)\left(-\dfrac{a_1x}{b_1y}\right)=-1$ or $\dfrac{aa_1}{bb_1}\dfrac{x^2}{y^2}=-1$ ...(v)

from (i) and (ii), $ax^2+by^2=a_1x^2+b_1y^2$

or $(a-a_1)x^2=(b_1-b)y^2$

$\therefore\ \dfrac{x^2}{y^2}=\dfrac{b_1-b}{a-a_1}$ ...(vi)

Putting the value of $\dfrac{x^2}{y^2}$ in (v), we get

$\dfrac{aa_1}{bb_1}\left(\dfrac{b_1-b}{a-a_1}\right)=-1$

or $\dfrac{a-a_1}{aa_1}=-\dfrac{b_1-b}{bb_1}=\dfrac{b-b_1}{bb_1}$ or $\dfrac{1}{a_1}-\dfrac{1}{a}=\dfrac{1}{b_1}-\dfrac{1}{b}$

**28.** **(4)** $\dfrac{dy}{dx}=x^2\ \Rightarrow$ Equation of tangent at $P_1(x_1,y_1)$ is $y-y_1=x_1^2(x_2-x_1)$

i.e. $x_2^3-x_1^3=3x_1^2(x_2-x_1^2)$

$\Rightarrow\ x_2^2+x_1^2+x_1x_2=3x_1^2$

$\Rightarrow\ 2x_1^2-x_1x_2-x_2^2=0\ \Rightarrow\ x_2=-2x_1$

Similarly $x_3=-2x_2=4x_1$

$x_4=-2x_3=-8x_1$

..................................

$x_{2n}=-2^{2n-1}x_1$

$3(y_1+y_2+y_3........+y_{2n})$

$=(x_1^3+x_2^3+x_3^3+.........+x_{2n}{}^3)+2n$

$=x_1^3\dfrac{((-8)^{2n}-1)}{-8-1}+2=-\dfrac{x_1^3}{9}(8^{2n}-1)+2n$

i.e. $y_1+y_2+y_3+....+y_{2n}$

$=-\left(\dfrac{x_1}{3}\right)^3(2^{6n}-1)+\left(\dfrac{2n}{3}\right),$

Now put $n=30$ and $x_1=2$ then

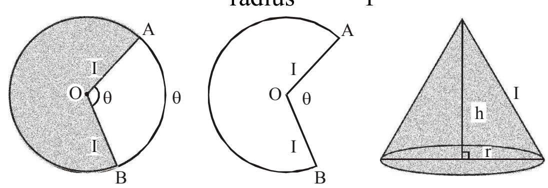
$y_1+y_2+y_3+......+y_{60}=-\left(\dfrac{2}{3}\right)^3\left(2^{180}-1\right)+20$

$\Rightarrow\ S=-\dfrac{2^{183}-8}{27}+20$ or $S+\left(\dfrac{2^{183}-8}{27}\right)=20$

$\Rightarrow$ 5k = 20 $\Rightarrow$ k = 4

**29.** **(9)** Sectorial area $AOB$ is removed and the remaining part be folded into a cone of height $h$ and radius $r$.

$\because\ \theta=\text{Angle}=\dfrac{ArcAB}{\text{radius}}=\dfrac{ArcAB}{1}=Arc\ AB$




circular sheet after removed part conical vessel

$2\pi r=2\pi-\theta$ and $r^2+h^2=1$

$\therefore$ volume of cone

$V=\dfrac{1}{3}\pi r^2h=\dfrac{1}{3}\pi r^2\sqrt{1-r^2}$

$\Rightarrow\ V^2=\dfrac{\pi^2r^4(1-r^2)}{9}$

Let, $y=V^2=\dfrac{\pi^2(r^4-r^6)}{9}$

$\therefore \dfrac{dy}{dr} = \dfrac{\pi^2}{9}(4r^3 - 6r^5)$ and

$\dfrac{d^2y}{dr^2} = \dfrac{\pi^2}{9}(12r^2 - 30r^4)$

For max or min of $y, \dfrac{dy}{dr} = 0 \qquad \therefore r = \sqrt{\left(\dfrac{2}{3}\right)}$

Then,

$$\left.\frac{d^2y}{dr^2}\right|_{r=\sqrt{2/3}} = \frac{\pi^2}{9}\left(12\cdot\frac{2}{3} - 30\cdot\frac{4}{9}\right) = -\frac{16}{27}\pi^2$$

$\therefore$ $y$ is maximum and hence $V$ is also maximum

at $r = \sqrt{\left(\dfrac{2}{3}\right)}$ ....(1)

$\therefore$ Removed sectorial area $= \dfrac{1}{2}.(1)^2.\theta$

$A_1 = \dfrac{1}{2}.\theta = \dfrac{1}{2}(2\pi - 2\pi r)$

$= \pi(1-r) = \pi\left(1 - \sqrt{\dfrac{2}{3}}\right)$ [from (1)]

Also, $A = \pi$

$$\therefore \frac{A}{A_1} = \frac{1}{1-\sqrt{\frac{2}{3}}} = \frac{1+\sqrt{\frac{2}{3}}}{1-\frac{2}{3}} = 3 + \sqrt{6} = m + \sqrt{n}$$

$\Rightarrow \quad m + n = 3 + 6 = 9.$

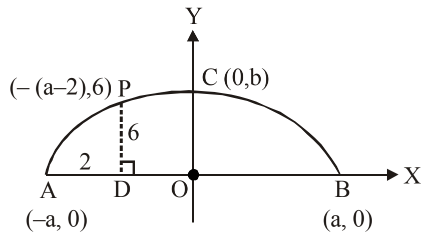
**30.** **(6)** Equation of the ellipse is $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$

The point $P(-(a-2), 6)$ lies on it.

$\therefore \dfrac{(a-2)^2}{a^2} + \dfrac{36}{b^2} = 1$

or $\dfrac{36}{b^2} = 1 - \dfrac{(a-2)^2}{a^2} = \dfrac{4(a-1)}{a^2}$

or $\dfrac{9}{b^2} = \dfrac{(a-1)}{a^2}$




or $b^2 = \dfrac{9a^2}{(a-1)} = 9\left\{a + 1 + \dfrac{1}{a-1}\right\}$

$\therefore \dfrac{d(b)^2}{da} = 9\left\{1 - \dfrac{1}{(a-1)^2}\right\}$ and $\dfrac{d^2(b^2)}{da^2} = \dfrac{18}{(a-1)^3}$

For extremum value of $b$, $\dfrac{db^2}{da} = 0$

$\Rightarrow$ we get $a = 2 \;\therefore\; b^2 = \dfrac{9(2)^2}{(2-1)} \Rightarrow b = 6\text{m}$

Hence greatest height of the arch is 6 m.

# CHAPTER 22 Indefinite Integration

**1.** **(b)** $\displaystyle\int \frac{\sin x}{\sin(x-\alpha)}dx = \int \frac{\sin(x-\alpha+\alpha)}{\sin(x-\alpha)}dx$

$\displaystyle = \int \frac{\sin(x-\alpha)\cos\alpha + \cos(x-\alpha)\sin\alpha}{\sin(x-\alpha)}dx$

$\displaystyle = \int \{\cos\alpha + \sin\alpha\cot(x-\alpha)\}dx$

$= (\cos\alpha)x + (\sin\alpha)\log\sin(x-\alpha) + C$

$\therefore \quad A = \cos\alpha, B = \sin\alpha$

**2.** **(d)** Let $g_n(x)$

$= 1 + x^2 + x^4 + .... + x^{2n} = \dfrac{x^{2n+2}-1}{x^2-1}$

So, $2x + 4x^3 + ..... + 2nx^{2n-1}$

$= h_n(x) = g'_n(x) = \dfrac{2x(nx^{2n+2} - (n+1)x^{2n} + 1)}{(x^2-1)^2}$

Now $f(x) = \displaystyle\lim_{n\to\infty} h_n(x) = \dfrac{2x}{(x^2-1)^2}$ as $0 < x < 1$

Thus $\displaystyle\int f(x)dx = \int \frac{2x}{(x^2-1)^2}dx$

$= -\dfrac{1}{x^2-1} = \dfrac{1}{1-x^2} + c$

**3.** **(c)** $\displaystyle\int\left(\frac{f(x)g'(x) - f'(x)g(x)}{f(x)g(x)}\right)\ln\left(\frac{g(x)}{f(x)}\right)dx$

Let $\dfrac{g(x)}{f(x)} = t$

$\Rightarrow \dfrac{f(x)g'(x) - g(x)f'(x)}{(f(x))^2}dx = dt;$

$\displaystyle\int \frac{\ln t}{t}dt = \frac{(\ln t)^2}{2} + C$

**4.** **(c)** We have $\displaystyle\int e^x \frac{1+nx^{n-1} - x^{2n}}{(1-x^n)\sqrt{1-x^{2n}}}dx$

$$= \int e^x \left[\sqrt{\frac{1+x^n}{1-x^n}} + \frac{nx^{n-1}}{(1-x^n)\sqrt{1-x^{2n}}}\right]dx$$

$$= e^x\sqrt{\frac{1+x^n}{1-x^n}} + C$$

Here, $f(x) = \sqrt{\dfrac{1+x^n}{1-x^n}}$, then

$$f'(x) = \frac{nx^{n-1}}{(1-x^n)\sqrt{1-x^{2n}}}$$

and $\displaystyle\int e^x(f(x)+f'(x))dx = e^x f(x) + C$

**5.** **(d)** $\displaystyle\lim_{n\to\infty} n^2\left(x^{1/n} - x^{1/(n+1)}\right)$

$$= \lim_{h\to 0}\frac{1}{h^2}\left(x^h - x^{\frac{h}{h+1}}\right)$$

$$= \lim_{h\to 0}\frac{1}{h^2}\left(x^{\left(h-\frac{h}{h+1}\right)} - 1\right)x^{\frac{h}{h+1}}$$

$$= \lim_{h\to 0}\left(\frac{x^{\frac{h^2}{h+1}} - 1}{\frac{h^2}{h+1}}\right)\cdot\left(\frac{1}{h+1}\right)x^{\frac{h}{h+1}} = \ln x.1.1$$

$\therefore\ f(x) = \ln x$

So, $I = \displaystyle\int xf(x)dx = \int x\ln x dx$

$$= \frac{x^2}{2}\ln x - \int\frac{x^2}{2}.\frac{1}{x}dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C$$

**6.** **(a)** $f(x) = \displaystyle\lim_{n\to\infty}\frac{x^{2n}-1}{x^{2n}+1} = -1 \quad (\because 0 < x < 1)$,

so $\displaystyle\int \sin^{-1}x(f(x))dx = -\int\sin^{-1}x\,dx$

$$= -[x\sin^{-1}x + \sqrt{1-x^2}] + C$$

**7.** **(b)** $I = \displaystyle\int x\left(\frac{\ln a^{a^{x/2}}}{3a^{5x/2}b^{3x}} + \frac{\ln b^{b^x}}{2a^{2x}b^{4x}}\right)dx$

$$= \int\frac{\ln a^{2x}b^{3x}}{6a^{2x}b^{3x}}dx$$

Let $a^{2x}b^{3x} = t.$

Then $t\ln a^2b^3dx = dt.$

Therefore,

$$I = \int\frac{1}{6\ln a^2b^3}\frac{\ln t}{t^2}dt$$

$$= \frac{1}{6\ln a^2b^3}\left(\frac{-\ln t}{t} - \int\frac{-1}{t^2}dt\right)$$

$$= -\frac{1}{6\ln a^2b^3}\left(\frac{\ln et}{t}\right) + k$$

$$= -\frac{1}{6\ln a^2b^3}\left(\frac{\ln a^{2x}b^{3x}e}{a^{2x}b^{3x}}\right) + k$$

**8.** **(c)** $I_n = \displaystyle\int(\sin x + \cos x)^n dx$

$$= \frac{(\sin x+\cos x)^{n-1}}{n}(\sin x - \cos x) + \frac{2(n-1)}{n}I_{n-2}$$

$$\Rightarrow I_5 = \frac{t^4}{5}(\sin x - \cos x) + \frac{8}{5}I_3 = \frac{t^4}{5}(\sin x - \cos x)$$

$$+\frac{8}{5}\left(\frac{t^2}{3}(\sin x - \cos x) + \frac{4}{3}I_1\right)$$

$$= \frac{t^4}{5}(\sin x - \cos x) + \frac{8}{15}t^2(\sin x - \cos x)$$

$$+\frac{32}{15}(\sin x - \cos x) + C$$

$$= \frac{(\sin x - \cos x)}{15}(3t^4 + 8t^2 + 32) + C;$$

where $t = (\sin x + \cos x)$.

**9.** **(b)** Let $I$ be the given integral. We can see that the derivative of $\log_e\left(\dfrac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\right)$ is $2\sec 2\theta$, so that using integration by parts we have

$$I = \frac{\sin 2\theta}{2}\log_e\left(\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\right)$$

$$-\int\frac{\sin 2\theta}{2}\cdot(2\sec 2\theta)d\theta$$

$$= \frac{\sin 2\theta}{2}\log_e\left(\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\right) - \int\tan 2\theta\,d\theta$$

$$= \frac{\sin 2\theta}{2}\log_e\left(\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\right) - \frac{1}{2}\log_e(\sec 2\theta) + c$$

Therefore

$f(x) = \log_e(\sec 2\theta).$

**10.** **(d)** We have,

$$\int \frac{1}{x}[\log_{ex} e \cdot \log_{e^2 x} \cdot \log_{e^3 x} e]\,dx$$

$$= \int \frac{1}{x\, log_e ex \, log_e e^2 x . log_e e^3 x} dx$$

$$= \int \frac{1}{x(\log_e e + \log_e x)(\log_e e^2 + \log_e x)(\log_e e^3 + \log_e x)} dx$$

$$= \int \frac{d(\log_e x)}{(1+\log_e x)(2+\log_e x)(3+\log_e x)}$$

$$= \int \frac{1}{(1+t)(2+t)(3+t)} dt, \text{ where } t = \log_e x$$

$$= \int \left( \frac{1}{2} \cdot \frac{1}{1+t} - \frac{1}{2+t} + \frac{1}{2} \cdot \frac{1}{3+t} \right) dt$$

$$= \frac{1}{2} \log|1+\log_e x| - \log|2+\log_e x| + \frac{1}{2}\log|3+\log_e x| + C$$

$$= \frac{1}{2}\log\{\log_e ex\} - \log\{\log_e e^2 x\} + \frac{1}{2}\log\{\log_e e^3 x\} + C$$

**11.** **(a)** $\mathrm{I} = \int \frac{\mathrm{dx}}{(\mathrm{x}-\beta)\sqrt{(\mathrm{x}-\alpha)(\beta-\mathrm{x})}}$

Put $\mathrm{x} = \alpha \sin^2\theta + \beta\cos^2\theta$

$\mathrm{dx} = 2(\alpha-\beta)\sin\theta\cos\theta\, d\theta$

Also, $(\mathrm{x}-\alpha) = (\beta-\alpha)\cos^2\theta$

$(\mathrm{x}-\beta) = (\alpha-\beta)\sin^2\theta$

$$\therefore \mathrm{I} = \int \frac{2(\alpha-\beta)\sin\theta\cos\theta\, d\theta}{(\alpha-\beta)\sin^2\theta(\beta-\alpha)\sin\theta\cos\theta}$$

$$= \frac{2}{\beta-\alpha}\int \frac{d\theta}{\sin^2\theta} = \frac{2}{\beta-\alpha}\int \mathrm{cosec}^2\theta\, d\theta$$

$$= \frac{2}{\beta-\alpha}(-\cot\theta) + c = \frac{2}{\alpha-\beta}\cot\theta + C$$

Now $\mathrm{x} = \alpha\sin^2\theta + \beta\cos^2\theta$

$\Rightarrow \mathrm{x}\,\mathrm{cosec}^2\theta = \alpha + \beta\cot^2\theta$

$\Rightarrow \mathrm{x}(1+\cot^2\theta) = \alpha + \beta\cot^2\theta$

$$\therefore \cot\theta = \sqrt{\frac{\mathrm{x}-\alpha}{\beta-\mathrm{x}}}\ ; \ \therefore I = \frac{2}{\alpha-\beta}\sqrt{\frac{x-\alpha}{\beta-x}} + C$$

**12.** **(a)** $I = -\int \frac{(\tan x - 1)\sec^2 x dx}{(\tan x+1)\sqrt{\tan^3 x + \tan^2 x + \tan x}} dx$

Put $\tan x = t$

$$I = -\int \frac{(t-1)}{(t+1)\sqrt{t^3+t^2+t}} dt$$

$$= -\int \frac{t^2-1}{(t^2+2t+1)\sqrt{t^3+t^2+t}} dt$$

$$= -\int \frac{1-\frac{1}{t^2}}{\left(t+2+\frac{1}{t}\right)\sqrt{t+1+\frac{1}{t}}} dt, \text{ put } 1+t+\frac{1}{t} = u^2$$

$$\therefore\ I = -\int \frac{2du}{1+u^2} = -2\tan^{-1} u + c,$$

where $u = \sqrt{1+\tan x + \frac{1}{\tan x}}$

**13.** **(c)** Let $ax^2 = b\sin^2\theta \Rightarrow x = \sqrt{\frac{b}{a}}\sin\theta$

$2ax\,dx = 2b\sin\theta\cos\theta\, d\theta$

$$\Rightarrow dx = \frac{2b\sin\theta\cos\theta}{2ax} d\theta = \sqrt{\frac{b}{a}}\cos\theta\, d\theta$$

$$I = \sqrt{\frac{b}{a}}\int \frac{\cos\theta\, d\theta}{\left(a + b.\frac{b}{a}\sin^2\theta\right).\sqrt{b}\cos\theta}$$

$$= \sqrt{a}\int \frac{d\theta}{(a^2+b^2\sin^2\theta)}$$

$$= \sqrt{a}\int \frac{\sec^2\theta\, d\theta}{(a^2\sec^2\theta + b^2\tan^2\theta)}$$

Putting $\tan\theta = z \Rightarrow \sec^2\theta\, d\theta = dz$ then

$$I = \sqrt{a}\int \frac{dz}{a^2(1+z^2)+b^2 z^2}$$

$$= \sqrt{a}\int \frac{dz}{(a^2+b^2)z^2+a^2}$$

$$= \frac{\sqrt{a}}{(a^2+b^2)}\int \frac{dz}{z^2 + \frac{a^2}{(a^2+b^2)}}$$

$$= \frac{\sqrt{a}}{a^2+b^2}.\frac{\sqrt{a^2+b^2}}{a}\tan^{-1}\left(\frac{z\sqrt{a^2+b^2}}{a}\right) + c$$

$$\therefore\ I = \frac{1}{\sqrt{a(a^2+b^2)}}\tan^{-1}\left(\frac{z\sqrt{a^2+b^2}}{a}\right) + c,$$

where $z = \tan\theta$.

**14.** **(b)** Given $\int f(x)\sin x\cos x dx$

$$= \frac{1}{2(b^2-a^2)}\ln f(x) + c$$

Differentiating both sides w.r.t.x then

$$f(x)\sin x\cos x=\frac{1}{2(b^2-a^2)}\cdot\frac{f'(x)}{f(x)}$$

$$\Rightarrow\ 2(b^2-a^2)\sin x\cos x=\frac{f'(x)}{\{f(x)\}^2}$$

$$\Rightarrow 2b^2\sin x\cos x-2a^2\sin x\cos x=\frac{f'(x)}{\{f(x)\}^2}$$

Integrating both side w.r.t. $x$ we get

$$-b^2\cos^2 x-a^2\sin^2 x=-\frac{1}{f(x)}$$

or $f(x)=\dfrac{1}{(a^2\sin^2 x+b^2\cos^2 x)}$

**15.** **(a)** We have $\dfrac{\sec 6\alpha}{\text{cosec}2\alpha}+\dfrac{\sec 18\alpha}{\text{cosec}6\alpha}+\dfrac{\sec 54\alpha}{\text{cosec}18\alpha}$

$$=\frac{\sin 2\alpha}{\cos 6\alpha}+\frac{\sin 6\alpha}{\cos 18\alpha}+\frac{\sin 18\alpha}{\cos 54\alpha}$$

Now, $\dfrac{\sin 2\alpha}{\cos 6\alpha}=\dfrac{1}{2}\dfrac{\sin 4\alpha}{\cos 2\alpha\cos 6\alpha}$

$$=\frac{1}{2}\left[\frac{\sin 6\alpha\cos 2\alpha-\cos 6\alpha\sin 2\alpha}{\cos 2\alpha\cos 6\alpha}\right]$$

$$=\frac{1}{2}(\tan 6\alpha-\tan 2\alpha)$$

Similarly, $\dfrac{\sin 6\alpha}{\cos 18\alpha}=\dfrac{1}{2}(\tan 18\alpha-\tan 6\alpha)$

and $\dfrac{\sin 18\alpha}{\cos 54\alpha}=\dfrac{1}{2}(\tan 54\alpha-\tan 18\alpha)$

Thus integral $=\dfrac{1}{2}\displaystyle\int(\tan 54\alpha-\tan 2\alpha)\,dx$

$$=\frac{1}{2}\left[\frac{\ell n|\sec 54\alpha|}{54}-\frac{\ell n|\sec 2\alpha|}{2}\right]+c$$

**16.** **(b)** Let

$$I_n=\int(\sin x+\cos x)^{n-1}(\sin x+\cos x)\,dx$$

Take, $u=(\sin x+\cos x)^{n-1}$

and $dv=(\sin x+\cos x)dx$

so that, $v=\sin x-\cos x$

Therefore, using integration by parts we have

$$I_n=(\sin x+\cos x)^{n-1}(\sin x-\cos x)$$
$$-\int(n-1)(\sin x+\cos x)^{n-2}$$
$$\times(\cos x-\sin x)(\sin x-\cos x)\,dx$$

$$=(\sin x+\cos x)^{n-1}(\sin x-\cos x)$$
$$+(n-1)\int(\sin x+\cos x)^{n-2}(\cos x-\sin x)^2\,dx$$

$$=(\sin x+\cos x)^{n-1}(\sin x-\cos x)$$
$$+(n-1)\int(\sin x+\cos x)^{n-2}[2-(\sin x+\cos x)^2]\,dx$$

$$=(\sin x+\cos x)^{n-1}(\sin x-\cos x)$$
$$+2(n-1)I_{n-2}-(n-1)I_n$$

Therefore,

$$I_n+(n-1)I_n-2(n-1)I_{n-2}$$
$$=(\sin x+\cos x)^{n-1}(\sin x-\cos x)$$

$$\Rightarrow nI_n-2(n-1)I_{n-2}$$
$$=(\sin x+\cos x)^{n-1}(\sin x-\cos x).$$

**17.** **(d)** We have,

$f'(x)=\ln(x^2+1)$

$$f(x)=\int\ln(x^2-1)\,dx$$

$$f(x)=x\ln(x^2-1)-\int\frac{2x^2}{x^2-1}dx$$

$$f(x)=x\ln(x^2-1)-2\int\left(\frac{x^2-1}{x^2-1}+\frac{1}{x^2-1}\right)dx$$

$$f(x)=x\ln(x^2-1)-2x-\ln\left(\frac{x-1}{x+1}\right)+C$$

$$f(2)=2\ln(3)-4-\ln\left(\frac{1}{3}\right)+C=0\quad[\because f(2)=0]$$

$\Rightarrow\quad C=4-3\ln 3$

$$\therefore\quad f(x)=x\ln(x^2-1)-2x-\ln\frac{(x-1)}{(x+1)}+4-3\ln 3$$

$\therefore$ the above function is defined for $S$ infinite $C$ values possible in set $S$ such that $f'(x)=\ln(x^2-1)$

**18.** **(1)** We have $\displaystyle\lim_{x\to 0}\left(1+x+\frac{f(x)}{x}\right)^{1/x}=e^3$

$$\Rightarrow\lim_{x\to 0}\left(1+x\left(1+\frac{f(x)}{x^2}\right)\right)^{1/x}=e^3$$

$$\Rightarrow\lim_{x\to 0}\left[\left(1+\frac{x^2+f(x)}{x}\right)^{\frac{x}{x^2+f(x)}}\right]^{\frac{x^2+f(x)}{x}\frac{1}{x}}=e^3$$

$$\Rightarrow\frac{x^2+f(x)}{x^2}=3\ \Rightarrow f(x)=2x^2$$

Therefore

$$\int f(x)\log_e x\,dx=2\int x^2\log_e x\,dx$$

$$=2\left[\frac{x^3}{3}\log_e x-\int\frac{x^3}{3}\cdot\frac{1}{x}dx\right]\text{(By Parts)}$$

$$=\frac{2}{3}x^3\log_e x-\frac{2}{9}x^3+c=\frac{2}{3}x^3\left(\log_e x-\frac{1}{3}\right)+c$$

$\Rightarrow\quad a+b=1$

**19.** **(0.60)** Let $I=\int\frac{(\cos^2 x+\sin 2x)}{(2\cos x-\sin x)^2}dx$

$=\int\frac{(\cos x+2\sin x)\cos x}{(2\cos x-\sin x)^2}dx$

Integrating by part , taking cos x as the first and $\frac{(\cos x+2\sin x)}{(2\cos x-\sin x)^2}$ as the second function, we have

$=\cos x\left\{\frac{1}{2\cos x-\sin x}\right\}-\int\frac{-\sin xdx}{(2\cos x-\sin x)}$

$=\cos x\left\{\frac{1}{2\cos x-\sin x}\right\}+\int\frac{-\sin x\ dx}{(2\cos x-\sin x)}$

$=\frac{\cos x}{(2\cos x-\sin x)}$

$+\int\frac{-\frac{1}{5}(2\cos x-\sin x)-\frac{2}{5}(-2\sin x-\cos x)}{(2\cos x-\sin x)}dx$

$\left[N^{\gamma}=\lambda Dr+\mu\frac{d}{dx}Dr\right]$

$=\frac{\cos x}{(2\cos x-\sin x)}-\frac{1}{5}$

$\int dx-\frac{2}{5}\int\frac{(-2\sin x-\cos x)}{2\cos x-\sin x}dx$

$=\frac{\cos x}{(2\cos x-\sin x)}-\frac{1}{5}x$

$-\frac{2}{5}ln|2\cos x-\sin x|+c$

$\Rightarrow\ |a+b|=0.60.$

**20.** **(2012)** Let $I=\int\frac{1-(\cot x)^{2010}}{\tan x+(\cot x)^{2011}}dx$

$=\int\frac{(\sin x)^{2010}-(\cos x)^{2010}}{(\sin x)^{2010}}\cdot\frac{(\sin x)^{2011}\cos x}{(\sin x)^{2012}+(\cos x)^{2012}}dx$

$=\int\frac{[(\sin x)^{2010}-(\cos x)^{2010}]\sin x\cos x}{(\sin x)^{2012}+(\cos x)^{2012}}dx$

Put $t=(\sin x)^{2012}+(\cos x)^{2012}$. Then

$dt=[2012(\sin x)^{2011}\cos x$

$+2012(\cos x)^{2011}(-\sin x)]dx$

$=2012[(\sin x)^{2010}-(\cos x)^{2010}](\sin x\cos x)\,dx$

Therefore

$I=\int\frac{1}{t}\left(\frac{1}{2012}\right)dt\ =\frac{1}{2012}\log_e|t|+c$

$=\frac{1}{2012}\log_e|(\sin x)^{2012}+(\cos x)^{2012}|+c$

**21.** **(3)** $I=\int\frac{\sin^3\frac{\theta}{2}}{\cos\frac{\theta}{2}\sqrt{\cos^3\theta+\cos^2\theta+\cos\theta}}d\theta$

$=\int\frac{\left(2\sin\frac{\theta}{2}.\cos\frac{\theta}{2}\right).\sin^2\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}\sqrt{\cos^3\theta+\cos^2\theta+\cos\theta}}d\theta$

$=\int\frac{2\sin^2\frac{\theta}{2}\sin\theta\,d\,\theta}{2\cos^2\frac{\theta}{2}\sqrt{\cos^3\theta+\cos^2\theta+\cos\theta}}$

Put $\cos\theta=t\ \Rightarrow\ -\sin\theta d\theta=dt$

Also $\cos\theta=2\cos^2\frac{\theta}{2}-1=1-2\sin^2\frac{\theta}{2}=t$

$\therefore\quad I=\int\frac{\frac{1-t}{2}(-dt)}{(1+t)\sqrt{t^3+t^2+t}}$

$=\frac{1}{2}\int\frac{(t^2-1)dt}{(t+1)^2\sqrt{t^3+t^2+t}}$

$=\frac{1}{2}\int\frac{\left(1-\frac{1}{t^2}\right)(dt)}{\left(t+\frac{1}{t}+2\right)\sqrt{t+\frac{1}{t}+1}}$

Put $t+\frac{1}{t}+1=u^2\ \Rightarrow\ \left(1-\frac{1}{t^2}\right)dt=2u\,du$

$I=\frac{1}{2}\int\frac{2udu}{\left(1+u^2\right)u}$

$=\tan^{-1}u=\tan^{-1}\sqrt{1+\frac{1}{t}+1}+c$

$=\tan^{-1}(\cos\theta+\sec\theta+1)^{1/2}+c$

So, $f(\theta)=\cos\theta+\sec\theta+1\geq 2+1=3$

**22.** **(2)** $f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$

$=\lim_{h\to 0}\frac{f\left(x\left(1+\frac{h}{x}\right)\right)-f(x)}{h}$

$=\lim_{h\to 0}\frac{f(x)\cdot f\left(1+\frac{h}{x}\right)-f(x)}{h}$

$(\because f(xy)=f(x)\cdot f(y)\ \forall\ x,y\in\mathbf{R})$

$=f(x)\lim_{h\to 0}\frac{f\left(1+\frac{h}{x}\right)-1}{h}$

$$= f(x)\left\{\lim_{h\to 0}\frac{1+\frac{h}{x}\left(1+g\left(\frac{h}{x}\right)\right)-1}{h}\right\}$$

$$= f(x)\cdot\left\{\lim_{h\to 0}\frac{1}{x}\left(1+g\left(\frac{h}{x}\right)\right)\right\}$$

As $h\to 0;\ \frac{h}{x}\to 0$

$\Rightarrow g\left(\frac{h}{x}\right)\to 0$ as $\lim_{x\to 0} g(x)=0$ (given)

$\therefore f'(x)=f(x)\cdot\frac{1}{x}\Rightarrow\frac{f(x)}{f'(x)}=x$

$\Rightarrow\int\frac{f(x)}{f'(x)}dx=\frac{x^2}{2}+C\Rightarrow k=2.$

**23.** **(1)** $I=\int\{\sin(100x+x)\cdot(\sin x)^{99}\}\,dx$

$=\int\{\sin(100x)\cos x+\cos 100x\sin x\}(\sin x)^{99}\,dx$

$=\int\underbrace{\sin(100x)}_{\mathrm{I}}\underbrace{\cos x\cdot(\sin x)^{99}}_{\mathrm{II}}\,dx$

$+\int\cos(100x)\cdot(\sin x)^{100}\,dx$

$=\frac{\sin(100x)(\sin x)^{100}}{100}$

$-\frac{100}{100}\int\cos(100x)(\sin x)^{100}\,dx$

$+\int\cos(100x)(\sin x)^{100}\,dx$

$=\frac{\sin(100x)(\sin x)^{100}}{100}+C$

$\Rightarrow\lambda=100,\ \mu=100\Rightarrow\frac{\lambda}{\mu}=\frac{100}{100}=1.$

**24.** **(8)** $\int\frac{dx}{(\cos x-\sin x)(1+\sin x\cos x)}$

$=2\int\frac{(\cos x-\sin x)dx}{(\cos x-\sin x)^2(2+(\sin x+\cos x)^2-1)}$

$=2\int\frac{(\cos x-\sin x)dx}{((\sin^2 x+\cos^2 x)-2\sin x\cos x)(1+(\sin x+\cos x)^2)}$

$=2\int\frac{(\cos x-\sin x)dx}{(2-(\sin x+\cos x)^2)(1+(\sin x+\cos x)^2)}$

$=2\int\frac{dt}{(2-t^2)(1+t^2)}$ where $t=\sin x+\cos x$

$=2\int\frac{dt}{(2-t^2)(1+t^2)}=\frac{2}{3}\left[\int\frac{1}{1+t^2}dt+\int\frac{dt}{2-t^2}\right]$

$=\frac{2}{3}\left[\tan^{-1}(t)+\frac{1}{2\sqrt{2}}\ln\left(\frac{\sqrt{2}+t}{\sqrt{2}-t}\right)\right]+C$

$\therefore A=\frac{2}{3},\ B=\frac{1}{3\sqrt{2}}$

$\therefore 12A+9\sqrt{2}B-3=12\cdot\frac{2}{3}+9\sqrt{2}\frac{1}{3\sqrt{2}}-3=8.$

**25.** **(4)** $I=\int\frac{\sin^8 x}{\cos x}dx=\int\frac{\sin^8 x\cdot\cos x}{\cos^2 x}dx$

$=\int\frac{\sin^8 x}{(1-\sin^2 x)}\cdot\cos x\,dx$

Put $\sin x=t,\ \cos x\,dx=dt$

$\therefore I=\int\frac{t^8}{1-t^2}dt=\int\frac{t^8-1+1}{1-t^2}dt$

$=\int\frac{(t^2-1)(t^2+1)(t^4+1)}{1-t^2}\,dt+\int\frac{1}{1-t^2}dt$

$=\int(t^2+1)(-1-t^4)\,dt+\int\frac{1}{1-t^2}dt$

$=\int(-t^6-t^4-t^2-1)\,dt+\int\frac{1}{1-t^2}dt$

$=\frac{-t^7}{7}-\frac{t^5}{5}-\frac{t^3}{3}-t+\frac{1}{2}\ln\left|\frac{1+t}{1-t}\right|+C$

$=\frac{-\sin^7 x}{7}-\frac{\sin^5 x}{5}-\frac{\sin^3 x}{3}$

$-\sin x+\frac{1}{2}\ln\left|\frac{1+\sin x}{1-\sin x}\right|+C$ ...(1)

Now, $\ln\left|\frac{\left(\sin\frac{x}{2}+\cos\frac{x}{2}\right)^2}{\left(\cos\frac{x}{2}-\sin\frac{x}{2}\right)^2}\right|$

$=2\ln\left|\frac{1+\tan x/2}{1-\tan x/2}\right|=2\ln\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right|$ ...(2)

$\therefore$ From (1) and (2)

$\therefore I=\frac{-\sin^7 x}{7}-\frac{\sin^5 x}{5}-\frac{\sin^3 x}{3}-\sin x$

$+\ln\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right|+C$

$=\frac{-\sin^7 x}{a}+\frac{\sin^5 x}{b}-\frac{\sin^3 x}{c}+\frac{\sin x}{d}$

$+\ln\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right|+C$ (given)

$\Rightarrow a=7,\ b=-5,\ c=3,\ d=-1$

$\Rightarrow(a+b+c+d)=(7-5+3-1)=4.$

CHAPTER 23

# Definite Integration

**1.** **(a)** Since the required function is a polynomial, the absciassae of the points of inflection can only be among the roots of the second derivative. Consequently

$p''(x) = ax(x-1)(x+1) = a(x^3 - x)$

Since at the point $x = 0$, $p'(0) = \tan 60° = \sqrt{3}$,

$$P'(x) = \int_0^x P''(x)dx + \sqrt{3} = a\left(\frac{x^4}{4} - \frac{x^2}{2}\right) + \sqrt{3}$$

hen, since $P(1) = 1$, we get

$P(x) =$

$$\int_1^x P'(x)dx + 1 = a\left(\frac{x^5}{20} - \frac{x^3}{6} + \frac{7}{60}\right) + \sqrt{3}(x-1) + 1$$

Since, $P(-1) = -1$ so, $a = \dfrac{60(\sqrt{3}-1)}{7}$

Thus

$$\int_0^1 P(x)dx = \int_0^1 \left[\frac{\sqrt{3}-1}{7}(3x^5 - 10x^3) + x\sqrt{3}\right]dx$$

$$= \frac{\sqrt{3}-1}{7}\left(\frac{x^6}{2} - \frac{5}{2}x^4\right) + \frac{x^2}{2}\sqrt{3}\Bigg|_0^1$$

$$= \frac{\sqrt{3}-1}{7}\left(\frac{1}{2} - \frac{5}{2}\right) + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{14} + \frac{2}{7}$$

**2.** **(d)** $I = \displaystyle\int_{-\pi}^{\pi} (\cos^2 px + \sin^2 qx - 2\cos px \sin qx)\, dx$

$\because$ $\sin^2 qx$, $\cos^2 px$ are even functions of $x$ and $\cos px$ . $\sin qx$ is an odd function.

$$\therefore \quad \int_{-\pi}^{\pi} \cos^2 px\, dx = 2\int_0^{\pi} \cos^2 px\, dx$$

$$\int_{-\pi}^{\pi} \sin^2 qx\, dx = 2\int_0^{\pi} \sin^2 qx\, dx$$

and $\displaystyle\int_{-\pi}^{\pi} \cos px \sin qx\, dx = 0$

$$\therefore \quad I = 2\int_0^{\pi} \cos^2 px\, dx + 2\int_0^{\pi} \sin^2 qx\, dx = 0$$

$$= 2\int_0^{\pi}\left(\frac{1+\cos 2px}{2}\right)dx + 2\int_0^{\pi}\left(\frac{1-\cos 2qx}{2}\right)dx$$

$$= \int_0^{\pi} (1+\cos 2px)\, dx + \int_0^{\pi} (1-\cos 2qx)\, dx$$

$$= \left[x + \frac{\sin 2px}{2p}\right]_0^{\pi} + \left[x - \frac{\sin 2qx}{2q}\right]_0^{\pi} = 2\pi$$

**3.** **(c)** $I = \displaystyle\int_0^{2\pi} e^{\cos\theta} \cos(\sin\theta)d\theta$ = Real part of

$$\int_0^{2\pi} e^{\cos\theta}\{\cos(\sin\theta) + i\sin(\sin\theta)\,d\theta$$

$$= \text{Real part of } \int_0^{2\pi} e^{\cos\theta} e^{i\sin\theta} d\theta = \text{Real part of}$$

$$\int_0^{2\pi} e^{\cos\theta + i\sin\theta} d\theta = \text{Real part of } \int_0^{2\pi} e^{e^{i\theta}} d\theta$$

= Real part of

$$\int_0^{2\pi}\left[1 + e^{i\theta} + \frac{e^{2i\theta}}{2!} + \frac{e^{3i\theta}}{3!} + \ldots\ldots\ldots\ldots\right]d\theta$$

= Real part of

$$\int_0^{2\pi}\left[1 + (\cos\theta + i\sin\theta) + \frac{1}{2!}(\cos 2\theta + i\sin 2\theta) + \ldots\right]d\theta$$

$$= \int_0^{2\pi}\left[1 + \cos\theta + \frac{1}{2!}\cos 2\theta + \ldots\ldots\right]d\theta$$

$$= \left[\theta + \sin\theta + \frac{\sin 2\theta}{2.2!} + \ldots\ldots\right]_0^{2\pi} = 2\pi$$

**4.** **(a)** Let $q = p + d, r = p + 2d, s = p + 3d$

$$\therefore f(x) = \begin{vmatrix} p + \sin x & p + d + \sin x & -2d + \sin x \\ p + d + \sin x & p + 2d + \sin x & -1 + \sin x \\ p + 2d + \sin x & p + 3d + \sin x & 2d + \sin x \end{vmatrix}$$

Applying $R \to R_1 + R_3 - 2R_2$, we get

$$f(x) = \begin{vmatrix} 0 & 0 & 2 \\ p + d + \sin x & p + 2d + \sin x & -1 + \sin x \\ p + 2d + \sin x & p + 3d + \sin x & 2d + \sin x \end{vmatrix}$$

$$= 2[(p + d + \sin x)(p + 3d + \sin x) - (p + 2d + \sin x)^2 = -2d^2$$

Given $\displaystyle\int_0^2 f(x)dx = -4$

$$\Rightarrow \int_0^2 (-2d^2)dx = -4 \quad d^2 = 1 \Rightarrow d = \pm 1$$

**5.** **(a)** $\int_0^{\pi/2} \frac{dx}{a^2\cos^2 x + b^2 \sin^2 x}$

$= \int_0^{\pi/2} \frac{\sec^2 x\, dx}{b^2 \tan^2 x + a^2} = \frac{1}{b^2}\int_0^{\pi/2} \frac{\sec^2 x\, dx}{\tan^2 x + \left(\frac{a}{b}\right)^2}$

$= \frac{1}{b^2}\int_0^{\infty} \frac{dt}{t^2 + \left(\frac{a}{b}\right)^2}$ (Putting $\tan x = t$)

$= \frac{1}{b^2} \times \frac{b}{a}\tan^{-1}\frac{bt}{a}\Big|_0^{\infty} = \frac{1}{ab}\left(\frac{\pi}{2}\right) = \frac{\pi}{2ab}$

Thus $\frac{\pi}{2ab} = \frac{\pi}{16} \Rightarrow ab = 8$

Now minimum value of $a\cos x + b\sin x$ is $-\sqrt{a^2+b^2}$, which is least if $a = b = 2\sqrt{2}$

$\Rightarrow -\sqrt{a^2+b^2} \geq -\sqrt{8+8} = -4$

**6.** **(a)** Let $f(x) = \frac{\sin x}{x}$

$\therefore f'(x) = \frac{x\cos x - \sin x}{x^2}$

$= \frac{(x - \tan x)\cos x}{x^2} < 0 \;\forall\, x \in \left[\frac{\pi}{4}, \frac{\pi}{3}\right]$

$\therefore f(x) = \frac{\sin x}{x}$ is decreases on the interval $\left[\frac{\pi}{4}, \frac{\pi}{3}\right]$

$\Rightarrow$ The least value of the function

$m = f\left(\frac{\pi}{3}\right) = \frac{\sin(\pi/3)}{(\pi/3)} = \frac{3\sqrt{3}}{2\pi}$

and the greastest value of the function

$M = f\left(\frac{\pi}{4}\right) = \frac{\sin(\pi/4)}{(\pi/4)} = \frac{2\sqrt{2}}{\pi}$

therefore

$\left(\frac{\pi}{3} - \frac{\pi}{4}\right)\frac{3\sqrt{3}}{2\pi} < \int_{\pi/4}^{\pi/3} \frac{\sin x}{x}dx < \left(\frac{\pi}{3} - \frac{\pi}{4}\right)\frac{2\sqrt{2}}{\pi}$

[Mean Value Theorem of Integral Calculus]

Hence, $\frac{\sqrt{3}}{8} < \int_{\pi/4}^{\pi/3} \frac{\sin x}{x}dx < \frac{\sqrt{2}}{6}$

**7.** **(d)** $f'(x) \geq f^3(x) + \frac{1}{f(x)} \Rightarrow \frac{f(x)f'(x)}{1+f^4(x)} \geq 1$

Integrating on the interval $(a, b)$, we get

$\int_a^b \frac{f(x)f'(x)}{1+f^4(x)}dx \geq \int_a^b dx$

$\Rightarrow \frac{1}{2}\tan^{-1} f^2(x)\Big|_a^b \geq b - a$

$\Rightarrow b - a \leq \frac{1}{2}\left[\lim_{x\to b^-} \tan^{-1} f^2(x) - \lim_{x \to a^+} \tan^{-1} f^2(x)\right]$

$= \frac{\pi}{24}$

**8.** **(b)** Given $\int_0^1 (1+\cos^8 x)(ax^2+bx+c)\,dx$

$= \int_0^2 (1+\cos^8 x)(ax^2+bx+c)\,dx$

$= \int_0^1 (1+\cos^8 x)(ax^2+bx+c)\,dx$

$+ \int_1^2 (1+\cos^8 x)(ax^2+bx+c)\,dx$

$\Rightarrow \int_1^2 (1+\cos^8 x)(ax^2+bx+c)\,dx = 0$

Now we know that if $\int_\alpha^\beta f(x)dx = 0$ then it means that $f(x)$ is + ve on some part of $(\alpha, \beta)$ and – ve on other part of $(\alpha, \beta)$.

But here $1 + \cos^8 x$ is always + ve,

$\therefore$ $ax^2 + bx + c$ is + ve on some part of $[1, 2]$ and – ve on other part $[1, 2]$

$\therefore ax^2 + bx + c = 0$ has at least one root in $(1, 2)$.

$\Rightarrow$ $ax^2 + bx + c = 0$ has at least one root in $(0,2)$.

**9.** **(c)** $f(x) = \int_x^{x^2} (t-1)dt$

$\Rightarrow f'(x) = (x^2-1)2x - (x-1) = (x-1)(2x^2+2x-1)$

$\therefore f'(\omega) = (\omega-1)(2\omega^2+2\omega-1)$

$= (\omega-1)\{2(-1)-1\} = 3(1-\omega)$

$\therefore |f'(\omega)| = 3|1-\omega|$

Now,

$|1-\omega|^2 = (1-\omega)(1-\overline{\omega}) = (1-\omega)(1-\omega^2)$

$= 1 - \omega - \omega^2 + \omega^3 = 3$

$\therefore |1-\omega| = \sqrt{3} \Rightarrow |f'(\omega)| = 3\sqrt{3}$

**10.** **(a)** We have

$\frac{d}{dx}\left[\int_{g(x)}^{h(x)} f(t)\,dt\right] = f(h(x))\,h'(x) - f(g(x)).g'(x)$

$= \frac{f(2)\times 2\times 2\times 1}{2\times\frac{\pi}{4}} = \frac{8}{\pi}f(2)$

Let $L = \lim_{x\to\frac{\pi}{4}} \frac{\int_2^{\sec^2 x} f(t)dt}{x^2 - \frac{\pi^2}{16}}$ $\left[\frac{0}{0}\text{ form}\right]$

On applying L' Hospital's rule, we get

$$L=\lim_{x\to\frac{\pi}{4}}\frac{\frac{d}{dx}\left[\int_2^{\sec^2 x} f(t)dt\right]}{\frac{d}{dx}\left(x^2-\frac{\pi^2}{16}\right)}$$

$$L=\lim_{x\to\frac{\pi}{4}}\frac{f(\sec^2 x).2\sec^2 x\tan x}{2x}$$

$$=\frac{f\left(\sec^2\frac{\pi}{4}\right).2.\sec^2\frac{\pi}{4}.\tan\frac{\pi}{4}}{2.\frac{\pi}{4}}=\frac{8}{\pi}.f(2)$$

**11. (c)** Let $I=\int_0^{\pi}(1-|\sin 8x)|)\,dx$

$\Rightarrow\ I=\int_0^{\pi}dx-\int_0^{\pi}|\sin 8x)|dx$

$\Rightarrow\ I=\pi-\int_0^{\pi 8\times 8}|\sin 8x|dx$

$\Rightarrow\ I=\pi-8\int_0^{\frac{\pi}{8}}\sin 8x\,dx$

[$\because$ sin $8x$ is periodic with period $\frac{\pi}{8}$ ]

$\Rightarrow\ I=\pi-8\left[\frac{-\cos 8x}{8}\right]_0^{\pi/8}$

$\Rightarrow\ I=\pi+(\cos\pi-\cos 0)=\pi-2$

**12. (d)** Given,

$\int_0^x f(x)dt=pf(x)$ ...(1)

Differentiating, we get,

$\Rightarrow\ f(x)=\text{p}f'(\text{x})$ ...(2)

$\Rightarrow\ \frac{f'(x)}{f(x)}=\frac{1}{p}$

Integrating, we get,

$\Rightarrow\ \log f(\text{x})=\frac{x}{p}+C$

$\Rightarrow\ f(\text{x})=Ae^{\text{x/p}}$ ...(3)

Putting $x=0$ in eqn. (i) we get,

$0=\text{p}f(\text{x})$ $\quad \text{p}f(0)=0$

**Case I** $\ f(0)=0,\ \text{p}\neq 0$

$\Rightarrow\ A=0$ (from 2)

$\Rightarrow\ f(x)=0$ $pf(0)=0$

$\Rightarrow\ f'(\text{x})=0$ (from (2))

$\Rightarrow$ there is no non-zero continuous to $f(\text{x})$.

**Case II** $\ p=0$

$\therefore\ \int_0^x f(t)dt=0,\forall\, x\in R$

If is possible when $f(\text{x})=0$.

Hence, $\forall\, p\in R$. There is no non-zero continous function.

Hence, $S\in R$.

**13. (a)** We have,

$$\text{L}=\lim_{x\to\infty}\sqrt{n}\int_0^1\frac{1}{(1+x^2)^n}dx$$

$\because\ (1+x^2)^n>1+nx^2$

$\Rightarrow\ \frac{1}{(1+x^2)^n}<\frac{1}{1+nx^2}$

$\Rightarrow\ \int_0^1\frac{dx}{(1+x^2)^n}<\int_0^1\frac{dx}{1+nx^2}$

$\Rightarrow\ \int_0^1\frac{dx}{(1+x^2)^n}<\frac{1}{\sqrt{n}}[\tan^{-1}\sqrt{n}x]_0^1$

$\Rightarrow\ \int_0^1\frac{dx}{(1+x^2)^n}<\frac{1}{\sqrt{n}}\tan^{-1}\sqrt{n}$

$\Rightarrow$ L

$$=\lim_{x\to\infty}\sqrt{n}\int_0^1\frac{1}{(1+x^2)^n}dx<\lim_{x\to\infty}\frac{\sqrt{n}}{\sqrt{n}}(\tan^{-1}\sqrt{n})$$

$\Rightarrow\ \text{L}<\tan^{-1}\infty=\frac{\pi}{2}$

$\therefore\ \frac{1}{2}<L<2\quad\left[\because\frac{\pi}{2}<2\right]$

**14. (a)** $f(x)=2\int_0^x t\,f(t)dt+1$ ...(i)

$\therefore\ f(0)=1$

Now, differentiate (i) w.r.t. $x$

$f'(x)=2xf(x)$

$\frac{f'(x)}{f(x)}=2x$

Integrate both sides, we get

$ln|f(x)|=x^2+c$ ...(ii)

$\because\ f(0)=1\ \Rightarrow\ c=0$

Then from eqn. (ii), $ln|f(x)|=x^2$

$\Rightarrow\ f(x)=e^{x^2}\ \Rightarrow f(1)=e$

**15. (a)** $\text{J}=\int_0^1\frac{x}{1+x^8}dx$

$\because\ 0<x^8<1$

$\Rightarrow\text{J}>\int_0^1\frac{x}{2}dx\Rightarrow\text{J}>\frac{1}{2}\left[\frac{x^2}{2}\right]_0^1\Rightarrow\text{J}>\frac{1}{4}$

$\text{J}<\int_0^1\frac{x}{1+0}dx\Rightarrow\text{J}<\left[\frac{x^2}{2}\right]_0^1\Rightarrow\text{J}<\frac{1}{2}$

Hence, (I) is true and (II) is false.

**16.** **(a)** It is given that $f(x)=|\sin x|$,

$g(x)=\int_0^x f(t).dt$

and $p(x)=g(x)-\frac{2}{\pi}x$

Now, $p(x+\pi)=g(x+\pi)-\frac{2}{\pi}(x+\pi)$

$=\int_0^{\pi+x} f(t)dt-\frac{2}{\pi}x-2$

$=\int_0^{\pi} f(t)dt+\int_{\pi}^{\pi+x} f(t)dt-\frac{2}{\pi}x-2$

$\because f(x)$ is periodic function with period $\pi$

$\therefore \int_{\pi}^{\pi+x} f(t)dt=\int_0^x f(t)dt$

$\Rightarrow p(x+\pi)=\int_0^{\pi}|\sin x|dx+g(x)-\frac{2}{\pi}x-2$

$=2+g(x)-\frac{2}{\pi}x-2$

$\Rightarrow p(x+\pi)=p(x)$ for all $x$

**17.** **(a)** If is given that, $x^2+(f(x))^2\le 1$

$\Rightarrow f(x)\le\sqrt{1-x^2}$

$\Rightarrow \int_0^1 f(x)dx\le\int_0^1\sqrt{1-x^2}dx$

$\Rightarrow \int_0^1 f(x)dx\le\left[\frac{x}{2}\sqrt{1-x^2}+\frac{1}{2}\sin^{-1}(x)\right]_0^1$

$\Rightarrow \int_0^1 f(x)dx\le\frac{\pi}{4} \Rightarrow f(x)=\sqrt{1-x^2}$

$\Rightarrow \int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}}\frac{f(x)}{1-x^2}dx=\int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}}\frac{\sqrt{1-x^2}}{(1-x^2)}dx$

$=\int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}}\frac{dx}{\sqrt{1-x^2}}$

$=\left[\sin^{-1}x\right]_{1/2}^{1/\sqrt{2}}=\frac{\pi}{4}-\frac{\pi}{6}=\frac{\pi}{12}$

**18.** **(16)** $\lim_{x\to 1}\frac{\int_4^{f(x)}2t\,dt}{x-1}=\lim_{x\to 1}\frac{(t^2)|_4^{f(x)}}{x-1}$

$=\lim_{x\to 1}\frac{(f(x))^2-4^2}{x-1}$

$=\lim_{x\to 1}\frac{2f(x).f'(x)}{1}$ (L-Hospital rule)

$=2f(1).f'(1)=2\times 4\times 2=16$

**19.** **(0)** $f(-x)=\begin{vmatrix}-\sin x & \sec x & x^2-1\\ -\operatorname{cosec} x & x\sin x & \cos x\\ -\tan x & x\tan x & x^2+1\end{vmatrix}$

$=-f(x)\Rightarrow f(x)$ is an odd function

$\Rightarrow \int_{-\pi/3}^{\pi/3} f(x)\,dx=0.$

**20.** **(0.30)** We have $\lim_{x\to 0}\frac{3\int_0^x e^{-t^2}dt-3x+x^3}{x^5}$

$\left(\text{form }\frac{0}{0}\right)$

Using L' Hospital's rule we get

$\lim_{x\to 0}\frac{3e^{-x^2}-3+3x^2}{5x^4}=\frac{3}{5}\lim_{x\to 0}\frac{e^{-x^2}-1+x^2}{x^4}$

$\left(\text{form }\frac{0}{0}\right)$

Again using L' Hospital's rule, we get

$\frac{3}{5}\lim_{x\to 0}\frac{e^{-x^2}.(-2x)-0+2x}{4x^3}$

$=\frac{3}{5}.\frac{2}{4}\lim_{x\to 0}\frac{-e^{-x^2}+1}{x^2}$ $\left(\text{form }\frac{0}{0}\right)$

Again using L' Hospital's rule we get

$\frac{3}{10}\lim_{x\to 0}\frac{-e^{-x^2}.(-2x)}{(2x)}=\frac{3}{10.}1=\frac{3}{10}=0.30$

**21.** **( 2)** $I(n)=\int_0^{\pi/2}\theta.\sin^n\theta\,d\theta.$

$\Rightarrow I(n)=\int_0^{\pi/2}\theta.\sin^{n-2}\theta(1-\cos^2\theta)d\theta$

$=I(n-2)-\int_0^{\pi/2}(\theta.\cos\theta).\cos\theta.\sin^{n-2}\theta d\theta$

$=I(n-2)-\theta.\cos\theta.\frac{\sin^{n-1}\theta}{n-1}\Big|_0^{\pi/2}$

$$+\int_0^{\pi/2}[\theta.(-\sin\theta)+\cos\theta].\ \frac{\sin^{n-1}\theta}{n-1}d\theta$$

$$=I(n-2)-\frac{1}{(n-1)}\int_0^{\pi/2}\theta.\sin^n\theta\,d\theta$$

$$+\frac{1}{n-1}\int_0^{\pi/2}\cos\theta.\ \sin^{n-1}\theta\,d\theta$$

$$=I(n-2)-\frac{1}{(n-1)}.I(n)+\frac{1}{(n-1)(n)}.\sin^n\theta\Big|_0^{\pi/2}$$

$$\Rightarrow\quad \frac{n}{n-1}I(n)=I(n-2)+\frac{1}{(n-1)(n)},$$

$$\Rightarrow\quad I(n)-I(n-2).\frac{n-1}{n}=\frac{1}{n^2}.$$

$$\Rightarrow\quad nI(n)-(n-1)I(n-2)=\frac{1}{n}$$

Put $n=2010$

$$\Rightarrow\quad 2010I(2010)-2009I(2008)=\frac{1}{2010}$$

$$\Rightarrow\quad [2010I(2010)\ \ -2009I(2008)]^{-1}=2010$$

**22.** **(0)** We have $x\int_0^x(1-t)f(t)dt=\int_0^x tf(t)dt$

Differentating both sides with respect to $x$ we get,

$$x(1-x)f(x)+\int_0^x(1-t)f(t)dt=xf(x)$$

$$\Rightarrow\quad x^2f(x)=\int_0^x(1-t)f(t)dt$$

Differentiating again with respect to $x$ on both sides, we get

$$x^2f'(x)+2xf(x)=(1-x)f(x)$$

$$\Rightarrow\quad \frac{f'(x)}{f(x)}=\frac{1-3x}{x^2}$$

Integrating both the sides, we get

$$\ell n\left|f(x)\right|=-\frac{1}{x}+3\ell nx+\lambda$$

$$\Rightarrow\quad \ell n\left[x^3\left|f(x)\right|\right]+\frac{1}{x}=\lambda\text{ and }f(1)=1$$

$$\Rightarrow\ \lambda=1$$

$$\Rightarrow\ \left|f(x)\right|=\frac{1}{x^3}e^{\left(1-\frac{1}{x}\right)}.$$

Thus $\lim\limits_{x\to\infty}f(x)=0$

**23.** **(0)** Using integration by parts

$$I=\left(-2\sin qt\frac{\cos pt}{p}\right)_0^1+\int_0^1 2q\cos qt.\frac{\cos pt}{p}dt$$

$$=-\frac{2}{p}\sin q\cos p+\frac{2q}{p}\int_0^1\cos qt.\cos pt dt$$

$$=-\frac{2}{p}\sin q\cos p+\frac{2q}{p}\left[\left(\cos qt\frac{\sin pt}{p}\right)_0^1+\frac{q}{p}\int_0^1\sin pt\sin qt dt\right]$$

$$=-\frac{2}{p}\sin q\cos p+\frac{2q}{p}\left(\frac{\sin p\cos q}{p}+\frac{q}{p}\frac{I}{2}\right)$$

$$\text{or, } I\left[1-\frac{q^2}{p^2}\right]=-\frac{2}{p}\sin q\cos p+\frac{2q}{p^2}\sin p\cos q$$

$$=\cos p\cos q\left[\frac{2q\tan p}{p^2}-\frac{2}{p}\tan q\right]$$

As $p$ and $q$ are different roots of equation, $\tan x=x$

$\Rightarrow$ $\tan p=p$ and $\tan q=q$

$\therefore$ $I=0$.

**24.** **(2)** $\int_0^3[\sqrt{x}]dx=\int_0^1[\sqrt{x}]dx+\int_1^3[\sqrt{x}]dx$

$$\left[\because\quad [\sqrt{x}]=\begin{cases}0, & \text{if } 0\le x<1\\ 1, & \text{if } 1\le x<3\end{cases}\right]$$

$$=\int_0^1 0\,dx+\int_1^3 1dx=[x]_1^3=2$$

**25.** **(7)** $\lim\limits_{x\to1}\frac{F(x)}{G(x)}=\frac{1}{14}\Rightarrow\lim\limits_{x\to1}\frac{\int_{-1}^x f(t)dt}{\int_{-1}^x t\left|f(f(t))\right|dt}$

$\because$ $\int_{-1}^1 f(t)dt=0$ and $\int_{-1}^1 t\left|f(f(t))\right|dt=0$

$f(t)$ being odd function

$\therefore$ Using $L$ Hospital's rule, we get

$$\lim_{x\to1}\frac{f(x)}{x\left|f(f(x))\right|}=\frac{1}{14}$$

$$\Rightarrow\quad \frac{f(1)}{\left|f(f(1))\right|}=\frac{1}{14}\Rightarrow\frac{1/2}{\left|f\left(\frac{1}{2}\right)\right|}=\frac{1}{14}$$

$$\Rightarrow\quad \left|f\left(\frac{1}{2}\right)\right|=7\Rightarrow f\left(\frac{1}{2}\right)=7$$

CHAPTER 24

# Application of Integrals

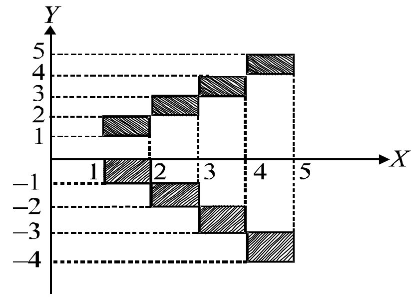
1. **(d)** Given $\int_1^b f(x)dx = \sqrt{b^2+1} - \sqrt{2}$

Differentiate with respect to b

$f(b) = \dfrac{b}{\sqrt{b^2+1}} \Rightarrow f(x) = \dfrac{x}{\sqrt{x^2+1}}$

2. **(b)** If $1 \le x < 2 \Rightarrow [x] = 1 \Rightarrow [y] = \pm 1$

$\Rightarrow y \in [-1, 0) \cup [1, 2)$




If $2 \le x < 3 \Rightarrow [x] = 2 \Rightarrow y = \pm 2$

$\Rightarrow y \in [-2, -1) \cup [2, 3)$

If $3 \le x < 4 \Rightarrow [x] = 3 \Rightarrow y = \pm 3$

$\Rightarrow y \in [-3, -2) \cup [3, 4)$

If $4 \le x < 5 \Rightarrow [x] = 4 \Rightarrow y = \pm 4$

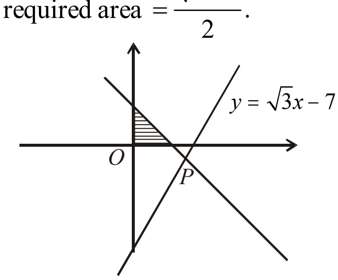
$\Rightarrow y \in [-4, -3) \cup [4, 5)$

Clearly, the required area consists of eight squares, each of area unity (see figure)

$\therefore$ Required area = 8 sq. units

3. **(a)** As point lies on the line. Locus of the point is straight line perpendicular to given line passing through $(2\sqrt{3}, -1)$ i.e. $y + \dfrac{x}{\sqrt{3}} = 1$

$\Rightarrow$ required area $= \dfrac{\sqrt{3} \times 1}{2}$.

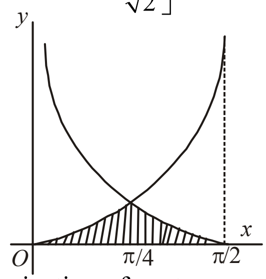



4. **(c)** Here $f(x) = \min\{|\tan x|, |\cot x|\}$

Required area $= 2\int_0^{\pi/4} \tan x dx + 2\int_{\pi/4}^{\pi/2} \cot x dx$

$= 2[ln|\sec x|_0^{\pi/4} + ln|\sin x|_{\pi/4}^{\pi/2}]$

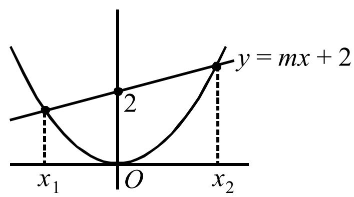
$= \left[ln\sqrt{2} + ln\,1 - ln\dfrac{1}{\sqrt{2}}\right] = 2[2ln\sqrt{2}] = ln\,4.$




5. **(d)** Eliminating $y$ from two equations,

We get $2mx + 4 = x^2 \Rightarrow x^2 - 2m - 4 = 0$

$x_1$ and $x_2$ are roots of this quadratic equation




$\therefore\ x_1 + x_2 = 2m$ and $x_1x_2 = -4$

Now, $\int_{x_1}^{x_2}\left(mx + 2 - \dfrac{x^2}{2}\right)dx$

$= \dfrac{m}{2}(x_2^2 - x_1^2) + 2(x_2 - x_1) - \dfrac{1}{6}(x_2^3 - x_1^3)$

$= (x_2 - x_1)\left[\dfrac{m}{2}(x_2 + x_1) + 2 - \dfrac{1}{6}(x_1^2 + x_1x_2 + x_2^2)\right]$

$= \sqrt{(x_2 + x_1)^2 - 4x_1x_2}$

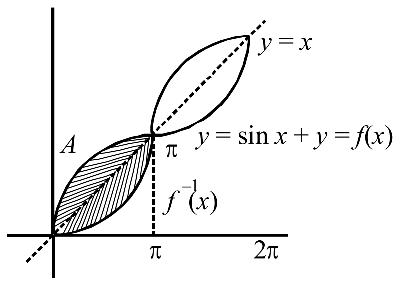
$\left[\dfrac{m}{2}(x_2 + x_1) + 2 - \dfrac{1}{6}\{(x_2 + x_1)^2 - x_2x_1\}\right]$

$= \sqrt{4m^2 + 16}\left[\dfrac{m^2 + 4}{3}\right]$

Clearly above is minumum if m = 0

6. **(d)** Required area $= 4A$, where

$A = \int_0^{\pi}(x + \sin x)dx - \int_0^{\pi} x dx$




$$A=\frac{\pi^2}{2}-\cos\pi+\cos 0-\frac{\pi^2}{2}=2 \text{ sq. units.}$$

**7.** **(b)** Let the equation of curve

$y^2(2a-x)=x^3$ ...(i)

and equation of line $x=2a$ ...(ii)

The given curve is symmetrical about x-axis and passes through origin.

From (i) we have, $y^2=\dfrac{x^3}{2a-x}$

But $\dfrac{x^3}{2a-x}<0$ for $x>2a$ and $x<0$

So, curve does not lie in the portion $x>2a$ and $x<0$, therefore curve lies in $0\le x\le 2a$.

$\therefore$ Area bounded by the curve and line

$$=\int_0^{2a} y\,dx=\int_0^{2a}\frac{x^{3/2}}{\sqrt{2a-x}}dx$$

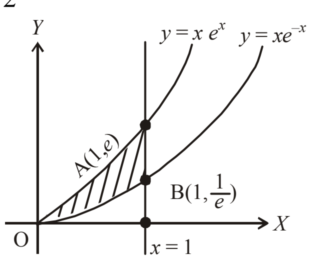
Put $x=2a\sin^2\theta$ and $dx=4a\sin\theta\cos\theta\,d\theta$

$$\therefore \quad I=\int_0^{\pi/2}8a^2\sin^4\theta\,d\theta=8a^2\left[\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}\right]$$

$$\left(\text{using }\int_0^{\pi/2}\sin^m x\cos^n x\,dx=\frac{\Gamma\frac{m+1}{2}\Gamma\frac{n+1}{2}}{2\Gamma\frac{m+n+2}{2}}\right)$$

$$=\frac{3\pi a^2}{2}\text{ sq. unit}$$

**8.** **(a)**




Given curves are $y=xe^x$ and $y=xe^{-x}$

Line $x=1$ meets the curves at $A(1,e)$ and $B\left(1,\frac{1}{e}\right)$.

Both the curves pass through origin.

$$\therefore \text{ Required area}=\int_0^1(xe^x-xe^{-x})dx$$

$$=\int_0^1 x(e^x-e^{-x})dx$$

$$=x(e^x+e^{-x})\Big|_0^1-\int_0^1(e^x+e^{-x})dx$$

$$=\left(e+\frac{1}{e}\right)-e^x\Big|_0^1+e^{-x}\Big|_0^1=\frac{2}{e}$$

**9.** **(a)** If both $|x+y-1|$ and $|2x+y-1|$ are positive, then $x+y-1+2x+y-1=1$

$\Rightarrow \quad 3x+2y=3$

$\Rightarrow \quad \dfrac{x}{1}+\dfrac{y}{3/2}=1$

If $|x+y-1|$ is positive and $|2x+y-1|$ is negative, then $(x+y-1)-(2x+y-1)=1$

$\Rightarrow -x-1=0 \Rightarrow x=-1$

If first negative and second positive, then

$-(x+y-1)+(2x+y-1)=1$

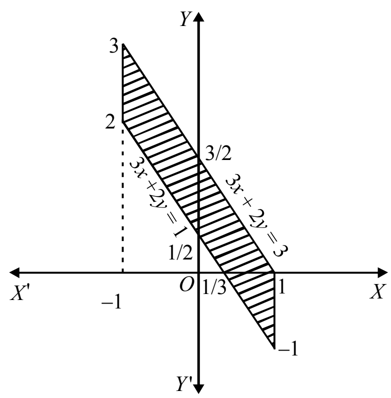
$\Rightarrow x-1=0$

$\Rightarrow x=1$

If both negative, then

$-(x+y-1)-(2x+y-1)=1 \Rightarrow -3x-2y+1=0$

$\Rightarrow \quad -3x-2y+1=0\Rightarrow \dfrac{x}{1/3}+\dfrac{y}{1/2}=1$




$\therefore$ Required area

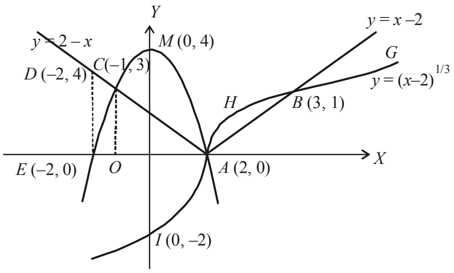
$$=\left(\frac{1}{2}\times 2\times 3\right)-\frac{1}{2}\left(\frac{4}{3}\times 2\right)+\frac{1}{2}\times 1\times\frac{2}{3}$$

$= 2$ sq units

**10.** **(c)** $y=4-x^2$ is the parabola $EMA$

$y=|x-2|$ is the pair of straight lines $DCA$ and $ABF$,

$y=(x-2)^{1/3}$ is the curve $IAHBG$




Thus $f(x) = \max.\ \{4 - x^2, |x-2|, (x-2)^{1/3}\}$ is

$$f(x) = \begin{cases} 2 - x & -2 \le x < -1 \\ 4 - x^2, & -1 \le x < 2 \\ (x-2)^{1/3}, & 2 \le x < 3 \\ x - 2, & 3 \le x \le 4 \end{cases}$$

Now, area bounded by the curve and x-axis.

$$= \int_{2}^{-1} (2-x)dx + \int_{-1}^{2} (4-x)dx + \int_{0}^{3} (x-2)^{1/3} dx$$

$$+ \int_{3}^{4} (x-2)dx$$

$$= \frac{7}{2} + 9 + \frac{3}{4} + \frac{3}{2} = \frac{59}{4} \text{ sq. units.}$$

**11. (b)** The given curves are

$$y = \sqrt{\frac{1+\sin x}{\cos x}} = \sqrt{\frac{1+\tan\frac{x}{2}}{1-\tan\frac{x}{2}}} \qquad ...(1)$$

and $$y = \sqrt{\frac{1-\sin x}{\cos x}} = \sqrt{\frac{1-\tan\frac{x}{2}}{1+\tan\frac{x}{2}}} \qquad ...(2)$$

$\therefore$ The area bounded by the above curves, by the lines $x = 0$ and $x = \dfrac{\pi}{4}$ is given by

$$A = \int_0^{\pi/4} \left( \sqrt{\frac{1+\tan\frac{x}{2}}{1-\tan\frac{x}{2}}} - \sqrt{\frac{1-\tan\frac{x}{2}}{1+\tan\frac{x}{2}}} \right) dx$$

$$= \int_0^{\pi/4} \frac{2\tan\frac{x}{2}}{\sqrt{1-\tan^2\frac{x}{2}}} dx$$

Let $\tan\dfrac{x}{2} = t \Rightarrow \dfrac{1}{2}\sec^2\dfrac{x}{2}dx = dt$

$$\Rightarrow dx = \frac{2}{1+t^2}dt$$

Also when $x \to 0$, $t \to 0$ and when $x \to \dfrac{\pi}{4}$,

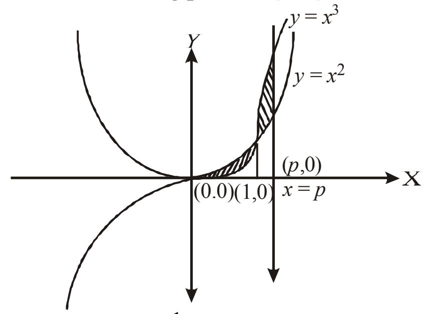
$t \to \tan\dfrac{\pi}{8}$

$$\therefore A = \int_0^{\tan\frac{\pi}{8}} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt = \int_0^{\sqrt{2}-1} \frac{4t}{(1+t^2)\sqrt{1-t^2}} dt$$

**12. (d)** Given curves are $y = x^2$ and $y = x^3$

Also, $x = 0$ and $x = p, p > 1$

Now, intersecting point is (1, 1)




$$\text{Required area} = \int_0^1 \left(x^2 - x^3\right)dx + \int_1^p \left(x^3 - x^2\right)dx$$

$$\frac{1}{6} = \left.\frac{x^3}{3} - \frac{x^4}{4}\right|_0^1 + \left.\frac{x^4}{4} - \frac{x^3}{3}\right|_1^p$$

$$\Rightarrow \quad \frac{1}{6} = \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{p^4}{4} - \frac{p^3}{3} - \frac{1}{4} + \frac{1}{3}\right)$$

$$\Rightarrow \quad \frac{1}{6} - \frac{1}{3} + \frac{1}{4} + \frac{1}{4} - \frac{1}{3} = \frac{3p^4 - 4p^3}{12}$$

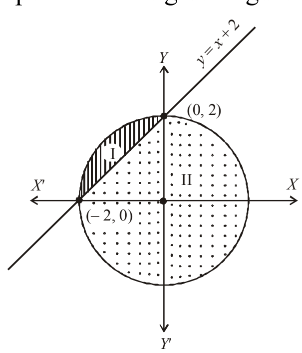
$$\Rightarrow \quad \frac{p^3(3p-4)}{12} = 0 \Rightarrow p^3(3p-4) = 0$$

$\Rightarrow p = 0$ or $\dfrac{4}{3}$

Since, it is given that $p > 1$ $\therefore$ $p$ can not be zero.

Hence, $p = \dfrac{4}{3}$

**13. (d)** Let I be the smaller portion and II be the greater portion of the given figure then,



Area of $I = \int_{-2}^{0}\left[\sqrt{4-x^2}-(x+2)\right]dx$

$= \left[\frac{x}{2}\sqrt{4-x^2}+\frac{4}{2}\sin^{-1}\left(\frac{x}{2}\right)\right]_{-2}^{0}-\left[\frac{x^2}{2}+2x\right]_{-2}^{0}$

$= \left[2\sin^{-1}(-1)\right]-\left[-\frac{4}{2}+4\right]=2\times\frac{\pi}{2}-2=\pi-2$

Now, area of II = Area of circle – area of I.

$= 4\pi-(\pi-2) = 3\pi+2$

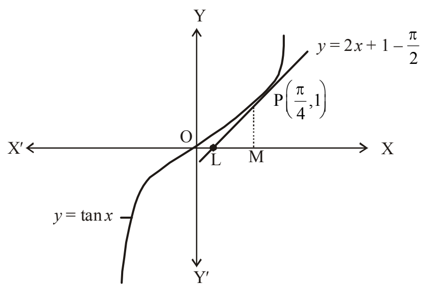
Hence, required ratio $= \frac{\text{area of I}}{\text{area of II}} = \frac{\pi-2}{3\pi+2}$

**14.** **(a)** The given curve is $y = \tan x$ ...(1)

when $x = \frac{\pi}{4}$, $y = 1$

Equation of tangent at P is

$y - 1 = \left(\sec^2\frac{\pi}{4}\right)\left(x-\frac{\pi}{4}\right)$




or $y = 2x + 1 - \frac{\pi}{2}$ ...(2)

Area of shaded region

= area of OPMO – $ar$ ($\Delta$PLM)

$= \int_0^{\frac{\pi}{4}}\tan x\,dx-\frac{1}{2}(\text{OM}-\text{OL})\text{PM}$

$= [\log\sec x]_0^{\frac{\pi}{4}}-\frac{1}{2}\left\{\frac{\pi}{4}-\frac{\pi-2}{4}\right\}\times 1$

$= \frac{1}{2}\left[\log 2-\frac{1}{2}\right]$ sq. unit

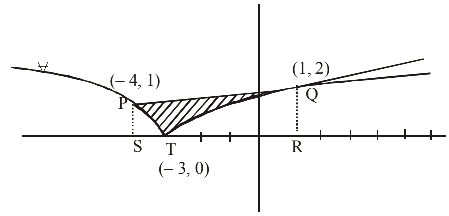
**15.** **(c)** $y \ge \sqrt{|x+3|} \Rightarrow y^2 = |x+3|$

$\Rightarrow y^2 = \begin{cases}-(x+3) \text{ if } x<-3\\ (x+3) \text{ if } x\ge -3\end{cases}$ ....(i)

Also $y \le \frac{x+9}{5}$ and $x \le 6$ ....(ii)

Solving (i) and (ii) we get intersection points as (1, 2), (6, 3), (– 4, 1), (–39, –6)

The graph of given region is as follows-




Required area

= Area (trap PQRS) – Area (PST + TQR)

$= \frac{1}{2}\times(1+2)\times 5-\left[\int_{-4}^{-3}\sqrt{-x-3}\,dx+\int_{-3}^{1}\sqrt{x+3}\,dx\right]$

$= \frac{15}{2}-\left[\left(\frac{2(-x-3)^{3/2}}{-3}\right)_{-4}^{-3}+\left(\frac{2(x+3)^{3/2}}{3}\right)_{-3}^{1}\right]$

$= \frac{15}{2}-\left[\frac{2}{3}+\frac{16}{3}\right] = \frac{15}{2}-6 = \frac{3}{2}$ sq. units

**16.** **(b)** Since,

$y = x^2+x+10$

$\Rightarrow \quad x^2+x+\frac{1}{4} = y-10+\frac{1}{4}$

$\Rightarrow \quad \left(x+\frac{1}{2}\right)^2 = \left(y-\frac{39}{4}\right)$

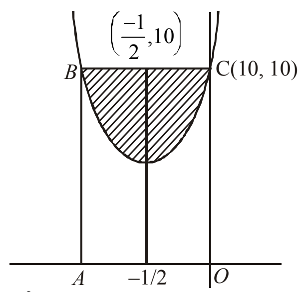
$\Rightarrow$ Latusrectum of the above parabola is 1.

$\therefore$ length of chord is also 1.

Area is maximum when chord is latusrectum.

Area of shaded reion =

Area of rectangle $OABC - 2\int_{-1/2}^{0}$ parabola




$= 10-2\int_{\frac{-1}{2}}^{0}(x^2+x+10)dx$

$= 10-2\left[\frac{x^3}{3}+\frac{x^2}{2}+10\right]_{\frac{-1}{2}}^{0}$

$= 10-2\left[\frac{-1}{24}+\frac{1}{8}-\frac{10}{2}\right]=\frac{1}{6}$

**17.** **(b)** $y=\dfrac{\sin x}{x}+e^x$ and $y=\dfrac{\sin x}{x}+\dfrac{x^2}{2}$

Required area of the region

$$=\int_1^2\left\{\frac{\sin x}{x}+e^x-\left(\frac{\sin x}{x}+\frac{x^2}{2}\right)\right\}dx$$

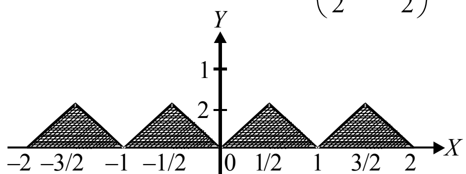
$$=\int_1^2\left(e^x-\frac{x^2}{2}\right)dx=\left[e^x-\frac{x^3}{6}\right]_1^2$$

$$=e^2-\frac{8}{6}-\left(e-\frac{1}{6}\right)=e^2-e-\frac{7}{6}$$

**18.** **(1)** The graph of $f(x)=\min\{x-[x],-x-[-x]\}$ is shown as in figure. Therefore

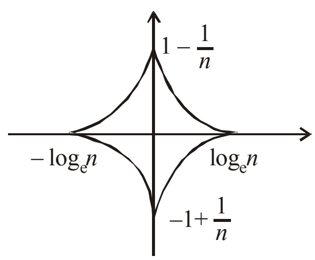
$$\int_{-2}^{2}\min\ \{x-[x],-\ x-[-x]\}dx$$

$=$ Area of shaded region $=4\left(\dfrac{1}{2}\times 1\times\dfrac{1}{2}\right)=1$




**19.** **(2)** The area of the region is

$$A=4\int_0^{\log_e n}\left(e^{-x}-\frac{1}{n}\right)dx=4\left(\frac{n-1-\log_e n}{n}\right)\quad ...(1)$$




Now, $n-2\ge 2\log_e n\Rightarrow 2n-2-2\log_e n\ge n$

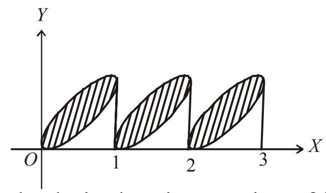
$$\Rightarrow\frac{n-1-\log_e n}{n}\ge\frac{1}{2}$$

$\therefore$ From (1), $A\ge 4\times\dfrac{1}{2}=2$

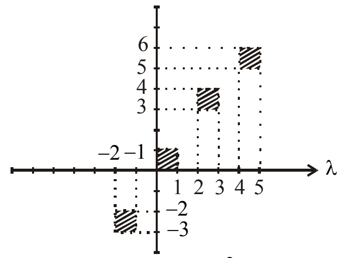
**20.** **(3.33)** Here both functions $f(x)$ and $g(x)$ are periodic. Thus required area

$$=10\left(\int_0^1(\sqrt{x}-x^2)dx\right)=10\left[\frac{x^{3/2}}{3/2}-\frac{x^3}{3}\right]_0^1$$

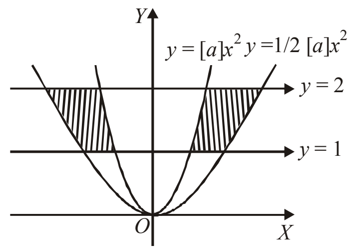
$$=\frac{10}{3}=3.33.$$




**21.** **(4)** The desired region consists of four disjoint squares. So the area $=1\times 4=4$ square unit.




**22.** **(1)** The curves $y=[a]\,x^2$ and $y=\dfrac{1}{2}[a]x^2$ represent parabolas which are symmetric about $y$-axis. The equation $y^2-3y+2=0$ gives a pair of straight lines $y=1, y=2$ which are parallel to $x$-axis. Thus the area bounded is shown as;




From above figure the required area

$$=2\int_{y=1}^{y=2}(x_2-x_1)\,dy=2\int_1^2\left(\sqrt{\frac{2y}{[a]}}-\sqrt{\frac{y}{[a]}}\right)dy$$

$$=\frac{2(\sqrt{2}-1)}{[a]}\int_1^2\sqrt{y}\ dy=\frac{2(\sqrt{2}-1)}{[a]}\cdot\frac{2}{3}.(y^{3/2})_1^2$$

$$=\frac{4}{3}\frac{(\sqrt{2}-1)}{[a]}.(2^{3/2}-1)$$

$$=\frac{4}{3}\frac{(5-\sqrt{2}-2\sqrt{2})}{[a]}=\frac{4}{3}\frac{(5-3\sqrt{2})}{[a]}$$

$\therefore$ Area is greatest, when $[a]$ is least, i.e., 1.

$\therefore$ Area is greatest, when $[a]=1$

$\Rightarrow\ a\in[1,2)$

**23.** **(2)** $f'(x) = \lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$

$= \lim_{h\to 0} \dfrac{f(x(1+h/x))-f(x)}{h}$

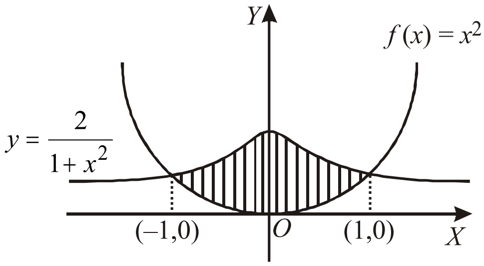
$= \dfrac{f(x)}{x} \lim_{h\to 0} \dfrac{f(1+h/x)-f(1)}{h/x}$

$= \dfrac{f(x)}{x} \cdot f'(1)$

$\therefore \quad f'(x) = \dfrac{2f(x)}{x}$ or $\dfrac{f'(x)}{f(x)} = \dfrac{2}{x}$

Integrating both sides, we get, $f(x) = cx^2$

Since $f(1) = 1$, $\therefore \quad c = 1$




So, $f(x) = x^2$

Now, $\dfrac{2}{1+x^2} = x^2 \Rightarrow x^4 + x^2 - 2 = 0$

$\Rightarrow \quad x^2 = 1 \quad \Rightarrow \quad x = \pm 1$

Required area $= 2\left[\int_0^1 \left(\dfrac{2}{1+x^2} - x^2\right) dx\right]$

$= 2\left[2\tan^{-1} x - \dfrac{x^3}{3}\right]_0^1 = 2\left[\dfrac{\pi}{2} - \dfrac{1}{3}\right]$

$= \left(\pi - \dfrac{2}{3}\right)$ sq. units.

**24.** **(0)** Since $y = f(x)$ has relative extremes at $x = \pm 2/\sqrt{3}$, these points are critical points, and hence they must be roots of $f'(x) = 0$ (clearly is differentiable everywhere). Thus

$f'(x) = a(x - 2/\sqrt{3})(x + 2/\sqrt{3})$

$= a(x^2 - 4/3)$

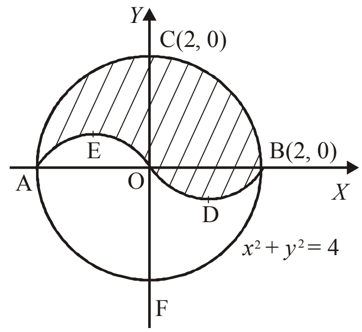
$\Rightarrow f(x) = a(x^3/3 - 4x/3) + b$. This passes through (0, 0) and (1, –2). So $\boldsymbol{b} = 0$ and $a(1/3 - 4/3) = -2 \Rightarrow a = 2$.

Thus $f(x) = \dfrac{2x}{3}(x^2 - 4)$. $f(x)$ meet the $x$-axis at (0, 0) (–2, 0) and (2, 0).

Since $f(-x) = -f(x)$, the curve $y = f(x)$ is symmetrical about the origin.

Also, as $a = 2, f'(x) > 0$ for $x < -2/\sqrt{3}$, $x > 2/\sqrt{3}$ and $f'(x) < 0$ for $-2/\sqrt{3} < x < 2/\sqrt{3}$.

Thus, shape of $y = f(x)$ is as shown in figure




Area of the region $ACBDOEA$

$= 2\pi - \int_{-2}^{0} |f(x)| dx + \int_0^2 f(x) dx$

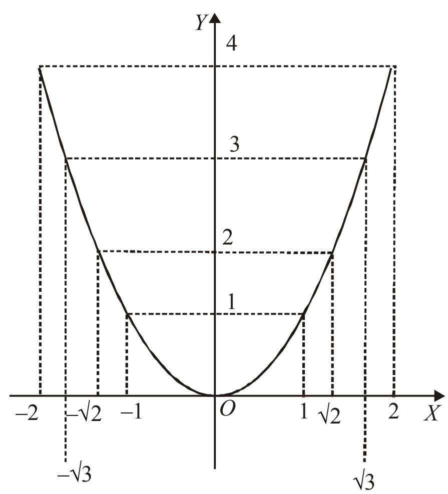
But $\int_{-2}^{0} |f(x)| dx = -\int_{-2}^{0} f(x) dx$

$= -\int_2^0 f(-t)(-1) dt = \int_0^2 f(t) dt$

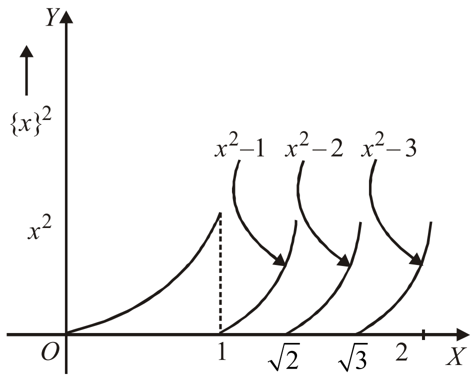
$\therefore \quad$ Area $(ACBDOEA) = 2\pi$

Thus, the area of the other region $AEODBFA = 2\pi$.

**25.** **(7)**



As we know that fractional part of any thing must lie between 0 and 1 thus




$\therefore$ Area of region bounded by $y = \{x\}^2$ between $x$-axis for the $x \in [0, 2]$ is

$$= \int_0^1 x^2 dx + \int_1^{\sqrt{2}} (x^2 - 1)\, dx + \int_{\sqrt{2}}^{\sqrt{3}} (x^2 - 2)\, dx + \int_{\sqrt{3}}^{2} (x^2 - 3)\, dx$$

$$A_0 = \sqrt{2} + \sqrt{3} - \frac{7}{3}$$

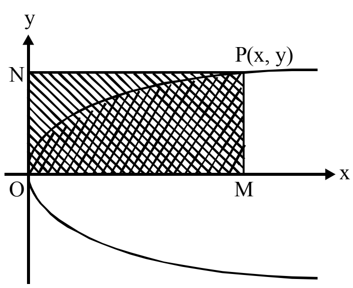
$\therefore$ Required area $= 2A_0 = 2\left(\sqrt{2} + \sqrt{3} - \frac{7}{3}\right)$

CHAPTER 25

# Differential Equation and its Applications

**1.** **(c)** Let $P(x, y)$ be the point on the curve passing through the origin $O(0, 0)$ and let $PN$ and $PM$ be the lines parallel to the $x$-and $y$-axis, respectively. If the equation of the curve is $y = y(x)$, the area $POM$ equals $\int_0^x y dx$ and the area PON equals $xy - \int_0^x y dx$. Assuming that $2(POM) = PON$, we therefore have

$$2\int_0^x y\, dx = xy - \int_0^x y\, dx \Rightarrow 3\int_0^x y\, dx = xy$$




Differentiating both sides of this gives

$$3y = x\frac{dy}{dx} + y \Rightarrow 2y = x\frac{dy}{dx} \Rightarrow \frac{dy}{y} = 2\frac{dx}{x}$$

$$\Rightarrow \log y = 2\log x + C \Rightarrow y = Cx^2$$

With $C$ being a constant. This solution represents a parabola. We will get a similar result if we have started instead with $2(PON) = POM$

**2.** **(c)** $f'(x) \geq f^3(x) + \frac{1}{f(x)}$

or $f'(x).f(x) \geq 1 + f^4(x)$

or $\frac{f(x).f'(x)}{1 + f^4(x)} \geq 1$

Integrating with respect to $x$, from $x = a$ to $x = b$

$$\frac{1}{2}\left(\tan^{-1}(f^2(x))\right)_a^b \geq b - a$$

or $(b - a) \leq \frac{1}{2}\left\{\lim_{x \to b^-} (\tan^{-1}(f^2(x))) - \lim_{x \to a^+} \left(\tan^{-1}(f^2(x))\right)\right\}$

$$\Rightarrow b - a \leq \frac{\pi}{24}$$

**3.** **(c)** Differentiating w.r.t. $x$ the given equation,

we have $\sqrt{1 - (f'(x))^2} = f(x)$

If $y = f(x)$, then we have,

$$1 - \left(\frac{dy}{dx}\right)^2 = y^2 \; \frac{dy}{dx} = \pm\sqrt{1 - y^2}$$

$$\Rightarrow \frac{dy}{\sqrt{1 - y^2}} = \pm dx$$

$\Rightarrow \sin^{-1} y = C \pm x \Rightarrow y = \sin(C \pm x)$

Since $f(0) = 0$ so $0 = \sin C \Rightarrow C = 0$

Thus $f(x) = \pm \sin x$

As $f(x) \geq 0$ for $x \in [0, 1]$ so we have $f(x) = \sin x$.

Since $\sin x < x \; \forall \, x > 0$, we get $f(x) < x$ for $0 < x \leq 1$. Thus $f(1/2) < 1/2$ and $f(1/3) < 1/3$.

**4.** **(a)** Put $y = x\sec\theta$

$\therefore \dfrac{dy}{dx} = x\sec\theta\tan\theta\dfrac{d\theta}{dx} + \sec\theta$

From the given equation,

$x^3\left(x\sec\theta\tan\theta\dfrac{d\theta}{dx} + \sec\theta\right)$

$= x^3\sec^3\theta + x^3\sec^2\theta\tan\theta$

Dividing both sides by $x^3\sec\theta$, we get

$x\tan\theta\dfrac{d\theta}{dx} + 1 = \sec^2\theta + \sec\theta\tan\theta$

$x\dfrac{d\theta}{dx} = (\tan\theta + \sec\theta)$

or $(\sec\theta - \tan\theta)d\theta = \dfrac{dx}{x}$

Integrating we get

$\ln(\sec\theta + \tan\theta) - \ln\sec\theta = \ln x + \ln c$

or $1 + \dfrac{\sqrt{y^2 - x^2}}{y} = cx \Rightarrow y + \sqrt{y^2 - x^2} = cxy$

where $c$ is an arbitrary constant.

**5.** **(a)** Put $x^2y^2 = z$

Given, $x^2 \cdot 2y\dfrac{dy}{dx} + y^2 \cdot 2x = \tan(x^2y^2)$

$\dfrac{d}{dx}(x^2y^2) = \tan(x^2y^2)$ put $x^2y^2 = z$

Now, given expression transform to

$\dfrac{dz}{dx} = \tan z \quad \therefore \int dx = \int \cot z\, dz$

$x = \ln(\sin z) + C$

when $x = 1,\ y = \sqrt{\dfrac{\pi}{2}} \Rightarrow z = \dfrac{\pi}{2} \quad \Rightarrow C = 1$

$\therefore \quad x = \ln\sin(x^2y^2) + 1 \therefore \ln\sin(x^2y^2) = x - 1$

$\sin(x^2y^2) = e^{x-1}$

**6.** **(b)** $x\left(\cos\dfrac{y}{x}\right)(y\,dx + x\,dy)$

$= y\sin\left(\dfrac{y}{x}\right)(x\,dy - y\,dx)$

Dividing both sides by $x^2\,dx$ we get

$\Rightarrow \left(\cos\dfrac{y}{x}\right)\left(\dfrac{y}{x} + \dfrac{dy}{dx}\right) = \dfrac{y}{x}\sin\left(\dfrac{y}{x}\right)\left(\dfrac{dy}{dx} - \dfrac{y}{x}\right)$

Which is homogeneous equation

Putting $y = vx$ we get $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

or $\cos v\left(v + v + x\dfrac{dv}{dx}\right) = v\sin v\left(x\dfrac{dv}{dx}\right)$

$\Rightarrow \quad 2v\cos v = x\dfrac{dv}{dx}(v\sin v - \cos v)$

Separating the variables, we get

$\Rightarrow \quad \dfrac{2dx}{x} = \left(\dfrac{v\sin v - \cos v}{v\cos v}\right)dv$

Integrating $2\ln x + \ln c = \ln\sec v - \ln v$

$\Rightarrow \quad cx^2 = \dfrac{\sec v}{v} \Rightarrow \sec v = cx^2 v$

$\Rightarrow \quad \sec\dfrac{y}{x} = cxy$

where '$c$' is an arbitrary constant.

**7.** **(c)** Differentiating $xy = C$, we get $y + xp = 0$

where $p = \dfrac{dy}{dx}$.

Replacing p by $\dfrac{p + \tan\pi/4}{1 - p\tan\pi/4} = \dfrac{p+1}{1-p}$,

we have $y + \dfrac{p+1}{1-p}x = 0$

$\Rightarrow \quad \dfrac{dy}{dx} = \dfrac{y/x + 1}{y/x - 1}$ Putting $y/x = v$, the last equation reduces to

$x\dfrac{dv}{dx} = \dfrac{v+1}{v-1} - v = -\dfrac{v^2 - 2v - 1}{v-1}$

$\Rightarrow 2\dfrac{dx}{x} = -\dfrac{2(v-1)}{v^2 - 2v - 1}dv$

$\Rightarrow \quad 2\ln x = -\ln(v^2 - 2v - 1) + \ln C$

$\Rightarrow \quad \ln x^2(y^2/x^2 - 2y/x - 1) = \ln C$

$\Rightarrow \quad y^2 - 2xy - x^2 = C$, which does not represent a pair of straight lines if $C \neq 0$.

**8.** **(a)** The given equation can be written as

$$\left(\frac{dx}{x} - \frac{dy}{y}\right) + \frac{(x^2dy - y^2dx)}{(x-y)^2} = 0$$

$$\Rightarrow \quad \left(\frac{dx}{x} - \frac{dy}{y}\right) + \frac{\left(\dfrac{dy}{y^2} - \dfrac{dx}{x^2}\right)}{\left(\dfrac{1}{y} - \dfrac{1}{x}\right)^2} = 0$$

$$\Rightarrow \quad \left(\frac{dx}{x} - \frac{dy}{y}\right) + \frac{\dfrac{dy}{y^2} - \dfrac{dx}{x^2}}{\left(\dfrac{1}{x} - \dfrac{1}{y}\right)^2} = 0$$

Integrating, we get: $\ln|x| - \ln|y| - \dfrac{1}{\left(\dfrac{1}{x} - \dfrac{1}{y}\right)} = c$

$$\Rightarrow \quad \ln\left|\frac{x}{y}\right| - \frac{xy}{y-x} = c \Rightarrow \ln\left|\frac{x}{y}\right| + \frac{xy}{x-y} = c$$

**9.** **(b)** Here, $\frac{a^2}{xy} \cdot \frac{dx}{dy} = \frac{x}{y} + \frac{y}{x} - 2$

$$\Rightarrow a^2 = \frac{dy}{dx}(x^2 + y^2 - 2xy)$$

$$\Rightarrow (x-y)^2 \cdot \frac{dy}{dx} = a^2,$$

put $x - y = v \therefore 1 - \frac{dy}{dx} = \frac{dv}{dx}$

$$\Rightarrow v^2\left(1 - \frac{dv}{dx}\right) = a^2$$

$$\Rightarrow v^2 - a^2 = v^2\frac{dv}{dx} \Rightarrow \frac{v^2\,dv}{v^2 - a^2} = dx$$

$$\Rightarrow \left(1 + \frac{a^2}{v^2 - a^2}\right)dv = dx$$

Integrating both the sides, we get

$$v + \frac{a}{2}\log\left|\frac{v-a}{v+a}\right| = x + C$$

$$\Rightarrow (x-y) + \frac{a}{2}\log\left(\frac{x-y-a}{x-y+a}\right) = x + C$$

$$\Rightarrow y + C = \frac{a}{2}\log\left(\frac{x-y-a}{x-y+a}\right)$$

It passes through $(3a, a)$

$$\Rightarrow a + C = \frac{a}{2}\log\left(\frac{1}{3}\right)$$

$$\Rightarrow C = -a + \frac{a}{2}\log\left(\frac{1}{3}\right) \Rightarrow C = -a\left(\frac{2 + \log 3}{2}\right)$$

$$\therefore y = \frac{a}{2}(2 + \log 3) + \log\left(\frac{x-y-a}{x-y+a}\right)$$

$$\Rightarrow \frac{x-y-a}{x-y+a} = e^{y-k}, \text{ where } k = \frac{a}{2}(2 + \log 3)$$

$$\therefore \frac{x-y}{a} = \frac{1 + e^{y-k}}{1 - e^{y-k}}$$

$$\Rightarrow x = y + a\left(\frac{1 + e^{y-k}}{1 - e^{y-k}}\right),$$

where $k = \frac{a}{2}(2 + \log 3)$

Which is required equation of curve.

Hence, (b) is the correct answer.

**10.** **(b)** The given equation can be written as

$$(y\,dx - x\,dy) + x\sqrt{xy}(x+y)dx + y\sqrt{xy}(x+y)dy = 0$$

$$\Rightarrow (y\,dx - x\,dy) + (x+y)\sqrt{xy}(x\,dx + y\,dy) = 0$$

$$\Rightarrow \frac{y\,dx - x\,dy}{y^2} + \left(\frac{x}{y} + 1\right) \cdot \sqrt{\frac{x}{y}}\,d\left(\frac{x^2 + y^2}{2}\right) = 0$$

$$\Rightarrow d\left(\frac{x}{y}\right) + d\left(\frac{x^2 + y^2}{2}\right)\left(\frac{x}{y} + 1\right)\sqrt{\frac{x}{y}} = 0$$

$$\Rightarrow d\left(\frac{x^2 + y^2}{2}\right) + \frac{d\left(\frac{x}{y}\right)}{\left(\frac{x}{y} + 1\right) \cdot \sqrt{\frac{x}{y}}} = 0$$

or $d\left(\frac{x^2 + y^2}{2}\right) + 2d\left(\tan^{-1}\left(\sqrt{\frac{x}{y}}\right)\right) = 0$

Integrating both the sides, we get

$$\Rightarrow \frac{x^2 + y^2}{2} + 2\tan^{-1}\sqrt{\frac{x}{y}} = C$$

Hence, (b) is the correct answer.

**11.** **(a)** $\frac{dy}{dx} = \frac{y^3}{e^{2x} + y^2}$

Dividing by $e^{2x}$: $\therefore \frac{dy}{dx} = \frac{y^3 e^{-2x}}{1 + y^2 e^{-2x}}$

$$\therefore dy + y^2 e^{-2x} dy = y^3 e^{-2x} dx$$

$$\therefore \int 2\frac{dy}{y} + \int 2(ye^{-2x}dy - y^2 e^{-2x}dx) = 0$$

$$\therefore \int 2\frac{dy}{y} + \int d(e^{-2x}y^2) + c = 0$$

$$\therefore 2\ln|y| + e^{-2x}y^2 = c$$

**12.** **(b)** Given curve is $a^{n-1}\,y = x^n$ ...(i)

Differentiating equation (i) w.r.t. $x$, we get

$$a^{n-1}\frac{dy}{dx} = nx^{n-1} \quad \text{...(ii)}$$

Eliminating a from equations using (i) & (ii), we get

$$\frac{x^n}{y}\frac{dy}{dx} = nx^{n-1} \quad \text{...(iii)}$$

Putting $-\frac{dx}{dy}$ in equation (iii) at the place of $\frac{dy}{dx}$, we get

$$-\frac{x^n}{y} \cdot \frac{dx}{dy} = nx^{n-1}, \Rightarrow \frac{-xdx}{ydy} = n,$$

$$\Rightarrow -xdy = ny\,dy,$$

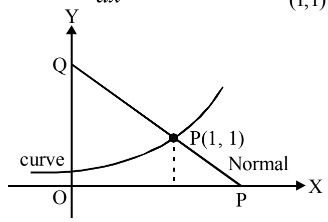
$$\Rightarrow -\frac{x^2}{2} + c = \frac{ny^2}{2}, \Rightarrow ny^2 + x^2 = \text{constant.}$$

**13.** **(c)** Given, equation of normal at $P(1,1)$ is $ay + x = a + 1$

$\therefore$ Slope of tangent at $P = a \Rightarrow \left(\dfrac{dy}{dx}\right)_{(1,1)} = a$

Given

$$\frac{dy}{dx} \propto y \;\Rightarrow\; \frac{dy}{dx} = ky \Rightarrow \left(\frac{dy}{dx}\right)_{(1,1)} = k = a$$




$$\frac{dy}{dx} = ay \Rightarrow \frac{dy}{y} = adx$$

(variable being separated)

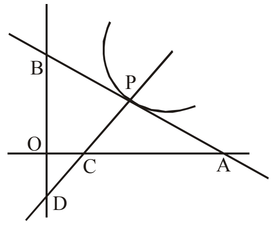
$\Rightarrow \ln y = ax + c$

It is passing through (1, 1) then $c = -a$

$\Rightarrow$ equation of the curve is $y = e^{a(x-1)}$

**14.** **(a)** Let $P$ be $(x, y)$. $C$ is $\left(x + y\dfrac{dy}{dx}, 0\right)$ and $B$ is $\left(0, y - x\dfrac{dy}{dx}\right)$. Centre of the circle through $O$, $C$, $P$ and $B$ has its centre at the mid-point of $BC$. Let it be $(\alpha, \beta)$ then

$$2\alpha = x + y\frac{dy}{dx} \text{ and } 2\beta = y - x\frac{dy}{dx}$$




Now, $(\alpha, \beta)$ lies on $y = x$

so, $y - x\dfrac{dy}{dx} = x + y\dfrac{dy}{dx} \Rightarrow \dfrac{dy}{dx} = \dfrac{y-x}{x+y}$

**15.** **(a)** $\dfrac{dy}{dx} - y\log_e 2 = 2^{\sin x}(\cos x - 1)\log_e 2$

This is linear differential equation

I.F. $= e^{-\log_e 2\int dx} = e^{-x\log_e 2} = 2^{-x}$

(i) Solution is

$$y2^{-x} = \int 2^{-x}2^{\sin x}(\cos x - 1)\log_e 2\, dx$$

(ii) put $\sin x - x = t \Rightarrow (\cos x - 1)dx = dt$

$\therefore y2^{-x} = \log_e 2\int 2^t\, dt$

$\therefore y2^{-x} = 2^t + c$

$\therefore y = 2^{x+t} + c2^x$

$\therefore y = 2^{\sin x} + c2^x$

**16.** **(c)** Rearranging the terms of equation, we get

$$\frac{dt}{dx} - t\frac{g'(x)}{g(x)} = -\frac{t^2}{g(x)}$$

$$\Rightarrow -\frac{1}{t^2}\frac{dt}{dx} + \frac{1}{t}\frac{g'(x)}{g(x)} = \frac{1}{g(x)} \quad \text{...(i)}$$

Let $z = \dfrac{1}{t} \Rightarrow -\dfrac{1}{t^2}\dfrac{dt}{dx} = \dfrac{dz}{dx}$

Thus, from (i) we obtain $\dfrac{dz}{dx} + \dfrac{g'(x)}{g(x)}z = \dfrac{1}{g(x)}$

which is clearly linear in $z$ and $\dfrac{dz}{dx}$ with

I.F. $= e^{\int \frac{g'(x)}{g(x)}dx} = e^{\log[g(x)]} = g(x)$

$\Rightarrow$ Thus, complete solution is

$$z \cdot g(x) = \int g(x) \cdot \frac{1}{g(x)}dx + C$$

$$\Rightarrow \frac{1}{t}g(x) = x + c \Rightarrow \frac{g(x)}{x+c} = t$$

**17.** **(a)** $1 = \lim\limits_{t \to x} \dfrac{t^2 f(x) - x^2 f(t)}{t - x}$ $\left(\dfrac{0}{0} \text{ form}\right)$

$$= \lim_{t \to x} \frac{2tf(x) - x^2 f'(t)}{1}$$

$\Rightarrow 2xf(x) - x^2 f'(x) = 1$

$\Rightarrow x^2\dfrac{dy}{dx} - 2xy = -1,\; y = f(x)$

$\Rightarrow \dfrac{dy}{dx} - \dfrac{2}{x}y = -\dfrac{1}{x^2}$

which is linear equation whose I.F.

$= e^{-2\log x} = \dfrac{1}{x^2}.$

Multiplying both sides of the last equation by I.F., we get

$$\frac{d}{dx}\left(\frac{y}{x^2}\right) = -\frac{1}{x^4} \Rightarrow \frac{y}{x^2} = \frac{1}{3x^3} + K.$$

Since $y = f(1) = 1$.

so $K = \dfrac{2}{3}$. Thus $y = \dfrac{1}{3x} + \dfrac{2}{3}x^2$

**18.** **(2)** The equation is $y = e^{x\frac{dy}{dx}} \Rightarrow x\dfrac{dy}{dx} = \ell\text{n } y$

**19.** **(0)** $(2ny + xy\log_e x)\,dx = x\log_e x\,dy$

$\Rightarrow \frac{dy}{y} = \left(\frac{2n}{x\log_e x} + 1\right)dx$

$\Rightarrow \log(y) = 2n\log|\log x| + x + c$ and $c = 0$

($\because$ curve passes through $(e, e^e)$)

$\therefore\ y = e^{x+\log(\log x)^{2n}} = e^x(\log x)^{2n}$

$\Rightarrow f(x) = e^x(\log x)^{2n}$

Now,

$$g(x) = \lim_{n\to\infty} f(x) = \begin{cases} \to\infty & \text{if} \quad x < \frac{1}{e} \\ 0 & \text{if} \quad \frac{1}{e} < x < e \\ \to\infty & \text{if} \quad x > e \end{cases}$$

$\therefore\ \int_{1/e}^{e} g(x)\,dx = 0$

**20.** **(2)** The given differential equation can be written as $\frac{dx}{dy} = xy[x^2\sin y^2 + 1]$

$\Rightarrow\ \frac{1}{x^3}\frac{dx}{dy} - \frac{1}{x^2}y = y\sin y^2$

This equation is reducible to linear equation, so putting $-\frac{1}{x^2} = u$, the last equation can be written as

$\frac{du}{dy} + 2uy = 2y\sin y^2$

The integrating factor of this equation is $e^{y^2}$.

So required solution is

$ue^{y^2} = \int 2y\sin y^2 . e^{y^2}\,dy + C$

$= \int(\sin t)e^t\,dt + C \qquad (t = y^2)$

$\Rightarrow ue^{y^2} = \frac{1}{2}e^{y^2}(\sin y^2 - \cos y^2) + C$

$\Rightarrow\ 2u = (\sin y^2 - \cos y^2) + Ce^{-y^2}$

$\Rightarrow\ 2 = x^2[\cos y^2 - \sin y^2 - 2Ce^{-y^2}]$

**21.** **(8)** Differentiating both sides w.r.t., x of the given equation

$$x.y(x) + \int_0^x y(t)dt.1 = (x+1)x.y(x) + \int_0^x ty(t)dt$$

or $\int_0^x y(t)dt = x^2y(x) + \int_0^x ty(t)dt$

Again differentiating both sides w.r.t. $x$

$y(x) = x^2y'(x) + y(x)2x + xy(x)$

or $(1-3x)y(x) = x^2y'(x)$

or $\frac{y'(x)}{y(x)} = \left(\frac{1}{x^2} - \frac{3}{x}\right)$

Integrating, we get

$\ell n\, y(x) = -\frac{1}{x} - 3\ell n\, x + \ell n\, c$

or $\ell n\left(\frac{x^3y(x)}{c}\right) = -\frac{1}{x}$ or $\frac{x^3y(x)}{c} = e^{-1/x}$

or $y(x) = \frac{ce^{-1/x}}{x^3}$

So, $y(1) = e \Rightarrow c = e^2 \quad \therefore\ y\left(\frac{1}{2}\right) = 8$

**22.** **(5)** Put $x = r\sec\theta$ and $y = r\tan\theta$ So,

$x^2 - y^2 = r^2 \qquad \ldots(1)$

and $\sin\theta = \frac{y}{x} \qquad \ldots(2)$

then differentiating (1) we get,

$2xdx - 2ydy = 2rdr$

or $xdx - ydy = rdr \qquad \ldots(3)$

and differentiaing (2) we get,

$\frac{xdy - ydx}{x^2} = \cos\theta d\theta$

or $xdy - ydx = x^2\cos\theta d\theta$

$= r^2\sec^2\theta\cos\theta d\theta = r^2\sec\theta d\theta \qquad \ldots(4)$

Substituting values from (3) and (4) in the given differetial equation, we get

$$\frac{rdr}{r^2\sec^2\theta\, d\theta} = \sqrt{\left(\frac{1+r^2}{r^2}\right)} = \frac{\sqrt{1+r^2}}{r}$$

or $\frac{dr}{\sqrt{(1+r^2)}} = \sec\theta\, d\theta$

Intergrating both sides,

$\ell n(r + \sqrt{(1+r^2)}) = \ell n(\sec\theta + \tan\theta) + \ell n\, c$

Where $c$ is an arbitrary constant.

or $(r + \sqrt{(1+r^2)}) = c(\sec\theta + \tan\theta)$

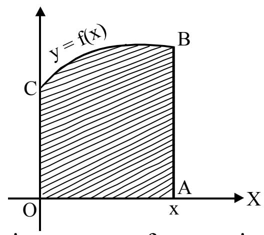
or $(\sqrt{(x^2-y^2)} + \sqrt{(1+x^2-y^2)}) = c\left(\frac{x+y}{\sqrt{(x^2-y^2)}}\right)$

**23.** **(1)** Area of curvilinear trapezoid

$OABCO = \int_0^x f(x)dx$ according to question

$\int_0^x f(x)dx \propto \{f(x)\}^{n+1}$

or $\int_0^x f(x)\,dx = k\{f(x)\}^{n+1}$




Where k is constant of proportionality.
Differentiating both sides w.r.t. x,

$f(x) = k(n+1)(f(x))^n f'(x)$

or $\{f(x)^{n-1}\}f'(x) = \dfrac{1}{k(n+1)'}$

Integrating both sides w.r.t. x,

$\dfrac{\{f(x)\}^n}{n} = \dfrac{x}{k(n+1)} + c$

Putting x = 0 $\dfrac{\{f(0)\}^n}{n} = 0 + c \Rightarrow 0 = 0 + c$

$(\because f(0) = 0)$

$\therefore \dfrac{\{f(x)\}^n}{n} = \dfrac{x}{k(n+1)}$

$\Rightarrow \{f(x)\}^n = \dfrac{nx}{k(n+1)}$ ....... (1)

Again putting x = 1 then $\{f(x)\}^n = \dfrac{n}{k(n+1)} = 1$

$(\because f(1) = 1)$

From (1), $(f(x))^n = x$ or $f(x) = x^{1/n}$

$\Rightarrow$ $(f(10))^n = 10 \Rightarrow k = 1$

**24.** **(7)** Given

$x^2 f'(x) f\left(\dfrac{1}{x}\right) - f(x) f'\left(\dfrac{1}{x}\right) = x^2 f'(x) - f'\left(\dfrac{1}{x}\right)$

$\Rightarrow x^2 f'(x)\left[f\left(\dfrac{1}{x}\right) - 1\right] = f'\left(\dfrac{1}{x}\right)[f(x) - 1]$

$\Rightarrow f'(x)\left\{f\left(\dfrac{1}{x}\right) - 1\right\} = \dfrac{1}{x^2} f'\left(\dfrac{1}{x}\right)\{f(x) - 1\}$

$\Rightarrow \dfrac{f'(x)}{f(x)-1} + \dfrac{f'\left(\dfrac{1}{x}\right)\left(-\dfrac{1}{x^2}\right)}{f\left(\dfrac{1}{x}\right) - 1} = 0$

Integrating, we get

$\ln|f(x) - 1| + \ln\left|f\left(\dfrac{1}{x}\right) - 1\right| = c$

$\Rightarrow f(x) - 1)\left(f\left(\dfrac{1}{x}\right) - 1\right) = e^c = k$

$\Rightarrow f(x) f\left(\dfrac{1}{x}\right) - f(x) - f\left(\dfrac{1}{x}\right) + 1 = k$

$\Rightarrow (f(1))^2 - f(1) - f(1) + 1 = k$

(substituting $x = 1$)

$\Rightarrow k = 1$ $(\because f(1) = 2,$ given$)$

$\Rightarrow f(x) f\left(\dfrac{1}{x}\right) = f(x) + f\left(\dfrac{1}{x}\right)$

$\Rightarrow f(x) = 1 \pm x^n$ (From functional equation)

Since $f(1) = 2$, we must have $f(x) = 1 + x^n$

Since $f(5) = 26$, $26 = 1 + 5^n \Rightarrow n = 2$

$\therefore f(x) = 1 + x^2$ $\therefore f(6) = 1 + (6)^2 = 37$

**25.** **(1)** Here,

$f'(x) = 1 + 2x^2 + \dfrac{2}{3}\cdot 4x^4 + \dfrac{2}{3}\cdot\dfrac{4}{5}\cdot 6x^6 + \ldots\infty$

$= 1 + x\left(\dfrac{d}{dx}(xf(x))\right)$

$\Rightarrow f'(x) = 1 + x|xf'(x) + f(x)|$

$\Rightarrow (1 - x^2) f'(x) = 1 + xf(x)$

$\Rightarrow \dfrac{dy}{dx} - \dfrac{x}{1-x^2}\cdot y = \dfrac{1}{x^2},$

I.F. $= e^{\int \frac{-x}{1-x^2}dx} = e^{\frac{1}{2}\log|1-x^2|} = \sqrt{1-x^2}$

$\therefore y\cdot\sqrt{1-x^2} = \int \dfrac{1}{1-x^2}\cdot\sqrt{1-x^2} + C$

$\Rightarrow y\sqrt{1-x^2} = \sin^{-1} x + C,$

as $f(0) = 0 \Rightarrow C = 0$

$\Rightarrow y = \dfrac{\sin^{-1} x}{\sqrt{1-x^2}}$

$\Rightarrow A = \int_{1/2}^{\sqrt{3}/2} \dfrac{\sin^{-1} x}{\sqrt{1-x^2}} dx$

$= \int_{\pi/6}^{\pi/3} t\,dt = \left(\dfrac{t^2}{2}\right)_{\frac{\pi}{6}}^{\frac{\pi}{2}} = \dfrac{1}{2}\left[\dfrac{\pi^2}{4} - \dfrac{\pi^2}{36}\right]$

$\therefore [4A] = 1.$

CHAPTER 26

# Vector Algebra

**1.** **(b)** $\vec{a}(x)$ and $\vec{b}(x)$ are collinear if and only if $\cos x = x$. Now let $f(x) = x - \cos x$, then $f'(x) = 1 + \sin x \geq 0$

$\Rightarrow$ $f(x)$ is increasing and hence $f(x) = 0$ for a unique value of $x$.

For $x \geq \frac{\pi}{3}$, $f(x) > 0$ and $x < \frac{\pi}{6}$, $f(x) < 0$. Thus $\cos x = x$, for a unique value of $x, x \in \left(\frac{\pi}{6}, \frac{\pi}{3}\right)$.

**2.** **(d)** Suppose that $\vec{a}, \vec{b}, \vec{c}$ are coplanar.

$$\Rightarrow \begin{vmatrix} \cos\alpha & 1 & 1 \\ 1 & \cos\beta & 1 \\ 1 & 1 & \cos\gamma \end{vmatrix} = 0$$

applying $R_2 \longrightarrow R_2 - R_1$ and $R_3 \longrightarrow R_1$

$$\text{or} \quad \begin{vmatrix} \cos\alpha & 1 & 1 \\ 1-\cos\alpha & \cos\beta - 1 & 1 \\ 1-\cos\alpha & 0 & \cos\gamma - 1 \end{vmatrix} = 0$$

or $\cos\alpha(\cos\beta - 1)(\cos\gamma - 1)$

$-(1-\cos\alpha)(\cos\gamma - 1) - (1-\cos\alpha)(\cos\beta - 1) = 0$

Dividing through out by

$(1-\cos\alpha)(1-\cos\beta)(1-\cos\gamma)$ ; we get

$$\frac{\cos\alpha}{1-\cos\alpha} + \frac{1}{1-\cos\beta} + \frac{1}{1-\cos\gamma} = 0$$

$$\Rightarrow \frac{-(1-\cos\alpha)+1}{(1-\cos\alpha)} + \frac{1}{(1-\cos\beta)} + \frac{1}{(1-\cos\gamma)} = 0$$

$$\Rightarrow -1 + \frac{1}{1-\cos\alpha} + \frac{1}{1-\cos\beta} + \frac{1}{1-\cos\gamma} = 0$$

$$\Rightarrow \frac{1}{1-\cos\alpha} + \frac{1}{1-\cos\beta} + \frac{1}{1-\cos\gamma} = 1$$

$$\Rightarrow \operatorname{cosec}^2\frac{\alpha}{2} + \operatorname{cosec}^2\frac{\beta}{2} + \operatorname{cosec}^2\frac{\gamma}{2} = 2$$

which is not possible as

$$\operatorname{cosec}^2\frac{\alpha}{2} \geq 1, \ \operatorname{cosec}^2\frac{\beta}{2} \geq 1, \ \operatorname{cosec}^2\frac{\gamma}{2} \geq 1.$$

So the vectors cannot be coplanar.

**3.** **(b)** a, b, c are distinct non-negative numbers and the vectors $a\hat{i} + a\hat{j} + c\hat{k}$, $\hat{i} + \hat{k}$ and $c\hat{i} + c\hat{j} + b\vec{k}$ are coplanar.

$$\therefore \begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} a & a & c-a \\ 1 & 0 & 0 \\ c & c & b-c \end{vmatrix}$$

Operating $C_3 \to C_3 - C_1$

Expanding along $R_2$, we get

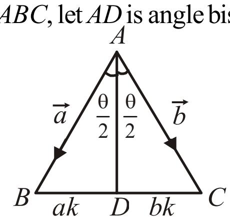
$$-\begin{vmatrix} a & c-a \\ c & b-c \end{vmatrix} = c\,(c-a) - a\,(b-c) = 0$$

$\Rightarrow c^2 - ac - ab + ac = 0$

$\Rightarrow c^2 = ab \Rightarrow a, c, b$ are in G.P.

$\therefore$ $c$ is the G.M. of $a$ and $b$.

**4.** **(c)** In $\Delta ABC$, let $AD$ is angle bisector of angle $A$.




$\therefore BD = ak,\ DC = bk \quad \therefore BC = (a+b)k$

Applying cosine formula, we have

$$\cos\theta = \frac{(AB)^2 + (AC)^2 - (BC)^2}{2(AB)(AC)}$$

$$= \frac{a^2 + b^2 - (a+b)^2 k^2}{2ab} \quad \text{...(1)}$$

Also in $\Delta ADC$ and $\Delta ABD$

$$\cos\frac{\theta}{2} = \frac{b^2+(AD)^2-b^2k^2}{2b\ AD} = \frac{a^2+(AD)^2-a^2k^2}{2a\ AD}$$

$$\Rightarrow (AD)^2 = ab(1-k^2)$$

$$= ab\left\{1-\frac{a^2+b^2-2ab\cos\theta}{(a+b)^2}\right\} \qquad \text{[from (1)]}$$

$$= \frac{4a^2b^2\cos^2\theta/2}{(a+b)^2} \Rightarrow AD = \frac{2ab\cos\theta/2}{(a+b)}$$

$$\therefore \overrightarrow{AD} = \pm\frac{(\vec{a}b+\vec{b}a)}{(a+b)} = \pm\frac{ab}{(a+b)}\left(\frac{\vec{a}}{a}+\frac{\vec{b}}{b}\right)$$

$$= \pm\frac{ab}{(a+b)}(\hat{a}+\hat{b})$$

$$\therefore \overrightarrow{AD} = \frac{\overrightarrow{AD}}{AD} = \pm\frac{(\hat{a}+\hat{b})}{2\cos\theta/2}$$

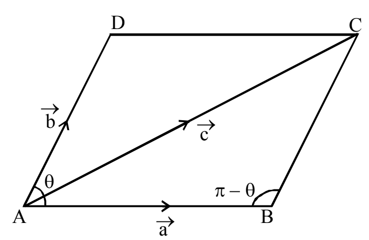
**5.** **(a)** Let $\overrightarrow{AB} = \vec{a}$ and $\overrightarrow{AD} = \vec{b}$ and $\overrightarrow{AC} = \vec{c}$ when $\vec{a}$, $\vec{b}$ and $\vec{c}$ are non-collinear coplanar vectors.

$\overrightarrow{DB} = \overrightarrow{AB} - \overrightarrow{AC} = \vec{a} - \vec{b}\cdot$

Now, $\overrightarrow{DB}.\overrightarrow{AB} = (\vec{a}-\vec{b}).(\vec{a}) = \vec{a}.\vec{a} - \vec{b}.\vec{a}$

$$a^2 - ab\cos\theta = a^2 - \frac{c^2-a^2-b^2}{2} = \frac{3a^2+b^2-c^2}{2}$$

$$\left[\therefore In\ \Delta ABC,\ \cos(\pi-\theta) = \frac{a^2+b^2-c^2}{2ab}\right]$$




**6.** **(a)** Given $|\vec{a}| = 1$, $|\vec{c}| = 1$ and $|\vec{b}| = 4$

$$\Rightarrow \vec{a}.\vec{c} = 1.1.\frac{1}{4} = \frac{1}{4}.$$

Again, $\vec{b} - 2\vec{c} = \lambda\vec{a} \Rightarrow \vec{a}.\vec{b} - 2\vec{a}.\vec{c} = \lambda a^2$

$$\Rightarrow \vec{a}.\vec{b} - 2.\frac{1}{4} = \lambda \ \Rightarrow\ \vec{a}.\vec{b} = \lambda + \frac{1}{2}.$$

Again,

$\vec{b} - 2\vec{c} = \lambda\vec{a} \Rightarrow \vec{b}.\vec{b} - 2\vec{b}.\vec{c} = \lambda\vec{b}.\vec{a}$

$$\Rightarrow 16 - 2\vec{b}.\vec{c} = \lambda\left(\lambda+\frac{1}{2}\right)$$

$$\Rightarrow \vec{b}.\vec{c} = 8 - \frac{\lambda^2}{2} - \frac{\lambda}{4}.$$

Furthermore,

$\vec{b} - 2\vec{c} = \lambda\vec{a} \Rightarrow \vec{b}.\vec{c} - 2\vec{c}.\vec{c} = \lambda\vec{a}.\vec{c}$

$$\Rightarrow 8 - \frac{\lambda^2}{2} - \frac{\lambda}{4} - 2(1) = \lambda\left(\frac{1}{4}\right)$$

$$\Rightarrow \lambda^2 + \lambda - 12 = 0 \Rightarrow \lambda = -4, 3.$$

**7.** **(b)** Let the triangle be $PQR$ with sides $\vec{p}$, $\vec{q}$, $\vec{r}$

Let $\vec{p} = \sqrt{3}(\hat{a}\times\vec{b})$ and $\vec{q} = \vec{b} - (\hat{a}.\ \vec{b})\ \hat{a}$

$$\therefore\ \vec{p}.\vec{q} = \sqrt{3}(\hat{a}\times\vec{b}).\{\vec{b}-(\hat{a}.\vec{b})\ \hat{a}\}$$

$$= \sqrt{3}[\hat{a}\ \vec{b}\ \vec{b}] - \sqrt{3}(\hat{a}.\vec{b})[\hat{a}\ \ \vec{b}\ \hat{a}] = 0$$

$$\Rightarrow \angle R = \frac{\pi}{2}.$$

Also $|\vec{p}| = \sqrt{3}(\hat{a}\times\vec{b}) = \sqrt{3}\,|\hat{a}|\,|\vec{b}|\sin\theta$

$$= \sqrt{3}\,|\vec{b}|\sin\theta,$$

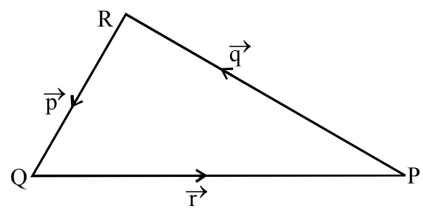
$\theta$ is the angle between $\hat{a}$ and $\vec{b}$ and $|\hat{a}| = 1$

and $|\vec{q}| = \sqrt{\{\vec{b}-(\hat{a}.\vec{b})\hat{a}\}.\{\vec{b}-((\hat{a}.\vec{b})\ \hat{a}\}}$

$$= \sqrt{|\vec{b}|^2 - 2(\hat{a}.\vec{b})^2 + (\hat{a}.\vec{b})^2(\hat{a})^2}$$

$$= \sqrt{|\vec{b}|^2 + |\vec{b}|^2\cos^2\theta . - 2|\vec{b}|^2\cos^2\theta}$$

$$= |\vec{b}|\sin\theta$$



$\therefore \quad \frac{|\vec{p}|}{|\vec{q}|} = \sqrt{3}$ . But $\frac{|\vec{p}|}{|\vec{q}|} = \tan P \Rightarrow P = \frac{\pi}{3}$

$\left[\because \angle R = \frac{\pi}{2}\right]$

$\therefore \quad P = \frac{\pi}{3} = \tan^{-1}(\sqrt{3})$ , $Q = \frac{\pi}{6} = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

and $R = \frac{\pi}{2} = \cot^{-1}(0)$ .

[**Note that** as soon as we get $R = \frac{\pi}{2}$ , only one option has the value of an angle $\frac{\pi}{2} = \cot^{-1}(0)$ , hence (b) is the correct answer].

**8.** **(c)** For the vector $\vec{a}$ and $\vec{b}$ to be inclined at an obtuse angle, we must have $\vec{a}.\vec{b} < 0$ for all $x \in (0, \infty)$

$\Rightarrow c(\log_2 x)^2 - 12 + 6c\log_2 x < 0$ for all $x \in (0, \infty)$

$\Rightarrow cy^2 + 6cy - 12 < 0$ for all $y \in R$, where $y = \log_2 x$

$\Rightarrow c < 0$ and $36c^2 + 48c < 0$ ,

(using $ax^2 + bx + c < 0, \forall x \in R$ if $a < 0$ and $D < 0$ )

$\Rightarrow c < 0$ and $c(3c+4) < 0 \Rightarrow c \in \left(-\frac{4}{3}, 0\right)$ .

**9.** **(b)** $\vec{p}.\vec{q} = ab + bc + ca$

$= \sqrt{a^2+b^2+c^2}\sqrt{b^2+c^2+a^2}\cos\theta$

$\Rightarrow \cos\theta = \frac{ab+bc+ca}{(a^2+b^2+c^2)}$ .

$\Rightarrow$ Now $(a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0$

$a^2 + b^2 + c^2 \geq ab + bc + ca$

$\Rightarrow \frac{ab+bc+ca}{a^2+b^2+c^2} \leq 1$.

Also

$(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca) \geq 0$

$\Rightarrow \frac{ab+bc+ca}{a^2+b^2+c^2} \geq -1/2$

$\Rightarrow -\frac{1}{2} \leq \cos\theta \leq 1 \quad \Rightarrow \theta \in [0, 2\pi/3]$.

**10.** **(c)** Vol. of paralleopiped formed by $\vec{u} = \hat{i} + a\hat{j} + \hat{k}, \vec{v} = \hat{j} + a\hat{k}, \vec{w} = a\hat{i} + \hat{k}$ is

$$V = [\vec{u}\,\vec{v}\,\vec{w}] = \begin{vmatrix} 1 & a & 1 \\ 0 & 1 & a \\ a & 0 & 1 \end{vmatrix}$$

$= 1(1-0) - a(0-a^2) + 1(0-a) = 1 + a^3 - a$

For $V$ to be min $\frac{dV}{da} = 0$

$\Rightarrow 3a^2 - 1 = 0 \quad \Rightarrow \quad a = \pm\frac{1}{\sqrt{3}}$ .

**11.** **(b)** We observe that

$\vec{a}.\vec{b}_1 = \vec{a}.\vec{b} - \left(\frac{\vec{b}.\vec{a}}{|\vec{a}|^2}\right)\vec{a}.\vec{a} = \vec{a}.\vec{b} - \vec{a}.\vec{b} = 0$

$\vec{a}.\vec{c}_2 = \vec{a}.\left(\vec{c} - \frac{\vec{c}.\vec{a}}{|\vec{a}|^2}\vec{a} - \frac{\vec{c}.\vec{b}_1}{|\vec{b}_1|^2}\vec{b}_1\right)$

$= \vec{a}.\vec{c} - \vec{c}.\frac{\vec{a}.\vec{c}}{|\vec{a}|^2}|\vec{a}|^2 - \frac{\vec{c}.\vec{b}_1}{|\vec{b}_1|^2}(\vec{a}.\vec{b}_1)$

$= \vec{a}.\vec{c} - \vec{a}.\vec{c} - 0 = 0 \quad [\because \vec{a}.\vec{b}_1 = 0]$

and $\vec{b}_1.\vec{c}_2 = \vec{b}_1.\left(\vec{c} - \frac{\vec{c}.\vec{a}}{|\vec{a}|^2}\vec{a} - \frac{\vec{c}.\vec{b}_1}{|\vec{b}_1|^2}\vec{b}_1\right)$

$= \vec{b}_1.\vec{c} - \frac{(\vec{c}.\vec{a})(\vec{b}_1.\vec{a})}{|\vec{a}|^2} - \frac{\vec{c}.\vec{b}_1}{|\vec{b}_1|^2}\vec{b}_1.\vec{b}_1$

$= \vec{b}_1.\vec{c} - 0 - b_1.\vec{c} = 0 \quad$ (Using $\vec{b}_1.\vec{a} = 0$ )

Hence $\vec{a}.\vec{b}_1 = \vec{a}.\vec{c}_2 = \vec{b}_1.\vec{c}_2 = 0$ .

$\Rightarrow (\vec{a}, \vec{b}_1, \vec{c}_2)$ is a set of orthogonal vectors.

**12.** **(d)** Let $x\hat{i} + y\hat{j} + z\hat{k}$ be the required unit vector.

Since $\hat{a}$ is perpendicular to $(2\hat{i} - \hat{j} + 2\hat{k})$ .

$\therefore \quad 2x - y + 2z = 0$ ......... (i)

Since vector $x\hat{i} + y\hat{j} + z\hat{k}$ is coplanar with the vector $\hat{i} + \hat{j} - \hat{k}$ and $2\hat{i} + 2\hat{j} - \hat{k}$ .

$\therefore \quad x\hat{i} + y\hat{j} + z\hat{k}$

$= p\,(\hat{i} + \hat{j} - \hat{k}) + q\,(2\hat{i} + 2\hat{j} - \hat{k})$, where $p$ and $q$ are some scalars.

$\Rightarrow \quad x\hat{i}+y\hat{j}+z\hat{k}$

$= (p+2q)\hat{i}+(p+2q)\hat{j}-(p+q)\hat{k}$

$\Rightarrow \quad x = p + 2q, y = p + 2q, z = -p - q$

Now from equation (i),

$2p + 4q - p - 2q - 2p - 2q = 0$

$\Rightarrow \quad -p = 0 \Rightarrow p = 0$

$\therefore \quad x = 2q, y = 2q, z = -q.$

Since vector $x\hat{i}+y\hat{j}+z\hat{k}$ is a unit vector, therefore

$|x\hat{i}+y\hat{j}+z\hat{k}| = 1 \Rightarrow \sqrt{x^2+y^2+z^2} = 1$

$\Rightarrow \quad x^2 + y^2 + z^2 = 1 \Rightarrow 4q^2 + 4q^2 + q^2 = 1$

$\Rightarrow \quad 9q^2 = 1 \Rightarrow q = \pm\frac{1}{3}$

When $q = \frac{1}{3}$, then $x = \frac{2}{3}, y = \frac{2}{3}, z = -\frac{1}{3}$.

When $q = -\frac{1}{3}$, then $x = -\frac{2}{3}, y = -\frac{2}{3}, z = \frac{1}{3}$.

Here required unit vector is $\frac{2}{3}\hat{i}+\frac{2}{3}\hat{j}-\frac{1}{3}\hat{k}$

or $-\frac{2}{3}\hat{i}-\frac{2}{3}\hat{j}+\frac{1}{3}\hat{k}$.

**13.** **(b)** Let $A$ be the first term and $D$ be the common difference of the corresponding $AP$. Then,

$\frac{1}{a} = A+(p-1)D, \frac{1}{b} = A+(q-1)D,$

$\frac{1}{c} = A+(r-1)D$

$\Rightarrow a^{-1}(q-r)+b^{-1}(r-p)+c^{-1}(p-q) = 0$

$\Rightarrow \vec{v}\cdot\vec{u} = 0 \Rightarrow \vec{u} \perp \vec{v}.$

Hence $\vec{u}$ and $\vec{v}$ are orthogonal vectors.

**14.** **(b)** We have $\vec{p}\cdot\vec{q} = 0$

$\Rightarrow (5\vec{a}-3\vec{b})\cdot(-\vec{a}-2\vec{b}) = 0$

$\Rightarrow 6|\vec{b}|^2 - 5|\vec{a}|^2 - 7\vec{a}\cdot\vec{b} = 0$ ...(1)

Also $\vec{r}\cdot\vec{s} = 0$

$\Rightarrow (-4\vec{a}-\vec{b})\ (-\vec{a}+\vec{b}) = 0$

$\Rightarrow 4|\vec{a}|^2 - |\vec{b}|^2 - 3\vec{a}\cdot\vec{b} = 0$ ...(2)

Now $\vec{x} = \frac{1}{3}(\vec{p}+\vec{r}+\vec{s})$

$= \frac{1}{3}(5\vec{a}-3\vec{b}-4\vec{a}-\vec{b}-\vec{a}+\vec{b}) = -\vec{b}$

and $\vec{y} = \frac{1}{5}(\vec{r}+\vec{s}) = \frac{1}{5}(-5\vec{a}) = -\vec{a}$

Angle between $\vec{x}$ and $\vec{y}$, i.e.,

$\cos\theta = \frac{\vec{x}\cdot\vec{y}}{|\vec{x}||\vec{y}|} = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$ ...(3)

From (1) and (2),

$|\vec{a}| = \sqrt{\frac{25}{19}}\sqrt{\vec{a}\cdot\vec{b}}$ and $|\vec{b}| = \sqrt{\frac{43}{19}}\sqrt{\vec{a}\cdot\vec{b}}$

$\therefore\ |\vec{a}||\vec{b}| = \frac{\sqrt{25\times 43}}{19}\cdot\vec{a}\cdot\vec{b}$

$\theta = \cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$

**15.** **(d)** Let $\vec{d}\cdot\vec{a} = (\cos y)[\vec{a}\,\vec{b}\,\vec{c}] = -\vec{d}\cdot(\vec{b}+\vec{c})$

[as $\vec{d}\cdot(\vec{a}+\vec{b}+\vec{c}) = 0$]

$\Rightarrow \cos y = -\frac{\vec{d}\cdot(\vec{b}+\vec{c})}{[\vec{a}\,\vec{b}\,\vec{c}]}.$ ...(1)

Similarly, $\sin x = -\frac{\vec{d}\cdot(\vec{a}+\vec{b})}{[\vec{a}\,\vec{b}\,\vec{c}]}$ ...(2)

$2 = -\frac{\vec{d}\cdot(\vec{a}+\vec{c})}{[\vec{a}\,\vec{b}\,\vec{c}]}.$ ...(3)

Adding above equations, we get

$\sin x + \cos y + 2 = 0$

$\Rightarrow \sin x + \cos y = -2$

$\Rightarrow \sin x = -1, \cos y = -1$

$\Rightarrow x = (4n-1)\pi/2,\ y = (2n+1);\ n \in \mathbf{Z}$

$x = (4n-1)\pi/2,\ y = (2n+1)\pi$.

Since we want minimum value of $x^2 + y^2$,

so $x = -\frac{\pi}{2}, y = \pm\pi \Rightarrow x^2 + y^2 = \frac{5\pi^2}{4}$.

**16.** **(d)** As angle between $\vec{a}$ and $\vec{b}$ is obtuse,

$\vec{a}\cdot\vec{b} < 0$

$\Rightarrow (2\lambda^2\hat{i}+4\lambda\hat{j}+\hat{k})\cdot(7\hat{i}-2\hat{j}+\lambda\hat{k}) < 0$

$\Rightarrow 14\lambda^2 - 8\lambda + \lambda < 0 \Rightarrow \lambda(2\lambda-1) < 0$

$\Rightarrow 0 < \lambda < \frac{1}{2}$ ...(1)

Angle between $\vec{b}$ and $\hat{k}$ is acute and less than $\frac{\pi}{6}$.

$\vec{b}\cdot\hat{k} = |\vec{b}|\cdot|\hat{k}|\cos\theta$

$\Rightarrow \lambda = \sqrt{53+\lambda^2}\,1\cdot\cos\theta$

$\Rightarrow \cos\theta = \frac{\lambda}{\sqrt{53+\lambda^2}}$ $\therefore$ $\theta < \frac{\pi}{6} \Rightarrow \cos\theta > \cos\frac{\pi}{6}$

$\Rightarrow \cos\theta > \frac{\sqrt{3}}{2} \Rightarrow \frac{\lambda}{\sqrt{53+\lambda^2}} > \frac{\sqrt{3}}{2}$

$\Rightarrow 4\lambda^2 - 3(53+\lambda^2) > 0$ $\Rightarrow \lambda^2 > 159$

$\Rightarrow \lambda < -\sqrt{159}$ ...(2)

From Eqs. (1) and (2), $\lambda = \phi$ $\therefore$ Domain of $\lambda$ is null set.

**17.** **(c)** Given,

$\cos\theta = (\vec{a}\times\hat{i})\cdot(\vec{b}\times\hat{i}) + (\vec{a}\times\hat{j})\cdot(\vec{b}\times\hat{j})$

$+(\vec{a}\times\hat{k})\cdot(\vec{b}\times\hat{k})$ ...(1)

Consider,

$(\vec{a}\times\hat{i})\cdot(\vec{b}\times\hat{i}) = [(\vec{a}\times\hat{i})\vec{b}\,\hat{i}] = ((\vec{a}\times\hat{i})\times\vec{b})\cdot\hat{i}$

$((\vec{a}\cdot\vec{b})\hat{i}) - ((\hat{i}\cdot\vec{b})\vec{a})\hat{i} = (a\cdot b)(\hat{i}\cdot\hat{i}) - (\hat{i}\cdot\hat{b})(a\cdot\hat{i})$

$= \vec{a}\cdot\vec{b} - \vec{a}_1\vec{b}_1$.

Similarly, $(\vec{a}\times\hat{j})\cdot(\vec{b}\times\hat{j}) = \vec{a}\cdot\vec{b} - \vec{a}_2\vec{b}_2$

and $(\vec{a}\times\hat{k})(\vec{b}\times\hat{k}) = \vec{a}\cdot\vec{b} - \vec{a}_3\vec{b}_3$

$\therefore$ From Eq. (i), we get

$\cos\theta = 3\vec{a}\cdot\vec{b} - (a_1b_1 + a_2b_2 + a_3b_3)$

$= 3\vec{a}\cdot\vec{b} - \vec{a}\cdot\vec{b}$

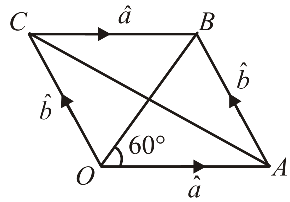
$\Rightarrow \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} = 2\vec{a}\cdot\vec{b} \Rightarrow |\vec{a}||\vec{b}| = \frac{1}{2}$.

Now, use $AM \geq GM$ on $|\vec{a}|, |\vec{b}|$

$\therefore \frac{|\vec{a}|+|\vec{b}|}{2} \geq (|\vec{a}|\cdot|\vec{b}|)^{\frac{1}{2}} \Rightarrow |\vec{a}|+|\vec{b}| \geq \frac{2}{\sqrt{2}}$

$\Rightarrow |\vec{a}|+|\vec{b}| \geq \sqrt{2}$.

**18.** **(3)**




From the figure, $\overrightarrow{OA} = \hat{a}$, $\overrightarrow{OC} = \hat{b}$

$\overrightarrow{OA} + \overrightarrow{OC} = \overrightarrow{OB} = \hat{a} + \hat{b}$.

Now $|\hat{a}+\hat{b}| = 1$ $\therefore$ $\Delta OAB$ is equilateral.

$\therefore$ In $\Delta AOC$, $\cos 120° = \frac{OA^2 + OC^2 - AC^2}{2OA\cdot OC}$

$\therefore -\frac{1}{2} = \frac{1+1-AC^2}{2} \therefore AC = |\vec{a}-\vec{b}| = \sqrt{3} = \sqrt{k}$.

$\Rightarrow$ $k = 3$.

**19.** **(3)** $(\hat{x}+\hat{y}+\hat{z})^2 \geq 0$

$\Rightarrow$ $3 + 2\Sigma\hat{x}.\hat{y} \geq 0$ $\Rightarrow$ $2\Sigma\hat{x}.\hat{y} \geq -3$

Now, $|\hat{x}+\hat{y}|^2 + |\hat{y}+\hat{z}|^2 + |\hat{z}+\hat{x}|^2$

$= 6 + 2\Sigma\hat{x}.\hat{y} \geq 6 + (-3)$

$\Rightarrow$ $|\hat{x}+\hat{y}|^2 + |\hat{y}+\hat{z}|^2 + |\hat{z}+\hat{x}|^2 \geq 3$.

**20.** **(3)** $\vec{F} = \alpha\hat{a} + \beta\hat{b} + \gamma\hat{c}$

$\therefore$ $\vec{F}\cdot(\hat{b}\times\hat{c}) = \alpha[\hat{a}\;\hat{b}\;\hat{c}]$ $\therefore$ $\alpha = \frac{\hat{F}\cdot(\hat{b}\times\hat{c})}{[\hat{a}\hat{b}\hat{c}]}$

Now, $\hat{b}\times\hat{c} = \hat{b}\times(\hat{a}\times\hat{b}) = (\hat{b}\cdot\hat{b})\hat{a} - (\hat{b}\cdot\hat{a})\hat{b}$

$= \hat{a} - \frac{1}{3}\hat{b}$

$[\hat{a}\hat{b}\hat{c}] = (\hat{a}\times\hat{b})^2 = |\hat{a}|^2|\hat{b}|^2 - (\hat{a}\cdot\hat{b})^2$

$= 1 - \frac{1}{9} = \frac{8}{9}$

$\therefore \alpha = \frac{9}{8}\left\{\hat{F}\cdot\hat{a} - \frac{1}{3}\hat{F}\cdot\hat{b}\right\}$ $\therefore k_1 = \frac{9}{8}, k_2 = \frac{3}{8}$

$\Rightarrow k_1 + k_2 = \frac{12}{8} = \frac{3}{2}$ $\therefore 2(k_1+k_2) = 3$.

**21.** **(5)** $\vec{a}.\vec{b} = 0 \Rightarrow x_1 + x_2 + x_3 = 0$

We have to obtain the number of integral solution of this equation $\Rightarrow$ Coefficient of

$x^0$ in $(x^{-3} + x^{-2} + x^{-1} + x^0 + x + x^2)^3$

$= \text{Coeff. of } x^0 \text{ in } \left(\dfrac{1 + x + x^2 + x^3 + x^4 + x^5}{x^3}\right)^3$

$= \text{Coeff. of } x^9 \text{ in } (1 - x^6)^3 (1 - x)^{-3}$

$= {}^{11}C_9 - 3.{}^5C_3 = 25.$

**22.** **(2)** $\overrightarrow{AB} = 2\hat{i} + \hat{j} + \hat{k},\ \overrightarrow{AC} = (t+1)\hat{i} + 0\hat{j} - \hat{k}$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 1 \\ t+1 & 0 & -1 \end{vmatrix}$$

$= -\hat{i} + (t+3)\hat{j} - (t+1)\hat{k}$

$= \sqrt{1 + (t+3)^2 + (t+1)^2} = \sqrt{2t^2 + 8t + 11}\,.$

Area of $\Delta ABC = \dfrac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$

$= \dfrac{1}{2}\sqrt{2t^2 + 8t + 1}$

Let $f(t) = \Delta^2 = \dfrac{1}{4}(2t^2 + 8t + 1)$

$f'(t) = 0 \Rightarrow t = -2$

At $t = -2$, $f''(t) > 0$.

So $\Delta$ is minimum at $t = -2$.

**23.** **(0)** Let $\vec{V_1} = (\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = (\vec{a} \times \vec{b}) \times \vec{p}$ (say)

$= (\vec{a} \cdot \vec{p})\vec{b} - (\vec{b} \cdot \vec{p})\vec{a} = \vec{a} \cdot (\vec{c} \times \vec{d})\vec{b} - \vec{b} \cdot (\vec{c} \times \vec{d})\vec{a}$

$= [\vec{a}\ \vec{c}\ \vec{d}]\vec{b} - [\vec{b}\ \vec{c}\ \vec{d}]\vec{a}$ ...(1)

Similarly $\vec{V_2} = (\vec{a} \times \vec{c}) \times (\vec{d} \times \vec{b}) = \vec{q} \times (\vec{d} \times \vec{b})$

$= [\vec{a}\ \vec{c}\ \vec{b}]\vec{d} - [\vec{a}\ \vec{c}\ \vec{d}]\vec{b}$ ...(2)

And $\vec{V_3} = (\vec{a} \times \vec{d}) \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{d}) \times \vec{r}$

$= [\vec{a}\ \vec{b}\ \vec{c}]\vec{d} - [\vec{d}\ \vec{b}\ \vec{c}]\vec{a}$ ...(3)

$\therefore \vec{A} = \vec{V_1} + \vec{V_2} + \vec{V_3} = -2[\vec{b}\ \vec{c}\ \vec{d}]\vec{a} \neq \vec{0}$ as $\vec{b}, \vec{c}, \vec{d}$ are non-coplanar and which is a vector parallel to $\vec{a}$

$\Rightarrow \vec{A} \times \vec{a} = -2[\vec{b}\ \vec{c}\ \vec{d}](\vec{a} \times \vec{a}) = \vec{0}$

$\Rightarrow |\vec{A} \times \vec{a}| = 0\,.$

**24.** **(2)** $\vec{v}_{n+1} - \vec{v}_n = \left(\begin{bmatrix} 0 & -1/2 \\ 1/2 & 0 \end{bmatrix}\right)^{n+1} \vec{v}_0$

$\vec{v}_2 - \vec{v}_1 = \begin{bmatrix} 0 & -1/2 \\ 1/2 & 0 \end{bmatrix} \vec{v}_0$

$\vec{v}_3 - \vec{v}_2 = \begin{bmatrix} 0 & -1/2 \\ 1/2 & 0 \end{bmatrix}^3 \vec{v}_0$

$\vec{v}_n - \vec{v}_{n-1} = \begin{bmatrix} 0 & -1/2 \\ 1/2 & 0 \end{bmatrix}^n \vec{v}_0$

Adding all the equations,

$\vec{v}_n - \vec{v}_0 = (A + A^2 + A^3 + \ldots\ldots A^n)\vec{v}_0$

where

$A = \begin{bmatrix} 0 & -1/2 \\ 1/2 & 0 \end{bmatrix} \Rightarrow \vec{v}_n = (I + A + A^2 + \ldots.)\vec{v}_0\,.$

**25.** **(7)** Each side subtends an angle of $2\pi/8 = \pi/4$ at the centre of the octagon. Let '$O$' be the centre of the octagon and $r$ the radius of the circum-circle of the octagon. Therefore

$\overrightarrow{OA_1} \times \overrightarrow{OA_2} = \left(r^2 \sin\dfrac{\pi}{4}\right)\vec{n}$

where $\vec{n}$ is the vector perpendicular to the plane of the polygon such that from the side of $\vec{n}$, the points $A_1, A_2, A_3, \ldots., A_n$ are in counterclock sense. Hence

$\overrightarrow{OA_2} \times \overrightarrow{OA_3} = \left(r^2 \sin\dfrac{\pi}{4}\right)\vec{n}$

$\overrightarrow{OA_3} \times \overrightarrow{OA_4} = \left(r^2 \sin\dfrac{\pi}{4}\right)\vec{n},$ ..... etc.

Therefore

$$\sum_{j=1}^{7} (\overrightarrow{OA_j} \times \overrightarrow{OA_{j+1}}) = 7\left(r^2 \sin\frac{\pi}{4}\right)\vec{n}$$

$= 7(\overrightarrow{OA_1} \times \overrightarrow{OA_2})\,.$

CHAPTER 27

# Three Dimensional Geometry

**1.** **(b)** The given equations are $3l + m + 5n = 0$ ...(i)

and $6mn - 2nl + 5lm = 0$ ...(ii)

From (i), we have $m = -3l - 5n$. Putting $m = -3l - 5n$ in (ii), we get

$$6(-3l - 5n)n - 2nl + 5l(-3l - 5n) = 0$$

$$\Rightarrow (n + l)(2n + l) = 0$$

$\Rightarrow$ either $l = -n$ or $l = -2n$.

If $l = -n$, then putting $l = -n$ in (i), we obtain $m = -2n$.

If $l = -2n$, then putting $l = -2n$ in (i), we obtain $m = n$.

Thus, the direction ratios of two lines are $-n, -2n, n$ and $-2n, n, n$ i.e., $1, 2, -1$ and $-2, 1, 1$. Hence, the direction cosines are

$\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}$ or $\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}$. The angle $\theta$ between the lines is given by

$$\cos\theta = \frac{1}{\sqrt{6}} \times \frac{-2}{\sqrt{6}} + \frac{2}{\sqrt{6}} \times \frac{1}{\sqrt{6}} + \frac{-1}{\sqrt{6}} \times \frac{1}{\sqrt{6}} = \frac{-1}{6}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{-1}{6}\right).$$

**2.** **(c)** It makes $\theta$ with x and y-axes.

$l = \cos\theta$, $m = \cos\theta$, $n = \cos(\pi - 2\theta)$

we have $l^2 + m^2 + n^2 = 1$

$\Rightarrow \cos^2\theta + \cos^2\theta + \cos^2(\pi - 2\theta) = 1$

$\Rightarrow 2\cos^2\theta + (-\cos 2\theta)^2 = 1$

$\Rightarrow 2\cos^2\theta - 1 + \cos^2 2\theta = 0$

$\Rightarrow \cos 2\theta - [1 + \cos 2\theta] = 0$

$\Rightarrow \cos 2\theta = 0$ or $\cos 2\theta = -1$

$\Rightarrow 2\theta = \pi/2$ or $2\theta = \pi$

$\Rightarrow \theta = \pi/4$ or $\theta = \frac{\pi}{2} \Rightarrow \theta = \left[\frac{\pi}{4}, \frac{\pi}{2}\right]$

**3.** **(b)** Let $OA$ and $OB$ be two lines with $DC$'s $l_1, m_1, n_1$ and $l_2, m_2, n_2$. Let $OA = OB = 1$. Then co-ordinates of $A$ and $B$ are $(l_1, m_1, n_1)$ and $(l_2, m_2, n_2)$ respectively. Let $OC\,Z$ be the bisector of $\angle AOB$ such that $C$ is the mid-point of $AB$ and so its co-ordinates are

$$\left(\frac{l_1 + l_2}{2}, \frac{m_1 + m_2}{2}, \frac{n_1 + n_2}{2}\right)$$

$\therefore$ $DR$'s of $OC$ are $\frac{l_1 + l_2}{2}, \frac{m_1 + m_2}{2}, \frac{n_1 + n_2}{2}$

$\therefore$ We have

$$OC = \sqrt{\left(\frac{l_1 + l_2}{2}\right)^2 + \left(\frac{m_1 + m_2}{2}\right)^2 + \left(\frac{n_1 + n_2}{2}\right)^2}$$

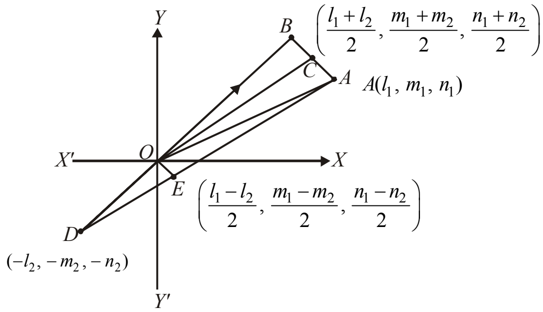
$$= \frac{1}{2}\sqrt{(l_1^2 + m_1^2 + n_1^2) + (l_2^2 + m_2^2 + n_2^2) + 2(l_1 l_2 + m_1 m_2 + n_1 n_2)}$$

$$= \frac{1}{2}\sqrt{2 + 2\cos\theta} \quad [\because \cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2]$$

$$= \frac{1}{2}\sqrt{2(1 + \cos\theta)} = \cos\left(\frac{\theta}{2}\right).$$

$\therefore$ $DC$s of $\overline{OC}$ are $\frac{l_1 + l_2}{2(OC)}, \frac{m_1 + m_2}{2(OC)}, \frac{n_1 + n_2}{2(OC)}$

i.e., $\frac{l_1 + l_2}{2\cos\theta/2}, \frac{m_1 + m_2}{2\cos\theta/2}, \frac{n_1 + n_2}{2\cos\theta/2}$




**4.** **(d)** The given lines are

$$x-1=\frac{y+3}{-\lambda}=\frac{z-1}{\lambda}=s \quad \text{..........(i)}$$

and $2x=y-1=\dfrac{z-2}{-1}=t$ .........(ii)

The lines are coplanar, if

$$\begin{vmatrix} 0-(-1) & -1-3 & -2-(-1) \\ 1 & -\lambda & \lambda \\ \frac{1}{2} & 1 & -1 \end{vmatrix}=0$$

$$C_2 \to C_2+C_3; \begin{vmatrix} 1 & -5 & -1 \\ 1 & 0 & \lambda \\ \frac{1}{2} & 0 & -1 \end{vmatrix}=0$$

$$\Rightarrow 5(-1-\frac{\lambda}{2})=0 \Rightarrow \lambda=-2$$

**5.** **(a)** Both the lines pass through origin.

Line $L_1$ is parallel to the vector

$$\vec{V}_1=(\cos\theta+\sqrt{3})\hat{\mathbf{i}}+(\sqrt{2}\sin\theta)\hat{\mathbf{j}}+(\cos\theta-\sqrt{3})\hat{\mathbf{k}}$$

and $L_2$ is parallel to the vector

$$\vec{V}_2=a\hat{\mathbf{i}}+b\hat{\mathbf{j}}+c\hat{\mathbf{k}}$$

$$\therefore \cos\alpha=\frac{\vec{V}_1\cdot\vec{V}_2}{|\vec{V}_1||\vec{V}_2|}$$

$$=\frac{a(\cos\theta+\sqrt{3})+(b\sqrt{2})\sin\theta+c(\cos\theta-\sqrt{3})}{\sqrt{a^2+b^2+c^2}\sqrt{(\cos\theta+\sqrt{3})^2+2\sin^2\theta+(\cos\theta-\sqrt{3})^2}}$$

$$=\frac{(a+c)\cos\theta+b\sqrt{2}\sin\theta+(a-c)\sqrt{3}}{\sqrt{a^2+b^2+c^2}\sqrt{2+6}}$$

In order that $\cos\alpha$ in independent of $\theta$

$a+c=0$ and $b=0$

$$\therefore \cos\alpha=\frac{2a\sqrt{3}}{a\sqrt{2}\cdot 2\sqrt{2}}=\frac{\sqrt{3}}{2} \Rightarrow \alpha=\frac{\pi}{6}$$

**6.** **(c)** Given one vertex $A(7, 2, 4)$ and line

$$\frac{x+6}{5}=\frac{y+10}{3}=\frac{z+14}{8}$$

General point on above line,

$B \equiv (5\lambda-6, 3\lambda-10, 8\lambda-14)$

Direction ratios of line $AB$ are

$< 5\lambda-13, 3\lambda-12, 8\lambda-18 >$

Direction ratios of line $BC$ are $<5, 3, 8>$

Since, angle between $AB$ and $BC$ is $\dfrac{\pi}{4}$.

$$\cos\frac{\pi}{4}=\frac{(5\lambda-3)5+3(3\lambda-12)+8(8\lambda-18)}{\sqrt{5^2+3^2+8^2}\cdot\sqrt{(5\lambda-13)^2+(3\lambda-12)^2+(8\lambda-18)^2}}$$

Squaring and solving, we have $\lambda=3, 2$.

Hence, equation of lines are

$$\frac{x-7}{2}=\frac{y-2}{-3}=\frac{z-4}{6} \text{ and}$$

$$\frac{x-7}{3}=\frac{y-2}{6}=\frac{z-4}{2}.$$

**7.** **(d)** $\Delta = \frac{1}{2}\left|(\hat{j}+\lambda\hat{k})\times(\hat{i}+\lambda\hat{k})\right|$

$= \frac{1}{2}\left|-\hat{k}+\lambda\hat{i}+\lambda\hat{j}\right| = \frac{1}{2}\sqrt{2\lambda^2+1}$

$\Rightarrow \frac{9}{4} \le \frac{1}{4}(2\lambda^2+1) \le \frac{33}{4}$

$\Rightarrow 4 \le \lambda^2 \le 16 \Rightarrow 2 \le |\lambda| \le 4.$

**8.** **(c)** The line has +ve and equal direction cosines, these are $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ or direction ratios are 1, 1, 1. Also the lines passes through $P$ $(2, -1, 2)$.

$\therefore$ Equation of line is

$$\frac{x-2}{1} = \frac{y+1}{1} = \frac{z-2}{1} = \lambda \text{ (say)}$$

Let $Q(\lambda+2, \lambda-1, \lambda+2)$ be a point on this line where it meets the plane

$$2x+y+z=9$$

Then $Q$ must satisfy the eq$^n$ of plane

i.e. $2(\lambda+2)+\lambda-1+\lambda+2=9 \Rightarrow \lambda=1$

$\therefore$ $Q$ has coordinates (3, 0, 3)

Hence the length of line segment

$= \sqrt{(2-3)^2+(-1-0)^2+(2-3)^2} = \sqrt{3}$

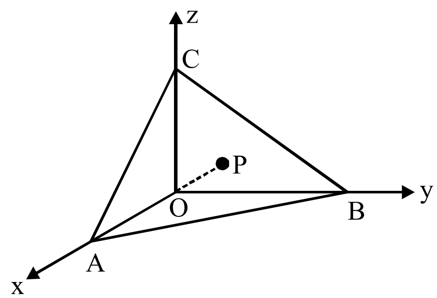
**9.** **(d)** Here $OP = \sqrt{h^2+k^2+l^2} = p$

$\therefore$ $DR$s of $OP$ are :

$$\frac{h}{\sqrt{h^2+k^2+l^2}}, \frac{k}{\sqrt{h^2+k^2+l^2}}, \frac{l}{\sqrt{h^2+k^2+l^2}}$$

or $\frac{h}{p}, \frac{k}{p}, \frac{l}{p}$

Since $OP$ is normal to the plane, therefore, equation of plane is




$$\frac{h}{p}x+\frac{k}{p}y+\frac{l}{p}z = p \quad \text{or} \quad hx+ky+lz = p^2$$

$$\therefore A\left(\frac{p^2}{h}, 0, 0\right), B\left(0, \frac{p^2}{k}, 0\right), C\left(0, 0, \frac{p^2}{l}\right)$$

Now, Area of $\Delta ABC$, $\Delta = \sqrt{A_{xy}^2+A_{yz}^2+A_{zx}^2}$

where, $A_{xy}^2$ is area of projection of $\Delta ABC$ on xy plane = area of $\Delta AOB$

Now, $A_{xy} = \frac{1}{2}\begin{vmatrix} p^2/h & 0 & 1 \\ 0 & p^2/k & 1 \\ 0 & 0 & 1 \end{vmatrix} = \frac{p^4}{2|hk|}$

Similarly, $A_{yz} = \frac{p^4}{2|kl|}$ and $A_{zx} = \frac{p^4}{2|lh|}$

$\therefore \Delta^2 = A_{xy}^2+A_{yz}^2+A_{zx}^2$, $\Delta = \frac{p^5}{2hkl}$

**10.** **(b)** A plane containing the given lines is

$2x+3y+4z-6+\lambda(x+y+z-3)=0$ … (i)

This plane is perpendicular to plane $z=0$

if $4+\lambda=0 \Rightarrow \lambda=-4$

So, the equation (i) becomes

$-2x-y+6=0 \Rightarrow 2x+y-6=0$ … (ii)

Equation of the projection will be the line of intersection of plane (2) and the plane z = 0. If the line has d.c. proportional to $\ell$, $m$, $n$ then $2\ell + m = 0$ and $n = 0$

$\Rightarrow \ell : m : n = 1 : -2 : 0$. Obviously (0, 6, 0) is a point on both the planes, hence lies on the line as well.

$\therefore$ Equation of the line is $\frac{x}{1} = \frac{y-6}{-2} = \frac{z}{0}$

**11. (d)** Let the eq$^n$ of variable plane be

$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ which meets the axes at

$A(a, 0, 0)$, $B(0, b, 0)$ and $C(0. 0, c)$.

$\therefore$ Centroid of $\Delta ABC$ is $\left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right)$

and it satisfies the relation

$$\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = k \Rightarrow \frac{9}{a^2} + \frac{9}{b^2} + \frac{9}{c^2} = k$$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{z^2} = \frac{k}{9} \quad \text{...(i)}$$

Also given that the distance of plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ from (0, 0, 0) is 1 unit.

$$\Rightarrow \frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}} = 1$$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = 1 \quad \text{...(ii)}$$

From (i) and (ii), we get $\frac{k}{9} = 1$ i.e. $k = 9$

**12. (b)** Equation of the plane containing the line

$\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ is

$a(x - 1) + b(y - 2) + c(z - 3) = 0$ ....... (i)

where $a.1 + b.2 + c.3 = 0$

i.e., $a + 2b + 3c = 0$ ....... (ii)

Since the plane (i) parallel to the line

$$\frac{x}{1} = \frac{y}{1} = \frac{z}{4}$$

$\therefore$ $a.1 + b.1 + c.4 = 0$

i.e., $a + b + 4c = 0$ ....... (iii)

From (ii) and (iii),

$$\frac{a}{8-3} = \frac{b}{3-4} = \frac{c}{1-2} = k \text{ (let)}$$

$\therefore$ $a = 5k$, $b = -k$, $c = -k$

On putting the value of $a$, $b$ and $c$ in equation (i),

$5(x - 1) - (y - 2) - (z - 3) = 0$

$\Rightarrow$ $5x - y - z = 0$ ....... (iv)

when $x = 1$, $y = 0$ and $z = 5$; then

L.H.S. of equation (iv) $= 5x - y - 2$

$= 5 \times 1 - 0 - 5$

$= 0$

= R.H.S. of equation (iv)

Hence coordinates of the point (1, 0, 5) satisfy the equation plane represented by equations (iv), Therefore the plane passes through the point (1,0,5).

**13. (c)** The position vectors of two given points are $a = \hat{i} - \hat{j} + 3\hat{k}$ and $b = 3\hat{i} + 3\hat{j} + 3\hat{k}$ and the equation of the given plane is

$r = (5\hat{i} + 2\hat{j} - 7\hat{k}) + 9 = 0$ or $r \cdot n + d = 0$

We have

$$a \cdot n + d = (\hat{i} - \hat{j} + 3\hat{k}) \cdot (5\hat{i} + 2\hat{j} - 7\hat{k}) + 9$$

$$= 5 - 2 - 21 + 9 < 0$$

and

$$b \cdot n + d = (3\hat{i} + 3\hat{j} + 3\hat{k}) \cdot (5\hat{i} + 2\hat{j} - 7\hat{k}) + 9$$

$$= 15 + 6 - 21 + 9 > 0$$

So, the points $a$ and $b$ are on the opposite sides of the plane.

**14. (c)** $2x^2 - 2y^2 + 4z^2 + 6xz + 2yz + 3xy = 0$

or $2x^2 + x(6z + 3y) - 2y^2 + 4z^2 + 2yz = 0$

$$\therefore x = \frac{-(6z+3y) \pm \sqrt{36z^2+9y^2+36yz-8(-2y^2+4z^2+2yz)}}{4}$$

$$\therefore x = \frac{-(6z+3y) \pm \sqrt{(2x+5y)^2}}{4}$$

$$\therefore x = \frac{-(6z+3y) \pm (2z+5y)}{4}$$

or $2x - y + 2z = 0,\ x + 2y + 2z = 0$

$\therefore$ Angle between planes

$$\theta = \cos^{-1}\frac{(2)(1)+(-1)(2)+(2)(2)}{\sqrt{(2)^2+(-1)^2+(2)^2}\sqrt{(1)^2+(2)^2+(2)^2}}$$

$$= \cos^{-1}\left(\frac{4}{9}\right)$$

**15. (b)** $PR : PQ = 1:3 \Rightarrow 3PR = PQ$

$$\underset{(-3,\,1,\,1)}{P} \overset{1}{\text{———}} R \overset{2}{\text{———}} \underset{(3,\,4,\,2)}{Q}$$

$\Rightarrow 3PR = PR + RQ \Rightarrow 2PR = RQ$

Therefore, $PR : RQ = 1 : 2$. Hence

$$R = \left(\frac{-6+3}{1+2}, \frac{2+4}{3}, \frac{2+2}{3}\right) = \left(-1, 2, \frac{4}{3}\right)$$

The normal to the required plane is

$\overline{PQ} = (6, 3, 1)$. Hence, the equation of the required plane is

$$6(x+1)+3(y-2)+1\left(z-\frac{4}{3}\right) = 0$$

$$\Rightarrow 18x+9y+3z-4=0$$

**16. (c)** $\sigma_1$ is perpendicular to $(\hat{i}+\hat{j}+\hat{k})$

$\sigma_2$ is perpendicular to $(a\hat{i}+b\hat{j}+c\hat{k})$

and $\sigma_3$ is perpendicular to

$(a^2\hat{i}+b^2\hat{j}+c^2\hat{k})$

Then, the planes are

$\sigma_1 : x+y+z=0$

$\sigma_2 : ax+by+cz=0$

$\sigma_3 : a^2x+b^2y+c^2z=0$

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$$

So, for unique solution, $\Delta \neq 0$

$\Rightarrow \Delta = (a-b)(b-c)(c-a) \neq 0$

$\Rightarrow a \neq b, b \neq c, c \neq a$

**17. (a)** The equation of plane is $\vec{r}.\vec{a} = 5$

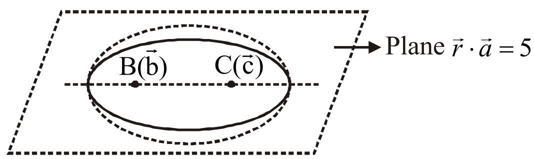
$\because |\vec{r}-\vec{b}| + |\vec{r}-\vec{c}| = 4$

$\Rightarrow$ sum of distances of a point $(\vec{r})$ from two fixed points with position vector $\vec{b}$ and $\vec{c}$ is constant.

$\Rightarrow$ such points lies on ellipsoid.

Now points with position vector $\vec{b}$ and $\vec{c}$ satisfies the equation of plane $\vec{r}.\vec{a} = 5$, then

$\vec{b}.\vec{a} = 5$ and $\vec{c}.\vec{a} = 5$




Area in the plane constitutes an ellipse

Distance between $\vec{b}$ and $\vec{c}$

$= 2 \times$ (semi major axis) $\times e = \sqrt{14}$

$2ae = \sqrt{14}$ ...(i)

Sum of distance = constant = major axis = 4

$2a = 4$ ...(ii)

From eq$^n$ (i) and (ii)

$e = \frac{\sqrt{14}}{4} \Rightarrow b = a\sqrt{(1-e^2)} = \frac{1}{\sqrt{2}}$ (semi minor axis)

Area of ellipse $= \pi.a.b.$

$$= \pi.2.\frac{1}{\sqrt{2}} = \sqrt{2}\pi \approx 4.443$$

**18.** **(3)** $L_1 : \frac{x-0}{1} = \frac{y+1}{1} = \frac{z-0}{1} = \lambda$

$$L_2 : \frac{x+1}{2} = \frac{y-0}{1} = \frac{z-0}{1} = \mu$$

Hence any point on $L_1$ and $L_2$ can be

$(\lambda, \lambda-1, \lambda)$ and $(2\mu-1, \mu, \mu)$, respectively.

According to the question,

$$\frac{2\mu-1-\lambda}{2} = \frac{\mu-\lambda+1}{1} = \frac{\mu-\lambda}{2}$$

On solving, we get $\mu = 1$ and $\lambda = 3$

$\therefore A = (3, 2, 3)$ $B = (1, 1, 1)$ $\therefore AB = 3.$

**19.** **(2)** The equation of any plane passing through given line is :

$$(x+y+2z-3)+\lambda(2x+3y+4z-4) =$$

$$\Rightarrow (1+2\lambda)x+(1+3\lambda)y+(2+4\lambda)z-(3+4\lambda) = 0 \quad \text{....(i)}$$

If this plane is parallel to z-axis whose direction cosines are 0, 0, 1; then the normal to the plane will be perpendicular to z-axis

$$\therefore (1+2\lambda)(0)+(1+3\lambda)(0)+(2+4\lambda)(1) = 0$$

$$\Rightarrow \lambda = -\frac{1}{2}$$

Put in eq (i), the required plane is

$$(x+y+2z-3)-\frac{1}{2}(2x+3y+4z-4) = 0$$

$$\Rightarrow y+2 = 0 \quad \text{...(ii)}$$

$\therefore$ S.D. = distance of any point say (0, 0, 0) on

z-axis from plane (ii) $= \frac{2}{\sqrt{(1)^2}} = 2$

**20.** **(7)**

**21.** **(7)** A plane containing line of intersection of the given planes is

$$x-y-z-4+\lambda(x+y+2z-4) = 0$$

i.e., $(\lambda+1)x+(\lambda-1)y+(2\lambda-1)z-4(\lambda+1) = 0$

vector normal to it

$$V = (\lambda+1)\hat{i}+(\lambda-1)\hat{j}+(2\lambda-1)\hat{k}$$

Now the vector along the line of intersection of the planes

$$2x+3y+z-1 = 0$$

and $x+3y+2z-2 = 0$ is given by

$$n = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 1 \\ 1 & 3 & 2 \end{vmatrix} = 3(\hat{i}-\hat{j}+\hat{k})$$

As $n$ is parallel to the plane (i), therefore,

$$n \cdot V = 0$$

$$(\lambda+1)-(\lambda-1)+(2\lambda-1) = 0$$

$$\Rightarrow 2+2\lambda-1 = 0 \Rightarrow \lambda = \frac{-1}{2}$$

Hence, the required plane is

$$\frac{x}{2}-\frac{3y}{2}-2z-2 = 0$$

$$\Rightarrow x-3y-4z-4 = 0$$

Hence, $|A+B+C-4| = 7$

**22.** **(7)** Under the given conditions the possible situation is $f(-2) = 2; f(0) = 3; f(1) = 1.$

{where $f(-2) = 1$ is false, $f(0) \neq 2$ is true and $f(1) \neq 1$ is false}. The triangle formed is with vertices

$A(-2, 1, 0), B(2, 1, 3)$ and $O(0, 0, 0)$

$$\text{Area of } \Delta AOB = \frac{1}{2}\left\| \begin{matrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 3 \\ -2 & 1 & 0 \end{matrix} \right\|$$

$= \frac{1}{2}|-3\hat{i} - 6\hat{j} + 4\hat{k}| = \frac{1}{2}\sqrt{61}$ square units

$= \frac{\sqrt{k}}{2}$; so $k = 61$.

**23.** **(6)** The required plane is of the form

$(x+2y+3z-4)+\lambda(2x+y-z+5)=0$

whose normal is $(1+2\lambda,\ 2+\lambda,\ 3-\lambda)$. This plane is perpendicular to the plane $5x+3y+6z+8=0$. So we have

$5(1+2\lambda)+3(2+\lambda)+6(3-\lambda)=0$

$\Rightarrow 7\lambda = -29 \Rightarrow \lambda = \frac{-29}{7}$

Therefore, the required plane is

$(x+2y+3z-4)-\frac{29}{7}(2x+y-z+5)=0$

$\Rightarrow 51x+15y-50z+173=0$

Comparing this with $ax+by+cz+173=0$ we get $a = 51$, $b = 15$, $c = -50$

so that, $b-9(a+c)=15-9=6$.

**24.** **(4)** Any point on parabola $z^2 = 4\alpha x$, $y = 0$ is given by $Q(\alpha t^2, 0, 2\alpha t)$

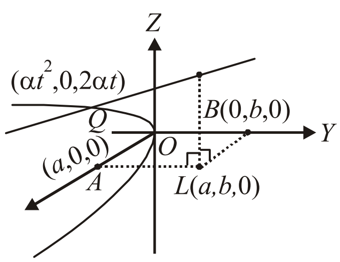
Now equation of line joining $P(a, b, c)$ and $Q$ $(\alpha t^2, 0, 2\alpha t)$ is given by

$\frac{x-a}{a-\alpha t^2} = \frac{y-b}{b-0} = \frac{z-c}{c-2\alpha t} = \lambda$ (say)

$\Rightarrow x = a+\lambda(a-\alpha t^2)$; $y = b+b\lambda$; $z = c+\lambda$;

$z = c+\lambda(c-\alpha t)$ by given condition point $Q$

lies on $(bz-cy)^2 = k\alpha(b-y)(bx-ay)$




$\Rightarrow [bc+b\lambda,\ (c-2\alpha t)-cb-cb\lambda]^2$

$= k\alpha(-b\lambda)(ba+b\lambda)(a-\alpha t^2)-(ab-ab\lambda)$

$\Rightarrow [bc+b\lambda c-2\alpha b\lambda t-cb-cb\lambda)^2$

$= -kb\alpha\lambda(ba+ab\lambda-ab\lambda t^2-ab-ab\lambda)$

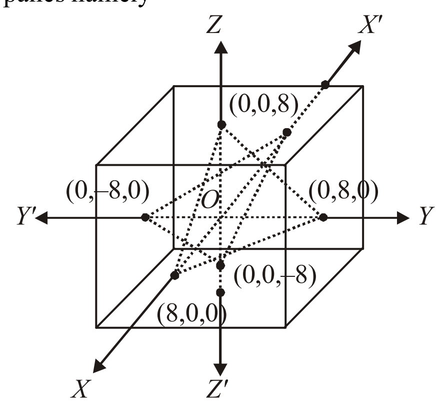
$\Rightarrow 4\alpha^2b^2\lambda^2t^2 = -kb\alpha\lambda(-\alpha b\lambda t^2)$

$\Rightarrow 4\alpha^2b^2\lambda^2t^2 = k^2b^2\alpha^2\lambda^2t^2 \Rightarrow k = 4.$

**25.** **(4)** $|x| \le 8 \Rightarrow x \in [-8, 8]$ similarly for $y$ and $z$.

This represent a cube of side 16 units with centre at origin.

Now, $-8 \le x+y+z \le 8$ gives space between two panes namely




$x+y+z+8=0$ and $x+y+z-8=0$ subject to the limits of $x,\ y,\ z \in [-8, 8]$

This will take out $\left(\frac{1}{4}+\frac{1}{4}\right)$ volume of the cube

$\Rightarrow$ The required volume $= \frac{1}{2}\times(16)^3 = 2018.$

$\Rightarrow t = 2048 \Rightarrow \frac{t}{512} = 4$

CHAPTER 28

# Probability-2

**1.** **(c)** Total number of functions from $A$ to $B$ is $(20)^{10}$.

The number of non-decreasing function is the number of non-negative integral solutions of the equation

$$x_1 + x_2 + \ldots + x_{20} = 10 = {}^{29}C_{10}$$

(Here $x_i$ is the pre-image of number i, i = 1, 2, 3, ....,20)

$$\Rightarrow \text{ the probability } = \frac{{}^{29}C_{19}}{(20)^{10}}$$

**2.** **(c)** When two dice is thrown then sample space has 6 × 6 = 36 elements so n(S) = 36.

Now consider the event of getting 9 is (3, 6), (4, 5), (5, 4) and (6, 3).

So probability of getting 9 when two dice is thrown is 4/36 = 1/9.

If Sanchita starts the game then the probability that she wins is

$$\left(\frac{1}{9}\right)+\left(\frac{8}{9}\right)\left(\frac{8}{9}\right)\left(\frac{1}{9}\right) +\ldots.. \infty =$$

$$\frac{\frac{1}{9}}{1-\frac{64}{81}} = \frac{1}{9}\times\frac{81}{17} = \frac{9}{17}$$

And if Raj starts the game then probability that Sanchita wins the game is 1 – 9/17 = 8/17

**3.** **(d)** As usual H denotes head and T denotes tail so that

$$P(H) = \frac{1}{2} = P(T)$$

Let $E_n$ denote the event that the score is n. One can easily see that

$E_n = (E_{n-2} \cap H) \cup (E_{n-1} \cap T)$

Therefore

$P_n = P(E_n)$

$= P(E_{n-2} \cap H) + P(E_{n-1} \cap T)$

$= P(E_{n-2})\, P(H) + P(E_{n-1}) P(T)$

$$= \frac{1}{2}(p_{n-2} + p_{n-1})$$

Now

$$p_n + \frac{1}{2}p_{n-1} = p_{n-1} + \frac{1}{2}p_{n-2}$$

$$= p_{n-2} + \frac{1}{2}p_{n-3}$$

$$= p_{n-3} + \frac{1}{2}p_{n-4}$$

................................

................................

$$= p_2 + \frac{1}{2}p_1$$

Since

$$p_1 = P(T) = \frac{1}{2}$$

and $p_2 = P((T \cap T) \cup H)$

$$= P(T)P(T) + P(H)$$

$$= \frac{1}{2}\times\frac{1}{2}+\frac{1}{2} = \frac{3}{4}$$

We have

$$p_n + \frac{1}{2}p_{n-1} = p_2 + \frac{1}{2}p_1 = \frac{3}{4}+\frac{1}{4} = 1$$

Therefore

$$p_n - \frac{2}{3} = \frac{1}{3} - \frac{1}{2}p_{n-1}$$

$$= -\frac{1}{2}\left(p_{n-1} - \frac{2}{3}\right) = \left(-\frac{1}{2}\right)^2\left(p_{n-2} - \frac{2}{3}\right)$$

$$= \left(-\frac{1}{2}\right)^3\left(p_{n-3} - \frac{2}{3}\right).....$$

$$= \left(\frac{-1}{2}\right)^{n-1}\left(p_1 - \frac{2}{3}\right)$$

$$= \left(-\frac{1}{2}\right)^{n-1}\left(\frac{1}{2}-\frac{2}{3}\right)$$

Hence

$$p_n = \frac{2}{3} + \left(-\frac{1}{2}\right)^{n-1}\left(-\frac{1}{6}\right) = \frac{2}{3} + \frac{1}{3}\left(-\frac{1}{2}\right)^n$$

$$= \frac{2^{n+1} + (-1)^n}{3.2^n}$$

**4.** **(c)** The number of determinants formed = 16. Observe that the determinant is non-zero when exactly once (–1) appears as shown

$$\begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} = 2 \Rightarrow 4 \text{ ways}$$

Similarly the determinant is non-zero when (–1) is used exactly three times as shown

$$\begin{vmatrix} -1 & -1 \\ -1 & 1 \end{vmatrix} = -2 \Rightarrow 4 \text{ ways.}$$

So non-zero determinant can be obtained in 8 ways. Similarly determinant will be zero in 8 determinants.

$$\Rightarrow \quad P(E) = \frac{1}{2}$$

**5.** **(b)** Let $E_1$, $E_2$ and $E_3$ be the events of the critics giving favourable remarks. Then

$$P(E_1) = \frac{5}{7},\ P(E_2) = \frac{4}{7} \text{ and } P(E_3) = \frac{3}{7}$$

Let E be the event that majority reviewed favourably. Therefore

$$E = (E_1 \cap E_2 \cap \bar{E}_3) \cup (\bar{E}_1 \cap E_2 \cap E_3) \cup (E_1 \cap \bar{E}_2 \cap E_3) \cup (E_1 \cap E_2 \cap E_3)$$

Hence P(E)

$$= P(E_1)P(E_2)P(\bar{E}_3) + P(\bar{E}_1)P(E_2)P(E_3) + P(E_1)P(\bar{E}_2)P(E_3) + P(E_1)P(E_2)P(E_3)$$

$$= \frac{5}{7}\times\frac{4}{7}\times\left(1-\frac{3}{7}\right) + \left(1-\frac{5}{7}\right)\times\frac{4}{7}\times\frac{3}{7} + \frac{5}{7} \times\left(1-\frac{4}{7}\right)\times\frac{3}{7} + \frac{5}{7}\times\frac{4}{7}\times\frac{3}{7}$$

$$= \frac{5}{7}\times\frac{4}{7}\times\frac{4}{7} + \frac{2}{7}\times\frac{4}{7}\times\frac{3}{7} + \frac{5}{7}\times\frac{3}{7}\times\frac{3}{7} + \frac{5}{7}\times\frac{4}{7}\times\frac{3}{7}$$

$$= \frac{80+24+45+60}{7\times7\times7} = \frac{209}{343}$$

**6.** **(d)**

**7.** **(b)** The graph of $y = 16x^2 + 8(a+5)x - 7a - 5$ is strictly above the $x$-axis

$\Rightarrow y > 0 \ \forall\ x \in \boldsymbol{R}$

$\Rightarrow 16x^2 + 8(a+5)x - 7a - 5 > 0 \ \forall\ x \in \boldsymbol{R}$

The above inequality holds if discriminant < 0

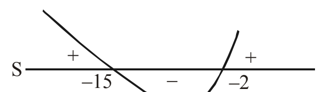
[$\because$ coefficient of $x^2 > 0$ ]

$\Rightarrow \ 64(a+5)^2 - 4.16(-7a-5) < 0$

$\Rightarrow \ a^2 + 17a + 30 < 0$

$\Rightarrow \ (a+2)(a+15) < 0$

$\Rightarrow \ -15 < a < -2$




Given $-20 \le a \le 0$ and favourable cases $-15 < a < -2$

$\therefore$ Required probability

$$= \frac{\text{length of interval } (-15, -2)}{\text{length of interval } (-20, 0)} = \frac{-2-(-15)}{0-(-20)} = \frac{13}{20}$$

**8.** **(a)** Let G, C, K, and A are events where G = Guess answer, C = Copy answer, K = Know the answer, and A = Answer correctly

P(G) = Probability that the Candidate Guess the answer = 1/3 (given)

P(C) = Probability that the Candidate Copy the answer = 1/6 (given)

P(K) = Probability that the Candidate Know the answer = ?

Now, G,C and K are mutually exclusive and exhaustive events

Therefore, P(G) + P(C) + P(K) = 1

P(K) = 1 – 1/3 – 1/6 = 1/2.

Say G(Candidate guesses) has occurred. As, there are four choices out of which only one is correct, then the probability that the candidate made a guess is 1/4.

$P(A/G) = 1/4$

$P(A/C) = 1/8$ ( given) then

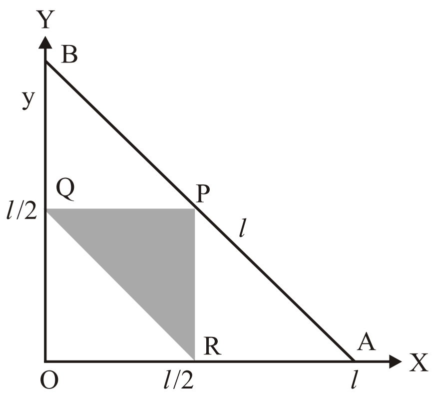
$P(A/K)$ = Answer correctly that the candidate know = 1

Applying Baye's Theorem we have,

$P(K/A) = [P(K).P(A/K)]/[P(G) \times P(A/G) + P(C) \times P(A/C) + P(K) \times P(A/K) = 24/29.$

**9.** **(b)** Let the sides be x, y, $l - (x + y)$

Since the triangle will be formed when sum of two sides is larger than the third.




i.e. $l - y > y, l - x > x$ and $x + y > l - (x + y)$

$$\Rightarrow \quad 0 < y < \frac{1}{2}, 0 < x < \frac{1}{2} \text{ and } \frac{1}{2} < (x + y) < 1$$

Hence required probability $= \dfrac{\Delta PQR}{\Delta OAB} = \dfrac{1}{4}$

(as OQ = 1/2 OB)

**10.** **(b)** Let x and y be the two quantities. When the sum of two non-negative quantities is fixed, the product will be maximum when they are equal.

So, the greatest product = xy = 10000

where x = y = 100

Now, $xy \geq \frac{3}{4} \times 10000$

$$\Rightarrow \quad xy \geq 7500 \Rightarrow x(200 - x) \geq 7500$$

$$\Rightarrow \quad x^2 - 200x + 7500 \leq 0$$

$$\Rightarrow \quad (x - 50)(x - 150) \leq 0 \quad \Rightarrow 50 \leq x \leq 150$$

So, favourable number of ways = 101

Total number of ways = 201

So, required probability $= \dfrac{101}{201}$

Hence, (b) is the correct answer.

**11.** **(c)** Let $E_1$ and $E_2$ be the events of the boy watching DOORDARSHAN and TEN SPORTS, respectively. It is given that

$$P(E_1) = \frac{1}{5} \text{ and } P(E_2) = \frac{4}{5}$$

Let E be the event of the boy falls asleep. Again by hypothesis

$$P(E/E_1) = \frac{3}{4} \text{ and } P(E/E_2) = \frac{1}{4}$$

Now,

$$E = E \cap (E_1 \cup E_2) = (E_1 \cap E) \cup (E_2 \cap E)$$

so that,

$$P(E) = P(E_1)P(E/E_1) + P(E_2)P(E/E_2)$$

By Bayes' theorem

$$P(E_1/E) = \frac{P(E_1)P(E/E_1)}{P(E_1)P(E/E_1) + P(E_2)P(E/E_2)}$$

$$= \frac{(1/5) \times (3/4)}{(1/5) \times (3/4) + (4/5) \times (1/4)} = \frac{3}{7}$$

**12.** **(c)** Here, $P_n = \alpha p^n, n \geq 1$

and $P_0 = 1 - \alpha p(1 + p + p^2 + ...)$

Consider the following events :

$E_j$ = There are j children in the family;

j = 0, 1, 2,....., n

A = There are exactly k boys in the family

We have, $P(E_j) = p_j = \alpha p^j; j = 0, 1, 2, ...., n$

$$\Rightarrow \quad P\left(\frac{A}{E_j}\right) = \frac{{}^jC_k}{2^j}, j \geq k$$

Now, $A = \bigcup_{j=k}^{\infty} (A \cap E_j)$

$$\Rightarrow \quad P(A) = P\left(\bigcup_{j=k}^{\infty} (A \cap E_j)\right)$$

$$\therefore \quad P(A) = \sum_{j=k}^{\infty} P(A \cap E_j) = \sum_{j=k}^{\infty} P(E_j) P\left(\frac{A}{E_j}\right)$$

$$= \sum_{j=k}^{\infty} \alpha p^j \left(\frac{{}^jC_k}{2^j}\right) = \alpha \sum_{j=k}^{\infty} \left(\frac{p}{2}\right)^j . {}^jC_k$$

$$= \alpha \sum_{j=k}^{\infty} {}^{k+r}C_r \left(\frac{p}{2}\right)^{k+r}$$

$$P(A) = \alpha \left(\frac{p}{2}\right)^k \sum_{r=0}^{\infty} {}^{k+r}C_r \left(\frac{p}{2}\right)^r$$

We have, ${}^{k+r}C_r = (-1)^{r-(k+1)}C_r$

$$\therefore \quad P(A) = \alpha \left(\frac{p}{2}\right)^k \sum_{r=0}^{\infty} {}^{-(k+1)}C_r \left(-\frac{p}{2}\right)^r$$

$$= \alpha \left(\frac{p}{2}\right)^k \left(1-\frac{p}{2}\right)^{-(k+1)} = \alpha \left(\frac{p}{2}\right)^k \cdot \frac{2^{k+1}}{(2-p)^{k+1}}$$

**13. (d)** The probability of showing an even number in a throw

$$= \frac{3}{6} = \frac{1}{2}$$

$\therefore$ Required probability

$$= {}^{2n+1}C_1 . \frac{1}{2} . \left(\frac{1}{2}\right)^{2n} + {}^{2n+1}C_3 . \left(\frac{1}{2}\right)^3 . \left(\frac{1}{2}\right)^{2n-2}$$

$$+ .... + {}^{2n+1}C_{2n+1} \left(\frac{1}{2}\right)^{2n+1}$$

$$= \left({}^{2n+1}C_1 + {}^{2n+1}C_3 + ..... + {}^{2n+1}C_{2n+1}\right)\left(\frac{1}{2}\right)^{2n+1}$$

$$= 2^{2n} . \left(\frac{1}{2}\right)^{2n+1} = \frac{1}{2}$$

**14. (a)** Let G = Event of good book
G' = Event of not a good book
E = Event of publication
Then

$$E = (G \cup G') \cap E = (G \cap E) \cup (G' \cap E)$$

Now, $P(E/G) = \frac{2}{3}$; $P(E/G') = \frac{1}{4}$

$$P(G) = \frac{1}{2} = P(G')$$

Therefore, $P(E) = \frac{1}{2} \times \frac{2}{3} + \frac{1}{2} \times \frac{1}{4} = \frac{11}{24}$

Further, X denotes the number of books published. Then P (at least one book will be published) $= P(X=1) + P(X=2)$

$$= {}^2C_1 \left(\frac{11}{24}\right)\left(\frac{13}{24}\right) + {}^2C_2 \left(\frac{11}{24}\right)^2 \left(\frac{13}{24}\right)^0$$

$$= 2 \times \frac{11}{24} \times \frac{13}{24} + \left(\frac{11}{24}\right)^2 = \frac{407}{576}$$

**15. (b)** Here, $p = \frac{1}{2}, n = 8$

$$\therefore \quad q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}$$

$\therefore$ The binomial distribution is $\left(\frac{1}{2} + \frac{1}{2}\right)^8$

Also, $|x-4| \le 2 \Rightarrow -2 \le x - 4 \le 2 \Rightarrow 2 \le x \le 6$

$\therefore \quad p(|x-4| \le 2) = p(x=2) + p(x=3) + p(x=4) + p(x=5) + p(x=6)$

$$= {}^8C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^6 + {}^8C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^5 + {}^8C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^4$$

$$+ {}^8C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^3 + {}^8C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^2$$

$$= \frac{{}^8C_2 + {}^8C_3 + {}^8C_4 + {}^8C_5 + {}^8C_6}{2^8}$$

$$= \frac{238}{256} = \frac{119}{128}$$

**16. (a)** If both have all 4 cards of the same color, then there are two possibilities at the end.

**Posibility 1:** Ravi holds 4 red cards and Rashmi 4 black cards.

Probability of this possibility = P(Ravi picks red in 1st pick 'AND' Rashmi picks black in 1st pick 'AND' Ravi picks red in 2nd pick' AND 'Rashmi pick black in 2nd pick')

All 4 picks are independent

$$= \frac{2}{4} \times \frac{2}{5} \times \frac{1}{4} \times \frac{1}{5} = \frac{1}{100}$$

**Posibility 2:** Ravi holds 4 black cards and Rashmi 4 red cards similarly probability

$$= \frac{1}{100}$$

Hence, required probability

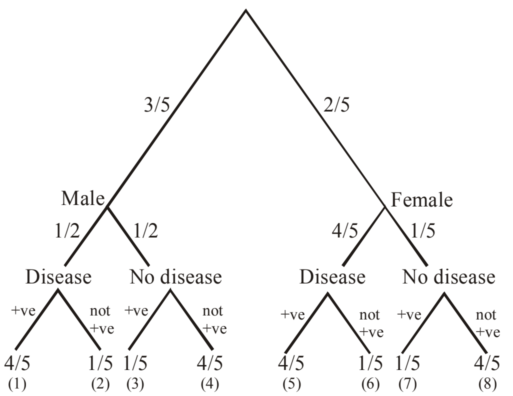
$$P = \frac{1}{100} + \frac{1}{100} = \frac{2}{100}$$

$$= \frac{1}{50} = 0.02 = 2\%$$

**17. (a)** We have 30 males and 20 females.

$$P(\text{males}) = \frac{30}{50} = \frac{3}{5}$$

$$P(\text{females}) = \frac{20}{50} = \frac{2}{5}$$




$$\text{Required probability} = \frac{(1)+(3)}{(1)+(3)+(5)+(7)}$$

$$= \frac{\frac{3}{5}\times\frac{1}{2}\times\frac{4}{5}+\frac{3}{5}\times\frac{1}{2}\times\frac{1}{5}}{\frac{3}{5}\times\frac{1}{2}\times\frac{4}{5}+\frac{3}{5}\times\frac{1}{2}\times\frac{1}{5}+\frac{2}{5}\times\frac{1}{5}\times\frac{4}{5}+\frac{2}{5}\times\frac{4}{5}\times\frac{1}{5}}$$

$$= \frac{\frac{15}{50}}{\frac{15}{50}+\frac{16}{125}} = \frac{75}{107}$$

**18. (0.50)** Let us assume that $A$ wins after $n$ deuces, $n = 0, 1, 2, 3$ ...........

Probability of a deuce $= \frac{2}{3}.\frac{2}{3} + \frac{1}{3}.\frac{1}{3} = \frac{5}{9}$

[$A$ wins his serve then $B$ wins his serve or $A$ loses his serve then $B$ also loses his serve]

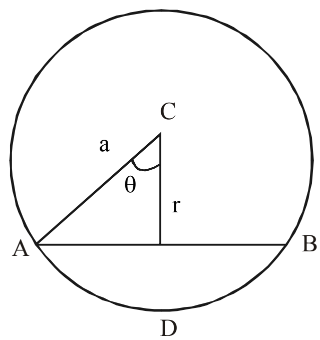
So, probability that '$A$' wins the game after $n$ deuces

$= \left(\frac{5}{9}\right)^n . \frac{2}{3}.\frac{1}{3}$ [After $n$th deuce, $A$ serves and wins then $B$ serves and loses]

$\therefore$ Required probability of '$A$' Winning the game

$$= \sum_{n=0}^{\infty}\left(\frac{5}{9}\right)^n . \frac{2}{3}.\frac{1}{3} = \frac{1}{1-\frac{5}{9}}.\frac{2}{9} = \frac{1}{2} = 0.50$$

**19. (0.25)** Let $l$ be the length of the chord $AB$ of the given circle of radius $a$ and $r$ be the distance of the mid point $D$ of the chord from the centre $C$, then $r = a\cos\theta$ and $l = 2a\sin\theta$. According to given condition :




$$\frac{2}{3}(2a) < 2a\sin\theta < \frac{5}{6}(2a)$$

$$\Rightarrow \frac{2}{3} < \sin\theta < \frac{5}{6} \quad \Rightarrow \frac{\sqrt{11}}{6} < \cos\theta < \frac{\sqrt{5}}{3}$$

$$\Rightarrow \frac{\sqrt{11}}{6}a < r < \frac{\sqrt{5}}{3}a$$

$\therefore$ The given condition is satisfied if the mid point of the chord lies within the region between the concentric circles of radii $\frac{\sqrt{11}}{6}a$ and $\frac{\sqrt{5}}{3}a$.

Hence, the required probability

$$= \frac{\pi\left(\frac{\sqrt{5}}{3}a\right)^2 - \pi\left(\frac{\sqrt{11}}{6}a\right)^2}{\pi a^2} = \frac{1}{4} = 0.25$$

**20.** **(0.60)** Let A denotes the event that the runner succeeds exactly 3 times out of five and B denotes the event that the runner succeeds on the first trial.

$$P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)}$$

But $P(B \cap A)$ = P (clearing succeeding in the first trial and exactly once in two other trials)
$P\,({}^4C_2\, p^2\,(1 - p^2)) = 6p^2\,(1 - p)^2$
and $P(A) = {}^5C_3\, p^3\,(1 - p)^2 = 10p^3\,(1 - p)^2$

Thus, $P\left(\frac{B}{A}\right) = \frac{6p^2(1-p)^2}{10p^3(1-p)^2} = \frac{3}{5} = 0.60$

**21.** **(6)** Let $x$ shell are fixed at point I. Define the following events

$E_1$ : The target is at point I $\Rightarrow P(E_1) = \frac{8}{9}$

$E_2$ : The target is at point II $\Rightarrow P(E_2) = \frac{1}{9}$

$A$ : The target is hit

The target will be hit if at least one shell hits the target.

$P(A/E_1)$ = 1 – None of the shells hit when the target is at point I = $1-\left(\frac{1}{2}\right)^x$ and

$$P(A/E_2) = 1-\left(\frac{1}{2}\right)^{21-x}$$

$$\therefore\quad P(A) = \frac{8}{9}\left[1-\left(\frac{1}{2}\right)^x\right] + \frac{1}{9}\left[1-\left(\frac{1}{2}\right)^{21-x}\right]$$

$$= 1-\frac{1}{9}\left[\left(\frac{1}{2}\right)^{x-3} + \left(\frac{1}{2}\right)^{21-x}\right]$$

For maximum probability; $\frac{dP(A)}{dx} = 0$

$$\Rightarrow -\frac{1}{9}\left[\left(\frac{1}{2}\right)^{21-x} ln2 - \left(\frac{1}{2}\right)^{x-3} ln2\right] = 0$$

$\Rightarrow x = 12$

Also, $\frac{d^2P(A)}{dx^2}$

$$= -\frac{1}{9}\left[\left(\frac{1}{2}\right)^{x-3}(ln2)^2 + \left(\frac{1}{2}\right)^{21-x}(ln2)^2\right] < 0$$

$\therefore$ $P(A)$ is maximum when $x = 12 \Rightarrow k = 6$

**22.** **(2)** Clearly all the solutions of f(x) = x are also solutions of f(f(x)) = x. First, we solve f(x) = x

$f(x) = x \Rightarrow x^2 - 3x + 3 = x$
$\Rightarrow x^2 - 4x + 3 = 0$
$\Rightarrow (x - 1)\,(x - 3) = 0$
$\Rightarrow x = 1, 3$

Therefore x = 1, 3 are also solutions of f(f(x)) = x. We want to seek if there are any more solutions of f(f(x)) = x other than 1 and 3

$f(f(x)) = x \Rightarrow f(x^2 - 3x + 3) = x$
$\Rightarrow (x^2 - 3x + 3)^2 - 3(x^2 - 3x + 3) + 3 = 0$
$\Rightarrow x^4 - 6x^3 + 12x^2 - 9x + 3 = 0$
$\Rightarrow (x^2 - 4x + 3)(x^2 - 2x + 1) = 0$
$\Rightarrow (x - 1)\,(x - 3)\,(x - 1)^2 = 0 \Rightarrow x = 1, 3$

In this case we have no additional solutions. Therefore the probability that x satisfies equation f(f(x)) = x is $\frac{2}{9}$. Therefore m = 2.

**23.** **(0)** Let $E_1$, $E_2$, $E_3$, $E_4$, $E_5$ and $E_6$ be the events of occurrence of 1, 2, 3, 4, 5 and 6 on the dice respectively and let E be the event of getting a sum of numbers equal to 9.

$$\therefore\quad P(E_1) = \frac{1-k}{6};\ P(E_2) = \frac{1+2k}{6};\ P(E_3) = \frac{1-k}{6}$$

$$P(E_4) = \frac{1+k}{6};\ P(E_5) = \frac{1-2k}{6};\ P(E_6) = \frac{1+k}{6}$$

and $\frac{1}{9} \le P(E) \le \frac{2}{9}$

Then, E ≡ {(3, 6), (6, 3), (4, 5), (5, 4)}

Hence, $P(E) = P(E_3E_6) + P(E_6E_3) + P(E_4E_5) + P(E_5E_4)$

$= P(E_3)P(E_6) + P(E_6)P(E_3) + P(E_4)P(E_5) + P(E_5)P(E_4)$

$= 2P(E_3)P(E_6) + 2P(E_4)P(E_5)$

[since $E_1, E_2, E_3, E_4, E_5$ and $E_6$ are independent]

$$= 2\left(\frac{1-k}{6}\right)\left(\frac{1+k}{6}\right) + 2\left(\frac{1+k}{6}\right)\left(\frac{1-2k}{6}\right)$$

$$= \frac{1}{18}[2 - k - 3k^2]$$

Since, $\frac{1}{9} \le P(E) \le \frac{2}{9}$

$\Rightarrow \quad \frac{1}{9} \le \frac{1}{18}\left[2 - k - 3k^2\right] \le \frac{2}{9}$

$\Rightarrow \quad 2 \le 2 - k - 3k^2 \le 4$

$\Rightarrow \quad 2 \le 2 - k - 3k^2$ and $2 - k - 3k^2 \le 4$

$\Rightarrow \quad 3k\left(k + \frac{1}{3}\right) \le 0$ and $3k^2 + k + 2 \ge 0$

$\Rightarrow \quad -\frac{1}{3} \le k \le 0$ and $k \in R$

$\therefore \quad -\frac{1}{3} \le k \le 0$

Hence, integral value of k is 0.

**24. (5)** The sportsman's chance of missing when $r = ja$ is

$$1 - \frac{a^2}{j^2a^2} = 1 - \frac{1}{j^2}\,(j = 2, 3, 4, 5, 6)$$

The animal escapes when the sportsman misses in all the five shots. Therefore the probability of animal escaping to jungle is

$$\prod_{j=2}^{6}\left(1 - \frac{1}{j^2}\right) = \left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{2^3}\right)\left(1 - \frac{1}{2^4}\right)\left(1 - \frac{1}{2^5}\right)\left(1 - \frac{1}{2^6}\right)$$

$$= \left[\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{5}\right)\left(1 - \frac{1}{6}\right)\right]$$

$$\times\left[\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{4}\right)\left(1 + \frac{1}{5}\right)\left(1 + \frac{1}{6}\right)\right]$$

$$= \left(\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \frac{5}{6}\right)\left(\frac{3}{2} \times \frac{4}{3} \times \frac{5}{4} \times \frac{6}{5} \times \frac{7}{6}\right)$$

$$= \frac{1}{6} \times \frac{7}{2} = \frac{7}{12} = \frac{p}{q}$$

Therefore, $q - p = 12 - 7 = 5$

**25. (3)** Let S be the sample space, then $n(S)$ = Total number of determinants that can be made with 0 and 1 $= 2 \times 2 \times 2 \times 2 = 16$

$\because \begin{vmatrix} a & b \\ c & d \end{vmatrix}$, each element can be replaced by two types

i.e., 0 and 1

and let E be the event that the determinant made is non-negative.

Also, E' be the event that the determinant is negative.

$$\therefore \quad E' = \left\{\begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix}, \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix}, \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}\right\}$$

$\therefore \quad P(E') = 3$

then $P(E') = \frac{n(E')}{n(S)} = \frac{3}{16}$

Hence, the required probability,

$P(E) = 1 - P(E') = 1 - \frac{3}{16} = \frac{13}{16} = \frac{m}{n}$ [given]

$\Rightarrow \quad m = 13$ and $n = 16$, then $n - m = 3$

CHAPTER 29

# Properties of Triangle

1. **(c)** Sum of the roots of the equation is given by

$$\sin A+\sin B=\frac{c(a+b)}{c^2}=\frac{a+b}{c}$$

$$=\frac{\sin A+\sin B}{\sin C}\ \sin C=1$$

$\Rightarrow$ the triangle is right angled.

2. **(c)** If $D$ is the diameter of the circumscribed circle of $\Delta ABC$, then $a = D \sin A$, $b = D \sin B$, $c = D \sin C$

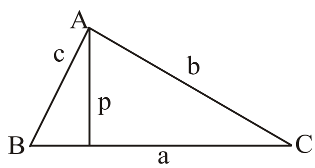
$$\therefore\ \frac{a^3+b^3+c^3}{\sum\sin^3 A}=7\Rightarrow\frac{D^3\left(\sum\sin^3 A\right)}{\sum\sin^3 A}=7$$

$\therefore\ D=\sqrt[3]{7}$.

Since no side of triangle can exceed the diameter of the circle, the maximum possible value of $a$ is $\sqrt[3]{7}$.

3. **(b)** Let the altitudes from $A$, $B$, $C$ be $p$, $q$, $r$ respectively. Then,
$p = b \sin C$, $q = c \sin A$, $r = a \sin B$

$\therefore p : q : r = b \sin C : c \sin A : a \sin B$




$= \sin B \sin C : \sin C \sin A : \sin A \sin B$

$$=\frac{1}{\sin A}:\frac{1}{\sin B}:\frac{1}{\sin C}$$

$\therefore$ $\sin A$, $\sin B$, $\sin C$ are in A.P.

4. **(d)** $\cos A=\dfrac{b^2+c^2-a^2}{2bc}$

$\Rightarrow c^2-(2b\cos A)c+b^2-a^2=0$.

It is a quadratic in c, whose roots are $c_1$ and $c_2$, so $c_1+c_2=2b\cos A$ and $c_1c_2=b^2-a^2$

$\therefore c_1^2+c_2^2-2c_1c_2\cos 2A$

$=(c_1+c_2)^2-2c_1c_2(1+\cos 2A)$

$=(2b\cos A)^2-2(b^2-a^2)(2\cos^2 A)$

$=4b^2\cos^2 A-4b^2\cos^2 A+4a^2\cos^2 A$

$= 4a^2 \cos^2 A$.

5. **(b)** $a$, $b$, $c$ are sides of a triangle

$\therefore\ a+b>c,\ b+c>a,\ c+a>b$

$\therefore\ a>|b-c|,\ b>|c-a|,\ c>|a-b|$ square and add

$a^2+b^2+c^2<2\,(ab+bc+ca)$

$\Rightarrow a^2+b^2+c^2+2\,(ab+bc+ac)<4\,(ab+bc+ca)$

$$\Rightarrow\ \frac{(a+b+c)^2}{ab+bc+ca}<4\Rightarrow P<4$$

Again $(a-b)^2+(b-c)^2+(c-a)^2\ge 0$

$$\Rightarrow\ \frac{(a+b+c)^2}{ab+bc+ca}\ge 3\Rightarrow P\ge 3$$

$\therefore\ 3\le P<4$ or $P\in[3,\,4)$

6. **(a)** We have,

$$\frac{b+c}{2}\ge\sqrt{bc}=\sqrt{\lambda^2}\Rightarrow b+c\ge 2\lambda$$

Now, $\dfrac{a}{\sin A}=\dfrac{b}{\sin B}=\dfrac{c}{\sin C}=\dfrac{b+c}{\sin B+\sin C}$

$$\Rightarrow\ \frac{a}{2\sin\frac{A}{2}\cos\frac{A}{2}}=\frac{b+c}{2\sin\frac{B+C}{2}\cos\frac{B-C}{2}}$$

$$\Rightarrow a=\frac{(b+c)\sin\frac{A}{2}}{\cos\left(\frac{B-C}{2}\right)}$$

$\because\ 0\le\cos\dfrac{B-C}{2}\le 1$ and

$b+c\ge 2\lambda\Rightarrow a\ge 2\lambda\sin\dfrac{A}{2}$.

**7.** **(c)** L.H.S. $= \frac{1}{2}(a^2(b+c-a)+b^2(c+a-b)+c^2(a+b-c)$

$= \frac{1}{2}(a(b^2+c^2-a^2)+b(c^2+a^2-b^2)+c(a^2+b^2-c^2))$

$= \frac{1}{2}(2abc\cos A + 2abc\cos B + 2abc\cos C)$

$= abc\left(1+4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)$

$= 4R\Delta\left(1+4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)$.

**8.** **(b)** $a\tan\theta + b\sec\theta = c$

$\Rightarrow b^2\sec^2\theta = (c-a\tan\theta)^2$

$\Rightarrow b^2(1+\tan^2\theta) = c^2 - 2ca\tan\theta + a^2\tan^2\theta$

$\Rightarrow (a^2-b^2)\tan^2\theta - 2ca\tan\theta + c^2 - b^2 = 0$ ...(i)

Roots of equation (i) are $\tan\alpha$ and $\tan\beta$, where $\alpha$ and $\beta$ are the two angles of the triangle.

We have $\tan\alpha + \tan\beta = \frac{2ca}{a^2-b^2}$ and

$\tan\alpha.\tan\beta = \frac{c^2-b^2}{a^2-b^2}$

$\therefore \tan(\alpha+\beta) = \frac{\frac{2ca}{a^2-b^2}}{1-\frac{c^2-b^2}{a^2-b^2}} = \frac{2ca}{a^2-c^2}$

$\therefore \tan\left(\pi - \frac{3\pi}{4}\right) = \frac{2ca}{a^2-c^2} \Rightarrow a^2-c^2 = 2ca$

**9.** **(b)** We know that $A+B+C = 180°$

$\Rightarrow \quad A+C-B = 180° - 2B$

Now $2ac\sin\left[\frac{1}{2}(A-B+C)\right]$

$= 2ac\sin(90° - B)$

$= 2ac\cos$

$B = \frac{2ac\,(a^2+c^2-b^2)}{2ac} = a^2+c^2-b^2$

**10.** **(c)** Let $a = \sin\alpha, b = \cos\alpha$ and

$c = \sqrt{1+\sin\alpha\cos\alpha}$

Clearly $a$ and $b < 1$ but $c > 1$ as $\sin\alpha > 0$ and $\cos\alpha > 0$

$\therefore c$ is the greatest side and greatest angle is $C$

$\therefore \cos C = \frac{a^2+b^2-c^2}{2ab}$

$= \frac{\sin^2\alpha + \cos^2\alpha - 1 - \sin\alpha\cos\alpha}{2\sin\alpha\cos\alpha} = -\frac{1}{2}$

$\therefore C = 120°$

**11.** **(d)** $\sin^2 A + \sin^2 C = 1001\sin^2 B$

$\Rightarrow a^2 + c^2 = 1001\,b^2$ (using sine rule)

Now, $\frac{2(\tan A + \tan C)\cdot\tan^2 B}{\tan A + \tan B + \tan C}$

$= \frac{2(\tan A + \tan C)\cdot\tan^2 B}{\tan A\cdot\tan B\cdot\tan C}$

$= 2\left(\frac{\cot A + \cot C}{\cot B}\right)$

$= \frac{2(\cos A\sin C + \sin A\cos C)}{\sin A\cdot\sin C\cdot\cos B}\sin B$

$= \frac{2\sin(\pi - B)\cdot\sin B}{\sin A\sin C\cos B} = \frac{2\sin^2 B}{\sin A\sin C\cos B}$

$= \frac{2\times 2b^2}{2ac\cdot\cos B} = \frac{2\times 2b^2}{a^2+c^2-b^2} = \frac{2\times 2b^2}{1000b^2} = \frac{1}{250}$

**12.** **(b)** $\frac{\sin A}{c\sin B} + \frac{\sin B}{c} + \frac{\sin C}{b} = \frac{c}{ab} + \frac{b}{ac} + \frac{a}{bc}$

or $\frac{a}{bc} + \frac{\sin B}{c} + \frac{\sin C}{b} = \frac{c}{ab} + \frac{b}{ac} + \frac{a}{bc}$

or $\frac{\sin B}{c} + \frac{\sin C}{b} = \frac{c}{ab} + \frac{b}{ac}$

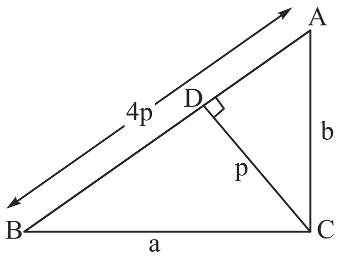
or $\frac{b\sin B + c\sin C}{bc} = \frac{c^2+b^2}{abc}$

or $a = \frac{b^2+c^2}{b\sin B + c\sin C}$

$= \frac{b(2R\sin B) + c(2R\sin C)}{b\sin B + c\sin C} = 2R$

$\Rightarrow \angle A = \frac{\pi}{2}$

**13. (a)**




$$\Delta = \frac{1}{2}ab = \frac{1}{2}p(4p) \Rightarrow ab = 4p^2$$

Also, $a^2 + b^2 = c^2 = 16p^2$

$\therefore \quad (a-b)^2 = a^2 + b^2 - 2ab = 8p^2$

Also, $(a+b)^2 = a^2 + b^2 + 2ab = 24p^2$

$$\tan\frac{A-B}{2} = \frac{a-b}{a+b}\cot\frac{C}{2} = \frac{1}{\sqrt{3}} \times 1$$

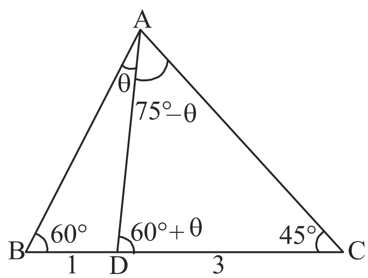
$$\text{or} \quad \frac{A-B}{2} = 30° \quad \text{or} \quad A - B = 60°$$

**14. (a)** In $\Delta BAD$, $\dfrac{BD}{\sin\theta} = \dfrac{AD}{\sin 60°}$

In $\Delta CAD$, $\dfrac{CD}{\sin(75° - \theta)} = \dfrac{AD}{\sin 45°}$

$$\Rightarrow \frac{BD}{\sin\theta}\sin 60° = \frac{CD\sin 45°}{\sin(75° - \theta)}$$

$$\Rightarrow \frac{\sin\theta}{\sin(75° - \theta)} = \frac{BD}{CD}\frac{\sin 60°}{\sin 45°} = \frac{1}{\sqrt{6}}$$




**15. (c)** $$\cos(A-B) = \frac{1 - \tan^2\left(\frac{A-B}{2}\right)}{1 + \tan^2\left(\frac{A-B}{2}\right)} = \frac{31}{32}$$

$$\Rightarrow \tan\left(\frac{A-B}{2}\right) = \frac{1}{3\sqrt{7}}$$

$$\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b}\cot\frac{C}{2} \Rightarrow \cos C = \frac{1}{8}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{1}{8} \Rightarrow c = 6$$

**16. (a)** $\angle A = \dfrac{\pi}{7}, \angle B = \dfrac{2\pi}{7}, \angle C = \dfrac{4\pi}{7}$

$(a^2 - b^2)(b^2 - c^2)(c^2 - a^2)$

$$= a^2b^2c^2\left(1 - \frac{b^2}{a^2}\right)\left(1 - \frac{c^2}{b^2}\right)\left(1 - \frac{a^2}{c^2}\right)$$

$= a^2b^2c^2$

$$\left(1 - \frac{\sin^2\frac{2\pi}{7}}{\sin^2\frac{\pi}{7}}\right)\left(1 - \frac{\sin^2\frac{4\pi}{7}}{\sin^2\frac{2\pi}{7}}\right)\left(1 - \frac{\sin^2\frac{\pi}{7}}{\sin^2\frac{4\pi}{7}}\right)$$

$= a^2b^2c^2$

**17. (a)** Since $4\sin A\cos B = 1$, so $A$ and $B$ can not be $\dfrac{\pi}{2}$

[As if $B = \dfrac{\pi}{2}$, then $\cos B = 0$ and if $A = \dfrac{\pi}{2}$, then $\tan A$ is not defined]

so, $C = \dfrac{\pi}{2} \Rightarrow B = \dfrac{\pi}{2} - A$

$$\Rightarrow 4\sin A\cos\left(\frac{\pi}{2} - A\right) = 1$$

$$\Rightarrow \sin^2 A = \frac{1}{4} \Rightarrow \sin A = \frac{1}{2} \Rightarrow A = \frac{\pi}{6} \Rightarrow B = \frac{\pi}{3}$$

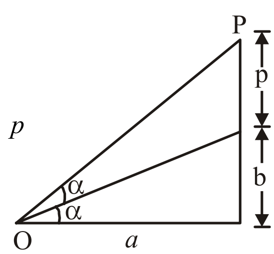
so angles are $\dfrac{\pi}{6}, \dfrac{\pi}{3}, \dfrac{\pi}{2}$ which are in A.P.

**18. (b)** $\tan\alpha = \dfrac{b}{a}, \tan 2\alpha = \dfrac{2(b/a)}{1-(b/a)^2} = \dfrac{p+b}{a}$

$$\Rightarrow \frac{2ba}{a^2 - b^2} = \frac{p+b}{a}$$

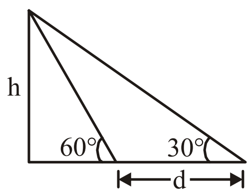
$$\Rightarrow \frac{2ba^2 - a^2b + b^3}{a^2 - b^2} = p$$

$$\Rightarrow p = \frac{b(a^2 + b^2)}{(a^2 - b^2)}$$



**19.** **(c)** $d = h\cot 30° - h\cot 60°$ and time = 3 min.

$$\therefore \text{Speed} = \frac{h(\cot 30° - \cot 60°)}{3} \text{ per minute}$$




It will travel distance $h\cot 60°$ in

$$\frac{h\cot 60° \times 3}{h(\cot 30° - \cot 60°)} = 1.5 \text{ minute}$$

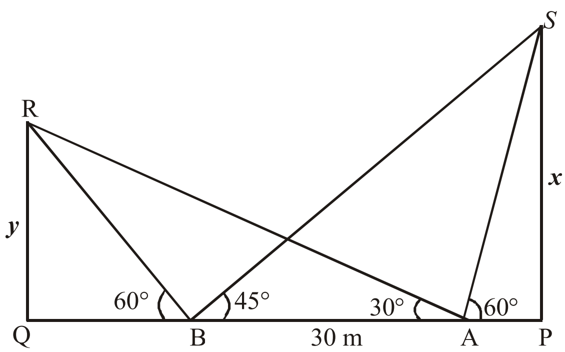
**20.** **(d)** Let $x$ and $y$ be the heights of the flagstaffs at $P$ and $Q$ respectively
Then,

$$AP = x\cot 60° = \frac{x}{\sqrt{3}},\ AQ = y\cot 30° = y\sqrt{3}$$

$$BP = x\cot 45° = x,\ BQ = y\cot 60° = \frac{y}{\sqrt{3}}$$

$$\Rightarrow AB = BP - AP = x - \frac{x}{\sqrt{3}} \qquad [\because AB = 30\ m]$$

$$\Rightarrow 30\sqrt{3} = (\sqrt{3}-1)x \Rightarrow x = 15(3+\sqrt{3})$$

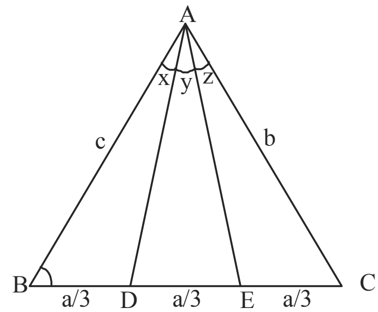



Similarly, $30 = y\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) \Rightarrow y = 15\sqrt{3}$

so that $PQ = BP + BQ = x + \frac{y}{\sqrt{3}}$

$$= 15(3+\sqrt{3}) + 15 = (60 + 15\sqrt{3})\text{m}$$

**21.** **(4)**




$$\frac{a}{3\sin x} = \frac{AD}{\sin B} \qquad ...(1)$$

$$\frac{2a}{3\sin(x+y)} = \frac{AE}{\sin B}$$

Dividing Eq. (1) by Eq. (2), we get ...(2)

$$\frac{\sin(x+y)}{2\sin x} = \frac{AD}{AE} \qquad ...(3)$$

$$\frac{2a}{3\sin(y+z)} = \frac{AD}{\sin C} \qquad ...(4)$$

$$\frac{a}{3\sin z} = \frac{AE}{\sin C}$$

Dividing Eq. (5) by Eq. (4), we get ...(5)

$$\frac{\sin(y+z)}{2\sin z} = \frac{AE}{AD}$$

Multiplying Eq. (3) and Eq. (6), we get ...(6)

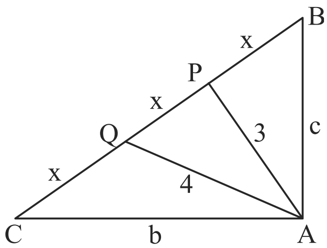
$$\therefore \quad \frac{\sin(x+y)\sin(y+z)}{\sin x \cdot \sin z} = 4$$

**22.** **(3)** The given equation is
$4\sin A\sin B + 4\sin B\sin C + 4\sin C\sin A = 9$
$\Rightarrow 2\cos(A-B) - 2\cos(A+B) + 2\cos(B-C)$
$-2\cos(B+C) + 2\cos(C-A) - 2\cos(C+A) = 9$
or $\quad 2[\cos(A-B) + \cos(B-C) + \cos(C-A)]$

$$= 9 - 2(\cos A + \cos B + \cos C) \geq 9 - 2 \times \frac{3}{2} = 6$$

or $\cos(A-B) + \cos(B-C) + \cos(C-A) \geq 3$
But $\cos(A-B) \leq 1, \cos(B-C) \leq 1,$
$\cos(C-A) \leq 1$
or $\quad \cos(A-B) = 1, \cos(B-C) = 1,$
$\cos(C-A) = 1$
Thus, $A = B = C$, i.e., triangle ABC is an equilateral triangle.
Hence, $\Delta = \sqrt{3}$.

**23. (2)**




Let BP = PQ = QC = x

In $\Delta$ABP, using cosine rule

$$9 = c^2 + x^2 - 2cx\cos B$$

But $\cos B = \dfrac{c}{3x}$

$$\Rightarrow \quad 9 = x^2 + \frac{c^2}{3} \quad ...(1)$$

Similarly using cosine rule is $\Delta$ACQ, we get

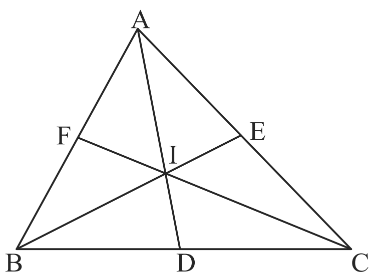
$$16 = x^2 + \frac{b^2}{3} \quad ...(2)$$

Adding (1) and (2), we get $25 = 2x^2 + \dfrac{b^2 + c^2}{3}$

$\therefore \quad 25 = 2x^2 + \dfrac{(3x)^2}{3} \quad \therefore 25 = 2x^2 + 3x^2$

$\therefore x^2 = 5 \ \therefore BC = 3x = 3\sqrt{5}$

**24. (2)**




$$\frac{ID}{AD} = \frac{ID}{\dfrac{2bc}{b+c}\cos\dfrac{A}{2}} = \frac{a(b+c)\sec\dfrac{A}{2}\cdot ID}{2abc}$$

Now $\dfrac{AI}{ID} = \dfrac{AB}{BD} = \dfrac{c}{\dfrac{ac}{b+c}} = \dfrac{b+c}{a}$

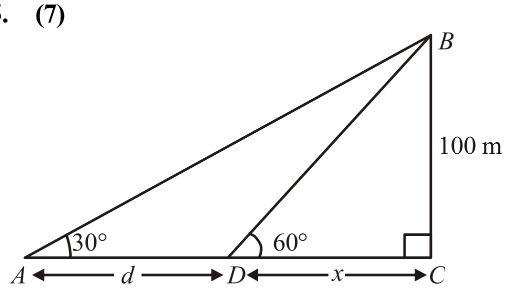
$$\therefore \quad \frac{ID}{AD} = \frac{ID}{AI + ID} = \frac{a}{a+b+c}$$

$$\therefore \quad \frac{ID}{AD} + \frac{IE}{BE} + \frac{IF}{CF} = \frac{a+b+c}{a+b+c} = 1$$

$$\therefore \quad a(b+c)\sec\frac{A}{2}\ ID + b(a+c)\sec\frac{B}{2}\ IE + c(a+b)\sec\frac{C}{2}\ IF = 2abc$$

$\therefore \quad k = 2$

**25. (7)**




In $\Delta CBD$, $\tan 60° = \dfrac{100}{x} \quad \Rightarrow x = \dfrac{100}{\sqrt{3}}$

In $\Delta ACB$, $\tan 30° = \dfrac{100}{x+d}$

$x + d = 100\sqrt{3}$.

$$\Rightarrow \quad d = 100\sqrt{3} - \frac{100}{\sqrt{3}} = \frac{200}{\sqrt{3}} = \frac{200\sqrt{3}}{3} m$$